6.0 ENGINEERED SAFETY FEATURES

Engineered safety features (ESF) systems protect the public in the event of an accidental release of radioactive fission products from the reactor coolant system (RCS), particularly as the result of a loss-of-coolant accident (LOCA). The safety features function to localize, control, mitigate, and terminate such accidents and to maintain radiation exposure levels to the public below applicable limits and guidelines (e.g., 10 CFR 100). The following systems are defined as ESF systems:

A. Containment Building (subsection 6.2.1)

   The containment building is a steel-lined, reinforced, prestressed concrete cylinder with a hemispherical dome and flat circular basemat. The building houses the nuclear steam supply system (NSSS) and is designed to minimize radioactive fission product release from the NSSS to the environs subsequent to postulated design basis accidents (DBAs).

B. Containment Spray System-Iodine Removal System (subsections 6.2.2 and 6.5.2)

   The containment spray system provides borated water spray for post-accident containment heat removal and pressure reduction. The iodine removal system provides chemical additives to the containment spray and/or recirculation system for post-accident iodine removal from the containment atmosphere.

C. Containment Fan Cooler System (subsection 6.2.2)

   The containment fan cooler system consists of redundant fan cooler units provided for post-accident containment atmosphere heat removal and pressure reduction.

D. Containment Isolation System (subsection 6.2.4)

   The containment isolation system provides for automatic containment isolation upon receipt of a containment isolation actuation signal. This system precludes the release of the containment atmosphere to the plant and the surroundings.

E. Combustible Gas Control System (subsection 6.2.5)

   The combustible gas control system consists of two hydrogen recombiner subsystems and two hydrogen monitoring subsystems. A post-LOCA cavity purge system is designed to prevent hydrogen pocketing in the reactor cavity. A manually initiated, non-ESF hydrogen purge subsystem is provided as a backup to the ESF subsystems. The system functions to maintain post-LOCA hydrogen concentrations below the combustible limit.

F. The Emergency Core Cooling System (ECCS) (section 6.3)

   The ECCS described in section 6.3 injects borated water into the RCS. This provides post-accident cooling of the core to limit core damage and fission product release and ensures adequate shutdown margin. The system also provides continuous long-term, post-accident cooling of the core by recirculation of borated water from the containment sump through the residual heat exchanger and back to the reactor core.

G. Habitability Systems (section 6.4 and subsection 9.4.1)

The control room heating, ventilation, and air-conditioning (HVAC) system is provided to protect control room personnel from post-accident airborne radioactivity.

H. ESF Filter Systems (subsection 6.5.1)

The fuel handling building post-accident exhaust system and the piping penetration filter exhaust system control fission product release resulting from postulated accidents. The control room HVAC system reduces radiation exposures to operating personnel in the control room.

I. Auxiliary Feedwater System (subsection 10.4.9)

The auxiliary feedwater system is provided to automatically supply feedwater to the steam generators for heat removal from the RCS during emergency conditions.

6.1 ENGINEERED SAFETY FEATURES MATERIALS

This section provides a discussion of the materials used in the fabrication of ESF components and of the material interactions that could potentially impair the operation of the ESF.

6.1.1 METALLIC MATERIALS

6.1.1.1 Materials Selection and Fabrication

Information on the selection and fabrication of the materials in the ESF of the plant, such as the ECCS, the containment heat removal systems, the combustible gas control system, and the containment spray systems is provided below. Materials for use in ESF are selected for their compatibility with the reactor coolant system (RCS) and containment spray solutions as described in Section III of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code, Articles NC-2160 and NC-3120.

6.1.1.1.1 Specifications for Principal Pressure-Retaining Materials

All pressure-retaining materials in ESF system components comply with the corresponding material specification permitted by ASME Section III, Division 1. The material specifications for pressure-retaining materials in each component of an ESF system meet the requirements of Article NC-2000 of ASME Section III, Class 2, for Quality Group B and Article ND-2000 of ASME Section III, Class 3, for Quality Group C components. Materials produced under American Society of Testing Materials (ASTM) designation are acceptable as complying with the corresponding ASME specification, provided the ASME specification is designated as being identical with the ASTM specification for the grade, class, or type produced and that the material is confirmed as complying with the ASME specification by a certified material test report or certification from the material manufacturer (Subarticle NA/NCA-1220). Containment penetration materials meet the requirements of Articles NC-2000 and NE-2000 of ASME Section III, Division 1. The quality groups assigned to each component are given in table

3.2.2-1. Principal pressure-retaining materials are indicated in table 6.1.1-1. Material specifications for equipment within the NSSS scope are provided in table 5.2.3-1.

6.1.1.1.2 Engineered Safety Features Construction Materials

The welding materials used for joining the ferritic base materials of the ESF conform to or are equivalent to ASME Material Specifications SFA 5.1, 5.2, 5.5, 5.17, 5.18, and 5.20. The welding materials used for joining nickel-chromium-iron alloy in similar base material combination and in dissimilar ferritic or austenitic base material combination conform to ASME Material Specifications SFA 5.11 and 5.14.

The welding materials used for joining the austenitic stainless steel base materials conform to ASME Material Specifications SFA 5.4 and 5.9. These materials are qualified to the requirements of the ASME Code, Section III and Section IX, and are used in procedures which have been qualified to these same rules. The methods utilized to control delta ferrite content in austenitic stainless steel weldments are discussed in section 1.9 and subsection 5.2.3.

Components in contact with borated water are fabricated of or clad with austenitic stainless steel or equivalent corrosion-resistant material. The integrity of the safety-related components of the ESF is maintained during all stages of component manufacture. Austenitic stainless steel is utilized in the final heat-treated condition as required by the respective ASME Code, Section II, material specification for the particular type or grade of alloy. Furthermore, austenitic stainless steel materials used in the ESF components are handled, protected, stored, and cleaned according to recognized and accepted methods which are designed to minimize contamination which could lead to stress corrosion cracking. These controls are stipulated in specifications which are discussed in subsection 5.2.3. Additional information concerning austenitic stainless steel, including the avoidance of sensitization and the prevention of intergranular attack, can be found in subsection 5.2.3. No cold-worked austenitic stainless steels having yield strengths greater than 90,000 psi are used for components of the ESF.

Materials utilized in ESF components within the containment that would be exposed to core cooling water and containment sprays in the event of a LOCA are listed in table 6.1.1-2. These components are manufactured primarily of stainless steel or other corrosion-resistant material. Protective coatings are applied on carbon steel equipment located inside the containment. (See subsection 6.1.2.)

To limit the generation of hydrogen within the containment, restrictions are placed on the use of aluminum, zinc, and mercury in the containment:

A. Aluminum is severely attacked by the alkaline containment spray solution, which results in the generation of gaseous hydrogen and the possible loss of structural integrity. The amount of aluminum present inside the containment is restricted to an essential minimum.

B. Boric acid spray reacts with zinc, oxidizing it and liberating hydrogen gas. The use of zinc in the containment is minimized to reduce generation of hydrogen.

   Table 6.2.5-6 contains a list of the amounts of aluminum and zinc which are expected to be present in the containment and which could potentially be exposed to a corrosive environment. These materials are listed by the system or component in which they are used, and an estimate of their expected corrosion rate is given. The use of mercury and mercuric compounds is prohibited inside the containment. Temporary use of fluorescent and high-pressure sodium lamps

is permitted during refueling outages/plant shutdowns during Modes 5 and 6 only. Usage during these times is administratively controlled.

6.1.1.1.3 Integrity of Safety-Related Components

The integrity of the materials of construction for ESF equipment when exposed to post-DBA conditions have been evaluated. Post-DBA conditions were conservatively represented by test conditions. The test program[1] considered spray and core cooling solutions of the design chemical compositions, as well as the design chemical compositions contaminated with corrosion and deterioration products which may be transferred to the solution during recirculation. The effects of sodium (free caustic), chlorine (chloride), and fluorine (fluoride) on austenitic stainless steels were considered. Based on the results of this investigation, as well as testing by Oak Ridge National Laboratory and others, the behavior of austenitic stainless steels in the post-DBA environment is acceptable. No cracking is anticipated on any equipment even in the presence of postulated levels of contaminants, provided the core cooling and spray solution Ph is maintained at an adequate level. The inhibitive properties of alkalinity (hydroxyl ion) against chloride cracking and the inhibitive characteristic of boric acid on fluoride cracking have been demonstrated.

The selection, procurement, testing, storage, and installation of all nonmetallic thermal insulation ensures that the leachable concentrations of chloride, fluoride, sodium, and silicate are in conformance with Regulatory Guide 1.36, Nonmetallic Thermal Insulation for Austenitic Stainless Steel.

Conformance with Regulatory Guide 1.36 is summarized in section 1.9.

Information is provided in section 1.9 concerning the degree of conformance with the following Regulatory Guides:

A. 1.31, Control of Ferrite Content in Stainless Steel Weld Metal.

B. 1.36, Nonmetallic Thermal Insulation for Austenitic Stainless Steel.

C. 1.37, Quality Assurance Requirements for Cleaning of Fluid Systems and Associated Components of Water-Cooled Nuclear Power Plants.

D. 1.44, Control of the Use of Sensitized Stainless Steel.

6.1.1.2 Composition, Compatibility, and Stability of Containment and Core Spray Coolants

The information given below is provided on the composition, compatibility, and stability of the core cooling water and the containment sprays of the ESF.

Supply for the containment sprays and ECCS is drawn from the refueling water storage tank. As described in sections 3.8 and 6.3, the refueling water storage tank is a stainless steel-lined concrete tank not subject to significant corrosive attack by the tank's contents. Trisodium phosphate for recirculation fluid pH adjustment is stored in baskets located in the containment.

The accumulator tanks, which store boric acid solution (1900-2600 ppm) for the accumulator portion of the safety injection system, are made of carbon steel and clad with stainless steel to ensure corrosion resistance.

The boron injection tank (Unit 1 only) is stainless steel. Because of the corrosion resistance of this material, significant corrosive attack on the vessel is not expected. The boron injection tank contains a boron concentration, as boric acid, of 0-2600 ppm.

6.1.1.3 Reference

1. Picone, L. F., and Whyte, D. D., "Behavior of Austenitic Stainless Steel in Post Hypothetical Loss-of-Coolant Environment," WCAP-7798-L (Proprietary), November 1971, and WCAP-7803 (Nonproprietary), December 1971.

6.1.2 ORGANIC MATERIALS

6.1.2.1 Protective Coatings

Certain coatings, which are in common industrial use, may deteriorate in the post-accident environment and may contribute substantial quantities of foreign solids and residue to the containment sump. Consequently, protective coatings used inside the containment, excluding components limited by size and/or exposed surface area, are demonstrated to withstand the design basis accident (DBA) conditions (subsection 3.11.B.1) and meet the intent of American National Standards Institute (ANSI) N101.2 (l972), Protective Coatings (Paints) for Light Water Nuclear Reactor Containment Facilities, as well as the recommendations of Regulatory Guide 1.54, Quality Assurance Requirements for Protective Coatings Applied to Water-Cooled Nuclear Power Plants. Information regarding conformance with Regulatory Guide 1.54 is provided in table 6.1.2-1 and further conformance information for nuclear steam supply system (NSSS) equipment has been submitted to the Nuclear Regulatory Commission (NRC) for review via reference 1 and accepted via reference 2.

A. Regulatory Guide 1.54 is imposed for items located within the containment building as follows:

   1. For shop priming of liner plate, structural steel, and fabricated shapes.
   2. For shop priming of fabricated pipes, tanks, heating, ventilation, and air-conditioning (HVAC) ducts, and equipment.
   3. For field finish painting of steel where called for in drawings and specifications.
   4. For surfacing of concrete where indicated in drawings and specifications.

B. Regulatory Guide 1.54 is implemented by requirements as follows:

   1. Use of specific coatings systems which are prequalified to ANSI N101.2.
   2. Surface preparation standards.
   3. Surface profile requirements.
   4. Application of the coating systems in accordance with instructions approved by the paint manufacturer.
   5. Inspections and nondestructive examinations.

6. Identification of all nonconformances. Coatings which do not conform with the VEGP position on Regulatory Guide 1.54 are limited in use and are evaluated on a case basis relative to impact on plant safety.

7. Certifications of compliance and/or documentation procedures to satisfy project requirements.

8. The vendor's procedures are subject to review prior to application, and the vendor's implementation of the specification requirements is monitored.

9. An inventory of unqualified coatings is maintained to ensure appropriate control of coatings inside containment.

C. Regulatory Guide 1.54 is not imposed for the following:

1. Surfaces to be insulated.

2. Surfaces "contained" within a cabinet or enclosure; for example, the interior surfaces of ducts.

3. Field repair to any small areas previously coated with a qualified coating system such as:

   a. Bolt heads, nuts, and miscellaneous fasteners.

   b. Damage resulting from spot, tack, or stud welding.

Field touchup and repair of large areas shall be in accordance with Regulatory Guide 1.54.

4. Small "production line" items such as small motors, handwheels, pipe supports, snubbers, electrical cabinets, control panels, loudspeakers, etc., where special painting requirements would be impracticable.

5. Stainless steel or galvanized surfaces.

6. Coating used for the banding of piping.

7. Concrete designated to receive a sealer coat only.

D. The majority of the coatings specified for use inside the containment are the inorganic type (ethyl silicate inorganic zinc). The mode of failure of inorganic zinc is powdering rather than blistering and delamination. This failure mode minimizes the accumulation of solid debris in the containment sumps. Any particles of appreciable size that do occur either settle out prior to reaching the sump screens or are trapped by the sump filter screens. The screen opening size (3/32 in.) for the Emergency Containment Cooling System (ECCS) Residual Heat Removal (RHR) Systems is smaller than the line piping, the RHR heat exchanger tubes, the branch line needle valves (used for throttling), pump running clearances, and clearances in the reactor core so particles that could potentially block the system are filtered out. (Refer to section 6.2 for a discussion of the sump design and consideration given to screen clogging.)

The screen opening size is 3/32-in. diameter by design for the Containment Spray (CS) System and the 3/32-in. diameter is smaller than the line piping, the spray nozzles, and pump running clearances, so particles that could potentially block the system are filtered out. (Refer to section 6.2 for a discussion of the sump design and consideration given to screen clogging.) After the new screens were installed, it was discovered that 124 holes (0.002% of total screen surface

area) in the plates had a hole > 3/32-in. diameter. No round holes were found to be > ¼-in. diameter in the screens and all hole areas had < ¼-in. length in the major axis through the minor axis. The CS nozzles have a 3/8-in. diameter hole; therefore, blockage in the system will not occur.

E. "N" AREAS

Areas within the containment are identified as "N" areas. Coatings used inside the containment, where required, are the prequalified coating systems. These coatings are prequalified to the intent of ANSI N101.2 and applicable portions of ANSI N5.12. Quality assurance and documentation requirements of ASME NQA-1-1994, as described in the SNC Quality Assurance Topical Report (QATR) and ANSI N101.4 (Class I) are enforced for both coating materials and applications procedures as discussed in table 6.1.2-1.

F. "D" AREAS

Areas outside the containments, but with potential contamination from radioactive sources, are identified as "D" areas. Coating materials specified for "D" areas are either the same or similar materials which are used in "N" areas, except the quality assurance program and the documentation requirements of ANSI do not apply (Class II).

G. "C" AREAS

Areas outside the containments, and not subject to potential contamination from radioactive sources are identified as "C" areas. "C" area coatings are standard commercial coatings formulated to withstand exposure to industrial environment and require minimum maintenance (Class II).

A coating schedule for items inside the containment is given in tables 6.1.2-2 and 6.1.2-3. Approximate paint film thickness and exposed surface area for major components and structures inside the containment are also provided. The painted areas of valve operators, miscellaneous parts on the reactor coolant pump drives, and instrumentation are considered insignificant. Exposed concrete in the containment is coated as indicated in table 6.1.2-2. The containment temperature profile used for the specification of coatings used inside containment is shown in figure 6.1.2-1.

Protective coatings for use on NSSS components in the reactor containment have been evaluated as to their suitability in post-DBA conditions. Tests have shown that the inorganic zinc, epoxy, and modified phenolic systems are the most desirable of the generic types evaluated for use inside containment. This evaluation[3] considers resistance to high temperature and chemical conditions anticipated during a loss-of-coolant accident, as well as high radiation resistance.

6.1.2.2 Other Organic Materials

A listing of other organic materials in the containment is included in table 6.1.2-4. The materials listed are not protective coatings applied to surfaces of nuclear facilities.

6.1.2.3 References

1. Letter NS-CE-1352, C. Eicheldinger (Westinghouse) to C. J. Heltemes, Jr. (NRC), dated February 1, 1977.

2. Letter, C. J. Heltemes, Jr. (NRC) to C. Eicheldinger (Westinghouse), dated April 27, 1977.

3. Picone, L. F., "Evaluation of Protective Coatings for use in Reactor Containment," WCAP-7198-L (Proprietary), April 1968, and WCAP-7825 (Nonproprietary), December 1971.

6.1.3 POST-ACCIDENT CHEMISTRY

Following a main steam line break or design basis loss-of-coolant accident, trisodium phosphate and boric acid solutions will be present in the containment sumps. Subsection 6.5.2 indicates the quantities of trisodium phosphate and boric acid that will be present in the containment after an accident. The pH control reduces the probability of chloride stress corrosion cracking of stainless steel and maximizes iodine retention in the sump solution.

TABLE 6.1.1-1 (SHEET 1 OF 2)

PRINCIPAL ESF PRESSURE-RETAINING MATERIALS

Component	Material
Piping/tubing	SA-53 Gr. B SA-106 Gr. B and C SA-155 Gr. 70 Class 1 and Gr. KC70 Class 1 SA-213, TP 304, 304L and 316 SA-249, TP 304L SA-312, TP 304 and 304L SA-333 Gr. 1 and 6 SA-335 Gr. P11 and P22 SA-376, TP 304 and 316 SB-111 Gr. CDA 706 SB-466 Gr. CDA 706
Fittings/flanges	SA-105 N SA-181 Gr. I and II SA-182, TP F304, F304L, F316, F316L SA-234 Gr. WPB, WPC, WPBW, and WPCW SA-403 WP 304, 304L, 304W, and 304LW SA-420 Gr. WPL6 SA-479, TP 304, 304L and 316
Plate	SA-240, TP 304, 304L and 316L SA-283 Gr. C SA-285 Gr. A and C SA-515 Gr. 70 SA-516 Gr. 70 SA-537 Class 1 SB-171 Gr. CDA 706
Bolting/nuts/studs	SA-193 Gr. B6, B7, B8, and B8M SA-194, Gr 2H, 4, 6, 7, 8H, 8M, and B8 SA-307 Gr. B SA-320 Gr. L7 SA-453 Gr. 660A and 660B SA-564 Gr. 630

TABLE 6.1.1-1 (SHEET 2 OF 2)

Component	Material
Castings	SA-216 Gr. WCB and WCC SA-217 Gr. WC9 SA-351 Gr. CF8M and CF3M SA-487 Gr. CA6NM SB-61 SB-62 Gr. CDA 836 SB-148 Gr. CA 952 ASTM-A276 TP 410
Forgings	SA-105 SA-182, TP F304, F304L, F316, and F316L; Gr. F11 and F22 SA-240 TP 304 and 316 SA-350 Gr. LF1 and LF2 SA-479, TP 304, 304L and 316
Bars	SA-479, TP 304, 316 and 410 Gr 316L and F316 SA-564 Gr. 630
Weld rod	SFA 5.1, E 6010 and E 7018 SFA 5.4, E 308-16, E 308L-16 and E 309 SFA 5.9, ER 308, ER 308L, and ER 309 SFA 5.17, EM 12K SFA 5.18, E 70S-2, E 70S-3, E 70S-4, E70S-6, and E70S-1B SFA 5.20, E 70T-1 and 70T-5

TABLE 6.1.1-2 (SHEET 1 OF 2)

PRINCIPAL ESF MATERIALS EXPOSED TO REACTOR COOLANT OR CONTAINMENT SPRAY

Component	Material
Piping/tubing	SA-106 Gr. B and C SA-155 Gr. KC70 Class 1 and 70 Class 1 SA-213, TP 304, 304L, and 316 SA-249, TP 304L SA-312, TP 304 and 304L SA-333 Gr. 1 and 6 SA-376, TP 304 and 316 SB-111 Gr. CDA 706 SB-466 Gr. CDA 706
Fittings/flanges	SA-105 N SA-181 Gr. I and II SA-182, TP F304, F304L, F316, and F316L SA-234 Gr. WPB, WPBW, WPCW, and WPC SA-403, WP 304, 304L, 304W, and 304LW SA-420 Gr. WPL6 SA-479, TP 304, 304L, and 316
Plate	SA-240, TP 304, 304L, and 316L SA-285 Gr. A and C SA-515 Gr. 70 SA-516 Gr. 70 SA-537 Class 1 SB-171 Gr. CDA 706 ASTM A 515 Gr. 70
Shapes	SA-36 ASTM A-36 ASTM A-500 Gr. B (Code Case N-71-10)
Bolts/nuts/studs/pins	SA-193 Gr. B6, B7, B8 and B8M SA-194 Gr. 2H, 8H, 8M, 7, 4, 6 and B8 SA-307 Gr. B SA-320 Gr. L7 SA-325 Type 1 SA-453 Gr. 660A and 660B

TABLE 6.1.1-2 (SHEET 2 OF 2)

Component	Material
	SA-564 Gr. 630 ASTM A 193 Gr. B7 ASTM A 194 Gr. 7 ASTM A 307 ASTM A 354 ASTM A 490
Bars	SA-479, TP 304, 316, 410, 316L, and F316 SA-564 Gr. 630 ASTM A 108 Gr. 1018 CW (Code Case N-71-5)
Forgings	SA-105 SA-182, TP F304, F304L, F316 and F316L; Gr F11 and F22 SA-240, TP 304 and 316 SA-350 Gr. LF1 and LF2 SA-479, TP 304, 304L and 316 ASTM A 668 Class C (Code Case N-71-5)
Castings	SA-216 Gr. WCB and WCC SA-217 Gr. WC9 SA-351 Gr. CF8M and CF3M SA-487 Gr. CA6NM SB-61 SB-62 Gr. CDA 836 SB-148 Gr. CA 952 ASTM A 276 TP 410 ASTM A-216 Gr. WCB
Cooling coil fins	SB-152 Gr. CDA 122

TABLE 6.1.2-1 (SHEET 1 OF 3)

REGULATORY GUIDE 1.54, REVISION 0, JUNE 1973, QUALITY ASSURANCE REQUIREMENTS FOR PROTECTIVE COATINGS APPLIED TO WATER-COOLED NUCLEAR POWER PLANTS

Regulatory Guide 1.54 Position	Position on Non-NSSS Components	Position on NSSS Components
The requirements and guidelines included in ANSI N101.4-1972, Quality Assurance for Protective Coatings Applied to Nuclear Facilities, for protective coatings applied to ferritic steels, aluminum, stainless steel, zinc-coated (galvanized) steel, concrete, or masonry surfaces of water-cooled nuclear power plants are generally acceptable and provide an adequate basis for complying with the pertinent quality assurance requirements of Appendix B to 10 CFR 50 subject to the following:		Westinghouse has developed an alternate approach to ANSI N101.4 for satisfying Reg. Guide 1.54 for the NSSS components inside containment. Stringent requirements are specified for the painting of major components in Westinghouse Process Specifications that are imposed on vendors by procurement documents. Large equipment must have coating systems qualified to meet ANSI N101.2 and requirements are defined for surface preparation, use of undercoating, and where applicable, inspection. Other major equipment is either fabricated from stainless steel or covered by insulation. For small items of equipment, conventional industry practices are applied. Details of this approach follow:
A. ANSI N101.4-1972 should be used in conjunction with ASME NQA-1-1994, Quality Assurance Requirements for Nuclear Facility Applications, as described in the SNC QATR.	A. The requirements of VEGP are that practical and adequate corrosion protection shall be provided for all surfaces being painted. All surfaces, however, are not coated with coating materials tested and accepted under ANSI N101.2 or ANSI N5.12 criteria, nor are all coatings documented as outlined under ANSI N101.4-1972. Coating materials used for items located outside of the containment are not documented, since removal of the coating from such items does not affect the safe shutdown of the facility. Coating materials used for items located within the containment shall meet, whenever possible, the requirements of ANSI N101.2 and selected portions of ANSI N5.12 and are documented in accordance with ANSI N101.4.	NSSS equipment located in the containment building is separated into four categories to identify the applicability of this regulatory guide to various types of equipment. These categories of equipment are as follows: Category 1 - Large equipment Category 2 - Intermediate equipment Category 3 - Small equipment Category 4 - Insulated/stainless steel equipment A discussion of each equipment category follows: Category 1 - Large Equipment The Category 1 equipment consists of the following: • Reactor coolant system supports. • Reactor coolant pumps (motor and motor stand). • Accumulator tanks. • Refueling machine. Since this equipment has a large surface area and is procured from only a few vendors, it is

TABLE 6.1.2-1 (SHEET 2 OF 3)

Regulatory Guide 1.54 Position	Position on Non-NSSS Components	Position on NSSS Components
B. Subdivision 2.7 of ANSI N101.4-1972, states that when references are made to other standards, these references shall imply the most recent or current editions of the referenced standards. The specific applicability or acceptability of referenced standards will be covered separately in other regulatory guides, where appropriate.	B. VEGP follows Steel Structures Painting Council (SSPC-1963 and 1971), ANSI N101.2-1972, and sections, where applicable, of ANSI N5.12-1974	possible to implement tight controls over these items. Stringent requirements are specified for protective coatings on this equipment through the use of a painting specification in the procurement documents. This specification defines requirements for: 1. Preparation of vendor procedures. 2. Use of specific coatings systems which are qualified to ANSI N101.2. 3. Surface preparation. 4. Application of the coating systems in accordance with the paint manufacturer's instructions. 5. Inspections and nondestructive examinations. 6. Exclusive of certain materials. 7. Identification of all nonconformance. 8. Certifications of compliance. The vendor's procedures are subject to review by engineering personnel, and the vendor's implementation of the specification requirements is monitored during quality assurance surveillance activities.
C. Subdivision 1.1.2 of ANSI N101.4-1972 states that quality assurance, as covered by this standard, comprises all those planned and systematic actions necessary to provide specified documentation and adequate confidence that shop or field coating work for nuclear facilities will perform satisfactorily in service. This statement should not be interpreted as implying that the end product of quality assurance actions is the production of specified documentation. The term "quality assurance" as used in ANSI N101.4-1972 should be considered to comprise all those planned and systematic actions necessary to provide adequate confidence that shop or field coating work for nuclear facilities will perform satisfactorily in service. In this connection it is emphasized that records and documents listed in subdivisions 7.4 through 7.8, and included in the standard, are suggested forms only. Alternate documentation consistent with the requirements of Appendix B to 10 CFR 50 is also considered acceptable.	C. Conform. Forms used by the project are similar to those of ANSI N101.4. Alternate means of documentation are also used to ensure the coating work is performed satisfactorily in service. The planned and systematic actions which are necessary to provide adequate confidence that shop or field coating work for nuclear facilities will perform satisfactorily in service are as follows. Regulatory Guide 1.54 is implemented as follows: 1. The use of specific coatings systems which are prequalified to ANSI N101.2 are specified. 2. Surface preparation standards are used which illustrate the surface preparation procedures used in ANSI N101.2. 3. Surface profile requirements are met. 4. Application of the coating systems are made in accordance with instructions approved by the paint manufacturer. 5. Inspections and nondestructive testing are performed. 6. Nonconformances are identified and evaluated as discussed in paragraph 6.1.2.1.B. 7. Certifications of compliance and/or documentation procedures are furnished to satisfy project requirements	This system of controls provides assurance that the protective coatings will properly adhere to the base metal during prolonged exposure to a post-accident environment present within the containment building. The Category 2 equipment consists of the following: • Seismic platform and tie rods. • Reactor internals lifting rig. • Head lifting rig. Since these items are procured from a large number of vendors, and individually have very small surface areas, it is not practical to enforce the complete set of stringent requirements which are applied to Category 1 items. Another painting specification is used in these procurement documents. This specification defines to the vendors the requirements for: 1. Use of specific coating systems which are qualified to ANSI N101.2. 2. Surface preparation. 3. Application of the coating systems in accordance with the paint manufacturer’s instructions.

TABLE 6.1.2-1 (SHEET 3 OF 3)

Regulatory Guide 1.54 Position	Position on Non-NSSS Components	Position on NSSS Components
D. Sections 3 and 4 of ANSI N101.4-1972 delineate quality assurance requirements for coating materials and surface preparation of substrates. Coatings and cleaning materials used with stainless steel should not be compounded from or treated with chemical compounds containing elements that could contribute to corrosion, intergranular cracking, or stress corrosion cracking. Examples of such chemical compounds are those containing chlorides, fluorides, lead, zinc, copper, sulfur, or mercury where such elements are leachable or where they could be released by breakdown of the chemical compounds under expected environmental conditions (e.g., by radiation). This limitation is not intended to prohibit the use of trichlorotrifluoroethane which meets the requirements of Military Specification MIL-C-81302b for cleaning or degreasing of austenitic stainless steel, provided adequate removal is ensured prior to painting.	D. Conform.	The vendor's compliance with the requirements is also checked during quality assurance surveillance activities in the vendor's plant. These measures of degree of assurance that the protective coatings will control provide a high adhere properly to the base metal and withstand the postulated accident environment within the containment building. Category 3 - Small Equipment Category 3 equipment consists of the following: • Transmitters. • Alarm horns. • Small instruments. • Valves. • Heat exchanger supports. These items are procured from several different vendors and are painted by the vendor in accordance with conventional industry practices. Because the total exposed surface area is very small, Westinghouse does not specify further requirements. Category 4 - Insulated or Stainless Steel Equipment Category 4 equipment consists of the following: • Steam generators - covered with wrapped insulation. • Pressurizer - covered with wrapped insulation. • Reactor pressure vessel - covered with rigid reflective insulation. • Reactor cooling piping - stainless steel. • Reactor coolant pump casings - stainless steel. Since Category 4 equipment is insulated or is stainless steel, no painted surface areas are exposed within the containment. Therefore, this regulatory guide is not applicable for Category 4 equipment.

TABLE 6.1.2-2 (SHEET 1 OF 10)

CONTAINMENT COMPONENTS - COATING SCHEDULE

Item	Material[a]	Thickness (in.)	Estimated Surface Area (ft^2)[b]	Tolerance Surface Area(%)	1st Coat[c]	Minimum Thickness (mils of) 1st Coat	2nd Coat	Minimum Thickness (mils) of 2nd Coat	3rd Coat	Minimum Thickness (mils) of 3rd Coat	4th Coat	Minimum Thickness (mils) of 4th Coat
Containment Liner Plate System[a]												
Dome	CS	0.25	30,800(1)	±5	Inorganic zinc	2.5	Epoxy-polyamide	3.0				
Cylinder shell	CS	0.25	68,612(1)	±5	Inorganic zinc	2.5						
Sumps	SS	0.19	220(1)	±5	NC							
Refueling canal walls and bottom	SS	0.25	8,115(1)	±5	NC							
Basemat	CS	0.25	14,423(2)	±5	Inorganic zinc	2.5						
Reactor cavity walls	CS	0.25	3,700(1)	±5	Inorganic zinc	2.5						
Reactor cavity bottom	CS	0.25	787(2)	±5	Inorganic zinc	2.5						
Primary Shield[a]												
Cylinder wall	CS	0.25	805(1)	±5	Inorganic zinc	2.5						
Cylinder wall and nozzles	CS	1.00	1,287(1)	±5	Inorganic zinc	2.5						
Nozzles	CS	0.50	54(1)	±5	Inorganic zinc	2.5						
Upper cone and nozzles	CS	0.38	71(1)	±5	Inorganic zinc	2.5						
Containment Locks and Hatch[a]												
Equipment Hatch												
Spherical head	CS	1.50	174(3)	+15 -5	Inorganic zinc	2.5						
Exposed hatch ring	CS	3.00	172(3)	+15 -5	Inorganic zinc	2.5						
Hatch structure (CBI details)	CS	0.50	3,250(4)	+15 -5	Inorganic zinc	2.5						

TABLE 6.1.2-2 (SHEET 2 OF 10)

Item	Material[a]	Thickness (in.)	Estimated Surface Area (ft^2)[b]	Tolerance Surface Area(%)	1st Coat[c]	Minimum Thickness (mils of) 1st Coat	2nd Coat	Minimum Thickness (mils) of 2nd Coat	3rd Coat	Minimum Thickness (mils) of 3rd Coat	4th Coat	Minimum Thickness (mils) of 4th Coat
Personnel Lock												
Barrel	CS	0.5	25(4)	+15 -5	Inorganic zinc	2.5						
Sleeve	CS	2.0	52(1)	+15 -5	Inorganic zinc	2.5						
Escape Lock												
Barrel	CS	0.5	25(4)	+15 -5	Inorganic zinc	2.5						
Sleeve	CS	2.0	33(1)	+15 -5	Inorganic zinc	2.5						
Containment Concrete[a]												
Dome												
Buttresses to 50° above horizontal	Concrete	72.0	2278(5)	±10	NC							
Buttresses from 50° to vertical	Concrete	48.0	1822(5)	±10	NC							
Dome area less buttress area	Concrete	45.0	26,700(5)	±10	NC							
Shell												
Buttresses	Concrete	72.0	5920(5)	±10	NC							
Shell	Concrete	45.0	62,692(5)	±10	NC							
Internal Structural Concrete[a]												
Basemat	Concrete	126.0	14,423(5)	±10	NC							
Reactor cavity walls	Concrete	96.0	3818(6)	±10	NC							
Reactor cavity slab	Concrete	96.0	787(6)	±10	NC							
Exterior refueling canal walls												

TABLE 6.1.2-2 (SHEET 3 OF 10)

Item	Material[a]	Thickness (in.)	Estimated Surface Area (ft^2)[b]	Tolerance Surface Area(%)	1st Coat[c]	Minimum Thickness (mils of) 1st Coat	2nd Coat	Minimum Thickness (mils) of 2nd Coat	3rd Coat	Minimum Thickness (mils) of 3rd Coat	4th Coat	Minimum Thickness (mils) of 4th Coat
5-ft 0-in. thick walls	Concrete	60.0	258(6)	±10	Epoxy[a] (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
7-ft 0-in. thick walls	Concrete	84.0	584(6)	±10	Epoxy[a] (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
4-ft 0-in. thick walls	Concrete	48.0	6067(6)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
Refueling canal slab	Concrete	48.0	1394(6)	±10	NC							
Filler slab at 171 ft 9 in.	Concrete	33.0	13,715(7)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	15.0	Epoxy (finish)	3.0
Reactor cavity filler slab	Concrete	12.0	787(1)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	15.0	Epoxy (finish)	3.0
Primary shield	Concrete	108.0	2014(8)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
Cavity access walls	Concrete	24.0	830(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
North/south beams at 199 ft 0 in.	Concrete	72.0	1849(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
Pressurizer walls	Concrete	30.0	4394(9)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
Secondary shield walls	Concrete	36.0	29,722(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
Air Shaft Concrete Walls												
Walls, Nos. 1, 2, and 3												
Below 181 ft 0 in.	Concrete	30.0	1114(9)	±10	Epoxy (sealer)	0.5						
Above 185 ft 0 in.	Concrete	36.0	3045(9)	±10	Epoxy (sealer)	0.5						
No. 4 airshaft	Concrete	36.0	1055(9)	±10	Epoxy (sealer)	0.5						
No. 1 and 2 ceiling slab	Concrete	48.0	229(9)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	15.0	Epoxy (finish)	3.0

TABLE 6.1.2-2 (SHEET 4 OF 10)

Item	Material[a]	Thickness (in.)	Estimated Surface Area (ft^2)[b]	Tolerance Surface Area(%)	1st Coat[c]	Minimum Thickness (mils of) 1st Coat	2nd Coat	Minimum Thickness (mils) of 2nd Coat	3rd Coat	Minimum Thickness (mils) of 3rd Coat	4th Coat	Minimum Thickness (mils) of 4th Coat
No. 3 ceiling slab	Concrete	48.0	216(9)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	15.0	Epoxy (finish)	3.0
Not used												
Operating deck slab - 220 ft 0 in.												
2-ft 0-in. slab, 180° to 360°	Concrete	24.0	1610(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	15.0	Epoxy (finish)	3.0
2-ft 9-in. slab at 90°	Concrete	33.0	625(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	15.0	Epoxy (finish)	3.0
3-ft 0-in. slab at 90° and 260°	Concrete	36.0	210(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	15.0	Epxoy (finish)	3.0
5-ft 0-in. slab at 0° and 180°	Concrete	60.0	940(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	15.0	Epoxy (finish)	3.0
Miscellaneous Walls												
1-ft 6-in. wall at 90°	Concrete	18.0	580(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
3-ft 0-in. wall	Concrete	36.0	1089(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
3-ft 0-in. wall - below R.F. canal	Concrete	36.0	229(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
Mass concrete	Concrete	176.0	678(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
Walls under R.F. canal	Concrete	36.0	315(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0

TABLE 6.1.2-2 (SHEET 5 OF 10)

Item	Material[a]	Thickness (in.)	Estimated Surface Area (ft^2)[b]	Tolerance Surface Area(%)	1st Coat[c]	Minimum Thickness (mils of) 1st Coat	2nd Coat	Minimum Thickness (mils) of 2nd Coat	3rd Coat	Minimum Thickness (mils) of 3rd Coat	4th Coat	Minimum Thickness (mils) of 4th Coat
Instrument walls	Concrete	30.0	1307(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
Stairs and Elevator Shaft Walls												
Stair No. 1												
Walls	Concrete	12.0	2206(9)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
Ceiling	Concrete	18.0	137(9)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
Stair No. 2												
1-ft 0-in. wall	Concrete	12.0	1400(9)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
1-ft 6-in. wall and ceiling	Concrete	18.0	578(9)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	10.0	Epoxy (finish)	3.0
Elevator												
Walls	Concrete	12.0	1391(9)	±10	Epoxy (sealer)	0.5						
Ceiling	Concrete	18.0	63(9)	±10	Epoxy (sealer)	0.5						
Miscellaneous pads	Concrete	24.0	200(4)	±10	Epoxy (sealer)	0.5	Epoxy (filler)	As needed to fill holes	Epoxy (surfacer)	15.0	Epoxy (finish)	3.0
Structural Steel												
Platform grating	CS	0.19	86,630(4)	±15	Galvanized							
Ladders and stairways	CS	0.22	1328(4)	±15	Inorganic zinc	2.5						

TABLE 6.1.2-2 (SHEET 6 OF 10)

Item	Material[a]	Thickness (in.)	Estimated Surface Area (ft^2)[b]	Tolerance Surface Area(%)	1st Coat[c]	Minimum Thickness (mils of) 1st Coat	2nd Coat	Minimum Thickness (mils) of 2nd Coat	3rd Coat	Minimum Thickness (mils) of 3rd Coat	4th Coat	Minimum Thickness (mils) of 4th Coat
Pipe whip restraints												
Light and medium capacity	CS	1.5	340(4)	±20	Inorganic zinc	2.5						
Heavy and extra heavy capacity	CS	3.0	15,670(4)	±20	Inorganic zinc	2.5						
Pressurizer support steel												
Pressurizer supports	CS	1.0	914(4)	±10	Inorganic zinc	2.5						
Grating	CS	0.19	5060(4)	±10	Galvanized							
Platform beams	CS	0.5	2588(4)	±10	Inorganic zinc	2.5						
Crossover leg support	CS	2.5	840(4)	±10	Inorganic zinc	2.5						
Platform at steam generator												
Grating	CS	0.19	3250(4)	+100 -20	Galvanized							
Support beams	CS	0.5	30,000(4)	+100 -20	Inorganic zinc	2.5						
Polar crane runway												
Brackets - 37	CS	1.25	2714(4)	±10	Inorganic zinc	2.5						
Rail	CS	1.5	980(4)	±10	Inorganic zinc	2.5						
Girders - 37	CS	1.25	14,950(4)	±10	Inorganic zinc	2.5						
Internal platform structure												
Columns	CS	1.2	14,625(4)	±10	Inorganic zinc	2.5						
Beams	CS	0.38	6194(4)	±15	Inorganic	2.5						

TABLE 6.1.2-2 (SHEET 7 OF 10)

Item	Material[a]	Thickness (in.)	Estimated Surface Area (ft^2)[b]	Tolerance Surface Area(%)	1st Coat[c]	Minimum Thickness (mils of) 1st Coat	2nd Coat	Minimum Thickness (mils) of 2nd Coat	3rd Coat	Minimum Thickness (mils) of 3rd Coat	4th Coat	Minimum Thickness (mils) of 4th Coat
	CS	0.5	13,096(4)	±15	zinc Inorganic zinc	2.5						
	CS	0.62	37,991(4)	±15	Inorganic zinc	2.5						
	CS	0.75	13,012(4)	±15	Inorganic zinc	2.5						
	CS	1.0	12,741(4)	±15	Inorganic zinc	2.5						
	CS	1.3	3141(4)	±15	Inorganic zinc	2.5						
Cable trays	CS	0.0613	20,697	-	Galvanized							
Cable tray support	CS	0.38	7900(4)	±20	Galvanized							
	CS	0.19	3500(4)	±20	Galvanized							
Conduit	CS	0.15	12,779	-	Galvanized							
Conduit Boxes	CS	0.105	6141	-	Galvanized							
HVAC ducting	SS	0.1	12,000	-	NC							
HVAC ducting bracing and hangers	CS	0.38	2000(4)	±20	Inorganic zinc	2.5						
Pipe supports (steel members)	CS	0.25	51,787	± 5	Inorganic zinc	2.5						
Pipe racks	CS	1.25	13,750	± 2	Inorganic zinc	2.5						
Snubbers	CS	0.5	2970	± 5	Inorganic zinc	2.5						
Spring hangers	CS	0.5	386	± 5	Inorganic zinc	2.5						
Uninsulated piping												
24 in.	SS	0.375	182	± 5	NC							
14 in.	SS	0.312	205	± 5	NC							
12 in.	SS	1.312	194	± 5	NC							
12 in.	SS	1.125	53	± 5								
12 in.	SS	0.688	387	± 5	NC							

TABLE 6.1.2-2 (SHEET 8 OF 10)

Item	Material[a]	Thickness (in.)	Estimated Surface Area (ft^2)[b]	Tolerance Surface Area(%)	1st Coat[c]	Minimum Thickness (mils of) 1st Coat	2nd Coat	Minimum Thickness (mils) of 2nd Coat	3rd Coat	Minimum Thickness (mils) of 3rd Coat	4th Coat	Minimum Thickness (mils) of 4th Coat
12 in.	SS	0.375	367	± 5	NC							
10 in.	CS	0.365	206	± 5	Inorganic zinc	2.5						
10 in.	SS	0.365	1509	± 5	NC							
8 in.	CS	0.322	832	± 5	Inorganic zinc	2.5						
8 in.	SS	0.322	4769	± 5	NC							
8 in.	SS	0.906	286	± 5	NC							
6 in.	SS	0.280	5752	± 5	NC							
6 in.	CS	0.280	1139	± 5	Inorganic zinc	2.5						
4 in.	CS	0.237	2530	± 5	Inorganic zinc	2.5						
4 in.	SS	0.337	9	± 5	Inorganic zinc	2.5						
4 in.	SS	0.12	2806	± 5	NC							
4 in.	SS	0.237	1289	± 5	NC							
4 in.	SS	0.531	210	± 5	NC							
3 in.	CS	0.216	119	± 5	Inorganic zinc	2.5						
3 in.	CS	0.438	203	± 5	Inorganic zinc	2.5						
3 in.	SS	0.12	58	± 5	NC							
3 in.	SS	0.216	1146	± 5	NC							
3 in.	SS	0.438	388	± 5	NC							
2 1/2 in.	CS	0.203	451	± 5	Inorganic zinc	2.5						
2 in.	CS	0.218	267	± 5	Inorganic zinc	2.5						
2 in.	CS	0.344	71	± 5	Inorganic zinc	2.5						
2 in.	CS	0.154	12	± 5	Inorganic zinc	2.5						
2 in.	SS	0.154	733	± 5	NC							

TABLE 6.1.2-2 (SHEET 9 OF 10)

Item	Material[a]	Thickness (in.)	Estimated Surface Area (ft^2)[b]	Tolerance Surface Area(%)	1st Coat[c]	Minimum Thickness (mils of) 1st Coat	2nd Coat	Minimum Thickness (mils) of 2nd Coat	3rd Coat	Minimum Thickness (mils) of 3rd Coat	4th Coat	Minimum Thickness (mils) of 4th Coat
2 in.	SS	0.344	512	± 5	NC							
2 in.	SS	0.218	25	± 5	NC							
2 in.	SS	0.109	21	± 5	NC							
1 1/2 in.	CS	0.145	36	± 5	Inorganic zinc	2.5						
1 1/2 in.	CS	0.2	421	± 5	Inorganic zinc	2.5						
1 1/2 in.	SS	0.2	48	± 5	NC							
1 1/2 in.	SS	0.281	463	± 5	NC							
1 1/2 in.	SS	0.145	627	± 5	NC							
1 in.	CS	0.133	35	± 5	Inorganic zinc	2.5						
1 in.	CS	0.179	783	± 5	Inorganic zinc	2.5						
1 in.	CS	0.25	14	± 5	Inorganic zinc	2.5						
1 in.	SS	0.133	443	± 5	NC							
1 in.	SS	0.179	72	± 5	NC							
1 in.	SS	0.25	190	± 5	NC							
3/4 in.	SS	0.154	303	± 5	NC							
3/4 in.	SS	0.218	407	± 5	NC							
3/4 in.	SS	0.113	956	± 5								
1/2 in.	CS	0.188	1	± 5	Inorganic zinc	2.5						
1/2 in.	SS	0.147	140	± 5	NC							
1/2 in.	SS	0.109	15	± 5	NC							
1/2 in.	SS	0.065	278	± 5	NC							
1/2 in.	SS	0.049	123	± 5	NC							

TABLE 6.1.2-2 (SHEET 10 OF 10)

Item	Material[a]	Thickness (in.)	Estimated Surface Area (ft^2)[b]	Tolerance Surface Area(%)	1st Coat[c]	Minimum Thickness (mils of) 1st Coat	2nd Coat	Minimum Thickness (mils) of 2nd Coat	3rd Coat	Minimum Thickness (mils) of 3rd Coat	4th Coat	Minimum Thickness (mils) of 4th Coat
3/8 in.	SS	0.065	38	± 5	NC							
Containment coolers	CS	0.25	20,500	+10	Inorganic zinc	2.5	Epoxy	3.0				
Containment auxiliary coolers	CS	0.25	4400	+10	Inorganic zinc	2.5	Epoxy	3.0				
Cavity cooling coils	CS	0.25	2200	+10	Inorganic zinc	2.5	Epoxy	3.0				
ESF fans	CS	0.25	200	+10	Inorganic zinc	2.5	Epoxy	3.0				

a. CS - carbon steel
SS - stainless steel

b. (1) Interior surface (liner plate) backed by concrete. Only one side directly exposed to containment environment. Surface area of exposed side given.
(2) Interior surface (liner plate) backed by concrete and covered with filler slab (concrete). Not directly exposed to containment environment. Surface area of one side given.
(3) Direct boundary between containment interior and exterior environment. Interior surface area given.
(4) Specified member (beams, walls, slabs, etc.) completely exposed (all sides) to the containment environment. Total exposed surface area given (e.g., for walls both sides are given; for wide flange shapes both sides of flanges and web are given).
(5) Exterior surface. One side directly exposed to exterior environment. Surface area of exposed side given.
(6) Interior surface covered with steel liner plate. Not directly exposed to containment environment. Interior surface area given.
(7) Interior surface. One side directly exposed to containment environment with basemat steel liner plate on opposite side. Exposed surface area given.
(8) Interior surface. One side directly exposed to containment environment with steel liner on opposite side (liner in turn exposed to containment environment). Surface area of directly exposed side given.
(9) Outer surface directly exposed to general containment environment with inner area almost entirely self-enclosed. Directly exposed outer surface area given.

c. NC denotes no coating.

d. This epoxy is applied as a wainscot which covers containment interior walls from elevations 171 ft 9 in. to 181 ft 9 in. and elevations 220 ft to 228 ft. The remainder of concrete walls are coated with 0.5 mils of epoxy only.

e. Although actual areas for individual categories may differ slightly from the values shown, the total surface areas are considered to be within the indicated tolerances.

TABLE 6.1.2-3

PROTECTIVE COATINGS ON WESTINGHOUSE-
SUPPLIED EQUIPMENT INSIDE CONTAINMENT

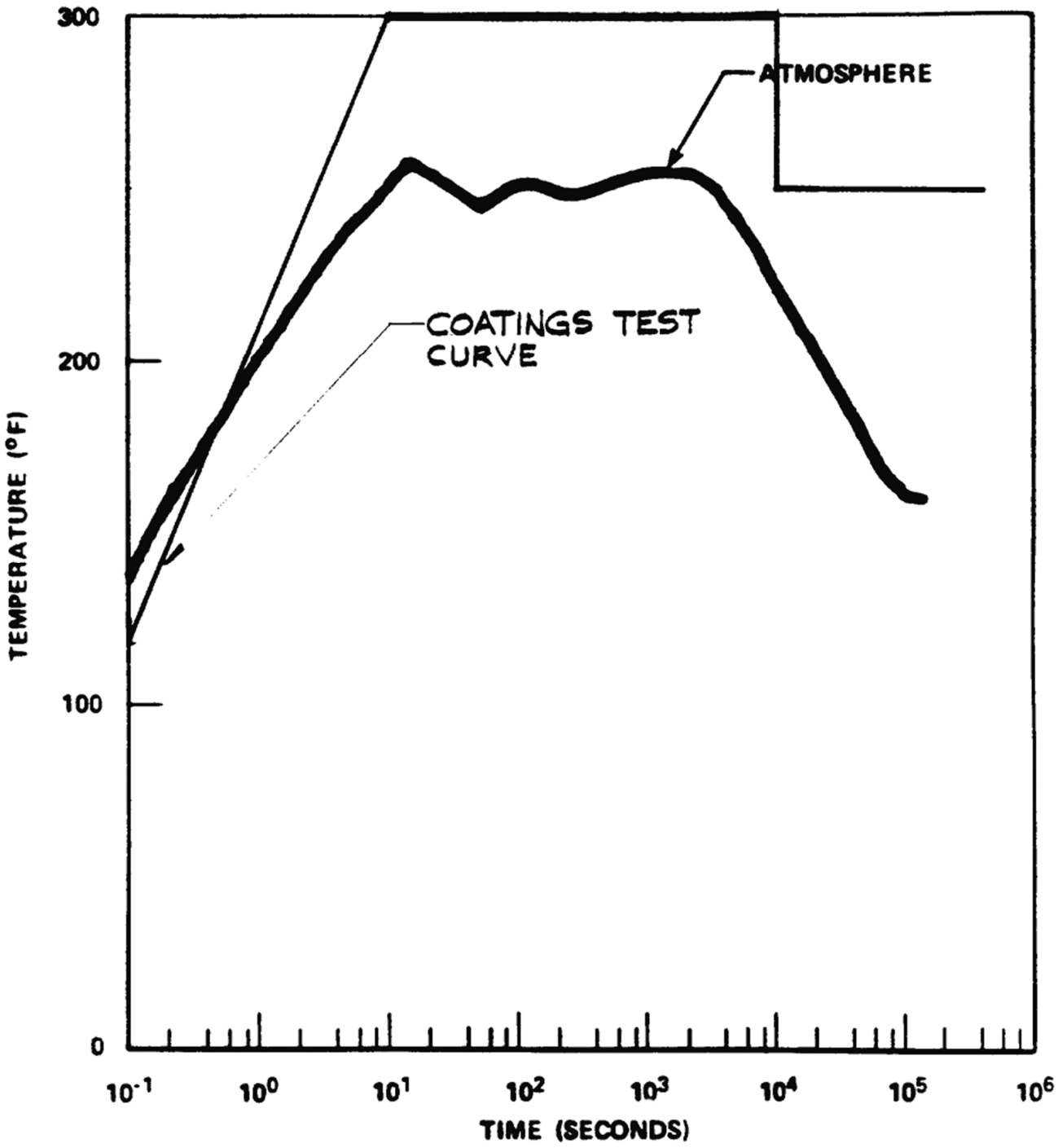
Component	Estimated Coated Surface Area (ft^2)
Reactor coolant system component supports	11,000[a]
Reactor coolant pump assemblies	5200
Accumulator tanks	5400
Refueling machine	3925
Other refueling equipment	2125
Remaining equipment (such as valves, auxiliary tanks and heat exchanger supports, transmitters, alarm horns, small instruments, etc.)	<1300

a. Primer only.

TABLE 6.1.2-4

OTHER ORGANIC MATERIALS

Item	Material	Quantity Enclosed (lb)	Quantity Exposed (lb)
Cable insulation	Ethylene propylene rubber and chlorosulphonated polyethylene	16,100	24,800
Heat shrink tubing	Raychem WCSF-N	650	-
Lug insulation	AMP special industries type PVF	200	-
Cable ties	Thomas and Bets Tefzel	100	500
Terminal blocks	Diallyl-phthalate long glass fiber fill	400	-



REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CONTAINMENT TEMPERATURE PROFILE
FOR POST–LOCA COATING TEST

FIGURE 6.1.2–1

6.2 CONTAINMENT SYSTEMS

The containment systems include the containment, the containment heat removal systems, the containment isolation system, and the containment combustible gas control system.

The design basis accident (DBA) is defined as the most severe of a spectrum of hypothetical loss-of-coolant accidents (LOCA) and high-energy line breaks within the containment. The ability of the containment systems to mitigate the consequences of a DBA depends upon the high reliability of these systems. This section provides the design criteria and evaluations to demonstrate that these systems function within the specified limits throughout the unit operating lifetime[a].

6.2.1 CONTAINMENT FUNCTIONAL DESIGN

6.2.1.1 Containment Structure

6.2.1.1.1 Design Bases

The containment system-is designed such that for all break sizes, up to and including the double-ended severance of a reactor coolant pipe or secondary system pipe, the containment peak pressure is below the design-pressure with an adequate margin, as presented in table 6.2.1-1.

The analyses presented in this section are based on assumptions that are conservative with respect to the containment and its heat removal systems; i.e., minimum heat removal, maximum containment pressure.

This capability is maintained by the containment system even assuming the worst single active failure affecting the operation of the emergency core cooling system (ECCS), containment spray system, and reactor containment fan coolers during the injection phase and the worst active or passive single failure during the recirculation phase. For primary system breaks, loss-of offsite power (LOSP) is assumed. For secondary system breaks, LOSP is not assumed, since this would reduce releases to the containment.

Paragraphs 6.2.1.3 and 6.2.1.4 present the mass and energy releases used in the design evaluation.

An evaluation was conducted to determine the impact of the uprating/T_{hot} reduction program conditions on the short term LOCA mass and energy releases used for the subcompartment analyses. The evaluation is discussed in paragraph 6.2.1.2.3.2.1.

6.2.1.1.2 Design Features

Containment design features are further described in the following:

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. In accordance with 10 CFR Part 54, appropriate aging management programs and activities have been initiated to manage the detrimental effects of aging to maintain functionality during the period of extended operation (see chapter 19).

A. A secondary shield wall constructed of a minimum of 3-ft-thick reinforced concrete, extends from the basement filler slab (171 ft 9 in.) to the operating deck (220 ft). The shield wall encloses the reactor coolant pumps, reactor vessel and its primary shield wall, steam generators up to 238 ft, the pressurizer up to 268 ft, and the refueling cavity. The secondary shield wall supports the operating deck, which together with the shield wall, prevents the containment liner from being impacted by potential internal missiles and the effects of-pipe whip. Refer to sections 3.5 and 3.6 for more detailed discussion.

B. The reactor containment is designed and constructed in accordance with the codes and standards discussed in subsections 3.8.1 and 3.8.3.

C. The VEGP design does not use a pressure-suppression-type containment.

D. Inasmuch as the reactor containment fan coolers are utilized during normal operation, inadvertent changes to the accident mode produce no significant effect upon containment internal pressure.

E. The only location inside the containment where water may be trapped and prevented from returning to the containment emergency sumps is the reactor cavity. Water can enter the cavity by flowing down through the ventilation openings surrounding the reactor cavity seal ring. Approximately 113,200 gal of water could collect in the reactor cavity. This would cause the static head for the residual heat removal (RHR) and spray pumps to decrease by about 1.31 ft. This decrease would not impair the operation of these pumps. Most of the water entering the refueling canal is returned to the containment emergency sumps via two, normally open, 12-in. drain lines.

F. The containment and subcompartment atmospheres are maintained during normal operation within prescribed pressure, temperature, and humidity limits by means of the nuclear service cooling water (NSCW) system which delivers 90°F water to the cooling coils within each containment fan cooler. Containment ventilation systems, such as the control rod drive mechanism booster fans and cooling fans, are used during normal operation and require no periodic testing to ensure functional capability.

6.2.1.1.3 Design Evaluation

The short-term pressure subcompartment analysis considers a LOSP. Consideration of single active failures is of no consequence, since none of the safety equipment functions during the initial seconds of the post-accident transient. The maximum calculated differential pressure in the steam generator compartment is 23.50 psid resulting from a 436 $in.^2$ in area guillotine rupture in the steam generator outlet nozzle. The maximum calculated differential pressure in the upper pressurizer cubical is 5.84 psid resulting from a spray line double-ended break. The maximum calculated differential pressure in the lower pressurizer cubicle is 20.7 psid from a surge line double-ended break. The containment subcompartment differential pressure analysis is described in detail in paragraph 6.2.1.2.

An evaluation was conducted to determine the impact of the uprating/T_{hot} reduction program conditions on the short term LOCA mass and energy releases used for the subcompartment analyses. The evaluation is discussed in paragraph 6.2.1.2.3.2.1.

The results of the pressure transient analysis of the containment for the LOCA are shown in figures 6.2.1-1 through 6.2.1-3. Containment temperature curves are presented in figures 6.2.1-

4 through 6.2.1-6. The cases examined in this analysis determine the effects of the full range of large reactor coolant break sizes up to and including a double-ended rupture. Cases illustrating the sensitivity to break location are also shown. All of these cases show that the containment pressure will remain below design pressure with margin. After the peak pressure is attained, the operation of the safeguards system reduces the containment pressure. At the end of the first day following the accident, the containment pressure has been reduced to a low value. The peak pressures are shown in table 6.2.1-1.

Calculation of containment pressure and temperature transients is accomplished by use of the digital computer code COCO.(1) The COCO code has been used and found acceptable to calculate containment pressure transients for the H. B. Robinson (docket number 50-261) and Zion (docket number 50-295) plants. Transient phenomena within the reactor coolant system (RCS) affect containment conditions by means of convective mass and energy transport through the pipe break.

For analytical rigor and convenience, the containment air-steam-water mixture is separated into two systems. The first system consists of the air-steam phase; the second is the water phase. Sufficient relationships to describe the transients are provided by the equations of conservation of mass and energy as applied to each system, together with appropriate boundary conditions. Since thermodynamic equations of state and conditions may vary during the transient, the equations have been derived for all possible cases of superheated or saturated steam and subcooled or saturated water. Switching between states is handled automatically by the code. The following are the major assumptions made in the analysis.

A. Discharge mass and energy flowrates through the RCS break are established from the analysis in paragraph 6.2.1.3.

B. For the steam line break analysis and the blowdown portion of the LOCA analysis, the discharge flow separates into steam and water phases at the breakpoint. The saturated water phase is at the total containment pressure, while the steam phase is at the partial pressure of the steam in the containment. For the post-blowdown portion of the LOCA analysis, steam and water releases are input separately.

C. Homogeneous mixing is assumed. The steam-air mixture and the water phase each have uniform properties. More specifically, thermal equilibrium between the air and steam is assumed. This does not imply thermal equilibrium between the steam-air mixture and water phase.

D. Air is taken as an ideal gas, while compressed water and steam tables are employed for water and steam thermodynamic properties.

E. For large steam line ruptures the saturation temperature at the partial pressure of the steam is used for heat transfer to the heat sinks and the fan coolers.

Paragraphs 6.2.1.3 and 6.2.1.4 present the mass and energy releases used for the analysis.

6.2.1.1.3.1 Initial Conditions. An analysis of containment response to the rupture of the RCS must start with knowledge of the initial conditions in the containment. The pressure, temperature, and humidity of the containment atmosphere prior to the postulated accident are specified in the analysis.

Also, values for the temperature of the NSCW and refueling water storage tank solution are assumed, along with the initial water inventory of the refueling water storage tank. All of these values are chosen conservatively, as shown in table 6.2.1-2.

Assumptions for containment analysis are shown in table 6.2.1-3. The assumed spray flowrate is based on one of two trains operating.

6.2.1.1.3.2 Heat Removal. The significant heat removal source during the early portion of the transient is structural heat removal. Provision is made in the containment pressure transient analysis for heat transfer through, and heat storage in, both interior and exterior walls. Every wall is divided into a large number of nodes. For each node, a conservation of energy equation expressed in finite-difference form accounts for transient conduction into and out of the node and temperature rise of the node. Tables 6.2.1-4 and 6.2.1-5 are summaries of the containment structural heat sinks used in the analysis.

The heat transfer coefficient to the containment structure is calculated by the code based primarily on the work of Tagami.[2] From this work, it was determined that the value of the heat transfer coefficient increases parabolically to peak value at the end of blowdown for LOCA and increases parabolically to peak at the time of steam line isolation. The value then decreases exponentially to a stagnant heat transfer coefficient which is a function of steam to air weight ratio.

Tagami presents a plot of the maximum value of h as a function of "coolant energy transfer speed," defined as follows:

$$\frac{\text{total coolant energy transferred into containment}}{(\text{containment volume})\,(\text{time interval to peak pressure}}$$

From this the maximum of h steel is calculated:

$$h_{max} = 75\left(\frac{E}{t_\rho V}\right)^{0.6} \qquad (1)$$

where:

h_{max} = Maximum value of h (Btu/h-ft^2-°F).

t_ρ = Time from start of accident to end of blowdown for LOCA and steam line isolation for secondary breaks (s).

V = Containment volume (ft^3).

E = Coolant energy--discharge (Btu).

The parabolic increase to the peak value is given by:

$$h_s = h_{max}\left(\frac{t}{t_\rho}\right)^{0.5} \quad (0 \le t \le t_\rho) \qquad (2)$$

where:

h_s = Heat transfer coefficient for steel (Btu/h-ft^2-°F).

t = Time from start of accident (s).

For concrete, the heat transfer coefficient is taken as 40% of the value calculated for steel.

The exponential decrease of the heat transfer coefficient is given by:

$$h_s = h_{stag} + (h_{max} - h_{stag})\, e^{-0.05(t-t_\rho)} \quad (t > t_\rho)$$

where:

h_{stag} = $2 + 50 X \quad (0 \leq X \leq 1.4)$.

h_{stag} = h for stagnant conditions (Btu/h-ft^2 °F).

x = Steam to air weight ratio in containment.

For a large break the safety features are quickly brought into operation. Because of the brief period of time required to depressurize the RCS, the safeguards are not a major influence on the blowdown peak pressure; however, they reduce the containment pressure after the blowdown and maintain a low long-term pressure. Also, although the containment structure is not as effective a heat sink as during the RCS blowdown, it still contributes significantly as a form of heat removal during the long-term cooling period.

During the injection phase of post-accident operation, the ECCS pumps water from the refueling water storage tank into the reactor vessel. Since this water enters the vessel at refueling water storage tank temperature, which is less than the temperature of the water in the vessel, it can absorb heat from the core until saturation temperature is reached. During the recirculation phase of operation, water is taken from the containment sump and cooled in the residual heat exchanger.

The cooled water is then pumped back to the reactor vessel to absorb more decay heat. The heat is removed from the residual heat exchanger by component cooling water.

Another containment heat removal system is the containment spray. During the injection phase of operation, the containment spray pumps draw water from the refueling water storage tank and spray it into the containment through nozzles mounted high above the operating deck. As the spray droplets fall, they absorb heat from the containment atmosphere. Since the water comes from the refueling water storage tank, the entire heat capacity of the spray from the refueling water storage tank temperature to the temperature of the containment atmosphere is available for energy absorption. During the recirculation phase of post-accident operation, water is drawn from the sump and sprayed into the containment atmosphere.

When a spray drop enters the hot, saturated, steam-air containment environment following a LOCA, the vapor pressure of the water at its surface is much less than the partial pressure of the steam in the atmosphere. Hence, there will be diffusion of steam to the drop surface and condensation on the drop. This mass flow will carry energy to the drop. Simultaneously, the temperature difference between the atmosphere and the drop will cause the drop temperature and vapor pressure to rise. The vapor pressure of the drop will eventually become equal to the partial pressure of the steam, and the condensation will cease. The temperature of the drop will essentially equal the temperature of the steam-air mixture.

The equations describing the temperature rise of a falling drop are as follows:

$$\frac{d}{dt}(Mu) = mh_g + q \qquad (3)$$

$$\frac{d}{dt}(Mu) = m \qquad (4)$$

where:

$q = h_c A\ (T_s - T)$.

$m = k_g A\ (P_s - P_v)$.

The coefficients of heat transfer (h_c) and mass transfer (k_g) are calculated from the Nusselt number for heat transfer, Nu' and the Nusselt number for mass transfer, Nu'.

Both Nu and Nu' may be calculated from the equations of Ranz and Marshall[3].

$$Nu \quad 2 + 0.6\,(Re)^{1/2}\,(Pr)^{1/3} \tag{5}$$

$$Nu' \quad 2 + 0.6\,(Re)^{1/2}\,(Sc)^{1/3} \tag{6}$$

Thus, equations 3 and 4 can be integrated numerically to find the internal energy and mass of the drop as a function of time as it falls through the atmosphere. Analysis shows that the temperature of the (mass) mean drop produced by the spray nozzles rises to a value within 99% of the bulk containment temperature in less than 2 s.

Drops of this size will reach temperature equilibrium with the steam-air containment atmosphere after falling through less than half the available spray fall height.

Detailed calculations of the heatup of spray drops in post-accident containment atmospheres by Parsly[4] show that drops of all sizes encountered in the containment spray reach equilibrium in a fraction of their residence time in a typical pressurized water reactor containment.

These results confirm the assumption that the containment spray will be 100% effective in removing heat from the atmosphere.

Nomenclature relevant to the above discussion is listed below:

A	=	Area.
h_c	=	Coefficient of heat transfer.
k_g	=	Coefficient of mass transfer.
h_g	=	Steam enthalphy.
M	=	Droplet mass.
m	=	Diffusion rate.
Nu	=	Nusselt number for heat transfer.
Nu'	=	Nusselt number for mass transfer.
P_s	=	Steam partial pressure.
P_v	=	Droplet vapor pressure.
Pr	=	Prandtl number.
q	=	Heat flowrate.
Re	=	Reynolds number
Sc	=	Schmidt number.
T_s	=	Droplet temperature.
T	=	Steam temperature.
t	=	Time.
u	=	Internal energy.

The reactor containment fan coolers are a final means of heat removal. The main aspects of a fan cooler from the heat removal standpoint are the fan and the banks of cooling coils. The fans draw the dense atmosphere through banks of finned cooling coils and mix the cooled steam-air mixture with the rest of the containment atmosphere. The coils are kept at a low temperature by

a constant flow of cooling water. Since this system does not use water from the refueling water storage tank, the mode of operation remains the same both before and after the spray system and ECCS change to the recirculation mode. Fan cooler heat removal performance is shown in figure 6.2.1-7.

6.2.1.1.3.3 Inadvertent Spray Actuation. In the event of inadvertent spray, the containment will depressurize until the air temperature is approximately equal to the spray temperature or the operator takes action to terminate the spray.

The COCO computer code was used to calculate the minimum pressure inside the containment. The following assumptions were made:

A. The spray flowrate is 6748 gal/min and is at 40°F.

B. The containment is initially at 120°F, 14.093 psi, and has 100% humidity.

C. Operator action to stop the spray occurs at 10 min.

The containment pressure reduces to 11.77 psia at 10 min into the transient. Thus, the peak differential pressure is 2.93 psi across the containment shell.

6.2.1.1.3.4 Inadvertent Purge Actuation. In the event of inadvertent containment purge exhaust actuation, the maximum differential pressure that can be drawn on the containment is 0.9 psi. The following assumptions were made:

A. The preaccess purge fan is assumed to be actuated because it has a larger capacity than the minipurge fan (subsection 9.4.6).

B. The preaccess purge fan is operating during mode 5 or 6. During modes 1 through 4, the preaccess purge penetrations are sealed closed.

C. There is no air supply or air leakage into the containment.

D. The containment is initially at 14.7 psia.

Consequently, the maximum differential pressure at the onset of a LOCA would be less than 0.9 psi.

In the VEGP minimum containment pressure analysis, an initial pressure of 14.7 psia (0.0 psig) has been assumed. This assumption is valid because it represents the containment pressure condition consistent with normal, full power operation of the plant. As a rule, parameters input to a 10 CFR 50.46 ECCS performance analysis correspond with expected, nominal plant conditions at 100% power operation. The large degree of conservatism present in 10 CFR 50.46 evaluation model analyses (as compared to historical best estimate results) was intended and does in fact cover uncertainties in the prevailing plant operating parameters at the time a postulated LOCA occurs. Therefore, utilization of atmospheric pressure, which is representative of the normal, full power containment condition, is appropriate for Vogtle and is consistent with the LOCA evaluation model philosophy.

Furthermore, at 100% power, the reactor coolant system is a significant heat source which heats and pressurizes air present in the containment. In the event that such pressurization occurs at Vogtle, containment purging will be conducted. In paragraph 6.2.1.5.8, a specific evaluation documents the penalty in calculated ECCS performance which is incurred when a LOCA occurs coincident with containment purging. This penalty is applied to the VEGP limiting case result. Since the very operation of the containment purge system is predicated upon a

high prevailing containment pressure, the consideration of containment purging being conducted at the time of a large break LOCA together with the assumption of 0 psig containment initial pressure accommodates any need for conservatism in the Vogtle ECCS performance analysis containment pressure computation.

Technical Specification limits on containment internal pressure cover operation through Mode 4. As discussed above, use of 0 psig is appropriate at the 100% power operation condition at which the VEGP ECCS performance analysis is conducted.

The 0 psig value of initial containment pressure is traditionally applied in 10 CFR 50.46 ECCS performance analyses.

In paragraph 6.2.1.5.8 is included an evaluation of purge operation during a double-ended cold leg guillotine break.

6.2.1.1.3.5 Accident Chronology. The accident chronology for the limiting LOCA is given in table 6.2.1-6.

6.2.1.2 Containment Subcompartments

6.2.1.2.1 Design Bases

Subcompartments within containment (principally, the reactor cavity, the steam generator compartments, and the pressurizer compartment) are designed to withstand the transient differential pressures and jet impingement forces of a postulated pipe break. Venting of these chambers is employed to keep the differential pressures within structural limits. In addition, restraints on the coolant pipes, reactor vessel, steam generators, etc., are designed to limit pipe whip effects and forces transmitted through component supports to ensure the integrity of subcompartments and the containment structure.

6.2.1.2.1.1 Summary of Subcompartment Pipe Break Analyses. The postulated breaks in the high-energy lines that are analyzed to determine the maximum differential pressure across the subcompartment walls are tabulated in table 6.2.1-8. The characteristics of the main coolant pipe ruptures are determined in accordance with the methods and criteria of subsection 3.6.2.

6.2.1.2.1.2 Reactor Vessel Cavity. The pipe restraints provided for the RCS limit the size of a break in the cavity region to 144 $in.^2$ The blowdown of coolant from a cold leg results in the highest pressure loadings on the reactor cavity walls and the reactor vessel. Accordingly, the analysis of the reactor cavity break assumes that the break occurs in a cold leg pipe.

A multi-compartment nodal model of the reactor cavity around the reactor vessel, reactor coolant piping nozzles, and connected volumes was developed in accordance with the guidelines presented in NUREG-0609. The subcompartment boundaries are set at planes of natural obstructions to flow.

The nodal network and the break mass/energy release rate (supplied by the nuclear steam supply system vendor) are input to the computer program COPDA, NE-699-D2, which performs

a stepwise calculation of the subcompartment pressures and temperatures as a function of time following the line break. Subcompartment pressures are used to calculate pressure differentials across the cavity structural members and force loadings on the reactor vessel.

In the development of the reactor cavity model, the following assumptions are made regarding the reactor vessel thermal insulation.

A. For the purpose of node volume calculation, all thermal insulation (uniform 4 in. thick) remains in place.

B. The upper reactor seal ring is completely plugged.

C. Fifty% of the annular space between the vessel and cavity wall is plugged along the plane perpendicular to the break location.

D. Fifty% of the remaining flow area in the instrument tunnel, after considering the steel of the first bottom mounted instrument support is plugged.

E. Access ports to annular inspection cavity are closed.

F. The nozzle penetrations into the cavity on the unaffected loops are completely plugged. The nozzle penetration into the cavity on the affected loop is open.

G. The nozzle penetrations to the steam generator compartments are open along the unaffected loops and plugged along the affected loop.

Figure 6.2.1-15 shows the resultant pressures in the central volumes shown in the nodal model in figure 6.2.1-16. The flow model is given in tables 6.2.1-9 and 6.2.1-10.

A summary of the forces and moments acting on the reactor vessel is included in table 6.2.1-73. Nodes 1 through 48 (figure 6.2.1-16) are those that contribute to F_x, F_g, F_{total}, M_X, and M . The magnitude of these loads is given at the axial centroids of the node groups. Uplift forces and the moment about the vessel axis are provided in table 6.2.1-74.

6.2.1.2.1.3 Steam Generator Compartment. For the analysis of the pressure transient in the steam generator compartment following a line break, the flow model, including control volumes, intercompartment flowpaths, and corresponding flow coefficients) is illustrated in figures 6.2.1-16A and 17 and tables 6.2.1-11, 11A through 6.2.1-22. The piping restraints on the reactor coolant loop restrict the motion of the piping following a pipe rupture in the primary loop. This restriction limits the size of the effective blowdown areas for postulated breaks in the steam generator compartments to an area less than, or equal to, a single-ended flow area of the pipe. The calculated steam generator compartment pressure response is shown in figures 6.2.1-17A through 6.2.1-23.

6.2.1.2.1.4 Pressurizer Compartment. For the analysis of the pressure transient following a pressurizer surge line or a spray (or relief) line break, the flow model of control volumes, intercompartment flowpaths, and corresponding flow coefficients are illustrated in figure 6.2.1-24 and tables 6.2.1-23 through 6.2.1-25. The calculated compartment pressure response is shown on figures 6.2.1-25 and 6.2.1-25a for the surge line break and the spray line break at the pressurizer nozzle, respectively.

6.2.1.2.2 Design Features

The effects of high-energy line breaks in the reactor cavity, steam generator compartments, and pressurizer compartment were analyzed to establish criteria for the structural design of the compartment walls. Plans and sections of these compartments are shown in drawings 1X2D01A001 and 1X2D01J015. The models of these compartments as used in the analyses are shown in figures 6.2.1-16, 6.2.1-16A, 17, and 6.2.1-24. The node volumes, flowpaths, flowpath areas, and flow coefficients for each of the compartments are tabulated in tables 6.2.1-9 through 6.2.1-25.

6.2.1.2.3 Design Evaluations

6.2.1.2.3.1 Analytical Model for Subcompartment Analyses. A digital computer program was used to perform the short-term compartment pressure transient analysis. This program is capable of handling up to 100 control volumes with a maximum of five flowpaths out of any compartment. The calculational methods used in this computer program are described in detail in BN-TOP-4, Rev. 1.

6.2.1.2.3.2 Analytical Model for Short Term Mass and Energy Releases. The computer models, which were used to develop the mass and energy release transients for the subcompartment pressurization analysis, are described in reference 9. Tables 6.2.1-26 through 6.2.1-28a provide tabulations of the mass and energy release rates versus time for the spectrum of breaks which were analyzed.

For the reactor vessel cavity, the design basis break is a double-ended rupture of the reactor vessel inlet pipe which is restrained to 144 $in.^2$ (table 6.2.1-26).

For the steam generator compartment, the following breaks are considered:

A. A double-ended reactor coolant pump outlet nozzle break restrained to 236 $in.^2$ (table 6.2.1-27, sheets 1 through 6).

B. A double-ended generator inlet nozzle break restrained to 306 $in.^2$ (table 6.2.1-27, sheets 7 through 12).

C. A double-ended break at the loop closure weld restrained to 336 $in.^2$ (table. 6.2.1-27, sheets 13 through 18).

D. A double-ended reactor coolant pump inlet nozzle break restrained to 336 $in.^2$ (table 6.2.1-27, sheets 19 through 24).

E. A double-ended steam generator outlet nozzle break restrained to 436 $in.^2$ (table 6.2.1-27, sheets 25 through 30).

F. A steam-generator inlet elbow split break with a break area equal to 763 $in.^2$ (table 6.2.1-27, sheets 31 through 36).

G. A double-ended main feedwater line break (table 6.2.1-26A).

For the pressurizer compartment, two break locations are considered:

A. A double-ended surge line break (table 6.2.1-28).

B. A double-ended spray line rupture (table 6.2.1-28a).

6.2.1.2.3.2.1 Effects of the Uprating/T_{hot} Reduction Programs on the LOCA Short Term Mass and Energy Releases. The short term mass and energy releases to containment, as a result of a LOCA, are described in paragraph 6.2.1.2.3.2. The blowdown mass and energy release rates are affected by the initial RCS temperature conditions. Since short term releases are linked directly to the critical mass flux, which increases with decreasing temperatures, the short term LOCA releases will increase due to the reductions in RCS coolant conditions associated with the uprating/T_{hot} reduction programs.

Releases currently described in paragraph 6.2.1.2.3.2 for the steam generator compartment have been evaluated for the uprating/T_{hot} reduction conditions. Based upon RCS temperature decreases of 15.0°F for the vessel outlet (from 618.2 to 603.2°F), 20.5°F for the vessel/core inlet (from 558.8 to 538.3°F), and 20.6°F for the steam generator outlet (from 558.6°F to 538.0 °F), the releases in table 6.2.1-27, sheets 1 through 36 can increase by as much as 12%. Even though the current analysis includes 10% margin to account for uncertainties in the analytical model, noding effects, and initial reactor coolant conditions, the 10% margin may not be adequate to offset the 12% increase. However, per reference 14, VEGP has been granted an exemption from a portion of the requirements of GDC 4. This exemption eliminates the need to consider the dynamic effects associated with postulated pipe breaks in the primary loop. Therefore, the releases currently in table 6.2.1-27, sheets 1 through 36 remain bounding for smaller nozzles attached to the RCS, whenever uprating/T_{hot} reduction effects are considered.

For the reactor vessel cavity, based upon RCS temperature decreases of 28.0°F for the vessel inlet, which is the maximum difference between the current data used in the paragraph 6.2.1.2.3.2 analysis and the uprating/T_{hot} reduction data, the releases in table 6.2.1-26 can increase by as much as 6%. The current analysis includes 10% margin to account for uncertainties, which is adequate to offset the 6% increase. Additionally, per reference 15, the average break opening area for the inlet break is 58 in.^{2}. The current releases, which are based upon a 144 in.^{2} break size, remain bounding considering uprating/T_{hot} reduction effects, whenever the actual break size based upon the structural evaluation is considered. Furthermore, per reference 14, VEGP has been granted an exemption from a portion of the requirements of GDC 4. This exemption eliminates the need to consider the dynamic effects associated with postulated pipe breaks in the primary loop.

For the pressurizer compartment, there are two break locations of importance; namely, the surge line and the spray line. For the surge line, based upon an RCS temperature decrease in the analysis of 20.7°F, the releases in table 6.2.1-28 can increase by as much as 16%. The current analysis, described in paragraph 6.2.1.2.3.2, includes 10% margin to account for uncertainties, which may not be adequate to offset this 16% increase. However, per references 16 and 17, VEGP has been granted an exemption from a portion of the requirements of GDC 4. This exemption eliminates the need to consider the dynamic effects associated with postulated rupture in the pressurizer surge line. For the spray line, the current releases are also expected to go up by 16%. However, upon closer inspection of the actual pipe sizes for the Vogtle spray line, and comparing to the current analysis break areas, it has been determined that the impact of the uprating/T_{hot} reduction is more than offset by the available margin in the break size, without the additional consideration of the 10% margin included in the current releases. Therefore, the releases in table 6.2.1-28a remain bounding for the spray line break.

In summary, an evaluation of the current short term LOCA mass and energy releases described in paragraph 6.2.1.2.3.2, utilized for the steam generator compartment, the reactor vessel cavity, and the pressurizer subcompartment, was conducted. It has been determined that the current short term LOCA mass and energy releases described in the previous section remain bounding for the uprating/T_{hot} reduction conditions whenever the margin in the current calculations are considered and whenever leak-before-break is appropriately credited.

6.2.1.2.3.2.2 Evaluation for MUR Power Uprate - LOCA Short-Term Mass and Energy Releases. Short-term LOCA mass and energy release calculations are performed to support the reactor cavity and loop subcompartment pressurization analyses (which includes the steam generator compartment and pressurizer compartment).

The analysis inputs that may potentially change with the uprate are the initial reactor coolant system (RCS) fluid temperatures. Since this event lasts for approximately 3 seconds, the single effect of power is not significant. The approved methodology for the short-term LOCA mass and energy analysis is documented in reference 9. The critical flow calculation employs appropriately defined critical flow correlations applied for fluid conditions at the break location. Most short-term blowdown transients are characterized by a peak mass and energy release rate that occurs during a subcooled condition. The Zaloudek correlation, which models this condition, is currently used in the short-term LOCA mass and energy release analyses. The Zaloudek correlation was used to conservatively evaluate the impact of the changes in the RCS inlet and outlet temperatures for the 2% uprate relative to those used in the current analysis of record.

The use of the lower temperatures maximizes the critical mass flux in the Zaloudek correlation. The analysis uses the minimum composite RCS T_{HOT}, T_{COLD} and steam generator outlet temperatures that are calculated for the MUR conditions. The critical flow correlation used in the mass and energy releases for this analysis will provide an increase in the mass and energy release for a slightly lower fluid temperature.

Unit 1 and Unit 2 have been approved for leak before-break (LBB) methods (reference 14).[a] This exemption eliminates the need to consider the dynamic effects associated with postulated RCS pipe breaks in the primary loop, eliminating the postulated primary system large pipe break from the subcompartment design basis. In addition, references 16 and 17 provide further justification for LBB application to eliminate surge line from the subcompartment design basis. Note, for Unit 2 only, reference 16 also approved LBB methods to eliminate, as a design basis, the accumulator and RHR piping. Therefore, the only break locations that need to be considered are the pressurizer spray line from the cold leg, the RHR line from the hot leg, and the smaller nozzles attached to the RCS.

Steam Generator Compartment

The licensing basis releases currently discussed in paragraph 6.2.1.2.3.2 for the steam generator compartment have been evaluated for the MUR uprate conditions. Based upon a comparison of the minimum hot full power RCS cold leg (T_{COLD}) temperature and the steam generator outlet temperature at the MUR uprate conditions, the temperature may be 0.7°F lower (537.6°F versus 538.3°F and 537.3°F versus 538.0°F) respectively, than that for the current licensing basis analysis. The minimum vessel outlet (T_{HOT}) temperature at the MUR uprate conditions is 603.8°F, which is 0.6°F higher (603.8°F versus 603.2°F). The differences in the temperatures presented are negligible and will have little effect on the mass and energy releases. In addition, based on the LBB exemption (reference 14), this exemption eliminates the need to consider the dynamic effects associated with postulated pipe break in the primary loop. In summary, the benefits of the decrease in mass and energy releases associated with the smaller primary system nozzle breaks, as compared to the larger RCS and larger RCS nozzle pipe breaks, more than offsets any penalty associated with possible increased releases which will result from decreased RCS coolant temperature. Therefore, the current licensing

[a] The leak-before-break analyses have been evaluated as time-limited aging analyses (TLAA) for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in paragraph 19.4.6.1.

basis (mass and energy releases in table 6.2.1-27) remains bounding for the smaller nozzles attached to the RCS whenever the MUR uprate effects are considered.

Reactor Vessel Cavity

For the reactor vessel cavity, a RCS temperature decrease of 0.7°F for the vessel inlet will be seen with the MUR uprate conditions. This is the maximum difference between the basis for the current licensing basis and the MUR uprate data. This decrease in temperature is considered negligible. However, the current analysis includes 10% margin to account for residual uncertainties in the analytical model, noding effects, and initial reactor coolant conditions. It has been determined that the impact of the MUR is more than offset by the available 10% margin included in the current releases to account for residual uncertainties. Additionally, per reference 15, the average break opening area for the inlet break is 58 $in.^2$. The current releases for the reactor cavity break, which are based upon a 144 $in.^2$ break size, remain bounding for the subcompartment pressurization analysis considering the MUR uprate effects whenever the actual break size based upon the structural evaluation of the reactor pressure vessel is considered. Furthermore, per reference 14, Units 1 and 2 have been granted an exemption from a portion of the requirements of GDC 4 relative to LBB. This exemption eliminates the need to consider the dynamic effects associated with postulated pipe breaks in the primary loop. Therefore, due to the conservatisms in the calculations for table 6.2.1-26 releases, it can be concluded that the releases presented in table 6.2.1-26 remain bounding when the MUR conditions are considered.

Pressurizer Compartment

For the pressurizer compartment, there are two break locations of importance: the surge line and the spray line. For the surge line, the RCS temperature will increase by 0.6°F as a result of the MUR, which results in a decrease in releases. In addition, per references 16 and 17, Units 1 and 2 have been granted an exemption from a portion of the requirements of GDC 4. Based on the LBB exemption, this exemption eliminates the need to consider the dynamic effects associated with postulated rupture in the pressurizer surge line. For the spray line, the RCS temperature reduction was determined to be 0.7°F, which is also considered negligible. The current analysis includes 10% margin to account for uncertainties in the analytical model, noding effects, and initial reactor coolant conditions. It has been determined that the impact of the MUR is more than offset by the available 10% margin included in the current releases to account for residual uncertainties. In addition, comparing the pipe size assumed in the current analysis of record versus the as-built piping; therefore, it has been determined that the impact of the reduction in temperature is more than offset by the available margin in the break size assumed in the analysis. In summary, the releases in table 6.2.1-28A remain bounding for the spray line break for application to the subcompartment pressurization analysis.

6.2.1.2.3.3 Initial Conditions for Subcompartment Pressure Analyses. The conditions tabulated in tables 6.2.1-9, 6.2.1-11 through 6.2.1-16, and 6.2.1-25 and the assumptions in paragraph 6.2.1.2.3.2 are used as the initial conditions for the subcompartment pressure analyses.

6.2.1.2.3.4 Flow Equation. The flow equations used for calculating the sonic and subsonic flow between nodes are fully described in BN-TOP-4, Rev 1. These flow equations, as

applied in the program, are based on the homogeneous equilibrium model. In addition, the application of the equations to the Nuclear Regulatory Commission (NRC) benchmark problems and to the evaluation of the Batelle Frankfurt experiments has developed conservative differential pressure between nodes.

6.2.1.2.3.5 Piping Systems. RCS, main steam, and main feedwater piping were considered in the containment subcompartment analysis; however, the limiting transient for each subcompartment resulted from a break in the encompassed RCS piping except that the main feedwater line break is the limiting transient for steam generator subcompartments above el 220 ft. There are no flow restrictions in the RCS piping. The dimensions of the pipe are tabulated in table 5.4.3-1, and the break areas are tabulated in table 6.2.1-8. For the reactor cavity and pressurizer compartments, the breaks are at the pipe to vessel nozzle weld. The break location in the cold leg pipe does not impact the analysis of the steam generator compartment node pressures.

6.2.1.2.3.6 Node Selection. The nodalization for the reactor cavity, steam generator, and pressurizer compartments is performed so that nodal boundaries are at the location of flow obstructions or geometry changes within the compartment. These discontinuities create pressure differentials across nodal boundaries; i.e., between adjacent nodes. Within each node, there are no discontinuities and hence negligible pressure gradients.

The COPDA computer code, consistent with BN-TOP-4, Rev 1, assumes stagnation within each node at every calculational time step, and requires that nodal boundaries be taken at significant flow restrictions. This, as well as the other guidelines and recommendations of section 3.2 of NUREG-0609, is followed. In light of this, and the fact that COPDA is an NRC approved code, no sensitivity studies were performed. This is consistent with section 3.2.1 of NUREG-0609. Furthermore, sensitivity studies would require the addition of nodes that are not at discontinuities which violates the requirements of the COPDA code and would lead to meaningless results.

Assumptions on insulation behavior are given in paragraph 6.2.1.2.1.2 for the reactor cavity analysis. This is in accordance with section 3.2.2.3 of NUREG-0609. The steam generator and pressurizer compartment analyses include a reduction in flow area and subcompartment volume due to insulation around piping and vessels. Subcompartment nodalization is shown in figures 6.2.1-16, 6.2.1-17, and 6.2.1-24. The COPDA computer code input data is summarized in tables 6.2.1-9 through 6.2.1-28.

6.2.1.2.3.7 Compartment Time Dependent Pressures. The time dependent pressures for those nodes that determine the structural design of the compartment walls are shown in the graphs of figures 6.2.1-15, 6.2.1-18, 6.2.1-19, 6.2.1-20, 6.2.1-21, 6.2.1-22, 6.2.1-23, and 6.2.1-25.

6.2.1.2.3.8 Vent Flowpath Flow Conditions. The node and flow characteristics for each of the subcompartments are tabulated in tables 6.2.1-9 through 6.2.1-25. The time dependent mass and energy flow conditions are provided in tables 6.2.1-26, 6.2.1-27, and 6.2.1-28.

6.2.1.2.3.9 Vent Flowpath Flow Coefficients. There are two orifice coefficients, C_v and C_g: C_v is a viscous loss coefficient, and C_g is the ratio of vena contract area to aperture area. The quantity C discussed below is obtained by multiplying C_v and C_g. The orifice coefficient is a function of the Reynolds number; however, for sufficiently high Reynolds numbers (greater than 100,000), C becomes independent of the Reynolds number.

The definition of the Reynolds number is:

$$\underline{Re} = \frac{SZ}{\gamma}$$

where:

s = Velocity (ft/s).

Z = Characteristic length; i.e., diameter (ft).

γ = Kinematic viscosity (ft^2/s).

Kinematic viscosity (γ) can be expected to have values of around 10^{-4}. Typical diameters of apertures are 10 to 30 ft. Velocities are 100 to 1000 ft/s. Therefore, Re will be very large and C will be independent of Re.

It is assumed throughout this section that flow coefficients obtained for single-phase flow are applicable for our two-phase situation and that the flow coefficients are constant with time.

The classification of openings and the development of head loss and flow coefficients are described in detail in BN-TOP-4, Rev 1.

6.2.1.3 Mass and Energy Release Analysis for Postulated Loss-of-Coolant Accidents

6.2.1.3.1 LOCA Mass and Energy Release Phases

The containment system receives mass and energy releases following a postulated rupture of the RCS. These releases continue through blowdown and post-blowdown.

The LOCA transient is typically divided into four phases:

A. Blowdown - which includes the period from accident initiation (when the reactor is at steady state operation) to the time that the RCS pressure reaches initial equilibrium with containment.

B. Refill - the period of time when the lower plenum is being filled by accumulator and safety injection water. At the end of blowdown, a large amount of water remains in the cold legs, downcomer, and lower plenum. To conservatively consider the refill period for the purpose of containment mass and energy releases, this water is instantaneously transferred to the lower plenum along with sufficient accumulator water to completely fill the lower plenum. This allows an uninterrupted release of mass and energy to containment. Thus, the refill period is conservatively neglected in the mass and energy release calculation.

C. Reflood - begins when the water from the lower plenum enters the core and ends when the core is completely quenched.

D. Post-Reflood (Froth) - describes the period following the reflood transient. For the pump suction break, a two-phase mixture exits the core, passes through the hot legs, and is superheated in the steam generators. After the broken loop steam generator cools, the break flow becomes two phase.

6.2.1.3.2 Break Size and Location

Generic studies have been performed with respect to the effect on the LOCA mass and energy releases relative to postulated break size. The double-ended guillotine break has been found to be limiting due to larger mass flow rates during the blowdown phase of the transient. During the reflood and post-reflood phases, the break size has little effect on the releases.

Three distinct locations in the reactor coolant system loop can be postulated for pipe rupture:

- Hot leg (between vessel and steam generator).
- Cold leg (between pump and vessel).
- Pump suction (between steam generator and pump).

The break location analyzed and described herein is the double-ended pump suction guillotine break (10.48 ft^2). Pump suction break mass and energy release have been calculated for the blowdown, reflood, and post-reflood phases of the LOCA.

The following information provides a discussion on each break location. The double ended hot leg guillotine has been shown in previous studies to result in the highest blowdown mass and energy release rates. Although the core flooding rate would be highest for this break location, the amount of energy released from the steam generator secondary side is minimal because the majority of the fluid which exits the core bypasses the steam generators in venting to containment. As a result, the reflood mass and energy releases are reduced significantly as compared to either the pump suction or cold leg break locations where the core exit mixture must pass through the steam generators before venting through the break.

For the hot leg break, there is no reflood peak as determined by generic studies (i.e., from the end of the blowdown period the releases would continually decrease). Therefore the reflood (and subsequent post-reflood) releases are not calculated for a hot leg break. The mass and energy releases for the hot leg break blowdown phase have been included in the scope of this containment integrity analysis.

The cold leg break location has also been found in previous studies to be much less limiting in terms of the overall containment peak pressure. The cold leg blowdown is faster than that of the pump suction break, and more mass is released into the containment. However, the core heat transfer is greatly reduced, and this results in a considerably lower energy release into containment. Studies have determined that the blowdown transient for the cold leg is, in general, less limiting than that for the pump suction break. During reflood, the flooding rate is greatly reduced and the energy release rate into the containment is reduced. Therefore, the cold leg break is not usually performed.

The pump suction break combines the effects of the relatively high core flooding rate, as in the hot leg break, and the addition of the stored energy in the steam generators. As a result, the pump suction break yields the highest energy flowrates during the post-blowdown period by including all of the available energy of the reactor coolant system in calculating the releases to containment. This break location has been determined to be the limiting break for typical dry

containment plants. The choice of this break location for VEGP as the limiting break is consistent with other dry containment plants for the post blowdown phase of the event.

The analysis of the limiting break location for a dry containment has been performed. The double-ended pump suction guillotine break has historically been considered to be the limiting break location for the post blowdown phase of the event, by virtue of its consideration of all energy sources present in the RCS. The analyses support the conclusions of the double-ended pump suction (DEPS) as the limiting break case for the post-blowdown period, considering both the minimum and maximum safety injection cases. This break location provides a mechanism for the release of the available energy in the reactor coolant system, including both the broken and intact loop steam generators.

6.2.1.3.3 Application of Single Failure Criteria

An analysis of the effects of the single failure criteria has been performed on the mass and energy release rates for the DEPS break.

For the DEPS results an inherent assumption in the generation of the mass and energy release is that offsite power is lost. This results in the actuation of the emergency diesel generators required to power the safety injection system. This is not an issue for the blowdown period which is limited by the double-ended hot leg (DEHL) break.

Two cases have been analyzed for the effects of a single failure. The DEPS case with both minimum and maximum safety injection for the 3579 MWt rerated conditions was analyzed.

An evaluation has been performed to support the MUR power uprate.

The analysis of record presently assumes a NSSS power of 3650 MWt (102% of 3579 MWt), which includes a bounding allowance for the MUR power uprate in conjunction with a reduced calorimetric uncertainty.

The limiting case for the VEGP is the minimum safeguards case. This was determined by prior generic and specific VEGP analyses. In the case of minimum safeguards, the single failure postulated to occur is the loss of an emergency diesel generator. This results in the loss of one pumped safety injection train and the containment safeguards components on that diesel, thereby minimizing the safety injection flow. The analysis further considers the safety injection pump head curves to be degraded by 5%. (See Westinghouse Letter GP-15580 for discussion of flow margins.) This results in the greatest reduction possible for the emergency core cooling system (ECCS) components. For the case analyzing maximum safety injection, a conservative assumption was made due to the availability of only the diesel train failure criteria. The maximum safety injection flows were modeled with the minimum containment safeguard components available. This applies to heat exchangers, fan coolers, and the containment spray system. This assumption would provide a bounding assessment in maximizing the mass release but minimizing the heat removal capability.

6.2.1.3.4 Mass and Energy Release Data

6.2.1.3.4.1 Significant Modeling Assumptions. The following items ensure that the mass and energy releases are conservatively calculated for maximum containment pressure:

A. Maximum expected operating temperature of the reactor coolant system.

B. Allowance in temperature for instrument error and dead band (+6.0°F).

C. Margin in volume of 3% (which is composed of 1.6% allowance for thermal expansion and 1.4% for uncertainty).

D. NSSS power of 3650 MWt (102% of 3579 MWt), which includes a bounding allowance for the MUR power uprate in conjunction with a reduced calorimetric uncertainty.

E. Allowance for calorimetric error is included in item D.

F. Conservative coefficients of heat transfer (i.e., steam generator primary/secondary heat transfer and reactor coolant system metal heat transfer).

G. Allowance in core stored energy effect of fuel densification.

H. Margin in core stored energy (+15%).

I. Allowance for RCS pressure uncertainty (+50 psi).

6.2.1.3.4.2 Blowdown Mass and Energy Release Data. The SATAN-VI code is used for computing the blowdown transient and is the same as that used for the ECCS calculation in reference 11. The methodology for the use of this model is described in reference 10.

Tables 6.2.1-29 and 6.2.1-30 present the calculated mass and energy releases for the blowdown phase of the break analyzed for the DEPS and DEHL breaks, respectively. The mass and energy release for the DEPS break and the DEHL break, given in tables 6.2.1-29 and 6.2.1-30, terminate 22.0 and 25.0 seconds, respectively, after the initiation of the postulated accident.

6.2.1.3.4.3 Reflood Mass and Energy Release Data. The WREFLOOD code is used for computing the reflood transient and is a modified version of that used in the ECCS calculation in reference 11. The methodology for the use of this model is described in reference 10.

An exception to the mass and energy evaluation model described in reference 10 is taken, in that steam/water mixing in the broken loop has been included in this analysis. This assumption is justified, supported by test data, and summarized as follows:

The model assumes a complete mixing condition (i.e., thermal equilibrium) for the steam/water interaction. The complete mixing process, however, is made up of two distinct physical processes. The first is a two phase interaction with condensation of steam by cold injection water. The second is a single phase mixing of condensate and injection water. Since the mass and energy of the steam released is the most important influence to the containment pressure transient, the steam condensation part of the mixing process is the only part that need be considered. (Any spillage directly heats only the sump).

The most applicable steam/water mixing test data has been reviewed for validation of the containment integrity reflood steam/water mixing model. This data is that generated in 1/3 scale tests (reference 12), which are the largest scale data available and thus most closely simulate the flow regimes and gravitational effects that would occur in a PWR. These tests were designed specifically to study the steam/water interaction for PWR reflood conditions.

From the entire series of 1/3 scale tests, a group corresponds almost directly to containment integrity reflood conditions. The injection flow rates for this group cover all phases and mixing conditions calculated during the reflood transient. The data from these tests were reviewed and

discussed in detail in reference 10. For all of these tests, the data clearly indicate the occurrence of very effective mixing with rapid steam condensation. The mixing model used in the containment integrity reflood calculation is therefore wholly supported by the 1/3 scale steam/water mixing data.

Additionally, the following justification is also noted. The limiting break for the containment integrity peak pressure analysis during the post-blowdown phase is the DEPS break. For this break, there are two flow paths available in the RCS by which mass and energy may be released to containment. One is through the outlet of the steam generator, the other via reverse flow through the reactor coolant pump. Steam, which is not condensed by ECCS injection in the intact RCS loops, passes around the downcomer and through the broken loop cold leg and pump in venting to containment. This steam also encounters ECCS injection water as it passes through the broken loop cold leg, complete mixing occurs, and a portion of it is condensed. It is this portion of steam which is condensed that is taken credit for in this analysis. This assumption is justified based upon the postulated break location and the actual physical presence of the ECCS injection nozzle. A description of the test and test results is contained in references 10 and 12.

The methodology previously discussed and described in reference 10 has been utilized and approved on the dockets for Catawba Units 1 and 2, McGuire Units 1 and 2, Sequoyah Units 1 and 2, Watts Bar Units 1 and 2, Millstone Unit 3, Beaver Valley Unit 2, and Surry Units 1 and 2.

Tables 6.2.1-31 and 6.2.1-32 present the calculated mass and energy release for the reflood phase of the DEPS break with minimum and maximum safety injection, respectively. A significantly higher discharge occurs during the period the accumulators are injecting (from 28.1 to 54.3 seconds for the minimum safety injection case and 28.1 to 54.1 seconds for maximum safety injection, as illustrated in tables 6.2.1-31 and 6.2.1-32).

The transient of the principal parameters during reflood is given in tables 6.2.1-33 and 6.2.1-34 for the minimum and maximum safety injection DEPS break cases.

6.2.1.3.4.4 Post-Reflood Mass and Energy Release Data. The FROTH code (reference 9) is used for computing the post-reflood transient. The methodology for the use of this model is described in reference 10. The mass and energy release rates calculated by the FROTH code are used in the containment analysis until the time of containment depressurization.

After depressurization, the mass and energy release from decay heat is based on the 1979 ANSI/ANS Standard, shown in reference 13, and the following input:

A. Decay heat sources considered are fission product decay and heavy element decay of U-239 and Np-239.

B. Decay heat power from fissioning isotopes other than U-235 is assumed to be identical to that of U-235.

C. Fission rate is constant over the operating history of maximum power level.

D. The factor accounting for neutron capture in fission products has been taken from Table 10 of ANSI/ANS 5.1-1979.

E. Operation time before shutdown is 3 years.

F. The total recoverable energy associated with one fission has been assumed to be 200 MeV/fission.

G. Two sigma uncertainty (2 times the standard deviation) has been applied to the fission product decay.

Tables 6.2.1-35 and 6.2.1-36 present the two phase post-reflood mass and energy release data for the double-ended pump suction break minimum and maximum safety injection cases.

6.2.1.3.5 Sources of Mass and Energy

The sources of mass considered in the LOCA mass and energy release analysis are given in tables 6.2.1-37, 6.2.1-38, and 6.2.1-39. These sources are the reactor coolant system, accumulators, and pumped safety injection.

The energy inventories considered in the LOCA mass and energy release analysis are given in tables 6.2.1-40, 6.2.1-41, and 6.2.1-42. The energy sources include:

- Reactor coolant system water.
- Accumulator water.
- Pumped injection water.
- Decay heat.
- Core stored energy.
- Reactor coolant system metal.
- Steam generator metal.
- Steam generator secondary energy.
- Secondary transfer of energy (feedwater into and steam out of the steam generator secondary).

In the mass and energy release data presented, no Zirc-water reaction heat was considered because the clad temperature did not rise high enough for the rate of the Zirc-water reaction heat to be of any significance.

System parameters needed to perform confirmatory analyses are provided in table 6.2.1-43.

The consideration of the various energy sources in the mass and energy release analysis provides assurance that all available sources of energy have been included in this analysis. Thus, the review guidelines presented in Standard Review Plan Section 6.2.1.3 have been satisfied.

The mass and energy inventories are presented at the following times, as appropriate:

- Time zero (initial conditions).
- End of blowdown time.
- End of refill time.
- End of reflood time.
- Time of full depressurizations.

- End of analysis.

The methods and assumptions used to release the various energy sources are given in reference 10, except as noted in paragraph 6.2.1.3.4.3, which has been approved as a valid evaluation model by the Nuclear Regulatory Commission.

6.2.1.4 Mass and Energy Release Analysis for Postulated Secondary System Pipe Ruptures Inside Containment

Steam line ruptures occurring inside a reactor containment structure may result in significant releases of high-energy fluid to the containment environment, possibly resulting in high containment temperatures and pressures. The quantitative nature of the releases following a steam line rupture is dependent upon the many possible configurations of the plant steam system and containment designs as well as the plant operating conditions and the size of the rupture. These variations make a reasonable determination of the single absolute worst case for both containment pressure and temperature evaluations following a steam break difficult. This section describes the methods used in determining the containment responses to a variety of postulated pipe breaks encompassing wide variations in plant operation, safety system performance, and break size. The spectrum of breaks analyzed is listed in table 6.2.1-60.

6.2.1.4.1 Significant Parameters Affecting Steam Line Break Mass and Energy Releases

There are four major factors that influence the release of mass and energy following a steam line break: steam generator fluid inventory, primary-to-secondary heat transfer, protective system operation, and the state of the secondary fluid blowdown. The following is a list of those plant variables that determine the influence of each of these factors:

A. Plant power level.

B. Main feedwater system design.

C. Auxiliary feedwater system design.

D. Postulated break type, size, and location.

E. Availability of offsite power.

F. Safety system failures.

G. Steam generator reverse heat transfer and RCS metal heat capacity.

The following is a discussion of each of these variables.

6.2.1.4.1.1 Plant Power Level. Steam line breaks can be postulated to occur with the plant in any operating condition ranging from hot shutdown to full power. Since steam generator mass decreases with increasing power level, breaks occurring at lower power generally result in a greater total mass release to the plant containment. However, because of increased energy storage in the primary plant, increased heat transfer in the steam generators, and the additional energy generation in the nuclear fuel, the energy release to the containment from breaks postulated to occur during power operation may be greater than for breaks occurring with the plant in a hot shutdown condition. Additionally, steam pressure and the dynamic conditions in the steam generators change with increasing power and have significant influence on both the

rate of blowdown and the amount of moisture entrained in the fluid leaving the break following a steam break event.

Because of the opposing effects of changing power level on steam line break releases, no single power level can be identified as a worst case initial condition for a steam line break event. Therefore, several different power levels spanning the operating range as well as the hot shutdown condition have been analyzed.

6.2.1.4.1.2 Main Feedwater System Design. The rapid depressurization which occurs following a rupture may result in large amounts of water being added to the steam generators through the main feedwater system. Rapid closing isolation valves are provided in the main feedwater lines to limit this effect. Also, the piping layout downstream of the isolation valves affects the volume in the feedwater lines that cannot be isolated from the steam generators. As the steam generator pressure decreases, some of the fluid in this volume will flash into the steam generator, providing additional secondary fluid which may exit out the rupture.

The feedwater addition which occurs prior to closing of the feedwater line isolation valves influences the steam generator blowdown in several ways. First, the rapid addition increases the amount of entrained water in large-break cases by lowering the bulk quantity of the steam generator inventory. Secondly, because the water entering the steam generator is subcooled, it lowers the steam pressure, thereby reducing the flowrate out of the break. Finally, the increased flowrate causes an increase in the heat transfer rate from the primary-to-secondary system, resulting in greater energy being released out the break. Since these are competing effects on the total mass and energy release, no worst case feedwater transient can be defined for all plant conditions. In the results presented, the worst effects of each variable have been used. For example, moisture entrainment for each break is calculated assuming conservatively small feedwater additions so that the entrained water is minimized. Determination of total steam generator inventory, however, is based on conservatively large feedwater additions, as explained in paragraph 6.2.1.4.3.2.

The unisolated feedwater line volumes between the steam generator and the isolation valves serve as a source for additional high-energy fluid to be discharged through the pipe break. This volume is accounted for in the mass and energy release data presented in paragraph 6.2.1.4.3.2.

6.2.1.4.1.3 Auxiliary Feedwater System Design. Within the first minute following a steam line break, the auxiliary feed system is initiated on any one of several protection system signals. The addition of auxiliary feedwater to the steam generators increases the secondary mass available for release to the containment, as well as increases the heat transferred to the secondary fluid.

The effects of the steam generator mass are realistically and conservatively modelled in the calculation described in paragraph 6.2.1.4.3.2 by assuming full auxiliary feedwater flow to the faulted steam generator starting at time zero and continuing until manually stopped by the plant operator for the small break cases and zero power cases. The split break cases assume full auxiliary feedwater (AFW) flow to the faulted loop starting at the time of Hi-1 containment signal and the large double-ended rupture case assumes full AFW flow to the faulted steam generator starting at the time a safety injection or low steam generator mass reactor trip signal is received.

6.2.1.4.1.4 Postulated Break Type, Size, and Location.

A. Postulated Break Type

Two types of postulated pipe ruptures are considered in evaluating steam line breaks.

First is a split rupture in which a hole opens at some point on the side of the steam pipe or steam header but does not-result in a complete severance of the pipe. A single, distinct break area is fed uniformly by all steam generators until steam line isolation occurs. The blowdown flowrates from the individual steam generators are interdependent, since fluid coupling exists between all steam lines. Because flow limiting orifices are provided in each steam generator, the largest possible split rupture can have an effective area prior to isolation that is no greater than the throat area of the flow restrictor times the number of plant primary coolant loops. Following isolation, the effective break area for the steam generator with the broken line can be no greater than the flow restrictor throat area.

The second break type is the double-ended guillotine rupture in which the steam pipe is completely severed and the ends of the break displace from each other. Guillotine ruptures are characterized by two distinct break locations, each of equal area, but being fed by different steam generators. The largest possible guillotine rupture can have an effective area per steam generator no greater than the throat area of one steam line flow restrictor.

The type of break influences the mass and energy releases to containment by altering both the nature of the steam blowdown from the piping in the steam plant and the effective break area fed by each steam generator prior to steam line isolation. For example, a double-ended rupture in a pipe having a cross-sectional area of 2.4 ft^2 would appear as a 1.4-ft^2 rupture to a single steam generator feeding one end of the break but would appear as a 0.8-ft^2 rupture to each of steam generators feeding the other end of the break.

B. Postulated Size

Break area is also important when evaluating steam line breaks. It controls the rate of releases to the containment, as well as exerts significant influence on the steam pressure decay and the amount of entrained water in the blowdown flow. The data presented in this section include releases for four break areas at each of four initial power levels. Included are three double-ended and one split rupture, as follows.

1. A full double-ended pipe rupture downstream of the steam line flow restrictor. For this case, the actual break area equals the cross-sectional area of the steam line, but the blowdown from the steam generator with the broken line is controlled by the flow restrictor throat area (1.4 ft^2). The reverse flow from the intact steam generators is controlled by the smaller of the pipe cross section, the steam stop valve seat area, or the total flow restrictor throat area in the intact loops. The reverse flow has been conservatively assumed to be controlled by the flow restrictors in each of the intact loop steam generators. Actually, the combined flow from the three steam generators must pass through an 18-in. (1.42-ft^2) line, which would greatly restrict the flow.

2. A small double-ended rupture at the steam generator nozzle having an area just larger than the area at which water entrainment occurs (i.e., with entrainment). Entrainment is assumed in the forward direction only. Dry steam blowdown is assumed to occur in the reverse direction.

3. A split break that represents the largest break which can neither generate a steam line isolation signal from the primary protection equipment nor result in moisture entrainment. Steam and feedwater line isolation signals are generated by high containment pressure signals for these cases. Being a split rupture, the effective area seen by the faulted steam generator increases by a factor of four, following steam line isolation. Conceivably, moisture entrainment could occur at that time. However, since steam line isolation for these breaks generally does not occur before 20 to 60 s, it is conservatively assumed that the pressure has decreased sufficiently in the affected steam generator to preclude any moisture carryover.

4. A small double-ended rupture at the steam generator nozzle having an area just smaller than that in which water entrainment occurs (i.e., without entrainment).

C. Postulated Break Location

Break location affects steam line blowdowns by virtue of the pressure losses which would occur in the length of piping between the steam generator and the break. The effect of the pressure loss is to reduce the effective break area- seen by the steam generator. Although this would reduce the rate of blowdown, it would not significantly change the total release of energy to the containment. Therefore, piping loss effects have been conservatively ignored in all blowdown results, except in the small double-ended ruptures in which moisture entrainment occurs. The effects of pipe friction are conservatively assumed to be sufficiently large in this case to prevent moisture entrainment in the reverse flow, thus minimizing water relief to the containment.

6.2.1.4.1.5 Availability of Offsite Power. The effects of the assumption of the availability of offsite power have been enveloped in the analysis. LOSP has been assumed where it delays the actuation of the containment heat removal systems (i.e., containment sprays and containment air coolers) due to the time required to start the emergency diesel generators. In these cases a 12-s diesel start time has been assumed. This includes the diesel initial sequencer loading step.

Offsite power has been assumed to be available where it maximizes the mass and energy released from the break due to:

A. The continued operation of the reactor coolant pumps, which maximizes the energy transferred from the RCS to the steam generator.

B. Continued operation of the feedwater pumps and actuation of the auxiliary feedwater system, which maximizes the steam generator inventories available for release.

6.2.1.4.1.6 Safety System Failures. In addition to assuming a LOSP, the following single active failures were considered.

- Loss of one emergency diesel.

- Failure of one main feedwater isolation valve.

The loss of one diesel results in the loss of one train of each of the containment heat removal systems. The analysis model conservatively accounted for the effects of the single failures by combining the failures into one bounding set of analyses. Two fast acting MSIVs are provided in each steam line. No more than one steam generator would experience an uncontrolled blowdown even if one of the MSIVs fails to close.

6.2.1.4.1.7 Steam Generator Reverse Heat Transfer and Reactor Coolant System Metal Heat Capacity. Once steam line isolation is complete, those steam generators in the intact steam loops become sources of energy that can be transferred to the steam generator with the broken line. This energy transfer occurs via the primary coolant. As the primary plant cools, the temperature of the coolant flowing in the steam generator tubes drops below the temperature of the secondary fluid in the intact units, resulting in energy being returned to the primary coolant. This energy is then available to be transferred to the steam generator with the broken steam line.

Similarly, the heat stored in the metal of the reactor coolant piping, the reactor vessel, and the reactor coolant pumps is transferred to the primary coolant as the plant cooldown progresses. This energy also is available to be transferred to the steam generator with the broken line.

The effects of both the RCS metal and the reverse steam generator heat transfer are included in the results presented in this document.

6.2.1.4.2 Description of Blowdown Model

A description of the blowdown model used is provided in reference 6. This reference is the basis for the data shown in tables 6.2.1-61 and 6.2.1-62.

6.2.1.4.3 Containment Response Analysis

The COCO computer code (reference 8), which is discussed in paragraph 6.2.1.1.3, was used to determine the containment responses following the postulated main steam line breaks (MSLB). The following assumptions were made to obtain these responses.

6.2.1.4.3.1 Initial Conditions. The initial containment conditions are the same as those used in the containment response analysis for the postulated RCS pipe ruptures. (See table 6.2.1-2.)

6.2.1.4.3.2 Mass and Energy Release Data. Using references 6 and 7 as a basis, mass and energy release data were developed to determine the containment pressure-temperature response for the spectrum of breaks analyzed. Tables 6.2.1-61 and 6.2.1-62 provide the mass and energy release data for the cases which result in the highest temperature and pressure. Table 6.2.1-64 provides specific plant data used for each case.

The rate of auxiliary feedwater addition represents the maximum runout flowrate to a fully depressurized steam generator. The value given for mass added by feedwater pumping assumes that no reduction in feedwater pump turbine speed occurs following a MSLB and prior to main feedwater isolation. Determination of feedwater flowrates prior to isolation assumed that the feedwater regulating valve in the broken loop goes wide open while those in the intact loops remain in their prebreak positions. Actual isolation is dependent on signals generated by the primary protection system. Feedwater isolation for the split breaks was based on the time required to reach the containment pressure setpoint that generates the isolation signal. For the split breaks and small double-ended rupture, feedwater was conservatively assumed to match the steam flow until the time of feedwater isolation.

6.2.1.4.3.3 Containment Pressure-Temperature Results. Figures 6.2.1-26 through 6.2.1-29 provide curves of the resultant containment pressure-temperature analyses for the cases producing the highest peak containment pressure and temperature. Table 6.2.1-65 summarizes the results of all the cases analyzed and indicates the times at which dryout occurs and the various containment pressure setpoints are reached. The sequence of events following a postulated MSLB is listed in tables 6.2.1-66 and 6.2.1-67 for worst pressure and temperature cases.

As illustrated in figure 6.2.1-26, case 16 results in a peak pressure of 36.5 psig. This case represents the peak calculated containment pressure for the spectrum of breaks analyzed. The containment vapor temperature profile versus time for this case is provided in figure 6.2.1-27.

It is important to note that the peak calculated pressure is coincident with the completion of the blowdown of the contents of the affected steam generator. In all cases, the peak calculated containment pressure demonstrates considerable margin below the containment design pressure.

As illustrated in figure 6.2.1-29, case 13 results in a peak vapor temperature of 303.1°F. This case represents the peak calculated containment vapor temperature for the spectrum of breaks analyzed. The containment pressure profile versus time for this case is provided in figure 6.2.1-28.

6.2.1.5 Minimum Containment Pressure Analysis for Performance Capability Studies on Emergency Core Cooling System

The containment backpressure and temperature and the containment wall-condensing heat transfer coefficient, used for the limiting case CD = 0.6 double-ended cold leg guillotine break for the (ECCS) analysis found in subsection 15.6.5, are presented in figures 6.2.1-30, 31, and 32. The containment backpressure is calculated using the methods and assumptions described in Westinghouse-Emergency Core Cooling System Evaluation Model -Summary, WCAP-8339, Appendix A. Input parameters including the containment initial conditions, net free containment volume, passive heat sink materials, thicknesses, surface areas, starting time, and number of containment cooling systems used in the analysis are described below.

6.2.1.5.1 Mass and Energy Release Data

The break mass and energy releases to containment during the blowdown and reflood portions of the limiting break transient are presented in tables 6.2.1-68 and 6.2.1-69, respectively.

The mathematical models which calculate the mass and energy releases to the containment are described in subsection 15.6.5. Since the requirements of Appendix K of 10 CFR 50 are very specific in regard to the modeling of the RCS during blowdown and since the models used are in conformance with Appendix K, no alterations to those models have been made in regard to the mass and energy releases. A break spectrum analysis is performed. (See the double-ended cold leg guillotines which affect the mass and energy released to the containment.) This effect is considered for each case analyzed. During refill, the mass and energy released to the containment is assumed to be zero, which minimizes the containment pressure. During reflood, the effect of steam-water mixing between the safety injection water and the steam flowing through the RCS intact loops reduces the available energy released to the containment vapor space and, therefore, tends to minimize containment pressure.

6.2.1.5.2 Initial Containment Internal Conditions

The following initial values were used in the analysis:

Containment pressure (psia)	14.7
Containment temperature (°F)	90
Refueling water storage tank temperature (°F)	40
NSCW temperature (°F)	40
Outside temperature (°F)	17

The containment initial conditions of 90°F and 14.7 psia are representatively low values anticipated during normal full-power operation. The initial relative humidity was conservatively assumed to be 99%.

6.2.1.5.3 Containment Volume

The volume used in the analysis is 2.95×10^6 ft^3 plus an additional amount to incorporate the effect of containment purge, resulting in a total containment volume of 3.20×10^6 ft^3.

6.2.1.5.4 Active Heat Sinks

The containment spray system and the containment fan coolers operate to remove heat from the containment. Pertinent data for these systems which were used in the analysis are presented in table 6.2.1-71. The heat removal capability of each fan cooler is presented in table 6.2.2-2.

Because the fan coolers use nuclear service cooling water (NSCW), the lowest normal NSCW temperature (40°F) was used in the analysis.

The containment sump temperature was not used in the analysis because the maximum peak cladding temperature occurs prior to initiation of the recirculation mode for the containment spray system. In addition, heat transfer between the sump water and the containment vapor space was not considered in the analysis.

6.2.1.5.5 Steam-Water Mixing

Water spillage rates from the broken loop accumulator are presented in table 6.2.1-70.

6.2.1.5.6 Passive Heat Sinks

The passive heat sinks used in the analysis, with their thermophysical properties, are given in table 6.2.1-72. The passive heat sinks and thermophysical properties were divided in compliance with Branch Technical Position CSB6-1, Minimum Containment Pressure Model for Pressurized-Water Reactor (PWR) ECCS Performance Evaluation.

6.2.1.5.7 Heat Transfer to Passive Heat Sinks

The condensing heat transfer coefficients used for heat transfer to the steel containment structures are given in figure 6.2.1-32 for the limiting break. The containment temperature transient for the limiting break is shown in figure 6.2.1-31.

6.2.1.5.8 Other Parameters

Operating containment purge at the onset of the double-ended cold leg guillotine break was shown to cause a drop in containment pressure of less than 0.19 psi. An increase in the containment volume was incorporated into the analysis, as discussed in paragraph 6.2.1.5.3. No other parameters have a substantial effect on the minimum containment pressure analysis.

6.2.1.5.9 Standard Review Plan Evaluation

The VEGP does not employ the heat transfer coefficients supplied in the Standard Review Plan.

The heat transfer coefficients were calculated in conformance with WCAP-8339, Appendix A, which has received Nuclear Regulatory Commission approval. This reference is for the COCO code and predates CSB6-1. Results using the heat transfer coefficients have been found acceptable in other plant applications.

6.2.1.6 Instrumentation Requirements

Adequate instrumentation is provided to monitor the conditions inside the containment and actuate the appropriate engineered safety features, should those conditions exceed the predetermined levels. The instruments measure the containment pressure, containment atmosphere radioactivity, purge exhaust effluent radioactivity, and containment hydrogen concentration.

The containment pressure is measured by four independent Q-class pressure transmitters and fed into the engineered safety features actuation system (ESFAS) as described in subsection 7.3.1. Upon detection of excessively high pressure inside the containment, the appropriate safety actuation signals are generated which automatically activate the necessary safety systems. These physically separated pressure transmitters are located outside the containment and connected to their sensors by filled and sealed hydraulic lines. Refer to section 7.3 for a detailed description.

The containment atmosphere radiation level is monitored by four independent Q-class area monitors located at the operating deck inside the containment building. The measurements of monitors RE-0002 and RE-0003 are continuously fed into the ESFAS logic and, in modes 1 through 4, cause the actuation of containment ventilation isolation (CVI) safety signals, should the measured radiation levels exceed their setpoints. In mode 6 during core alterations or movement of irradiated fuel assemblies in containment, two channels of radiation monitors are required operable to provide input to control room alarms to ensure prompt operator action to manually close the containment purge and exhaust valves. In addition, the containment purge exhaust air radiation is measured by a non-Q-class, three-channel airborne effluent monitor located outside the containment in the purge exhaust duct. Its isolated output is also fed into the ESFAS logic and, in modes 1 through 4, causes actuation of the CVI signal when the monitored radiation level exceeds the setpoint. In mode 6 during core alterations or movement of irradiated fuel assemblies, only operable radiation monitors are required to alert the operators of the need for containment ventilation isolation. Manual isolation using individual valve hand switches following a radiation alarm is the means for isolating containment in the event of a fuel handling accident during shutdown. For detailed information on the containment area radiation monitors, effluent monitor, and the ESFAS operation, refer to subsections 12.3.4 and 11.5.2 and section 7.3, respectively.

The containment hydrogen concentration is measured by two redundant non-Q-class hydrogen monitors as described in subsection 6.2.5. The readouts and alarms are provided in the control room to facilitate manual actuation of safety-related hydrogen control systems by the operator, should it become necessary.

6.2.1.7 References

1. Bordelon, E. T., “Murphy Containment Pressure Analysis Code (COCO),” WCAP-8326, July 1974.
2. Tagami, Takashi, "Interim Report on Safety Assessments and Facilities, Establishment Project in Japan for Period Ending June, 1965 (No. 1)."
3. Ranz, E. W., and Marshall, W. R., Jr., "Evaporation for Drops." Chemical Engineering Progress, 48141-146, March 1952.
4. Parsly, L. F., "Spray Tests at the Nuclear Safety Pilot Plant," in "Nuclear Safety Program Annual Progress Report for Period Ending December 31, 1971," ORNL 4647, p. 182, Oak Ridge National Laboratory, 1971.
5. Shepard, R. M., et al. "Westinghouse Mass and Energy Release Data for Containment Design," WCAP-8264-PA, June 1975 (Proprietary) and WCAP-8312-A, Revision June 1975 (Nonproprietary).
6. Land, R. E., "Mass and Energy Releases Following a Steam Line Rupture," WCAP-8822, (Proprietary) and WCAP-8860 (Nonproprietary), September 1976.
7. Burnett, T. W. T., et al., "LOFTRAN Code Description," WCAP-7907-A, April 1984.
8. Bordelon, F. M., and Murphy, E. T., “Containment Pressure Analysis Code (COCO),” WCAP-8326, June 1974.
9. Shepard, R. M., et al., "Westinghouse Mass and Energy Release Data for Containment Design," WCAP-8264-P-A, June 1975 (Proprietary), and WCAP-8312-A, Revision 2, August 1975 (Nonproprietary).

10. WCAP-10325-A, "Westinghouse LOCA Mass and Energy Release Model for Containment Design - March 1979 Version," April 1979.

11. "Westinghouse ECCS Evaluation Model - 1981 Version," WCAP-9220-P-A, Rev. 1, February 1982 (Proprietary), WCAP-9221-A, Rev. 1.

12. EPRI 294-2, "Mixing of Emergency Core Cooling Water with Steam 1/3 Scale Test and Summary," (WCAP-8423), Final Report, June 1975.

13. ANSI/ANS-5.1-1979, "American National Standard for Decay Heat Power in Light Water Reactors," August 1979.

14. NUREG-1137, "Safety Evaluation Report related to the Operation of Vogtle Electric Generating Plant, Units 1 and 2," Docket Nos. 50-424 and 50-425, Page 3-25, June 1985.

15. WCAP-10033, "Dynamic Analysis of Reactor Pressure Vessel For Postulated Loss-of-Coolant Accidents: Alvin W. Vogtle Nuclear Plant Units 1 and 2," pages 3-22, 3-23, 4-4, and 5-1, December 1981.

16. NUREG-1137, Supplement 7, "Safety Evaluation Report related to the Operation of Vogtle Electric Generating Plant, Units 1 and 2," Docket Nos. 50-424 and 50-425, January 1988.

17. Letter from T. A. Reed, of the NRC, to W. G. Hairston III, of the Georgia Power Company, "Vogtle Unit 1 Safety Evaluation on Pressurizer Surge Line Thermal Stratification (NRC Bulletin 88-11) (TAC 72178)," April 12, 1990.

6.2.2 CONTAINMENT HEAT REMOVAL SYSTEMS

The functional performance objective of the containment heat removal system as an engineered safety features (ESF) system is to reduce the containment temperature and pressure following a loss-of-coolant accident (LOCA) or main steam line break (MSLB) accident inside the containment by removing thermal energy from the containment atmosphere. These cooling systems also serve to limit offsite radiation levels by reducing the pressure differential between the containment atmosphere and the external environment, thereby diminishing the driving force for leakage of fission products from the containment to the atmosphere. The containment heat removal systems include the containment cooling system and the containment spray system. The containment cooling system described here also functions during normal operation to maintain a suitable atmosphere for the equipment located within the containment.

6.2.2.1 Containment Cooling System

6.2.2.1.1 Design Bases

6.2.2.1.1.1 Safety Design Bases.

A. The containment cooling system is designed to withstand the effects of natural phenomena such as earthquakes, winds, tornadoes, or floods.

B. The containment cooling system is automatically placed in operation on receipt of a safety injection (SI) signal following a LOCA or MSLB accident.

C. The containment cooling system is designed so that a single failure of any active component, assuming loss of offsite power, cannot impair the capability of the system to perform its safety function.

The capability of isolating components, systems, or piping is provided, if required, so that the system's safety function will not be compromised.

D. Active components of the containment cooling system are capable of being tested during plant operation. Provisions are made for inspection of major components at appropriate times specified in American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code Section XI.

E. The containment cooling system components required to mitigate the consequences of an accident are designed to remain functional in the accident environment and to withstand the dynamic effect of the accident.

F. The containment cooling system in conjunction with the containment spray system is capable of removing sufficient thermal energy and subsequent decay heat from the containment atmosphere following the postulated LOCA or MSLB accident to maintain the containment pressure below design values.

G. The containment cooling system is designed and fabricated to codes consistent with Regulatory Guide 1.26 as described in table 3.2.2-1 and Seismic Category 1 in accordance with Regulatory Guide 1.29. The power supply and control functions are in accordance with Regulatory Guide 1.32.

6.2.2.1.1.2 Power Generation Design Bases. The containment cooling system is designed to limit the ambient containment air temperature during normal plant operation to 120°F with any four of the eight coolers operating. Normal operation and the power generation design bases are discussed in section 9.4.

6.2.2.1.2 System Design

6.2.2.1.2.1 General Description. The containment cooling units are designed to the codes and standards identified in table 3.2.2-1; flood design is discussed in section 3.4; missile protection is discussed in section 3.5. Protection against dynamic effects associated with the postulated rupture of piping is discussed in section 3.6. Environmental design and equipment qualification is discussed in section 3.11. The actuation system is discussed in section 7.3.

6.2.2.1.2.2 System Description. The containment cooling system consists of eight separate 25% fan cooler units inside the containment. The system piping and instrumentation diagram is given in drawing 1X4DB212. The location and arrangement of the fan coolers are indicated in drawings 1X4DJ4103, 1X4DJ4113, 1X4DJ4123 and 1X4DJ4133. The system is designed to permit periodic testing and inspection as discussed in paragraph 6.2.2.1.4. Table 6.2.2-1 provides a tabulation of the design and performance data for each containment cooling unit. The containment fan coolers reject heat to the nuclear service cooling water (NSCW) system, which is described in subsection 9.2.1.

6.2.2.1.2.3 Component Description. Each 25%-capacity fan cooling unit consists of a vane axial fan, a fan motor, copper-nickel cooling coils with copper fins, a carbon steel housing, round metal ducting, and a concrete discharge duct. The 8 units are located at el 238 ft in the containment building and discharge to the bottom of the containment building through four concrete ducts. Each fan unit has two speeds of operation, high speed for normal operation and low speed for post-accident operation. For purposes of heat removal from the containment atmosphere, any combination of four fans in slow speed is sufficient.

The containment cooling system is separated into two trains consisting of four fan cooler units each. The two trains are supplied cooling water and electrical power from the corresponding train of the NSCW system and the Class 1E electrical power system.

Each concrete air supply duct contains three large outlets located above the maximum calculated containment flood elevation to supply cool air to the containment. The air cooling unit enclosures and concrete air supply duct remain intact following a design basis accident (DBA). There is a backdraft damper located at the inlet of each concrete air supply duct to prevent overpressurizing the ducting and coolers. The capability to remain intact is discussed in relation to the hydrogen mixing function of the containment cooling system in paragraph 6.2.5.2.1. Plan and elevation drawings of the containment showing the routing of airflow guidance ductwork are given in drawings 1X4DJ4103, 1X4DJ4113, 1X4DJ4123 and 1X4DJ4133. The fans and motors are designed to operate in the containment post-accident environment. The heat removal capability of the containment fan coolers versus containment temperature for the maximum NSCW inlet temperature is provided in table 6.2.2-2.

6.2.2.1.2.4 System Operation. The containment cooling system is an ESF system that is in use during normal plant operation with four fans operating at high speed. The containment isolation valves on the NSCW line to the containment air cooling units are normally open. System ESF operation is initiated automatically upon receipt of an SI signal. Upon receipt of an SI signal, all fans are restarted at low speed. The basis for the setpoint for the automatic initiation of the cooling system is the diesel start time (12 + 0.5 s) and the load sequencer for the fan coolers (30.5 s). The containment cooling system is fully operational within 40.5 s following receipt of an SI signal. The containment air cooling units can be stopped and started from the control room and from the shutdown panels.

6.2.2.1.3 Safety Evaluation

A. The safety-related portions of the containment cooling system are located in the containment building. This building is designed to withstand the effects of natural phenomena such as earthquakes, winds, tornadoes, or floods.

B. Operation of the containment cooling system is initiated automatically following the receipt of an SI signal. Use of this signal provides a reliable indication that a LOCA or a MSLB accident has occurred inside containment. Operation of the containment air coolers may also be manually initiated from the control room and from the shutdown panels. A detailed description of the actuation system is contained in section 7.3.

C. Two trains, each containing four air cooling units, are supplied from redundant emergency power sources and redundant NSCW trains. Failure of any component in one train will not affect the operability of the other train. The containment analyses of LOCA and MSLB accidents were performed in

subsection 6.2.1, assuming the availability of one of two containment air cooling trains. A failure modes and effects analysis of the containment cooling system is presented in table 6.2.2-3.

D. Capability is provided to periodically test the entire startup sequence of the containment air cooling system. Active components can be tested periodically during plant operation to verify operability. The entire system can be inspected during unit shutdown. Additional information is contained in section 3.1, paragraph 6.2.2.2.4, and the Technical Specifications.

E. The containment air cooling units are tested and demonstrated to perform in a simulated MSLB and LOCA environment. The units are located in a manner to minimize the effects of jet impingement and pipe whip in case of a high-energy line break.

F. The analyses show that the containment cooling system in conjunction with the containment spray system is capable of removing sufficient heat energy and subsequent decay heat from the containment atmosphere to ensure the accident peak pressure is below the containment design pressure. Accident analyses assume the occurrence of a single failure that results in the loss of one air cooling train and one containment spray train. Containment cooler heat removal capacity is provided in table 6.2.2-2.

   Containment accident analysis assumes a constant NSCW temperature equal to the highest anticipated system temperature (95°F) to maximize the calculated containment peak pressure. The assumptions used in calculating this temperature are discussed in subsections 9.2.1 and 9.2.5.

   Curves showing heat removal rates of the containment air cooling system, and containment total pressure, and temperatures as a function of time for minimum ESF performance are given in the figures of subsection 6.2.1.

   Location of the containment air cooling units in different quadrants of the containment, the difference in elevation between suction and discharge points, and the significant flowrate developed (table 6.2.2-1) ensure adequate circulation in the containment following a LOCA which prevents the formation of localized high temperature air pockets or areas of high combustible gas concentration. The mixing capability of this system is supplemented by the containment spray system and natural convection. Additional information is contained in subsection 6.2.5 and section 9.4.

G. The containment cooling system is designed to Seismic Category 1 requirements as specified in section 3.2.

6.2.2.1.4 Testing and Inspection

Fans are tested and rated by the manufacturer in accordance with the standards of the Air Moving and Conditioning Association.

The containment air cooling units are tested and/or analyzed by the manufacturer to ensure operation in the post-accident environmental conditions indicated in section 3.11 and following a safe shutdown earthquake (SSE).

The preoperational testing of the containment air cooling system, as well as its components, demonstrates the initial capability of the equipment. Written test procedures establish minimum acceptance values for all tests. See section 14.2 for details of the preoperational test program.

Following completion of the preoperational integrated leakage rate tests, fans are operated at reduced speed, and the motor currents are monitored with temporarily attached ammeters to verify satisfactory operation in denser than normal atmospheres.

The fans are run and monitored during plant shutdown in accordance with the planned maintenance program.

6.2.2.1.5 Instrumentation Requirements

Containment cooling system controls and instrumentation are discussed in sections 7.3 and 7.5, respectively. The system is designed to function automatically following receipt of an SI signal. Fans and cooling water supply can also be controlled remotely from the control room and from the shutdown panels.

The status of the containment cooling system is displayed in the control room. The NSCW and emergency power supplies are discussed in subsection 9.2.1 and section 8.3, respectively.

6.2.2.2 Containment Spray System

6.2.2.2.1 Design Bases

6.2.2.2.1.1 Safety Design Bases.

A. The containment spray system is designed to withstand the effects of natural phenomena such as earthquakes.

B. The containment spray system is automatically placed in operation on receipt of two out of four containment pressure (high-3) signals.

C. The containment spray system is designed so that a single failure of any active component, assuming loss of offsite power, cannot impair the capability of the system to perform its safety function during the injection phase.

A single active or passive failure cannot impair the capability of the system to perform its safety function during the recirculation phase.

D. Active components of the containment spray system are capable of being tested during plant operation. Provisions are made for inspection of major components at appropriate times specified in ASME Boiler and Pressure Vessel Code, Section XI.

E. The containment spray system components are designed to remain functional during the accident environment and to withstand the dynamic effect of the accident.

F. The containment spray system in conjunction with the containment cooling system is capable of removing sufficient thermal energy and subsequent decay

heat from the containment atmosphere following the postulated LOCA or MSLB accident to maintain the containment pressure below design values.

G. The containment spray system is designed and fabricated to codes consistent with Regulatory Guide 1.26 as described in table 3.2.2-1 and Seismic Category 1 in accordance with Regulatory Guide 1.29. The power supply and control functions are in accordance with Regulatory Guide 1.32.

6.2.2.2.1.2 Power Generation Design Bases. The containment spray system has no power generation design bases.

6.2.2.2.2 System Design

6.2.2.2.2.1 General Description. The containment spray system is designed to the codes and standards identified in table 3.2.2-1; flood design is discussed in section 3.4; missile protection is discussed in section 3.5. Protection against dynamic effects associated with the postulated rupture of piping is discussed in section 3.6. Environmental design and equipment qualification is discussed in section 3.11. The actuation system is discussed in section 7.3.

6.2.2.2.2.2 System Description. The containment spray system, shown schematically in drawing 1X4DB131, consists of two pumps, spray ring headers and spray nozzles, valves, and connecting piping. Initially, water from the refueling water storage tank (RWST) is used for the containment spray followed by water recirculated from the containment emergency sump. The recirculated spray is mixed with trisodium phosphate in the containment sump region.

At the RWST empty level alarm the operator should initiate manual switchover of the containment spray pumps to the recirculation mode of operation. See table 6.3.2-7 for a summary of the necessary manual actions for switchover. Adequate transfer allowance is provided to allow the operator to perform the switchover sequence without securing the containment spray pumps. The total amount of borated refueling water injected into the containment by the charging, safety injection, residual heat removal, and containment spray pumps will provide a sump pH of 7.5 or above when mixed with the contents of the trisodium phosphate baskets.

A single failure can occur which may prevent the switchover of one of the two trains. However, a single failure cannot prevent the switchover of both trains simultaneously. The containment pressure transient analysis shows that only one of the two redundant spray trains is necessary to prevent containment pressure from reaching the containment design point. Thus, even if one train is not available following the switchover, the remaining operating train is sufficient to control containment pressure, assuming that four of the eight containment fan coolers are also in operation.

A gas accumulation monitoring and trending process for the Vogtle Unit 1 and 2 ECCS and containment spray systems has been established to meet the requirements of NRC Generic Letter 2008-01.

6.2.2.2.2.3 Component Description. The mechanical components of the containment spray system are described in this section. Component design parameters are given in table 6.2.2-4. Parts of the system in contact with borated water are stainless steel or an equivalent corrosion-resistant material.

Corrosion tests have been performed on the materials that the spray would come in contact with, e.g., the paint on the inside of the containment structure. (Tests are detailed in WCAP-7825 and NUREG CR-3803.) These tests have shown that no significant amount of corrosion products is produced. Those corrosion products or any chemical precipitation of appreciable size that does occur is trapped by the sump filter screen. The screen size is smaller than the line piping, residual heat removal heat exchanger tubes, and the spray nozzles, so that particles which could potentially block the system will be filtered out. The spray nozzle material (stainless steel, SA351) was chosen for its resistance to corrosion. Tests have been performed on this material in the same type of environment that the nozzle would see during spray actuation. (Corrosion tests of austenitic stainless steel are detailed in WCAP-7803 and WCAP-11611.) The resulting corrosion levels were very low.

Stress corrosion does not present a problem since it only becomes a factor under a combination of conditions of high stress levels at high temperatures for extended periods of time. The stress levels in the pumps during operation are relatively low; the working temperature of the pumps is less than 212°F.

The pumps are located outside the containment. The external surfaces of the pumps, as well as the pump motors, are not subjected to the corrosive atmosphere of the spray solution. Only the internal stainless steel surfaces of the pumps are exposed to a corrosive atmosphere.

6.2.2.2.2.3.1 Refueling Water Storage Tank. This tank serves as a source of emergency borated cooling water for injection and containment spray. It is normally used to fill the refueling canal for refueling operations. During all other plant operating periods, it is aligned to the suction of the emergency core cooling pumps and the containment spray pumps. The tank is a concrete tank lined with type 304 stainless steel plates. The nominal tank volume is 715,000 gal. The contents of this tank are protected from freezing by a sludge mixing system which includes an electric circulation heater. Lines and appurtenances to the RWST serving a safety-related function are heat traced as necessary to prevent freezing.

6.2.2.2.2.3.2 Containment Spray Pumps. The containment spray pumps are of the horizontal centrifugal type, driven by electric motors powered from the emergency buses.

The design head of the pumps is sufficient to continue at rated capacity with a minimum level in the RWST against a head equivalent to the sum of the design pressure of the containment, the head to the uppermost nozzles, line losses, and nozzle pressure losses. The containment spray system is designed so that adequate net positive suction head (NPSH) is provided to the containment spray pumps, in accordance with Regulatory Guide 1.1.

To demonstrate that adequate NPSH is provided for the containment spray pumps, it is only necessary to demonstrate that the NPSH is adequate under the worst limiting conditions. NPSH for the containment spray pumps is evaluated for both the injection and recirculation modes of operation for the DBA. The recirculation mode of operation gives the limiting NPSH requirement for the containment spray pumps, and the NPSH available is determined from the following equation:

$$NPSH_{actual} = (h)_{containment\ pressure} - (h)_{vapor\ pressure} + (h)_{static\ head} - (h)_{loss}$$

To evaluate the adequacy of the available NPSH, the debris generation from a high energy line break within the containment and the resultant impact on the containment spray pump performance was evaluated using the following.

A. The minimum flood level inside the containment based on the RWST water discharged during injection and switchover to recirculation, plus the water volume of 3 accumulator tanks, is at 177 ft 0 in. The minimum flood level also takes into account the flooding of the reactor cavity/incore instrument tunnel via the penetrations in the primary shield.

B. The containment spray pump suction elevation is 121 ft 5 in.

C. The calculated maximum line losses which include losses through pipe fittings, valves, entrance and exit are 10.3 ft at 3200 gal/min, containment spray pump design flow.

D. The quantity of insulation debris generated by the double-ended rupture of the RCS hot leg at its connection to the steam generator, the limiting case, is 432 ft^3.

E. Hydraulic model studies performed on a scale model of the VEGP containment emergency sump configuration showed that the approach velocities to the four sumps during two train operation was essentially the same. Therefore, it is assumed that the volume of debris transported and deposited on one sump screen is one quarter of the total debris generated by the postulated pipe break event. For one RHR and CS pump train operating the volume of debris transported and deposited on one CS sump screen is 41.6% of the total debris.

   However, the volume of debris transported to the sumps is also evaluated as a pump flow-weighted ratio. The most limiting case is reflected in paragraph 6.2.2.2.2.3.2.

F. The maximum containment spray emergency sump screen head loss, assuming 41.6% of the total debris generated is evenly deposited on the sump screen, is less than 16.65 ft.

G. The vapor pressure of the pumped liquid is assumed to be in equilibrium with the containment ambient pressure (i.e., no credit is taken for subcooling of the sump fluid) for sump fluid temperatures of equal to or greater than 211°F:

$$h_{containment\ ambient\ pressure} = h_{vapor\ pressure}$$

The NPSH equation for the recirculation mode when suction is taken from the containment emergency sump becomes:

$$NPSH_{available} = h_{static\ head} - h_{line\ loss} - h_{sump\ screen\ loss}$$

Using the above values, the calculated containment spray pump available NPSH is greater than 36 ft. The required NPSH at 3200 gal/min is 19.5 ft. Therefore, adequate available NPSH margin is provided for proper containment spray pump operation.

Design parameters for these pumps are presented in table 6.2.2-4.

6.2.2.2.2.3.3 Spray Nozzles. The hollow cone spray nozzles are not subject to clogging by particles less than 1/4 in. in size and produce a drop size spectrum with a mean diameter of less than 700 mm at 40 psi differential pressure. During spray recirculation operation, the water is screened through perforated plates with 3/32-in. diameter holes (a small%age – 124 holes – are larger than

3/32-in. diameter but < ¼-in. diameter – see section 6.1.2.) before leaving the containment emergency sump. With the spray pump operating at design conditions and the containment at design pressure, the pressure drop provided across the nozzles exceeds 40 psi. The spray nozzles used in the construction of the containment spray system are designed to withstand differential pressures in excess of those expected to occur as a result of an accident.

The spray nozzles are stainless steel and have a 3/8-in.-diameter orifice.

6.2.2.2.2.3.4 Spray Additive Tank. The spray additive tank has been abandoned in place.

6.2.2.2.2.3.5 Spray Additive Eductors. The spray additive eductors serve only to maintain the spray system pressure boundary integrity.

6.2.2.2.2.3.6 Piping, Valves, and Containment Sumps. The piping used in the construction of the containment spray system is designed to withstand differential pressures in excess of those expected to occur as a result of an accident. The containment sump recirculation lines that run from the containment sump to the containment spray pumps are enclosed within guard pipes from the containment emergency sump floor to the first valve outside the containment. The pipe and guard pipe are capable of withstanding containment pressure and temperature. The guard pipe prevents leakage from the containment if the recirculation line ruptures.

The two containment ECCS sumps are designed in accordance with the requirements of Regulatory Guide 1.82. These sumps are located in a manner that protects them from the effects of high-energy line breaks, and they are separated from each other. The elevation of the containment emergency sumps are selected to allow optimum use of the available coolant. The sump intakes are protected by vertically stacked disk screens. The size of the opening in the screens is based on the minimum flow area through the components that receive coolant from the emergency sumps.

Analyses were conducted to ensure that effects such as reduction of NPSH and screen blockage will not result in degraded pump or system performance. The screens are installed in a manner to facilitate inspection of the structures and pump suction intakes.

There are two Containment Spray (CS) sumps. A screen is installed on each sump. The CS screens are composed of four stacks of 14 disks that are 30-in. long by 30-in. wide by ~40-in. high, four of which provide 590 ft^2 of perforated plate area and 133 ft^2 of circumscribed surface area per sump. Each typical screen disk is a welded assembly of two perforated plates and their structural support components. The screens are designed to withstand the loading for the largest postulated debris pieces, types, and amount. The plate-hole (perforation) diameter of the screen is 3/32 in. (a small%age - 124 - holes are larger than 3/32-in.diameter but none are larger than ¼-in. diameter – see section 6.1.2.) The screen is mounted over the containment sump. Because of these dimensions, it is not considered credible that a screen can plug sufficiently to impede pump suction. In the remote case that particles did traverse the screen, they would pass through the piping pumps and valves as well as the 3/8-in. diameter containment spray nozzle openings without difficulty. The screens bolt to the floor and may be removed by unbolting individual screen sections for inspection during shutdown periods, or the dedicated inspection port may be used.

Water sprayed in the refueling canal from the containment sprays may escape back to the elevation of the emergency sump through two 12-in. drain pipes located at the lowest point of

the refueling canal. The water passes from the canal to a passageway on the containment floor. Spray water from 33 out of 171 nozzles during 1-train operation or a maximum of 66 out of a total of 342 nozzles from 2 containment spray trains in operation falls into the refueling canal for a maximum canal fill rate of approximately 500 gal/min and 1000 gal/min, respectively. This represents less than 20% of the total spray rate in either case.

The drain layout is such that each 12-in. drain line is capable of passing approximately 2000 gal/min; therefore, there is no danger of starving the sump via the refueling canal. The drain piping is isolated during refueling and left open during normal reactor operation. Plant refueling procedures ensure that these drain pipes are opened after refueling prior to plant startup.

One containment spray pump is provided for each train. A single failure therefore leaves one of the two trains in service. The containment spray pumps are located within compartments sealed by watertight doors; a postulated rupture in one train cannot flood the other.

The first of the two motor-operated valves in series on the recirculation lines are totally enclosed in protective chambers, ensuring that all liquid escaping from a damaged or leaky valve does not escape to the outer environment. All materials that can come in contact with recirculation fluid are austenitic stainless steel. The spray headers are located in the proximity of the containment liner of the dome. The spray headers are anchored to the concrete through the liner plate. Each system is designed for SSE and appropriate thermal and dead weight loading conditions. The spray system is designed for maximum coverage in the containment, with the nozzles located and oriented so that the spray will not be blocked by structures or equipment.

6.2.2.2.2.3.7 Motors for Pumps and Valves. The motors for the containment spray system components will be designed in accordance with specifications discussed for the motors in the SI system. (See section 6.3.)

6.2.2.2.2.3.8 Electrical Supply. Details of the emergency bus power sources are discussed in chapter 8.

6.2.2.2.2.4 System Operation. The spray system is actuated by a signal initiated manually from the control room or automatically on coincidence of two of four containment pressure (high-3) signals. These signals start the containment spray pumps and open the discharge valves to the spray headers.

During all modes of operation except refueling, the suction of the pumps is normally aligned to the RWST. The spray pumps continue to draw a suction on the RWST until the later stages of the injection phase. After the ECCS is realigned from injection to recirculation, and when the RWST level reaches empty, the spray pump suction is remote-manually shifted to the containment emergency sumps.

6.2.2.2.2.5 pH Control. Containment spray could be operated for a short period of time and recirculation will never occur. This is during the secondary line break or inadvertent containment spray actuation events. Since the containment spray can be operated only in the injection phase without any recirculation, the spray will be acidic (4.5 pH). If containment spray is actuated and terminated prior to recirculation, a controlled cleanup and inspection of equipment in containment should begin within 5 days of the event. This is necessary to assess and recover from the effects of the acidic containment sprays on equipment in containment prior to reaching unacceptable conditions.

The initial containment spray pH will be approximately 4.5 during the injection phase which can last up to approximately 4.5 hours. After switchover to the recirculation phase, the spray pH will be between 7.5 and 10.5. If the containment spray system is actuated during a LOCA, its operation should continue in the injection phase and through switchover and into the recirculation phase. Operation of the containment spray system in the recirculation mode is necessary to ensure that the containment is resprayed with the solution (7.5 to 10.5 pH) from the emergency sumps. Once this respray occurs, then the containment sprays can be terminated, provided that containment pressures have already decreased to acceptable levels.

The pH of the sump solution will be adjusted to greater than 7.5 within 8 hours of the accident (LOCA) initiation to counteract the buildup of chloride concentrations to critical levels. This is considered a conservatively short period in which to make the pH adjustment, even with the potential for rapidly increasing sump chloride concentrations. The materials in containment are qualified for long term exposure to a high pH solution and will not be adversely affected by short term exposure to a low pH solution.

A summary of procedural requirements for containment spray operation follows:

A. If containment spray is actuated and terminated prior to recirculation, a controlled cleanup and inspection of equipment in containment should begin within 5 days of the event.

B. If containment spray was actuated and has been operated in the recirculation mode, then the operation in the recirculation mode is required for a duration of the 1.5 hours prior to the termination of containment spray to ensure desired pH of the spray solution is reached.

C. If containment spray is actuated and a primary LOCA is indicated, continuous spray for a minimum duration of 2 hours is required. This should include or be followed by operation in the recirculation mode for at least 1.5 hours.

6.2.2.2.3 Safety Evaluation

A. The components of the containment spray system are located inside the Category 1 auxiliary and containment buildings, except for the RWST, which is a Category 1 concrete tank. These buildings and this tank are designed to withstand the effects of natural phenomena such as earthquakes, tornadoes, or floods.

B. Operation of the containment spray system is automatically initiated by a coincident two-out-of-four containment (high-3) pressure signal. Use of this safety-related signal provides a reliable indication of a DBA occurring inside the containment. Operation of the spray system may also be manually initiated from the control room. A detailed description of the actuation system is contained in section 7.3.

C. Two trains, each containing a spray pump, spray eductor, and spray header, are supplied borated water from the RWST. The spray pump and motor-operated valves in each train are powered from the corresponding train of onsite emergency power. Failure of any active component in one train will not affect the operability of the redundant train. The containment analyses of LOCA and MSLB accidents performed in subsection 6.2.1 assume the availability of only one train of the containment spray system. The results of a failure modes and effects analysis are provided in table 6.2.2-5.

D. The containment spray system components are qualified to operate in the LOCA and MSLB environments through testing or analysis. Sections 3.9, 3.10, and 3.11 discuss the qualification of mechanical and electrical components.

E. The analyses described in subsection 6.2.1 show that the spray system in conjunction with the containment coolers is capable of removing sufficient heat energy and subsequent decay heat from the containment atmosphere to ensure the accident peak pressure is below the containment design pressure. The analyses assume the operation of only one train of containment coolers and containment spray.

F. The containment spray system is designed to Category 1 and Quality Group B requirements as shown in table 3.2.2-1.

As stated in 1.9.26.2, Westinghouse classifies components according to ANSI N 18.2A-1975. The spray additive tanks have been abandoned in place.

The minimum fall path of water droplets is conservatively assumed to be the distance from the lowest spray ring to the operating deck. Heat transfer calculations presented in chapter 15 show that essentially all spray droplets reach thermal equilibrium at containment design temperature and pressure in a distance considerably less than the minimum fall path. For a detailed description of the analytical methods and models used to assess the performance capability of the containment heat removal systems, refer to the containment integrity analysis presented in chapter 15.

6.2.2.2.4 Testing and Inspection

6.2.2.2.4.1 Containment Spray System.

6.2.2.2.4.1.1 Inspections. The containment spray system is designed to permit periodic determination of proper functioning to demonstrate system readiness as specified in the Technical Specifications.

The pressure containing systems are inspected for leaks from pump seals, valve packings, flange joints, and safety valves during system testing. The components of the system outside the containment are accessible for leaktightness inspection during periodic flow tests.

Examinations/inspections of the pressure retaining piping welds should be performed in accordance with the requirements of applicable codes and standards.

6.2.2.2.4.1.2 Preoperational Testing. The objective of preoperational testing is to:

A. Demonstrate that the spray nozzles in the containment spray header are clear of obstructions by passing air through the test connections.

B. Verify that the proper sequencing of valves and pumps occurs on initiating the containment spray signal, and demonstrate the proper operation of remotely operated valves.

C. Verify the operation of the spray pumps; each pump is run at minimum flow, and the flow is directed through the normal path back to the RWST. During this time, the miniflow is measured to verify required flow.

6.2.2.2.4.1.3 Operational Testing. The objective of the periodic testing is to:

A. Verify that the proper sequencing of valves and pumps occurs on initiation of the containment spray signal, and demonstrate the proper operation of remotely operated valves.

B. Verify the operation of the spray pumps; each pump is run at minimum flow, and the flow is directed through normal path back to the RWST.

C. Automatic transfer of powered components to the emergency diesel generators can be demonstrated during an integrated system test conducted when the plant is cooled down and the residual heat removal system is in operation.

6.2.2.2.4.2 Containment Spray Additive Subsystem (Abandoned in place).

6.2.2.2.5 Instrumentation Application

The status of the containment spray system is displayed in the control room. Using a combination of alarms and monitor lights, the operator is alerted to a maloperation of this equipment both during normal operation and post-accident conditions.

Operation of the containment spray system is demonstrated by monitoring spray water temperature and RWST level. Locally mounted pressure indicators are provided at the spray pumps' suction and discharge to verify pump performance.

The activation signal generating equipment fully meets Institute of Electrical and Electronic Engineers Standard 279 as to operation, diversity, separation of power supplies (diesels, instrument power), etc.; details are discussed in chapter 7.

6.2.2.3 References

1. Deleted.

2. Deleted.

6.2.3 SECONDARY CONTAINMENT FUNCTIONAL DESIGN

Based on the performance of the fission product removal and control systems discussed in section 6.5 and the acceptable radiological consequences following a loss-of-coolant accident or fuel handling accident inside the containment presented in chapter 15.0, a secondary containment is not required for VEGP.

6.2.4 CONTAINMENT ISOLATION SYSTEM

The containment isolation system consists of the piping, valves, and actuators required to isolate the containment following a loss-of-coolant accident (LOCA), steam line rupture, fuel handling accident inside the containment, small breaks in the reactor coolant system (RCS), or releases of radioactivity from systems within the containment. The design of the containment isolation system satisfies the requirements of TMI Action Plan Task II.E.4.2 as described in the following paragraphs.

6.2.4.1 Design Bases

Protection of the containment isolation system from wind and tornado effects is discussed in section 3.3. Flood design is discussed in section 3.4. Missile protection is discussed in section 3.5. Protection against dynamic effects associated with the postulated rupture of piping is discussed in section 3.6. Environmental design is discussed in section 3.11.

6.2.4.1.1 Safety Design Bases

A. In the event of a LOCA, the containment isolation system provides isolation of lines penetrating the containment which are not required for operation of the engineered safety features (ESF) systems to minimize the release of radioactive materials to the atmosphere.

B. Upon failure of a main steam line, the containment isolation system isolates the steam generators as required to prevent excessive cooldown of the RCS or overpressurization of the containment.

C. To control the release of radioactivity to the outside atmosphere, the containment isolation system isolates the containment atmosphere following a fuel handling accident inside the containment.

During refueling operations, manual containment ventilation isolation (CVI) is permitted, along with open personnel and emergency air lock doors. Manual CVI capability, using individual valve hand switches, is performed during these refueling operations as described in paragraph 6.2.4.5.

D. The containment isolation system is designed in accordance with 10 CFR 50, Appendix A, General Design Criterion 54.

E. Each line which penetrates the containment and which either is a part of the reactor coolant pressure boundary (RCPB) or connects directly to the containment atmosphere or does not meet the requirements for a closed system as defined in item F below, except instrument sensing lines, is provided with containment isolation valves in accordance with 10 CFR 50, Appendix A, General Design Criteria 55 and 56.

F. Each line which penetrates the containment and is neither part of the RCPB nor connected directly to the atmosphere of the containment and which satisfies the requirements of a closed system is provided a containment isolation valve in accordance with 10 CFR 50, Appendix A, General Design Criterion 57. A closed system is not a part of the RCPB nor connected directly to the atmosphere of the containment and meets the following additional requirements:

1. The system is protected against missiles and the effects of high energy line break.
2. The system is designed to Seismic Category 1 requirements.
3. The system is designed to American Society of Mechanical Engineers Section III, Class 2 requirements.
4. The system is designed to withstand temperatures at least equal to the containment design temperature.
5. The system is designed to withstand the external pressure from the containment structural acceptance test.

6. The system is designed to withstand the design basis accident transient and environment.

G. The containment pressure transmitters and reactor vessel level instrumentation system (RVLIS) are designed in accordance with Nuclear Regulatory Commission (NRC) Regulatory Guide 1.141. Six containment pressure sensors are provided as sealed systems with bellow seals inside the containment, liquid filled capillaries between the seals, and the sensing element outside containment. RVLIS consists of six level sensors and has bellow seals inside the containment, liquid filled capillaries between the seals, and a secondary isolator seal outside the containment between the containment penetration and the transmitter. These instrument lines are closed systems both inside and outside containment, are designed to withstand the containment pressure and temperature conditions following a loss of coolant accident, and are designed to withstand dynamic effects.

H. The containment isolation system is designed to remain functional following a safe shutdown earthquake (SSE).

6.2.4.1.2 Power Generation Design Basis

The containment isolation system as a whole has no power generation design basis. Power generation design bases associated with individual components of the containment isolation system are discussed in the section describing the system of which they are an integral part.

6.2.4.2 System Description

6.2.4.2.1 General Description

Each piping system which penetrates the containment is provided with containment isolation features which serve to minimize the release of fission products following a design basis accident. Provisions are made to allow for passage of emergency fluid through the boundary following a postulated accident. Figure 6.2.4-1 provides the arrangement for each piping penetration. NRC Standard Review Plan 6.2.4 and Regulatory Guide 1.141 provide acceptable alternative arrangements to the explicit arrangements given in General Design Criteria 55, 56, and 57. Each penetration is designed so that in the event that a single failure is postulated, the containment integrity is maintained. Table 6.2.4-1 lists each penetration and provides a summary of the containment penetration/isolation valve information.

For those systems which have automatic isolation valves or for which remote-manual isolation is provided, paragraph 6.2.4.5 describes the power supply and associated actuation system. Power-operated (air, motor, electrohydraulic, or solenoid) containment isolation valves have position indication in the control room.

Two modes of valve actuation are considered in table 6.2.4-1. The actuation signal which occurs directly as a result of the event initiating containment isolation is designated as the primary actuation signal. The post-accident valve position is a consequence of the primary actuation signal. If a change in valve position is required at any time following primary actuation, a secondary actuation signal is generated which places the valve in an alternative position. The closure times for automatic containment isolation valves are provided in table 6.2.4-1. Containment isolation valves required to be operable by Technical Specification

3.6.3, Containment Isolation Valves, are demonstrated operable with isolation times as shown in FSAR table 6.2.4-2.

The containment purge system is designed in accordance with Branch Technical Position CSB 6-4 as described in table 9.4.6-4. As described in subsection 9.4.6, the 14-in. minipurge lines may be open during normal plant operation and are provided with isolation valves capable of 5-s closure against the peak calculated containment pressure following a LOCA. The 24-in. purge lines are open only during a cold shutdown condition and are provided with an isolation valve capable of 10-s closure. An analysis of the radiological consequences and the effect on the containment backpressure due to the release of containment atmosphere are discussed in chapter 15 and paragraph 6.2.1.5, respectively.

In the event of a LOCA, the secondary shield wall and other protective features prevent any missiles or high energy line break effects from damaging or degrading the performance capability of the containment isolation system. Sections 3.5 and 3.6 discuss in detail the missiles and pipe break effects, and section 3.8 discusses the internal structures, including the secondary shield wall. The actuators for power-operated containment isolation valves inside the containment are located above the maximum anticipated containment water level. In addition, lines associated with those penetrations which are considered closed systems inside the containment are protected from the effects of a LOCA.

Provisions are made to ensure that closure of the containment isolation valves is not inhibited by entrapped debris in the valve body. For the majority of the systems, the fluid is demineralized water; thus, process fluid quality does not affect valve operation. For containment minipurge lines, screens are provided in the lines inboard of the isolation valves. For the containment sump lines, including the containment emergency sump, screens are provided to prevent large debris from entering the system.

Other defined bases for containment isolation are provided in NRC Standard Review Plan 6.2.4 and Regulatory Guide 1.141. Conformance with Regulatory Guide 1.141 is provided to the extent specified in this section and in subsections 6.2.5 and 6.2.6. For the emergency core cooling system (ECCS) and containment spray system penetrations, the acceptability of the alternative arrangement relies upon provisions for the detection of possible leakage from these lines outside the containment. Subsection 9.3.3 describes the leak detection provisions that have been made in the plant drainage system. Other provisions, such as containment water level and system flow, temperature, and pressure instrumentation may be used by the operator.

The containment penetrations associated with the secondary side of the steam generators are not subject to General Design Criterion 57. The valves associated with these penetrations do not receive a containment isolation signal and are not credited with effecting containment isolation in the safety analyses. The barriers against fission product release to the environment are the steam generator tubes and the piping associated with the steam generators.

In addition to containment penetration isolation, table 6.2.4-1 also contains systems which are required for post-LOCA mitigation. Since these systems, such as the ECCS, perform additional safety-related functions, they are associated with ESF and are so indicated in table 6.2.4-1.

6.2.4.2.2 Component Description

Codes and standards applicable to the piping and valves associated with containment isolation are listed in table 3.2.2-1. Containment penetrations are classified as Quality Group B and Seismic Category 1.

Section 3.11 provides the post-LOCA environment that is used to qualify the operability of power-operated isolation valves located inside the containment.

The containment penetrations are designed to meet the stress requirements of NRC Branch Technical Position MEB 3-1 and the classification and inspection requirements of NRC Branch Technical Position APCSB 3-1, as described in section 3.6. Section 3.8 discusses the interface between the piping system and the containment liner.

6.2.4.2.3 System Operation

During normal operation, many penetrations are not isolated. Lines which are not required for the passage of emergency fluids are automatically isolated upon receipt of isolation signals, as discussed in paragraphs 6.2.4.3 and 6.2.4.5 and chapter 7. Essential lines which penetrate the containment are closed loops within the containment or provide flow paths into or out of the RCS and can be isolated by remote-manual operation when dictated by the emergency system functional requirements. Lines not in use during power operation are normally closed and remain closed under administrative control during reactor operation.

Upon detection of abnormal radioactivity levels indicative of a fuel handling accident during refueling or other release, the isolation valves in the containment purge system are closed to minimize any fission product release to the environment.

6.2.4.3 Design Evaluation

Safety evaluations are lettered to correspond to the safety design bases.

A. Containment isolation signals automatically isolate process lines which are nonessential as identified in table 6.2.4-1. Nonessential lines are those lines which are not required to mitigate or limit an accident and which, if required at all, would be required for long term recovery only; e.g., days or weeks following an accident.

   Lines which are required to mitigate an accident or which, if unavailable, could increase the magnitude of the event are designated as essential lines. Table 6.2.4-1 identifies the associated line as essential or nonessential and shows the automatic isolation signal for each penetration, if applicable.

   The containment isolation system utilizes diversity in the parameters sensed for the initiation of containment isolation. The two redundant train-oriented containment isolation phase A signals (CIA-A, CIA-B) are initiated on receipt of any of the following signals:

   1. Any signal initiating a safety injection:

      - Manual safety injection actuation.
      - High containment pressure (high-1).
      - Low steam line pressure.
      - Low pressurizer pressure.

   2. Manual containment isolation actuation.

B. Upon failure of a main steam line, the steam generators are isolated to prevent excessive cooldown of the RCS or overpressurization of the containment.

The two redundant train-oriented steam line isolation signals (SLI-A, SLI-B) are initiated upon receipt of any of the following signals:

1. High steam line pressure rate.
2. Low steam line pressure.
3. Containment high-2 pressure.
4. Manual actuation.

For main steam line breaks resulting in a high steam line pressure rate or containment high-2 pressure signal, only the main steam line isolation valves (MSIVs) and MSIV bypass valves are shut to prevent excessive cooldown of the RCS. When the main steam line break causes a low steam line pressure signal, a safety injection signal (followed by containment isolation) is generated as well as the steam line isolation signal.

The main steam line isolation valves, MSIV bypass valves, and piping are designed to prevent uncontrolled blowdown from more than one steam generator. The main steam line isolation valves and MSIV bypass valves will shut fully within 5 s after SLI is initiated. The blowdown rate is restricted by steam flow restrictors located within the steam generator outlet steam nozzles in each blowdown path. For main steam line breaks upstream of an isolation valve, uncontrolled blowdown from more than one steam generator is prevented by the isolation valves in the unaffected steam lines and by the isolation valve in the affected line. For main steam line breaks downstream of an isolation valve, blowdown from more than one steam generator is prevented by the main steam isolation valves on each main steam line.

Failure of any one of the above components relied upon to prevent uncontrolled blowdown of more than one steam generator will not permit a second steam generator blowdown to occur. Piping restraints and pipe whip barriers between the main steam lines prevent a rupture in one line from causing a blowdown from more than one steam generator. No single active component failure will result in the failure of more than one main steam isolation valve to operate. Redundant main steam isolation signals, described in section 7.3, are fed to redundant parallel activation cylinder vent valves and redundant series actuation cylinder air supply valves to ensure isolation valve closure in the event of a single isolation signal failure.

The effect on the RCS after a steam line break resulting in single steam generator blowdown and the offsite radiation exposure after a steam line break outside containment are discussed in detail in chapter 15. The containment pressure transient following a main steam line break inside containment is discussed in section 6.2.

C. The containment purge system is automatically isolated following an abnormal release of radioactivity in the containment by either of two redundant train-oriented containment ventilation isolation signals (CVI-A, CVI-B) generated upon receipt of any of the following:

1. Any signal resulting in a safety injection.
2. Containment high area radiation.
3. Containment high radioactive air particulate.

4. Containment high radioactive gas.
5. Containment high iodine concentration.
6. Manual actuation of either containment spray or containment isolation phase A.

The preaccess purge supply and exhaust valves in the 24-in. lines, which are only open in the cold shutdown condition, are designed to shut in less than 10 s. The minipurge line isolation valves, which may be open during normal operation, shut in less than 5 s.

D. The containment isolation system is designed in accordance with 10 CFR 50, Appendix A, General Design Criterion 54. Leakage detection capabilities and the leakage detection test program are discussed in subsection 6.2.6. Valve operability tests are also discussed in subsection 3.9.6. Redundancy of valves and reliability of the isolation system are ensured by conformance with the other safety design bases stated in section 6.2. Redundancy and reliability of the actuation system are covered in section 7.3.

The use of motor-operated valves which fail as is upon loss of actuating power in lines penetrating the containment is based upon the consideration of what valve position ensures the greatest plant safety. Furthermore, each of these valves that fails as is provided with redundant backup valves to ensure that no single failure will prevent the system as a whole from performing its isolation function; e.g., a check valve inside the containment and motor-operated valve outside the containment or two motor-operated valves in series, each powered from a separate ESF bus.

E. Lines which penetrate the containment and which either are part of the RCPB, connect directly to the containment atmosphere, or do not meet the requirements for a closed system, except instrument sensing lines, are provided with one of the following valve arrangements conforming to the requirements of 10 CFR 50, Appendix A, General Design Criteria 55 and 56, as follows:

1. One locked closed isolation valve inside and one locked closed isolation valve outside containment.
2. One automatic isolation valve inside and one locked closed isolation valve outside containment.
3. One locked closed isolation valve inside and one automatic isolation valve outside containment. (A simple check valve is not used as the automatic isolation valve outside containment.)
4. One automatic isolation valve inside and one automatic isolation valve outside containment. (A simple check valve is not used as the automatic isolation valve outside containment.)

Isolation valves outside containment are located as close to the containment as practical, and upon loss of actuating power, air-operated automatic isolation valves fail closed.

F. Each line which penetrates the containment and is neither part of the RCPB nor connected directly to the containment atmosphere and satisfies the requirements of a closed system has at least one containment isolation valve which is either automatic, locked closed, or capable of remote-manual operation. The valve is outside the containment and located as close to the containment as practical. A

simple check valve is not used as the automatic isolation valve. This design is in compliance with 10 CFR 50, Appendix A, General Design Criterion 57.

G. Instrument lines penetrating the containment and the containment pressure instrument lines are designed in accordance with NRC Regulatory Guide 1.141.

H. The containment isolation system is designed in accordance with Seismic Category 1 requirements as specified in section 3.2. The components (and supporting structures) of any system, equipment, or structure which is non-Seismic Category 1 and whose collapse could result in loss of a required function of the containment isolation system through either impact or flooding are analytically checked to determine that they will not collapse when subjected to seismic loading resulting from an SSE.

Air-operated isolation valves fail in the shut position upon loss of air if they are not required to operate after a design basis accident. Containment isolation system valves required to be operated after a design basis accident are powered by the Class 1E electric power system.

6.2.4.4 Tests and Inspections

Preoperational testing is described in chapter 14. The containment isolation system is testable through the operational sequence that is postulated to take place following an accident, including operation of applicable portions of the protection system and the transfer between normal and standby power sources.

The piping and valves associated with the containment penetration are designed and located to permit preservice and inservice inspection in accordance with ASME Section XI, as discussed in section 6.6.

Each line penetrating the containment is provided with testing features to allow containment leak rate tests in accordance with 10 CFR 50, Appendix J, as discussed in subsection 6.2.6.

6.2.4.5 Instrumentation Application

The generation of CIA or CVI signals which automatically isolate the appropriate containment isolation valves is described in section 7.3.

For those valves for which automatic closure is not desired, based on the system safety function, remote-manual operation is available from the control room.

Containment isolation valves which are equipped with power operators and which are automatically actuated may also be controlled individually by positioning hand switches in the control room. Also, in the case of certain valves with actuators, a manual override of an automatic isolation signal is installed to permit manual control of the associated valve. The override control function can be performed only subsequent to resetting of the actuation signal; that is, deliberate manual action is required to change the position of containment isolation valves in addition to resetting the original actuation signal. The design does not allow ganged reopening of the containment isolation valves. Reopening of the isolation valves must be performed on a valve-by-valve basis, or on a line-by-line basis. Safety injection signals take precedence over manual overrides of other isolation signals, for example, a safety injection signal causes isolation valve closure even though the high radiation signal is being overridden by the operator. Overrides are input to the system status monitoring panel, described in subsection 7.5.5. Containment isolation valves with power operators are provided with

open/closed indication, which is displayed in the control room. The valve mechanism also provides a local, mechanical indication of valve position.

In mode 6 during core alterations and movement of irradiated fuel assemblies inside containment, automatic or system-level manual initiation CVI capability no longer applies. During these refueling operations, manual containment ventilation isolation is permitted, along with open personnel and emergency air lock doors. Manual CVI capability, using individual valve hand switches, is performed during these refueling operations. The following conditions apply:

- One personnel air lock door and one emergency air lock door must be operable, and
- At least 23 feet of water is maintained above the reactor vessel flange, and
- A designated individual is available to close the doors. The emergency air lock will not normally be open during core alterations or fuel movement inside containment. Therefore, in the event the emergency air lock is open at the same time the personnel air lock is open, a separate individual shall be responsible for closing the emergency air lock (within 15 minutes) in addition to the individual designated to close the personnel air lock.

All power supplies and control functions necessary for containment isolation are Class 1E, as described in chapters 7 and 8.

6.2.5 COMBUSTIBLE GAS CONTROL IN CONTAINMENT

Technical Specifications (TS) Amendment Number 134/113 eliminated the requirements regarding containment hydrogen recombiners and relaxed the requirements for hydrogen monitors. The hydrogen recombiners and monitors have been deleted from the TS. The hydrogen monitors are included in the post accident monitoring instrument program.

Following a loss-of-coolant accident (LOCA), hydrogen may be produced inside the reactor containment by radiolysis of the core and sump solutions, by corrosion of aluminum and zinc, by reaction of the Zircaloy fuel cladding with water, and by release of the hydrogen dissolved in the reactor coolant and contained in the pressurizer vapor space. To ensure that the containment hydrogen concentration is maintained at a level low enough to preclude endangering containment integrity, a combustible gas control system is provided. This subsection describes the systems that are provided in accordance with General Design Criterion 41 to control the buildup of hydrogen within the containment.

Five mechanisms for monitoring and controlling hydrogen inside the containment are considered in the VEGP design:

- Hydrogen recombiners.
- Post-LOCA containment hydrogen purge.
- Post-LOCA cavity hydrogen purge.
- Containment hydrogen monitoring.
- Containment hydrogen mixing.

6.2.5.1 Design Bases

6.2.5.1.1 Electric Hydrogen Recombiners

The following design bases apply to the electric hydrogen recombiners:

A. The recombiners are designed to sustain all normal and accident loads including safe shutdown earthquake and pressure transients from a design basis LOCA.

B. The recombiners are designed for a lifetime of 40 years, consistent with that of the plant.[a]

C. All materials used in the recombiners are selected to be compatible with the environmental conditions inside the reactor containment during normal operation or during accident conditions.

D. Process capacity is such that the containment hydrogen concentration will not exceed 4 volume% based on the Nuclear Regulatory Commission (NRC) TID release model as indicated in Regulatory Guide 1.7.

E. Two redundant electric hydrogen recombiners are provided to meet the single-failure criterion.

6.2.5.1.2 Post-LOCA Containment Hydrogen Purge System

A. The containment post-LOCA purge exhaust system constitutes a 100% single train backup to the hydrogen recombiner. It is designed as Seismic Category 1 except for ductwork inside the containment upstream of the containment isolation valve and ductwork at the filter inlet.

B. The post-LOCA containment pressure provides the motive force to purge the containment.

C. This system may be used post-LOCA to maintain hydrogen levels in the containment below 4 volume% in conjunction with a portable air compressor through the Seismic Category 1 portion of the service air piping.

D. The redundant motor-operated isolation valves located inside the containment are in parallel and powered from the 480-V, Class 1E buses. All other system components are powered from the normal ac bus. The outside isolation valve is locked closed and manually controlled by the operator. The isolation valves inside the containment have position indication in the control room and are normally closed.

E. The system, including the isolation valves and the filtration unit, is designed to be manually started for operation.

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years, resulting in a plant operating life of 60 years. In accordance with 10 CFR Part 54, appropriate aging management programs and activities have been initiated to manage the detrimental effects of aging to maintain functionality during the period of extended operation (see chapter 19).

6.2.5.1.3 Post-LOCA Cavity Hydrogen Purge System

A. The post-LOCA cavity purge system prevents hydrogen pocketing in the reactor cavity after a LOCA by supplying air to the reactor cavity for dilution.

B. The post-LOCA cavity purge system is capable of accomplishing its function with the single failure of an active component.

C. The post-LOCA cavity purge system is capable of performing its function with Class 1E power, each redundant train being connected to separate Class 1E safety buses.

D. The system is designed to automatically start upon receipt of a safety injection signal.

E. The system is designed to meet Seismic Category 1 requirements.

6.2.5.1.4 Containment Hydrogen Monitoring System

A. The hydrogen monitoring system is designed as a Class 1E, Seismic Category 1 system. It is designed to retain its integrity and operability following a design basis accident (DBA).

B. All materials and equipment required by this system are selected to be compatible with the environmental conditions anticipated during accident operation and are suitable for a lifetime consistent with that of the plant[a].

C. The system samples containment air, providing the means to measure the containment hydrogen concentration and to alert the operator in the event that a high hydrogen concentration is detected, in accordance with the requirements of Regulatory Guide 1.7.

D. The hydrogen monitoring system consists of two identical units that are completely independent of each other and are powered from independent Class 1E power sources. Assuming a single failure and compensatory operator actions from the control room, capability is available to monitor the hydrogen concentration in the containment.

E. Proper shielding and other provisions are incorporated into the design to ensure that personnel exposure does not exceed the limits of General Design Criterion 19 and that the required radiological analysis can be performed on the containment air sample.

6.2.5.1.5 Containment Hydrogen Mixing

The following design bases apply to mechanisms or systems for mixing of hydrogen-bearing gases inside the reactor containment:

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years, resulting in a plant operating life of 60 years. In accordance with 10 CFR Part 54, appropriate aging management programs and activities have been initiated to manage the detrimental effects of aging to maintain functionality during the period of extended operation (see chapter 19).

A. Local hydrogen concentrations inside the reactor containment shall be maintained at less than 4 volume%.

B. Any active systems required for mixing containment air should meet the same redundancy, environmental, seismic, and quality requirements as the hydrogen recombiner system as described in paragraph 6.2.5.1.1.

6.2.5.2 System Design

6.2.5.2.1 Electric Hydrogen Recombiners

The applicable codes and standards used in the design of the electric hydrogen recombiner are listed in table 6.2.5-1, and a typical electrical recombiner is shown in figure 6.2.5-1.

Each recombiner system consists of a control panel located in the control building, a power supply cabinet located on level B of the control building, and a recombiner located above the operating deck at el 261 ft in the containment. There are no moving parts or controls inside the containment. Heating air within the unit causes airflow by natural convection. The recombiner is a completely passive device.

To regulate the power supply to the recombiner, the power supply cabinet located in the control building contains an isolation transformer and a controller. This equipment will not be exposed to the post-LOCA environment. The controls for the power supply are located in the control building beside the power supply panel and are manually actuated.

Each hydrogen recombiner consists of the following components:

A. A preheater section, consisting of a shroud placed around the central heaters to take advantage of heat conduction through the central walls, for preheating incoming air.

B. An orifice plate to regulate the rate of airflow through the unit.

C. A heater section, consisting of four banks of metal-sheathed electric resistance heaters, to heat the air flowing through it to hydrogen-oxygen recombination temperatures.

D. An exhaust chamber which mixes and dilutes the hot effluent with containment air to lower the temperature of the discharge stream.

E. An outer enclosure to protect the unit from impingement by containment spray.

The hydrogen recombiner has no need for external services except electrical power.

The containment atmosphere is heated within the recombiner in a vertical duct, causing it to rise by natural convection. As it rises, replacement air is drawn through intake louvers downward through a preheater section which will temper the air and lower its relative humidity. The preheated air then flows through an orifice plate, sized to maintain a 100-sf^3/min flowrate, to the heater section. The airflow is heated to a temperature above 1150°F, the reaction temperature for the hydrogen-oxygen reaction. Any free hydrogen present reacts with atmospheric oxygen to form water vapor. After passing through the heater section, the flow enters a mixing section, which is a louvered chamber where the hot gases are mixed and cooled with containment atmosphere before the gases are discharged directly into the containment. The air discharge louvers are located on three sides of the recombiner. To avoid short-circuiting previously processed air, no discharge louvers are located on the intake side of the recombiner.

Tests have verified that the hydrogen-oxygen recombination is not a catalytic surface effect associated with the heaters (paragraph 6.2.5.4) but occurs due to the increased temperature of the process gases. As the phenomenon is not a catalytic effect, saturation of the unit cannot occur.

Two recombiners are provided to meet the requirements for redundancy and independence. Each recombiner is powered from a separate safeguard bus and is provided with a separate power panel and control panel. This system and the other safety-related subsystems are not interdependent.

The unit is manufactured of corrosion-resistant, high-temperature material. The electric hydrogen recombiner uses commercial-type electric resistance heaters sheathed with Incoloy-800, which is an excellent corrosion-resistant material for this service. The recombiner heaters operate at significantly lower power densities than similar heaters used in commercial practice.

The recombiner is operated manually from a control panel located in the control building. The recombiner, power supply panel, and control panel are shown schematically in figure 6.2.5-2. The power panel for the recombiner contains an isolation transformer and a controller to regulate power into the recombiner. This equipment is not exposed to the post-LOCA containment environment.

To control the recombination process, the correct power input to bring the recombiner above the threshold temperature for recombination is set on the controller. The correct power required for recombination depends upon containment atmosphere conditions and is determined when recombiner operation is required. A thermocouple readout instrument is also provided in the control panel to monitor temperatures in the recombiner.

6.2.5.3 System Design

6.2.5.3.1 Containment Hydrogen Purge System.

The post-LOCA containment hydrogen purge system is provided as a backup means of controlling hydrogen inside the containment. It provides a means of purging the hydrogen from the containment and is intended as a backup to the hydrogen recombiner system. The exhaust filters and exhaust duct to plant vent are designed as Seismic Category 1.

The system consists of an exhaust penetration line and a filtered exhaust system; it is shown in drawings 1X4DB213-1 and 1X4DB213-2. Design data for principal system components are presented in table 6.2.5-3. The containment isolation valves and interconnecting piping are Seismic Category 1; all other portions of the system are Seismic Category 2.

The purge exhaust filter unit includes in the direction of airflow:

- A demister.
- Electrical heating coil.
- High-efficiency particulate air (HEPA) prefilter.
- A 4-in. charcoal adsorber.
- A HEPA afterfilter.

The hydrogen purge exhaust filter unit is located in the equipment building, and the hydrogen purge intake point is located in the containment dome. The ductwork is fastened to the inside of the containment and routed through a containment penetration. The flowrate through the filter unit is 500 ft^3/min.

The system is actuated manually. The operator unlocks and opens the manual isolation valve located outside the containment. From the control room, the operator opens the remote-manual isolation valves located inside the containment. The outward purge flow is due to the pressure differential existing between the containment atmosphere and the environment. Periodically, the operator will dilute the containment atmosphere by charging air into the containment via the instrument and service air system.

6.2.5.3.2 Post-LOCA Cavity Purge System

The post-LOCA cavity purge system is designed to prevent hydrogen pocketing in the reactor cavity following a LOCA by supplying air to the reactor cavity for dilution of the hydrogen released in the cavity area. The system fans take a suction on the atmosphere within the generator compartments and discharge the air into the cavity above (el 193 ft 2 in.) and below (169 ft 9 in.) the reactor vessel nozzles. The air flows out of the cavity along the reactor vessel nozzles or through the ventilation openings surrounding the seal ring.

The system consists of two 100%-capacity fans. Each fan is powered from an independent Class 1E power supply. The fans automatically start on a safety injection signal. There is an independent discharge pipe which routes the dilution air from the fans to the cavity. A failure modes and effects analysis is provided in table 6.2.5-2. Distribution of the air in the cavity area is accomplished via common discharge headers. All portions of the system are designed to Seismic Category 1 requirements.

Design data for principal components are provided in table 6.2.5-4. The system is schematically shown in drawing 1X4DB214-2.

6.2.5.3.3 Containment Hydrogen Monitoring System

Each redundant hydrogen monitoring train in the hydrogen monitoring system consists of a hydrogen analyzer and two associated sample lines with solenoid-operated isolation valves inside and outside the containment. These sampling lines are designed to be free of water traps (runs where liquid could accumulate) and are equipped with sufficient heat tracing to prevent condensation from the sample being supplied to the analyzers.

After the sample has been analyzed, it is returned to the containment. The analyzers are located in accessible areas outside the containment. The hydrogen monitoring subsystem piping is in accordance with the criteria of Regulatory Guide 1.26, Quality Group B. Solenoid-operated isolation valves are arranged to obtain samples from two locations within the containment for each train. The operator may select either of these sampling points from the main control room.

The operation of the hydrogen gas analyzer is based on the measurement of thermal conductivity of the gaseous containment atmosphere sample. The thermal conductivity of the gas mixture changes in proportion to the changes in the concentration of the individual gas constituents of the mixture. The thermal conductivity of hydrogen is far greater (approximately seven times the thermal conductivity of air) than any other gases or vapors expected to be present. This operation of the hydrogen monitoring system is not limited due to radiation,

moisture, or temperature expected at the equipment location. The monitors are designed to function under design pressure conditions of -2 to 60 psig.

The containment hydrogen monitors are aligned for operation within 60 minutes after initiating safety injection following a LOCA. Accurate indication of hydrogen concentration is available within 30 min of initiating flow through the monitors. This is accomplished by operating the monitors in standby during normal plant operation. Therefore, indication of containment hydrogen concentration is available to the operators within 90 minutes of initiating safety injection following a LOCA.

The range of the monitors is 0 to 10 volume% with an accuracy of ±5.0% of scale.

The output signal of the hydrogen monitors is indicated and alarmed locally as well as indicated, recorded, and alarmed in the control room. In addition to the high hydrogen alarm, a common malfunction alarm is located in the control room to indicate loss of power, low gas pressure, low analyzer chamber temperature, analyzer cell failure, or high hydrogen concentration.

Design data for principal system components are presented in table 6.2.5-5. The system is schematically shown in drawings 1X4DB213-1 and 1X4DB213-2.

The hydrogen monitoring system meets the requirements of TMI Action Plan Task II.F.1 with the clarification that accurate indication of containment hydrogen concentration is available to the operators within 90 minutes of initiating safety injection following a LOCA.

6.2.5.3.4 Containment Hydrogen Mixing

Hydrogen mixing is facilitated by the containment fan coolers, which take suction from above the operating deck and discharge to the lower levels of the containment. Functional descriptions of the containment coolers are provided in subsections 6.2.2 and 9.4.6. A flow diagram for the containment coolers is provided in drawings 1X4DB251-1, 1X4DB252, and 1X4DB253-1.

In addition, the post-LOCA cavity purge system described in paragraph 6.2.5.2.3 is available for hydrogen mixing.

6.2.5.4 Design Evaluation

6.2.5.4.1 Hydrogen Production and Accumulation

6.2.5.4.1.1 Zirconium-Water Reaction. A major source of hydrogen immediately following a LOCA is caused by the reaction of the Zircaloy fuel cladding with water. The extent of the zirconium-water reaction depends upon the effectiveness of the emergency core cooling systems (ECCS). An evaluation of the VEGP ECCS shows the zirconium-water reaction to be less than 0.3%.

Zirconium reacts with steam according to the following equation:

$$Zr + 2\ H_2O \rightarrow Zr\ O_2 + 2\ H_2 + Heat$$

The hydrogen produced is calculated as follows:

$$\frac{2\ \text{lb}-\text{mole}\ H_2/\text{lb}-\text{mole}\ Zr}{91.22\ \text{lb}\ Zr/\text{lb}-\text{mole}\ Zr} = \frac{0.022\ \text{lb}-\text{mole}\ H_2}{\text{lb}\ Zr}$$

The NRC model suggested in Regulatory Guide 1.7 and Standard Review Plan 6.2.5 conservatively assumes a 1.5% zirc-water reaction (five times the maximum amount calculated in the ECCS evaluations).(1) There are approximately 45,914 lb of zirconium metal in the reactor core. The hydrogen produced by the reaction of 689 lb of zirconium is 15.15 lb-moles. This hydrogen is assumed to be immediately released to the containment atmosphere.

6.2.5.4.1.2 Radiolysis Core and Sump Solutions. Water radiolysis is a complex process involving reactions of numerous intermediates. However, the overall radiolytic process may be described by the equation:

$$H_2O \leftrightarrow H_2 + \frac{1}{2} O_2$$

An extensive program was conducted by Westinghouse to investigate the radiolytic decomposition of the core cooling solution following the DBA. During the investigation it became apparent that post-accident conditions in the containment create two distinct radiolytic environments. One environment exists inside the reactor vessel, where radiolysis can occur when energy emitted by decaying fission products in the fuel is absorbed by the solution pumped through the reactor to cool the core. The other environment exists outside the reactor vessel, in the containment sump solution, where radiolysis can also occur when decay energy emitted by dissolved fission products is absorbed by the sump solution. The two basic differences between the core environment and the sump environment that affect the rate of hydrogen production are the rate of energy absorption and the type of flow regime. The results of these investigations are discussed in reference 1.

The rate of hydrogen production by radiolysis depends upon the rate of energy absorption by the solution. A detailed analysis of energy deposition in the reactor core where decaying fission products are retained in the fuel shows that beta radiation (which represents roughly 50% of the total decay energy) is emitted at an energy level too low to permit its penetration of the fuel and cladding. As a result, roughly 50% of the total decay energy emitted by fission products in the fuel is absorbed by the fuel and cladding and therefore does not contribute significantly to the rate of energy absorption by the water. Furthermore, approximately 7% of the gamma energy is absorbed by the core solution; the rest is absorbed by the fuel, cladding, or other core components.

In the containment sump, where fission products are assumed to be dissolved in the sump solution, energy is emitted directly to the solution. Since the depth of the sump is relatively large compared to the penetrating capability of even gamma energy, effectively 100% of the decay energy of the fission products dissolved in solution is absorbed by the solution. The other significant difference between the core and sump environment is the type of flow regime to which the products of radiolysis are exposed.

Radiolytic decomposition of water is a reversible reaction. In the core, where the products of radiolysis are continuously flushed away by the circulation of cooling solutions, there is little chance for hydrogen and oxygen to accumulate. Consequently, recombination of hydrogen and oxygen is assumed not to occur because significant quantities of the two reactants are not available. The sump, however, is a relatively deep and static environment, where the products of radiolysis are removed by molecular diffusion. Experimental tests simulating sump conditions demonstrate that there is significant reverse reaction in the sump. Hence, there is an apparent reduction in the quantity of hydrogen produced per unit energy absorbed.

The results of Westinghouse and Oak Ridge National Laboratory studies indicate maximum hydrogen yields of 0.44 molecules per 100 eV for core radiolysis and 0.3 molecules per 100 eV for sump radiolysis. The results of these studies are published in references 2, 3, and 4. This analysis, based on the conservative recommendations of Regulatory Guide 1.7 and Standard Review Plan 6.2.5, assumes a hydrogen yield of 0.5 molecules per 100 eV of energy absorbed for both core and sump radiolysis.

The rate of hydrogen gas production from radiolysis has been evaluated using the methodology presented in Appendix A to Standard Review Plan 6.2.5, and the calculational assumptions detailed in Regulatory Guide 1.7. Table 6.2.5-6 provides a summary of the assumptions made in the analysis.

6.2.5.4.1.3 Corrosion of Metals and Paints in Containment. Following a LOCA, hydrogen may be produced inside the containment by corrosion of aluminum and zinc.

Extensive corrosion testing has been conducted to determine the behavior and compatibility of various materials with alkaline borate solution.[7][8][9] Metals tested included Zircaloy, Inconel, aluminum alloys, cupronickel alloys, carbon steel, galvanized carbon steel, and copper. The tests showed that only aluminum and zinc will corrode at a rate that will significantly add to the hydrogen accumulation in the containment atmosphere.

Aluminum is found in the containment as aluminum metal components. Zinc will be in the form of either galvanized steel, zinc metal, or zinc-based paint.

Aluminum corrosion may be described by the overall reaction:

$$2\,Al + 3\,H_2O \rightarrow Al_2O_3 + 3\,H_2$$

Three moles of hydrogen gas are produced for every two moles of aluminum that is oxidized. Approximately 0.0556 lb-moles of hydrogen gas are produced for each pound of aluminum corroded.

The corrosion of zinc may be described by the overall reaction:

$$Zn + H_2O \rightarrow Zn(OH)_2 + H_2$$

One mole of hydrogen gas is produced for each mole of zinc that is oxidized. Approximately 0.0153 lb-moles of hydrogen gas are produced for each pound of zinc corroded.

The time-temperature cycle (table 6.2.5-9) considered in the calculation of aluminum and zinc corrosion is a representation of the postulated post-accident containment temperature transient. The corrosion rates at the various temperatures are shown in table 6.2.5-10. With these corrosion rates and the baseline aluminum and zinc inventory given in table 6.2.5-6, the contribution of aluminum and zinc corrosion to the hydrogen accumulation in the containment following the DBA was calculated. No credit was taken for the protective shielding effects of insulation or enclosures; i.e., complete and continuous immersion in spray was assumed.

Calculations based on Regulatory Guide 1.7 are performed by increasing the aluminum corrosion rate during the final interval of the post-accident containment temperature transient (table 6.2.5-9) to 200 mils/year. The aluminum and zinc corrosion rates earlier in the accident sequence are shown in table 6.2.5-10.

In order to determine an allowable margin of increase for the zinc and aluminum inventories, the hydrogen accumulation in the containment following a LOCA was calculated for different cases of varying zinc and aluminum inventories. Three cases were evaluated. They included the case where no recombiners were in operation, the case where one recombiner is in operation starting at day 2, and the case where one recombiner is in operation at 3.5 v/o of hydrogen.

Based on the zinc and aluminum inventories provided by table 6.2.5-6, it was determined that the bounding case is 3 times the baseline zinc inventory or 3 times the baseline aluminum inventory or any combination thereof which satisfies the following equation:

$$\frac{N_{al}}{O_{al}} + \frac{N_{zn} + (N_{zbp} - O_{zbp})}{O_{zn}} \leq 3.8$$

where:

N_{al} = new total surface area of aluminum (ft^2)
O_{al} = baseline surface area of aluminum
O_{al} = 3607 ft^2 (table 6.2.5-6)

N_{zn} = new total surface area of zinc (ft^2)
O_{zn} = baseline surface area of zinc
O_{zn} = 182,053 ft^2 (table 6.2.5-6)

N_{zbp} = new total surface area of zinc-based paint (ft^2)
O_{zbp} = baseline surface area of zinc-based paint
O_{zbp} = 680,722 ft^2 (table 6.2.5-6)

Future additions of aluminum and zinc beyond those quantities identified in table 6.2.5-6 are documented and used in this equation to ensure that the limits of hydrogen generation are not exceeded.

The above equation is valid if at least one recombiner is turned on at a hydrogen concentration of 3.03 v/o (2 days), or less, following a LOCA. It is dependent on surface area only and not dependent on the total mass of zinc and aluminum in containment.

Although 4.0 v/o is the acceptance limit, the maximum H_2 v/o should not exceed 3.3 v/o to account for instrument inaccuracies.

6.2.5.4.1.4 Hydrogen in the Primary Coolant. During normal operation of the plant, hydrogen is dissolved in the reactor coolant and is also contained in the pressurizer vapor space. Following a LOCA, this hydrogen is assumed to be immediately released to the containment atmosphere. The maximum equilibrium quantity of hydrogen from this source is 1600 sf^3.

The pressurizer vapor space hydrogen is based on the following:

A. Reactor coolant hydrogen concentration of 50 cm^3 (STP)/kg of coolant.

B. Normal pressurizer heaters turned on 50% of the time and all of the heat going to the boiling water.

C. Bypass spray rate of 1 gal/min.

D. Normal liquid level in pressurizer (60%).

E. Pressurizer relief valves closed.

6.2.5.4.1.5 Hydrogen Mixing. Experiments (references 10 through 16) demonstrate that for the period of high hydrogen evolution during and following blowdown, bulk turbulence and natural convective transport will be available to distribute and diffuse hydrogen throughout the containment.

Following this period, long-term mixing within and between the lower volumes of the containment and the region above the operating deck will be provided by the containment sprays (if operating) and the emergency containment coolers. The reactor vessel head vent system will provide for the release of any concentrated hydrogen from the primary loop at a controlled rate and in a location so as to allow complete dispersion due to the natural diffusion tendencies of hydrogen and by augmented means such as the containment cooler discharge, which is ducted so as to maintain hydrogen concentration equilibrium between the upper and lower containment regions through forced convection.

6.2.5.4.1.6 Conclusions. Figure 6.2.5-5 shows bulk containment volume% hydrogen versus time following a LOCA. The results from the post-LOCA hydrogen generation calculation have determined that for the three cases of varying aluminum/zinc inventory with a single 95% efficient, 100-sf^3/min recombiner starting operation on the second day following a LOCA or when the containment hydrogen concentration reaches 3.03 v/o, the hydrogen concentration peaks below 3.5 v/o and is maintained well below 4.0 v/o, thus showing ample margin in the hydrogen control system.

The integrated hydrogen production for each source is shown in figure 6.2.5-7.

6.2.5.5 Tests and Inspections

6.2.5.5.1 Electric Hydrogen Recombiners

The electric hydrogen recombiners underwent extensive testing in the Westinghouse development program. These tests encompassed the initial analytical studies, laboratory proof-of-principal tests, and full-scale prototype testing. The full-scale prototype tests included the effects of:

- Varying hydrogen concentrations.
- Alkaline spray atmosphere.
- Steam effects.
- Convection currents.
- Seismic effects.

A detailed discussion of these tests is provided in references 17 through 24.

6.2.5.5.2 Post-LOCA Containment Hydrogen Purge System

Safety-related equipment is qualified by the vendor to meet the codes and standards required by the system classification. Functional testing is performed after installation but prior to plant startup to verify the system performance capability. Periodic testing of the system components will be performed in accordance with plant procedures.

6.2.5.5.3 Post-LOCA Cavity Hydrogen Purge System

Safety-related equipment is qualified by the vendor to meet the codes and standards required by the system classification. Functional testing is performed after installation but prior to plant startup to verify the system performance capability.

6.2.5.5.4 Post-Accident Hydrogen Monitoring System

Equipment for this system is vendor qualified to meet the codes and standards required by the system classification. Functional and preoperational testing is performed after installation and prior to plant startup to verify the system performance capability. Periodic testing of the system and the isolation and sample selector valves will be performed in accordance with plant procedures.

6.2.5.6 Instrumentation Requirements

6.2.5.6.1 Electric Hydrogen Recombiner

The recombiners do not require any instrumentation inside the containment for proper operation after a LOCA. The recombiners are started manually after a LOCA. The sampling system is used to obtain containment atmosphere samples that indicate when the recombiners or the venting system should be actuated. Control measures can be initiated when the hydrogen concentration reaches 3 volume%.

6.2.5.6.2 Post-LOCA Containment Hydrogen Purge System

Instrumentation and controls for this system are located outside of the containment in the equipment building or in the control room. Control switches and status indication for the fans, isolation valves, and control valves are provided in the control room. Indication monitoring the operation of the filter exhaust unit is also provided in the control room.

6.2.5.6.3 Post-LOCA Cavity Hydrogen Purge System

Control switches for manual operation of the fans and low-flow alarms for each discharge line are provided on the HVAC panel in the control room.

6.2.5.6.4 Containment Hydrogen Monitoring System

The control switches for the sample selector valves and containment isolation valves are located on the process control panel in the control room. Operation of the hydrogen analyzers is controlled remotely from the main control board. Hydrogen concentration is both indicated and recorded on the main control board.

6.2.5.7 Materials

The materials of construction for the hydrogen control systems are selected for their compatibility with the post-LOCA environment.

The major structural components of the hydrogen recombiners are manufactured from 300-Series stainless steel. Incoloy-800 is used for the heater sheaths and for other parts such as the heat duct, which operates at high temperature.

There are no radiolytic or pyrolytic composition products from these materials.

6.2.5.8 References

1. NRC Regulations:

   Regulatory Guide 1.7, Rev. 2, November 1978.

   Standard Review Plan 6.2.5, "Combustible Gas Control in Containment," July 1981.

   Branch Technical Position CSB 6-2, "Control of Combustible Gas Concentration in Containment Following a Loss-of-Coolant Accident."

2. Fletcher, W. D., Bell, M. J., and Picone, L. F., "Post-LOCA Hydrogen Generation in PWR Containments," Nuclear Technology 10, pp 420-427, 1971.

3. Zittel, H. E., and Row, T. H., "Radiation and Thermal Stability of Spray Solutions," Nuclear Technology 10, pp 436-443, 1971.

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9. Deleted.

10. Alan R. Barton Nuclear Plant, Preliminary Safety Analysis Report, paragraph 6.2.5.3.3.

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23. Wilson, J. F., "Electric Hydrogen Recombiner IEEE 323-1974 Qualification," WCAP-7709-L, Supplement 6 (Proprietary), and WCAP-7820, Supplement 6 (Nonproprietary), October 1976.

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6.2.6 CONTAINMENT LEAKAGE TESTING[a]

The reactor containment, containment penetrations, and containment isolation barriers are designed to permit periodic leakage rate testing as required by 10 CFR 50, Appendix A, General Design Criteria (GDC) 52, 53, and 54. The containment leak test requirements are outlined and the acceptance criteria for such tests are established in 10 CFR 50, Appendix J, Option B. The objective of the leakage rate testing is to ensure that the leakage from the containment is within the limits set by the Technical Specifications.

Compliance with 10 CFR 50, Appendix J, Option B, Types A, B, and C, testing is discussed in paragraphs 6.2.6.1, 6.2.6.2, 6.2.6.3, and 6.2.6.4.

6.2.6.1 Containment Integrated Leakage Rate Test (Type A Test)

The design leakage rate (L_d) for the containment is 0.2% free volume per day for the first 24 h. The actual leakage rate will be determined by using the methods and requirements of 10 CFR 50, Appendix J, Option B, for Type A tests.

[a] The 10 CFR 50 Appendix J Program is credited as a license renewal aging management program (see subsection 19.2.29).

The acceptance criteria specified in Appendix J for the integrated leakage rate test (ILRT) includes a margin for possible deterioration of the containment leakage integrity during the service intervals between tests. The measured leakage rate (L_{am}) will be less than 0.75 of the maximum allowable leakage rate value L_a.

6.2.6.1.1 ILRT Pretest Requirements

Several pretest requirements are to be met before the ILRT is performed. A general inspection of the accessible nonconcrete interior and exterior surfaces of the containment structures and components for any evidence of structural deterioration which may affect either the containment structural integrity or leaktightness will be made. The visual examination of containment concrete surfaces intended to fulfill the requirements of 10 CFR 50, Appendix J, Option B testing, will be performed in accordance with the requirements of and frequency specified by ASME Section XI Code, Subsection IWL, except where relief has been authorized by the NRC. At the discretion of the licensee, the containment concrete visual examinations may be performed during either power operation, e.g., performed concurrently with other containment inspection-related activities such as tendon testing, or during a maintenance/refueling outage. Any evidence of structural deterioration will be evaluated and corrected if necessary before the Type A test is performed.

Systems that are required to maintain the plant in a safe condition during the test, such as the nuclear service cooling water lines to the containment air coolers, are operable in their normal mode and need not be vented.

Systems that are normally filled with water and operating under post-accident conditions, such as the containment heat removal system, need not be vented or drained. Systems which are not vented to containment or drained during Type A tests are identified in table 6.2.6-1.

The steam generator tubes and shell and the associated piping systems passing through the containment liner are considered an extension of the containment. Therefore, the secondary side of the steam generator and connecting systems are not vented to the containment atmosphere. However, since the secondary side of the steam generator is a valid leak path, the main steam lines are vented outside of containment. The penetrations associated with the secondary side of the steam generator are identified in figure 6.2.4-1.

The reactor coolant drain tank, pressurizer relief tank, and accumulator tanks are vented to the containment atmosphere. This is done to protect the tanks from the external pressure of the test and to preclude leakage to or from the tanks which could affect the accuracy of the test results.

An equipment protection list is provided in the ILRT procedure to identify any components that cannot withstand the ILRT test pressure.

During preoperational testing, a structural integrity test (SIT) is performed. The SIT is a pressure test conducted to verify that the containment structural response due to the induced load is consistent with the predicted behavior. Paragraph 3.8.1.7 describes the SIT deflection measurements and concrete crack inspections.

Following the SIT, the preoperational ILRT is performed.

6.2.6.1.2 ILRT Test Method

The ILRT will be conducted in accordance with 10 CFR 50 Appendix J, Option B. A short duration ILRT may be performed in accordance with Bechtel's Topical Report BN-TOP-1. The

test procedure used during the preoperational Type A test is described in chapter 14. Drawing 1X4DB132 shows the test arrangement for a Type A test. Two permanent flowthrough tubes have been added in a spare electrical penetration (electrical penetration number 31) to be utilized for the flow verification and pressure sensing lines. After completion of the ILRT, the flowthrough tubes are sealed and the spare electrical penetration is restored to its original configuration and Type B tested. For penetrations which are exempt from Type B or C tests the leakage testing is accomplished by the Type A test, as noted in table 6.2.4-1.

Containment dry bulb temperature, pressure, and dewpoint temperature are periodically monitored during the test. These data are analyzed as they are taken so that the leakage rate and its statistical significance are known as the test progresses. Once the leakage rate has been found with sufficient accuracy, a known additional leak is imposed and the measurements are continued, giving additional verification of the leakage rate.

6.2.6.2 Containment Penetration Leakage Rate Tests (Type B Tests)

Containment penetrations whose design incorporates resilient seals, gaskets, or sealant compounds; airlocks and lock-door seals; equipment and access hatch seals; and electrical canister and modular type penetrations receive a preoperational and periodic Type B leakage rate test in accordance with 10 CFR 50, Appendix J, Option B.

Electrical penetrations are of modular- or canister-type design (66 modular and 6 canister) and their leakage testing provisions were designed and initially tested to meet the requirements of Institute of Electrical and Electronics Engineers (IEEE) 317. Each of the 66 modular-type electrical penetrations consists of a single header plate sealed to a nozzle on the containment exterior by double O-rings with interspace connection. Feedthrough modules carrying conductors are sealed into the header plate by a metal compression fitting assembly with interspace connection. The feedthrough conductors are sealed to the feedthrough module by double, high-temperature, thermoplastic seals with interspace connection. The seal interspaces (header plate O-rings, feedthrough module compression fittings, and thermoplastic seals) are pressurized with nitrogen; the pressure is periodically monitored to detect leakage. Each of the canister-type electrical penetrations is similar in design to the modular type, except that the canister type extends through the length of the nozzle with header plates at each end, through which the conductor modules are sealed and terminate in electrical ceramic bushings. Each of the 72 electrical penetrations is provided with a local pressure gauge.

Expansion bellows utilized on the fuel transfer tube penetration accommodate relative movement between the refueling pool liner and the containment penetration and do not form part of the containment pressure boundary. Expansion bellows utilized on the containment emergency sump suction line guard pipes accommodate relative movement between the auxiliary and fuel handling buildings with respect to the containment and do not form part of the containment pressure boundary.

The equipment hatch, personnel lock, and escape hatch doors are fitted with double seals with an interspace test connection.

The test connection for the equipment hatch is located inside the containment. Test connections for the personnel lock and escape hatch doors are located such that testing is accomplished without entering the containment. Two pressure gauges are provided, one inside the lock which penetrates the bulkhead at the inner airlock door to measure containment pressure and one located outside the airlock to penetrate the bulkhead at the outer airlock door to read lock pressure. The handwheel shafts are provided with double seals and test connections where the shafts penetrate the airlock bulkheads.

The containment leak rate penetrations (table 6.2.4-1, penetrations 64A, 64B, 68, and 87) are provided with blind flanges that have testable double O-ring seals and are Type B tested. The eddy current/sludge lancing penetrations (table 6.2.4-1, penetrations 5, 55, and 90) are provided with blind flanges that have O-ring type joints per ANSI B16.5 with valve test connections and are Type B tested.

Type B tests are conducted at calculated peak post-LOCA containment internal pressure (P_a). The acceptance criteria and leakage rate limits are given in the Technical Specifications. Test methods and equipment are described in paragraph 6.2.6.3.

6.2.6.3 Containment Isolation Valve Leakage Rate Tests

Containment isolation valves are Type C tested in accordance with 10 CFR 50, Appendix J, Option B.

The process piping, instrumentation tubing, and personnel access penetrations are listed in table 6.2.4-1. Figure 6.2.4-1 shows the location of all test vent and drain connections and the normal direction of flow.

The CIVs for each piping penetration and process isolation valve for those piping penetrations where no GDC is applicable are tabulated in table 6.2.4-1, together with the test type.

Type B and C tests are performed by local pressurization utilizing either the pressure decay or flowmeter method. For the pressure decay method, the test volume is pressurized with air or nitrogen to at least P_a. The rate of decay of pressure of the known free air test volume is monitored to calculate the leakage rate. In the flowmeter method, pressure is maintained in the test volume by makeup air, nitrogen, or water (if applicable) through a calibrated flowmeter. The flowmeter fluid flowrate is the isolation valve leakage rate.

The leakage will be measured in the same direction as would occur in an accident, unless it can be determined that leakage measured in a different direction will provide an equivalent or more conservative result. The test medium used for pressurization is determined by the valve's post-accident condition. Valves which could be exposed to the containment atmosphere subsequent to an accident will be tested with air or nitrogen. Valves which are in lines designated to be filled with a liquid for at least 30 days subsequent to an accident may be leakage-rate tested with water.

Type C testing of the safety injection lines, residual heat removal lines, high head safety injection lines of the chemical and volume control system, RCP seal injection lines, the containment emergency sump lines to the residual heat removal and containment spray pumps, and nuclear service cooling water lines to and from the containment fan coolers is not performed. The justification for this is that these valves are either normally open at the time of a LOCA or are opened at some time after the accident to effect immediate and long term core cooling. These systems are closed systems outside containment except for the NSCW system which is a closed system inside containment, designed and constructed to ASME III, Class 2 and Seismic Category I requirements, and as such they do not constitute a potential containment atmosphere leak path during or following a loss-of-coolant accident with a single active failure of a system component. Should the valves leak slightly when closed, the fluid seal within the pipe or the closed piping system outside/inside containment would preclude release of containment atmosphere to the environs. Furthermore, inservice testing and inspection of these isolation valves and the associated piping system outside the containment is performed periodically under the inservice inspection requirements of ASME XI as described in subsection 3.9.6 and section 6.6. During normal operation, the systems are water filled, and

degradation of valves or piping is readily detected. Containment penetrations not vented to containment or drained during Type A testing are identified in table 6.2.6-1.

The steam generator and associated secondary system piping that form the primary barrier to the outside, much the same as the containment liner plate, are subjected to Type A test as shown in table 6.2.4-1. The barriers against fission product release to the environment are the steam generator tubes and piping associated with the steam generators.

Containment pressure monitoring lines are considered an extension of the containment boundary, and therefore the isolation valves are not Type C tested.

Isolation valves will be positioned to their post accident position by the normal method with no accompanying adjustments. Exercising valves for the purpose of improving leakage performance shall not be permitted.

For larger test volumes, a pressure decay method may be utilized to determine the leakage rate. The makeup flow rate or the pressure decay methods or other proven techniques may be used to determine the leakage.

The total leakage rate for Type B and C tests must be less than 0.6 L_a.

The criteria for determining the direction in which the test pressure is applied to the isolation valves are as follows:

A. Gate Valves

1. Parallel Disc

a. Test in the design basis accident (DBA) direction.

b. Testing can be performed between the discs if a test connection or drain is provided in the valve design.

2. Flexible Wedge

a. Test in the DBA direction.

b. Testing can be performed between the wedge sections if a test connection or drain is provided in the valve design.

3. Solid Wedge

a. Test in the DBA direction.

B. Globe Valves

If the DBA flow direction is over the disc (flow to close), the valve may be tested in the reverse direction. However, if the DBA flow direction is under the disc (flow to open), then the valve must be tested in this direction.

C. Butterfly Valves

Test in the DBA direction for Type C tests.

D. Flanges

Test in either direction.

The leakage rate test acceptance criteria for penetrations and isolation valves subject to Type B and C tests are given in the Technical Specifications.

6.2.6.4 Scheduling and Reporting of Periodic Tests

Type A, B, and C tests are conducted at the intervals specified in the containment leakage testing program as specified in the Technical Specifications. These intervals are in accordance with 10 CFR 50, Appendix J, Option B, with approved exceptions.

A post outage report will be prepared, presenting the results of the previous cycle's Type B and Type C tests; and Type A, Type B, and Type C tests if performed during that outage. Sufficient documentation will be collected and retained so that the effectiveness of the implementation of the containment leakage testing program can be reviewed and determined. The technical contents of the report will be available onsite for NRC review.

The preoperational test report contains a schematic of the leakage measuring system, instrumentation used, supplemental test method, test program, and analysis and interpretation of the leakage test data for the Type A test.

6.2.6.5 Special Testing Requirements

VEGP does not have a subatmospheric containment or a secondary containment, hence there are no special testing requirements beyond those delineated in paragraphs 6.2.6.1 through 6.2.6.4.

6.2.7 FRACTURE PREVENTION OF CONTAINMENT PRESSURE BOUNDARY

In accordance with General Design Criterion (GDC) 51, the reactor containment pressure boundary is designed with sufficient margin to ensure that under operating, maintenance, testing, and design basis accident conditions, its ferritic materials behave in a nonbrittle manner and the probability of rapidly propagating fracture is minimized.

For VEGP, the reactor containment pressure boundary components with ferritic materials are:

- Containment liner plate.
- Containment penetration sleeve assemblies.
- Equipment hatch.
- Personnel lock.
- Escape lock.
- Flued heads.
- Containment isolation boundary piping.
- Containment isolation boundary valves.

6.2.7.1 Design Bases

A. In accordance with GDC 1, the containment pressure boundary is designed to quality standards commensurate with the safety function performed.

B. In accordance with GDC 16, the containment pressure boundary is designed to provide a barrier against the uncontrolled release of radioactivity to the environment.

6.2.7.2 Specifications for Ferritic Materials

A. Containment Liner Plate

Since the containment liner plate is only 1/4 in. thick, it is exempt from fracture toughness testing. However, liner plate greater than 1/4 in. thick (referred to as thickened liner plate) satisfies the fracture toughness test requirements presented in NE-2320 of Subsection NE of the following code editions and addenda:

1. All thickened liner plate up to the basemat liner plate at el 169 ft 0 in., 1971 edition through summer 1973 addenda.
2. All thickened liner plate above el 169 ft 0 in., 1974 edition through summer 1975 addenda.

B. Penetration Sleeve Assemblies

The penetration sleeve assemblies are composed of the sleeve, thickened liner plate (as reinforcing), anchor rings and stiffeners, gusset plates, and cap plates for spare penetrations. The penetration sleeve assemblies for the electrical penetrations contain a weld neck flange with captive nuts attached to receive the electrical header plates. Sleeves are fabricated from SA-516 Grade 70 plates and weld neck flanges are steel forgings of SA-105 material. The impact requirements for fracture toughness for penetration sleeve assemblies in the containment shell are in accordance with American Society of Mechanical Engineers (ASME) Section III, Division 1, Subsection NE, 1974 edition through summer 1975 addenda. Penetration sleeves in the containment basemat are in accordance with ASME Section III, Division 1, Subsection NE, 1971 edition through summer 1973 addenda.

C. Equipment Hatch, Personnel Lock, and Escape Lock

These items are fracture toughness tested in accordance with ASME Section III, Division 1, Subsection NE, 1974 edition through summer 1975 addenda.

D. Flued Heads

The flued heads are fracture toughness tested in accordance with ASME Section III, Division 1, Subsection NC, 1977 through summer 1978 addenda.

E. Containment Pressure Boundary Piping

Those portions of the containment pressure boundary piping comprising the main steam and feedwater system are designed and fracture toughness tested in accordance with ASME Section III, Division 1, Subsection NC, 1974 edition through summer 1975 addenda. The remaining containment pressure boundary piping is designed in accordance with ASME III, 1974 edition through summer

1975 addenda, without fracture toughness testing. The fracture toughness of these items is further addressed in paragraph 6.2.7.4.

The configuration of containment pressure boundary piping is shown in figure 6.2.4-1.

F. Containment Pressure Boundary Valves

Pressure boundary materials of the main steam isolation valves and main feedwater isolation valves are fracture toughness tested in accordance with Subarticle NC-2300 of ASME Section III, Division 1, Subsection NC, 1977 edition. For other containment pressure boundary valves in the balance of plant scope, the specific code edition and addenda to which the valves were procured did not require fracture toughness testing. These valves were procured in accordance with the 1974 edition up to and including the summer 1975 addenda of the ASME Code. The fracture toughness of these items is further addressed in paragraph 6.2.7.4.

Containment isolation valves within the nuclear steam supply system scope are fabricated of austenitic stainless steel or carbon steel and therefore need not be fracture toughness tested.

6.2.7.3 Documentation

A. ASME Code Data Reports

ASME Code Data Reports are prepared by the cognizant organization which applies the appropriate "N" type symbol to certify that the design, fabrication, installation, inspection, testing, and stamping are in accordance with ASME III for containment pressure boundary systems.

B. Certified Material Test Reports

Certified material test reports, when applicable, are provided to document the actual results of required chemical analyses, tests, examinations, and weld repairs of materials comprising the containment pressure boundary.

C. Other Documents

Drawings and related supplemental information are available as applicable to convey information necessary to fabricate, install, test, etc., the containment pressure boundary piping system.

6.2.7.4 Evaluation

The containment isolation boundary piping and balance of plant containment isolation boundary valves were procured to meet the minimum requirements of the 1974 edition through summer 1975 addenda of the ASME Code, which did not require impact testing. The specific requirements of NX-2300 of this code effective date are that the Design Specification shall state whether or not impact testing is required. Due to the exclusion criteria of subparagraph NX-2300 of the code, the piping and valves discussed in Sections 6.2.7.2.E and F did not require impact testing.

The containment pressure boundary piping and valves meet or exceed the minimum ASME Code requirements for design, materials, fabrication, examination, testing, and code stamping.

TABLE 6.2.1-1

CONTAINMENT DESIGN LIMITS AND CALCULATED
CONTAINMENT PEAK PRESSURE AND TEMPERATURE

Break	Peak Pressure (psig)	Available Margin (psi)	Peak Temperature (°F)
Primary Side Ruptures			
Double-ended pump suction, maximum safety injection	35.9[b]	16.1[b]	247[c]
Double-ended pump suction, minimum safety injection	35.9[b]	16.1[b]	247[c]
Double-ended hot leg	36.5	15.5	250
Containment Design Pressure			
52 psig -3 psig			
Containment Atmosphere Design Temperature			
381°F[a]			

a. A peak containment atmosphere temperature of 381°F was used in calculating the thermal gradients across the containment wall.

b. Per LDCR 2005051, the pressure is being increased by 1.3 psig from 34.6 psig, and the margin is correspondingly reduced temporarily until re-evaluated during a future analysis requiring re-evaluation of the containment mass and energy releases.

c. Per LDCR 2005051, there will be an insignificant increase in peak temperature. However, the peak temperature remains well below the design temperature of 381 °F.

TABLE 6.2.1-2

ASSUMPTIONS FOR CONTAINMENT ANALYSIS - PART 1

Service water temperature (°F)		95
Refueling water temperature (°F)		130
Refueling water storage tank volume (gal) (deliverable volume)		710,700
Initial Containment		
Temperature (°F)		120
Initial pressure (psia)		17.7
Initial relative humidity (%)		20
	Net free volume (ft^3)	2.75×10^6

TABLE 6.2.1-3

ASSUMPTIONS FOR CONTAINMENT ANALYSIS - PART 2

	Accident Analysis	Design
Fan coolers		
Number operating	4	8
Flowrate per fan cooler (ft^3/min)	43,500 (slow speed)	97,000 (full speed)
Heat removal capacity per fan cooler (Btu/h)	8.15×10^5	2.605×10^6
Spray pumps		
Number available	1	2
Flowrate per pump (gal/min)	2597	3200

TABLE 6.2.1-4 (SHEET 1 of 2)

CONTAINMENT STRUCTURAL HEAT SINKS

Passive Heat Sinks

Wall Description	Heat Transfer Area (ft^2)	Material	Thickness (ft)
1. Dome	29260	Epoxy	0.00025
		Inorganic zinc paint	0.0002083
		Carbon steel	0.02083
		Concrete	3.30
2. Shell	65181	Inorganic zinc paint	0.0002083
		Carbon steel	0.02083
		Concrete	3.75
3. Miscellaneous interior concrete	15971	Epoxy	0.00154
		Concrete	2.82
4. Primary shields	2106	Inorganic zinc paint	0.0002083
		Carbon steel	0.05795
		Epoxy	0.001125
		Concrete	1.5
5. Mechanical equipment	209	Stainless steel	0.0158
6 Refueling canal wall	6455	Stainless steel	0.02083
		Epoxy	0.001125
		Concrete	4.133
7. Miscellaneous interior concrete	4693	Epoxy	0.000042
		Concrete	2.893
8. Miscellaneous interior concrete	16147	Epoxy	0.01125
		Concrete	2.371
9. Miscellaneous interior concrete	58883	Epoxy	0.001125
		Concrete	1.4705

TABLE 6.2.1-4 (SHEET 2 of 2)

Wall Description		Heat Transfer Area (ft^2)	Material	Thickness (ft)
10.	Refueling canal slab	1255	Stainless steel	0.02083
			Concrete	4.0
11.	Structural steel	192088	Galvanizing	0.00033
			Carbon steel	0.0141
12.	Structural steel	183286	Inorganic zinc paint	0.000208
			Carbon steel	0.0263
13.	Piping	72630	Inorganic zinc paint	0.000208
				0.03676
			Carbon steel	0.03676
14.	Piping	34158	Stainless steel	0.0163
15.	Piping	1395	Stainless steel	0.0340
16.	Mechanical equipment	98158	Inorganic zinc paint	0.000208
			Carbon steel	0.142
17.	Mechanical equipment	24750	Inorganic zinc paint	0.000208
			Carbon steel	0.0208
18.	Miscellaneous steel	40599	Inorganic zinc paint	0.000208
			Carbon steel	0.01288
19.	Basement	13702	Inorganic zinc paint	0.000208
			Carbon steel	0.02083
			Concrete	10.5
20.	Reactor cavity	4263	Inorganic zinc paint	0.000208
			Carbon steel	0.02083
			Concrete	8.0

TABLE 6.2.1-5

THERMOPHYSICAL PROPERTIES OF CONTAINMENT HEAT SINKS

Material	Thermal Conductivity (Btu/h-ft-°F)	Volumetric Heat Capacity (Btu/ft^3-°F)
Paint	0.9	20.0
Carbon steel	28.0	52.5
Stainless steel	9.4	60.1
Concrete	0.65	28.8
Galvanizing	64.8	40.9
Epoxy	0.97	20.0

TABLE 6.2.1-6

ACCIDENT SEQUENCE FOR DOUBLE-ENDED PUMP SUCTION BREAK
MINIMUM SAFETY INJECTION

	Event	Time of Occurrence (s)
1.	Accumulators begin injecting	14.5
2.	Containment peak pressure	17.6
3.	End of blowdown	22.0
4.	Safety injection begins	43.3
5.	Accumulator injection stopped	54.3
6.	Fan coolers start	100.8
7.	Spray start	108.0
8.	End of reflood	225.8
9.	Recirculation, injection	3952.0
10.	Recirculation, spray	5000.0

TABLE 6.2.1-7

DELETED

TABLE 6.2.1-8

POSTULATED SUBCOMPARTMENT PIPE BREAKS

Subcompartment	High-Energy Line	Break Area ($in.^2$)	Maximum ΔP (psid)
Reactor cavity	1201-009-27.5 in.	144	192.90
Steam generator (pipe break below el 220 ft)	1201-008-31 in.	436	23.50
Steam generator (pipe break above el 220 ft)	1305-064-16 in.	144.5	6.77
Pressurizer	1201-053-14 in.	308	20.7

TABLE 6.2.1-9 (SHEET 1 OF 6)

REACTOR CAVITY FLOW MODEL: NODE CHARACTERISTICS
(INITIAL CONDITIONS REMAIN THE SAME)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psi)	Humidity (%)		
1	Approximately 1/8 of the upper inspection annulus.	3.625	28.4	120	13.2	25	192.89	102.95
2	Approximately 1/8 of the upper inspection annulus.	3.625	28.22	120	13.2	25	83.17	102.286
3	Approximately 1/8 of the upper inspection annulus.	3.625	28.034	120	13.2	25	55.42	101.624
4	Approximately 1/8 of the upper inspection annulus.	3.625	28.22	120	13.2	25	50.29	102.286
5	Approximately 1/8 of the upper inspection annulus.	3.625	28.4	120	13.2	25	50.46	102.95
6	Approximately 1/8 of the upper inspection annulus.	3.625	28.22	120	13.2	25	192.63	102.286
7	Approximately 1/8 of the upper inspection annulus.	3.625	28.034	120	13.2	25	100.85	101.624
8	Approximately 1/8 of the upper inspection annulus.	3.625	28.22	120	13.2	25	63.71	102.286
9	Approximately 1/8 of the lower inspection annulus.	2.083	34.20	120	13.2	25	193.0	71.245
10	Approximately 1/8 of the lower inspection annulus.	2.083	33.88	120	13.2	25	83.04	70.57

TABLE 6.2.1-9 (SHEET 2 OF 6)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psi)	Humidity (%)		
11	Approximately 1/8 of the lower inspection annulus.	2.083	33.56	120	13.2	25	55.64	69.91
12	Approximately 1/8 of the lower inspection annulus.	2.083	33.88	120	13.2	25	50.56	70.57
13	Approximately 1/8 of the lower inspection annulus.	2.083	34.20	120	13.2	25	50.6	71.245
14	Approximately 1/8 of the lower inspection annulus.	2.083	33.88	120	13.2	25	192.45	70.57
15	Approximately 1/8 of the lower inspection annulus.	2.083	33.56	120	13.2	25	100.91	69.91
16	Approximately 1/8 of the lower inspection annulus.	2.083	33.88	120	13.2	25	64.26	70.57
17	Approximately 1/8 of the upper inner cavity.	7.125	1.287	120	13.2	25	159.86	9.169
18	Approximately 1/8 of the upper inner cavity.	7.125	1.33	120	13.2	25	135.98	9.447
19	Approximately 1/8 of the upper inner cavity.	7.125	1.365	120	13.2	25	127.72	9.7265
20	Approximately 1/8 of the upper inner cavity.	7.125	1.326	120	13.2	25	124.98	9.447
21	Approximately 1/8 of the upper inner cavity.	7.125	1.287	120	13.2	25	124.99	9.169
22	Approximately 1/8 of the upper inner cavity.	7.125	1.33	120	13.2	25	159.75	9.447

TABLE 6.2.1-9 (SHEET 3 OF 6)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psi)	Humidity (%)		
23	Approximately 1/8 of the upper inner cavity.	7.125	1.365	120	13.2	25	139.1	9.7265
24	Approximately 1/8 of the upper inner cavity.	7.125	1.326	120	13.2	25	128.38	9.447
25	Approximately 1/8 of the lower inner cavity.	4.72	0.99	120	13.2	25	177.58	4.678
26	Approximately 1/8 of the lower inner cavity.	4.72	1.05	120	13.2	25	98.84	4.96
27	Approximately 1/8 of the lower inner cavity.	4.72	1.11	120	13.2	25	56.69	5.24
28	Approximately 1/8 of the lower inner cavity.	4.72	1.051	120	13.2	25	52.15	4.96
29	Approximately 1/8 of the lower inner cavity.	4.72	0.99	120	13.2	25	52.41	4.678
30	Approximately 1/8 of the lower inner cavity.	4.72	1.051	120	13.2	25	177.48	4.96
31	Approximately 1/8 of the lower inner cavity.	4.72	1.11	120	13.2	25	102.44	5.24
32	Approximately 1/8 of the lower inner cavity.	4.72	1.051	120	13.2	25	65.39	4.96
33	Approximately 1/8 of the upper cavity annulus.	6.45	3.101	120	13.2	25	117.24	20.005
34	Approximately 1/8 of the upper cavity annulus.	6.45	3.101	120	13.2	25	94.04	20.005

TABLE 6.2.1-9 (SHEET 4 OF 6)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psi)	Humidity (%)		
35	Approximately 1/8 of the upper cavity annulus.	6.45	3.101	120	13.2	25	54.65	20.005
36	Approximately 1/8 of the upper cavity annulus.	6.45	3.101	120	13.2	25	51.09	20.005
37	Approximately 1/8 of the upper cavity annulus.	6.45	3.101	120	13.2	25	50.69	20.005
38	Approximately 1/8 of the upper cavity annulus.	6.45	3.101	120	13.2	25	117.29	20.005
39	Approximately 1/8 of the upper cavity annulus.	6.45	3.101	120	13.2	25	98.57	20.005
40	Approximately 1/8 of the upper cavity annulus.	6.45	3.101	120	13.2	25	64.37	20.005
41	Approximately 1/8 of the lower cavity annulus.	6.45	3.101	120	13.2	25	76.9	20.005
42	Approximately 1/8 of the lower cavity annulus. annulus.	6.45	3.101	120	13.2	25	74.16	20.005
43	Approximately 1/8 of the lower cavity annulus.	6.45	3.101	120	13.2	25	48.21	20.005
44	Approximately 1/8 of the lower cavity annulus.	6.45	3.101	120	13.2	25	46.97	20.005
45	Approximately 1/8 of the lower cavity annulus.	6.45	3.101	120	13.2	25	47.11	20.005
46	Approximately 1/8 of the lower cavity annulus.	6.45	3.101	120	13.2	25	76.94	20.005

TABLE 6.2.1-9 (SHEET 5 OF 6)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psi)	Humidity (%)		
47	Approximately 1/8 of the lower cavity annulus.	6.45	3.101	120	13.2	25	74.93	20.005
48	Approximately 1/8 of the lower cavity annulus.	6.45	3.101	120	13.2	25	51.3	20.005
49	Lower RX Cavity region el 160 ft 10 3/4 in. to 169 ft 4 1/2 in.	8.48	121.81	120	13.2	25	12.4	1032.93
50	Lower RX cavity region el 156 ft to 160 ft 10 3/4 in.	4.90	274.66	120	13.2	25	11.91	1345.85
51	Lower RX cavity; near guide tube; supports region 156 ft.	4.75	301.30	120	13.2	25	11.25	1431.16
52	Located between first and second guide tube supports in lower RX cavity.	7.75	303.9	120	13.2	25	10.59	2355.23
53	Incore instrumentation tunnel.	10.0	70.60	120	13.2	25	10.2	705.96
54	Instrumentation tunnel horizontal passageway.	10.0	100.85	120	13.2	25	9.71	1008.52
55	Instrumentation tunnel access shaft	15.52	52.96	120	13.2	25	9.69	821.894
56	Instrumentation tunnel access shaft	23.48	22.58	120	13.2	25	9.68	530.23
57	Instrumentation tunnel	11.0	83.42	120	13.2	25	8.43	917.62
58	Instrumentation tunnel	8.0	94.44	120	13.2	25	6.74	755.48

TABLE 6.2.1-9 (SHEET 6 OF 6)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psi)	Humidity (%)		
59	Instrumentation tunnel.		94.44	120	13.2	25	5.88	755.48
60	Instrumentation tunnel.	8.0	117.30	120	13.2	25	5.01	938.39
61	Instrumentation tunnel.	9.0	112.39	120	13.2	25	4.34	1011.51
62	Instrumentation tunnel.	8.0	88.09	120	13.2	25	3.68	704.69
63	South steam generator (SG) compartment.	NA	NA	120	13.2	25	2.25	79,242.0
64	North SG compartment.	NA	NA	120	13.2	25	2.147	79,242.0
65	Free containment.	NA	NA	120	13.2	25	0.97	2.75E+06

TABLE 6.2.1-10 (SHEET 1 OF 3)

REACTOR CAVITY FLOW MODEL: FLOW CHARACTERISTICS

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
1	1-2	x		6.5	0	0	1.0	0.231	1.231	0.7725
2	1-6		*	6.5	0	0	1.0	0.231	1.231	0.7725
3	1-17	x		0.743	0	0	1.0	0.48	1.48	0.957
4	1-9		*	18.524	0	0	1.0	0.188	1.188	0.124
5	1-63	x		3.31	0	0	1.0	0.45	1.45	2.77
6	2-3	x		6.25	0	0	1.0	0.24	1.24	0.778
7	2-10		*	18.28	0	0	1.0	0.19	1.19	0.125
8	2-63	x		6.45	0	0	1.0	0.40	1.40	1.26
9	3-4		*	6.25	0	0	1.0	0.24	1.24	0.778
10	3-11		*	18.0	0	0	1.0	0.197	1.197	0.125
11	3-63	x		6.24	0	0	1.0	0.40	1.40	0.715
12	4-5		*	6.5	0	0	1.0	0.231	1.231	0.7725
13	4-12		*	18.28	0	0	1.0	0.19	1.19	0.125
14	4-63	x		6.45	0	0	1.0	0.40	1.40	1.26
15	5-8		*	6.5	0	0	1.0	0.231	1.231	0.7725
16	5-13		*	18.28	0	0	1.0	0.188	1.188	0.125
17	5-63	x		3.31	0	0	1.0	0.45	1.45	2.77
18	5-64	x		3.31	0	0	1.0	0.45	1.45	2.77
19	6-7	x		6.25	0	0	1.0	0.24	1.24	0.778
20	6-14		*	18.28	0	0	1.0	0.19	1.19	0.125
21	6-22	x		0.743	0	0	1.0	0.48	1.48	0.957
22	6-64	x		3.12	0	0	1.0	0.45	1.45	1.43
23	7-8	x		6.25	0	0	1.0	0.24	1.24	0.778
24	7-15		*	18.0	0	0	1.0	0.197	1.197	0.125
25	7-64	x		6.24	0	0	1.0	0.40	1.40	0.715
26	8-16		*	18.28	0	0	1.0	0.19	1.19	0.125
27	8-64	x		6.45	0	0	1.0	0.40	1.40	1.26
28	9-10	x		4.16	0	0	1.0	0.286	1.286	1.12
29	9-14		*	4.16	0	0	1.0	0.286	1.286	1.12
30	9-25		*	0.743	0	0	1.0	0.48	1.48	0.983
31	9-25		*	2.71	0	1.12	1.0	0.5	2.62	3.00
32	9-33	x		0.347	0	0	1.0	0.5	1.5	6.32
33	9-63	x		2.34	0	0	1.0	0.45	1.45	3.93
34	10-11	x		6.25	0	0	1.0	0.18	1.18	1.13
35	10-26		*	2.71	0	1.12	1.0	0.5	2.62	3.00
36	10-34		*	0.347	0	0	1.0	0.5	1.5	6.32
37	10-63	x		4.48	0	0	1.0	0.4	1.4	2.08
38	11-12		*	6.25	0	0	1.0	0.18	1.18	1.13
39	11-27		*	2.71	0	1.12	1.0	0.5	2.62	3.00
40	11-35		*	0.347	0	0	1.0	0.5	1.5	6.32
41	11-63	x		4.288	0	0	1.0	0.4	1.4	1.04
42	12-13		*	4.16	0	0	1.0	0.286	1.286	1.12
43	12-28		*	2.71	0	1.12	1.0	0.5	2.62	3.00
44	12-36		*	0.347	0	0	1.0	0.5	1.5	6.32
45	12-63	x		4.48	0	0	1.0	0.4	1.40	1.36
46	13-16		*	4.16	0	0	1.0	0.286	1.286	1.12
47	13-29		*	2.71	0	1.12	1.0	0.5	2.62	3.00
48	13-37		*	0.347	0	0	1.0	0.5	1.50	6.32

TABLE 6.2.1-10 (SHEET 2 OF 3)

Vent Path	Node Number (From-To)	Description of Flow Choked	Description of Flow Unchoked	Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
49	13-63	x		2.144	0	0	1.0	0.45	1.45	2.08
50	13-64	x		2.34	0	0	1.0	0.45	1.45	3.93
51	14-15	x		6.25	0	0	1.0	0.18	1.18	1.13
52	14-30		*	0.743	0	0	1.0	0.48	1.48	0.983
53	14-30		*	2.71	0	1.12	1.0	0.5	2.62	3.00
54	14-38	x		0.347	0	0	1.0	0.5	1.50	6.32
55	14-64	x		2.144	0	0	1.0	0.45	1.45	2.08
56	15-16	x		6.25	0	0	1.0	0.18	1.18	1.13
57	15-31		*	2.71	0	1.12	1.0	0.5	2.62	3.00
58	15-39		*	0.347	0	0	1.0	0.5	1.50	6.32
59	15-64	x		4.288	0	0	1.0	0.4	1.40	1.04
60	16-32		*	2.71	0	1.12	1.0	0.5	2.62	3.00
61	16-40		*	0.347	0	0	1.0	0.5	1.50	6.32
62	16-64	x		4.48	0	0	1.0	0.4	1.40	1.36
63	17-18	x		0.91	0	0	1.0	0.18	1.18	4.8
64	17-22		*	0.91	0	0	1.0	0.18	1.18	4.8
65	17-25	x		0.38	0	0	1.0	0.22	1.22	5.15
66	18-19		*	0.975	0	0	1.0	0.16	1.16	4.662
67	18-26		*	0.316	0	0	1.0	0.26	1.26	4.93
68	19-20		*	0.975	0	0	1.0	0.16	1.16	4.662
69	19-27	x		0.25	0	0	1.0	0.31	1.31	4.74
70	20-21		*	0.91	0	0	1.0	0.18	1.18	4.8
71	20-28	x		0.316	0	0	1.0	0.26	1.26	4.93
72	21-24		*	0.91	0	0	1.0	0.18	1.18	4.8
73	21-29	x		0.38	0	0	1.0	0.22	1.22	5.15
74	22-23	x		0.975	0	0	1.0	0.16	1.16	4.662
75	22-30	x		0.316	0	0	1.0	0.26	1.26	4.93
76	23-24		*	0.975	0	0	1.0	0.16	1.16	4.662
77	23-31		*	0.25	0	0	1.0	0.31	1.31	4.74
78	24-32	x		0.316	0	0	1.0	0.26	1.26	4.93
79	25-26	x		0.427	0	0	1.0	0.274	1.274	9.3
80	25-30		*	0.427	0	0	1.0	0.274	1.274	9.3
81	25-33	x		1.254	0.178	0	1.0	0.032	1.21	3.43
82	26-27	x		0.494	0	0	1.0	0.24	1.24	9.03
83	26-34		*	1.254	0.178	0	1.0	0.032	1.21	3.43
84	27-28		*	0.494	0	0	1.0	0.24	1.24	9.03
85	27-35		*	1.254	0.178	0	1.0	0.032	1.21	3.43
86	28-29		*	0.427	0	0	1.0	0.274	1.274	9.3
87	28-36		*	1.254	0.178	0	1.0	0.032	1.21	3.43
88	29-32		*	0.427	0	0	1.0	0.274	1.274	9.3
89	29-37		*	1.254	0.178	0	1.0	0.032	1.21	3.43
90	30-31	x		0.494	0	0	1.0	0.24	1.24	9.03
91	30-38	x		1.254	0.178	0	1.0	0.032	1.21	3.43
92	31-32	x		0.494	0	0	1.0	0.24	1.24	9.03
93	31-39		*	1.254	0.178	0	1.0	0.032	1.21	3.43
94	32-40		*	1.254	0.178	0	1.0	0.032	1.21	3.43
95	33-34		*	1.226	0.2085	0	- 1.0	0	1.21	5.38
96	33-38		*	1.226	0.2085	0	1.0	0	1.21	5.38
97	33-41	x		1.25	0.21	0	1.0	0	1.21	5.154
98	34-35	x		0.613	0.21	0	1.0	0.25	1.46	5.38

TABLE 6.2.1-10 (SHEET 3 OF 3)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
99	34-42		*	1.25	0.21	0	1.0	0	1.21	5.154
100	35-36		*	1.226	0.21	0	1.0	0	1.21	5.38
101	35-43		*	1.25	0.21	0	1.0	0	1.21	5.154
102	36-37		*	1.226	0.21	0	1.0	0	1.21	5.38
103	36-44		*	1.25	0.21	0	1.0	0	1.21	5.154
104	37-40		*	1.226	0.21	0	1.0	0	1.21	5.38
105	37-45		*	1.25	0.21	0	1.0	0	1.21	5.154
106	38-39		*	1.226	0.21	0	1.0	0	1.21	5.38
107	38-46	x		1.25	0.21	0	1.0	0	1.21	5.154
108	39-47	x		1.25	0.21	0	1.0	0	1.21	5.154
109	40-39	x		0.613	0.21	0	1.0	0.25	1.46	5.38
110	40-48		*	1.25	0.21	0	1.0	0	1.21	5.154
111	41-42		*	1.226	0.21	0	1.0	0	1.21	5.384
112	41-46		*	1.226	0.21	0	1.0	0	1.21	5.384
113	41-49	x		1.254	0.12	0	1.0	0	1.12	2.63
114	42-43	x		0.613	0.21	0	1.0	0.25	1.46	5.38
115	42-49	x		1.254	0.12	0	1.0	0	1.12	2.63
116	43-44		*	1.226	0.21	0	1.0	0	1.21	5.384
117	43-49	x		1.254	0.12	0	1.0	0	1.12	2.63
118	44-45		*	1.226	0.21	0	1.0	0	1.21	5.384
119	44-49	x		1.254	0.12	0	1.0	0	1.12	2.63
120	45-48		*	1.226	0.21	0	1.0	0	1.21	5.384
121	45'49	x		1.254	0.12	0	1.0	0	1.12	2.63
122	46-47		*	1.226	0.21	0	1.0	0	1.21	5.384
123	46-49	x		1.254	0.12	0	1.0	0	1.12	2.63
124	47-49	x		1.254	0.12	0	1.0	0	1.12	2.63
125	48-47		*	0.613	0.21	0	1.0	0.25	1.46	5.384
126	48-49	x		1.254	0.12	0	1.0	0.0	1.12	2.63
127	49-50		*	93.49	0	0	1.0	0.083	1.083	0.069
128	50-51		*	84.92	0	0	1.0	0.225	1.225	0.045
129	51-52		*	101.86	0	0.5	1.0	0.25	1.75	0.033
130	52-53		*	103.56	0	0	1.0	0.223	1.223	0.1
131	53-54		*	94.34	0	0	1.0	0.13	1.13	0.063
132	54-55		*	47.66	0	1.12	1.0	0.29	2.41	0.177
133	54-57		*	59.66	0	0	1.0	0.266	1.266	0.106
134	55-56		*	22.58	0.525	0	1.0	0.05	1.575	0.777
135	57-58		*	71.73	0	1.12	1.0	0.14	2.26	0.082
136	58-59		*	71.73	0	0	1.0	0.141	1.141	0.08
137	59-60		*	71.73	0	0	1.0	0.141	1.141	0.08
138	60-61		*	82.08	0	0	1.0	0.155	1.155	0.076
139	61-62		*	82.46	0	0	1.0	0.153	1.153	0.064
140	62-65		*	47.25	0	0	1.0	0.301	1.301	0.0264
141	63-64		*	248.0	0	0	1.0	0.5	1.5	0.11
142	63-65		*	452.0	0	0	1.0	0.45	1.45	0.071
143	64-65		*	452.0	0	0	1.0	0.45	1.45	0.071

TABLE 6.2.1-11 (SHEET 1 OF 7)

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW EL. 220 ft):
NODE CHARACTERISTICS LOOP CLOSURE WELD (336-in.^{2} BREAK)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
1	Interface of two halves of SG compartment.	6.56	222.97	120	13.2	25	7.11	1316.42
2	Between cold leg and wall; from el 171 ft 9 in. to 187 ft.	15.25	176.0	120	13.2	25	13.58	2283.14
3	Between RCP 4, SG 4, and wall; from el 171 ft 9 in. to 187 ft.	15.25	132.24	120	13.2	25	15.89	1661.50
4	By SG 4; from el 171 ft 9 in. to 183 ft.	11.25	83.6	120	13.2	25	13.65	809.05
5	Adjacent to SG 4; from el 171 ft 9 in. to 183 ft.	11.25	95.78	120	13.2	25	13.27	895.04
6	Around SG 4; from el 171 ft 9 in. to 195 ft.	23.25	165.73	120	13.2	25	13.24	3124.98
7	Between hot, cold and suction legs; from el 171 ft 9 in. to 187 ft.	15.25	289.03	120	13.2	25	14.96	3721.19
8	Around SG 1; from el 171 ft 9 in. to 195 ft.	23.25	163.51	120	13.2	25	12.76	3078.62
9	Adjacent to SG 1; from el 171 ft 9 in. to 183 ft.	11.25	95.31	120	13.2	25	12.89	890.21
10	By SG 1; from el 171 ft 9 in. to 183 ft.	11.25	86.44	120	13.2	25	10.93	837.79

TABLE 6.2.1-11 (SHEET 2 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
11	Between RCP 1, SG 1, and wall; from el 171 ft 9 in. to 187 ft.	15.25	137.8	120	13.2	25	7.62	1737.79
12	Between hot, cold, and suction legs 1; from el 171 ft 9 in. to 187 ft.	15.25	289.64	120	13.2	25	11.11	3729.58
13	Between hot leg 1 and wall; from el 171 ft 9 in. to 187 ft.	15.25	219.7	120	13.2	25	7.60	2815.04
14	Interface of two halves of SG compartment; between quadrants 1 and 2.	20.25	131.71	120	13.2	25	5.85	2400.44
15	Over node 4; from el 183 to 195 ft.	12.0	83.6	120	13.2	25	13.52	766.38
16	Over node 5; from el 183 to 195 ft.	12.0	95.78	120	13.2	25	13.28	892.25
17	Over node 9 in quadrant 1; from el 183 to 195 ft.	12.0	95.31	120	13.2	25	12.88	887.09
18	Over node 10 in quadrant 1; from el 183 to 195 ft.	12.0	86.44	120	13.2	25	10.96	796.98
19	Over node 2; from el 187 to 195 ft.	8.0	179.22	120	13.2	25	13.56	1251.37
20	Over node 3; from el 187 to 195 ft.	8.0	113.1	120	13.2	25	15.9	814.29
21	Over-node 7; from el 187 to 195 ft.	8.0	229.18	120	13.2	25	14.93	1581.52
22	Over node 12; from el 187 to 195 ft.	8.0	236.11	120	13.2	25	10.26	1631.41

TABLE 6.2.1-11 (SHEET 3 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
23	Over node 11; from el 187 to 195 ft.	8.0	110.51	120	13.2	25	8.56	795.67
24	Over node 13; from el 187 to 195 ft.	8.0	223.84	120	13.2	25	8.61	1572.59
25	Around RCP 4; from el 195 to 200 ft.	5.0	89.33	120	13.2	25	13.01	402.0
26	Between RCP 4 and wall; from el 195 to 200 ft.	5.0	31.62	120	13.2	25	13.11	142.3
27	Between SG 4 and RCP 4; from el 195 to 200 ft.	5.0	89.14	120	13.2	25	13.2	348.52
28	Between SG 4 and HVAC shaft; from el 195 to 200 ft.	5.0	32.94	120	13.2	25	12.77	145.4
29	Between SG 4 and wall; from el 195 to 200 ft.	5.0	83.81	120	13.2	25	12.06	324.23
30	Between SG 4 and concrete beam; from el 195 to 200 ft.	5.0	176.1	120	13.2	25	12.06	668.07
31	Adjacent to node 32; from el 195 to 200 ft.	5.0	102.29	120	13.2	25	12.73	460.28
32	Opposite node 27; from el 195 to 200 ft.	5.0	197.63	120	13.2	25	12.98	788.23
33	Between SG 1 and concrete beam; from el 195 to 290 ft.	5.0	176.84	120	13.2	25	11.55	671.4
34	Between SG 1 and wall; from el 195 to 200 ft.	5.0	81.91	120	13.2	25	11.14	315.67

TABLE 6.2.1-11 (SHEET 4 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
35	Between SG 1 and HVAC shaft; from el 195 to 200 ft.	5.0	33.08	120	13.2	25	10.68	146.0
36	Between SG 1 and RCP 1; from el 195 to 200 ft.	5.0	99.23	120	13.2	25	8.59	393.9
37	Between RCP 1 and wall; from el 195 to 200 ft.	5.0	37.59	120	13.2	25	8.60	169.15
38	Around RCP 1; from el 195 to 200 ft.	5.0	114.17	120	13.2	25	8.61	513.74
39	Opposite node 36; from el 195 to 200 ft.	5.0	225.29	120	13.2	25	9.12	912.67
40	Adjacent to node 39; from el 195 to 200 ft.	5.0	103.1	120	13.2	25	10.38	463.95
41	Between RCP 4 and northern wall of quadrant; from el 200 to 220 ft.	20.0	100.1	120	13.2	25	11.38	1796.2
42	Between RCP 4 and SW wall of quadrant; from el 200 to 220 ft.	20.0	35.3	120	13.2	25	11.42	635.37
43	Between HVAC shaft, RCP 4, and SG 4; from el 200 to 220 ft.	20.0	144.38	120	13.2	25	11.53	2577.68
44	Between SG 4 and southern wall of quadrant; from el 200 to 215 ft.	15.0	31.7	120	13.2	25	11.50	419.36
45	Between node 44 and quadrant 1; from el 200 to 207 ft.	7.0	111.91	102	13.2	25	11.46	697.96

TABLE 6.2.1-11 (SHEET 5 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
46	Over node 45; from el 207 to 215 ft.	8.0	110.59	120	13.2	25	11.27	796.22
47	Between node 48 and quadrant 1; from el 200 to 215 ft.	15.0	225.12	120	13.2	25	11.49	3015.61
48	Between SG 4 and northern wall, adjacent to node 49; from el 200 to 215 ft.	15.0	93.84	120	13.2	25	11.58	1258.89
49	Adjacent to node 43; between RCP 4 and northern wall; from el 200 to 220 ft.	20.0	124.07	120	13.2	25	11.53	2214.86
50	Between node 58 and quadrant 4; from el 200 to 215 ft.	15.0	224.85	120	13.2	25	10.89	3012.0
51	Between node 53 and quadrant 4; from el 200 to 207 ft.	7.0	111.55	120	13.2	25	11.07	695.7
52	Over node 51; from el 207 to 215 ft.	8.0	110.91	120	13.2	25	10.96	798.54
53	Between SG 1 and southern wall of quadrant 1; from el 200 to 215 ft.	15.0	30.43	120	13.2	25	10.8	402.25
54	Between HVAC shaft, RCP 1, and SG 1; from el 200 to 220 ft.	20.0	147.88	120	13.2	25	8.55	2640.6

TABLE 6.2.1-11 (SHEET 6 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
55	Between RCP 1 and SE wall of quadrant 1; from el 200 to 220 ft.	20.0	35.91	120	13.2	25	8.33	646.32
56	Between RCP 1 and northern wall of quadrant 1; from el 200 to 220 ft.	20.0	110.65	120	13.2	25	8.36	1986.14
57	Adjacent to node 54; between RCP 1 and northern wall; from el 200 to 220 ft.	20.0	139.29	120	13.2	25	8.66	2488.86
58	Between SG 1 and northern wall, adjacent to node 57; from el 200 to 215 ft.	15.0	100.43	120	13.2	25	9.40	1347.85
59	NW node of left half of SG 4 doghouse; from el 215 to 238 ft.	23.0	50.73	120	13.2	25	3.75	918.0
60	SW node of left half of SG 4 dog-house; from el 215 to 229 ft.	14.0	52.98	120	13.2	25	4.41	596.4
61	Over node 60; from el 229 to 238 ft.	9.0	52.84	120	13.2	25	3.28	404.2
62	SE node of left half of SG 4 dog-house; from el 215 to 238 ft.	23.0	79.51	120	13.2	25	3.88	1356.2
63	NE node of left half of SG 4 dog-house; from el 215 to 238 ft.	23.0	79.87	120	13.2	25	3.87	1234.0

TABLE 6.2.1-11 (SHEET 7 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
64	Symmetric of node 63 in quadrant 1.	23.0	79.87	120	13.2	25	3.75	1234.0
65	Symmetric of node 62 in quadrant 1.	23.0	79.51	120	13.2	25	3.76	1356.2
66	Symmetric of node 60 in quadrant 1.	14.0	52.98	120	13.2	25	4.12	596.4
67	Symmetric of node 61 in quadrant 1.	9.0	52.84	120	13.2	25	3.20	404.2
68	Symmetric of node 59 in quadrant 1.	23.0	50.73	120	13.2	25	3.60	918.0
69	HVAC shaft 4	-	-	120	13.2	25	11.41	838.6
70	HVAC shaft 1.	-	-	120	13.2	25	7.91	838.6
71	HVAC duct 4.	-	-	120	13.2	25	8.64	1231.6
72	HVAC duct 1.	-	-	120	13.2	25	6.06	1231.6
73	Quadrants 2 and 3 of SG compartment.	-	-	120	13.2	25	3.06	8.36E+04
74	Containment atmosphere.	-	-	120	13.2	25	2.74	2.75E+06

TABLE 6.2.1-11A

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK ABOVE EL. 220 ft):
NODE CHARACTERISTICS

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
1	Quadrant 1 from el 216 ft to 238 ft	12	149.7	120	14.7	20	8.06	1796
2	Quadrant 2 from el 229 ft to 238 ft	9	45.4	120	14.7	20	5.31	409
3	Quadrant 3 from el 229 ft to 238 ft	9	48.2	120	14.7	20	5.23	434
4	Quadrant 4 from el 229 ft to 238 ft	9	73.0	120	14.7	20	6.22	657
5	Quadrant 2 from el 216 ft to 229 ft	13	54.4	120	14.7	20	6.65	707
6	Quadrant 3 from el 216 ft to 229 ft	13	60.4	120	14.7	20	6.56	785
7	Quadrant 4 from el 216 ft to 229	13	79.9	120	14.7	20	6.90	1039
8	Quadrant 2 from el 200 ft to 216 ft	16	76.3	120	14.7	20	6.58	1221
9	Quadrant 3 from el 200 ft to 216 ft	16	166.56	120	14.7	20	6.49	2665
10	Quadrant 4 from el 200 ft to 216 ft	16	91.3	120	14.7	20	6.58	1461
11	Quadrant 1 from el 200 ft to 216 ft	16	186.13	120	14.7	20	6.60	2978
12	Containment Atmosphere	-	-	120	14.7	20	3.29	2.75×10^6

TABLE 6.2.1-12 (SHEET 1 OF 7)

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW EL 220 ft):
NODE CHARACTERISTICS STEAM GENERATOR INLET NOZZLE (306-in.^{2} BREAK)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature ($^\circ$F)	Pressure (psia)	Humidity (%)		
1	Interface of two halves of SG compartment.	6.56	222.97	120	13.2	25	3.08	1316.42
2	Between cold leg and wall; from el 171 ft 9 in. to 187 ft.	15.25	176.0	120	13.2	25	7.43	2283.14
3	Between RCP 4, SG 4, and wall; from el 171 ft 9 in. to 187 ft.	15.25	132.24	120	13.2	25	7.91	1661.50
4	By SG 4; from el 171 ft 9 in. to 183 ft.	11.25	83.6	120	13.2	25	8.21	809.05
5	Adjacent to SG 4; from el 171 ft 9 in. to 183 ft.	11.25	95.78	120	13.2	25	8.48	895.04
6	Around SG 4; from el 171 ft 9 in. to 195 ft.	23.25	165.73	120	13.2	25	9.0	3124.98
7	Between hot, cold, and suction legs; from el 171 ft 9 in. to 187 ft.	15.25	189.03	120	13.2	25	9.0	3721.19
8	Around SG 1; from el 171 ft 9 in. to 195 ft.	23.25	163.51	120	13.2	25	8.24	3078.62
9	Adjacent to SG 1; from el 171 ft 9 in. to 183 ft.	11.25	95.31	120	13.2	25	8.38	890.21
10	By SG 1; from el 171 ft 9 in. to 183 ft.	11.25	86.44	120	13.2	25	6.35	837.79

TABLE 6.2.1-12 (SHEET 2 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
11	Between RCP 1, SG 1, and wall; from el 171 ft 9 in. to 187 ft.	15.25	137.8	120	13.2	25	3.90	1737.79
12	Between hot, cold, and suction legs 1; from el 171 ft 9 in. to 187 ft.	15.25	289.64	120	13.2	25	6.57	3729.58
13	Between hot leg 1 and wall; from el 171 ft 9 in. to 187 ft.	15.25	219.7	120	13.2	25	3.95	2815.04
14	Interface of two halves of SG compartment; between quadrants 1 and 2.	20.25	131.71	120	13.2	25	2.71	2400.44
15	Over node 4; from el 183 to 195 ft.	12.0	83.6	120	13.2	25	8.19	766.38
16	Over node 5; from el 183 to 195 ft.	12.0	95.78	120	13.2	25	8.46	892.25
17	Over node 9 in quadrant 1; from el 183 to 195 ft.	12.0	95.31	120	13.2	25	8.82	887.09
18	Over node 10 in quadrant 1; from el 183 to 195 ft.	12.0	86.44	120	13.2	25	6.25	796.98
19	Over node 2; from el 187 to 195 ft.	8.0	179.22	120	13.2	25	7.43	1251.37
20	Over node 3; from el 187 to 195 ft.	8.0	113.1	120	13.2	25	7.98	814.29
21	Over node 7; from el 187 to 195 ft.	8.0	229.18	120	13.2	25	8.92	1581.52
22	Over node 12; from el 187 to 195 ft.	8.0	236.11	120	13.2	25	5.82	1631.41

TABLE 6.2.1-12 (SHEET 3 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
23	Over node 11; from el 187 to 195 ft.	8.0	110.51	120	13.2	25	4.66	795.67
24	Over node 13; from el 187 to 195 ft.	8.0	223.84	120	13.2	25	4.54	1572.59
25	Around RCP 4; from el 195 to 200 ft.	5.0	89.33	120	13.2	25	7.19	402.0
26	Between RCP 4 and wall; from el 195 to 200 ft.	5.0	31.62	120	13.2	25	7.16	142.3
27	Between SG 4 and RCP 4; from el 195 to 200 ft.	5.0	89.14	120	13.2	25	7.18	348.52
28	Between SG 4 and HVAC shaft; from el 195 to 200 ft.	5.0	32.94	120	13.2	25	7.11	145.4
29	Between SG 4 and wall; from el 195 to 200 ft.	5.0	83.81	120	13.2	25	6.90	324.23
30	Between SC 4 and concrete beam; from el 195 to 200 ft.	5.0	176.1	120	13.2	25	6.94	668.07
31	Adjacent to node 32; from el 195 to 200 ft.	5.0	102.29	120	13.2	25	7.13	460.28
32	Opposite node 27; from el 195 to 200 ft.	5.0	197.63	120	13.2	25	7.18	788.23
33	Between SG 1 and concrete beam; from el 195 to 200 ft.	5.0	176.84	120	13.2	25	6.70	671.4
34	Between SG 1 and wall; from el 195 to 200 ft.	5.0	81.91	120	13.2	25	6.43	315.67

TABLE 6.2.1-12 (SHEET 4 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
35	Between SG 1 and HVAC shaft; from el 195 to 200 ft.	5.0	33.08	120	13.2	25	6.0	146.0
36	Between SG 1 and RCP 1; from el 195 to 200 ft.	5.0	99.23	120	13.2	25	4.67	393.9
37	Between RCP 1 and wall; from el 195 to 200 ft.	5.0	37.59	120	13.2	25	4.54	169.15
38	Around RCP 1; from el 195 to 200 ft.	5.0	114.17	120	13.2	25	4.53	513.74
39	Opposite node 36; from el 195 to 200 ft.	5.0	225.29	120	13.2	25	4.87	912.67
40	Adjacent to node 39; from el 195 to 200 ft.	5.0	103.1	120	13.2	25	5.85	463.95
41	Between RCP 4 and northern wall of quadrant; from el 200 to 220 ft.	20.0	100.1	120	13.2	25	6.12	1796.2
42	Between RCP 4 and SW wall of quadrant; from el 200 to 220 ft.	20.0	35.3	120	13.2	25	6.10	635.37
43	Between HVAC shaft, RCP 4, and SG 4. from el 200 to 220 ft.	20.0	144.38	120	13.2	25	6.22	2577.68
44	Between SG 4 and southern wall of quadrant; from el 200 to 215 ft.	15.0	31.7	120	13.2	25	6.23	419.36
45	Between node 44 and quadrant 1; from el 200 to 207 ft.	7.0	111.91	102	13.2	25	6.39	697.96

TABLE 6.2.1-12 (SHEET 5 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
46	Over node 45; from el 207 to 215 ft.	8.0	110.59	120	13.2	25	6.19	796.22
47	Between node 48 and quadrant 1; from el 200 to 215 ft.	15.0	225.12	120	13.2	25	6.30	3015.61
48	Between SG 4 and northern wall, adjacent to node 49; from el 200 to 215 ft.	15.0	93.84	120	13.2	25	6.30	1258.89
49	Adjacent to node 43; between RCP 4 and northern wall; from el 200 to 220 ft.	20.0	124.07	120	13.2	25	6.22	2214.86
50	Between node 58 and quadrant 4; from el 200 to 215 ft.	15.0	224.85	120	13.2	25	6.03	3012.0
51	Between node 53 and quadrant 4; from el 200 to 207 ft.	7.0	111.55	120	13.2	25	6.23	695.7
52	Over node 51; from el 207 to 215 ft.	8.0	110.91	120	13.2	25	6.07	798.54
53	Between SG 1 and southern wall of quadrant 1; from el 200 to 215 ft.	15.0	30.43	120	13.2	25	5.97	402.25
54	Between HVAC shaft, RCP 1, and SG 1; from el 200 to 220 ft.	20.0	147.88	120	13.2	25	4.38	2640.6

TABLE 6.2.1-12 (SHEET 6 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
55	Between RCP 1 and SE wall of quadrant 1; from el 200 to 220 ft.	20.0	35.91	120	13.2	25	4.23	646.32
56	Between RCP 1 and northern wall of quadrant 1; from el 200 to 220 ft.	20.0	110.65	120	13.2	25	4.27	1986.14
57	Adjacent to node 54; between RCP 1 and northern wall; from el 200 to 220 ft.	20.0	139.29	120	13.2	25	4.47	2488.86
58	Between SG 1 and northern wall, adjacent to node 57; from el 200 to 215 ft.	15.0	100.43	120	13.2	25	4.99	1347.85
59	NW node of left half of SG 4 doghouse; from el 215 to 238 ft.	23.0	50.73	120	13.2	25	2.10	918.0
60	SW node of left half of SG 4 doghouse; from el 215 to 229 ft.	14.0	52.98	120	13.2	25	2.33	596.4
61	Over node 60; from el 229 to 238 ft.	9.0	52.84	120	13.2	25	1.94	404.2
62	SE node of left half of SG 4 dog-house; from el 215 to 238 ft.	23.0	79.51	120	13.2	25	2.17	1356.2
63	NE node of left half of SG 4 dog-house; from el 215 to 238 ft.	23.0	79.87	120	13.2	25	2.15	1234.0

TABLE 6.2.1-12 (SHEET 7 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
64	Symmetric of node 63 in quadrant 1.	23.0	79.87	120	13.2	25	2.11	1234.0
65	Symmetric of node 62 in quadrant 1.	23.0	79.51	120	13.2	25	2.11	1356.2
66	Symmetric of node 60 in quadrant 1.	14.0	52.98	120	13.2	25	2.24	596.4
67	Symmetric of node 61 in quadrant 1.	9.0	52.84	120	13.2	25	1.92	404.2
68	Symmetric of node 59 in quadrant 1.	23.0	50.73	120	13.2	25	2.05	918.0
69	HVAC shaft 4	-	-	120	13.2	25	6.06	838.6
70	HVAC shaft 1.	-	-	120	13.2	25	3.98	838.6
71	HVAC duct 4.	-	-	120	13.2	25	3.99	1231.6
72	HVAC duct 1.	-	-	120	13.2	25	3.0	1231.6
73	Quadrants 2 and 3 of SG compartment.	-	-	120	13.2	25	1.85	8.36E+04
74	Containment atmosphere.	-	-	120	13.2	25	1.75	2.75E+06

TABLE 6.2.1-13 (SHEET 1 OF 7)

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW EL 220 ft):
NODE CHARACTERISTICS STEAM GENERATOR OUTLET NOZZLE (436-in.^{2} BREAK)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
1	Interface of two halves of SG compartment.	6.56	222.97	120	13.2	25	8.71	1316.42
2	Between cold leg and wall; from el 171 ft 9 in. to 187 ft.	15.25	176.0	120	13.2	25	16.79	2283.14
3	Between RCP 4, SG 4, and wall; from el 171 ft 9 in. to 187 ft.	15.25	132.24	120	13.2	25	21.33	1661.50
4	By SG 4; from el 171 ft 9 in. to183 ft.	11.25	83.6	120	13.2	25	17.26	809.05
5	Adjacent to SG 4; from el 171 ft 9 in. to 183 ft.	11.25	95.78	120	13.2	25	16.41	895.04
6	Around SG 4; from el 171 ft 9 in. to 195 ft.	23.25	165.73	120	13.2	25	16.27	3124.98
7	Between hot, cold, and suction legs; from el 171 ft 9 in. to 187 ft.	15.25	189.03	120	13.2	25	17.45	3721.19
8	Around SG 1; from el 171 ft 9 in. to 195 ft.	23.25	163.51	120	13.2	25	15.72	3078.62
9	Adjacent to SG 1; from el 171 ft 9 in. to 183 ft.	11.25	95.31	120	13.2	25	15.85	890.21
10	By SG 1; from el 171 ft 9 in. to 183 ft.	11.25	86.44	120	13.2	25	15.98	837.79

TABLE 6.2.1-13 (SHEET 2 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
11	Between RCP 1, SG 1, and wall; from el 171 ft 9 in. to 187 ft.	15.25	137.8	120	13.2	25	9.83	1737.79
12	Between hot, cold, and suction legs 1; from el 171 ft 9 in. to 187 ft.	15.25	289.64	120	13.2	25	13.90	3729.58
13	Between hot leg 1 and wall; from el 171 ft 9 in. to 187 ft.	15.25	219.7	120	13.2	25	9.81	2815.04
14	Interface of two halves of SG compartment; between quadrants 1 and 2.	20.25	131.71	120	13.2	25	7.53	2400.44
15	Over node 4; from el 183 to 195 ft.	12.0	83.6	120	13.2	25	17.17	766.38
16	Over node 5; from el 183 to 195 ft.	12.0	95.78	120	13.2	25	16.39	892.25
17	Over node 9 in quadrant 1; from el 183 to 195 ft.	12.0	95.31	120	13.2	25	15.90	887.09
18	Over node 10 in quadrant 1; from el 183 to 195 ft.	12.0	86.44	120	13.2	25	15.94	796.98
19	Over node 2; from el 187 to 195 ft.	8.0	179.22	120	13.2	25	16.83	1251.37
20	Over node 3; from el 187 to 195 ft.	8.0	113.1	120	13.2	25	24.79	814.29
21	Over-node 7; from el 187 to 195 ft.	8.0	229.18	120	13.2	25	17.57	1581.52
22	Over node 12; from el 187 to 195 ft.	8.0	236.11	120	13.2	25	12.96	1631.41

TABLE 6.2.1-13 (SHEET 3 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
23	Over node 11; from el 187 to 195 ft.	8.0	110.51	120	13.2	25	11.15	795.67
24	Over node 13; from el 187 to 195 ft.	8.0	223.84	120	13.2	25	11.08	1572.59
25	Around RCP 4; from el 195 to 200 ft.	5.0	89.33	120	13.2	25	16.26	402.0
26	Between RCP 4 and wall; from el 195 to 200 ft.	5.0	31.62	120	13.2	25	16.54	142.3
27	Between SG 4 and RCP 4; from el 195 to 200 ft.	5.0	89.14	120	13.2	25	17.76	348.52
28	Between SG 4 and HVAC shaft; from el 195 to 200 ft.	5.0	32.94	120.	13.2	25	16.72	145.4
29	Between SG 4 and wall; from el 195 to 200 ft.	5.0	83.81	120	13.2	25	15.26	324.23
30	Between SG 4 and concrete beam; from el 195 to200 ft.	5.0	176.1	120	13.2	25	15.22	668.07
31	Adjacent to node 32; from el 195 to 200 ft.	5.0	102.29	120	13.2	25	15.87	460.28
32	Opposite node 27; from el 195 to200 ft.	5.0	197.63	120	13.2	25	16.20	788.23
33	Between SG 1 and concrete beam; from el 195 to 200 ft.	5.0	176.84	120	13.2	25	14.66	671.4
34	Between SG 1 and wall; from el195 to 200 ft.	5.0	81.91	120	13.2	25	14.71	315.67

TABLE 6.2.1-13 (SHEET 4 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
35	Between SG 1 and HVAC shaft; from el 195 to 200 ft.	5.0	33.08	120	13.2	25	14.36	146.0
36	Between SG 1 and RCP 1; from el195 to 200 ft.	5.0	99.23	120	13.2	25	11.14	393.9
37	Between RCP 1 and wall; from el 195 to 200 ft.	5.0	37.59	120	13.2	25	11.18	169.15
38	Around RCP 1; from el 195 to 200 ft.	5.0	114.17	120	13.2	25	11.10	513.74
39	Opposite node 36; from el 195 to200 ft.	5.0	225.29	120	13.2	25	11.55	912.67
40	Adjacent to node 39; from el 195 to 200 ft.	5.0	103.1	120	13.2	25	13.20	463.95
41	Between RCP 4 and northern wall of quadrant; from el 200 to 220 ft.	20.0	100.1	120	13.2	25	14.58	1796.2
42	Between RCP 4 and SW wall of quadrant; from el 200 to 220 ft.	20.0	35.3	120	13.2	25	14.54	635.37
43	Between HVAC shaft, RCP 4, and SG 4. from el 200 to 220 ft.	20.0	144.38	120	13.2	25	14.71	2577.68
44	Between SG 4 and southern wall of quadrant; from el 200 to 215 ft.	15.0	31.7	120	13.2	25	14.87	419.36
45	Between node 44 and quadrant 1; from el 200 to 207 ft.	7.0	111.91	102	13.2	25	14.69	697.96

TABLE 6.2.1-13 (SHEET 5 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
46	Over node 45; from el 207 to 215 ft.	8.0	110.59	120	13.2	25	14.48	796.22
47	Between node 48 and quadrant 1; from el 200 to 215 ft.	15.0	225.12	120	13.2	25	14.63	3015.61
48	Between SG 4 and northern wall, adjacent to node 49; from el 200 to 215 ft.	15.0	93.84	120	13.2	25	14.74	1258.89
49	Adjacent to node 43; between RCP 4 and northern wall; from el 200 to 220 ft.	20.0	124.07	120	13.2	25	14.7	2214.86
50	Between node 58 and quadrant 4; from el 200 to 215 ft.	15.0	224.85	120 °	13.2	25	13.96	3012.0
51	Between node 53 and quadrant 4; from el 200 to 207 ft.	7.0	111.55	120	13.2	25	14.37	695.7
52	Over node 51; from el 207 to 215 ft.	8.0	110.91	120	13.2	25	14.19	798.54
53	Between SG 1 and southern wall of quadrant 1; from el 200 to 215 ft.	15.0	30.43	120	13.2	25	14.13	402.25
54	Between HVAC shaft, RCP 1, and SG 1,; from el 200 to 220 ft.	20.0	147.88	120	13.2	25	11.02	2640.6

TABLE 6.2.1-13 (SHEET 6 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
55	Between RCP 1 and SE wall of quadrant 1; from el 200 to 220 ft.	20.0	35.91	120	13.2	25	10.75	646.32
56	Between RCP 1 and northern wall of quadrant 1; from el 200 to 220 ft.	20.0	110.65	120	13.2	25	10.77	1986.14
57	Adjacent to node 54; between RCP 1 and northern wall; from el 200 to 220 ft.	20.0	139.29	120	13.2	25	11.12	2488.86
58	Between SG 1 and northern wall, adjacent to node 57; from el 200 to 215 ft.	15.0	100.43	120	13.2	25	12.04	1347.85
59	NW node of left half of SG 4 doghouse; from el 215 to 238 ft.	23.0	50.73	120	13.2	25	4.45	918.0
60	SW node of left half of SG 4 dog house; from el 215 to 229 ft.	14.0	52.98	120	13.2	25	5.28	596.4
61	Over node 60; from el 229 to 238 ft.	9.0	52.84	120	13.2	25	3.85	404.2
62	SE node of left half of SG 4 dog house; from el 215 to 238 ft.	23.0	79.51	120	13.2	25	4.62	1356.2
63	NE node of-left half of SG 4 dog house; from el 215 to 238 ft.	23.0	79.87	120	13.2	25	4.60	1234.0

TABLE 6.2.1-13 (SHEET 7 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
64	Symmetric of node 63 in quadrant 1.	23.0	79.87	120	13.2	25	4.47	1234.0
65	Symmetric of node 62 in quadrant 1.	23.0	79.51	120	13.2	25	4.48	1356.2
66	Symmetric of node 60 in quadrant 1.	14.0	52.98	120	13.2	25	4.96	596.4
67	Symmetric of node 61 in quadrant 1.	9.0	52.84	120	13.2	25	3.76	404.2
68	Symmetric of node 59 in quadrant 1.	23.0	50.73	120	13.2	25	4.28	918.0
69	HVAC shaft 4	-	-	120	13.2	25	14.63	838.6
70	HVAC shaft 1.	-	-	120	13.2	25	10.20	838.6
71	HVAC duct 4.	-	-	120	13.2	25	11.34	1231.6
72	HVAC duct 1.	-	-	120	13.2	25	7.88	1231.6
73	Quadrants 2 and 3 of SG compartment.	-	-	120	13.2	25	3.58	8.36E+04
74	Containment atmosphere.	-	-	120	13.2	25	3.15	2.75E+06

TABLE 6.2.1-14 (SHEET 1 OF 7)

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW EL. 220 ft):
NODE CHARACTERISTICS REACTOR COOLANT PUMP OUTLET NOZZLE (236-in.^{2} BREAK)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature ($^\circ$F)	Pressure (psia)	Humidity (%)		
1	Interface of two halves of SG compartment.	6.56	222.97	120	13.2	25	7.62	1316.42
2	Between cold leg and wall; from el 171 ft 9 in. to 187 ft.	15.25	176.0	120	13.2	25	14.32	2283.14
3	Between RCP 4, SG 4, and wall; from el 171 ft 9 in. to 187 ft.	15.25	132.24	120	13.2	25	13.42	1661.50
4	By SG 4; from el 171 ft 9 in. to 183 ft.	11.25	83.6	120	13.2	25	12.51	809.05
5	Adjacent to SG 4; from el 171 ft 9 in. to 183 ft.	11.25	95.78	120	13.2	25	12.36	895.04
6	Around SG 4; from el 171 ft 9 in. to 195 ft.	23.25	165.73	120	13.2	25	12.34	3124.98
7	Between hot, cold, and suction legs; from el 171 ft 9 in. to 187 ft.	15.25	189.03	120	13.2	25	14.33	3721.19
8	Around SG 1; from el 171 ft 9 in. to 195 ft.	23.25	163.51	120	13.2	25	11.92	3078.62
9	Adjacent to SG 1; from el 171 ft 9 in. to 183 ft.	11.25	95.31	120	13.2	25	12.0	890.21
10	By SG 1; from el 171 ft 9 in. to 183 ft.	11.25	86.44	120	13.2	25	10.56	837.79

TABLE 6.2.1-14 (SHEET 2 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
11	Between RCP 1, SG 1, and wall; from el 171 ft 9 in. to 187 ft.	15.25	137.8	120	13.2	25	7.35	1737.79
12	Between hot, cold, and suction legs 1; from el 171 ft 9 in. to 187 ft.	15.25	289.64	120	13.2	25	10.44	3729.58
13	Between hot leg 1 and wall; from el 171 ft 9 in. to 187 ft.	15.25	219.7	120	13.2	25	7.34	2815.04
14	Interface of two halves of SG compartment; between quadrants 1 and 2.	20.25	131.71	120	13.2	25	5.79	2400.44
15	Over node 14; from el 183 to 195 ft.	12.0	83.6	120	13.2	25	12.41	766.38
16	Over node 5; from el 183 to 195 ft.	12.0	95.78	120	13.2	25	12.32	892.25
17	Over node 9 in quadrant 1; from el 183 to 195 ft.	12.0	95.31	120	13.2	25	12.0	887.09
18	Over node 10 in quadrant 1; from el 183 to 195 ft.	12.0	86.44	120	13.2	25	10.55	796.98
19	0ver node 2; from el 187 to 195 ft.	8.0	179.22	120	13.2	25	14.37	1251.37
20	Over node 3; from el 187 to 195 ft.	8.0	113.1	120	13.2	25	13.37	814.29
21	Over node 7; from el 187 to 1'95 ft.	8.0	229.18	120	13.2	25	14.25	1581.52
22	Over node 12; from el 187 to 195 ft.	8.0	236.11	120	13.2	25	9.69	1631.41

TABLE 6.2.1-14 (SHEET 3 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
23	Over node 11; from el 187 to 195 ft.	8.0	110.51	120	13.2	25	8.25	795.67
24	Over node 13; from el 187 to 195 ft.	8.0	223.84	120	13.2	25	8.25	1572.59
25	Around RCP 14; from el 195 to 200 ft.	5.0	89.33	120	13.2	25	13.08	402.0
26	Between RCP 4 and wall; from el 195 to 200 ft.	5.0	31.62	120	13.2	25	13.04	142.3
27	Between SG 4 and RCP 4; from el 195 to 200 ft.	5.0	89.14	120	13.2	25	12.47	348.52
28	Between SG 4 and HVAC shaft; from el 195 to 200 ft.	5.0	32.94	120	13.2	25	12.0	145.14
29	Between SG 4 and wall; from el 195 to 200 ft.	5.0	83.81	120	13.2	25	11.41	324.23
30	Between SG 4 and concrete beam; from el 195 to 200 ft.	5.0	176.1	120	13.2	25	11.43	668.07
31	Adjacent to node 32; from el 195 to 200 ft.	5.0	102.29	120	13.2	25	12.24	460.28
32	Opposite node 27; from el 195 to 200 ft.	5.0	197.63	120	13.2	25	12.63	788.23
33	Between SG 1 and concrete beam; from el 195 to 200 ft.	5.0	176.84	120	13.2	25	10.87	671.14
34	Between SG 1 and wall; from el 195 to 200 ft.	5.0	81.91	120	13.2	25	10.69	315.67

TABLE 6.2.1-14 (SHEET 4 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
35	Between SG 1 and HVAC shaft; from el 195 to 200 ft.	5.0	33.08	120	13.2	25	10.22	146.0
36	Between SG 1 and RCP 1; from el 195 to 200 ft.	5.0	99.23	120	13.2	25	8.25	393.9
37	Between RCP 1 and wall; from el 195 to 200 ft.	5.0	37.59	120	13.2	25	8.23	169.15
38	Around RCP 1; from el 195 to 200 ft.	5.0	114.17	120	13.2	25	8.23	513.74
39	Opposite node 36; from el 195 to 200 ft.	5.0	225.29	120	13.2	25	8.60	912.67
40	Adjacent to node 39; from el 195 to 200 ft.	5.0	103.1	120	13.2	25	9.83	463.95
41	Between RCP 4 and northern wall of quadrant; from el 200 to 220 ft.	20.0	100.1	120	13.2	25	10.94	1796.2
42	Between RCP 4 and SW wall of quadrant; from el 200 to 220 ft.	20.0	35.3	120	13.2	25	10.9	635.37
43	Between HVAC shaft, RCP 4, and SG 4. from el 200 to 220 ft.	20.0	144.38	120	13.2	25	10.99	2577.68
44	Between SG 4 and southern wall of quadrant; from el 200 to 215 ft.	15.0	31.7	120	13.2	25	10.95	419.36
45	Between node 44 and quadrant 1; from el 200 to 207 ft.	7.0	111.91	102	13.2	25	10.93	697.96

TABLE 6.2.1-14 (SHEET 5 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
46	Over node 45; from el 207 to 215 ft.	8.0	110.59	120	13.2	25	10.74	796.22
47	Between node 48 and quadrant 1; from el 200 to 215 ft.	15.0	225.12	120	13.2	25	10.95	3015.61
48	Between SG 4 and northern wall, adjacent to node 49; from el 200 to 215 ft.	15.0	93.84	120	13.2	25	11.07	1258.89
49	Adjacent to node 43; between RCP 4 and northern wall; from el 200 to 220 ft.	20.0	214.07	120	13.2	25	11.0	2214.86
50	Between node 58 and quadrant 4; from el 200 to 215 ft.	15.0	224.85	120	13.2	25	10.36	3012.0
51	Between node 53 and quadrant 4; from el 200 to 207 ft.	7.0	111.55	120	13.2	25	10.58	695.7
52	Over node 51; from el 207 to 215 ft.	8.0	110.91	120	13.2	25	10.46	798.54
53	Between SG 1 and southern wall of quadrant 1; from el 200 to 215 ft.	15.0	30.43	120	13.2	25	10.32	402.25
54	Between HVAC shaft, RCP 1, and SG 1; from el 200 to 220 ft.	20.0	147.88	120	13.2	25	8.21	2640.6

TABLE 6.2.1-14 (SHEET 6 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
55	Between RCP 1 and SE wall of quadrant 1; from el 200 to 220 ft.	20.0	35.91	120	13.2	25	7.98	646.32
56	Between RCP 1 and northern wall of quadrant 1; from el 200 to 220 ft.	20.0	110.65	120	13.2	25	8.0	1986.14
57	Adjacent to node 54; between RCP 1 and northern wall; from el 200 to 220 ft.	20.0	139.29	120	13.2	25	8.26	2488.86
58	Between SG 1 and northern wall, adjacent to node 57; from el 200 to 215 ft.	15.0	100.43	120	13.2	25	8.95	1347.85
59	NW node of left half of SG 4 doghouse; from el 215 to 238 ft.	23.0	50.73	120	13.2	25	4.01	918.0
60	SW node of left half of SG 4 dog-house; from el 215 to 229 ft.	14.0	52.98	120	13.2	25	4.60	596.4
61	Over node 60; from el 229 to 238 ft.	9.0	52.84	120	13.2	25	3.59	404.2
62	SE node of left half of SG 4 dog-house; from el 215 to 238 ft.	23.0	79.51	120	13.2	25	4.13	1356.2
63	NE node of left half of SG 4 dog-house; from el 215 to 238 ft.	23.0	79.87	120	13.2	25	4.12	1234.0

TABLE 6.2.1-14 (SHEET 7 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
64	Symmetric of node 63 in quadrant 1.	23.0	79.87	120	13.2	25	4.0	1234.0
65	Symmetric of node 62 in quadrant 1.	23.0	79.51	120	13.2	25	4.01	1356.2
66	Symmetric of node 60 in quadrant 1.	14.0	52.98	120	13.2	25	4.33	596.4
67	Symmetric of node 61 in quadrant 1.	9.0	52.84	120	13.2	25	3.51	404.2
68	Symmetric of node 59 in quadrant 1.	23.0	50.73	120	13.2	25	3.86	918.0
69	HVAC shaft 4		-	120	13.2	25	10.82	838.6
70	HVAC shaft 1.	-	-	120	13.2	25	7.63	838.6
71	HVAC duct 4.	-	-	120	13.2	25	8.27	1231.6
72	HVAC duct 1.	-	-	120	13.2	25	5.96	1231.6
73	Quadrants 2 and 3 of SG compartment.	-	-	120	13.2	25	3.41	8.36E+04
74	Containment atmosphere.	-	-	120	13.2	25	3.10	2.75E+06

TABLE 6.2.1-15 (SHEET 1 OF 7)

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW EL. 220 ft):
NODE CHARACTERISTICS REACTOR COOLANT PUMP INLET NOZZLE (336-in.^{2} BREAK)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
1	Interface of two halves of SG compartment.	6.56	222.97	120	13.2	25	7.13	1316.42
2	Between cold leg and wall; from el 171 ft 9 in. to 187 ft.	15.25	176.0	120	13.2	25	13.64	2283.14
3	Between RCP 4, SG 4, and wall; from el 171 ft 9 in. to 187 ft.	15.25	132.24	120	13.2	25	15.95	1661.50
4	By SG 4; from el 171 ft 9 in. to 183 ft.	11.25	83.6	120	13.2	25	13.71	809.05
5	Adjacent to SG 4; from el 171 ft 9 in. to 183 ft.	11.25	95.78	120	13.2	25	13.33	895.04
6	Around SG 4; from el 171 ft 9 in. to 195 ft.	23.25	165.73	120	13.2	25	13.33	3124.98
7	Between hot, cold, and suction legs; from el 171 ft 9 in. to 187 ft.	15.25	189.03	120	13.2	25	15.02	3721.19
8	Around SG 1; from el 171 ft 9 in. to 195 ft.	23.25	163.51	120	13.2	25	12.83	3078.62
9	Adjacent to SG 1; from el 171 ft 9 in. to 183 ft.	11.25	95.31	120	13.2	25	12.96	890.21
10	By SG 1; from el 171 ft 9 in. to 183 ft.	11.25	86.44	120	13.2	25	11.06	837.79

TABLE 6.2.1-15 (SHEET 2 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
11	Between RCP 1, SG 1, and wall; from el 171 ft 9 in. to 187 ft.	15.25	137.8	120	13.2	25	7.76	1737.79
12	Between hot, cold, and suction legs 1; from el 171 ft 9 in. to 187 ft.	15.25	289.64	120	13.2	25	11.19	3729.58
13	Between hot leg 1 and wall; from el 171 ft 9 in. to 187 ft.	15.25	219.7	120	13.2	25	7.75	2815.04
14	Interface of two halves of SG compartment between quadrants 1 and 2.	20.25	131.71	120	13.2	25	5.96	2400.44
15	Over node 14; from el 183 to 195 ft.	12.0	83.6	120	13.2	25	13.58	766.38
16	Over node 5; from el 183 to 195 ft.	12.0	95.78	120	13.2	25	13.32	892.25
17	Over node 9 in quadrant 1; from el 183 to 195 ft.	12.0	95.31	120	13.2	25	12.94	887.09
18	Over node 10 in quadrant 1; from el 183 to 195 ft.	12.0	86.144	120	13.2	25	11.06	796.98
19	Over node 2; from el 187 to 195 ft.	8.0	179.22	120	13.2	25	13.63	1251.37
20	Over node 3; from el 187 to 195 ft.	8.0	113.1	120	13.2	25	16.06	814.29
21	Over node 7; from el 187 to 195 ft.	8.0	229.18	120	13.2	25	14.98	1581.52
22	Over node 12; from el 187 to 195 ft.	8.0	236.11	120	13.2	25	10.33	1631.41

TABLE 6.2.1-15 (SHEET 3 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
23	Over node 11; from el 187 to 195 ft.	8.0	110.51	120	13.2	25	8.73	795.67
24	Over node 13; from el 187 to 195 ft.	8.0	223.84	120	13.2	25	8.75	1572.59
25	Around RCP 4; from el 195 to 200 ft.	5.0	89.33	120	13.2	25	13.10	402.0
26	Between RCP 4 and wall; from el 195 to 200 ft.	5.0	31.62	120	13.2	25	13.15	142.3
27	Between SG 4 and RCP 4; from el 195 to 200 ft.	5.0	89.14	120	13.2	25	13.31	348.52
28	Between SG 4 and HVAC shaft; from el 195 to 200 ft.	5.0	32.94	120	13.2	25	12.84	145.14
29	Between SG 4 and wall; from el 195 to 200 ft.	5.0	83.81	120	13.2	25	12.11	324.23
30	Between SG 4 and concrete beam; from el 195 to 200 ft.	5.0	176.1	120	13.2	25	12.14	668.07
31	Adjacent to node 32; from el 195 to 200 ft.	5.0	102.29	120	13.2	25	12.82	460.28
32	Opposite node 27; from el 195 to 200 ft.	5.0	197.63	120	13.2	25	13.06	788.23
33	Between SG 1 and concrete beam; from el 195 to 200 ft.	5.0	176.84	120	13.2	25	11.63	671.14
34	Between SG 1 and wall; from el 195 to 200 ft.	5.0	81.91	120	13.2	25	11.24	315.67

TABLE 6.2.1-15 (SHEET 4 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
35	Between SG 1 and HVAC shaft; from el 195 to 200 ft.	5.0	33.08	120	13.2	25	10.75	146.0
36	Between SG 1 and RCP 1; from el 195 to 200 ft.	5.0	99.23	120	13.2	25	8.75	393.9
37	Between RCP 1 and wall; from el 195 to 200 ft.	5.0	37.59	120	13.2	25	8.83	169.15
38	Around RCP 1; from el 195 to 200 ft.	5.0	114.17	120	13.2	25	8.76	513.74
39	Opposite node 36; from el 195 to 200 ft.	5.0	225.29	120	13.2	25	9.13	912.67
40	Adjacent to node 39; from el 195 to 200 ft.	5.0	103.1	120	13.2	25	10.46	463.95
41	Between RCP 4 and northern wall of quadrant; from el 200 to 220 ft.	20.0	100.1	120	13.2	25	11.48	1796.2
42	Between RCP 4 and SW wall of quadrant; from el 200 to 220 ft.	20.0	35.3	120	13.2	25	11.42	635.37
43	Between HVAC shaft, RCP 4, and SG 4. from el 200 to 22O ft.	20.0	144.38	120	13.2	25	11.56	2577.68
44	Between SG 4 and southern wall of quadrant; from el 200 to 215 ft.	15.0	31.7	120	13.2	25	11.67	419.36
45	Between node 44 and quadrant 1; from el 200 to 207 ft.	7.0	111.91	102	13.2	25	11.57	697.96

TABLE 6.2.1-15 (SHEET 5 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
46	Over node 45; from el 207 to 215 ft.	8.0	110.59	120	13.2	25	11.37	796.22
47	Between node 48 and quadrant 1; from el 200 to 215 ft.	15.0	225.12	120	13.2	25	11.56	3015.61
48	Between SG 4 and northern wall, adjacent to node 49; from el 200 to 215 ft.	15.0	93.84	120	13.2	25	11.67	1258.89
49	Adjacent to node 43; between RCP 4 and northern wall; from el 200 to 220 ft.	20.0	124.07	120	13.2	25	11.58	2214.86
50	Between node 58 and quadrant 4; from el 200 to 215 ft.	15.0	224.85	120	13.2	25	10.97	3012.0
51	Between node 53 and quadrant 4; from el 200 to 207 ft.	7.0	111.55	120	13.2	25	11.14	695.7
52	Over node 51; from el 207 to 215 ft.	8.0	110.91	120	13.2	25	11.03	798.54
53	Between SG 1 and southern wall of quadrant 1; from el 200 to 215 ft.	15.0	30.43	120	13.2	25	10.88	402.25
54	Between HVAC shaft, RCP 1, and SG 1; from el 200 to 220 ft.	20.0	147.88	120	13.2	25	8.69	2640.6

TABLE 6.2.1-15 (SHEET 6 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
55	Between RCP 1 and SE wall of quadrant 1; from el 200 to 220 ft.	20.0	35.91	120	13.2	25	8.46	646.32
56	Between RCP 1 and northern wall of quadrant 1; from el 200 to 220 ft.	20.0	110.65	120	13.2	25	8.48	1986.14
57	Adjacent to node 54; between RCP 1 and northern wall; from el 200 to 220 ft.	20.0	139.29	120	13.2	25	8.76	2488.86
58	Between SG 1 and northern wall, adjacent to node 57; from el 200 to 215 ft.	15.0	100.43	120	13.2	25	9.49	1347.85
59	NW node of left half of SG 4 doghouse; from el 215 to 238 ft.	23.0	50.73	120	13.2	25	3.76	918.0
60	SW node of left half of SG 14 doghouse; from el 215 to 229 ft.	14.0	52.98	120	13.2	25	4.42	596.4
61	Over node 60; from el 229 to 238 ft.	9.0	52.84	120	13.2	25	3.29	1404.2
62	SE node of left half of SG 14 dog-house; from el 215 to 238 ft.	23.0	79.51	120	13.2	25	3.89	1356.2
63	NE node of left half of SG 4 doghouse; from el 215 to 238 ft.	23.0	79.87	120	13.2	25	3.88	1234.0

TABLE 6.2.1-15 (SHEET 7 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
64	Symmetric of node 63 in quadrant 1.	23.0	79.87	120	13.2	25	3.76	1234.0
65	Symmetric of node 62 in quadrant 1.	23.0	79.51	120	13.2	25	3.77	1356.2
66	Symmetric of node 60 in quadrant 1.	14.0	52.98	120	13.2	25	4.13	596.14
67	Symmetric of node 61 in quadrant 1.	9.0	52.84	120	13.2	25	3.20	404.2
68	Symmetric of node 59 in quadrant 1.	23.0	50.73	120	13.2	25	3.60	918.0
69	HVAC shaft 4		-	120	13.2	25	11.45	838.6
70	HVAC shaft 1.	-	-	120	13.2	25	8.04	838.6
71	HVAC duct 4.	-	-	120	13.2	25	8.72	1231.6
72	HVAC duct 1.	-	-	120	13.2	25	6.20	1231.6
73	Quadrants 2 and 3 of SG compartment.	-	-	120	13.2	25	3.06	8.36E+O14
74	Containment atmosphere.	-	-	120	13.2	25	2.73	2.75E+O6

TABLE 6.2.1-16 (SHEET 1 OF 7)

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW EL 220 ft):
NODE CHARACTERISTICS STEAM GENERATOR INLET ELBOW (763-in.^{2} BREAK)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
1	Interface of two halves of SG compartment.	6.56	222.97	120	13.2	25	10.54	1316.42
2	Between cold leg and wall; from el 171 ft 9 in. to 187 ft.	15.25	176.0	120	13.2	25	19.63	2283.14
3	Between RCP 4, SG 4, and wall; from el 171 ft 9 in. to 187 ft.	15.25	132.24	120	13.2	25	20.52	1661.50
4	By SG 4; from el 171 ft 9 in. to 183 ft.	11.25	83.6	120	13.2	25	20.97	809.05
5	Adjacent to SG 4; from el 171 ft 9 in. to 183 ft.	11.25	95.78	120	13.2	25	21.34	895.04
6	Around SG 4; from el 171 ft 9 in. to 195 ft.	23.25	165.73	120	13.2	25	22.13	3124.98
7	Between hot, cold, and suction legs; from el 171 ft 9 in. to 187 ft.	15.25	189.03	120	13.2	25	22.16	3721.19
8	Around SG 1; from el 171 ft 9 in. to 195 ft.	23.25	163.51	120	13.2	25	20.93	3078.62
9	Adjacent to SG 1; from el 171 ft 9 in. to 183 ft.	11.25	95.31	120	13.2	25	21.12	890.21
10	By SG 1; from el 171 ft 9 in. to 183 ft.	11.25	86.44	120	13.2	25	17.50	837.79

TABLE 6.2.1-16 (SHEET 2 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
11	Between RCP 1, SG 1, and wall; from el 171 ft 9 in. to 187 ft.	15.25	137.8	120	13.2	25	12.69	1737.79
12	Between hot, cold, and suction legs 1; from el 171 ft 9 in. to 187 ft.	15.25	289.64	120	13.2	25	18.17	3729.58
13	Between hot leg 1 and wall; from el 171 ft 9 in. to 187 ft.	15.25	219.7	120	13.2	25	12.66	2815.04
14	Interface of two halves of SG compartment; between quadrants 1 and 2.	20.25	131.71	120	13.2	25	9.83	2400.44
15	Over node 4; from el 183 to 195 ft.	12.0	83.6	120	13.2	25	20.90	766.38
16	Over node 5; from el 183 to 195 ft.	12.0	95.78	120	13.2	25	21.22	892.25
17	Over node 9 in quadrant 1; from el 183 to 195 ft.	12.0	95.31	120	13.2	25	21.03	887.09
18	Over node 10 in quadrant 1; from el 183 to 195 ft.	12.0	86.44	120	13.2	25	17.52	796.98
19	Over node 2; from el 187 to 195 ft.	8.0	179.22	120	13.2	25	19.66	1251.37
20	Over node 3; from el 187 to 195 ft.	8.0	113.1	120	13.2	25	20.54	814.29
21	Over node 7; from el 187 to 195 ft.	8.0	229.18	120	13.2	25	22.03	1581.52
22	Over node 12; from el 187 to 195 ft.	8.0	236.11	120	13.2	25	16.74	1631.41

TABLE 6.2.1-16 (SHEET 3 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
23	Over node 11; from el 187 to 195 ft.	8.0	110.51	120	13.2	25	14.13	795.67
24	Over node 13; from el 187 to 195 ft.	8.0	223.84	120	13.2	25	14.21	1572.59
25	Around RCP 4; from el 195 to 200 ft.	5.0	89.33	120	13.2	25	19.22	402.0
26	Between RCP 4 and wall; from el 195 to 200 ft.	5.0	31.62	120	13.2	25	19.26	142.3
27	Between SG 4 and RCP 4; from el 195 to 200 ft.	5.0	89.14	120	19.28	25	19.28	348.52
28	Between SG 4 and HVAC shaft; from el 195 to 200 ft.	5.0	32.94	120	13.2	25	19.24	145.14
29	Between SG 4 and wall; from el 195 to 200 ft.	5.0	83.81	120	13.2	25	18.83	324.23
30	Between SG 4 and concrete beam; from el 195 to 200 ft.	5.0	176.1	120	13.2	25	18.93	668.07
31	Adjacent to node 32; from el 195 to 200 ft.	5.0	102.29	120	13.2	25	19.31	460.28
32	Opposite node 27; from el 195 to 200 ft. 200 ft.	5.0	197.63	120	13.2	25	19.34	788.23
33	Between SG 1 and concrete beam; from el 195 to 200 ft.	5.0	176.84	120	13.2	25	18.44	671.4
34	Between SG 1 and wall; from el 195 to 200 ft.	5.0	81.91	120	13.2	25	17.86	315.67

TABLE 6.2.1-16 (SHEET 4 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
35	Between SG 1 and HVAC shaft; from el 195 to 200 ft.	5.0	33.08	120	13.2	25	17.03	146.0
36	Between SG 1 and RCP 1; from el 195 to 200 ft.	5.0	99.23	120	13.2	25	14.21	393.9
37	Between RCP 1 and wall; from el 195 to 200 ft.	5.0	37.59	120	13.2	25	14.41	169.15
38	Around RCP 1; from el 195 to 200 ft.	5.0	114.17	120	13.2	25	14.22	513.74
39	Opposite node 36; from el 195 to 200 ft.	5.0	225.29	120	13.2	25	14.93	912.67
40	Adjacent to node 39; from el 195 to 200 ft.	5.0	103.1	120	13.2	25	16.82	463.95
41	Between RCP 4 and northern wall of quadrant; from el 200 to 220 ft.	20.0	100.1	120	13.2	25	17.55	1796.2
42	Between RCP 4 and SW wall of quadrant; from el 200 to 220 ft.	20.0	35.3	120	13.2	25	17.52	635.37
43	Between HVAC shaft, RCP 4, and SG 4; from el 200 to 220 ft.	20.0	144.38	120	13.2	25	17.72	2577.68
44	Between SG 4 and southern wall of quadrant; from el 200 to 215 ft.	15.0	31.7	120	13.2	25	17.72	419.36
45	Between node 44 and quadrant 1; from el 200 to 207 ft.	7.0	111.91	102	13.2	25	17.94	697.96

TABLE 6.2.1-16 (SHEET 5 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
46	Over node 45; from el 207 to 215 ft.	8.0	110.59	120	13.2	25	17.63	796.22
47	Between node 48 and quadrant 1; from el 200 to 215 ft.	15.0	225.12	120	13.2	25	17.82	3015.61
48	Between SG 4 and northern wall, adjacent to node 49; from el 200 to 215 ft.	15.0	93.84	120	13.2	25	17.82	1258.89
49	Adjacent to node 43; between RCP 4 and northern wall; from el 200 to 220 ft.	20.0	124.07	120	13.2	25	17.73	2214.86
50	Between node 58 and quadrant 4; from el 200 to 215 ft.	15.0	224.85	120	13.2	25	17.28	3012.0
51	Between node 53 and quadrant 4; from el 200 to 207 ft.	7.0	111.55	120	13.2	25	17.71	695.7
52	Over node 51; from el 207 to 215 ft.	8.0	110.91	120	13.2	25	17.38	798.54
53	Between SG 1 and southern wall of quadrant 1; from el 200 to 215 ft.	15.0	30.43	120	13.2	25	17.20	402.25
54	Between HVAC shaft, RCP 1, and SG 1; from el 200 to 220 ft.	20.0	147.88	120	13.2	25	14.08	2640.6

TABLE 6.2.1-16 (SHEET 6 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
55	Between RCP 1 and SE wall of quadrant 1; from el 200 to 220 ft.	20.0	35.91	120	13.2	25	13.78	646.32
56	Between RCP 1 and northern wall of quadrant 1; from el 200 to 220 ft.	20.0	110.65	120	13.2	25	13.82	1986.114
57	Adjacent to node 54; between RCP 1 and northern wall; from el 200 to 220 ft.	20.0	139.29	120	13.2	25	14.23	2488.86
58	Between SG 1 and northern wall, adjacent to node 57; from el 200 to 215 ft.	15.0	100.43	120	13.2	25	15.26	1347.85
59	NW node of left half of SG 4 doghouse; from el 215 to 238 ft.	23.0	50.73	120	13.2	25	5.65	918.0
60	SW node of left half of SG 4 dog-house; from el 215 to 229 ft.	14.0	52.98	120	13.2	25	6.62	596.4
61	Over node 60; from el 229 to 238 ft.	9.0	52.84	120	13.2	25	4.92	404.2
62	SE node of left half of SG 4 dog-house; from el 215 to 238 ft.	23.0	79.51	120	13.2	25	5.85	1356.2
63	NE node of left half of SG 4 dog-house; from el 215 to 238 ft.	23.0	79.87	120	13.2	25	5.83	1234.0

TABLE 6.2.1-16 (SHEET 7 OF 7)

Volume No.	Description	Height (ft)	Cross-Sectional Area (ft^2)	Initial Conditions			Calc Peak Pressure Differential (psig)	Net Free Volume (ft^3)
				Temperature (°F)	Pressure (psia)	Humidity (%)		
64	Symmetric of node 63 in quadrant 1.	23.0	79.87	120	13.2	25	5.69	1234.0
65	Symmetric of node 62 in quadrant 1.	23.0	79.51	120	13.2	25	5.71	1356.2
66	Symmetric of node 60 in quadrant 1.	14.0	52.98	120	13.2	25	6.30	596.4
67	Symmetric of node 61 in quadrant 1.	9.0	52.84	120	13.2	25	14.83	404.2
68	Symmetric of node 59 in quadrant 1.	23.0	50.73	120	13.2	25	5.47	918.0
69	HVAC shaft 14	-	-	120	13.2	25	17.45	838.6
70	HVAC shaft 1.	-	-	120	13.2	25	13.11	838.6
71	HVAC duct 14.	-	-	120	13.2	25	13.69	1231.6
72	HVAC duct 1.	-	-	120	13.2	25	10.24	1231.6
73	Quadrants 2 and 3 of SG compartment.	-		120	13.2	25	14.63	8.36E+04
74	Containment atmosphere.	-	-	120	13.2	25	14.07	2.75E+06

TABLE 6.2.1-16A

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK ABOVE EL 220 ft):
FLOW CHARACTERISTICS

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
1	2-1		X	22.8	0	0	1	0.41	1.41	0.329
2	1-4		X	39.1	0	0	1	0.41	1.41	0.234
3	1-5		X	52.4	0	0	1	0.41	1.41	0.097
4	1-7		X	75.8	0	0	1	0.41	1.41	0.070
5	1-11		X	23.2	0	0	1	0.35	1.35	0.286
6	1-12		X	73.0	2.04	0	1	0.20	3.24	0.139
7	2-3		X	30.5	0	0	1	0.41	1.41	0.777
8	2-5		X	32.6	0	0	1	0.16	1.16	0.211
9	2-12		X	45.4	0	0	1	0.02	1.02	0.099
10	3-4		X	22.8	0	0	1	0.41	1.41	0.637
11	3-6		X	34.3	0	0	1	0.16	1.16	0.199
12	3-12		X	48.26	0	0	1	0.02	1.02	0.089
13	4-7		X	49.1	0	0	1	0.18	1.18	0.131
14	4-12		X	73.0	2.04	0	1	0.20	3.24	0.059
15	5-6		X	63.5	0	0	1	0.41	1.41	0.186
16	5-8		X	13.44	0	0	1	0.38	1.38	0.302
17	6-7		X	52.4	0	0	1	0.41	1.41	0.155
18	6-9		X	19.06	0	0	1	0.35	1.35	0.256
19	7-10		X	19.12	0	0	1	0.38	1.38	0.219
20	8-9		X	86.0	0	0	1	0.29	1.29	0.139
21	8-11		X	178.6	0	0	1	0.29	1.29	0.054
22	9-10		X	52.1	0	0	1	0.29	1.29	0.168
23	10-11		X	93.7	0	0	1	0.41	1.41	0.126
24	9-12		X	29.5	0	0	1	0.45	1.45	0.198

TABLE 6.2.1-17 (SHEET 1 OF 5)

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW EL. 220 ft): FLOW CHARACTERISTICS
STEAM GENERATOR INLET ELBOW (763-in.^{2} BREAK)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
1	1-74		X	20.25	0	0	1.0	0.38	1.38	0.096
2	1-73		X	97.08	0	0	1.0	0	1.0	0.079
3	20-19		X	29.58	0	0	1.0	0.27	1.27	0.295
4	23-24		X	29.58	0	0	1.0	0.27	1.27	0.295
5	2-1		X	92.1	0	0	1.0	0.25	1.25	0.122
6	3-2		X	63.12	0	0	1.0	0.24	1.24	0.151
7	11-13		X	63.12	0	0	1.0	0.24	1.24	0.151
8	7-2		X	94.0	0	0	1.0	0.27	1.27	0.0623
9	13-12		X	125.8	0	0	1.0	0.27	1.27	0.0623
10	21-19		X	64.65	0	0	1.0	0.20	1.20	0.136
11	22-24		X	80.0	0	0	1.0	0.20	1.20	0.134
12	3-7		X	47.57	0	0	1.0	0.41	1.41	0.0698
13	11-12		X	47.57	0	0	1.0	0.41	1.41	0.0698
14	20-21		X	58.72	0	0	1.0	0.29	1.29	0.297
15	23-22		X	58.72	0	0	1.0	0.29	1.29	0.297
16	3-4		X	63.65	0	0	1.0	0.28	1.28	0.103
17	11-10		X	63.65	0	0	1.0	0.28	1.28	0.103
18	20-15		X	36.93	0	0	1.0	0.25	1.25	0.192
19	23-18		X	36.93	0	0	1.0	0.25	1.25	0.192
20	7-6		X	129.19	0	0	1.0	0.27	1.27	0.0509
21	12-8		X	129.19	0	0	1.0	0.27	1.27	0.0509
22	21-6		X	70.58	0	0	1.0	0.26	1.26	0.1303
23	22-8		X	70.58	0	0	1.0	0.26	1.26	0.1303
24	4-5		X	99.53	0	0	1.0	0.07	1.07	0.1265
25	10-9		X	99.53	0	0	1.0	0.07	1.07	0.1265
26	15-16		X	81.3	0	0	1.0	0.17	1.17	0.1093
27	18-17		X	81.3	0	0	1.0	0.17	1.17	0.1093
28	5-6		X	113.9	0	0	1.0	0.25	1.25	0.08965
29	9-8		X	113.9	0	0	1.0	0.25	1.25	0.08965
30	16-6		X	94.2	0	0	1.0	0.29	1.29	0.153
31	17-8		X	94.2	0	0	1.0	0.29	1.29	0.153
32	6-8		X	247.93	0	0	1.0	0.16	1.16	0.0310
33	5-9		X	76.37	0	0	1.0	0.17	1.17	0.1071
34	16-17		X	68.31	0	0	1.0	0.22	1.22	0.115
35	13-14		X	92.65	0	0	1.0	0.30	1.30	0.086
36	24-14		X	13.7	0	0	1.0	0.38	1.38	0.163
37	14-73		X	44.86	0	0	1.0	0.40	1.40	0.1063
38	14-74		X	50.6	0	0	1.0	0.35	1.35	0.42
39	19-25		X	80.58	0	0	1.0	0.05	1.05	0.0798
40	24-38		X	80.58	0	0	1.0	0.05	1.05	0.0798
41	19-26		X	28.52	0	0	1.0	0.18	1.18	0.185
42	24-37		X	28.52	0	0	1.0	0.18	1.18	0.185

TABLE 6.2.1-17 (SHEET 2 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
43	19-32		X	53.45	0	0	1.0	0	1.0	0.112
44	24-39		X	53.45	0	0	1.0	0	1.0	0.112
45	20-27		X	62.58	0	0	1.0	0.15	1.15	0.0794
46	23-36		X	62.58	0	0	1.0	0.15	1.15	0.0794
47	21-32		X	81.5	0	0	1.0	0.27	1.27	0.0389
48	22-39		X	81.5	0	0	1.0	0.27	1.27	0.0389
49	21-31		X	57.19	0	0	1.0	0.22	1.22	0.0778
50	22-40		X	57.19	0	0	1.0	0.22	1.22	0.0778
51	15-28		X	12.58	0	0	1.0	0.30	1.30	0.2903
52	18-35		X	12.58	0	0	1.0	0.30	1.30	0.2903
53	15-29		X	21.43	0	0	1.0	0.16	1.16	0.2396
54	18-34		X	21.43	0	0	1.0	0.16	1.16	0.2396
55	4-15		X	66.88	0	0	1.0	0.10	1.10	0.161
56	10-18		X	66.88	0	0	1.0	0.10	1.10	0.161
57	5-16		X	76.62	0	0	1.0	0.10	1.10	0.1378
58	9-17		X	76.62	0	0	1.0	0.10	1.10	0.1378
59	6-30		X	80.58	0	0	1.0	0.26	1.26	0.113
60	8-33		X	80.58	0	0	1.0	0.26	1.26	0.113
61	16-29		X	11.0	0	0	1.0	0.40	1.40	0.195
62	17-34		X	11.0	0	0	1.0	0.40	1.40	0.195
63	16-30		X	6.89	0	0	1.0	0.37	1.37	0.321
64	17-33		X	6.89	0	0	1.0	0.37	1.37	0.321
65	2-19		X	107.33	0	0	1.0	0.20	1.20	0.0711
66	13-24		X	107.33	0	0	1.0	0.20	1.20	0.0711
67	3-20		X	85.83	0	0	1.0	0.18	1.18	0.102
68	11-23		X	85.83	0	0	1.0	0.18	1.18	0.102
69	7-21		X	174.0	0	0	1.0	0.20	1.20	0.0482
70	12-22		X	174.0	0	0	1.0	0.20	1.20	0.0482
71	25-26		X	18.42	0	0	1.0	0.74	1.74	0.43
72	38-37		X	18.42	0	0	1.0	0.74	1.74	0.43
73	25-32		X	34.77	0	0	1.0	0.39	1.39	0.30
74	38-39		X	34.77	0	0	1.0	0.39	1.39	0.30
75	26-27		X	17.66	0	0	1.0	0.74	1.74	0.44
76	37-36		X	17.66	0	0	1.0	0.74	1.74	0.44
77	27-32		X	29.7	0	0	1.0	0.28	1.28	0.36
78	36-39		X	29.7	0	0	1.0	0.28	1.28	0.36
79	27-28		X	26.72	0	0	1.0	0.20	1.20	0.31
80	36-35		X	26.72	0	0	1.0	0.20	1.20	0.31
81	32-31		X	68.93	0	0	1.0	0.10	1.10	0.13
82	39-40		X	68.93	0	0	1.0	0.10	1.10	0.13
83	31-30		X	58.55	0	0	1.0	0.34	1.34	0.15
84	40-33		X	58.55	0	0	1.0	0.34	1.34	0.15
85	28-29		X	13.75	0	0	1.0	0.56	1.56	0.55

TABLE 6.2.1-17 (SHEET 3 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
86	35-34		X	13.75	0	0	1.0	0.56	1.56	0.55
87	29-30		X	26.7	0	0	1.0	0.23	1.23	0.35
88	34-33		X	26.7	0	4	1.0	0.23	1.23	0.35
89	29-34		X	4.56	0	0	1.0	0.38	1.38	1.04
90	30-33		X	34.72	0	0	1.0	0.35	1.35	0.23
91	25-41		X	51.3	0	0	1.0	0.21	1.21	0.154
92	38-56		X	51.3	0	0	1.0	0.21	1.21	0.154
93	26-42		X	18.36	0	0	1.0	0.21	1.21	0.434
94	37-55		X	18.36	0	0	1.0	0.21	1.21	0.434
95	27-43		X	54.96	0	0	1.0	0.22	1.22	0.138
96	36-54		X	54.96	0	0	1.0	0.22	1.22	0.138
97	32-49		X	58.97	0	0	1.0	0.25	1.25	0.119
98	39-57		X	58.97	0	0	1.0	0.25	1.25	0.119
99	32-43		X	9.55	0	0	1.0	0.29	1.29	0.615
100	39-54		X	9.55	0	0	1.0	0.29	1.29	0.615
101	32-48		X	46.3	0	0	1.0	0.13	1.13	0.17
102	39-58		X	46.3	0	0	1.0	0.13	1.13	0.17
103	31-48		X	15.89	0	0	1.0	0.10	1.10	0.56
104	40-58		X	15.89	0	0	1.0	0.10	1.10	0.56
105	31-47		X	50.26	0	0	1.0	0.13	1.13	0.16
106	40-50		X	50.26	0	0	1.0	0.13	1.13	0.16
107	28-43		X	7.65	0	0	1.0	0.27	1.27	0.814
108	35-54		X	7.65	0	0	1.0	0.27	1.27	0.814
109	28-44		X	9.16	0	0	1.0	0.18	1.18	0.76
110	35-53		X	9.16	0	0	1.0	0.18	1.18	0.76
111	30-45		X	26.8	0	0	1.0	0.09	1.09	0.201
112	33-51		X	26.8	0	0	1.0	0.09	1.09	0.201
113	30-47		X	116.2	0	0	1.0	0.10	1.10	0.076
114	33-50		X	116.2	0	0	1.0	0.10	1.10	0.076
115	29-45		X	56.71	0	0	1.0	0.10	1.10	0.16
116	34-51		X	56.71	0	0	1.0	0.10	1.10	0.16
117	41-74	X		43.29	0	0	1.0	0.26	1.26	0.123
118	56-74	X		43.29	0	0	1.0	0.26	1.26	0.123
119	42-74	X		30.16	0	0	1.0	0.04	1.04	0.338
120	55-74	X		30.16	0	0	1.0	0.04	1.04	0.338
121	43-74	X		9.63	0	0	1.0	0.47	1.47	0.0794
122	54-74	X		9.63	0	0	1.0	0.47	1.47	0.0794
123	49-74	X		15.82	0	0	1.0	0.43	1.43	0.095
124	57-74	X		15.82	0	0	1.0	0.43	1.43	0.095
125	50-64	X		15.1	0	0	1.0	0.46	1.46	0.276
126	47-63	X		15.1	0	0	1.0	0.46	1.46	0.276
127	52-65	X		21.55	0	0	1.0	0.39	1.39	0.259
128	46-62	X		21.55	0	0	1.0	0.39	1.39	0.259
129	53-66	X		8.63	0	0	1.0	0.35	1.35	0.475
130	44-60	X		8.63	0	0	1.0	0.35	1.35	0.475

TABLE 6.2.1-17 (SHEET 4 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
131	54-66		X	10.05	0	0	1.0	0.46	1.46	0.465
132	43-60	X		10.05	0	0	1.0	0.46	1.46	0.465
133	58-68	X		9.95	0	0	1.0	0.44	1.44	0.374
134	48-59	X		9.95	0	0	1.0	0.44	1.44	0.374
135	45-46		X	91.35	0	0	1.0	0.04	1.04	0.0828
136	51-52		X	91.35	0	0	1.0	0.04	1.04	0.0828
137	41-42		X	81.18	0	0	1.0	0.25	1.25	0.094
138	56-55		X	81.18	0	0	1.0	0.25	1.25	0.094
139	41-49		X	128.07	0	0	1.0	0.23	1.23	0.056
140	56-57		X	128.07	0	0	1.0	0.23	1.23	0.056
141	42-43		X	78.89	0	0	1.0	0.08	1.08	0.128
142	55-54		X	78.89	0	0	1.0	0.08	1.08	0.128
143	43-49		X	107.19	0	0	1.0	0.24	1.24	0.0893
144	54-57		X	107.19	0	0	1.0	0.24	1.24	0.0893
145	43-48		X	85.73	0	0	1.0	0.33	1.33	0.0482
146	54-58		X	85.73	0	0	1.0	0.33	1.33	0.0482
147	43-44		X	95.48	0	0	1.0	0.31	1.31	0.127
148	54-53		X	95.48	0	0	1.0	0.31	1.31	0.127
149	49-58		X	109.22	0	0	1.0	0.27	1.27	0.0406
150	57-58		X	109.22	0	0	1.0	0.27	1.27	0.0406
151	48-47		X	178.11	0	0	1.0	0.53	1.53	0.053
152	58-50		X	178.11	0	0	1.0	0.53	1.53	0.053
153	44-45		X	22.12	0	0	1.0	0.56	1.56	0.308
154	53-51		X	22.12	0	0	1.0	0.56	1.56	0.308
155	44-46		X	29.81	0	0	1.0	0.50	1.50	0.276
156	53-52		X	29.81	0	0	1.0	0.50	1.50	0.276
157	45-47		X	27.88	0	0	1.0	0.61	1.61	0.186
158	51-50		X	27.88	0	0	1.0	0.61	1.61	0.186
159	46-47		X	53.57	0	0	1.0	0.55	1.55	0.173
160	52-50		X	53.57	0	0	1.0	0.55	1.55	0.173
161	47-50		X	175.22	0	0	1.0	0.16	1.16	0.065
162	45-51		X	35.04	0	0	1.0	0.25	1.25	0.26
163	46-52		X	58.39	0	0	1.0	0.14	1.14	0.229
164	59-60		X	21.0	0	0	1.0	0.27	1.27	0.60
165	68-66		X	21.0	0	0	1.0	0.27	1.27	0.60
166	59-61		X	13.5	0	0	1.0	0.35	1.35	0.937
167	68-67		X	13.5	0	0	1.0	0.35	1.35	0.937
168	60-62		X	16.38	0	0	1.0	0.18	1.18	0.77
169	66-65		X	16.38	0	0	1.0	0.18	1.18	0.77
170	61-62		X	10.53	0	0	1.0	0	1.0	1.2
171	67-65		X	10.53	0	0	1.0	0	1.2	1.2
172	62-63		X	0.59	0	0	1.0	0.13	1.13	0.41
173	65-64		X	30.59	0	0	1.0	0.13	1.13	0.41
174	63-59		X	23.0	0	0	1.0	0.16	1.16	0.55

TABLE 6.2.1-17 (SHEET 5 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
175	64-68		X	23.0	0	0	1.0	0.16	1.16	0.55
176	59-74		X	38.1	0	0	1.0	0.08	1.08	0.276
177	68-74		X	38.1	0	0	1.0	0.08	1.08	0.276
178	61-74		X	42.0	0	0	1.0	0.03	1.03	0.11
179	67-74		X	42.0	0	0	1.0	0.03	1.03	0.11
180	62-74		X	47.25	0	0	1.0	0.10	1.10	0.215
181	65-74		X	47.25	0	0	1.0	0.10	1.10	0.215
182	63-74		X	29.4	0	0	1.0	0.23	1.23	0.236
183	64-74		X	29.4	0	0	1.0	0.23	1.23	0.236
184	60-61		X	15.54	0	0	1.0	0.32	1.32	0.181
185	66-67		X	15.54	0	0	1.0	0.32	1.32	0.181
186	3-69		X	21.6	0	0.90	1.0	0.50	2.40	1.256
187	11-70		X	21.6	0	0.90	1.0	0.50	2.40	1.256
188	20-69		X	5.4	0	0.90	1.0	0.50	2.40	1.273
189	23-70		X	5.4	0	0.90	1.0	0.50	2.40	1.273
190	69-43		X	75.6	0	0	1.0	0.35	1.35	0.0869
191	70-54		X	75.6	0	0	1.0	0.35	1.35	0.0869
192	69-74	X		7.2	0	0	1.0	0.44	1.44	0.776
193	70-74	X		7.2	0	0	1.0	0.44	1.44	0.776
194	69-71		X	38.88	0.012	0	1.0	0.44	1.45	0.98
195	70-72		X	33.88	0.012	0	1.0	0.44	1.45	0.98
196	71-74	X		39.26	0.045	1.38	1.0	0.45	2.88	0.795
197	72-74		X	39.26	0.045	1.38	1.0	0.45	2.88	0.795
198	73-74		X	391.01	0	0	1.0	0.29	1.29	0.008

TABLE 6.2.1-18 (SHEET 1 OF 5)

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW EL. 220 ft): FLOW CHARACTERISTICS
STEAM GENERATOR INLET NOZZLE (306-in.^{2} BREAK)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
1	1-74		X	20.25	0	0	1.0	0.38	1.38	0.096
2	1-73		X	97.08	0	0	1.0	0	1.0	0.079
3	20-19		X	29.58	0	0	1.0	0.27	1.27	0.295
4	23-24		X	29.58	0	0	1.0	0.27	1.27	0.295
5	2-1		X	92.1	0	0	1.0	0.25	1.25	0.122
6	3-2		X	63.12	0	0	1.0	0.24	1.24	0.151
7	11-13		X	63.12	0	0	1.0	0.24	1.24	0.151
8	7-2		X	94.0	0	0	1.0	0.27	1.27	0.0623
9	13-12		X	125.8	0	0	1.0	0.27	1.27	0.0623
10	21-19		X	64.65	0	0	1.0	0.20	1.20	0.136
11	22-24		X	80.0	0	0	1.0	0.20	1.20	0.134
12	3-7		X	47.57	0	0	1.0	0.41	1.41	0.0698
13	11-12		X	47.57	0	0	1.0	0.41	1.41	0.0698
14	20-21		X	58.72	0	0	1.0	0.29	1.29	0.297
15	23-22		X	58.72	0	0	1.0	0.29	1.29	0.297
16	3-4		X	63.65	0	0	1.0	0.28	1.28	0.103
17	11-10		X	63.65	0	0	1.0	0.28	1.28	0.103
18	20-15		X	36.93	0	0	1.0	0.25	1.25	0.192
19	23-18		X	36.93	0	0	1.0	0.25	1.25	0.192
20	7-6		X	129.19	0	0	1.0	0.27	1.27	0.0509
21	12-8		X	129.19	0	0	1.0	0.27	1.27	0.0509
22	21-6		X	70.58	0	0	1.0	0.26	1.26	0.1303
23	22-8		X	70.58	0	0	1.0	0.26	1.26	0.1303
24	4-5		X	99.53	0	0	1.0	0.07	1.07	0.1265
25	10-9		X	99.53	0	0	1.0	0.07	1.07	0.1265
26	15-16		X	81.3	0	0	1.0	0.17	1.17	0.1093
27	18-17		X	81.3	0	0	1.0	0.17	1.17	0.1093
28	5-6		X	113.9	0	0	1.0	0.25	1.25	0.08965
29	9-8		X	113.9	0	0	1.0	0.25	1.25	0.08965
30	16-6		X	94.2	0	0	1.0	0.29	1.29	0.153
31	17-8		X	94.2	0	0	1.0	0.29	1.29	0.153
32	6-8		X	247.93	0	0	1.0	0.16	1.16	0.0310
33	5-9		X	76.37	0	0	1.0	0.17	1.17	0.1071
34	16-17		X	68.31	0	0	1.0	0.22	1.22	0.115
35	13-14		X	92.65	0	0	1.0	0.30	1.30	0.086
36	24-14		X	13.7	0	0	1.0	0.38	1.38	0.163
37	14-73		X	44.86	0	0	1.0	0.40	1.40	0.1063
38	14-74		X	50.6	0	0	1.0	0.35	1.35	0.42
39	19-25		X	80.58	0	0	1.0	0.05	1.05	0.0798
40	24-38		X	80.58	0	0	1.0	0.05	1.05	0.0798
41	19-26		X	28.52	0	0	1.0	0.18	1.18	0.185
42	24-37		X	28.52	0	0	1.0	0.18	1.18	0.185
43	19-32		X	53.45	0	0	1.0	0	1.0	0.112
44	24-39		X	53.45	0	0	1.0	0	1.0	0.112
45	20-27		X	62.58	0	0	1.0	0.15	1.15	0.0794
46	23-36		X	62.58	0	0	1.0	0.15	1.15	0.0794
47	21-32		X	81.5	0	0	1.0	0.27	1.27	0.0389

TABLE 6.2.1-18 (SHEET 2 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
48	22-39		X	81.5	0	0	1.0	0.27	1.27	0.0389
49	21-31		X	57.19	0	0	1.0	0.22	1.22	0.0778
50	22-40		X	57.19	0	0	1.0	0.22	1.22	0.0778
51	15-28		X	12.58	0	0	1.0	0.30	1.30	0.2903
52	18-35		X	12.58	0	0	1.0	0.30	1.30	0.2903
53	15-29		X	21.43	0	0	1.0	0.16	1.16	0.2396
54	18-34		X	21.43	0	0	1.0	0.16	1.16	0.2396
55	4-15		X	66.88	0	0	1.0	0.10	1.10	0.161
56	10-18		X	66.88	0	0	1.0	0.10	1.10	0.161
57	5-16		X	76.62	0	0	1.0	0.10	1.10	0.1378
58	9-17		X	76.62	0	0	1.0	0.10	1.10	0.1378
59	6-30		X	80.58	0	0	1.0	0.26	1.26	0.113
60	8-33		X	80.58	0	0	1.0	0.26	1.26	0.113
61	16-29		X	11.0	0	0	1.0	0.40	1.40	0.195
62	17-34		X	11.0	0	0	1.0	0.40	1.40	0.195
63	16-30		X	6.89	0	0	1.0	0.37	1.37	0.321
64	17-33		X	6.89	0	0	1.0	0.37	1.37	0.321
65	2-19		X	107.33	0	0	1.0	0.20	1.20	0.0711
66	13-24		X	107.33	0	0	1.0	0.20	1.20	0.0711
67	3-20		X	85.83	0	0	1.0	0.18	1.18	0.102
68	11-28		X	85.83	0	0	1.0	0.18	1.18	0.102
69	7-21		X	174.0	0	0	1.0	0.20	1.20	0.0482
70	12-22		X	174.0	0	0	1.0	0.20	1.20	0.0482
71	25-26		X	18.42	0	0	1.0	0.74	1.74	0.43
72	38-37		X	18.42	0	0	1.0	0.74	1.74	0.43
73	25-32		X	34.77	0	0	1.0	0.39	1.39	0.30
74	38-39		X	34.77	0	0	1.0	0.39	1.39	0.30
75	26-27		X	17.66	0	0	1.0	0.74	1.74	0.44
76	37-36		X	17.66	0	0	1.0	0.74	1.74	0.44
77	27-32		X	29.7	0	0	1.0	0.28	1.28	0.36
78	36-39		X	29.7	0	0	1.0	0.28	1.28	0.36
79	27-28		X	26.72	0	0	1.0	0.20	1.20	0.31
80	36-35		X	26.72	0	0	1.0	0.20	1.20	0.31
81	32-31		X	68.93	0	0	1.0	0.10	1.10	0.13
82	39-40		X	68.93	0	0	1.0	0.10	1.10	0.13
83	31-30		X	58.55	0	0	1.0	0.34	1.34	0.15
84	40-33		X	58.55	0	0	1.0	0.34	1.34	0.15
85	28-29		X	13.75	0	0	1.0	0.56	1.56	0.55
86	35-34		X	13.75	0	0	1.0	0.56	1.56	0.55
87	29-30		X	26.7	0	0	1.0	0.23	1.23	0.35
88	34-33		X	26.7	0	0	1.0	0.23	1.23	0.35
89	29-34		X	4.56	0	0	1.0	0.38	1.38	1.04
90	30-33		X	34.72	0	0	1.0	0.35	1.35	0.23
91	25-41		X	51.3	0	0	1.0	0.21	1.21	0.154
92	38-56		X	51.3	0	0	1.0	0.21	1.21	0.154
93	26-42		X	18.36	0	0	1.0	0.21	1.21	0.434
94	37-55		X	18.36	0	0	1.0	0.21	1.21	0.434
95	27-43		X	54.96	0	0	1.0	0.22	1.22	0.138
96	36-54		X	54.96	0	0	1.0	0.22	1.22	0.138
97	32-49		X	58.97	0	0	1.0	0.25	1.25	0.119

TABLE 6.2.1-18 (SHEET 3 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
98	39-57		X	58.97	0	0	1.0	0.25	1.25	0.119
99	32-43		X	9.55	0	0	1.0	0.29	1.29	0.615
100	39-54		X	9.55	0	0	1.0	0.29	1.29	0.615
101	32-48		X	46.3	0	0	1.0	0.13	1.13	0.17
102	39-58		X	46.3	0	0	1.0	0.13	1.13	0.17
103	31-48		X	15.89	0	0	1.0	0.10	1.10	0.56
104	40-58		X	15.89	0	0	1.0	0.10	1.10	0.56
105	31-47		X	50.26	0	0	1.0	0.13	1.13	0.16
106	40-50		X	50.26	0	0	1.0	0.13	1.13	0.16
107	28-43		X	7.65	0	0	1.0	0.27	1.27	0.814
108	35-54		X	7.65	0	0	1.0	0.27	1.27	0.814
109	28-44		X	9.16	0	0	1.0	0.18	1.18	0.76
110	35-53		X	9.16	0	0	1.0	0.18	1.18	0.76
111	30-45		X	26.8	0	0	1.0	0.09	1.09	0.201
112	33-51		X	26.8	0	0	1.0	0.09	1.09	0.201
113	30-47		X	116.2	0	0	1.0	0.10	1.10	0.076
114	33-50		X	116.2	0	0	1.0	0.10	1.10	0.076
115	29-45		X	56.71	0	0	1.0	0.10	1.10	0.16
116	34-51		X	56.71	0	0	1.0	0.10	1.10	0.16
117	41-74		X	43.29	0	0	1.0	0.26	1.26	0.123
118	56-74		X	43.29	0	0	1.0	0.26	1.26	0.123
119	42-74		X	30.16	0	0	1.0	0.04	1.04	0.338
120	55-74		X	30.16	0	0	1.0	0.04	1.04	0.338
121	43-74		X	9.63	0	0	1.0	0.47	1.47	0.0794
122	54-74		X	9.63	0	0	1.0	0.47	1.47	0.0794
123	49-74		X	15.82	0	0	1.0	0.43	1.43	0.095
124	57-74		X	15.82	0	0	1.0	0.43	1.43	0.095
125	50-64		X	15.1	0	0	1.0	0.46	1.46	0.276
126	47-63		X	15.1	0	0	1.0	0.46	1.46	0.276
127	52-65		X	21.55	0	0	1.0	0.39	1.39	0.259
128	46-62		X	21.55	0	0	1.0	0.39	1.39	0.259
129	53-66		X	8.63	0	0	1.0	0.35	1.35	0.475
130	44-60		X	8.63	0	0	1.0	0.35	1.35	0.475
131	54-66		X	10.05	0	0	1.0	0.46	1.46	0.465
132	43-60		X	10.05	0	0	1.0	0.46	1.46	0.465
133	58-68		X	9.95	0	0	1.0	0.44	1.44	0.374
134	48-59		X	9.95	0	0	1.0	0.44	1.44	0.374
135	45-46		X	91.35	0	0	1.0	0.04	1.04	0.0828
136	51-52		X	91.35	0	0	1.0	0.04	1.04	0.0828
137	41-42		X	81.18	0	0	1.0	0.25	1.25	0.094
138	56-55		X	81.18	0	0	1.0	0.25	1.25	0.094
139	41-49		X	128.07	0	0	1.0	0.23	1.23	0.056
140	56-57		X	128.07	0	0	1.0	0.23	1.23	0.056
141	42-43		X	78.89	0	0	1.0	0.08	1.08	0.128
142	55-54		X	78.89	0	0	1.0	0.08	1.08	0.128
143	43-49		X	107.19	0	0	1.0	0.24	1.24	0.0893
144	54-57		X	107.19	0	0	1.0	0.24	1.24	0.0893
145	43-48		X	85.73	0	0	1.0	0.33	1.33	0.0482
146	54-58		X	85.73	0	0	1.0	0.33	1.33	0.0482
147	43-44		X	95.48	0	0	1.0	0.31	1.31	0.127

TABLE 6.2.1-18 (SHEET 4 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
148	54-53		X	95.48	0	0	1.0	0.31	1.31	0.127
149	49-58		X	109.22	0	0	1.0	0.27	1.27	0.0406
150	57-58		X	109.22	0	0	1.0	0.27	1.27	0.0406
151	48-47		X	178.11	0	0	1.0	0.53	1.53	0.053
152	58-50		X	178.11	0	0	1.0	0.53	1.53	0.053
153	44-45		X	22.12	0	0	1.0	0.56	1.56	0.308
154	53-51		X	22.12	0	0	1.0	0.56	1.56	0.308
155	44-46		X	29.81	0	0	1.0	0.50	1.50	0.276
156	53-52		X	29.81	0	0	1.0	0.50	1.50	0.276
157	45-47		X	27.88	0	0	1.0	0.61	1.61	0.186
158	51-50		X	27.88	0	0	1.0	0.61	1.61	0.186
159	46-47		X	53.57	0	0	1.0	0.55	1.55	0.173
160	52-50		X	53.57	0	0	1.0	0.55	1.55	0.173
161	47-50		X	175.22	0	0	1.0	0.16	1.16	0.065
162	45-51		X	35.04	0	0	1.0	0.25	1.25	0.26
163	46-52		X	58.39	0	0	1.0	0.14	1.14	0.229
164	59-60		X	21.0	0	0	1.0	0.27	1.27	0.60
165	68-66		X	21.0	0	0	1.0	0.27	1.27	0.60
166	59-61		X	13.5	0	0	1.0	0.35	1.35	0.937
167	68-67		X	13.5	0	0	1.0	0.35	1.35	0.937
168	60-62		X	16.38	0	0	1.0	0.18	1.18	0.77
169	66-65		X	16.38	0	0	1.0	0.18	1.18	0.77
170	61-62		X	10.53	0	0	1.0	0	1.0	1.2
171	67-65		X	10.53	0	0	1.0	0	1.0	1.2
172	62-63		X	30.59	0	0	1.0	0.13	1.13	0.41
173	65-64		X	30.59	0	0	1.0	0.13	1.13	0.41
174	63-59		X	23.0	0	0	1.0	0.16	1.16	0.55
175	64-68		X	23.0	0	0	1.0	0.16	1.16	0.55
176	59-74		X	38.1	0	0	1.0	0.08	1.08	0.276
177	68-74		X	38.1	0	0	1.0	0.08	1.08	0.276
178	61-74		X	42.0	0	0	1.0	0.03	1.03	0.11
179	67-74		X	42.0	0	0	1.0	0.03	1.03	0.11
180	62-74		X	47.25	0	0	1.0	0.10	1.10	0.215
181	65-74		X	47.25	0	0	1.0	0.10	1.10	0.215
182	63-74		X	29.4	0	0	1.0	0.23	1.23	0.236
183	64-74		X	29.4	0	0	1.0	0.23	1.23	0.236
184	60-61		X	15.54	0	0	1.0	0.32	1.32	0.181
185	66-67		X	15.54	0	0	1.0	0.32	1.32	0.181
186	3-69		X	21.6	0	0.09	1.0	0.50	2.40	1.256
187	11-70		X	21.6	0	0.90	1.0	0.50	2.40	1.256
188	20-69		X	5.4	0	0.90	1.0	0.50	2.40	1.273
189	23-70		X	5.4	0	0.90	1.0	0.50	2.40	1.273
190	69-43		X	75.6	0	0	1.0	0.35	1.35	0.0869
191	70-54		X	75.6	0	0	1.0	0.35	1.35	0.0869
192	69-74		X	7.2	0	0	1.0	0.44	1.44	0.776
193	70-74		X	7.2	0	0	1.0	0.44	1.44	0.776
194	69-71		X	33.88	0.012	0	1.0	0.44	1.45	0.98
195	70-72		X	33.88	0.012	0	1.0	0.44	1.45	0.98
196	71-74		X	39.26	0.045	1.38	1.0	0.45	2.88	0.795
197	72-74		X	39.26	0.045	1.38	1.0	0.45	2.88	0.795

TABLE 6.2.1-18 (SHEET 5 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
198	73-74		X	391.01	0	0	1.0	0.29	1.29	0.008

TABLE 6.2.1-19 (SHEET 1 OF 5)

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW EL. 220 ft): FLOW CHARACTERISTICS
STEAM GENERATOR OUTLET NOZZLE (436-in.^{2} BREAK)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
1	1-74		X	20.25	0	0	1.0	0.38	1.38	0.096
2	1-73		X	97.08	0	0	1.0	0	1.0	0.079
3	20-19	X		29.58	0	0	1.0	0.27	1.27	0.295
4	23-24		X	29.58	0	0	1.0	0.27	1.27	0.295
5	2-1		X	92.1	0	0	1.0	0.25	1.25	0.122
6	3-2		X	63.12	0	0	1.0	0.24	1.24	0.151
7	11-13		X	63.12	0	0	1.0	0.24	1.24	0.151
8	7-2		X	94.0	0	0	1.0	0.27	1.27	0.0623
9	13-12		X	125.8	0	0	1.0	0.27	1.27	0.0623
10	21-19		X	64.65	0	0	1.0	0.20	1.20	0.136
11	22-24		X	80.0	0	0	1.0	0.20	1.20	0.134
12	3-7		X	47.57	0	0	1.0	0.41	1.41	0.0698
13	11-12		X	47.57	0	0	1.0	0.41	1.41	0.0698
14	20-21	X		58.72	0	0	1.0	0.29	1.29	0.297
15	23-22		X	58.72	0	0	1.0	0.29	1.29	0.297
16	3-4		X	63.65	0	0	1.0	0.28	1.28	0.103
17	11-10		X	63.65	0	0	1.0	0.28	1.28	0.103
18	20-15	X		36.93	0	0	1.0	0.25	1.25	0.192
19	23-18		X	36.93	0	0	1.0	0.25	1.25	0.192
20	7-6		X	129.19	0	0	1.0	0.27	1.27	0.0509
21	12-8		X	129.19	0	0	1.0	0.27	1.27	0.0509
22	21-6		X	70.58	0	0	1.0	0.26	1.26	0.1303
23	22-8		X	70.58	0	0	1.0	0.26	1.26	0.1303
24	4-5		X	99.53	0	0	1.0	0.07	1.07	0.1265
25	10-9		X	99.53	0	0	1.0	0.07	1.07	0.1265
26	15-16		X	81.3	0	0	1.0	0.17	1.17	0.1093
27	18-17		X	81.3	0	0	1.0	0.17	1.17	0.1093
28	5-6		X	113.9	0	0	1.0	0.25	1.25	0.08965
29	9-8		X	113.9	0	0	1.0	0.25	1.25	0.08965
30	16-6		X	94.2	0	0	1.0	0.29	1.29	0.153
31	17-8		X	94.2	0	0	1.0	0.29	1.29	0.153
32	6-8		X	247.93	0	0	1.0	0.16	1.16	0.0310
33	5-9		X	76.37	0	0	1.0	0.17	1.17	0.1071
34	16-17		X	68.31	0	0	1.0	0.22	1.22	0.115
35	13-14		X	92.65	0	0	1.0	0.30	1.30	0.086
36	24-14		X	13.7	0	0	1.0	0.38	1.38	0.163
37	14-73		X	44.86	0	0	1.0	0.40	1.40	0.1063
38	14-74		X	50.6	0	0	1.0	0.35	1.35	0.42
39	19-25		X	80.58	0	0	1.0	0.05	1.05	0.0798
40	24-38		X	80.58	0	0	1.0	0.05	1.05	0.0798
41	19-26		X	28.52	0	0	1.0	0.18	1.18	0.185
42	24-37		X	28.52	0	0	1.0	0.18	1.18	0.185
43	19-32		X	53.45	0	0	1.0	0	1.0	0.112
44	24-39		X	53.45	0	0	1.0	0	1.0	0.112
45	20-27		X	62.58	0	0	1.0	0.15	1.15	0.0794
46	23-36		X	62.58	0	0	1.0	0.15	1.15	0.0794
47	21-32		X	81.5	0	0	1.0	0.27	1.27	0.0389

TABLE 6.2.1-19 (SHEET 2 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
48	22-39		X	81.5	0	0	1.0	0.27	1.27	0.0389
49	21-31		X	57.19	0	0	1.0	0.22	1.22	0.0778
50	22-40		X	57.19	0	0	1.0	0.22	1.22	0.0778
51	15-28		X	12.58	0	0	1.0	0.30	1.30	0.2903
52	18-35		X	12.58	0	0	1.0	0.30	1.30	0.2903
53	15-29		X	21.43	0	0	1.0	0.16	1.16	0.2396
54	18-34		X	21.43	0	0	1.0	0.16	1.16	0.2396
55	4-15		X	66.88	0	0	1.0	0.10	1.10	0.161
56	10-18		X	66.88	0	0	1.0	0.10	1.10	0.161
57	5-16		X	76.62	0	0	1.0	0.10	1.10	0.1378
58	9-17		X	76.62	0	0	1.0	0.10	1.10	0.1378
59	6-30		X	80.58	0	0	1.0	0.26	1.26	0.113
60	8-33		X	80.58	0	0	1.0	0.26	1.26	0.113
61	16-29		X	11.0	0	0	1.0	0.40	1.40	0.195
62	17-34		X	11.0	0	0	1.0	0.40	1.40	0.195
63	16-30		X	6.89	0	0	1.0	0.37	1.37	0.321
64	17-33		X	6.89	0	0	1.0	0.37	1.37	0.321
65	2-19		X	107.33	0	0	1.0	0.20	1.20	0.0711
66	13-24		X	107.33	0	0	1.0	0.20	1.20	0.0711
67	3-20		X	85.83	0	0	1.0	0.18	1.18	0.102
68	11-23		X	85.83	0	0	1.0	0.18	1.18	0.102
69	7-21		X	174.0	0	0	1.0	0.20	1.20	0.0482
70	12-22		X	174.0	0	0	1.0	0.20	1.20	0.0482
71	25-26		X	18.42	0	0	1.0	0.74	1.74	0.43
72	38-37		X	18.42	0	0	1.0	0.74	1.74	0.43
73	25-32		X	34.77	0	0	1.0	0.39	1.39	0.30
74	38-39		X	34.77	0	0	1.0	0.39	1.39	0.30
75	26-27		X	17.66	0	0	1.0	0.74	1.74	0.44
76	37-36		X	17.66	0	0	1.0	0.74	1.74	0.44
77	27-32		X	29.7	0	0	1.0	0.28	1.28	0.36
78	36-39		X	29.7	0	0	1.0	0.28	1.28	0.36
79	27-28		X	26.72	0	0	1.0	0.20	1.20	0.31
80	36-35		X	26.72	0	0	1.0	0.20	1.20	0.31
81	32-31		X	68.93	0	0	1.0	0.10	1.10	0.13
82	39-40		X	68.93	0	0	1.0	0.10	1.10	0.13
83	31-30		X	58.55	0	0	1.0	0.34	1.34	0.15
84	40-33		X	58.55	0	0	1.0	0.34	1.34	0.15
85	28-29		X	13.75	0	0	1.0	0.56	1.56	0.55
86	35-34		X	13.75	0	0	1.0	0.56	1.56	0.55
87	29-30		X	26.7	0	0	1.0	0.23	1.23	0.35
88	34-33		X	26.7	0	0	1.0	0.23	1.23	0.35
89	29-34		X	4.56	0	0	1.0	0.38	1.38	1.04
90	30-33		X	34.72	0	0	1.0	0.35	1.35	0.23
91	25-41		X	51.3	0	0	1.0	0.21	1.21	0.154
92	38-56		X	51.3	0	0	1.0	0.21	1.21	0.154
93	26-42		X	18.36	0	0	1.0	0.21	1.21	0.434
94	37-55		X	18.36	0	0	1.0	0.21	1.21	0.434
95	27-43		X	54.96	0	0	1.0	0.22	1.22	0.138
96	36-54		X	54.96	0	0	1.0	0.22	1.22	0.138
97	32-49		X	58.97	0	0	1.0	0.25	1.25	0.119

TABLE 6.2.1-19 (SHEET 3 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
98	39-57		X	58.97	0	0	1.0	0.25	1.25	0.119
99	32-43		X	9.55	0	0	1.0	0.29	1.29	0.615
100	39-54		X	9.55	0	0	1.0	0.29	1.29	0.615
101	32-48		X	46.3	0	0	1.0	0.13	1.13	0.17
102	39-58		X	46.3	0	0	1.0	0.13	1.13	0.17
103	31-48		X	15.89	0	0	1.0	0.10	1.10	0.56
104	40-58		X	15.89	0	0	1.0	0.10	1.10	0.56
105	31-47		X	50.26	0	0	1.0	0.13	1.13	0.16
106	40-50		X	50.26	0	0	1.0	0.13	1.13	0.16
107	28-43		X	7.65	0	0	1.0	0.27	1.27	0.814
108	35-54		X	7.65	0	0	1.0	0.27	1.27	0.814
109	28-44		X	9.16	0	0	1.0	0.18	1.18	0.76
110	35-53		X	9.16	0	0	1.0	0.18	1.18	0.76
111	30-45		X	26.8	0	0	1.0	0.09	1.09	0.201
112	33-51		X	26.8	0	0	1.0	0.09	1.09	0.201
113	30-47		X	116.2	0	0	1.0	0.10	1.10	0.076
114	33-50		X	116.2	0	0	1.0	0.10	1.10	0.076
115	29-45		X	56.71	0	0	1.0	0.10	1.10	0.16
116	34-51		X	56.71	0	0	1.0	0.10	1.10	0.16
117	41-74	X		43.29	0	0	1.0	0.26	1.26	0.123
118	56-74		X	43.29	0	0	1.0	0.26	1.26	0.123
119	42-74	X		30.16	0	0	1.0	0.04	1.04	0.338
120	55-74		X	30.16	0	0	1.0	0.04	1.04	0.338
121	43-74	X		9.63	0	0	1.0	0.47	1.47	0.0794
122	54-74		X	9.63	0	0	1.0	0.47	1.47	0.0794
123	49-74	X		15.82	0	0	1.0	0.43	1.43	0.095
124	57-74		X	15.82	0	0	1.0	0.43	1.43	0.095
125	50-64	X		15.1	0	0	1.0	0.46	1.46	0.276
126	47-63	X		15.1	0	0	1.0	0.46	1.46	0.276
127	52-65	X		21.55	0	0	1.0	0.39	1.39	0.259
128	46-62	X		21.55	0	0	1.0	0.39	1.39	0.259
129	53-66		X	8.63	0	0	1.0	0.35	1.35	0.475
130	44-60		X	8.63	0	0	1.0	0.35	1.35	0.475
131	54-66		X	10.05	0	0	1.0	0.46	1.46	0.465
132	43-60		X	10.05	0	0	1.0	0.46	1.46	0.465
133	58-68		X	9.95	0	0	1.0	0.44	1.44	0.374
134	48-59	X		9.95	0	0	1.0	0.44	1.44	0.374
135	45-46		X	91.35	0	0	1.0	0.04	1.04	0.0828
136	51-52		X	91.35	0	0	1.0	0.04	1.04	0.0828
137	41-42		X	81.18	0	0	1.0	0.25	1.25	0.094
138	56-55		X	81.18	0	0	1.0	0.25	1.25	0.094
139	41-49		X	128.07	0	0	1.0	0.23	1.23	0.056
140	56-57		X	128.07	0	0	1.0	0.23	1.23	0.056
141	42-43		X	78.89	0	0	1.0	0.08	1.08	0.128
142	55-54		X	78.89	0	0	1.0	0.08	1.08	0.128
143	43-49		X	107.19	0	0	1.0	0.24	1.24	0.0893
144	54-57		X	107.19	0	0	1.0	0.24	1.24	0.0893
145	43-48		X	85.73	0	0	1.0	0.33	1.33	0.0482
146	54-58		X	85.73	0	0	1.0	0.33	1.33	0.0482
147	43-44		X	95.48	0	0	1.0	0.31	1.31	0.127

TABLE 6.2.1-19 (SHEET 4 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
148	54-53		X	95.48	0	0	1.0	0.31	1.31	0.127
149	49-58		X	109.22	0	0	1.0	0.27	1.27	0.0406
150	57-58		X	109.22	0	0	1.0	0.27	1.27	0.0406
151	48-47		X	178.11	0	0	1.0	0.53	1.53	0.053
152	58-50		X	178.11	0	0	1.0	0.53	1.53	0.053
153	44-45		X	22.12	0	0	1.0	0.56	1.56	0.308
154	53-51		X	22.12	0	0	1.0	0.56	1.56	0.308
155	44-46		X	29.81	0	0	1.0	0.50	1.50	0.276
156	53-52		X	29.81	0	0	1.0	0.50	1.50	0.276
157	45-47		X	27.88	0	0	1.0	0.61	1.61	0.186
158	51-50		X	27.88	0	0	1.0	0.61	1.61	0.186
159	46-47		X	53.57	0	0	1.0	0.55	1.55	0.173
160	52-50		X	53.57	0	0	1.0	0.55	1.55	0.173
161	47-50		X	175.22	0	0	1.0	0.16	1.16	0.065
162	45-51		X	35.04	0	0	1.0	0.25	1.25	0.26
163	46-52		X	58.39	0	0	1.0	0.14	1.14	0.229
164	59-60		X	21.0	0	0	1.0	0.27	1.27	0.60
165	68-66		X	21.0	0	0	1.0	0.27	1.27	0.60
166	59-61		X	13.5	0	0	1.0	0.35	1.35	0.937
167	68-67		X	13.5	0	0	1.0	0.35	1.35	0.937
168	60-62		X	16.38	0	0	1.0	0.18	1.18	0.77
169	66-65		X	16.38	0	0	1.0	0.18	1.18	0.77
170	61-62		X	10.53	0	0	1.0	0	1.0	1.2
171	67-65		X	10.53	0	0	1.0	0	1.0	1.2
172	62-63		X	30.59	0	0	1.0	0.13	1.13	0.41
173	65-64		X	30.59	0	0	1.0	0.13	1.13	0.41
174	63-59		X	23.0	0	0	1.0	0.16	1.16	0.55
175	64-68		X	23.0	0	0	1.0	0.16	1.16	0.55
176	59-74		X	38.1	0	0	1.0	0.08	1.08	0.276
177	68-74		X	38.1	0	0	1.0	0.08	1.08	0.276
178	61-74		X	42.0	0	0	1.0	0.03	1.03	0.11
179	67-74		X	42.0	0	0	1.0	0.03	1.03	0.11
180	62-74		X	47.25	0	0	1.0	0.10	1.10	0.215
181	65-74		X	47.25	0	0	1.0	0.10	1.10	0.215
182	63-74		X	29.4	0	0	1.0	0.23	1.23	0.236
183	64-74		X	29.4	0	0	1.0	0.23	1.23	0.236
184	60-61		X	15.54	0	0	1.0	0.32	1.32	0.181
185	66-67		X	15.54	0	0	1.0	0.32	1.32	0.181
186	3-69		X	21.6	0	0.90	1.0	0.50	2.40	1.256
187	11-70		X	21.6	0	0.90	1.0	0.50	2.40	1.256
188	20-69	X		5.4	0	0.90	1.0	0.50	2.40	1.273
189	23-70		X	5.4	0	0.90	1.0	0.50	2.40	1.273
190	69-43		X	75.6	0	0	1.0	0.35	1.35	0.0869
191	70-54		X	75.6	0	0	1.0	0.35	1.35	0.0869
192	69-74	X		7.2	0	0	1.0	0.44	1.44	0.776
193	70-74		X	7.2	0	0	1.0	0.44	1.44	0.776
194	69-71		X	33.88	0.012	0	1.0	0.44	1.45	0.98
195	70-72		X	33.88	0.012	0	1.0	0.44	1.45	0.98
196	71-74		X	39.26	0.045	1.38	1.0	0.45	2.88	0.795
197	72-74		X	39.26	0.045	1.38	1.0	0.45	2.88	0.795

TABLE 6.2.1-19 (SHEET 5 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
198	73-74		X	391.01	0	0	1.0	0.29	1.29	0.008

TABLE 6.2.1-20 (SHEET 1 OF 5)

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW EL. 220 ft): FLOW CHARACTERISTICS
REACTOR COOLANT PUMP OUTLET NOZZLE (236-in.^{2} BREAK)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
1	1-74		X	20.25	0	0	1.0	0.38	1.38	0.096
2	1-73		X	97.08	0	0	1.0	0	1.0	0.079
3	20-19		X	29.58	0	0	1.0	0.27	1.27	0.295
4	23-24		X	29.58	0	0	1.0	0.27	1.27	0.295
5	2-1		X	92.1	0	0	1.0	0.25	1.25	0.122
6	3-2		X	63.12	0	0	1.0	0.24	1.24	0.151
7	11-13		X	63.12	0	0	1.0	0.24	1.24	0.151
8	7-2		X	94.0	0	0	1.0	0.27	1.27	0.0623
9	13-12		X	125.8	0	0	1.0	0.27	1.27	0.0623
10	21-19		X	64.65	0	0	1.0	0.20	1.20	0.136
11	22-24		X	80.0	0	0	1.0	0.20	1.20	0.134
12	3-7		X	47.57	0	0	1.0	0.41	1.41	0.0698
13	11-12		X	47.57	0	0	1.0	0.41	1.41	0.0698
14	20-21		X	58.72	0	0	1.0	0.29	1.29	0.297
15	23-22		X	58.72	0	0	1.0	0.29	1.29	0.297
16	3-4		X	63.65	0	0	1.0	0.28	1.28	0.103
17	11-10		X	63.65	0	0	1.0	0.28	1.28	0.103
18	20-15		X	36.93	0	0	1.0	0.25	1.25	0.192
19	23-18		X	36.93	0	0	1.0	0.25	1.25	0.192
20	7-6		X	129.19	0	0	1.0	0.27	1.27	0.0509
21	12-8		X	129.19	0	0	1.0	0.27	1.27	0.0509
22	21-6		X	70.58	0	0	1.0	0.26	1.26	0.1303
23	22-8		X	70.58	0	0	1.0	0.26	1.26	0.1303
24	4-5		X	99.53	0	0	1.0	0.07	1.07	0.1265
25	10-9		X	99.53	0	0	1.0	0.07	1.07	0.1265
26	15-16		X	81.3	0	0	1.0	0.17	1.17	0.1093
27	18-17		X	81.3	0	0	1.0	0.17	1.17	0.1093
28	5-6		X	113.9	0	0	1.0	0.25	1.25	0.08965
29	9-8		X	113.9	0	0	1.0	0.25	1.25	0.08965
30	16-6		X	94.2	0	0	1.0	0.29	1.29	0.153
31	17-8		X	94.2	0	0	1.0	0.29	1.29	0.153
32	6-8		X	247.93	0	0	1.0	0.16	1.16	0.0310
33	5-9		X	76.37	0	0	1.0	0.17	1.17	0.1071
34	16-17		X	68.31	0	0	1.0	0.22	1.22	0.115
35	13-14		X	92.65	0	0	1.0	0.30	1.30	0.086
36	24-14		X	13.7	0	0	1.0	0.38	1.38	0.163
37	14-73		X	44.86	0	0	1.0	0.40	1.40	0.1063
38	14-74		X	50.6	0	0	1.0	0.35	1.35	0.42
39	19-25		X	80.58	0	0	1.0	0.05	1.05	0.0798
40	24-38		X	80.58	0	0	1.0	0.05	1.05	0.0798
41	19-26		X	28.52	0	0	1.0	0.18	1.18	0.185
42	24-37		X	28.52	0	0	1.0	0.18	1.18	0.185

TABLE 6.2.1-20 (SHEET 2 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
43	19-32		X	53.45	0	0	1.0	0	1.0	0.112
44	24-39		X	53.45	0	0	1.0	0	1.0	0.112
45	20-27		X	62.58	0	0	1.0	0.15	1.15	0.0794
46	23-36		X	62.58	0	0	1.0	0.15	1.15	0.0794
47	21-32		X	81.5	0	0	1.0	0.27	1.27	0.0389
48	22-39		X	81.5	0	0	1.0	0.27	1.27	0.0389
49	21-31		X	57.19	0	0	1.0	0.22	1.22	0.0778
50	22-40		X	57.19	0	0	1.0	0.22	1.22	0.0778
51	15-28		X	12.58	0	0	1.0	0.30	1.30	0.2903
52	18-35		X	12.58	0	0	1.0	0.30	1.30	0.2903
53	15-29		X	21.43	0	0	1.0	0.16	1.16	0.2396
54	18-34		X	21.43	0	0	1.0	0.16	1.16	0.2396
55	4-15		X	66.88	0	0	1.0	0.10	1.10	0.161
56	10-18		X	66.88	0	0	1.0	0.10	1.10	0.161
57	5-16		X	76.62	0	0	1.0	0.10	1.10	0.1378
58	9-17		X	76.62	0	0	1.0	0.10	1.10	0.1378
59	6-30		X	80.58	0	0	1.0	0.26	1.26	0.113
60	8-33		X	80.58	0	0	1.0	0.26	1.26	0.113
61	16-29		X	11.0	0	0	1.0	0.40	1.40	0.195
62	17-34		X	11.0	0	0	1.0	0.40	1.40	0.195
63	16-30		X	6.89	0	0	1.0	0.37	1.37	0.321
64	17-33		X	6.89	0	0	1.0	0.37	1.37	0.321
65	2-19		X	107.33	0	0	1.0	0.20	1.20	0.0711
66	13-24		X	107.33	0	0	1.0	0.20	1.20	0.0711
67	3-20		X	85.83	0	0	1.0	0.18	1.18	0.102
68	11-23		X	85.83	0	0	1.0	0.18	1.18	0.102
69	7-21		X	174.0	0	0	1.0	0.20	1.20	0.0482
70	12-22		X	174.0	0	0	1.0	0.20	1.20	0.0482
71	25-26		X	18.42	0	0	1.0	0.74	1.74	0.43
72	38-37		X	18.42	0	0	1.0	0.74	1.74	0.43
73	25-32		X	34.77	0	0	1.0	0.39	1.39	0.30
74	38-39		X	34.77	0	0	1.0	0.39	1.39	0.30
75	26-27		X	17.66	0	0	1.0	0.74	1.74	0.44
76	37-36		X	17.66	0	0	1.0	0.74	1.74	0.44
77	27-32		X	29.7	0	0	1.0	0.28	1.28	0.36
78	36-39		X	29.7	0	0	1.0	0.28	1.28	0.36
79	27-28		X	26.72	0	0	1.0	0.20	1.20	0.31
80	36-35		X	26.72	0	0	1.0	0.20	1.20	0.31
81	32-31		X	68.93	0	0	1.0	0.10	1.10	0.13
82	39-40		X	68.93	0	0	1.0	0.10	1.10	0.13
83	31-30		X	58.55	0	0	1.0	0.34	1.34	0.15
84	40-33		X	58.55	0	0	1.0	0.34	1.34	0.15
85	28-29		X	13.75	0	0	1.0	0.56	1.56	0.55
86	35-34		X	13.75	0	0	1.0	0.56	1.56	0.55

TABLE 6.2.1-20 (SHEET 3 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
87	29-30		X	26.7	0	0	1.0	0.23	1.23	0.35
88	34-33		X	26.7	0	0	1.0	0.23	1.23	0.35
89	29-34		X	4.56	0	0	1.0	0.38	1.38	1.04
90	30-33		X	34.72	0	0	1.0	0.35	1.35	0.23
91	25-41		X	51.3	0	0	1.0	0.21	1.21	0.154
92	38-56		X	51.3	0	0	1.0	0.21	1.21	0.154
93	26-42		X	18.36	0	0	1.0	0.21	1.21	0.434
94	37-55		X	18.36	0	0	1.0	0.21	1.21	0.434
95	27-43		X	54.96	0	0	1.0	0.22	1.22	0.138
96	36-54		X	54.96	0	0	1.0	0.22	1.22	0.138
97	32-49		X	58.97	0	0	1.0	0.25	1.25	0.119
98	39-57		X	58.97	0	0	1.0	0.25	1.25	0.119
99	32-43		X	9.55	0	0	1.0	0.29	1.29	0.615
100	39-54		X	9.55	0	0	1.0	0.29	1.29	0.615
101	32-48		X	46.3	0	0	1.0	0.13	1.13	0.17
102	39-58		X	46.3	0	0	1.0	0.13	1.13	0.17
103	31-48		X	15.89	0	0	1.0	0.10	1.10	0.56
104	40-58		X	15.89	0	0	1.0	0.10	1.10	0.56
105	31-47		X	50.26	0	0	1.0	0.13	1.13	0.16
106	40-50		X	50.26	0	0	1.0	0.13	1.13	0.16
107	28-43		X	7.65	0	0	1.0	0.27	1.27	0.814
108	35-54		X	7.65	0	0	1.0	0.27	1.27	0.814
109	28-44		X	9.16	0	0	1.0	0.18	1.18	0.76
110	35-53		X	9.16	0	0	1.0	0.18	1.18	0.76
111	30-45		X	26.8	0	0	1.0	0.09	1.09	0.201
112	33-51		X	26.8	0	0	1.0	0.09	1.09	0.201
113	30-47		X	116.2	0	0	1.0	0.10	1.10	0.076
114	33-50		X	116.2	0	0	1.0	0.10	1.10	0.076
115	29-45		X	56.71	0	0	1.0	0.10	1.10	0.16
116	34-51		X	56.71	0	0	1.0	0.10	1.10	0.16
117	41-74		X	43.29	0	0	1.0	0.26	1.26	0.123
118	56-74		X	43.29	0	0	1.0	0.26	1.26	0.123
119	42-74		X	30.16	0	0	1.0	0.04	1.04	0.338
120	55-74		X	30.16	0	0	1.0	0.04	1.04	0.338
121	43-74		X	9.63	0	0	1.0	0.47	1.47	0.0794
122	54-74		X	9.63	0	0	1.0	0.47	1.47	0.0794
123	49-74		X	15.82	0	0	1.0	0.43	1.43	0.095
124	57-74		X	15.82	0	0	1.0	0.43	1.43	0.095
125	50-64		X	15.1	0	0	1.0	0.46	1.46	0.276
126	47-63		X	15.1	0	0	1.0	0.46	1.46	0.276
127	52-65		X	21.55	0	0	1.0	0.39	1.39	0.259
128	46-62		X	21.55	0	0	1.0	0.39	1.39	0.259
129	53-66		X	8.63	0	0	1.0	0.35	1.35	0.475
130	44-60		X	8.63	0	0	1.0	0.35	1.35	0.475

TABLE 6.2.1-20 (SHEET 4 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
131	54-66		X	10.05	0	0	1.0	0.46	1.46	0.465
132	43-60		X	10.05	0	0	1.0	0.46	1.46	0.465
133	58-68		X	9.95	0	0	1.0	0.44	1.44	0.374
134	48-59		X	9.95	0	0	1.0	0.44	1.44	0.374
135	45-46		X	91.35	0	0	1.0	0.04	1.04	0.0828
136	51-52		X	91.35	0	0	1.0	0.04	1.04	0.0828
137	41-42		X	81.18	0	0	1.0	0.25	1.25	0.094
138	56-55		X	81.18	0	0	1.0	0.25	1.25	0.094
139	41-49		X	128.07	0	0	1.0	0.23	1.23	0.056
140	56-57		X	128.07	0	0	1.0	0.23	1.23	0.056
141	42-43		X	78.89	0	0	1.0	0.08	1.08	0.128
142	55-54		X	78.89	0	0	1.0	0.08	1.08	0.128
143	43-49		X	107.19	0	0	1.0	0.24	1.24	0.0893
144	54-57		X	107.19	0	0	1.0	0.24	1.24	0.0893
145	43-48		X	85.73	0	0	1.0	0.33	1.33	0.0482
146	54-58		X	85.73	0	0	1.0	0.33	1.33	0.0482
147	43-44		X	95.48	0	0	1.0	0.31	1.31	0.127
148	54-53		X	95.48	0	0	1.0	0.31	1.31	0.127
149	49-58		X	109.22	0	0	1.0	0.27	1.27	0.0406
150	57-58		X	109.22	0	0	1.0	0.27	1.27	0.0406
151	48-47		X	178.11	0	0	1.0	0.53	1.53	0.053
152	58-50		X	178.11	0	0	1.0	0.53	1.53	0.053
153	44-45		X	22.12	0	0	1.0	0.56	1.56	0.308
154	53-51		X	22.12	0	0	1.0	0.56	1.56	0.308
155	44-46		X	29.81	0	0	1.0	0.50	1.50	0.276
156	53-52		X	29.81	0	0	1.0	0.50	1.50	0.276
157	45-47		X	27.88	0	0	1.0	0.61	1.61	0.186
158	51-50		X	27.88	0	0	1.0	0.61	1.61	0.186
159	46-47		X	53.57	0	0	1.0	0.55	1.55	0.173
160	52-50		X	53.57	0	0	1.0	0.55	1.55	0.173
161	47-50		X	175.22	0	0	1.0	0.16	1.16	0.065
162	45-51		X	35.04	0	0	1.0	0.25	1.25	0.26
163	46-52		X	58.39	0	0	1.0	0.14	1.14	0.229
164	59-60		X	21.0	0	0	1.0	0.27	1.27	0.60
165	68-66		X	21.0	0	0	1.0	0.27	1.27	0.60
166	59-61		X	13.5	0	0	1.0	0.35	1.35	0.937
167	68-67		X	13.5	0	0	1.0	0.35	1.35	0.937
168	60-62		X	16.38	0	0	1.0	0.18	1.18	0.77
169	66-65		X	16.38	0	0	1.0	0.18	1.18	0.77
170	61-62		X	10.53	0	0	1.0	0	1.0	1.2
171	67-65		X	10.53	0	0	1.0	0	1.0	1.2
172	62-63		X	30.59	0	0	1.0	0.13	1.13	0.41
173	65-64		X	30.59	0	0	1.0	0.13	1.13	0.41
174	63-59		X	23.0	0	0	1.0	0.16	1.16	0.55

TABLE 6.2.1-20 (SHEET 5 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
175	64-68		X	23.0	0	0	1.0	0.16	1.16	0.55
176	59-74		X	38.1	0	0	1.0	0.08	1.08	0.276
177	68-74		X	38.1	0	0	1.0	0.08	1.08	0.276
178	61-74		X	42.0	0	0	1.0	0.03	1.03	0.11
179	67-74		X	42.0	0	0	1.0	0.03	1.03	0.11
180	62-74		X	47.25	0	0	1.0	0.10	1.10	0.215
181	65-74		X	47.25	0	0	1.0	0.10	1.10	0.215
182	63-74		X	29.4	0	0	1.0	0.23	1.23	0.236
183	64-74		X	29.4	0	0	1.0	0.23	1.23	0.236
184	60-61		X	15.54	0	0	1.0	0.32	1.32	0.181
185	66-67		X	15.54	0	0	1.0	0.32	1.32	0.181
186	3-69		X	21.6	0	0.90	1.0	0.50	2.40	1.256
187	11-70		X	21.6	0	0.90	1.0	0.50	2.40	1.256
188	20-69		X	5.4	0	0.90	1.0	0.50	2.40	1.273
189	23-70		X	5.4	0	0.90	1.0	0.50	2.40	1.273
190	69-43		X	75.6	0	0	1.0	0.35	1.35	0.0869
191	70-54		X	75.6	0	0	1.0	0.35	1.35	0.0869
192	69-74		X	7.2	0	0	1.0	0.44	1.44	0.776
193	70-74		X	7.2	0	0	1.0	0.44	1.44	0.776
194	69-71		X	33.88	0.012	0	1.0	0.44	1.45	0.98
195	70-72		X	33.88	0.012	0	1.0	0.44	1.45	0.98
196	71-74		X	39.26	0.045	1.38	1.0	0.45	2.88	0.795
197	72-74		X	39.26	0.045	1.38	1.0	0.45	2.88	0.795
198	73-74		X	391.01	0	0	1.0	0.29	1.29	0.008

TABLE 6.2.1-21 (SHEET 1 OF 5)

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW EL. 220 ft): FLOW CHARACTERISTICS
REACTOR COOLANT PUMP INLET NOZZLE (336-in.^{2} BREAK)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
1	1-74		X	20.25	0	0	1.0	0.38	1.38	0.096
2	1-73		X	97.08	0	0	1.0	0	1.0	0.079
3	20-19		X	29.58	0	0	1.0	0.27	1.27	0.295
4	23-24		X	29.58	0	0	1.0	0.27	1.27	0.295
5	2-1		X	92.1	0	0	1.0	0.25	1.25	0.122
6	3-2		X	63.12	0	0	1.0	0.24	1.24	0.151
7	11-13		X	63.12	0	0	1.0	0.24	1.24	0.151
8	7-2		X	94.0	0	0	1.0	0.27	1.27	0.0623
9	13-12		X	125.8	0	0	1.0	0.27	1.27	0.0623
10	21-19		X	64.65	0	0	1.0	0.20	1.20	0.136
11	22-24		X	80.0	0	0	1.0	0.20	1.20	0.134
12	3-7		X	47.57	0	0	1.0	0.41	1.41	0.0698
13	11-12		X	47.57	0	0	1.0	0.41	1.41	0.0698
14	20-21		X	58.72	0	0	1.0	0.29	1.29	0.297
15	23-22		X	58.72	0	0	1.0	0.29	1.29	0.297
16	3-4		X	63.65	0	0	1.0	0.28	1.28	0.103
17	11-10		X	63.65	0	0	1.0	0.28	1.28	0.103
18	20-15		X	36.93	0	0	1.0	0.25	1.25	0.192
19	23-18		X	36.93	0	0	1.0	0.25	1.25	0.192
20	7-6		X	129.19	0	0	1.0	0.27	1.27	0.0509
21	12-8		X	129.19	0	0	1.0	0.27	1.27	0.0509
22	21-6		X	70.58	0	0	1.0	0.26	1.26	0.1303
23	22-8		X	70.58	0	0	1.0	0.26	1.26	0.1303
24	4-5		X	99.53	0	0	1.0	0.07	1.07	0.1265
25	10-9		X	99.53	0	0	1.0	0.07	1.07	0.1265
26	15-16		X	81.3	0	0	1.0	0.17	1.17	0.1093
27	18-17		X	81.3	0	0	1.0	0.17	1.17	0.1093
28	5-6		X	113.9	0	0	1.0	0.25	1.25	0.08965
29	9-8		X	113.9	0	0	1.0	0.25	1.25	0.08965
30	16-6		X	94.2	0	0	1.0	0.29	1.29	0.153
31	17-8		X	94.2	0	0	1.0	0.29	1.29	0.153
32	6-8		X	247.93	0	0	1.0	0.16	1.16	0.0310
33	5-9		X	76.37	0	0	1.0	0.17	1.17	0.1071
34	16-17		X	68.31	0	0	1.0	0.22	1.22	0.115
35	13-14		X	92.65	0	0	1.0	0.30	1.30	0.086
36	24-14		X	13.7	0	0	1.0	0.38	1.38	0.163
37	14-73		X	44.86	0	0	1.0	0.40	1.40	0.1063
38	14-74		X	50.6	0	0	1.0	0.35	1.35	0.42
39	19-25		X	80.58	0	0	1.0	0.05	1.05	0.0798
40	24-38		X	80.58	0	0	1.0	0.05	1.05	0.0798
41	19-26		X	28.52	0	0	1.0	0.18	1.18	0.185
42	24-37		X	28.52	0	0	1.0	0.18	1.18	0.185
43	19-32		X	53.45	0	0	1.0	0	1.0	0.112
44	24-39		X	53.45	0	0	1.0	0	1.0	0.112
45	20-27		X	62.58	0	0	1.0	0.15	1.15	0.0794
46	23-36		X	62.58	0	0	1.0	0.15	1.15	0.0794
47	21-32		X	81.5	0	0	1.0	0.27	1.27	0.0389

TABLE 6.2.1-21 (SHEET 2 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
48	22-39		X	81.5	0	0	1.0	0.27	1.27	0.0389
49	21-31		X	57.19	0	0	1.0	0.22	1.22	0.0778
50	22-40		X	57.19	0	0	1.0	0.22	1.22	0.0778
51	15-28		X	12.58	0	0	1.0	0.30	1.30	0.2903
52	18-35		X	12.58	0	0	1.0	0.30	1.30	0.2903
53	15-29		X	21.43	0	0	1.0	0.16	1.16	0.2396
54	18-34		X	21.43	0	0	1.0	0.16	1.16	0.2396
55	4-15		X	66.88	0	0	1.0	0.10	1.10	0.161
56	10-18		X	66.88	0	0	1.0	0.10	1.10	0.161
57	5-16		X	76.62	0	0	1.0	0.10	1.10	0.1378
58	9-17		X	76.62	0	0	1.0	0.10	1.10	0.1378
59	6-30		X	80.58	0	0	1.0	0.26	1.26	0.113
60	8-33		X	80.58	0	0	1.0	0.26	1.26	0.113
61	16-29		X	11.0	0	0	1.0	0.40	1.40	0.195
62	17-34		X	11.0	0	0	1.0	0.40	1.40	0.195
63	16-30		X	6.89	0	0	1.0	0.37	1.37	0.321
64	17-33		X	6.89	0	0	1.0	0.37	1.37	0.321
65	2-19		X	107.33	0	0	1.0	0.20	1.20	0.0711
66	13-24		X	107.33	0	0	1.0	0.20	1.20	0.0711
67	3-20		X	85.83	0	0	1.0	0.18	1.18	0.102
68	11-23		X	85.83	0	0	1.0	0.18	1.18	0.102
69	7-21		X	174.0	0	0	1.0	0.20	1.20	0.0482
70	12-22		X	174.0	0	0	1.0	0.20	1.20	0.0482
71	25-26		X	18.42	0	0	1.0	0.74	1.74	0.43
72	38-37		X	18.42	0	0	1.0	0.74	1.74	0.43
73	25-32		X	34.77	0	0	1.0	0.39	1.39	0.30
74	38-39		X	34.77	0	0	1.0	0.39	1.39	0.30
75	26-27		X	17.66	0	0	1.0	0.74	1.74	0.44
76	37-36		X	17.66	0	0	1.0	0.74	1.74	0.44
77	27-32		X	29.7	0	0	1.0	0.28	1.28	0.36
78	36-39		X	29.7	0	0	1.0	0.28	1.28	0.36
79	27-28		X	26.72	0	0	1.0	0.20	1.20	0.31
80	36-35		X	26.72	0	0	1.0	0.20	1.20	0.31
81	32-31		X	68.93	0	0	1.0	0.10	1.10	0.13
82	39-40		X	68.93	0	0	1.0	0.10	1.10	0.13
83	31-30		X	58.55	0	0	1.0	0.34	1.34	0.15
84	40-33		X	58.55	0	0	1.0	0.34	1.34	0.15
85	28-29		X	13.75	0	0	1.0	0.56	1.56	0.55
86	35-34		X	13.75	0	0	1.0	0.56	1.56	0.55
87	29-30		X	26.7	0	0	1.0	0.23	1.23	0.35
88	34-33		X	26.7	0	0	1.0	0.23	1.23	0.35
89	29-34		X	4.56	0	0	1.0	0.38	1.38	1.04
90	30-33		X	34.72	0	0	1.0	0.35	1.35	0.23
91	25-41		X	51.3	0	0	1.0	0.21	1.21	0.154
92	38-56		X	51.3	0	0	1.0	0.21	1.21	0.154
93	26-42		X	18.36	0	0	1.0	0.21	1.21	0.434
94	37-55		X	18.36	0	0	1.0	0.21	1.21	0.434
95	27-43		X	54.96	0	0	1.0	0.22	1.22	0.138
96	36-54		X	54.96	0	0	1.0	0.22	1.22	0.138
97	32-49		X	58.97	0	0	1.0	0.25	1.25	0.119

TABLE 6.2.1-21 (SHEET 3 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
98	39-57		X	58.97	0	0	1.0	0.25	1.25	0.119
99	32-43		X	9.55	0	0	1.0	0.29	1.29	0.615
100	39-54		X	9.55	0	0	1.0	0.29	1.29	0.615
101	32-48		X	46.3	0	0	1.0	0.13	1.13	0.17
102	39-58		X	46.3	0	0	1.0	0.13	1.13	0.17
103	31-48		X	15.89	0	0	1.0	0.10	1.10	0.56
104	40-58		X	15.89	0	0	1.0	0.10	1.10	0.56
105	31-47		X	50.26	0	0	1.0	0.13	1.13	0.16
106	40-50		X	50.26	0	0	1.0	0.13	1.13	0.16
107	28-43		X	7.65	0	0	1.0	0.27	1.27	0.814
108	35-54		X	7.65	0	0	1.0	0.27	1.27	0.814
109	28-44		X	9.16	0	0	1.0	0.18	1.18	0.76
110	35-53		X	9.16	0	0	1.0	0.18	1.18	0.76
111	30-45		X	26.8	0	0	1.0	0.09	1.09	0.201
112	33-51		X	26.8	0	0	1.0	0.09	1.09	0.201
113	30-47		X	116.2	0	0	1.0	0.10	1.10	0.076
114	33-50		X	116.2	0	0	1.0	0.10	1.10	0.076
115	29-45		X	56.71	0	0	1.0	0.10	1.10	0.16
116	34-51		X	56.71	0	0	1.0	0.10	1.10	0.16
117	41-74	X		43.29	0	0	1.0	0.26	1.26	0.123
118	56-74		X	43.29	0	0	1.0	0.26	1.26	0.123
119	42-74	X		30.16	0	0	1.0	0.04	1.04	0.338
120	55-74		X	30.16	0	0	1.0	0.04	1.04	0.338
121	43-74	X		9.63	0	0	1.0	0.47	1.47	0.0794
122	54-74		X	9.63	0	0	1.0	0.47	1.47	0.0794
123	49-74	X		15.82	0	0	1.0	0.43	1.43	0.095
124	57-74		X	15.82	0	0	1.0	0.43	1.43	0.095
125	50-64		X	15.1	0	0	1.0	0.46	1.46	0.276
126	47-63		X	15.1	0	0	1.0	0.46	1.46	0.276
127	52-65		X	21.55	0	0	1.0	0.39	1.39	0.259
128	46-62		X	21.55	0	0	1.0	0.39	1.39	0.259
129	53-66		X	8.63	0	0	1.0	0.35	1.35	0.475
130	44-60		X	8.63	0	0	1.0	0.35	1.35	0.475
131	54-66		X	10.05	0	0	1.0	0.46	1.46	0.465
132	43-60		X	10.05	0	0	1.0	0.46	1.46	0.465
133	58-68		X	9.95	0	0	1.0	0.44	1.44	0.374
134	48-59		X	9.95	0	0	1.0	0.44	1.44	0.374
135	45-46		X	91.35	0	0	1.0	0.04	1.04	0.0828
136	51-52		X	91.35	0	0	1.0	0.04	1.04	0.0828
137	41-42		X	81.18	0	0	1.0	0.25	1.25	0.094
138	56-55		X	81.18	0	0	1.0	0.25	1.25	0.094
139	41-49		X	128.07	0	0	1.0	0.23	1.23	0.056
140	56-57		X	128.07	0	0	1.0	0.23	1.23	0.056
141	42-43		X	78.89	0	0	1.0	0.08	1.08	0.128
142	55-54		X	78.89	0	0	1.0	0.08	1.08	0.128
143	43-49		X	107.19	0	0	1.0	0.24	1.24	0.0893
144	54-57		X	107.19	0	0	1.0	0.24	1.24	0.0893
145	43-48		X	85.73	0	0	1.0	0.33	1.33	0.0482
146	54-58		X	85.73	0	0	1.0	0.33	1.33	0.0482
147	43-44		X	95.48	0	0	1.0	0.31	1.31	0.127

TABLE 6.2.1-21 (SHEET 4 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
148	54-53		X	95.48	0	0	1.0	0.31	1.31	0.127
149	49-58		X	109.22	0	0	1.0	0.27	1.27	0.0406
150	57-58		X	109.22	0	0	1.0	0.27	1.27	0.0406
151	48-47		X	178.11	0	0	1.0	0.53	1.53	0.053
152	58-50		X	178.11	0	0	1.0	0.53	1.53	0.053
153	44-45		X	22.12	0	0	1.0	0.56	1.56	0.308
154	53-51		X	22.12	0	0	1.0	0.56	1.56	0.308
155	44-46		X	29.81	0	0	1.0	0.50	1.50	0.276
156	53-52		X	29.81	0	0	1.0	0.50	1.50	0.276
157	45-47		X	27.88	0	0	1.0	0.61	1.61	0.186
158	51-50		X	27.88	0	0	1.0	0.61	1.61	0.186
159	46-47		X	53.57	0	0	1.0	0.55	1.55	0.173
160	52-50		X	53.57	0	0	1.0	0.55	1.55	0.173
161	47-50		X	175.22	0	0	1.0	0.16	1.16	0.065
162	45-51		X	35.04	0	0	1.0	0.25	1.25	0.26
163	46-52		X	58.39	0	0	1.0	0.14	1.14	0.229
164	59-60		X	21.0	0	0	1.0	0.27	1.27	0.60
165	68-66		X	21.0	0	0	1.0	0.27	1.27	0.60
166	59-61		X	13.5	0	0	1.0	0.35	1.35	0.937
167	68-67		X	13.5	0	0	1.0	0.35	1.35	0.937
168	60-62		X	16.38	0	0	1.0	0.18	1.18	0.77
169	66-65		X	16.38	0	0	1.0	0.18	1.18	0.77
170	61-62		X	10.53	0	0	1.0	0	1.0	1.2
171	67-65		X	10.53	0	0	1.0	0	1.0	1.2
172	62-63		X	30.59	0	0	1.0	0.13	1.13	0.41
173	65-64		X	30.59	0	0	1.0	0.13	1.13	0.41
174	63-59		X	23.0	0	0	1.0	0.16	1.16	0.55
175	64-68		X	23.0	0	0	1.0	0.16	1.16	0.55
176	59-74		X	38.1	0	0	1.0	0.08	1.08	0.276
177	68-74		X	38.1	0	0	1.0	0.08	1.08	0.276
178	61-74		X	42.0	0	0	1.0	0.03	1.03	0.11
179	67-74		X	42.0	0	0	1.0	0.03	1.03	0.11
180	62-74		X	47.25	0	0	1.0	0.10	1.10	0.215
181	65-74		X	47.25	0	0	1.0	0.10	1.10	0.215
182	63-74		X	29.4	0	0	1.0	0.23	1.23	0.236
183	64-74		X	29.4	0	0	1.0	0.23	1.23	0.236
184	60-61		X	15.54	0	0	1.0	0.32	1.32	0.181
185	66-67		X	15.54	0	0	1.0	0.32	1.32	0.181
186	3-69		X	21.6	0	0.90	1.0	0.50	2.40	1.256
187	11-70		X	21.6	0	0.90	1.0	0.50	2.40	1.256
188	20-69		X	5.4	0	0.90	1.0	0.50	2.40	1.273
189	23-70		X	5.4	0	0.90	1.0	0.50	2.40	1.273
190	69-43		X	75.6	0	0	1.0	0.35	1.35	0.0869
191	70-54		X	75.6	0	0	1.0	0.35	1.35	0.0869
192	69-74	X		7.2	0	0	1.0	0.44	1.44	0.776
193	70-74		X	7.2	0	0	1.0	0.44	1.44	0.776
194	69-71		X	33.88	0.012	0	1.0	0.44	1.45	0.98
195	70-72		X	33.88	0.012	0	1.0	0.44	1.45	0.98
196	71-74		X	39.26	0.045	1.38	1.0	0.45	2.88	0.795
197	72-74		X	39.26	0.045	1.38	1.0	0.45	2.88	0.795

TABLE 6.2.1-21 (SHEET 5 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
198	73-74		X	391.01	0	0	1.0	0.29	1.29	0.008

TABLE 6.2.1-22 (SHEET 1 OF 5)

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW EL. 220 ft): FLOW CHARACTERISTICS
LOOP CLOSURE WELD (336-in.^{2} BREAK)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
1	1-74		X	20.25	0	0	1.0	0.38	1.38	0.096
2	1-73		X	97.08	0	0	1.0	0	1.0	0.079
3	20-19		X	29.58	0	0	1.0	0.27	1.27	0.295
4	23-24		X	29.58	0	0	1.0	0.27	1.27	0.295
5	2-1		X	92.1	0	0	1.0	0.25	1.25	0.122
6	3-2		X	63.12	0	0	1.0	0.24	1.24	0.151
7	11-13		X	63.12	0	0	1.0	0.24	1.24	0.151
8	7-2		X	94.0	0	0	1.0	0.27	1.27	0.0623
9	13-12		X	125.8	0	0	1.0	0.27	1.27	0.0623
10	21-19		X	64.65	0	0	1.0	0.20	1.20	0.136
11	22-24		X	80.0	0	0	1.0	0.20	1.20	0.134
12	3-7		X	47.57	0	0	1.0	0.41	1.41	0.0698
13	11-12		X	47.57	0	0	1.0	0.41	1.41	0.0698
14	20-21		X	58.72	0	0	1.0	0.29	1.29	0.297
15	23-22		X	58.72	0	0	1.0	0.29	1.29	0.297
16	3-4		X	63.65	0	0	1.0	0.28	1.28	0.103
17	11-10		X	63.65	0	0	1.0	0.28	1.28	0.103
18	20-15		X	36.93	0	0	1.0	0.25	1.25	0.192
19	23-18		X	36.93	0	0	1.0	0.25	1.25	0.192
20	7-6		X	129.19	0	0	1.0	0.27	1.27	0.0509
21	12-8		X	129.19	0	0	1.0	0.27	1.27	0.0509
22	21-6		X	70.58	0	0	1.0	0.26	1.26	0.1303
23	22-8		X	70.58	0	0	1.0	0.26	1.26	0.1303
24	4-5		X	99.53	0	0	1.0	0.07	1.07	0.1265
25	10-9		X	99.53	0	0	1.0	0.07	1.07	0.1265
26	15-16		X	81.3	0	0	1.0	0.17	1.17	0.1093
27	18-17		X	81.3	0	0	1.0	0.17	1.17	0.1093
28	5-6		X	113.9	0	0	1.0	0.25	1.25	0.08965
29	16-6		X	94.2	0	0	1.0	0.29	1.29	0.153
31	17-8		X	94.2	0	0	1.0	0.29	1.29	0.153
32	6-8		X	247.93	0	0	1.0	0.16	1.16	0.0310
33	5-9		X	76.37	0	0	1.0	0.17	1.17	0.1071
34	16-17		X	68.31	0	0	1.0	0.22	1.22	0.115
35	13-14		X	92.65	0	0	1.0	0.30	1.30	0.086
36	24-14		X	13.7	0	0	1.0	0.38	1.38	0.163
37	14-73		X	44.86	0	0	1.0	0.40	1.40	0.1063
38	14-74		X	50.6	0	0	1.0	0.35	1.35	0.42
39	19-25		X	80.58	0	0	1.0	0.05	1.05	0.0798
40	24-38		X	80.58	0	0	1.0	0.05	1.05	0.0798
41	19-26		X	28.52	0	0	1.0	0.18	1.18	0.185
42	24-37		X	28.52	0	0	1.0	0.18	1.18	0.185
43	19-32		X	53.45	0	0	1.0	0	1.0	0.112
44	24-39		X	53.45	0	0	1.0	0	1.0	0.112
45	20-27		X	62.58	0	0	1.0	0.15	1.15	0.0794
46	23-36		X	62.58	0	0	1.0	0.15	1.15	0.0794
47	21-32		X	81.5	0	0	1.0	0.27	1.27	0.0389

TABLE 6.2.1-22 (SHEET 2 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
48	22-39		X	81.5	0	0	1.0	0.27	1.27	0.0389
49	21-31		X	57.19	0	0	1.0	0.22	1.22	0.0778
50	22-40		X	57.19	0	0	1.0	0.22	1.22	0.0778
51	15-28		X	12.58	0	0	1.0	0.30	1.30	0.2903
52	18-35		X	12.58	0	0	1.0	0.30	1.30	0.2903
53	15-29		X	21.43	0	0	1.0	0.16	1.16	0.2396
54	18-34		X	21.43	0	0	1.0	0.16	1.16	0.2396
55	4-15		X	66.88	0	0	1.0	0.10	1.10	0.161
56	10-18		X	66.88	0	0	1.0	0.10	1.10	0.161
57	5-16		X	76.62	0	0	1.0	0.10	1.10	0.1378
58	9-17		X	76.62	0	0	1.0	0.10	1.10	0.1378
5	6-30		X	80.58	0	0	1.0	0.26	1.26	0.113
60	8-33		X	80.58	0	0	1.0	0.26	1.26	0.113
61	16-29		X	11.0	0	0	1.0	0.40	1.40	0.195
62	17-34		X	11.0	0	0	1.0	0.40	1.40	0.195
63	16-30		X	6.89	0	0	1.0	0.37	1.37	0.321
64	17-33		X	6.89	0	0	1.0	0.37	1.37	0.321
65	2-19		X	107.33	0	0	1.0	0.20	1.20	0.0711
66	13-24		X	107.33	0	0	1.0	0.20	1.20	0.0711
67	3-20		X	85.83	0	0	1.0	0.18	1.18	0.102
68	11-23		X	85.83	0	0	1.0	0.18	1.18	0.102
69	7-21		X	174.0	0	0	1.0	0.20	1.20	0.0482
70	12-22		X	174.0	0	0	1.0	0.20	1.20	0.0482
71	25-26		X	18.42	0	0	1.0	0.74	1.74	0.43
72	38-37		X	18.42	0	0	1.0	0.74	1.74	0.43
73	25-32		X	34.77	0	0	1.0	0.39	1.39	0.30
74	38-39		X	34.77	0	0	1.0	0.39	1.39	0.30
75	26-27		X	17.66	0	0	1.0	0.74	1.74	0.44
76	37-36		X	17.66	0	0	1.0	0.74	1.74	0.44
77	27-32		X	29.7	0	0	1.0	0.28	1.28	0.36
78	36-39		X	29.7	0	0	1.0	0.28	1.28	0.36
79	27-28		X	26.72	0	0	1.0	0.20	1.20	0.31
80	36-35		X	26.72	0	0	1.0	0.20	1.20	0.31
81	32-31		X	68.93	0	0	1.0	0.10	1.10	0.13
82	39-40		X	68.93	0	0	1.0	0.10	1.10	0.13
83	31-30		X	58.55	0	0	1.0	0.34	1.34	0.15
84	40-33		X	58.55	0	0	1.0	0.34	1.34	0.15
85	28-29		X	13.75	0	0	1.0	0.56	1.56	0.55
86	35-34		X	13.75	0	0	1.0	0.56	1.56	0.55
87	29-30		X	26.7	0	0	1.0	0.23	1.23	0.35
88	34-33		X	26.7	0	0	1.0	0.23	1.23	0.35
89	29-34		X	4.56	0	0	1.0	0.38	1.38	1.04
90	30-33		X	34.72	0	0	1.0	0.35	1.35	0.23
91	25-41		X	51.3	0	0	1.0	0.21	1.21	0.154
92	38-56		X	51.3	0	0	1.0	0.21	1.21	0.154
93	26-42		X	18.36	0	0	1.0	0.21	1.21	0.434
94	37-55		X	18.36	0	0	1.0	0.21	1.21	0.434
95	27-43		X	54.96	0	0	1.0	0.22	1.22	0.138
96	36-54		X	54.96	0	0	1.0	0.22	1.22	0.138
97	32-49		X	58.97	0	0	1.0	0.25	1.25	0.119

TABLE 6.2.1-22 (SHEET 3 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
98	39-57		X	58.97	0	0	1.0	0.25	1.25	0.119
99	32-43		X	9.55	0	0	1.0	0.29	1.29	0.615
100	39-54		X	9.55	0	0	1.0	0.29	1.29	0.615
101	32-48		X	46.3	0	0	1.0	0.13	1.13	0.17
102	39-58		X	46.3	0	0	1.0	0.13	1.13	0.17
103	31-48		X	15.89	0	0	1.0	0.10	1.10	0.56
104	40-58		X	15.89	0	0	1.0	0.10	1.10	0.56
105	31-47		X	50.26	0	0	1.0	0.13	1.13	0.16
106	40-50		X	50.26	0	0	1.0	0.13	1.13	0.16
107	28-43		X	7.65	0	0	1.0	0.27	1.27	0.814
108	35-54		X	7.65	0	0	1.0	0.27	1.27	0.814
109	28-44		X	9.16	0	0	1.0	0.18	1.18	0.76
110	35-53		X	9.16	0	0	1.0	0.18	1.18	0.76
111	30-45		X	26.8	0	0	1.0	0.09	1.09	0.201
112	33-51		X	26.8	0	0	1.0	0.09	1.09	0.201
113	30-47		X	116.2	0	0	1.0	0.10	1.10	0.076
114	33-50		X	116.2	0	0	1.0	0.10	1.10	0.076
115	29-45		X	56.71	0	0	1.0	0.10	1.10	0.16
116	34-51		X	56.71	0	0	1.0	0.10	1.10	0.16
117	41-74	X		43.29	0	0	1.0	0.26	1.26	0.123
118	56-74		X	43.29	0	0	1.0	0.26	1.26	0.123
119	42-74	X		30.16	0	0	1.0	0.04	1.04	0.338
120	55-74		X	30.16	0	0	1.0	0.04	1.04	0.338
121	43-74	X		9.63	0	0	1.0	0.47	1.47	0.0794
122	54-74		X	9.63	0	0	1.0	0.47	1.47	0.0794
123	49-74	X		15.82	0	0	1.0	0.43	1.43	0.095
124	57-74		X	15.82	0	0	1.0	0.43	1.43	0.095
125	50-64		X	15.1	0	0	1.0	0.46	1.46	0.276
126	47-63		X	15.1	0	0	1.0	0.46	1.46	0.276
127	52-65		X	21.55	0	0	1.0	0.39	1.39	0.259
128	46-62		X	21.55	0	0	1.0	0.39	1.39	0.259
129	53-66		X	8.63	0	0	1.0	0.35	1.35	0.475
130	44-60		X	8.63	0	0	1.0	0.35	1.35	0.475
131	54-66		X	10.05	0	0	1.0	0.46	1.46	0.465
132	43-60		X	10.05	0	0	1.0	0.46	1.46	0.465
133	58-68		X	9.95	0	0	1.0	0.44	1.44	0.374
134	48-59		X	9.95	0	0	1.0	0.44	1.44	0.374
135	45-46		X	91.35	0	0	1.0	0.04	1.04	0.0828
136	51-52		X	91.35	0	0	1.0	0.04	1.04	0.0828
137	41-42		X	81.18	0	0	1.0	0.25	1.25	0.094
138	56-55		X	81.18	0	0	1.0	0.25	1.25	0.094
139	41-49		X	128.07	0	0	1.0	0.23	1.23	0.056
140	56-57		X	128.07	0	0	1.0	0.23	1.23	0.056
141	42-43		X	78.89	0	0	1.0	0.08	1.08	0.128
142	55-54		X	78.89	0	0	1.0	0.08	1.08	0.128
143	43-49		X	107.19	0	0	1.0	0.24	1.24	0.0893
144	54-57		X	107.19	0	0	1.0	0.24	1.24	0.0893
145	43-48		X	85.73	0	0	1.0	0.33	1.33	0.0482
146	54-58		X	85.73	0	0	1.0	0.33	1.33	0.0482
147	43-44		X	95.48	0	0	1.0	0.31	1.31	0.127

TABLE 6.2.1-22 (SHEET 4 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
148	54-53		X	95.48	0	0	1.0	0.31	1.31	0.127
149	49-58		X	109.22	0	0	1.0	0.27	1.27	0.0406
150	57-58		X	109.22	0	0	1.0	0.27	1.27	0.0406
151	48-47		X	178.11	0	0	1.0	0.53	1.53	0.053
152	58-50		X	178.11	0	0	1.0	0.53	1.53	0.053
153	44-45		X	22.12	0	0	1.0	0.56	1.56	0.308
154	53-51		X	22.12	0	0	1.0	0.56	1.56	0.308
155	44-46		X	29.81	0	0	1.0	0.50	1.50	0.276
156	53-52		X	29.81	0	0	1.0	0.50	1.50	0.276
157	45-47		X	27.88	0	0	1.0	0.61	1.61	0.186
158	51-50		X	27.88	0	0	1.0	0.61	1.61	0.186
159	46-47		X	53.57	0	0	1.0	0.55	1.55	0.173
160	52-50		X	53.57	0	0	1.0	0.55	1.55	0.173
161	47-50		X	175.22	0	0	1.0	0.16	1.16	0.065
162	45-51		X	35.04	0	0	1.0	0.25	1.25	0.26
163	46-52		X	58.39	0	0	1.0	0.14	1.14	0.229
164	59-60		X	21.0	0	0	1.0	0.27	1.27	0.60
165	68-66		X	21.0	0	0	1.0	0.27	1.27	0.60
166	59-61		X	13.5	0	0	1.0	0.35	1.35	0.937
167	68-67		X	13.5	0	0	1.0	0.35	1.35	0.937
168	60-62		X	16.38	0	0	1.0	0.18	1.18	0.77
169	66-65		X	16.38	0	0	1.0	0.18	1.18	0.77
170	61-62		X	10.53	0	0	1.0	0	1.0	1.2
171	67-65		X	10.53	0	0	1.0	0	1.0	1.2
172	62-63		X	30.59	0	0	1.0	0.13	1.13	0.41
173	65-64		X	30.59	0	0	1.0	0.13	1.13	0.41
174	63-59		X	23.0	0	0	1.0	0.16	1.16	0.55
175	64-68		X	23.0	0	0	1.0	0.16	1.16	0.55
176	59-74		X	38.1	0	0	1.0	0.08	1.08	0.276
177	68-74		X	38.1	0	0	1.0	0.08	1.08	0.276
178	61-74		X	42.0	0	0	1.0	0.03	1.03	0.11
179	67-74		X	42.0	0	0	1.0	0.03	1.03	0.11
180	62-74		X	47.25	0	0	1.0	0.10	1.10	0.215
181	65-74		X	47.25	0	0	1.0	0.10	1.10	0.215
182	63-74		X	29.4	0	0	1.0	0.23	1.23	0.236
183	64-74		X	29.4	0	0	1.0	0.23	1.23	0.236
184	60-61		X	15.54	0	0	1.0	0.32	1.32	0.181
185	66-67		X	15.54	0	0	1.0	0.32	1.32	0.181
186	3-69		X	21.6	0	0.90	1.0	0.50	2.40	1.256
187	11-70		X	21.6	0	0.90	1.0	0.50	2.40	1.256
188	20-69		X	5.4	0	0.90	1.0	0.50	2.40	1.273
189	23-70		X	5.4	0	0.90	1.0	0.50	2.40	1.273
190	69-43		X	75.6	0	0	1.0	0.35	1.35	0.0869
191	70-54		X	75.6	0	0	1.0	0.35	1.35	0.0869
192	69-74	X		7.2	0	0	1.0	0.44	1.44	0.776
193	70-74		X	7.2	0	0	1.0	0.44	1.44	0.776
194	69-71		X	33.88	0.012	0	1.0	0.44	1.45	0.98
195	70-72		X	33.88	0.012	0	1.0	0.44	1.45	0.98
196	71-74		X	39.26	0.045	1.38	1.0	0.45	2.88	0.795
197	72-74		X	39.26	0.045	1.38	1.0	0.45	2.88	0.795

TABLE 6.2.1-22 (SHEET 5 OF 5)

Vent Path	Node Number (From-To)	Description of Flow		Flow Area (ft^2)	Friction (K)	Turning and Obstruction (K)	Expansion (K)	Contraction (K)	Total (K_T)	L/A (ft^{-1})
		Choked	Unchoked							
198	73-74		X	391.01	0	0	1.0	0.29	1.29	0.008

TABLE 6.2.1-23 (SHEET 1 OF 2)

PRESSURIZER COMPARTMENT MODEL: FLOW CHARACTERISTICS (SPRAY LINE)
LOSS COEFFICIENT (K)

Vent Parameters	Node 5 (From-To)	Description of Flow Choked	Description of Flow Unchoked	Area (ft^2)	Turning Friction (ft^2-h)	and Obstruction	Expansion	Contraction	Total	L/A (ft^{-1})
1	1-28		X	172.0	0	0	1.0	0.82	1.82	0.027
2	2-1		X	27.4	0	0	1.0	0.23	1.23	0.190
3	2-3		X	18.1	0	0	1.0	0.32	1.32	0.157
4	3-1		X	23.6	0	0	1.0	0.29	1.29	0.201
5	3-4		X	25.0	0	0	1.0	0.29	1.29	0.114
6	4-1		X	26.8	0	0	1.0	0.26	1.26	0.153
7	4-5		X	26.2	0	0	1.0	0.32	1.32	0.119
8	5-1		X	33.4	0	0	1.0	0.23	1.23	0.143
9	5-2		X	27.5	0	0	1.0	0.38	1.38	0.105
10	6-2		X	16.4	0	0	1.0	0.29	1.29	0.319
11	6-7		X	20.5	0	0	1.0	0.38	1.38	0.180
12	7-3		X	12.7	0	0	1.0	0.35	1.35	0.352
13	7-8		X	30.0	0	0	1.0	0.32	1.32	0.126
14	8-4		X	19.2	0	0	1.0	0.32	1.32	0.253
15	8-9		X	33.0	0	0	1.0	0.32	1.32	0.129
16	9-5		X	19.9	0	0	1.0	0.35	1.35	0.231
17	9-6		X	35.0	0	0	1.0	0.29	1.29	0.117
18	10-6		X	16.1	0	0	1.0	0.35	1.35	0.303
19	10-11		X	14.0	0	0	1.0	0.38	1.38	0.213
20	10-28		X	12.7	0	0	1.06	0.26	2.32	0.517
21	11-7		X	13.5	0	0	1.0	0.35	1.35	0.335
22	11-12		X	25.2	0	0	1.0	0.32	1.32	0.150
23	12-8		X	19.2	0	0	1.0	0.32	1.32	0.241
24	12-13		X	27.8	0	0	1.0	0.32	1.32	0.154
25	13-9		X	22.9	0	0	1.0	0.32	1.32	0.219
26	13-10		X	29.4	0	0	1.0	0.32	1.32	0.140
27	14-10		X	27.9	0	0	1.0	0.21	1.21	0.225
28	14-15		X	9.5	0	0	1.0	0.35	1.35	0.350
29	15-11		X	24.9	0	0	1.0	0.26	1.26	0.248
30	15-16		X	15.3	0	0	1.0	0.21	1.21	0.449
31	16-12		X	32.5	0	0	1.0	0.21	1.21	0.178
32	16-17		X	16.5	0	0	1.0	0.32	1.32	0.255
33	17-13		X	37.3	0	0	1.0	0.21	1.21	0.163
34	17-14		X	17.8	0	0	1.0	0.29	1.29	0.410
35	18-14		X	14.5	0	0	1.0	0.35	1.35	0.161
36	18-19		X	51.0	0	0	1.0	0.26	1.26	0.100
37	19-15		X	10.5	0	0	1.0	0.35	1.35	0.178
38	19-20		X	23.0	0	0	1.0	0.32	1.32	0.330
39	20-16		X	24.2	0	0	1.0	0.32	1.32	0.186
40	20-21		X	29.3	0	0	1.0	0.32	1.32	0.143
41	21-17		X	26.0	0	0	1.0	0.32	1.32	0.170
42	21-18		X	31.5	0	0	1.0	0.29	1.29	0.152
43	22-18		X	33.0	0	0	1.0	0.32	1.32	0.154

TABLE 6.2.1-23 (SHEET 2 OF 2)

Vent Parameters	Node 5 (From-To)	Description of Flow		Flow Area (ft^2)	Friction (ft^2-h)	Turning and Obstruction	Expansion	Contraction	Total	L/A (ft^{-1})
		Choked	Unchoked							
44	22-23		X	51.8	0	0	1.0	0.26	1.26	0.099
45	22-28		X	0.20	0	0	1.0	0.49	1.49	12.5
46	23-19		X	29.2	0	0	1.0	0.32	1.32	0.167
47	23-24		X	27.5	0	0	1.0	0.32	1.32	0.162
48	24-20		X	19.2	0	0	1.0	0.32	1.32	0.238
49	24-25		X	29.7	0	0	1.0	0.32	1.32	0.141
50	25-21		X	24.9	0	0	1.0	0.32	1.32	0.218
51	25-22		X	32.0	0	0	1.0	0.29	1.29	0.149
52	25-28		X	1.23	0	0	1.0	0.49	1.49	2.08
53	26-22		X	26.1	0	0	1.0	0.35	1.35	0.099
54	26-23		X	27.0	0	0	1.0	0.32	1.32	0.104
55	26-24		X	13.4	0	0	1.0	0.35	1.35	0.140
56	26-25		X	16.4	0	0	1.0	0.35	1.35	0.130
57	26-27		X	161.5	0	0	1.0	0.26	1.26	0.043
58	26-28		X	5.62	0	0	1.0	0.49	1.49	0.532
59	27-28		X	304.3	0	0	1.0	0.23	1.23	0.024

TABLE 6.2.1-24 (SHEET 1 OF 2)

PRESSURIZER COMPARTMENT FLOW CHARACTERISTIC MODE (SURGE LINE)
LOSS COEFFICIENT (K)

Vent Parameters	Node 5 (From-To)	Description of Flow		Flow Area (ft^2)	Friction (ft^2-h)	Turning and Obstruction	Expansion	Contraction	Total	L/A (ft^{-1})
		Choked	Unchoked							
1	1-28		X	172.0	0	0	1.0	0.82	1.82	0.027
2	2-1		X	27.4	0	0	1.0	0.23	1.23	0.190
3	2-3		X	18.1	0	0	1.0	0.32	1.32	0.157
4	3-1		X	23.6	0	0	1.0	0.29	1.29	0.201
5	3-4		X	25.0	0	0	1.0	0.29	1.29	0.114
6	4-1		X	26.8	0	0	1.0	0.26	1.26	0.153
7	4-5		X	26.2	0	0	1.0	0.32	1.32	0.119
8	5-1		X	33.4	0	0	1.0	0.23	1.23	0.143
9	5-2		X	27.5	0	0	1.0	0.38	1.38	0.105
10	6-2		X	16.4	0	0	1.0	0.29	1.29	0.319
11	6-7		X	20.5	0	0	1.0	0.38	1.38	0.180
12	7-3		X	12.7	0	0	1.0	0.35	1.35	0.352
13	7-8		X	30.0	0	0	1.0	0.32	1.32	0.126
14	8-4		X	19.2	0	0	1.0	0.32	1.32	0.253
15	8-9		X	33.0	0	0	1.0	0.32	1.32	0.129
16	9-5		X	19.9	0	0	1.0	0.35	1.35	0.231
17	9-6		X	35.0	0	0	1.0	0.29	1.29	0.117
18	10-6		X	16.1	0	0	1.0	0.35	1.35	0.303
19	10-11		X	14.0	0	0	1.0	0.38	1.38	0.213
20	10-24	X		12.7	0	1.06	1.0	0.26	2.32	0.517
21	11-7		X	13.5	0	0	1.0	0.35	1.35	0.335
22	11-12		X	25.2	0	0	1.0	0.32	1.32	0.150
23	12-8		X	19.2	0	0	1.0	0.32	1.32	0.241
24	12-13		X	27.8	0	0	1.0	0.32	1.32	0.154
25	13-9		X	22.9	0	0	1.0	0.32	1.32	0.219
26	13-10		X	29.4	0	0	1.0	0.32	1.32	0.140
27	14-10		X	27.9	0	0	1.0	0.21	1.21	0.225
28	14-15		X	9.5	0	0	1.0	0.35	1.35	0.350
29	15-11		X	24.9	0	0	1.0	0.26	1.26	0.248
30	15-16		X	15.3	0	0	1.0	0.21	1.21	0.449
31	16-12		X	32.5	0	0	1.0	0.21	2.21	0.178
32	16-17		X	16.5	0	0	1.0	0.32	1.32	0.255
33	17-13		X	37.3	0	0	1.0	0.21	1.21	0.163
34	17-14		X	17.8	0	0	1.0	0.29	1.29	0.410
35	18-14		X	14.5	0	0	1.0	0.35	1.35	0.161
36	18-19		X	51.0	0	0	1.0	0.26	1.26	0.100
37	19-15		X	10.5	0	0	1.0	0.35	1.35	0.178
38	19-20		X	23.0	0	0	1.0	0.32	1.32	0.330
39	20-16		X	24.2	0	0	1.0	0.32	1.32	0.186
40	20-21		X	29.3	0	0	1.0	0.32	1.32	0.143
41	21-17		X	26.0	0	0	1.0	0.32	1.32	0.170
42	21-18		X	31.5	0	0	1.0	0.29	1.29	0.152

TABLE 6.2.1-24 (SHEET 2 OF 2)

Vent Parameters	Node 5 (From-To)	Description of Flow		Flow Area (ft^2)	Friction (ft^2-h)	Turning and Obstruction	Expansion	Contraction	Total	L/A (ft^{-1})
		Choked	Unchoked							
43	22-18		X	33.0	0	0	1.0	0.32	1.32	0.154
44	22-23		X	51.8	0	0	1.0	0.26	1.26	0.099
45	22-28	X		0.2	0	0	1.0	0.49	1.49	12.5
46	23-19		X	29.2	0	0	1.0	0.32	1.32	0.167
47	23-24		X	27.5	0	0	1.0	0.32	1.32	0.162
48	24-20		X	19.2	0	0	1.0	0.32	1.32	0.238
49	24-25		X	29.7	0	0	1.0	0.32	1.32	0.141
50	25-21		X	24.9	0	0	1.0	0.32	1.32	0.218
51	25-22		X	32.0	0	0	1.0	0.29	1.29	0.149
52	25-28	X		1.23	0	0	1.0	0.49	1.49	2.08
53	26-22		X	26.1	0	0	1.0	0.35	1.35	0.099
54	26-23		X	27.0	0	0	1.0	0.32	1.32	0.104
55	26-24		X	13.4	0	0	1.0	0.35	1.35	0.140
56	26-25		X	16.4	0	0	1.0	0.35	1.35	0.130
57	26-27	X		161.5	0	0	1.0	0.26	1.26	0.043
58	26-28	X		5.62	0	0	1.0	0.49	1.49	0.532
59	27-28		X	304.3	0	0	1.0	0.23	1.23	0.024

TABLE 6.2.1-25 (SHEET 1 OF 4)

PRESSURIZER COMPARTMENT MODEL: NODE CHARACTERISTICS

Node	Description	Height (ft)	Cross-Sectional Area (ft)	Net Volume (ft^3)	Initial Conditions			Calc Peak Pressure Differential (psi)
					Temperature (°F)	Pressure (psia)	Humidity (%)	
1	In pressurizer compartment between el 252 ft 4 in. and 265 ft.	12.67	215.7	2596	120	13.2	50.0	3.15
2	Northwest quadrant in pressurizer compartment between el 242 ft 0 in. and 252 ft 4 in.	10.33	49.3	400	120	13.2	50.0	5.71
3	Northeast quadrant in pressurizer compartment between el 242 ft 0 in. and 252 ft 4 in.	10.33	46.3	367	120	13.2	50.0	5.84
4	Southeast quadrant in pressurizer compartment between el 242 ft 0 in. and 252 ft 4 in.	10.33	58.1	485	120	13.2	50.0	5.70
5	Southwest quadrant in pressurizer compartment between el 242 ft 0 in. and 252 ft 4 in.	10.33	62.0	524	120	13.2	50.0	5.64
6	Northwest quadrant in pressurizer compartment between el 231 ft 0 in. and 242 ft 0 in.	11.0	49.3	350	120	13.2	50.0	7.41
7	Northeast quadrant in pressurizer compartment between el 231 ft 0 in. and 242 ft 0 in.	11.0	46.3	317	120	13.2	50.0	7.46

TABLE 6.2.1-25 (SHEET 2 OF 4)

Node	Description	Height (ft)	Cross-Sectional Area (ft)	Net Volume (ft^3)	Initial Conditions Temperature (°F)	Initial Conditions Pressure (psia)	Initial Conditions Humidity (%)	Calc Peak Pressure Differential (psi)
8	Southeast quadrant in pressurizer compartment between el 231 ft 0 in. and 242 ft 0 in.	11.0	58.1	441	120	13.2	50.0	7.36
9	Southwest quadrant in pressurizer compartment between el 231 ft 0 in. and 242 ft 0 in.	11.0	62.0	480	120	13.2	50.0	7.44
10	Northwest quadrant of pressurizer compartment between el 221 ft 7 7/8 in. and 231 ft 0 in.	9.34	49.3	296	120	13.2	50.0	10.5
11	Northeast quadrant of pressurizer compartment between el 221 ft 7 7/8 in. and 231 ft 0 in.	9.34	46.3	269	120	13.2	50.0	10.7
12	Southeast quadrant of pressurizer compartment between el 221 ft 7 7/8 in. and 231 ft 0 in.	9.34	58.1	374	120	13.2	50.0	10.7
13	Southwest quadrant of pressurizer compartment between el 221 ft 7 7/8 in. and 231 ft 0 in.	9.34	62.0	409	120	13.2	50.0	10.6
14	Northwest quadrant of pressurizer compartment between el 216 ft 0 in. and 221 ft 7 7/8 in.	5.66	49.3	180	120	13.2	50.0	11.8
15	Northeast quadrant of pressurizer compartment between el 216 ft 0 in. and 221 ft 7 7/8 in.	5.66	46.3	163	120	13.2	50.0	11.7

TABLE 6.2.1-25 (SHEET 3 OF 4)

Node	Description	Height (ft)	Cross-Sectional Area (ft)	Net Volume (ft^3)	Initial Conditions Temperature (°F)	Initial Conditions Pressure (psia)	Initial Conditions Humidity (%)	Calc Peak Pressure Differential (psi)
16	Southeast quadrant of pressurizer compartment between el 216 ft 0 in. and 221 ft 7 7/8 in.	5.66	58.1	227	120	13.2	50.0	11.9
17	Southwest quadrant of pressurizer compartment between el 216 ft 0 in. and 221 ft 7 7/8 in.	5.66	62.0	244	120	13.2	50.0	11.9
18	Northwest quadrant in pressurizer compartment between el 206 ft 0 in. and 216 ft 0 in.	10.0	81.3	622	120	13.2	50.0	15.8
19	Northeast quadrant in pressurizer compartment between el 206 ft 0 in. and 216 ft 0 in.	10.0	76.2	573	120	13.2	50.0	15.8
20	Southeast quadrant in pressurizer compartment between el 206 ft 0 in. and 216 ft 0 in.	10.0	58.1	401	120	13.2	50.0	15.4
21	Southwest quadrant in pressurizer compartment between el 206 ft 0 in. and 216 ft 0 in.	10.0	62.0	438	120	13.2	50.0	15.4
22	Northwest quadrant in pressurizer compartment between el 195 ft 9 3/4 in. and 206 ft 0 in.	10.19	81.3	633	120	13.2	50.0	17.3
23	Northeast quadrant in pressurizer compartment between el 195 ft 9 3/4 in. and 206 ft 0 in.	10.19	76.2	584	120	13.2	50.0	17.2

TABLE 6.2.1-25 (SHEET 4 OF 4)

Node	Description	Height (ft)	Cross-Sectional Area (ft)	Net Volume (ft^3)	Initial Conditions Temperature (°F)	Initial Conditions Pressure (psia)	Humidity (%)	Calc Peak Pressure Differential (psi)
24	Southeast quadrant in pressurizer compartment between el 195 ft 9 3/4 in. and 206 ft 0 in.	10.19	58.1	409	120	13.2	50.0	17.1
25	Southwest quadrant in pressurizer compartment between el 195 ft 9 3/4 in. and 306 ft 0 in.	10.19	62.0	445	120	13.2	50.0	17.2
26	Inside pressurizer compartment between el 185 ft 0 in. and 195 ft 9 3/4 in.	10.81	277.7	2419	120	13.2	50.0	20.7
27	Inside pressurizer compartment between el 171 ft 9 in. and 185 ft 0 in.	13.25	277.7	3496	120	13.2	50.0	6.51
28	Containment building free volume.	--	--	2.75E+06	120	13.2	50.0	--

a. Note that this value includes allowances for volumes occupied by large equipment as well as a 5 percent reduction to account for smaller objects.

b. Results are for a 308-in. surge line break except nodes 1 through 5, which are for a spray line break at the pressurizer nozzle.

TABLE 6.2.1-26 (SHEET 1 OF 7)

REACTOR CAVITY TIME DEPENDENT
FLOW CONDITIONS

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
0.00000,	0.0,	0.,
0.00250,	1.3586147E+4,	7.6261951E+6,
0.00500,	1.7940995E+4,	1.0071084E+7,
0.00752,	2.0648819E+4,	1.1591079E+7,
0.01001,	2.2913458E+4,	1.2857206E+7,
0.01251,	2.4062034E+4,	1.3493659E+7,
0.01501,	2.3856512E+4,	1.3362610E+7,
0.01751,	2.6598904E+4,	1.4911117E+7,
0.02003,	2.7096058E+4,	1.5177837E+7,
0.02253,	2.6427985E+4,	1.4785699E+7,
0.02507,	2.6171937E+4,	1.4631801E+7,
0.02756,	2.6566208E+4,	1.4848680E+7,
0.03005,	2.6676953E+4,	1.4905002E+7,
0.03252,	2.7098266E+4,	1.5139974E+7,
0.03510,	2.7742642E+4,	1.5502190E+7,
0.03751,	2.8169787E+4,	1.5741567E+7,
0.04005,	2.8575125E+4,	1.5968826E+7,
0.04257,	2.8940040E+4,	1.6173308E+7,
0.04508,	2.9015063E+4,	1.6212095E+7,
0.04753,	2.8845152E+4,	1.6111615E+7,
0.05000,	2.8677447E+4,	1.6013188E+7,
0.05251,	2.8523145E+4,	1.5923076E+7,
0.05510,	2.8416210E+4,	1.5860480E+7,
0.05752,	2.8345423E+4,	1.5818843E+7,
0.06003,	2.8188576E+4,	1.5728214E+7,
0.06257,	2.7840963E+4,	1.5529144E+7,
0.06505,	2.7390292E+4,	1.5271875E+7,
0.06759,	2.7024295E+4,	1.5063533E+7,
0.07005,	2.6934059E+4,	1.5012643E+7,
0.07261,	2.7117701E+4,	1.5117598E+7,
0.07501,	2.7372228E+4,	1.5262759E+7,
0.07759,	2.7520339E+4,	1.5346859E+7,
0.08005,	2.7439861E+4,	1.5300528E+7,
0.08255,	2.7153041E+4,	1.5136868E+7,
0.08502,	2.6724728E+4,	1.4893012E+7,
0.08761,	2.6197717E+4,	1.4593675E+7,
0.09002,	2.5681851E+4,	1.4301168E+7,
0.09253,	2.5203708E+4,	1.4030333E+7,
0.09502,	2.4835535E+4,	1.3822267E+7,
0.09754,	2.4578990E+4,	1.3677537E+7,
0.10013,	2.4419910E+4,	1.3588081E+7,
0.10253,	2.4355729E+4,	1.3552374E+7,

TABLE 6.2.1-26 (SHEET 2 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
0.10500,	2.4365128E+4,	1.3558416E+7,
0.10751,	2.4455309E+4,	1.3610180E+7,
0.11010,	2.4625670E+4,	1.3707282E+7,
0.11255,	2.4845949E+4,	1.3832514E+7,
0.11503,	2.5086307E+4,	1.3969057E+7,
0.12010,	2.5466453E+4,	1.4184861E+7,
0.12260,	2.5551384E+4,	1.4232951E+7,
0.12513,	2.5567152E+4,	1.4241650E+7,
0.12761,	2.5523282E+4,	1.4216391E+7,
0.13014,	2.5427294E+4,	1.4161608E+7,
0.13265,	2.5276425E+4,	1.4075754E+7,
0.13507,	2.5071667E+4,	1.3959456E+7,
0.13754,	2.4813584E+4,	1.3813099E+7,
0.14005,	2.4539029E+4,	1.3657649E+7,
0.14258,	2.4313941E+4,	1.3530409E+7,
0.14509,	2.4165958E+4,	1.3446888E+7,
0.14765,	2.4092923E+4,	1.3405844E+7,
0.15005,	2.4058579E+4,	1.3386724E+7,
0.15256,	2.4033247E+4,	1.3372693E+7,
0.15502,	2.3993516E+4,	1.3350446E+7,
0.15762,	2.3909192E+4,	1.3302862E+7,
0.16014,	2.3778645E+4,	1.3229134E+7,
0.16258,	2.3624748E+4,	1.3142275E+7,
0.16515,	2.3480926E+4,	1.3061202E+7,
0.16764,	2.3386045E+4,	1.3007825E+7,
0.17012,	2.3341080E+4,	1.2982715E+7,
0.17261,	2.3337016E+4,	1.2980780E+7,
0.17506,	2.3360386E+4,	1.2994312E+7,
0.17760,	2.3404110E+4,	1.3019393E+7,
0.18014,	2.3465021E+4,	1.3054108E+7,
0.18255,	2.3537716E+4,	1.3095460E+7,
0.18503,	2.3632725E+4,	1.3149368E+7,
0.18760,	2.3744212E+4,	1.3212592E+7,
0.19013,	2.3871936E+4,	1.3284937E+7,
0.19261,	2.4004296E+4,	1.3359858E+7,
0.19513,	2.4135817E+4,	1.3434247E+7,
0.19765,	2.4247269E+4,	1.3497285E+7,
0.20012,	2.4332207E+4,	1.3545267E+7,
0.20255,	2.4395464E+4,	1.3580953E+7,
0.20503,	2.4443214E+4,	1.3607826E+7,
0.20755,	2.4475122E+4,	1.3625711E+7,
0.21005,	2.4480973E+4,	1.3628774E+7,
0.21252,	2.4448240E+4,	1.3609927E+7,
0.21513,	2.4370781E+4,	1.3565733E+7,
0.21757,	2.4275725E+4,	1.3511689E+7,

TABLE 6.2.1-26 (SHEET 3 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
0.22002,	2.4182117E+4,	1.3458565E+7,
0.22262,	2.4107319E+4,	1.3416180E+7,
0.22513,	2.4069692E+4,	1.3394923E+7,
0.22754,	2.4059667E+4,	1.3389325E+7,
0.23021,	2.4062445E+4,	1.3390990E+7,
0.23263,	2.4067014E+4,	1.3393646E+7,
0.23504,	2.4070423E+4,	1.3395637E+7,
0.23752,	2.4078331E+4,	1.3400188E+7,
0.24011,	2.4100562E+4,	1.3412891E+7,
0.24259,	2.4140469E+4,	1.3435579E+7,
0.24509,	2.4200933E+4,	1.3469938E+7,
0.24750,	2.4265283E+4,	1.3506417E+7,
0.25003,	2.4321768E+4,	1.3538422E+7,
0.25261,	2.4353615E+4,	1.3556377E+7,
0.25505,	2.4344223E+4,	1.3550864E+7,
0.25757,	2.4290066E+4,	1.3519957E+7,
0.26001,	2.4201115E+4,	1.3459386E+7,
0.26258,	2.4078015E+4,	1.3399541E+7,
0.26505,	2.3945320E+4,	1.3324386E+7,
0.26754,	2.3810797E+4,	1.3248271E+7,
0.27004,	2.3687905E+4,	1.3178772E+7,
0.27253,	2.3586351E+4,	1.3121419E+7,
0.27511,	2.3502079E+4,	1.3073899E+7,
0.27752,	2.3455387E+4,	1.3047656E+7,
0.28006,	2.3438152E+4,	1.3038153E+7,
0.28252,	2.3453951E+4,	1.3047366E+7,
0.28511,	2.3503649E+4,	1.3075784E+7,
0.28760,	2.3577850E+4,	1.3117988E+7,
0.29004,	2.3669172E+4,	1.3169845E+7,
0.29253,	2.3774953E+4,	1.3229813E+7,
0.29506,	2.3886183E+4,	1.3292821E+7,
0.29756,	2.3996260E+4,	1.3355146E+7,
0.30005,	2.4097381E+4,	1.3412378E+7,
0.30261,	2.4184345E+4,	1.3461533E+7,
0.30501,	2.4247804E+4,	1.3497363E+7,
0.30766,	2.4289171E+4,	1.3520597E+7,
0.31011,	2.4303098E+4,	1.3528275E+7,
0.31262,	2.4292119E+4,	1.3521774E+7,
0.31508,	2.4261540E+4,	1.3504208E+7,
0.31755,	2.4219339E+4,	1.3480093E+7,
0.32003,	2.4174469E+4,	1.3454522E+7,
0.32255,	2.4138798E+4,	1.3434171E+7,
0.32507,	2.4117832E+4,	1.3422282E+7,
0.32755,	2.4111291E+4,	1.3418580E+7,
0.33000,	2.4108026E+4,	1.3416724E+7,

TABLE 6.2.1-26 (SHEET 4 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
0.33255,	2.4097899E+4,	1.3410959E+7,
0.33503,	2.4079157E+4,	1.3400306E+7,
0.33753,	2.4059258E+4,	1.3389024E+7,
0.34000,	2.4048690E+4,	1.3383058E+7,
0.34262,	2.4055783E+4,	1.3387132E+7,
0.34511,	2.4075206E+4,	1.3398197E+7,
0.34758,	2.4094905E+4,	1.3409386E+7,
0.35002,	2.4103796E+4,	1.3414398E+7,
0.35256,	2.4096385E+4,	1.3410127E+7,
0.35504,	2.4075163E+4,	1.3398034E+7,
0.35762,	2.4045807E+4,	1.3381330E+7,
0.36008,	2.4017017E+4,	1.3364990E+7,
0.36254,	2.3991591E+4,	1.3350596E+7,
0.36508,	2.3972836E+4,	1.3340000E+7,
0.36762,	2.3966208E+4,	1.3336299E+7,
0.37000,	2.3971003E+4,	1.3339075E+7,
0.37253,	2.3979807E+4,	1.3344103E+7,
0.37510,	2.3980967E+4,	1.3344761E+7,
0.37760,	2.3966909E+4,	1.3336760E+7,
0.38008,	2.3941814E+4,	1.3322495E+7,
0.38251,	2.3916742E+4,	1.3308297E+7,
0.38502,	2.3903255E+4,	1.3300691E+7,
0.38579,	2.3908591E+4,	1.3303782E+7,
0.39003,	2.3926500E+4,	1.3313992E+7,
0.39251,	2.3950989E+4,	1.3327913E+7,
0.39509,	2.3969493E+4,	1.3338405E+7,
0.39751,	2.3976395E+4,	1.3342291E+7,
0.40006,	2.3979176E+4,	1.3343952E+7,
0.40261,	2.3986081E+4,	1.3347733E+7,
0.40508,	2.3999851E+4,	1.3355527E+7,
0.40770,	2.4020182E+4,	1.3367041E+7,
0.41005,	2.4041523E+4,	1.3379124E+7,
0.41280,	2.4061453E+4,	1.3390382E+7,
0.41514,	2.4079853E+4,	1.3400786E+7,
0.41763,	2.4100740E+4,	1.3412580E+7,
0.42010,	2.4144685E+4,	1.3426111E+7,
0.42256,	2.4148532E+4,	1.3439591E+7,
0.42511,	2.4166027E+4,	1.3449429E+7,
0.42760,	2.4173080E+4,	1.3453348E+7,
0.43009,	2.4172381E+4,	1.3452864E+7,
0.43256,	2.4169703E+4,	1.3451262E+7,
0.43504,	2.4168789E+4,	1.3450676E+7,
0.43778,	2.4169276E+4,	1.3450904E+7,
0.44017,	2.4165751E+4,	1.3448854E+7,
0.44263,	2.4154330E+4,	1.3442313E+7,

TABLE 6.2.1-26 (SHEET 5 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
0.44509,	2.4135122E+4,	1.3431355E+7,
0.44753,	2.4113758E+4,	1.3419208E+7,
0.45012,	2.4094979E+4,	1.3408512E+7,
0.45263,	2.4084151E+4,	1.3402379E+7,
0.45510,	2.4078610E+4,	1.3399245E+7,
0.45766,	2.4072112E+4,	1.3395561E+7,
0.46015,	2.4061009E+4,	1.3389283E+7,
0.46261,	2.4044888E+4,	1.3380159E+7,
0.46507,	2.4028418E+4,	1.3370837E+7,
0.46754,	2.4014861E+4,	1.3363172E+7,
0.47015,	2.4007389E+4,	1.3358971E+7,
0.47259,	2.4007662E+4,	1.3359161E+7,
0.47505,	2.4013697E+4,	1.3362618E+7,
0.47760,	2.4023721E+4,	1.3368333E+7,
0.48009,	2.4036380E+4,	1.3375569E+7,
0.48262,	2.4052248E+4,	1.3384600E+7,
0.48509,	2.4074545E+4,	1.3397290E+7,
0.48760,	2.4107246E+4,	1.3415863E+7,
0.49013,	2.4148401E+4,	1.3439192E+7,
0.49260,	2.4192806E+4,	1.3464343E+7,
0.49504,	2.4232365E+4,	1.3486726E+7,
0.49750,	2.4262844E+4,	1.3503931E+7,
0.50009,	2.4280483E+4,	1.3513829E+7,
0.51004,	2.4229105E+4,	1.3484245E+7,
0.52019,	2.4120069E+4,	1.3422231E+7,
0.53012,	2.3944743E+4,	1.3322982E+7,
0.54004,	2.3980354E+4,	1.3343661E+7,
0.55002,	2.4145342E+4,	1.3437383E+7,
0.56008,	2.4221418E+4,	1.3480274E+7,
0.57014,	2.4209329E+4,	1.3473144E+7,
0.58011,	2.4144979E+4,	1.3436505E+7,
0.59000,	2.4107243E+4,	1.3415227E+7,
0.60006,	2.4159685E+4,	1.3445105E+7,
0.61010,	2.4217996E+4,	1.3478169E+7,
0.62013,	2.4261479E+4,	1.3502731E+7,
0.63013,	2.4256432E+4,	1.3499721E+7,
0.64003,	2.4203746E+4,	1.3469774E+7,
0.65008,	2.4162745E+4,	1.3446618E+7,
0.66009,	2.4159154E+4,	1.3444706E+7,
0.67000,	2.4187283E+4,	1.3460781E+7,
0.68000,	2.4231721E+4,	1.3486025E+7,
0.69011,	2.4263450E+4,	1.3503981E+7,
0.70012,	2.4273260E+4,	1.3509453E+7,
0.71012,	2.4252390E+4,	1.3503212E+7,
0.72015,	2.4246713E+4,	1.3494349E+7,

TABLE 6.2.1-26 (SHEET 6 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
0.73014,	2.4255830E+4,	1.3499597E+7,
0.74019,	2.4284243E+4,	1.3515760E+7,
0.75002,	2.4295893E+4,	1.3522331E+7,
0.76012,	2.4285562E+4,	1.3516424E+7,
0.77013,	2.4270343E+4,	1.3507794E+7,
0.78001,	2.4266732E+4,	1.3505797E+7,
0.79003,	2.4279302E+4,	1.3513005E+7,
0.80011,	2.4295891E+4,	1.3522447E+7,
0.81004,	2.4310716E+4,	1.3530822E+7,
0.82001,	2.4318677E+4,	1.3535287E+7,
0.83012,	2.4318126E+4,	1.3534984E+7,
0.84019,	2.4323296E+4,	1.3537931E+7,
0.85007,	2.4333920E+4,	1.3543954E+7,
0.86001,	2.4343134E+4,	1.3549166E+7,
0.87001,	2.4345149E+4,	1.3550301E+7,
0.88004,	2.4338762E+4,	1.3546668E+7,
0.89006,	2.4335464E+4,	1.3544826E+7,
0.90005,	2.4335806E+4,	1.3545058E+7,
0.91005,	2.4340583E+4,	1.3547769E+7,
0.92020,	2.4348287E+4,	1.3552160E+7,
0.93008,	2.4355085E+4,	1.3556021E+7,
0.94000,	2.4360745E+4,	1.3559249E+7,
0.95007,	2.4367196E+4,	1.3562914E+7,
0.96009,	2.4374109E+4,	1.3566815E+7,
0.97015,	2.4378996E+4,	1.3569608E+7,
0.98005,	2.4379731E+4,	1.3570001E+7,
0.99007,	2.4378418E+4,	1.3569255E+7,
1.00010,	2.4375674E+4,	1.3567753E+7,
1.05019,	2.4380088E+4,	1.3570353E+7,
1.10005,	2.4382999E+4,	1.3572130E+7,
1.15002,	2.4383322E+4,	1.3572555E+7,
1.20009,	2.4377813E+4,	1.3569670E+7,
1.25007,	2.4374905E+4,	1.3568424E+7,
1.30007,	2.4378444E+4,	1.3570840E+7,
1.35018,	2.4377156E+4,	1.3570652E+7,
1.40010,	2.4370716E+4,	1.3567578E+7,
1.45004,	2.4382101E+4,	1.3574759E+7,
1.50024,	2.4361620E+4,	1.3563940E+7,
1.55003,	2.4349789E+4,	1.3558156E+7,
1.60012,	2.4323449E+4,	1.3544249E+7,
1.65001,	2.4297197E+4,	1.3530488E+7,
1.70009,	2.4265055E+4,	1.3513472E+7,
1.75003,	2.4230524E+4,	1.3495182E+7,
1.80007,	2.4195149E+4,	1.3476460E+7,
1.85006,	2.4154185E+4,	1.3454644E+7,

TABLE 6.2.1-26 (SHEET 7 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
1.90005,	2.4110518E+4,	1.3431345E+7,
1.95001,	2.4061980E+4,	1.3405353E+7,
2.00002,	2.4006998E+4,	1.3375804E+7,
10.0000,	2.4006998E+4,	1.3375804E+7,

TABLE 6.2.1-26A

STEAM GENERATOR COMPARTMENT TIME DEPENDENT FLOW CONDITIONS
(16 IN. FEEDWATER LINE CONDITION)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
0.0	1.3371E4	425.11
5.0	1.3371E4	425.11

TABLE 6.2.1-27 (SHEET 1 OF 36)

STEAM GENERATOR COMPARTMENT TIME DEPENDENT FLOW CONDITIONS
(236-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.00000,	1.0904410E+04,	6.0480220E+06,
.00101,	2.3103719E+04,	1.2714696E+07,
.00202,	2.2381056E+04,	1.2345418E+07,
.00301,	2.5906342E+04,	1.4299363E+07,
.00402,	2.5436400E+04,	1.4035341E+07,
.00501,	2.5656111E+04,	1.4159121E+07,
.00602,	2.5357243E+04,	1.3989857E+07,
.00701,	2.4931699E+04,	1.3752451E+07,
.00800,	2.4571808E+04,	1.3551184E+07,
.00904,	2.4187011E+04,	1.3336488E+07,
.01002,	2.3880219E+04,	1.3165588E+07,
.01103,	2.3651012E+04,	1.3037901E+07,
.01202,	2.3496965E+04,	1.2952142E+07,
.01303,	2.3396491E+04,	1.2896245E+07,
.01401,	2.3336370E+04,	1.2862813E+07,
.01500,	2.3332478E+04,	1.2843994E+07,
.01602,	2.3286653E+04,	1.2835217E+07,
.01702,	2.3282699E+04,	1.2833050E+07,
.01801,	2.3290222E+04,	1.2837292E+07,
.01901,	2.3318349E+04,	1.2853066E+07,
.02001,	2.3382726E+04,	1.2889117E+07,
.02100,	2.3507313E+04,	1.2958857E+07,
.02200,	2.3716174E+04,	1.3075720E+07,
.02301,	2.4020877E+04,	1.3246185E+07,
.02402,	2.4407558E+04,	1.3462450E+07,
.02502,	2.4805974E+04,	1.3685261E+07,
.02603,	2.5179215E+04,	1.3893997E+07,
.02702,	2.5476048E+04,	1.4059991E+07,
.02801,	2.5688755E+04,	1.4178854E+07,
.02900,	2.5815616E+04,	1.4249685E+07,
.03001,	2.5872569E+04,	1.4281352E+07,
.03100,	2.5871120E+04,	1.4280349E+07,
.03202,	2.5831980E+04,	1.4258255E+07,
.03300,	2.5797867E+04,	1.4238755E+07,
.03403,	2.5729553E+04,	1.4200369E+07,
.03502,	2.5666245E+04,	1.4164838E+07,
.03601,	2.5615037E+04,	1.4136130E+07,
.03701,	2.5584268E+04,	1.4118888E+07,
.03802,	2.5576045E+04,	1.4114292E+07,
.03900,	2.5593825E+04,	1.4124256E+07,

TABLE 6.2.1-27 (SHEET 2 OF 36)

(236-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.04000,	2.5636989E+04,	1.4148441E+07,
.04101,	2.5709142E+04,	1.4189612E+07,
.04201,	2.5803415E+04,	1.4242315E+07,
.04300,	3.9073270E+04,	2.1622832E+07,
.04400,	3.6123909E+04,	1.9959244E+07,
.04500,	3.7514640E+04,	2.0734863E+07,
.04600,	3.7123722E+04,	2.0519158E+07,
.04701,	3.7595199E+04,	2.0781432E+07,
.04801,	3.7990734E+04,	2.1004792E+07,
.04901,	3.8569457E+04,	2.1328762E+07,
.05000,	3.9090779E+04,	2.1621351E+07,
.05101,	3.9522644E+04,	2.1863598E+07,
.05203,	3.9924193E+04,	2.2089344E+07,
.05301,	4.0328238E+04,	2.2316935E+07,
.05402,	4.0780930E+04,	2.2571622E+07,
.05502,	4.1058245E+04,	2.2726524E+07,
.05602,	4.1028097E+04,	2.2707867E+07,
.05701,	4.0827711E+04,	2.2594210E+07,
.05801,	4.0614117E+04,	2.2473446E+07,
.05902,	4.0306748E+04,	2.2298999E+07,
.06005,	3.9705141E+04,	2.1958902E+07,
.06101,	3.9019798E+04,	2.1573731E+07,
.06203,	3.8524030E+04,	2.1296608E+07,
.06306,	3.8264535E+04,	2.1151257E+07,
.06402,	3.8043324E+04,	2.1027233E+07,
.06506,	3.7528857E+04,	2.0963996E+07,
.06609,	3.8047881E+04,	2.1031529E+07,
.06710,	3.8248523E+04,	2.1144413E+07,
.06810,	3.8431685E+04,	2.1247369E+07,
.06904,	3.8577465E+04,	2.1329199E+07,
.07010,	3.8681749E+04,	2.1387566E+07,
.07103,	3.8716321E+04,	2.1406737E+07,
.07203,	3.8694016E+04,	2.1393940E+07,
.07308,	3.8610314E+04,	2.1346583E+07,
.07404,	3.8477661E+04,	2.1271803E+07,
.07508,	3.8286475E+04,	2.1164259E+07,
.07601,	3.8094753E+04,	2.1056554E+07,
.07705,	3.7869887E+04,	2.0930259E+07,
.07808,	3.7625546E+04,	2.0793040E+07,
.07903,	3.7377536E+04,	2.0653886E+07,
.08003,	3.7108876E+04,	2.0503174E+07,
.08109,	3.6815615E+04,	2.0338789E+07,

TABLE 6.2.1-27 (SHEET 3 OF 36)

(236-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.08203,	3.6570906E+04,	2.0201650E+07,
.08307,	3.6322997E+04,	2.0062868E+07,
.08403,	3.6146874E+04,	1.9964492E+07,
.08504,	3.6011558E+04,	1.9888990E+07,
.08601,	3.5921555E+04,	1.9838855E+07,
.08704,	3.5854749E+04,	1.9801639E+07,
.08815,	3.5799896E+04,	1.9771041E+07,
.08902,	3.5757645E+04,	1.9747418E+07,
.09014,	3.5703170E+04,	1.9716932E+07,
.09105,	3.5652585E+04,	1.9688585E+07,
.09211,	3.5585871E+04,	1.9651180E+07,
.09305,	3.5518181E+04,	1.9613209E+07,
.09403,	3.5440371E+04,	1.9569576E+07,
.09505,	3.5365526E+04,	1.9527655E+07,
.09610,	3.5293384E+04,	1.9487297E+07,
.09704,	3.5246775E+04,	1.9461321E+07,
.09814,	3.5215885E+04,	1.9444199E+07,
.09909,	3.5206027E+04,	1.9438841E+07,
.10014,	3.5208081E+04,	1.9440142E+07,
.10502,	3.5235098E+04,	1.9455575E+07,
.11015,	3.5216755E+04,	1.9445382E+07,
.11517,	3.5019113E+04,	1.9334344E+07,
.12014,	3.4879507E+04,	1.9256199E+07,
.12506,	3.4768055E+04,	1.9193846E+07,
.13009,	3.4649578E+04,	1.9127645E+07,
.13501,	3.4580271E+04,	1.9389133E+07,
.14013,	3.4693845E+04,	1.9153331E+07,
.14508,	3.4881703E+04,	1.9258839E+07,
.15008,	3.4994292E+04,	1.9322028E+07,
.15511,	3.4990962E+04,	1.9320158E+07,
.16007,	3.4968263E+04,	1.9307534E+07,
.16503,	3.4964778E+04,	1.9305667E+07,
.17010,	3.4993582E+04,	1.9321916E+07,
.17505,	3.5092374E+04,	1.9377433E+07,
.18005,	3.5274019E+04,	1.9479423E+07,
.18511,	3.5445015E+04,	1.9575282E+07,
.19012,	3.5513330E+04,	1.9613385E+07,
.19511,	3.5476643E+04,	1.9592519E+07,
.20000,	3.5389047E+04,	1.9543169E+07,
.21012,	3.5255198E+04,	1.9468035E+07,
.22007,	3.5256683E+04,	1.9468911E+07,

TABLE 6.2.1-27 (SHEET 4 OF 36)

(236-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.23004,	3.5231527E+04,	1.9454787E+07,
.24009,	3.5109052E+04,	1.9386133E+07,
.25004,	3.5068150E+04,	1.9363427E+07,
.26004,	3.5121658E+04,	1.9393549E+07,
.27000,	3.5159026E+04,	1.9414539E+07,
.28002,	3.5246849E+04,	1.9463869E+07,
.29009,	3.5606682E+04,	1.9665651E+07,
.30013,	3.6110831E+04,	1.9947367E+07,
.31010,	3.5883817E+04,	1.9819724E+07,
.32015,	3.5731357E+04,	1.9733879E+07,
.33008,	3.5406465E+04,	1.9552039E+07,
.34008,	3.5403076E+04,	1.9550425E+07,
.35014,	3.5428013E+04,	1.9564578E+07,
.36003,	3.5367341E+04,	1.9530647E+07,
.37004,	3.5316627E+04,	1.9502441E+07,
.38005,	3.5402063E+04,	1.9550400E+07,
.39001,	3.5506771E+04,	1.9609110E+07,
.40001,	3.5628132E+04,	1.9677021E+07,
.41008,	3.5647173E+04,	1.9687547E+07,
.42013,	3.5587259E+04,	1.9653881E+07,
.43005,	3.5518685E+04,	1.9615449E+07,
.44011,	3.5458889E+04,	1.9582016E+07,
.45002,	3.5423570E+04,	1.9562332E+07,
.46007,	3.5389368E+04,	1.9543272E+07,
.47014,	3.5360551E+04,	1.9527243E+07,
.48003,	3.5344742E+04,	1.9518465E+07,
.49004,	3.5331713E+04,	1.9511278E+07,
.50014,	3.5464944E+04,	1.9586157E+07,
.51004,	3.5455419E+04,	1.9580555E+07,
.52016,	3.5171002E+04,	1.9421101E+07,
.53013,	3.4956311E+04,	1.9301100E+07,
.54020,	3.4967429E+04,	1.9307806E+07,
.55003,	3.5173577E+04,	1.9423487E+07,
.56006,	3.5249947E+04,	1.9466290E+07,
.57007,	3.5274526E+04,	1.9480020E+07,
.58008,	3.5192533E+04,	1.9434100E+07,
.59009,	3.5173286E+04,	1.9423484E+07,
.60004,	3.5207910E+04,	1.9443066E+07,
.61004,	3.5263128E+04,	1.9474124E+07,
.62003,	3.5289369E+04,	1.9488901E+07,
.63008,	3.5280693E+04,	1.9484094E+07,

TABLE 6.2.1-27 (SHEET 5 OF 36)

(236-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.64011,	3.5205381E+04,	1.9441981E+07,
.65018,	3.5185255E+04,	1.9430942E+07,
.66010,	3.5247094E+04,	1.9465823E+07,
.67011,	3.5294402E+04,	1.9492475E+07,
.68006,	3.5270057E+04,	1.9478927E+07,
.69010,	3.5224067E+04,	1.9453332E+07,
.70027,	3.5205546E+04,	1.9443190E+07,
.71002,	3.5205036E+04,	1.9443150E+07,
.72011,	3.5204349E+04,	1.9443007E+07,
.73008,	3.5216969E+04,	1.9450347E+07,
.74015,	3.5248956E+04,	1.9468535E+07,
.75007,	3.5278490E+04,	1.9485325E+07,
.76011,	3.5294514E+04,	1.9494545E+07,
.77010,	3.5324933E+04,	1.9511865E+07,
.78003,	3.5323636E+04,	1.9511366E+07,
.80009,	3.5349418E+04,	1.9526430E+07,
.81007,	3.5364223E+04,	1.9535023E+07,
.82009,	3.5337390E+04,	1.9520270E+07,
.83007,	3.5315749E+04,	1.9508490E+07,
.84002,	3.5304547E+04,	1.9502577E+07,
.85010,	3.5315673E+04,	1.9509200E+07,
.86007,	3.5398275E+04,	1.9555915E+07,
.87000,	3.5422401E+04,	1.9569725E+07,
.88005,	3.5471641E+04,	1.9597672E+07,
.89000,	3.5514959E+04,	1.9622268E+07,
.90011,	3.5544815E+04,	1.9639309E+07,
.91008,	3.5574072E+04,	1.9656051E+07,
.92015,	3.5609336E+04,	1.9676169E+07,
.93004,	3.5644462E+04,	1.9696218E+07,
.94007,	3.5681054E+04,	1.9717092E+07,
.95006,	3.5708010E+04,	1.9732566E+07,
.96004,	3.5732460E+04,	1.9746650E+07,
.97008,	3.5767399E+04,	1.9766649E+07,
.98003,	3.5808071E+04,	1.9789841E+07,
.99019,	3.5852729E+04,	1.9815310E+07,
1.00012,	3.5895623E+04,	1.9839774E+07,
1.05008,	3.6112647E+04,	1.9963381E+07,
1.10007,	3.6338443E+04,	2.0092128E+07,
1.15002,	3.6608642E+04,	2.0245910E+07,
1.20007,	3.6888439E+04,	2.0405162E+07,
1.25001,	3.7127241E+04,	2.0541579E+07,

TABLE 6.2.1-27 (SHEET 6 OF 36)

(236-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
1.30009,	3.7127043E+04,	2.0656265E+07,
1.35002,	3.7484280E+04,	2.0747071E+07,
1.40001,	3.7596272E+04,	2.0812441E+07,
1.45005,	3.7694652E+04,	2.0870059E+07,
1.50008,	3.7762278E+04,	2.0910243E+07,
1.55002,	3.7934417E+04,	2.1009055E+07,
1.60009,	3.7995308E+04,	2.1045152E+07,
1.65015,	3.8086014E+04,	2.1098323E+07,
1.70001,	3.8172222E+04,	2.1148978E+07,
1.73004,	3.8252964E+04,	2.1196588E+07,
1.80009,	3.8321222E+04,	2.1237084E+07,
1.85004,	3.8383561E+04,	2.1274274E+07,
1.90007,	3.8435015E+04,	2.1305277E+07,
1.95004,	3.8437624E+04,	2.1308836E+07,
2.00011,	3.8421768E+04,	2.1302004E+07,
2.05008,	3.8369901E+04,	2.1274851E+07,
2.10006,	3.8189733E+04,	2.1175694E+07,
2.15010,	3.8119992E+04,	2.1139027E+07,
2.20004,	3.7912908E+04,	2.1024926E+07,
2.25002,	3.7738727E+04,	2.0929322E+07,
2.30000,	3.7497730E+04,	2.0796234E+07,
2.35011,	3.7281923E+04,	2.0677130E+07,
2.40001,	3.7071816E+04,	2.0561347E+07,
2.45002,	3.7082174E+04,	2.0570251E+07,
2.50011,	3.6948336E+04,	2.0497962E+07,
2.55008,	3.6903695E+04,	2.0476484E+07,
2.60014,	3.6908560E+04,	2.0483188E+07,
2.65001,	3.6817126E+04,	2.0435988E+07,
2.70001,	3.6667266E+04,	2.0355691E+07,
2.75018,	3.6307184E+04,	2.0157719E+07,
2.80003,	3.5923918E+04,	1.9946870E+07,
2.85001,	3.5715709E+04,	1.9835248E+07,
2.90018,	3.5330716E+04,	1.9623353E+07,
2.95009,	3.4981705E+04,	1.9432201E+07,
3.00005,	3.4718072E+04,	1.9289429E+07,

TABLE 6.2.1-27 (SHEET 7 OF 36)

(306-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.00000,	1.0904418E+04,	6.9546146E+06,
.00101,	2.6783296E+04,	1.7817432E+07,
.00201,	2.6059905E+04,	1.6564051E+07,
.00301,	2.6630723E+04,	1.6923731E+07,
.00401,	2.6299962E+04,	1.6714564E+07,
.00500,	2.6431184E+04,	1.6797431E+07,
.00602,	2.6349622E+04,	1.6749019E+07,
.00700,	2.6305987E+04,	1.6768774E+07,
.00802,	2.6365680E+04,	1.6755933E+07,
.00903,	2.6380583E+04,	1.6765424E+07,
.01003,	2.6309531E+04,	1.6771220E+07,
.01101,	2.6428212E+04,	1.6748873E+07,
.01203,	2.6455432E+04,	1.6813421E+07,
.01302,	2.6431622E+04,	1.6830145E+07,
.01403,	2.6494736E+04,	1.6830535E+07,
.01502,	2.6507040E+04,	1.6846432E+07,
.01605,	2.6532537E+04,	1.6862820E+07,
.01703,	2.6614296E+04,	1.6919424E+07,
.01804,	2.6873875E+04,	1.7032259E+07,
.01901,	2.7467003E+04,	1.7463110E+07,
.02004,	2.8525431E+04,	1.8142452E+07,
.02103,	2.9703593E+04,	1.8893017E+07,
.02204,	3.0491958E+04,	1.9401975E+07,
.02304,	3.0380016E+04,	1.9461919E+07,
.02403,	3.0268393E+04,	1.9254922E+07,
.02502,	2.9002195E+04,	1.9006799E+07,
.02607,	2.9394309E+04,	1.8822112E+07,
.02707,	2.9400495E+04,	1.8702756E+07,
.02807,	2.9276976E+04,	1.8618401E+07,
.02906,	2.9210971E+04,	1.9581326E+07,
.03005,	2.9233801E+04,	1.8590523E+07,
.03107,	2.9300991E+04,	1.8689301E+07,
.03207,	2.9433417E+04,	1.8719351E+07,
.03305,	2.9508009E+04,	1.8818687E+07,
.03400,	2.9757005E+04,	1.8927234E+07,
.03507,	2.9977287E+04,	1.9068795E+07,
.03603,	3.0194624E+04,	1.9208524E+07,
.03700,	3.0444230E+04,	1.9369022E+07,
.03803,	3.0748400E+04,	1.9364717E+07,
.03906,	3.1197470E+04,	1.9795836E+07,
.04000,	3.1497312E+04,	2.0047030E+07,

TABLE 6.2.1-27 (SHEET 8 OF 36)

(306-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.04100,	3.1024106E+04,	2.0256915E+07,
.04205,	3.2108183E+04,	2.0435911E+07,
.04305,	3.2203201E+04,	2.0499450E+07,
.04406,	3.2156609E+04,	2.0468593E+07,
.04507,	3.2011837E+04,	2.8374677E+07,
.04602,	3.1845034E+04,	2.0266726E+07,
.04700,	3.1651549E+04,	2.0141733E+07,
.04808,	3.1498506E+04,	2.0042877E+07,
.04909,	3.1359642E+04,	1.9953230E+07,
.05001,	3.1253206E+04,	1.9004520E+07,
.05109,	3.1149257E+04,	1.9817440E+07,
.05206,	3.1072705E+04,	1.9760037E+07,
.05300,	3.1004020E+04,	1.9723740E+07,
.05401,	3.0954130E+04,	1.9691557E+07,
.05501,	3.0912015E+04,	1.9664420E+07,
.05611,	3.0660596E+04,	1.9644190E+07,
.05711,	3.0067966E+04,	1.9626102E+07,
.05811,	3.0871682E+04,	1.9630555E+07,
.05914,	3.0091361E+04,	1.9651626E+07,
.06007,	3.0920740E+04,	1.9670236E+07,
.06130,	3.0951744E+04,	1.9690274E+07,
.06200,	3.0975342E+04,	1.9705433E+07,
.06308,	3.0977493E+04,	1.9706693E+07,
.06401,	3.0958439E+04,	1.9694300E+07,
.06504,	3.0916964E+04,	1.9667593E+07,
.06605,	3.0876672E+04,	1.9641563E+07,
.06700,	3.0837718E+04,	1.9616490E+07,
.06803,	3.0817279E+04,	1.9603306E+07,
.06903,	3.0811027E+04,	1.9599425E+07,
.07008,	3.0821414E+04,	1.9606100E+07,
.07106,	3.0043773E+04,	1.9620637E+07,
.07201,	3.0874702E+04,	1.9640660E+07,
.07206,	3.0915265E+04,	1.9666707E+07,
.07409,	3.0958322E+04,	1.9694572E+07,
.07507,	3.1001246E+04,	1.9722270E+07,
.07607,	3.1044705E+04,	1.9750317E+07,
.07715,	3.1000957E+04,	1.9770337E+07,
.07814,	3.1123120E+04,	1.9882146E+07,
.07900,	3.1157700E+04,	1.9823092E+07,
.08009,	3.1183245E+04,	1.9839430E+07,
.08107,	3.1200280E+04,	1.9850340E+07,

TABLE 6.2.1-27 (SHEET 9 OF 36)

(306-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.08206,	3.1203954E+04,	1.9852600E+07,
.08308,	3.1193945E+04,	1.9846042E+07,
.08404,	3.1170566E+04,	1.9838001E+07,
.08501,	3.1134922E+04,	1.9807821E+07,
.08601,	3.1087116E+04,	1.9776942E+07,
.08709,	3.1029503E+04,	1.9739000E+07,
.08804,	3.0974094E+04,	1.9704012E+07,
.08901,	3.0917695E+04,	1.9667644E+07,
.09089,	3.0853531E+04,	1.9626200E+07,
.09110,	3.8796140E+04,	1.9569278E+07,
.09201,	3.0743409E+04,	1.9555347E+07,
.09301,	3.0685573E+04,	1.9518020E+07,
.09409,	3.0624348E+04,	1.9478562E+07,
.09501,	3.0573333E+04,	1.9445674E+07,
.09600,	3.0516623E+04,	1.9489168E+07,
.09713,	3.0452875E+04,	1.9368057E+07,
.09814,	3.0400141E+04,	1.9334105E+07,
.09912,	3.0350250E+04,	1.9301972E+07,
.10012,	3.0300946E+04,	1.9270230E+07,
.10516,	3.0004490E+04,	1.9131006E+07,
.11013,	2.9947775E+04,	1.9043465E+07,
.11505,	2.9896911E+04,	1.9019056E+07,
.12002,	2.9827896E+04,	1.8967496E+07,
.12505,	2.9681357E+04,	1.8873476E+07,
.13003,	2.9477222E+04,	1.8742507E+07,
.13504,	2.9200877E+04,	1.8616757E+07,
.14007,	2.9125707E+04,	1.8517710E+07,
.14509,	2.8981100E+04,	1.8425514E+07,
.15006,	2.8811361E+04,	1.8217077E+07,
.15504,	2.8594534E+04,	1.8178459E+07,
.16002,	2.8344040E+04,	1.8010209E+07,
.16506,	2.8067722E+04,	1.7041673E+07,
.17004,	2.7764467E+04,	1.7647852E+07,
.17501,	2.7423570E+04,	1.7429901E+07,
.18001,	2.7069411E+04,	1.7283536E+07,
.18504,	2.6715030E+04,	1.6977152E+07,
.19002,	2.6493250E+04,	1.6778063E+07,
.19500,	2.6124672E+04,	1.6682797E+07,
.20018,	2.5841963E+04,	1.6419880E+07,
.21000,	2.5138580E+04,	1.5970348E+07,
.22013,	2.4375517E+04,	1.5482003E+07,

TABLE 6.2.1-27 (SHEET 10 OF 36)

(306-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.23016,	2.3823925E+04,	1.5130751E+07,
.24014,	2.3369722E+04,	1.4049362E+07,
.25007,	2.2929182E+04,	1.4559762E+07,
.26013,	2.2691429E+04,	1.4408545E+07,
.27011,	2.2497616E+04,	1.4205460E+07,
.28001,	2.2356313E+04,	1.4196081E+07,
.29006,	2.2180437E+04,	1.4004375E+07,
.30001,	2.2073196E+04,	1.4016719E+07,
.31009,	2.1944465E+04,	1.3935290E+07,
.32009,	2.1820878E+04,	1.3857087E+07,
.33014,	2.1706070E+04,	1.3734560E+07,
.34090,	2.1612880E+04,	1.3725889E+07,
.35014,	2.1490487E+04,	1.3648525E+07,
.36013,	2.1391424E+04,	1.3586161E+07,
.37011,	2.1260616E+04,	1.3583657E+07,
.38014,	2.1181301E+04,	1.3454095E+07,
.39882,	2.1112816E+04,	1.3411754E+07,
.40019,	2.1013401E+04,	1.3349771E+07,
.41010,	2.0924470E+04,	1.3294601E+07,
.42003,	2.0869091E+04,	1.3261205E+07,
.43008,	2.0889126E+04,	1.3224993E+07,
.44008,	2.0723134E+04,	1.3172146E+07,
.45083,	2.0620172E+04,	1.3113971E+07,
.46002,	2.0515965E+04,	1.3044064E+07,
.47010,	2.0475523E+04,	1.3920890E+07,
.48024,	2.0433069E+04,	1.2995489E+07,
.49003,	2.0389661E+04,	1.2971487E+07,
.50003,	2.0340102E+04,	1.2940015E+07,
.51002,	2.0306981E+04,	1.2924773E+07,
.52010,	2.0267614E+04,	1.2902050E+07,
.53003,	2.0235873E+04,	1.2885929E+07,
.54001,	2.0205732E+04,	1.2869978E+07,
.55014,	2.0176703E+04,	1.2854603E+07,
.56001,	2.0151311E+04,	1.2841861E+07,
.57001,	2.0126861E+04,	1.2020991E+07,
.58013,	2.0095633E+04,	1.2012343E+07,
.59003,	2.0062537E+04,	1.2794139E+07,
.60003,	2.0030307E+04,	1.2776336E+07,
.61809,	2.0002602E+04,	1.2761137E+07,
.62004,	1.9977344E+04,	1.2747300E+07,
.63007,	1.9955914E+04,	1.2735706E+07,

TABLE 6.2.1-27 (SHEET 11 OF 36)

(306-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.64002,	1.9937743E+04,	1.2723896E+07,
.65013,	1.9919953E+04,	1.2716322E+07,
.66007,	1.9892660E+04,	1.2649427E+07,
.67006,	1.9861153E+04,	1.2679012E+07,
.68007,	1.9846173E+04,	1.2678675E+07,
.69009,	1.9830751E+04,	1.2660895E+07,
.70009,	1.9816368E+04,	1.2651831E+07,
.71012,	1.9805505E+04,	1.2644140E+07,
.72004,	1.9795400E+04,	1.2637627E+07,
.73002,	1.9786282E+04,	1.2635125E+07,
.74012,	1.9778482E+04,	1.2623013E+07,
.75003,	1.9772434E+04,	1.2610364E+07,
.76010,	1.9767230E+04,	1.2613063E+07,
.77003,	1.9762966E+04,	1.2608096E+07,
.78023,	1.9759570E+04,	1.2603391E+07,
.79001,	1.9757060E+04,	1.2598978E+07,
.80001,	1.9753590E+04,	1.2594963E+07,
.81010,	1.9755388E+04,	1.2591514E+07,
.82005,	1.9755886E+04,	1.2588207E+07,
.83011,	1.9756346E+04,	1.2584745E+07,
.84102,	1.9756783E+04,	1.2583036E+07,
.85005,	1.9758443E+04,	1.2577762E+07,
.86809,	1.9763788E+04,	1.2576771E+07,
.87017,	1.9772159E+04,	1.2577829E+07,
.88007,	1.9780558E+04,	1.2578600E+07,
.89004,	1.9780392E+04,	1.2578731E+07,
.90001,	1.9797611E+04,	1.2580141E+07,
.91004,	1.9837670E+04,	1.2581783E+07,
.92005,	1.9817255E+04,	1.2583110E+07,
.93003,	1.9828011E+04,	1.2565229E+07,
.94097,	1.9841004E+04,	1.2563790E+07,
.95039,	1.9854040E+04,	1.2592694E+07,
.96003,	1.9869443E+04,	1.2597593E+07,
.97010,	1.9884994E+04,	1.2602041E+07,
.98004,	1.9899603E+04,	1.2607500E+07,
.99014,	1.9913539E+04,	1.2611643E+07,
1.00014,	1.9927001E+04,	1.2616035E+07,
1.05808,	2.0003035E+04,	1.2648624E+07,
1.10006,	2.0001506E+04,	1.2666790E+07,
1.15083,	2.0159717E+04,	1.2683963E+07,

TABLE 6.2.1-27 (SHEET 12 OF 36)

(306-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
1.23001,	2.0229722E+04,	1.2703628E+07,
1.25006,	2.0271963E+04,	1.2714947E+07,
1.30002,	2.0329809E+04,	1.2733996E+07,
1.35006,	2.0445870E+04,	1.2730469E+07,
1.40006,	2.0595390E+04,	1.2069199E+07,
1.45001,	2.0797697E+04,	1.2973763E+07,
1.50001,	2.2597603E+04,	1.4130098E+07,
1.55006,	2.1727914E+04,	1.3531673E+07,
1.60003,	2.2033142E+04,	1.3603610E+07,
1.65008,	2.2569311E+04,	1.3902740E+07,
1.70001,	2.7069624E+04,	1.6559945E+07,
1.75004,	2.9992324E+04,	1.8156965E+07,
1.83007,	3.1111373E+04,	1.8901704E+07,
1.85009,	3.8141272E+04,	1.8262443E+07,
1.90010,	3.8279306E+04,	1.8347891E+07,
1.95302,	3.8503591E+04,	1.8471000E+07,
2.00011,	3.8724291E+04,	1.8593102E+07,
2.05007,	3.1050715E+04,	1.8793001E+07,
2.10001,	3.1246237E+04,	1.8937338E+07,
2.15006,	3.1427902E+04,	1.9017557E+07,
2.20014,	3.1528101E+04,	1.9070014E+07,
2.25007,	3.1406946E+04,	1.8997013E+07,
2.30001,	3.1327452E+04,	1.9951002E+07,
2.35006,	3.1352832E+04,	1.8979205E+07,
2.40002,	3.1247389E+04,	1.8921466E+07,
2.45090,	3.1046631E+04,	1.8000368E+07,
2.50003,	3.0660505E+04,	1.8564644E+07,
2.55022,	3.0143320E+04,	1.8247507E+07,
2.60003,	2.9549638E+04,	1.7070145E+07,
2.65004,	2.8990574E+04,	1.7535716E+07,
2.70011,	2.8403195E+04,	1.7166309E+07,
2.75007,	2.8107991E+04,	1.6999214E+07,
2.80009,	2.7602489E+04,	1.6695309E+07,
2.85005,	2.7334948E+04,	1.6590445E+07,
2.90004,	2.7149613E+04,	1.6473829E+07,
2.95002,	2.6945061E+04,	1.6376299E+07,
3.00010,	2.6740150E+04,	1.6273687E+07,

TABLE 6.2.1-27 (SHEET 13 OF 36)

(336-in.^{2} CLOSURE WELD BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.00000,	1.0904410E+04,	6.0400220E+06,
.00102,	3.1315332E+04,	1.7242679E+07,
.00200,	2.7940955E+04,	1.5372641E+07,
.00300,	2.9058594E+04,	1.6005069E+07,
.00401,	2.9324630E+04,	1.6141002E+07,
.00501,	2.0732274E+04,	1.5817640E+07,
.00601,	2.9150326E+04,	1.6049104E+07,
.00703,	2.9082265E+04,	1.6010471E+07,
.00801,	2.9371131E+04,	1.6173599E+07,
.00901,	3.0350271E+04,	1.6720200E+07,
.01002,	3.2337123E+04,	1.7030637E+07,
.01102,	3.4808617E+04,	1.9207799E+07,
.01200,	3.5977677E+04,	1.9858237E+07,
.01306,	3.5252689E+04,	1.9437759E+07,
.01402,	3.4082369E+04,	1.8706010E+07,
.01503,	3.3472383E+04,	1.8449149E+07,
.01606,	3.3290402E+04,	1.8340323E+07,
.01703,	3.3200217E+04,	1.8298546E+07,
.01801,	3.3225546E+04,	1.8313396E+07,
.01900,	3.3391040E+04,	1.8406700E+07,
.02004,	3.3702763E+04,	1.8580301E+07,
.02101,	3.4041621E+04,	1.8769631E+07,
.02203,	3.4247080E+04,	1.8882856E+07,
.02386,	3.4120032E+04,	1.8810176E+07,
.02406,	3.3755622E+04,	1.8606049E+07,
.02503,	3.3380219E+04,	1.8397173E+07,
.02602,	3.3167359E+04,	1.8279444E+07,
.02705,	3.3119492E+04,	1.8253315E+07,
.02805,	3.3145091E+04,	1.8268110E+07,
.02901,	3.3190096E+04,	1.8292848E+07,
.03007,	3.3253045E+04,	1.8327994E+07,
.03101,	3.3313912E+04,	1.8361057E+07,
.03200,	3.3353834E+04,	1.8303881E+07,
.03302,	3.3343114E+04,	1.8377561E+07,
.03408,	3.3272446E+04,	1.8337869E+07,
.03507,	3.3183214E+04,	1.8200103E+07,
.03602,	3.3123953E+04,	1.8255267E+07,
.03707,	3.3116108E+04,	1.8251151E+07,
.03805,	3.3150502E+04,	1.8270439E+07,

TABLE 6.2.1-27 (SHEET 14 OF 36)

(336-in.^{2} CLOSURE WELD BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.03901,	3.3197302E+04,	1.8296529E+07,
.04005,	3.3242461E+04,	1.8321653E+07,
.04103,	3.3275084E+04,	1.8340223E+07,
.04206,	3.3299100E+04,	1.8353138E+07,
.04304,	3.3307218E+04,	1.8357510E+07,
.04401,	3.3505517E+04,	1.8470440E+07,
.04502,	3.4449731E+04,	1.8992921E+07,
.04601,	3.5262496E+04,	1.9440836E+07,
.04702,	3.5879893E+04,	1.9779923E+07,
.04806,	3.6236755E+04,	1.9972897E+07,
.04900,	3.6269583E+04,	1.9985931E+07,
.05004,	3.6060590E+04,	1.9867704E+07,
.05102,	3.5924401E+04,	1.9795340E+07,
.05201,	3.6126195E+04,	1.9912190E+07,
.05306,	3.6575781E+04,	2.0161982E+07,
.05406,	3.6849965E+04,	2.0310129E+07,
.05506,	3.6897666E+04,	2.0333794E+07,
.05602,	3.6882889E+04,	2.0331701E+07,
.05705,	3.6999489E+04,	2.0392556E+07,
.05805,	3.7219065E+04,	2.0515462E+07,
.05900,	3.7476960E+04,	2.0658312E+07,
.06006,	3.7707201E+04,	2.0794494E+07,
.06103,	3.7832928E+04,	2.0852950E+07,
.06203,	3.7925042E+04,	2.0904186E+07,
.06364,	3.8068676E+04,	2.0985074E+07,
.06406,	3.8264791E+04,	2.1094051E+07,
.06503,	3.8403280E+04,	2.1169090E+07,
.06604,	3.8394878E+04,	2.1161822E+07,
.06701,	3.8260995E+04,	2.1085868E+07,
.06800,	3.8071294E+04,	2.0980414E+07,
.06906,	3.7910358E+04,	2.0892004E+07,
.07002,	3.7831074E+04,	2.0848952E+07,
.07103,	3.7803983E+04,	2.0834629E+07,
.07206,	3.7811991E+04,	2.0839661E+07,
.07308,	3.7856688E+04,	2.0865381E+07,
.07402,	3.7957543E+04,	2.0922723E+07,
.07507,	3.8165873E+04,	2.1040090E+07,
.07608,	3.8451658E+04,	2.1200016E+07,
.07702,	3.8764481E+04,	2.1374287E+07,
.07808,	3.9132485E+04,	2.1579074E+07,

TABLE 6.2.1-27 (SHEET 15 OF 36)

(336-in.^{2} CLOSURE WELD BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.07906,	3.9492063E+04,	2.1779484E+07,
.08001,	3.9879446E+04,	2.1995774E+07,
.08103,	4.0353849E+04,	2.2260672E+07,
.08211,	4.0908980E+04,	2.2570087E+07,
.08308,	4.1412318E+04,	2.2850120E+07,
.26011,	4.0316515E+04,	2.2239430E+07,
.27014,	4.0058097E+04,	2.2097363E+07,
.28016,	3.9818692E+04,	2.1967568E+07,
.29008,	3.9879564E+04,	2.2005338E+07,
.30006,	4.0093220E+04,	2.2127686E+07,
.31003,	4.0207611E+04,	2.2194375E+07,
.32005,	4.0201445E+04,	2.2194242E+07,
.33030,	4.0216076E+04,	2.2206028E+07,
.34000,	4.0219714E+04,	2.2211666E+07,
.35014,	4.0145695E+04,	2.2174008E+07,
.36008,	3.9994375E+04,	2.2093549E+07,
.37012,	3.9893347E+04,	2.2041597E+07,
.38021,	3.9907754E+04,	2.2054342E+07,
.39002,	3.9952948E+04,	2.2083866E+07,
.40002,	3.9935496E+04,	2.2078383E+07,
.41003,	3.9923464E+04,	2.2076291E+07,
.42007,	3.9978870E+04,	2.2111915E+07,
.43000,	4.0047123E+04,	2.2154536E+07,
.44009,	4.0054216E+04,	2.2163031E+07,
.45011,	4.0115169E+04,	2.2202630E+07,
.46005,	4.0211520E+04,	2.2260985E+07,
.47009,	4.0184188E+04,	2.2250703E+07,
.48008,	4.0117447E+04,	2.2218402E+07,
.49011,	4.0013181E+04,	2.2165700E+07,
.50003,	4.0008453E+04,	2.2168665E+07,
.51004,	4.0050068E+04,	2.2197559E+07,
.52010,	4.0110299E+04,	2.2236690E+07,
.53007,	4.0097367E+04,	2.2234944E+07,
.54009,	4.0068467E+04,	2.2224382E+07,
.55008,	4.0018860E+04,	2.2202150E+07,
.56003,	3.9933661E+04,	2.2160059E+07,
.57006,	3.9836338E+04,	2.2111598E+07,
.58006,	3.9806384E+04,	2.2100931E+07,
.59003,	3.9786229E+04,	2.2095666E+07,
.60003,	3.9758272E+04,	2.2086180E+07,

TABLE 6.2.1-27 (SHEET 16 OF 36)

(336-in.^{2} CLOSURE WELD BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.61011,	3.9733403E+04,	2.2078365E+07,
.62019,	3.9687180E+04,	2.2058478E+07,
.63005,	3.9633316E+04,	2.2034454E+07,
.64013,	3.9595001E+04,	2.2019153E+07,
.65008,	3.9544578E+04,	2.1996974E+07,
.66007,	3.9478791E+04,	2.1966303E+07,
.67001,	3.9415717E+04,	2.1937149E+07,
.68024,	3.9345246E+04,	2.1903947E+07,
.69004,	3.9272848E+04,	2.1869592E+07,
.70003,	3.9209415E+04,	2.1860313E+07,
.71001,	3.9153229E+04,	2.1815160E+07,
.72008,	3.9101670E+04,	2.1792552E+07,
.73012,	3.9048780E+04,	2.1769073E+07,
.74000,	3.8988046E+04,	2.1741096E+07,
.75012,	3.8920540E+04,	2.1709414E+07,
.76006,	3.8849061E+04,	2.1675378E+07,
.77002,	3.8768716E+04,	2.1636330E+07,
.78001,	3.8683025E+04,	2.1594241E+07,
.79014,	3.8595052E+04,	2.1550884E+07,
.80009,	3.8507172E+04,	2.1507415E+07,
.81009,	3.8420234E+04,	2.1464492E+07,
.82009,	3.8379224E+04,	2.1447517E+07,
.83007,	3.8367035E+04,	2.1446716E+07,
.84007,	3.8347948E+04,	2.1441847E+07,
.85001,	3.8302612E+04,	2.1422006E+07,
.86003,	3.8221354E+04,	2.1381845E+07,
.87009,	3.8138299E+04,	2.1340767E+07,
.88011,	3.8079445E+04,	2.1313324E+07,
.89010,	3.8030601E+04,	2.1291442E+07,
.90001,	3.7979574E+04,	2.1268167E+07,
.91006,	3.7917809E+04,	2.1238829E+07,
.92001,	3.7850827E+04,	2.1206416E+07,
.93010,	3.7789108E+04,	2.1176988E+07,
.94008,	3.7734890E+04,	2.1151639E+07,
.95003,	3.7684520E+04,	2.1128412E+07,
.96004,	3.7635988E+04,	2.1106175E+07,
.97008,	3.7584676E+04,	2.1082300E+07,
.98008,	3.7531491E+04,	2.1057291E+07,
.99001,	3.7479837E+04,	2.1033066E+07,

TABLE 6.2.1-27 (SHEET 17 OF 36)

(336-in.^{2} CLOSURE WELD BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
1.00009,	3.7430349E+04,	2.1010098E+07,
1.05001,	3.7616906E+04,	2.0996744E+07,
1.10010,	3.7766230E+04,	2.0901135E+07,
1.15010,	3.8045396E+04,	2.1157066E+07,
1.20002,	3.8355584E+04,	2.1341372E+07,
1.25003,	3.8657749E+04,	2.1520398E+07,
1.30009,	3.8968707E+04,	2.1704303E+07,
1.35004,	3.9164171E+04,	2.1023411E+07,
1.40001,	3.9307973E+04,	2.1912914E+07,
1.45005,	3.9326743E+04,	2.1932111E+07,
1.50008,	3.9363693E+04,	2.1961100E+07,
1.55004,	3.9457534E+04,	2.2021794E+07,
1.60008,	3.9468121E+04,	2.2031990E+07,
1.65008,	3.9500104E+04,	2.2061126E+07,
1.70019,	3.9585397E+04,	2.2072423E+07,
1.75005,	3.9431749E+04,	2.2039800E+07,
1.80008,	3.9341064E+04,	2.1997493E+07,
1.85010,	3.9261865E+04,	2.1961693E+07,
1.90003,	3.9296701E+04,	2.1991160E+07,
1.95006,	3.9235303E+04,	2.1994017E+07,
2.00088,	3.9153504E+04,	2.1931664E+07,
2.05010,	3.8974005E+04,	2.1042375E+07,
2.10010,	3.8742276E+04,	2.1723844E+07,
2.15027,	3.8304543E+04,	2.1490275E+07,
2.20002,	3.7877215E+04,	2.1261769E+07,
2.25003,	3.7703755E+04,	2.1177745E+07,
2.30003,	3.7377102E+04,	2.1004598E+07,
2.35003,	3.7050456E+04,	2.0831472E+07,
2.40012,	3.6771210E+04,	2.0686139E+07,
2.45007,	3.6533469E+04,	2.0565091E+07,
2.50003,	3.6394909E+04,	2.0503197E+07,
2.55003,	3.5981609E+04,	2.0286802E+07,
2.60006,	3.5565120E+04,	2.0060275E+07,
2.65008,	3.5170092E+04,	1.9360301E+07,
2.70007,	3.4761140E+04,	1.9641536E+07,
2.75001,	3.4506170E+04,	1.9511926E+07,
2.80004,	3.4050260E+04,	1.9272390E+07,
2.85006,	3.3531241E+04,	1.8903699E+07,
2.90035,	3.2924740E+04,	1.8661662E+07,
2.95008,	3.2356095E+04,	1.8353048E+07,
3.00006,	3.1975899E+04,	1.8153529E+07,

TABLE 6.2.1-27 (SHEET 18 OF 36)

(336-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.00000,	1.0904410E+04,	6.0480220E+06,
.00102,	3.1319824E+04,	1.7245649E+07,
.00201,	2.7946215E+04,	1.5375841E+07,
.00302,	2.9121894E+04,	1.6040584E+07,
.00400,	2.9399851E+04,	1.6183312E+07,
.00503,	2.8826792E+04,	1.5870642E+07,
.00603,	2.9234675E+04,	1.6096228E+07,
.00703,	3.0946665E+04,	1.7044444E+07,
.00804,	3.0758539E+04,	1.6933443E+07,
.00902,	2.9383448E+04,	1.6176691E+07,
.01003,	2.9135851E+04,	1.6040374E+07,
.01102,	2.9233906E+04,	1.6095455E+07,
.01200,	2.9381316E+04,	1.6175652E+07,
.01302,	2.9281523E+04,	1.6121855E+07,
.01400,	2.9575649E+04,	1.6206387E+07,
.01500,	3.0328683E+04,	1.6706998E+07,
.01600,	3.1745495E+04,	1.7497768E+07,
.01701,	3.3610875E+04,	1.8537603E+07,
.01802,	3.5095205E+04,	1.9361516E+07,
.01905,	3.5434097E+04,	1.9544472E+07,
.02005,	3.4878420E+04,	1.9230944E+07,
.02103,	3.4073856E+04,	1.8782276E+07,
.02205,	3.3541485E+04,	1.8487222E+07,
.02301,	3.3323030E+04,	1.8366196E+07,
.02402,	3.3191577E+04,	1.8278652E+07,
.02504,	3.3082371E+04,	1.8232257E+07,
.02601,	3.3002613E+04,	1.8187853E+07,
.02706,	3.2932539E+04,	1.8148856E+07,
.02806,	3.2888128E+04,	1.8124209E+07,
.02902,	3.2876176E+04,	1.8117719E+07,
.03001,	3.2912300E+04,	1.8138147E+07,
.03103,	3.3023916E+04,	1.8200783E+07,
.03205,	3.3225583E+04,	1.8313675E+07,
.03302,	3.3480035E+04,	1.8455733E+07,
.03407,	3.3738721E+04,	1.8599652E+07,
.03503,	3.3867245E+04,	1.8670587E+07,
.03601,	3.3840843E+04,	1.8654958E+07,
.03708,	3.3661255E+04,	1.8554098E+07,
.03803,	3.3438771E+04,	1.8429893E+07,
.03902,	3.3247515E+04,	1.8323650E+07,

TABLE 6.2.1-27 (SHEET 19 OF 36)

(336-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.04001,	3.3149606E+04,	1.8269451E+07,
.04102,	3.4858290E+04,	1.9234289E+07,
.04204,	3.6596807E+04,	2.0194634E+07,
.04300,	3.7811404E+04,	2.0861141E+07,
.04403,	3.8537083E+04,	2.1255744E+07,
.04505,	3.8789200E+04,	2.1389356E+07,
.04602,	3.8713281E+04,	2.1342235E+07,
.04705,	3.8376772E+04,	2.1151524E+07,
.04804,	3.7912998E+04,	2.0892494E+07,
.04904,	3.7450575E+04,	2.0636400E+07,
.05005,	3.7085790E+04,	2.0435341E+07,
.05106,	3.6835150E+04,	2.0297233E+07,
.05203,	3.6663090E+04,	2.0201827E+07,
.05305,	3.6510859E+04,	2.0117200E+07,
.05408,	3.6418075E+04,	2.0066715E+07,
.05503,	3.6484002E+04,	2.0105915E+07,
.05604,	3.6795543E+04,	2.0282122E+07,
.05702,	3.7285754E+04,	2.0555863E+07,
.05801,	3.7915465E+04,	2.0848865E+07,
.05903,	3.8213922E+04,	2.1066913E+07,
.06005,	3.8376449E+04,	2.1153361E+07,
.06105,	3.8307867E+04,	2.1111641E+07,
.06205,	3.8019257E+04,	2.0947908E+07,
.06300,	3.7570319E+04,	2.0696225E+07,
.06402,	3.6975766E+04,	2.0365317E+07,
.06508,	3.6374761E+04,	2.0032690E+07,
.06608,	3.5905768E+04,	1.9774418E+07,
.06702,	3.5615671E+04,	1.9615745E+07,
.06808,	3.5480589E+04,	1.9543488E+07,
.06905,	3.5539272E+04,	1.9578348E+07,
.07000,	3.5753754E+04,	1.9699418E+07,
.07103,	3.6140128E+04,	1.9915977E+07,
.07202,	3.6666822E+04,	2.0210557E+07,
.07308,	3.7386655E+04,	2.0612760E+07,
.07404,	3.8159480E+04,	2.1043976E+07,
.07506,	3.9061995E+04,	2.1546568E+07,
.07607,	3.9923806E+04,	2.2025178E+07,
.07705,	4.0654471E+04,	2.2430038E+07,
.07800,	4.1238952E+04,	2.2753206E+07,
.07905,	4.1713347E+04,	2.3014682E+07,

TABLE 6.2.1-27 (SHEET 20 OF 36)

(336-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.08005,	4.2016658E+04,	2.3181200E+07,
.08105,	4.2198966E+04,	2.3280508E+07,
.08210,	4.2279767E+04,	2.3323496E+07,
.08310,	4.2281321E+04,	2.3322819E+07,
.08404,	4.2239756E+04,	2.3298599E+07,
.08501,	4.2177672E+04,	2.3263276E+07,
.08611,	4.2104329E+04,	2.3221916E+07,
.08704,	4.2052709E+04,	2.3193003E+07,
.08807,	4.2020811E+04,	2.3175363E+07,
.08909,	4.2025770E+04,	2.3178449E+07,
.09009,	4.2065480E+04,	2.3200843E+07,
.09110,	4.2124772E+04,	2.3233807E+07,
.09211,	4.2172432E+04,	2.3259759E+07,
.09310,	4.2175533E+04,	2.3260295E+07,
.09411,	4.2105578E+04,	2.3219849E+07,
.09511,	4.1961821E+04,	2.3138422E+07,
.09603,	4.1778332E+04,	2.3035338E+07,
.09706,	4.1547717E+04,	2.2906509E+07,
.09803,	4.1339159E+04,	2.2790489E+07,
.09907,	4.1143922E+04,	2.2682407E+07,
.10005,	4.1030160E+04,	2.2619962E+07,
.10504,	4.1277515E+04,	2.2760967E+07,
.11006,	4.1684568E+04,	2.2984817E+07,
.11501,	4.0861846E+04,	2.2522864E+07,
.12002,	4.0110925E+04,	2.2107457E+07,
.12508,	4.0271513E+04,	2.2200062E+07,
.13011,	4.0672513E+04,	2.2423415E+07,
.13509,	4.0462782E+04,	2.2304174E+07,
.14009,	4.0129215E+04,	2.2119836E+07,
.14508,	4.0128710E+04,	2.2120539E+07,
.15011,	4.0036140E+04,	2.2068352E+07,
.15500,	3.9694514E+04,	2.1878076E+07,
.16006,	3.9472130E+04,	2.1755828E+07,
.16505,	3.9703270E+04,	2.1886290E+07,
.17002,	3.9968584E+04,	2.2033203E+07,
.17509,	3.9622431E+04,	2.1838486E+07,
.18011,	3.9104870E+04,	2.1550930E+07,
.18516,	3.9013925E+04,	2.1502278E+07,
.19010,	3.9202980E+04,	2.1608665E+07,
.19500,	3.9485145E+04,	2.1766258E+07,

TABLE 6.2.1-27 (SHEET 21 OF 36)

(336-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.20002,	3.9794566E+04,	2.1938705E+07,
.21006,	4.0042339E+04,	2.2075738E+07,
.22010,	4.0034978E+04,	2.2071672E+07,
.23003,	4.0069032E+04,	2.2090788E+07,
.24000,	3.9921154E+04,	2.2008323E+07,
.25003,	3.9753266E+04,	2.1915694E+07,
.26002,	3.9661005E+04,	2.1865181E+07,
.27003,	3.9609295E+04,	2.1837809E+07,
.28007,	3.9784416E+04,	2.1936634E+07,
.29010,	3.9855442E+04,	2.1977128E+07,
.30007,	3.9896288E+04,	2.2001107E+07,
.31006,	3.9954873E+04,	2.2035214E+07,
.32012,	3.9985827E+04,	2.2053889E+07,
.33004,	3.9951025E+04,	2.2036286E+07,
.34005,	3.9962792E+04,	2.2044922E+07,
.35017,	3.9932590E+04,	2.2030225E+07,
.36001,	3.9882825E+04,	2.2004858E+07,
.37009,	3.9819271E+04,	2.1971978E+07,
.38001,	3.9741062E+04,	2.1931179E+07,
.39010,	3.9729809E+04,	2.1928009E+07,
.40014,	3.9790731E+04,	2.1965091E+07,
.41008,	3.9848676E+04,	2.2000469E+07,
.42008,	3.9901479E+04,	2.2033191E+07,
.43006,	3.9943806E+04,	2.2059947E+07,
.44011,	3.9928071E+04,	2.2054514E+07,
.45004,	3.9890825E+04,	2.2037290E+07,
.46002,	3.9879296E+04,	2.2034773E+07,
.47001,	4.0099196E+04,	2.2161891E+07,
.48010,	4.0052539E+04,	2.2139260E+07,
.49004,	4.0009473E+04,	2.2119273E+07,
.50003,	3.9967072E+04,	2.2100057E+07,
.51013,	4.0033119E+04,	2.2141239E+07,
.52000,	3.9986071E+04,	2.2119181E+07,
.53008,	3.9995917E+04,	2.2129457E+07,
.54009,	4.0062363E+04,	2.2171146E+07,
.55003,	4.0939414E+04,	2.2162845E+07,
.56000,	3.9982026E+04,	2.2135457E+07,
.57007,	3.9916363E+04,	2.2183738E+07,
.58002,	3.9855869E+04,	2.2074846E+07,
.59007,	3.9797104E+04,	2.2047191E+07,

TABLE 6.2.1-27 (SHEET 22 OF 36)

(336-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.60011,	3.9801350E+04,	2.2054765E+07,
.61007,	3.9813400E+04,	2.2066753E+07,
.62001,	3.9806850E+04,	2.2068233E+07,
.63011,	3.9785117E+04,	2.2061235E+07,
.64010,	3.9745377E+04,	2.2044375E+07,
.65010,	3.9683897E+04,	2.2015240E+07,
.66002,	3.9627585E+04,	2.1989164E+07,
.67003,	3.9582612E+04,	2.1969491E+07,
.68012,	3.9535443E+04,	2.1948660E+07,
.69009,	3.9490536E+04,	2.1929156E+07,
.70008,	3.9446800E+04,	2.1909657E+07,
.71006,	3.9396414E+04,	2.1887528E+07,
.72007,	3.9348911E+04,	2.1866617E+07,
.73009,	3.9308340E+04,	2.1849517E+07,
.74006,	3.9266575E+04,	2.1831721E+07,
.75003,	3.9221422E+04,	2.1812018E+07,
.76008,	3.9167478E+04,	2.1787392E+07,
.77027,	3.9101610E+04,	2.1756044E+07,
.78014,	3.9029067E+04,	2.1720941E+07,
.79004,	3.8954031E+04,	2.1684527E+07,
.80006,	3.8880815E+04,	2.1649063E+07,
.81003,	3.8810827E+04,	2.1615375E+07,
.82008,	3.8743384E+04,	2.1583084E+07,
.83008,	3.8703083E+04,	2.1566116E+07,
.84013,	3.8698525E+04,	2.1569190E+07,
.85007,	3.8686489E+04,	2.1567963E+07,
.86008,	3.8646193E+04,	2.1550798E+07,
.87003,	3.8596282E+04,	2.1522539E+07,
.88036,	3.8509193E+04,	2.1484667E+07,
.89009,	3.8441697E+04,	2.1452192E+07,
.90005,	3.8394362E+04,	2.1431046E+07,
.91010,	3.8355403E+04,	2.1414617E+07,
.92009,	3.8312887E+04,	2.1396098E+07,
.93010,	3.8262021E+04,	2.1372841E+07,
.94004,	3.8201289E+04,	2.1343908E+07,
.95009,	3.8138982E+04,	2.1314158E+07,
.96010,	3.8085254E+04,	2.1288159E+07,
.97009,	3.8037122E+04,	2.1267278E+07,
.98005,	3.7991015E+04,	2.1246553E+07,
.99008,	3.7942798E+04,	2.1224593E+07,

TABLE 6.2.1-27 (SHEET 23 OF 36)

(336-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
1.00007,	3.7889110E+04,	2.1199669E+07,
1.05003,	3.7810879E+04,	2.1179703E+07,
1.10019,	3.7798934E+04,	2.1196647E+07,
1.15003,	3.8081679E+04,	2.1163255E+07,
1.20005,	3.8384633E+04,	2.1332920E+07,
1.25003,	3.8697696E+04,	2.1518565E+07,
1.30003,	3.8991710E+04,	2.1693715E+07,
1.35000,	3.9204347E+04,	2.1823086E+07,
1.40002,	3.9341273E+04,	2.1909702E+07,
1.45010,	3.9359873E+04,	2.1929688E+07,
1.50011,	3.9377289E+04,	2.1948093E+07,
1.55009,	3.9503057E+04,	2.2027510E+07,
1.60007,	3.9505578E+04,	2.2037144E+07,
1.65004,	3.9515997E+04,	2.2051235E+07,
1.70006,	3.9488035E+04,	2.2044140E+07,
1.75012,	3.9404968E+04,	2.2006330E+07,
1.80012,	3.9322910E+04,	2.1968669E+07,
1.85000,	3.9242225E+04,	2.1931719E+07,
1.90002,	3.9276580E+04,	2.1988992E+07,
1.95007,	3.9252680E+04,	2.1956123E+07,
2.00004,	3.9113078E+04,	2.1887882E+07,
2.05024,	3.8941004E+04,	2.1801685E+07,
2.10008,	3.8699189E+04,	2.1676477E+07,
2.15006,	3.8304963E+04,	2.1465963E+07,
2.20010,	3.7870398E+04,	2.1232248E+07,
2.25001,	3.7731360E+04,	2.1165724E+07,
2.30003,	3.7417022E+04,	2.0998164E+07,
2.35015,	3.7105293E+04,	2.0831994E+07,
2.40009,	3.6817342E+04,	2.0680490E+07,
2.45005,	3.6586350E+04,	2.0562003E+07,
2.50012,	3.6424999E+04,	2.0485987E+07,
2.55001,	3.6005489E+04,	2.0265259E+07,
2.60002,	3.5606846E+04,	2.0055890E+07,
2.65010,	3.5240474E+04,	1.9862977E+07,
2.70000,	3.4855162E+04,	1.9657410E+07,
2.75000,	3.4606376E+04,	1.9530965E+07,
2.80001,	3.4147934E+04,	1.9285313E+07,
2.85006,	3.3611093E+04,	1.8996154E+07,
2.90003,	3.2983430E+04,	1.8657303E+07,
2.95005,	3.2431145E+04,	1.8159907E+07,
3.00012,	3.2121901E+04,	1.8199162E+07,

TABLE 6.2.1-27 (SHEET 24 OF 36)

(436-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.00000,	.	.
.00101,	1.0912112E+04,	6.0279802E+06,
.00202,	1.5876922E+04,	8.7501640E+06,
.00301,	1.9107228E+04,	1.0528134E+07,
.00401,	2.2394932E+04,	1.2352347E+07,
.00501,	2.6556709E+04,	1.4651538E+07,
.00601,	2.9317134E+04,	1.6155227E+07,
.00701,	3.0770794E+04,	1.6953810E+07,
.00804,	3.2095162E+04,	1.7683326E+07,
.00902,	3.3063797E+04,	1.8214192E+07,
.01003,	3.3630048E+04,	1.8519761E+07,
.01101,	3.3690832E+04,	1.8547935E+07,
.01202,	3.3572936E+04,	1.8481671E+07,
.01301,	3.3414537E+04,	1.8392567E+07,
.01402,	3.3201386E+04,	1.8274230E+07,
.01500,	3.3292380E+04,	1.8216214E+07,
.01600,	3.3335582E+04,	1.8356246E+07,
.01704,	3.4004110E+04,	1.8727901E+07,
.01804,	3.4615891E+04,	1.9061515E+07,
.01904,	3.4722628E+04,	1.9109429E+07,
.02006,	3.3412646E+04,	1.8379770E+07,
.02106,	3.2920608E+04,	1.8115117E+07,
.02201,	3.2594040E+04,	1.7928432E+07,
.02301,	3.2123172E+04,	1.7672891E+07,
.02402,	3.1960616E+04,	1.7582280E+07,
.02500,	3.1616771E+04,	1.7391027E+07,
.02604,	3.1391551E+04,	1.7269437E+07,
.02703,	3.1250702E+04,	1.7190452E+07,
.02801,	3.1060788E+04,	1.7086540E+07,
.02903,	3.1177394E+04,	1.7155856E+07,
.03005,	3.1416908E+04,	1.7284245E+07,
.03102,	3.1233809E+04,	1.7178519E+07,
.03202,	3.0877682E+04,	1.6981724E+07,
.03304,	3.0644644E+04,	1.6856355E+07,
.03401,	3.0690733E+04,	1.6883908E+07,
.03504,	3.0785240E+04,	1.6935342E+07,
.03605,	3.0786569E+04,	1.6934025E+07,
.03703,	3.0631717E+04,	1.6847147E+07,
.03800,	3.0466928E+04,	1.6756951E+07,
.03901,	3.0358633E+04,	1.6698084E+07,

TABLE 6.2.1-27 (SHEET 25 OF 36)

(436-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.04000,	3.0438252E+04,	1.6745592E+07,
.04102,	3.0846188E+04,	1.6973687E+07,
.04201,	3.1253101E+04,	1.7193824E+07,
.04301,	3.1106960E+04,	1.7106070E+07,
.04400,	3.0710999E+04,	1.6888803E+07,
.04502,	3.0570958E+04,	1.6815748E+07,
.04600,	3.0688257E+04,	1.6881790E+07,
.04700,	3.0701911E+04,	1.6885873E+07,
.04801,	3.0574827E+04,	1.6818120E+07,
.04902,	3.0832885E+04,	1.6964301E+07,
.05000,	3.1189824E+04,	1.7162053E+07,
.05101,	3.1912778E+04,	1.7569750E+07,
.05200,	3.3570213E+04,	1.8500191E+07,
.05301,	3.6325123E+04,	2.0032669E+07,
.05402,	3.8686112E+04,	2.1316538E+07,
.05502,	3.9264176E+04,	2.1613079E+07,
.05602,	3.9164552E+04,	2.1560199E+07,
.05703,	3.9631396E+04,	2.1830694E+07,
.05804,	4.0664118E+04,	2.2406067E+07,
.05902,	4.1526297E+04,	2.2876487E+07,
.06004,	4.2026121E+04,	2.3149620E+07,
.06103,	4.2469172E+04,	2.3395779E+07,
.06203,	4.2856047E+04,	2.3606824E+07,
.06301,	4.2978961E+04,	2.3670597E+07,
.06401,	4.3026975E+04,	2.3700508E+07,
.06502,	4.3549054E+04,	2.4001981E+07,
.06600,	4.4721753E+04,	2.4661505E+07,
.06702,	4.6168579E+04,	2.5462312E+07,
.06807,	4.7325363E+04,	2.0096691E+07,
.06906,	4.8123725E+04,	2.6536014E+07,
.07004,	4.8775747E+04,	2.6895365E+07,
.07103,	4.9224183E+04,	2.7137644E+07,
.07203,	4.9270169E+04,	2.7153251E+07,
.07305,	4.8890924E+04,	2.6936090E+07,
.07400,	4.8385139E+04,	2.6654382E+07,
.07500,	4.7968558E+04,	2.6391960E+07,
.07606,	4.7563277E+04,	2.6202327E+07,
.07704,	4.7314610E+04,	2.6064344E+07,
.07806,	4.7036794E+04,	2.5908764E+07,

TABLE 6.2.1-27 (SHEET 26 OF 36)

(436-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.07904,	4.6723180E+04,	2.5733981E+07,
.08000,	4.6426196E+04,	2.5570341E+07,
.08106,	4.6243806E+04,	2.5472215E+07,
.08202,	4.6246547E+04,	2.5476682E+07,
.08303,	4.6390383E+04,	2.5558327E+07,
.08402,	4.6584489E+04,	2.5665636E+07,
.08504,	4.6699180E+04,	2.5726088E+07,
.08606,	4.6613903E+04,	2.5674637E+07,
.08701,	4.6324981E+04,	2.5511111E+07,
.08803,	4.5908154E+04,	2.5279153E+07,
.08901,	4.5514602E+04,	2.5062319E+07,
.09005,	4.5170201E+04,	2.4872751E+07,
.09101,	4.4878381E+04,	2.4710995E+07,
.09203,	4.4550294E+04,	2.4529427E+07,
.09300,	4.4258533E+04,	2.4369254E+07,
.09402,	4.4041145E+04,	2.4250486E+07,
.09504,	4.3883324E+04,	2.4163816E+07,
.09602,	4.3748654E+04,	2.4089902E+07,
.09704,	4.3731915E+04,	2.4085325E+07,
.09803,	4.4055327E+04,	2.4270897E+07,
.09906,	4.4593085E+04,	2.4568299E+07,
.10004,	4.4893727E+04,	2.4729042E+07,
.10506,	4.4915203E+04,	2.4738296E+07,
.11009,	4.5843578E+04,	2.5265301E+07,
.11508,	4.7961126E+04,	2.6441304E+07,
.12006,	4.8641610E+04,	2.6811261E+07,
.12511,	4.8600261E+04,	2.6791935E+07,
.13003,	4.8897292E+04,	2.6954703E+07,
.13507,	4.7822446E+04,	2.6353890E+07,
.14007,	4.6760026E+04,	2.5769811E+07,
.14510,	4.6613100E+04,	2.5690052E+07,
.15004,	4.5690755E+04,	2.5177113E+07,
.15504,	4.5251619E+04,	2.4940772E+07,
.16009,	4.5889095E+04,	2.5299141E+07,
.16521,	4.6090261E+04,	2.5411637E+07,
.17003,	4.6472870E+04,	2.5629872E+07,
.17512,	4.7251129E+04,	2.6064692E+07,
.18005,	4.7225150E+04,	2.6048738E+07,
.18515,	4.6856876E+04,	2.5847037E+07,
.19008,	4.6705900E+04,	2.5764932E+07,
.19500,	4.6273001E+04,	2.5525670E+07,

TABLE 6.2.1-27 (SHEET 27 OF 36)

(436-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.20008,	4.6134180E+04,	2.5453781E+07,
.21005,	4.6386621E+04,	2.5599180E+07,
.22011,	4.7146235E+04,	2.6028554E+07,
.23005,	4.7015697E+04,	2.5959264E+07,
.24013,	4.6347097E+04,	2.5591888E+07,
.25001,	4.6162333E+04,	2.5496921E+07,
.26009,	4.6307811E+04,	2.5584812E+07,
.27006,	4.6570319E+04,	2.5736626E+07,
.28006,	4.6444756E+04,	2.5672954E+07,
.29009,	4.6083901E+04,	2.5479119E+07,
.30002,	4.6141090E+04,	2.5517881E+07,
.31009,	4.6525927E+04,	2.5739030E+07,
.32012,	4.6713708E+04,	2.5849861E+07,
.33009,	4.6460700E+04,	2.5714637E+07,
.34001,	4.6141225E+04,	2.5543500E+07,
.35007,	4.6109222E+04,	2.5533279E+07,
.36007,	4.6227538E+04,	2.5606394E+07,
.37007,	4.6178311E+04,	2.5586123E+07,
.38005,	4.5918254E+04,	2.5448585E+07,
.39016,	4.5767288E+04,	2.5372179E+07,
.40008,	4.8571189E+04,	2.5438048E+07,
.41016,	4.6015706E+04,	2.5526360E+07,
.42012,	4.6063981E+04,	2.5527136E+07,
.43008,	4.5902058E+04,	2.5477926E+07,
.44004,	4.5873316E+04,	2.5469601E+07,
.45003,	4.5921183E+04,	2.5504168E+07,
.46003,	4.5928820E+04,	2.5516775E+07,
.47005,	4.5863777E+04,	2.5489062E+07,
.48007,	4.5822618E+04,	2.5474726E+07,
.49020,	4.5868611E+04,	2.5509101E+07,
.50011,	4.5941758E+04,	2.5558760E+07,
.51002,	4.5968058E+04,	2.5582434E+07,
.52003,	4.6069171E+04,	2.5648956E+07,
.53004,	4.6064521E+04,	2.5654282E+07,
.54004,	4.6024512E+04,	2.5640936E+07,
.55004,	4.6017318E+04,	2.5645755E+07,
.56007,	4.5951087E+04,	2.5618073E+07,
.57003,	4.5875425E+04,	2.5585545E+07,
.58003,	4.5836519E+04,	2.5573007E+07,
.59001,	4.5786252E+04,	2.5554223E+07,
.60015,	4.5761354E+04,	2.5550060E+07,

TABLE 6.2.1-27 (SHEET 28 OF 36)

(436-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.61002,	4.5679475E+04,	2.5513841E+07,
.62006,	4.5614735E+04,	2.5487332E+07,
.63001,	4.5580814E+04,	2.5477627E+07,
.64010,	4.5537959E+04,	2.5463058E+07,
.65006,	4.5472967E+04,	2.5436110E+07,
.66004,	4.5368041E+04,	2.5386841E+07,
.67008,	4.5276146E+04,	2.5344907E+07,
.68008,	4.5212833E+04,	2.5318834E+07,
.69010,	4.5146649E+04,	2.5291273E+07,
.70013,	4.5063020E+04,	2.5253774E+07,
.71008,	4.4970006E+04,	2.5211126E+07,
.72015,	4.4899570E+04,	2.5180997E+07,
.73015,	4.4842223E+04,	2.5157915E+07,
.74007,	4.4768907E+04,	2.5125831E+07,
.75000,	4.4675265E+04,	2.5082213E+07,
.76013,	4.4575080E+04,	2.5034875E+07,
.77005,	4.4486279E+04,	2.4993566E+07,
.78003,	4.4393089E+04,	2.4949759E+07,
.79011,	4.4280814E+04,	2.4895178E+07,
.80002,	4.4159534E+04,	2.4835323E+07,
.81009,	4.4047896E+04,	2.4780782E+07,
.82019,	4.4002782E+04,	2.4763851E+07,
.83017,	4.3971608E+04,	2.4754636E+07,
.84015,	4.3897477E+04,	2.4720952E+07,
.85012,	4.3807403E+04,	2.4677902E+07,
.86011,	4.3719832E+04,	2.4636057E+07,
.87001,	4.3641735E+04,	2.4599452E+07,
.88002,	4.3538419E+04,	2.4548494E+07,
.89016,	4.3425841E+04,	2.4492077E+07,
.90029,	4.3323151E+04,	2.4441194E+07,
.91012,	4.3237202E+04,	2.4399583E+07,
.92001,	4.3197013E+04,	2.4383525E+07,
.93007,	4.3135609E+04,	2.4354941E+07,
.94009,	4.3075249E+04,	2.4326592E+07,
.95014,	4.3008400E+04,	2.4294520E+07,
.96017,	4.2928460E+04,	2.4254898E+07,
.97012,	4.2855367E+04,	2.4218973E+07,
.98005,	4.2795246E+04,	2.4190366E+07,
.99007,	4.2780821E+04,	2.4187831E+07,

TABLE 6.2.1-27 (SHEET 29 OF 36)

(436-in.^{2} BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
1.00003,	4.2752246E+04,	2.4177108E+07,
1.05004,	4.2539695E+04,	2.4082305E+07,
1.10007,	4.2334300E+04,	2.3988517E+07,
1.15004,	4.2209823E+04,	2.3939257E+07,
1.20025,	4.2149730E+04,	2.3926104E+07,
1.25006,	4.2092436E+04,	2.3914261E+07,
1.30021,	4.2110966E+04,	2.3946258E+07,
1.35002,	4.2078652E+04,	2.3949385E+07,
1.40027,	4.1966932E+04,	2.3905238E+07,
1.45004,	4.1784741E+04,	2.3818295E+07,
1.50000,	4.1727473E+04,	2.3801830E+07,
1.55013,	4.1699789E+04,	2.3801814E+07,
1.60003,	4.1558067E+04,	2.3735546E+07,
1.65008,	4.1435150E+04,	2.3679377E+07,
1.70000,	4.1483172E+04,	2.3722962E+07,
1.75006,	4.1504535E+04,	2.3754789E+07,
1.80008,	4.1483365E+04,	2.3718907E+07,
1.85002,	4.1227437E+04,	2.3642725E+07,
1.90002,	4.1016295E+04,	2.3550974E+07,
1.95014,	4.0870107E+04,	2.3501156E+07,
2.00003,	4.0455307E+04,	2.3295283E+07,
2.05006,	3.9934443E+04,	2.3025753E+07,
2.10007,	3.9753225E+04,	2.2954260E+07,
2.15010,	3.9290013E+04,	2.2718398E+07,
2.20016,	3.8805140E+04,	2.2466812E+07,
2.25009,	3.8502236E+04,	2.2320490E+07,
2.30029,	3.8306991E+04,	2.2241263E+07,
2.35003,	3.7919102E+04,	2.2052489E+07,
2.40001,	3.7768094E+04,	2.2012574E+07,
2.45017,	3.7325121E+04,	2.1803745E+07,
2.50005,	3.6821854E+04,	2.1557137E+07,
2.55005,	3.6306581E+04,	2.1301116E+07,
2.60014,	3.5600723E+04,	2.0928827E+07,
2.65026,	3.4826145E+04,	2.0513480E+07,
2.70004,	3.4058488E+04,	2.0100456E+07,
2.75012,	3.3204439E+04,	1.9641103E+07,
2.80013,	3.2227744E+04,	1.9115998E+07,
2.85005,	3.1113386E+04,	1.8509041E+07,
2.90000,	3.0258151E+04,	1.8052239E+07,
2.95002,	3.0030674E+04,	1.7966679E+07,
3.00007,	2.9782520E+04,	1.7874253E+07,

TABLE 6.2.1-27 (SHEET 30 OF 36)

(763-in.^{2} HOT LEG BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.00000,	0. ,	0. ,
.00102,	7.7823700E+03,	4.9504728E+06,
.00201,	1.1086818E+04,	7.0489849E+06,
.00303,	2.2100842E+04,	1.4051387E+07,
.00403,	3.4238857E+04,	2.1774422E+07,
.00500,	4.1741684E+04,	2.6547063E+07,
.00602,	4.6481946E+04,	2.9556339E+07,
.00702,	4.9110254E+04,	3.1218089E+07,
.00800,	5.0377887E+04,	3.2012324E+07,
.00901,	5.0849559E+04,	3.2301064E+07,
.01003,	5.0878346E+04,	3.2311861E+07,
.01104,	5.0761880E+04,	3.2234808E+07,
.01202,	5.0695144E+04,	3.2192830E+07,
.01301,	5.0769941E+04,	3.2242396E+07,
.01400,	5.1006142E+04,	3.2394472E+07,
.01503,	5.1391505E+04,	3.2640344E+07,
.01601,	5.1837302E+04,	3.2923221E+07,
.01701,	5.2309698E+04,	3.3221666E+07,
.01801,	5.2748453E+04,	3.3497754E+07,
.01903,	5.3127817E+04,	3.3735439E+07,
.02003,	5.3411771E+04,	3.3911858E+07,
.02102,	5.3587249E+04,	3.4019292E+07,
.02202,	5.3662466E+04,	3.4063267E+07,
.02300,	5.3650130E+04,	3.4052182E+07,
.02403,	5.3565754E+04,	3.3995967E+07,
.02504,	5.3436329E+04,	3.3912156E+07,
.02601,	5.3290526E+04,	3.3819075E+07,
.02702,	5.1347931E+04,	3.3729261E+07,
.02802,	5.3036867E+04,	3.3660527E+07,
.02904,	5.2971948E+04,	3.3622112E+07,
.03000,	5.2968743E+04,	3.3623478E+07,
.03100,	5.3029110E+04,	3.3665533E+07,
.03201,	5.3153722E+04,	3.3748506E+07,
.03302,	5.3331104E+04,	3.3864763E+07,
.03402,	5.3549493E+04,	3.4006787E+07,
.03501,	5.3794178E+04,	3.4165108E+07,
.03601,	5.4062063E+04,	3.4337820E+07,
.03702,	5.4337548E+04,	3.4515029E+07,

TABLE 6.2.1-27 (SHEET 31 OF 36)

(763-in.^{2} HOT LEG BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.03801,	5.4603859E+04,	3.4686046E+07,
.03903,	5.4871375E+04,	3.4857557E+07,
.04002,	5.5115012E+04,	3.5013532E+07,
.04104,	5.5345808E+04,	3.5161050E+07,
.04203,	5.5556895E+04,	3.5295796E+07,
.04302,	5.5742463E+04,	3.5414074E+07,
.04402,	5.5914907E+04,	3.5523738E+07,
.04500,	5.6061661E+04,	3.5616908E+07,
.04602,	5.6195171E+04,	3.5701406E+07,
.04701,	5.6304293E+04,	3.5770270E+07,
.04800,	5.6400251E+04,	3.5830544E+07,
.04906,	5.6479238E+04,	3.5879874E+07,
.05003,	5.6538079E+04,	3.5916299E+07,
.05101,	5.6580392E+04,	3.5942104E+07,
.05202,	5.6607447E+04,	3.5958098E+07,
.05301,	5.6618964E+04,	3.5964131E+07,
.05403,	5.6614906E+04,	3.5960166E+07,
.05507,	5.6595249E+04,	3.5948240E+07,
.05604,	5.6563240E+04,	3.5924545E+07,
.05706,	5.6517173E+04,	3.5893887E+07,
.05808,	5.6459413E+04,	3.5855845E+07,
.05903,	5.6395002E+04,	3.5813728E+07,
.06000,	5.6323849E+04,	3.5767426E+07,
.06105,	5.6238056E+04,	3.5711826E+07,
.06204,	5.6152107E+04,	3.5656354E+07,
.06303,	5.6061848E+04,	3.5598461E+07,
.06406,	5.5969187E+04,	3.5539262E+07,
.06503,	5.5882016E+04,	3.5483752E+07,
.06600,	5.5797599E+04,	3.5430181E+07,
.06705,	5.5711644E+04,	3.5375809E+07,
.06803,	5.5631219E+04,	3.5325235E+07,
.06907,	5.5559086E+04,	3.5280130E+07,
.07002,	5.5492411E+04,	3.5238611E+07,
.07103,	5.5432132E+04,	3.5201303E+07,
.07201,	5.5378454E+04,	3.5168222E+07,
.07301,	5.5330161E+04,	3.5138690E+07,
.07402,	5.5287428E+04,	3.5112743E+07,
.07501,	5.5249761E+04,	3.5090042E+07,
.07601,	5.5215621E+04,	3.5069674E+07,
.07701,	5.5185947E+04,	3.5052144E+07,

TABLE 6.2.1-27 (SHEET 32 OF 36)

(763-in.^{2} HOT LEG BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.07800,	5.5515956E+04,	3.5036721E+07,
.07900,	5.5135988E+04,	3.5023107E+07,
.08000,	5.5114523E+04,	3.5010852E+07,
.08100,	5.5095190E+04,	3.4999922E+07,
.08201,	5.5077046E+04,	3.4989734E+07,
.08301,	5.5059845E+04,	3.4980098E+07,
.08402,	5.5042363E+04,	3.4970280E+07,
.08501,	5.5024404E+04,	3.4960147E+07,
.08600,	5.5005332E+04,	3.4949397E+07,
.08701,	5.4984427E+04,	3.4937654E+07,
.08803,	5.4961479E+04,	3.4924813E+07,
.08902,	5.4937813E+04,	3.4911638E+07,
.09001,	5.4912757E+04,	3.4897802E+07,
.09100,	5.4887504E+04,	3.4884017E+07,
.09201,	5.4862924E+04,	3.4870828E+07,
.09303,	5.4839475E+04,	3.4859563E+07,
.09402,	5.4818668E+04,	3.4848096E+07,
.09503,	5.4801182E+04,	3.4839804E+07,
.09601,	5.4787478E+04,	3.4833887E+07,
.09700,	5.4776706E+04,	3.4829916E+07,
.09802,	5.4768772E+04,	3.4827716E+07,
.09902,	5.4762804E+04,	3.4826644E+07,
.10002,	5.4757413E+04,	3.4825796E+07,
.10505,	5.4634156E+04,	3.4756575E+07,
.11004,	5.3966335E+04,	3.4334414E+07,
.11506,	5.2981994E+04,	3.3711323E+07,
.12000,	5.2125082E+04,	3.3169628E+07,
.12507,	5.1385350E+04,	3.2700539E+07,
.13006,	5.0623676E+04,	3.2217691E+07,
.13502,	4.9952734E+04,	3.1794294E+07,
.14002,	4.9624244E+04,	3.1590996E+07,
.14507,	4.9659208E+04,	3.1619886E+07,
.15002,	4.9826967E+04,	3.1730674E+07,
.15504,	4.9958804E+04,	3.1816492E+07,
.16014,	4.9974844E+04,	3.1827172E+07,
.16512,	4.9873952E+04,	3.1762947E+07,
.17007,	4.9676572E+04,	3.1637817E+07,
.17500,	4.9424401E+04,	3.1478882E+07,
.18002,	4.9169537E+04,	3.1319320E+07,
.18507,	4.8977379E+04,	3.1200412E+07,
.19008,	4.8891947E+04,	3.1149864E+07,

TABLE 6.2.1-27 (SHEET 33 OF 36)

(763-in.^{2} HOT LEG BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.19512,	4.8905230E+04,	3.1162286E+07,
.20017,	4.8971499E+04,	3.1208073E+07,
.21011,	4.9132117E+04,	3.1316312E+07,
.22004,	4.9118318E+04,	3.1311984E+07,
.23006,	4.8895594E+04,	3.1176020E+07,
.24001,	4.8547872E+04,	3.0963726E+07,
.25009,	4.8202471E+04,	3.0756576E+07,
.26004,	4.7929553E+04,	3.0596316E+07,
.27003,	4.7702439E+04,	3.0462393E+07,
.28009,	4.7508889E+04,	3.0345842E+07,
.29002,	4.7371087E+04,	3.0261785E+07,
.30001,	4.7306930E+04,	3.0223088E+07,
.31008,	4.7280472E+04,	3.0207649E+07,
.32012,	4.7230590E+04,	3.0177807E+07,
.33006,	4.7094913E+04,	3.0094690E+07,
.34005,	4.6887690E+04,	2.9968310E+07,
.35010,	4.6650874E+04,	2.9824836E+07,
.36013,	4.6424525E+04,	2.9688533E+07,
.37004,	4.6213540E+04,	2.9562593E+07,
.38012,	4.6028987E+04,	2.9452992E+07,
.39005,	4.5905610E+04,	2.9381484E+07,
.40008,	4.5854401E+04,	2.9355651E+07,
.41013,	4.5863015E+04,	2.9367338E+07,
.42003,	4.5897394E+04,	2.9394808E+07,
.43010,	4.5915956E+04,	2.9411982E+07,
.44013,	4.5895521E+04,	2.9403662E+07,
.45009,	4.5822802E+04,	2.9361067E+07,
.46003,	4.5702115E+04,	2.9287550E+07,
.47007,	4.5560837E+04,	2.9201025E+07,
.48012,	4.5440327E+04,	2.9128036E+07,
.49005,	4.5370502E+04,	2.9087580E+07,
.50001,	4.5356383E+04,	2.9082777E+07,
.51009,	4.5385682E+04,	2.9105652E+07,
.52009,	4.5432669E+04,	2.9139564E+07,
.53005,	4.5468619E+04,	2.9166079E+07,
.54006,	4.5468395E+04,	2.9169117E+07,
.55003,	4.5416172E+04,	2.9138892E+07,
.56000,	4.5321254E+04,	2.9081899E+07,
.57006,	4.5210656E+04,	2.9015687E+07,
.58001,	4.5115650E+04,	2.8960172E+07,

TABLE 6.2.1-27 (SHEET 34 OF 36)

(763-in.^{2} HOT LEG BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
.59004,	4.5044293E+04,	2.8920623E+07,
.60012,	4.5010443E+04,	2.8906032E+07,
.61010,	4.5013950E+04,	2.8915961E+07,
.62013,	4.5038708E+04,	2.8939886E+07,
.63009,	4.5059370E+04,	2.8961219E+07,
.64014,	4.5060179E+04,	2.8969927E+07,
.65011,	4.5035184E+04,	2.8962020E+07,
.66013,	4.4986898E+04,	2.8939247E+07,
.67009,	4.4925663E+04,	2.8908217E+07,
.68003,	4.4865426E+04,	2.8877950E+07,
.69012,	4.4818243E+04,	2.8855888E+07,
.70008,	4.4791555E+04,	2.8846866E+07,
.71002,	4.4780332E+04,	2.8847766E+07,
.72000,	4.4772711E+04,	2.8850961E+07,
.73002,	4.4760797E+04,	2.8851406E+07,
.74007,	4.4741266E+04,	2.8847037E+07,
.75002,	4.4712307E+04,	2.8836656E+07,
.76001,	4.4673720E+04,	2.8820326E+07,
.77004,	4.4627231E+04,	2.8799190E+07,
.78001,	4.4576132E+04,	2.8775296E+07,
.79005,	4.4521680E+04,	2.8749623E+07,
.80009,	4.4469922E+04,	2.8726022E+07,
.81002,	4.4425386E+04,	2.8707297E+07,
.82009,	4.4385236E+04,	2.8691873E+07,
.83004,	4.4346941E+04,	2.8677848E+07,
.84013,	4.4306106E+04,	2.8662665E+07,
.85007,	4.4261121E+04,	2.8644991E+07,
.86005,	4.4201427E+04,	2.8618225E+07,
.87006,	4.4147923E+04,	2.8595735E+07,
.88001,	4.4100598E+04,	2.8577514E+07,
.89009,	4.4061089E+04,	2.8564640E+07,
.90003,	4.4041189E+04,	2.8564429E+07,
.91005,	4.4031122E+04,	2.8570789E+07,
.92010,	4.4021949E+04,	2.8577993E+07,
.93001,	4.4004878E+04,	2.8580150E+07,
.94003,	4.3975310E+04,	2.8574674E+07,
.95000,	4.3932837E+04,	2.8561112E+07,
.96007,	4.3877626E+04,	2.8539750E+07,
.97011,	4.3813505E+04,	2.8512862E+07,
.98001,	4.3746579E+04,	2.8434177E+07,
.99004,	4.3679310E+04,	2.8455659E+07,

TABLE 6.2.1-27 (SHEET 35 OF 36)

(763-in.^{2} HOT LEG BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
1.00009,	4.3616582E+04,	2.8430269E+07,
1.05004,	4.3362851E+04,	2.8343817E+07,
1.10009,	4.3106996E+04,	2.8203634E+07,
1.15007,	4.2656312E+04,	2.8059859E+07,
1.20002,	4.2276635E+04,	2.7914949E+07,
1.25002,	4.1996846E+04,	2.7844326E+07,
1.30006,	4.1778160E+04,	2.7808761E+07,
1.35006,	4.1502658E+04,	2.7729192E+07,
1.40003,	4.1145142E+04,	2.7590021E+07,
1.45007,	4.0719791E+04,	2.7403645E+07,
1.50011,	4.0278464E+04,	2.7196215E+07,
1.55009,	3.9836748E+04,	2.6978928E+07,
1.60001,	3.9415183E+04,	2.6769679E+07,
1.65012,	3.8998375E+04,	2.6559661E+07,
1.70012,	3.8602806E+04,	2.6359988E+07,
1.75011,	3.8221231E+04,	2.6165738E+07,
1.80006,	3.7854297E+04,	2.5973809E+07,
1.85007,	3.7502521E+04,	2.5782014E+07,
1.90000,	3.7155071E+04,	2.5583127E+07,
1.95010,	3.6808801E+04,	2.5377270E+07,
2.00009,	3.6453550E+04,	2.5162290E+07,
2.05014,	3.6085701E+04,	2.4941049E+07,
2.10011,	3.5697676E+04,	2.4713441E+07,
2.15005,	3.5290806E+04,	2.4482273E+07,
2.20000,	3.4873172E+04,	2.4253393E+07,
2.25004,	3.4442934E+04,	2.4025041E+07,
2.30012,	3.4020636E+04,	2.3801494E+07,
2.35007,	3.3630302E+04,	2.3589986E+07,
2.40002,	3.3267885E+04,	2.3389694E+07,
2.45004,	3.2900597E+04,	2.3195920E+07,
2.50007,	3.2674711E+04,	2.3007341E+07,
2.55013,	3.2398991E+04,	2.2818065E+07,
2.60007,	3.2112088E+04,	2.2622375E+07,
2.65003,	3.1797713E+04,	2.2417317E+07,
2.70013,	3.1441037E+04,	2.2198094E+07,
2.75000,	3.1057741E+04,	2.1971109E+07,
2.80009,	3.0669535E+04,	2.1741814E+07,
2.85004,	3.0303664E+04,	2.1519357E+07,

TABLE 6.2.1-27 (SHEET 36 OF 36)

(763-in.^{2} HOT LEG BREAK)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)
2.90009,	2.9952884E+04,	2.1304776E+07,
2.95003,	2.9588947E+04,	2.1034229E+07,
3.00005,	2.9263868E+04,	2.0871117E+07,

TABLE 6.2.1-28 (SHEET 1 OF 7)

PRESSURIZER COMPARTMENT MASS AND ENERGY RELEASE
PRESSURIZER SURGE LINE BREAK

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
0.00000	0.00	0.00	0.00
0.00251	5.6681148E+04	1.1296008E+07	677.17
0.00501	1.6556361E+04	1.1212058E+07	677.21
0.00752	1.6699069E+04	1.1302997E+07	676.86
0.01002	1.9033506E+04	1.2830006E+07	674.07
0.01250	2.2089365E+04	1.4828262E+07	671.29
0.01501	2.1648161E+04	1.4533929E+07	671.37
0.01754	2.1247911E+04	1.4268193E+07	671.51
0.02002	2.0465838E+04	1.3752132E+07	671.96
0.02250	2.0393611E+04	1.3704347E+07	671.99
0.02505	2.0706044E+04	1.3907231E+07	671.65
0.02752	2.0931729E+04	1.4053966E+07	671.42
0.03001	2.0998600E+04	1.4096217E+07	671.29
0.03258	2.0967919E+04	1.4074876E+07	671.26
0.03503	2.0990414E+04	1.4088700E+07	671.20
0.03750	2.1019840E+04	1.4107187E+07	671.14
0.04009	2.1062241E+04	1.4134287E+07	671.07
0.04259	2.1156624E+04	1.4195514E+07	670.97
0.04512	2.1160405E+04	1.4197306E+07	670.94
0.04761	2.1098863E+04	1.4156324E+07	670.95
0.05009	2.1066994E+04	1.4134990E+07	670.95
0.05264	2.1095761E+04	1.4153509E+07	670.92
0.05505	2.1085426E+04	1.4146331E+07	670.91
0.05751	2.1000054E+04	1.4089897E+07	670.95
0.06008	2.0863697E+04	1.4000108E+07	671.03
0.06255	2.0694171E+04	1.3888722E+07	671.14
0.06512	2.0509657E+04	1.3767663E+07	671.28
0.06759	2.0407265E+04	1.3700589E+07	671.36
0.07002	2.0418448E+04	1.3708034E+07	671.36
0.07250	2.0481072E+04	1.3749121E+07	671.31
0.07507	2.0519777E+04	1.3774417E+07	671.28
0.07754	2.0521037E+04	1.3775092E+07	671.27
0.08008	2.0488129E+04	1.3753315E+07	671.28
0.08255	2.0410939E+04	1.3702503E+07	671.33
0.08504	2.0305825E+04	1.3633481E+07	671.41
0.08755	2.0203846E+04	1.3566573E+07	671.48
0.09006	2.0124810E+04	1.3514710E+07	671.54
0.09261	2.0067276E+04	1.3476990E+07	671.59
0.09501	2.0049927E+04	1.3465634E+07	671.61
0.09751	2.0091047E+04	1.3492584E+07	671.57
0.10011	2.0190095E+04	1.3557427E+07	671.49

TABLE 6.2.1-28 (SHEET 2 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
0.10259	2.0322401E+04	1.3644022E+07	671.38
0.10513	2.0470869E+04	1.3741182E+07	671.26
0.10762	2.0613302E+04	1.3834228E+07	671.13
0.11001	2.0698652E+04	1.3889843E+07	671.05
0.11251	2.0710177E+04	1.3896965E+07	671.02
0.11504	2.0648143E+04	1.3855823E+07	671.04
0.11753	2.0539220E+04	1.3783979E+07	671.11
0.12009	2.0392131E+04	1.3687301E+07	671.21
0.12255	2.0235027E+04	1.3584155E+07	671.32
0.12505	2.0076107E+04	1.3479844E+07	671.44
0.12762	1.9907462E+04	1.3369294E+07	671.57
0.13001	1.9721330E+04	1.3247406E+07	671.73
0.13253	1.9515669E+04	1.3112767E+07	671.91
0.13513	1.9325524E+04	1.2988402E+07	672.09
0.13760	1.9143891E+04	1.2869713E+07	672.26
0.14000	1.8989530E+04	1.2768843E+07	672.41
0.14263	1.8858030E+04	1.2682997E+07	672.55
0.14517	1.8757129E+04	1.2617164E+07	672.66
0.14757	1.8681814E+04	1.2568006E+07	672.74
0.15014	1.8626059E+04	1.2531659E+07	672.80
0.15265	1.8590127E+04	1.2508250E+07	672.84
0.15504	1.8569023E+04	1.2494497E+07	672.87
0.15764	1.8551204E+04	1.2482868E+07	672.89
0.16005	1.8525873E+04	1.2466307E+07	672.91
0.16250	1.8477290E+04	1.2434527E+07	672.96
0.16508	1.8394683E+04	1.2380511E+07	673.05
0.16765	1.8282083E+04	1.2306962E+07	673.17
0.17002	1.8146081E+04	1.2218190E+07	673.32
0.17258	1.7994507E+04	1.2119261E+07	673.50
0.17510	1.7851733E+04	1.2026175E+07	673.67
0.17754	1.7718757E+04	1.1939508E+07	673.83
0.18004	1.7607714E+04	1.1867137E+07	673.97
0.18257	1.7516255E+04	1.1807583E+07	674.09
0.18510	1.7440413E+04	1.1758207E+07	674.19
0.18753	1.7384799E+04	1.1721993E+07	674.27
0.19006	1.7331275E+04	1.1687159E+07	674.34
0.19258	1.7283259E+04	1.1655900E+07	674.40
0.19513	1.7239302E+04	1.1627267E+07	674.46
0.19757	1.7196585E+04	1.1599461E+07	674.52
0.20000	1.7153255E+04	1.1571232E+07	674.58
0.20257	1.7114304E+04	1.1545858E+07	674.63
0.20515	1.7075604E+04	1.1520650E+07	674.68
0.20764	1.7047204E+04	1.1502146E+07	674.72
0.21013	1.7030892E+04	1.1491517E+07	674.75

TABLE 6.2.1-28 (SHEET 3 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
0.21262	1.7026041E+04	1.1488355E+07	674.75
0.21509	1.7031337E+04	1.1491802E+07	674.74
0.21755	1.7044630E+04	1.1500441E+07	674.73
0.22014	1.7064281E+04	1.1513204E+07	674.70
0.22261	1.7085144E+04	1.1526752E+07	674.67
0.22518	1.7104103E+04	1.1539031E+07	674.64
0.22757	1.7120939E+04	1.1549913E+07	674.61
0.23002	1.7133999E+04	1.1558321E+07	674.58
0.23261	1.7139862E+04	1.1562022E+07	674.57
0.23511	1.7135943E+04	1.1559323E+07	674.57
0.23756	1.7121402E+04	1.1549700E+07	674.58
0.24007	1.7091493E+04	1.1530055E+07	674.61
0.24253	1.7044464E+04	1.1499260E+07	674.66
0.24509	1.6981075E+04	1.1457784E+07	674.74
0.24752	1.6909137E+04	1.1410815E+07	674.83
0.25009	1.6816993E+04	1.1350678E+07	674.95
0.25256	1.6725891E+04	1.1291225E+07	675.07
0.25505	1.6633374E+04	1.1230921E+07	675.20
0.25751	1.6542717E+04	1.1171824E+07	675.33
0.26005	1.6466594E+04	1.1122207E+07	675.44
0.26251	1.6403107E+04	1.1080847E+07	675.53
0.26504	1.6352876E+04	1.1048104E+07	675.61
0.26751	1.6321725E+04	1.1027787E+07	675.65
0.27009	1.6296620E+04	1.1011400E+07	675.69
0.27265	1.6276846E+04	1.0998473E+07	675.71
0.27509	1.6255893E+04	1.0984763E+07	675.74
0.27751	1.6228524E+04	1.0966874E+07	675.78
0.28003	1.6185426E+04	1.0938740E+07	675.84
0.28253	1.6127011E+04	1.0900616E+07	675.92
0.28512	1.6053276E+04	1.0852567E+07	676.03
0.28755	1.5970294E+04	1.0798521E+07	676.16
0.29001	1.5898962E+04	1.0752073E+07	676.28
0.29256	1.5832910E+04	1.0709102E+07	676.38
0.29509	1.5829868E+04	1.0707076E+07	676.38
0.29751	1.5825312E+04	1.0704075E+07	676.39
0.30009	1.5808551E+04	1.0693108E+07	676.41
0.30265	1.5808165E+04	1.0692807E+07	676.41
0.30517	1.5803688E+04	1.0689818E+07	676.41
0.30753	1.5753769D+04	1.0657243E+07	676.49
0.31006	1.5744724E+04	1.0651197E+07	676.49
0.31256	1.5753012E+04	1.0656385E+07	676.47
0.31501	1.5760820E+04	1.0661304E+07	676.44
0.31767	1.5737615E+04	1.0646068E+07	676.47
0.32014	1.5735895E+04	1.0644859E+07	676.47

TABLE 6.2.1-28 (SHEET 4 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
0.32273	1.5734566E+04	1.0643956E+07	676.47
0.32509	1.5732595E+04	1.0642618E+07	676.47
0.32753	1.5729905E+04	1.0640794E+07	676.47
0.33024	1.5726344E+04	1.0638378E+07	676.47
0.33251	1.5723018E+04	1.0636121E+07	676.47
0.33526	1.5718891E+04	1.0633317E+07	676.47
0.33752	1.5715349E+04	1.0630912E+07	676.47
0.34007	1.5711537E+04	1.0628319E+07	676.47
0.34274	1.5707870E+04	1.0625820E+07	676.46
0.34506	1.5704632E+04	1.0623609E+07	676.46
0.34770	1.5701190E+04	1.0621254E+07	676.46
0.35025	1.5698041E+04	1.0619096E+07	676.46
0.35261	1.5695247E+04	1.0617177E+07	676.46
0.35516	1.5692136E+04	1.0615038E+07	676.46
0.35762	1.5688914E+04	1.0612827E+07	676.45
0.36011	1.5685466E+04	1.0610463E+07	676.45
0.36259	1.5681810E+04	1.0607961E+07	676.45
0.36518	1.5677911E+04	1.0605294E+07	676.45
0.36758	1.5674219E+04	1.0602767E+07	676.45
0.37034	1.5670037E+04	1.0599906E+07	676.44
0.37267	1.5666639E+04	1.0597581E+07	676.44
0.37504	1.5663437E+04	1.0595386E+07	676.44
0.37757	1.5660250E+04	1.0593197E+07	676.44
0.38027	1.5657086E+04	1.0591016E+07	676.44
0.38277	1.5654322E+04	1.0589105E+07	676.43
0.38516	1.5651744E+04	1.0587321E+07	676.43
0.38754	1.5649144E+04	1.0585526E+07	676.43
0.39027	1.5646820E+04	1.0583914E+07	676.43
0.39262	1.5644725E+04	1.0582456E+07	676.42
0.39522	1.5642789E+04	1.0581100E+07	676.42
0.39760	1.5641070E+04	1.0579890E+07	676.42
0.40004	1.5639449E+04	1.0578741E+07	676.41
0.40257	1.5637925E+04	1.0577652E+07	676.41
0.40505	1.5636591E+04	1.0576689E+07	676.41
0.40753	1.5635413E+04	1.0575828E+07	676.40
0.41013	1.5634337E+04	1.0575027E+07	676.40
0.41251	1.5633562E+04	1.0574432E+07	676.39
0.41504	1.5632975E+04	1.0573960E+07	676.39
0.41775	1.4532551E+04	1.0573584E+07	676.38
0.42001	1.5632366E+04	1.0573381E+07	676.38
0.42268	1.5632315E+04	1.0573254E+07	676.37
0.42508	1.5632378E+04	1.0573212E+07	676.37
0.42781	1.5632537E+04	1.0573215E+07	676.36
0.43027	1.5632708E+04	1.0573240E+07	676.35

TABLE 6.2.1-28 (SHEET 5 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
0.43267	1.5632877E+04	1.0573264E+07	676.35
0.43529	1.5633025E+04	1.0573264E+07	676.34
0.43752	1.4533096E+04	1.0573227E+07	676.34
0.44015	1.5633088E+04	1.0573125E+07	676.33
0.44255	1.5632964E+04	1.0572955E+07	676.32
0.44523	1.5632658E+04	1.0572657E+07	676.32
0.44756	1.5632172E+04	1.0572247E+07	676.31
0.45019	1.5631366E+04	1.0571613E+07	676.31
0.45237	1.5630249E+04	1.0570770E+07	676.30
0.45517	1.5629049E+04	1.0569885E+07	676.30
0.45757	1.5627623E+04	1.0568848E+07	676.29
0.46022	1.5625881E+04	1.0567590E+07	676.29
0.46289	1.5623874E+04	1.0566152E+07	676.28
0.46514	1.5622119E+04	1.0564900E+07	676.28
0.46777	1.5619955E+04	1.0563361E+07	676.27
0.47036	1.5617741E+04	1.0561792E+07	676.27
0.47288	1.5615520E+04	1.0560223E+07	676.26
0.47503	1.5613565E+04	1.0558844E+07	676.26
0.47757	1.5611133E+04	1.0557133E+07	676.26
0.48011	1.5608643E+04	1.0555384E+07	676.25
0.48252	1.5606080E+04	1.0553588E+07	676.25
0.48512	1.5603492E+04	1.0551775E+07	676.24
0.48758	1.5600916E+04	1.0549973E+07	676.24
0.49009	1.5598319E+04	1.0548154E+07	676.24
0.49273	1.5595641E+04	1.0546276E+07	676.23
0.49505	1.5593323E+04	1.0544647E+07	676.23
0.49786	1.5590573E+04	1.0542711E+07	676.22
0.50029	1.5588315E+04	1.0541119E+07	676.22
0.51010	1.5579889E+04	1.0535146E+07	676.20
0.52041	1.5571914E+04	1.0529460E+07	676.18
0.53013	1.5564820E+04	1.0524389E+07	676.17
0.54025	1.5557270E+04	1.0518995E+07	676.15
0.55044	1.5550075E+04	1.0513839E+07	676.13
0.56029	1.5544285E+04	1.0509628E+07	676.11
0.57048	1.5539811E+04	1.0506279E+07	676.09
0.58035	1.5537036E+04	1.0504053E+07	676.07
0.59012	1.5535838E+04	1.0502863E+07	676.04
0.60013	1.5535793E+04	1.0502401E+07	676.01
0.61039	1.5536152E+04	1.0502172E+07	675.98
0.62027	1.5535777E+04	1.0501463E+07	675.95
0.63022	1.5533739E+04	1.0499646E+07	675.93
0.64028	1.5529494E+04	1.0496323E+07	675.90
0.65009	1.5523642E+04	1.0491943E+07	675.87
0.66010	1.5516674E+04	1.0486820E+07	675.84

TABLE 6.2.1-28 (SHEET 6 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
0.67006	1.5509527E+04	1.0481657E+07	675.82
0.68002	1.5503246E+04	1.0476939E+07	675.79
0.69043	1.5497613E+04	1.0472703E+07	675.76
0.70003	1.5493333E+04	1.0469413E+07	675.74
0.71008	1.5489729E+04	1.0466564E+07	675.71
0.72062	1.5486462E+04	1.0463919E+07	675.68
0.73049	1.5483628E+04	1.0461579E+07	675.65
0.74006	1.5481145E+04	1.0459476E+07	675.63
0.75052	1.5478710E+04	1.0457353E+07	675.60
0.76039	1.5476578E+04	1.0455447E+07	675.57
0.77046	1.5474438E+04	1.0453521E+07	675.53
0.78005	1.5472213E+04	1.0451554E+07	675.50
0.79034	1.5469339E+04	1.0449119E+07	675.47
0.80022	1.5465740E+04	1.0446217E+07	675.44
0.81004	1.5461051E+04	1.0442584E+07	675.41
0.82013	1.5455070E+04	1.0438084E+07	675.38
0.83054	1.5447988E+04	1.0432852E+07	675.35
0.84013	1.5441170E+04	1.0427849E+07	675.33
0.85057	1.5433935E+04	1.0422535E+07	675.30
0.86010	1.5427803E+04	1.0418005E+07	675.27
0.87064	1.5421686E+04	1.0413445E+07	675.25
0.88049	1.5416519E+04	1.0409555E+07	675.22
0.89029	1.5411736E+04	1.0405923E+07	675.19
0.90062	1.5406860E+04	1.0402204E+07	675.17
0.91043	1.5402219E+04	1.0398661E+07	675.14
0.92014	1.5397541E+04	1.0395092E+07	675.11
0.93032	1.5392543E+04	1.0391286E+07	675.09
0.94013	1.5387606E+04	1.0387534E+07	675.06
0.95020	1.5382356E+04	1.0383562E+07	675.03
0.96044	1.5376745E+04	1.0379341E+07	675.00
0.97022	1.5371025E+04	1.0375073E+07	674.98
0.98023	1.5364719E+04	1.0370408E+07	674.95
0.99000	1.5358118E+04	1.0365561E+07	674.92
1.00024	1.5350854E+04	1.0360257E+07	674.90
1.05009	1.5319038E+04	1.0336848E+07	674.77
1.10020	1.5295400E+04	1.0318842E+07	674.64
1.15027	1.5270563E+04	1.0299964E+07	674.50
1.20065	1.5244981E+04	1.0280607E+07	674.36
1.25022	1.5225611E+04	1.0265425E+07	674.22
1.30001	1.5204826E+04	1.0249226E+07	674.08
1.35008	1.5181682E+04	1.0231482E+07	673.94
1.40047	1.5158958E+04	1.0214058E+07	673.80
1.45008	1.5135654E+04	1.0196263E+07	673.66
1.50052	1.5108628E+04	1.0175997E+07	673.52

TABLE 6.2.1-28 (SHEET 7 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
1.55020	1.5082453E+04	1.0156373E+07	673.39
1.60001	1.5056220E+04	1.0136713E+07	673.26
1.65041	1.5029192E+04	1.0116506E+07	673.12
1.70057	1.5003251E+04	1.0097044E+07	672.99
1.75042	1.4978790E+04	1.0078568E+07	672.86
1.80008	1.4954574E+04	1.0060259E+07	672.72
1.85001	1.4910178E+04	1.0027829E+07	672.55
1.90030	1.4937484E+04	1.0045448E+07	672.50
1.95350	1.4846071E+04	9.9803806E+06	672.26
2.00022	1.4821801E+04	9.9621501E+06	672.13

TABLE 6.2.1-28A (SHEET 1 OF 7)

PRESSURIZER COMPARTMENT
MASS AND ENERGY RELEASE
PRESSURIZER SPRAY LINE BREAK

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
0.00000	0.00	0.00	0.00
0.00251	5.5520269E+03	3.4074591E+06	613.73
0.00502	5.7566695E+03	3.5214197E+06	611.71
0.00751	5.6923083E+03	3.4814317E+06	611.60
0.01002	5.6156477E+03	3.4348228E+06	611.65
0.01251	5.5820416E+03	3.4132128E+06	611.46
0.01502	5.6059056E+03	3.4246446E+06	610.90
0.01755	5.9216506E+03	3.6028096E+06	608.41
0.02003	6.0469291E+03	3.6722760E+06	607.30
0.02255	6.0807799E+03	3.6897530E+06	606.79
0.02505	6.0942687E+03	3.6961327E+06	606.49
0.02754	6.2322326E+03	3.7737113E+06	605.52
0.03004	6.3772227E+03	3.8554807E+06	604.57
0.03259	6.4620881E+03	3.9026378E+06	603.93
0.03507	6.4752834E+03	3.9088990E+06	603.66
0.03753	6.5151813E+03	3.9305146E+06	603.29
0.04002	6.5143568E+03	3.9287741E+06	603.09
0.04251	6.4438043E+03	3.8870628E+06	603.22
0.04511	6.3427181E+03	3.8281043E+06	603.54
0.04763	6.2773775E+03	3.7899141E+06	603.74
0.05005	6.2551735E+03	3.7764886E+06	603.74
0.05263	6.2524046E+03	3.7741758E+06	603.64
0.05518	6.2668754E+03	3.7818297E+06	603.46
0.05769	6.2974621E+03	3.7987461E+06	603.22
0.06003	6.3249883E+03	3.8139264E+06	602.99
0.06259	6.3350379E+03	3.8191169E+06	602.86
0.06510	6.3288577E+03	3.8150521E+06	602.80
0.06750	6.3276385E+03	3.8139641E+06	602.75
0.07005	6.3455081E+03	3.8238003E+06	602.60
0.07250	6.3655132E+03	3.8354407E+06	602.44
0.07503	6.3688606E+03	3.8363851E+06	602.37
0.07765	6.3404989E+03	3.8197353E+06	602.43
0.08003	6.2872107E+03	3.7888759E+06	602.63
0.08254	6.2175863E+03	3.7487047E+06	602.92
0.08512	6.1352700E+03	3.7014579E+06	603.31
0.08757	6.0708749E+03	3.6645461E+06	603.63
0.09001	6.0409120E+03	3.6473208E+06	603.77
0.09257	6.0539679E+03	3.6546606E+06	603.68
0.09509	6.1075851E+03	3.6851210E+06	603.33
0.09759	6.1876085E+03	3.7306914E+06	602.93
0.10002	6.2735718E+03	3.7796317E+06	602.47
0.10257	6.3664980E+03	3.8325765E+06	601.99

TABLE 6.2.1-28A (SHEET 2 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
0.10507	6.4429047E+03	3.8761465E+06	601.61
0.10758	6.4965895E+03	3.9067176E+06	601.35
0.11001	6.5270618E+03	3.9240262E+06	601.19
0.11253	6.5344748E+03	3.9281326E+06	601.14
0.11510	6.5204882E+03	3.9199857E+06	601.18
0.11761	6.4843044E+03	3.8991168E+06	601.32
0.12009	6.4325725E+03	3.8693579E+06	601.53
0.12256	6.3726344E+03	3.8349813E+06	601.79
0.12518	6.3034597E+03	3.7953523E+06	602.11
0.12761	6.2383094E+03	3.7580645E+06	602.42
0.13012	6.1820566E+03	3.7258948E+06	602.70
0.13264	6.1395327E+03	3.7015958E+06	602.91
0.13504	6.1127110E+03	3.6862746E+06	603.05
0.13756	6.0998879E+03	3.6789327E+06	603.11
0.14010	6.0988357E+03	3.6783086E+06	603.12
0.14267	6.1080626E+03	3.6835439E+06	603.06
0.14510	6.1240390E+03	3.6926378E+06	602.97
0.14758	6.1442581E+03	3.7041585E+06	602.87
0.15011	6.1667410E+03	3.7169743E+06	602.75
0.15260	6.1867163E+03	3.7283579E+06	602.64
0.15502	6.1992384E+03	3.7354879E+06	602.57
0.15755	6.2031636E+03	3.7377038E+06	602.55
0.16011	6.1939722E+03	3.7324198E+06	602.59
0.16265	6.1745723E+03	3.7213319E+06	602.69
0.16510	6.1423746E+03	3.7029436E+06	602.85
0.16751	6.1097626E+03	3.6843379E+06	603.02
0.17014	6.0750772E+03	3.6645786E+06	603.22
0.17253	6.0482471E+03	3.6492888E+06	603.36
0.17512	6.0289841E+03	3.6383136E+06	603.47
0.17760	6.0193184E+03	3.6328088E+06	603.52
0.18008	6.0197962E+03	3.6330808E+06	603.52
0.18258	6.0276833E+03	3.6375701E+06	603.48
0.18501	6.0400469E+03	3.6446001E+06	603.41
0.18766	6.0528858E+03	3.6518945E+06	603.33
0.19013	6.0646210E+03	3.6585557E+06	603.26
0.19254	6.0708106E+03	3.6620600E+06	603.22
0.19500	6.0717703E+03	3.6625777E+06	603.21
0.19754	6.0668335E+03	3.6597328E+06	603.24
0.20010	6.0570812E+03	3.6541472E+06	603.29
0.20274	6.0452208E+03	3.6473601E+06	603.35
0.20509	6.0330499E+03	3.6404113E+06	603.41
0.20760	6.0241617E+03	3.6353420E+06	603.46
0.21005	6.0191035E+03	3.6324522E+06	603.49
0.21354	6.0197127E+03	3.6327939E+06	603.48

TABLE 6.2.1-28A (SHEET 3 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
0.21510	6.0257956E+03	3.6362513E+06	603.45
0.21754	6.0357699E+03	3.6419285E+06	603.39
0.22016	6.0480859E+03	3.6489364E+06	603.32
0.22256	6.0625560E+03	3.6571731E+06	603.24
0.22504	6.0749929E+03	3.6642500E+06	603.17
0.22753	6.0866417E+03	3.6708838E+06	603.10
0.23007	6.0951080E+03	3.6757012E+06	603.06
0.23259	6.1001899E+03	3.6785898E+06	603.05
0.23508	6.1011008E+03	3.6791040E+06	603.02
0.23753	6.0973306E+03	3.6769487E+06	603.04
0.24023	6.0876426E+03	3.6714129E+06	603.09
0.24255	6.0753055E+03	3.6643848E+06	603.16
0.24503	6.0564102E+03	3.6536132E+06	603.26
0.24761	6.0364519E+03	3.6422456E+06	603.38
0.25009	6.0184816E+03	3.6320047E+06	603.48
0.25280	6.0028077E+03	3.6230838E+06	603.56
0.25504	5.9931687E+03	3.6175960E+06	603.62
0.25759	5.9900445E+03	3.6158252E+06	603.64
0.26004	5.9938317E+03	3.6179873E+06	603.62
0.26255	6.0039391E+03	3.6237539E+06	603.56
0.26501	6.0188653E+03	3.6322491E+06	603.48
0.26759	6.0373297E+03	3.6427624E+06	603.37
0.27011	6.0551031E+03	3.6528848E+06	603.27
0.27258	6.0711548E+03	3.6620230E+06	603.18
0.27506	6.0821145E+03	3.6682550E+06	603.12
0.27752	6.0879002E+03	3.6715416E+06	603.09
0.28003	6.0867031E+03	3.6708478E+06	603.09
0.28252	6.0784821E+03	3.6661507E+06	603.14
0.28504	6.0651856E+03	3.6585621E+06	603.21
0.28754	6.0434821E+03	3.6461959E+06	603.33
0.29006	6.0196102E+03	3.6325934E+06	603.46
0.29255	5.9947341E+03	3.6184236E+06	603.60
0.29520	5.9687892E+03	3.6036635E+06	603.75
0.29758	5.9450244E+03	3.5901412E+06	603.89
0.30004	5.9252478E+03	3.5788953E+06	604.01
0.30284	5.9052411E+03	3.5675125E+06	604.13
0.30504	5.8913813E+03	3.5596397E+06	604.21
0.30760	5.8782215E+03	3.5521680E+06	604.29
0.31001	5.8700234E+03	3.5475147E+06	604.34
0.31256	5.8639105E+03	3.5440436E+06	604.38
0.31501	5.8607978E+03	3.5422734E+06	604.40
0.31753	5.8599636E+03	3.5417959E+06	604.41
0.32004	5.8608011E+03	3.5422668E+06	604.40
0.32254	5.8629988E+03	3.5435071E+06	604.38
0.32505	5.8656482E+03	3.5450037E+06	604.37

TABLE 6.2.1-28A (SHEET 4 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
0.32784	5.8686407E+03	3.5466905E+06	604.35
0.33009	5.8714918E+03	3.5482967E+06	604.33
0.33273	5.8741682E+03	3.5498034E+06	604.31
0.33508	5.8771774E+03	3.5514986E+06	604.29
0.33756	5.8802735E+03	3.5532429E+06	604.26
0.34009	5.8841700E+03	3.5554426E+06	604.24
0.34260	5.8884789E+03	3.5578797E+06	604.21
0.34507	5.8930780E+03	3.5604779E+06	604.18
0.34755	5.8975952E+03	3.5630314E+06	604.15
0.35027	5.9014291E+03	3.5651946E+06	604.12
0.35257	5.9039809E+03	3.5666279E+06	604.11
0.35506	5.9044769E+03	3.5668906E+06	604.10
0.35763	5.9024245E+03	3.5657013E+06	604.11
0.36011	5.8983668E+03	3.5633740E+06	604.13
0.36265	5.8905550E+03	3.5589069E+06	604.17
0.36502	5.8818991E+03	3.5539585E+06	604.22
0.36763	5.8702901E+03	3.5473402E+06	604.29
0.37011	5.8585785E+03	3.5406607E+06	604.35
0.37259	5.8478945E+03	3.5345723E+06	604.42
0.37512	5.8384907E+03	3.5292171E+06	604.47
0.37757	5.8315699E+03	3.5252690E+06	604.51
0.38001	5.8283142E+03	3.5234085E+06	604.53
0.38255	5.8284433E+03	3.5234795E+06	604.53
0.38504	5.8325211E+03	3.5257837E+06	604.50
0.38764	5.8405465E+03	3.5303363E+06	604.45
0.39003	5.8503172E+03	3.5358907E+06	604.39
0.39255	5.8632583E+03	3.5432350E+06	604.31
0.39507	5.8769944E+03	3.5510359E+06	604.23
0.39752	5.8901754E+03	3.5585146E+06	604.14
0.40002	5.9026209E+03	3.5655867E+06	604.07
0.40266	5.9140894E+03	3.5720883E+06	604.00
0.40506	5.9222442E+03	3.5767163E+06	603.95
0.40761	5.9272907E+03	3.5795690E+06	603.91
0.41010	5.9287654E+03	3.5803886E+06	603.90
0.41261	5.9266016E+03	3.5791393E+06	603.91
0.41515	5.9207163E+03	3.5757700E+06	603.94
0.41756	5.9120146E+03	3.5708002E+06	603.99
0.42010	5.9007105E+03	3.5643532E+06	604.05
0.42264	5.8873278E+03	3.5567264E+06	604.13
0.42510	5.8733889E+03	3.5487826E+06	604.21
0.42776	5.8596323E+03	3.5409490E+06	604.30
0.43002	5.8458384E+03	3.5330901E+06	604.38
0.43273	5.8331063E+03	3.5258467E+06	604.45
0.43505	5.8221311E+03	3.5196022E+06	604.52

TABLE 6.2.1-28A (SHEET 5 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
0.43751	5.8130794E+03	3.5144558E+06	604.58
0.44011	5.8051154E+03	3.5099211E+06	604.63
0.44254	5.7994338E+03	3.5066831E+06	604.66
0.44512	5.7943623E+03	3.5037894E+06	604.69
0.44756	5.7908671E+03	3.5017915E+06	604.71
0.45006	5.7884098E+03	3.5003828E+06	604.72
0.45255	5.7868114E+03	3.4994613E+06	604.73
0.45501	5.7860305E+03	3.4990035E+06	604.73
0.45755	5.7859404E+03	3.4989362E+06	604.73
0.46008	5.7865023E+03	3.4992415E+06	604.72
0.46255	5.7876727E+03	3.4998863E+06	604.71
0.46505	5.7891053E+03	3.5006850E+06	604.70
0.46756	5.7908624E+03	3.5016617E+06	604.69
0.47006	5.7926250E+03	3.5026427E+06	604.67
0.47253	5.7940101E+03	3.5034098E+06	604.66
0.47514	5.7949974E+03	3.5039478E+06	604.65
0.47751	5.7952080E+03	3.5040456E+06	604.65
0.48017	5.7945401E+03	3.5036413E+06	604.65
0.48257	5.7927826E+03	3.5026177E+06	604.65
0.48509	5.7901457E+03	3.5010969E+06	604.66
0.48757	5.7864539E+03	3.4989748E+06	604.68
0.49011	5.7818731E+03	3.4963467E+06	604.71
0.49257	5.7768716E+03	3.4934807E+06	604.74
0.49505	5.7713451E+03	3.4903136E+06	604.77
0.49760	5.7657954E+03	3.4871343E+06	604.80
0.50011	5.7603602E+03	3.4840234E+06	604.83
0.51009	5.7491923E+03	3.4775926E+06	604.88
0.52009	5.7609480E+03	3.4841862E+06	604.79
0.53006	5.7844690E+03	3.4974703E+06	604.63
0.54005	5.7987721E+03	3.5055003E+06	604.52
0.55000	5.7970116E+03	3.5044053E+06	604.52
0.56006	5.7863331E+03	3.4982404E+06	604.57
0.57002	5.7741241E+03	3.4912139E+06	604.63
0.58007	5.7631540E+03	3.4848951E+06	604.69
0.59036	5.7548799E+03	3.4801119E+06	604.72
0.60004	5.7492097E+03	3.4768060E+06	604.75
0.61010	5.7426034E+03	3.4729615E+06	604.77
0.62004	5.7325199E+03	3.4671373E+06	604.82
0.63014	5.7211947E+03	3.4605989E+06	604.87
0.64006	5.7162332E+03	3.4576773E+06	604.89
0.65003	5.7176191E+03	3.4583568E+06	604.86
0.66012	5.7204578E+03	3.4598599E+06	604.82
0.67003	5.7223865E+03	3.4608454E+06	604.79
0.68006	5.7257890E+03	3.4626698E+06	604.75

TABLE 6.2.1-28A (SHEET 6 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
0.69002	5.7317272E+03	3.4659330E+06	604.69
0.70010	5.7366181E+03	3.4686004E+06	604.64
0.71004	5.7366207E+03	3.4684965E+06	604.62
0.72003	5.7320245E+03	3.4657852E+06	604.64
0.73012	5.7251012E+03	3.4617556E+06	604.66
0.74006	5.7172129E+03	3.4571775E+06	604.70
0.75003	5.7094890E+03	3.4526916E+06	604.73
0.76011	5.7035213E+03	3.4491984E+06	604.75
0.77015	5.6997968E+03	3.4469781E+06	604.75
0.78006	5.6960407E+03	3.4447350E+06	604.76
0.79006	5.6906934E+03	3.4415867E+06	604.77
0.80005	5.6856760E+03	3.4386253E+06	604.79
0.81021	5.6845992E+03	3.4379041E+06	604.78
0.82005	5.6883128E+03	3.4399046E+06	604.73
0.83010	5.6942403E+03	3.4431581E+06	604.67
0.84013	5.6985107E+03	3.4454743E+06	604.63
0.85000	5.6994327E+03	3.4458899E+06	604.60
0.86014	5.6970829E+03	3.4444488E+06	604.60
0.87001	5.6929918E+03	3.4420216E+06	604.61
0.88010	5.6890670E+03	3.4396914E+06	604.61
0.89006	5.6860123E+03	3.4378527E+06	604.62
0.90008	5.6830007E+03	3.4360407E+06	604.62
0.91009	5.6783432E+03	3.4332928E+06	604.63
0.92011	5.6715216E+03	3.4293170E+06	604.66
0.93018	5.6647085E+03	3.4253454E+06	604.68
0.94003	5.6629152E+03	3.4242285E+06	604.68
0.95003	5.6673076E+03	3.4266195E+06	604.63
0.96010	5.6722227E+03	3.4293052E+06	604.58
0.97011	5.6735480E+03	3.4299524E+06	604.55
0.98001	5.6715705E+03	3.4287262E+06	604.55
0.99004	5.6708125E+03	3.4281936E+06	604.53
1.00005	5.6717795E+03	3.4286424E+06	604.51
1.05002	5.6542380E+03	3.4181930E+06	604.54
1.10009	5.6578617E+03	3.4197733E+06	604.43
1.15011	5.6427925E+03	3.4107498E+06	604.44
1.20005	5.6426635E+03	3.4102125E+06	604.36
1.25004	5.6325971E+03	3.4040406E+06	604.35
1.30006	5.6245751E+03	3.3990528E+06	604.32
1.35009	5.6163063E+03	3.3939444E+06	604.30
1.40023	5.6056208E+03	3.3874824E+06	604.30
1.45004	5.5990362E+03	3.3833512E+06	604.27
1.50006	5.5888967E+03	3.3772132E+06	604.27
1.55004	5.5798754E+03	3.3717233E+06	604.26
1.60010	5.5703163E+03	3.3659471E+06	604.26

TABLE 6.2.1-28A (SHEET 7 OF 7)

Time (s)	Mass Flow (lb/s)	Energy Flow (Btu/s)	Average Enthalpy (Btu/lb)
1.65004	5.5670531E+03	3.3637635E+06	604.23
1.70008	5.5583218E+03	3.3584925E+06	604.23
1.75012	5.5601180E+03	3.3591994E+06	604.16
1.80010	5.5588433E+03	3.3581573E+06	604.11
1.85006	5.5580249E+03	3.3573797E+06	604.06
1.90001	5.5513019E+03	3.3532687E+06	604.05
1.95012	5.5450010E+03	3.3494269E+06	604.04
2.00000	5.5358334E+03	3.3439853E+06	604.06

TABLE 6.2.1-29

BLOWDOWN MASS AND ENERGY RELEASES
DOUBLE-ENDED PUMP SUCTION

Time	Break Path No. 1 Flow		Break Path No. 2 Flow	
(s)	(lbm/s)	Thousand (Btu/s)	(lbm/s)	Thousand (Btu/s)
0.0000	0.0	0.0	0.0	0.0
0.0503	42452.7	23717.3	23884.1	13269.7
0.100	42477.6	23781.0	22007.5	12268.7
0.250	43738.3	24766.2	24698.8	13790.4
0.450	45532.2	26371.7	23441.7	13118.1
0.700	44738.7	26657.0	20526.6	11504.7
1.30	37135.2	23390.3	18674.9	10501.9
2.50	26961.5	18469.6	18221.0	10254.8
2.80	22569.4	15708.3	17541.1	9880.5
3.40	19872.4	13967.1	16413.6	9263.4
4.30	18259.2	12735.7	14784.9	8363.0
5.25	17440.3	11908.7	13610.2	7705.3
5.50	17631.0	11941.9	14324.4	8111.6
5.75	17013.2	11406.1	14341.5	8120.9
6.25	13138.8	9669.6	13750.2	7783.1
7.00	17339.3	12063.4	13198.4	7473.1
7.75	19110.4	12856.5	12481.5	7069.1
8.25	25395.3	16695.2	11754.4	6654.1
9.00	25795.2	16541.7	10565.4	5976.8
9.50	25151.6	15997.0	9857.1	5568.7
10.0	23378.7	14868.0	9265.5	5221.3
10.3	11791.1	7528.6	9398.9	5305.8
10.5	8184.8	5542.7	9303.3	5279.1
10.8	7267.3	5102.7	9492.4	5392.7
11.3	8748.9	5693.9	9678.6	5491.4
11.5	8904.4	6167.3	9314.0	5277.8
11.8	6683.6	5609.3	9465.3	5376.3
13.0	6218.1	5121.7	8558.9	4905.3
15.0	4748.8	4248.4	6899.7	3906.8
16.0	3841.2	3967.5	6259.4	3268.9
17.8	1792.5	2239.4	3863.0	1754.3
18.5	1391.3	1750.4	2980.7	1162.3
19.3	884.7	1118.0	3387.0	1209.2
20.3	286.0	363.9	722.3	259.2
22.0	0.0	0.0	0.0	0.0

TABLE 6.2.1-30

BLOWDOWN MASS AND ENERGY RELEASES
DOUBLE-ENDED HOT LEG GUILLOTINE

Time	Break Path No. 1 Flow		Break Path No. 2 Flow	
(s)	(lbm/s)	Thousand (Btu/s)	(lbm/s)	Thousand (Btu/s)
0.0000	0.0	0.0	0.0	0.0
0.0507	38706.4	25536.9	28739.9	18622.6
0.100	42320.7	27947.5	28199.4	18281.1
0.200	36861.1	24360.0	23814.9	15352.6
0.350	35128.6	23213.6	20763.4	13095.8
0.750	33808.0	22364.6	18400.5	10939.9
1.30	31743.0	21475.4	17941.3	10257.0
2.10	28464.4	19660.5	18631.7	10405.6
3.20	24775.4	17144.8	18081.1	10086.5
4.20	23104.0	15666.3	16889.2	9499.1
4.70	23182.3	15399.6	16063.0	9078.4
5.00	23877.7	15568.4	15239.8	8640.2
5.25	17131.5	12528.8	14889.5	8469.0
6.00	18769.3	12994.2	13227.6	7574.8
6.50	20547.3	13761.0	12435.3	7146.3
6.75	27819.8	18329.9	12036.4	6926.3
7.00	29507.8	19265.7	11669.9	6724.5
7.25	29030.5	18672.1	11298.0	6518.5
7.75	29436.5	18610.1	10299.7	5966.2
8.75	28906.2	18096.5	8432.4	4957.7
9.25	27483.4	17015.2	7583.8	4509.8
9.50	18993.4	11798.8	7184.9	4303.1
9.75	16879.4	10322.2	6812.6	4112.8
10.0	9611.9	7793.2	6495.8	3954.9
10.3	10818.6	7971.5	6269.8	3844.4
10.8	10674.0	7894.7	5971.6	3685.0
12.5	10937.3	7822.4	4928.9	3150.3
13.0	8893.1	6668.0	4565.7	2976.4
14.0	7453.4	6124.6	3710.3	2579.7
14.8	5067.7	4829.4	2957.8	2212.7
17.5	1500.8	1780.3	1361.6	1486.4
18.5	817.9	1017.8	1091.9	1296.4
21.0	76.3	99.2	96.8	122.3
24.3	630.3	776.0	252.8	318.8
25.0	0.0	0.0	0.0	0.0

TABLE 6.2.1-31

REFLOOD MASS AND ENERGY RELEASES
DOUBLE-ENDED PUMP SUCTION - MINIMUM SAFETY INJECTION

Time	Break Path No. 1 Flow		Break Path No. 2 Flow	
(s)	(lbm/s)	Thousand (Btu/s)	(lbm/s)	Thousand (Btu/s)
22.0	0.0	0.0	0.0	0.0
23.0	0.0	0.0	0.0	0.0
23.1	89.1	105.2	0.0	0.0
23.4	22.5	26.6	0.0	0.0
24.5	78.5	92.6	0.0	0.0
27.1	147.3	173.9	0.0	0.0
28.1	195.5	230.9	1135.8	166.3
28.8	456.0	540.8	4373.5	687.1
29.1	471.9	559.8	4495.8	721.1
30.1	470.7	558.5	4480.6	727.1
31.1	462.4	548.5	4404.8	718.4
35.1	427.9	507.3	4081.8	678.3
39.1	397.7	471.3	3787.7	640.7
43.1	372.0	440.6	3529.2	607.4
44.1	397.3	470.8	3810.3	638.0
46.1	386.4	457.7	3701.2	623.8
47.1	381.2	451.5	3648.9	617.0
49.1	371.3	439.8	3548.6	603.8
53.1	353.3	418.3	3363.3	579.5
54.1	297.6	352.1	2675.8	502.8
54.3	292.6	346.1	2674.8	495.6
55.1	511.4	607.1	347.3	288.1
56.1	516.2	612.9	349.2	291.2
60.1	456.5	541.5	322.0	254.7
64.1	406.8	482.1	299.3	224.9
68.1	364.0	431.1	280.0	199.7
69.1	354.3	419.5	275.7	194.0
77.1	288.6	341.4	246.5	156.3
85.1	241.0	284.8	225.8	129.9
99.1	189.8	224.2	204.0	102.6
111.1	167.9	198.2	194.9	91.3
125.1	156.7	185.0	190.3	85.6
141.1	153.4	181.0	188.8	83.7
167.1	155.3	183.4	189.2	84.2
225.8	165.4	195.3	192.5	88.2

TABLE 6.2.1-32

REFLOOD MASS AND ENERGY RELEASES
DOUBLE-ENDED PUMP SUCTION - MAXIMUM SAFETY INJECTION

Time	Break Path No. 1 Flow		Break Path No. 2 Flow	
(s)	(lbm/s)	Thousand (Btu/s)	(lbm/s)	Thousand (Btu/s)
22.0	0.0	0.0	0.0	0.0
23.0	0.0	0.0	0.0	0.0
23.1	89.1	105.2	0.0	0.0
23.4	22.5	26.6	0.0	0.0
24.5	78.5	92.6	0.0	0.0
27.1	147.3	173.9	0.0	0.0
28.1	195.5	230.9	1135.8	166.3
28.8	456.0	540.8	4373.5	687.1
29.1	471.9	559.8	4495.8	721.1
30.1	470.7	558.5	4480.6	727.1
31.1	462.4	548.5	4404.8	718.4
35.1	427.9	507.3	4081.8	678.3
39.1	397.7	471.3	3787.7	640.7
43.1	372.0	440.6	3529.2	607.4
44.1	435.3	516.2	4209.5	683.2
46.1	424.4	503.1	4104.6	669.4
50.1	404.7	479.5	3911.0	643.8
54.1	387.2	458.7	3736.3	620.6
55.2	367.4	435.1	474.6	196.0
56.2	370.6	438.9	467.4	196.4
58.2	368.9	436.9	471.6	196.2
66.2	361.9	428.6	488.5	195.5
81.2	346.5	410.2	524.2	194.9
87.2	338.7	400.9	541.0	195.1
101.2	315.6	373.4	586.5	196.4
109.2	298.7	353.4	617.3	198.0
117.2	278.8	329.7	652.2	200.6
125.2	255.1	301.6	691.9	204.4
127.2	248.5	293.8	702.6	205.6
135.2	218.8	258.6	750.0	211.4
143.2	181.4	214.3	807.3	219.8
145.2	170.3	201.1	823.9	222.6
147.2	165.9	195.8	831.1	222.9
157.2	161.5	190.7	840.8	220.6
189.2	148.1	174.8	870.9	213.9

TABLE 6.2.1-33

PRINCIPAL PARAMETERS DURING REFLOOD
DOUBLE-ENDED PUMP SUCTION - MINIMUM SAFETY INJECTION

Time (s)	Flooding Temp (°F)	Flooding Rate (in./s)	Carryover Fraction	Core Height (ft)	Downcomer Height (ft)	Flow Fraction	Injection Total (lbm/s)	Injection Accumulator (lbm/s)	Injection Spill (lbm/s)	Enthalpy (Btu/lbm)
22.0	240.8	0.000	0.000	0.00	0.00	0.250	0.0	0.0	0.0	0.00
22.8	236.2	21.570	0.000	0.62	1.50	0.000	7488.3	7488.3	0.0	99.54
23.0	233.1	23.702	0.000	1.10	1.37	0.000	7406.8	7406.8	0.0	99.54
23.4	231.8	2.948	0.130	1.34	2.00	0.217	7305.1	7305.1	0.0	99.54
23.5	231.7	2.968	0.151	1.36	2.30	0.243	7279.5	7279.5	0.0	99.54
24.3	231.2	2.596	0.312	1.50	4.76	0.328	7065.0	7065.0	0.0	99.54
25.1	230.8	2.520	0.414	1.61	7.06	0.347	6888.4	6888.4	0.0	99.54
28.8	229.3	4.517	0.643	2.00	16.05	0.587	5728.4	5728.4	0.0	99.54
29.1	229.1	4.542	0.654	2.04	16.06	0.586	5632.9	5632.9	0.0	99.54
30.1	228.5	4.349	0.681	2.16	16.07	0.584	5468.5	5468.5	0.0	99.54
33.5	227.0	3.903	0.722	2.50	16.07	0.577	5052.1	5052.1	0.0	99.54
39.7	226.1	3.476	0.745	3.00	16.07	0.561	4467.3	4467.3	0.0	99.54
43.1	228.2	3.315	0.749	3.25	16.07	0.552	4206.3	4206.3	0.0	99.54
44.1	226.3	3.450	0.751	3.32	16.07	0.566	4523.8	4042.5	0.0	99.37
46.7	226.5	3.349	0.753	3.50	16.07	0.561	4356.7	3872.3	0.0	99.36
54.3	228.0	2.807	0.754	4.00	16.07	0.510	3214.7	2712.7	0.0	99.29
55.1	228.3	4.027	0.759	4.06	15.85	0.623	453.4	0.0	0.0	97.96
61.1	230.6	3.488	0.759	4.52	13.91	0.617	468.2	0.0	0.0	97.96
68.7	235.2	2.874	0.756	5.00	12.19	0.607	485.5	0.0	0.0	97.96
78.1	241.9	2.331	0.753	5.50	10.87	0.592	498.6	0.0	0.0	97.96
89.6	249.9	1.901	0.749	6.00	10.09	0.572	507.2	0.0	0.0	97.96
105.1	258.2	1.594	0.746	6.57	9.87	0.549	511.9	0.0	0.0	97.96
118.5	263.9	1.476	0.746	7.00	10.07	0.538	513.5	0.0	0.0	97.96
135.1	269.8	1.421	0.750	7.50	10.53	0.533	514.1	0.0	0.0	97.96
152.2	275.0	1.408	0.756	8.00	11.07	0.533	514.2	0.0	0.0	97.96
157.1	276.4	1.407	0.758	8.14	11.24	0.534	514.2	0.0	0.0	97.96
171.1	280.0	1.409	0.763	8.53	11.70	0.536	514.1	0.0	0.0	97.96
188.0	283.8	1.416	0.769	9.00	12.26	0.539	513.9	0.0	0.0	97.96
207.1	287.5	1.423	0.777	9.51	12.87	0.542	513.6	0.0	0.0	97.96
225.8	290.6	1.431	0.784	10.00	13.46	0.545	513.4	0.0	0.0	97.96

TABLE 6.2.1-34

PRINCIPAL PARAMETERS DURING REFLOOD
DOUBLE-ENDED PUMP SUCTION - MAXIMUM SAFETY INJECTION

Time (s)	Flooding Temp (°F)	Flooding Rate (in./s)	Carryover Fraction	Core Height (ft)	Downcomer Height (ft)	Flow Fraction	Injection Total (lbm/s)	Injection Accumulator (lbm/s)	Injection Spill (lbm/s)	Enthalpy (Btu/lbm)
22.0	240.8	0.000	0.000	0.00	0.00	0.250	0.0	0.0	0.0	0.00
22.8	236.2	21.570	0.000	0.62	1.50	0.000	7488.3	7488.3	0.0	99.54
23.0	233.1	23.702	0.000	1.10	1.37	0.000	7406.8	7406.8	0.0	99.54
23.4	231.8	2.948	0.130	1.34	2.00	0.217	7305.1	7305.1	0.0	99.54
23.5	231.7	2.968	0.151	1.36	2.30	0.243	7279.5	7279.5	0.0	99.54
24.3	231.2	2.596	0.312	1.50	4.76	0.328	7065.0	7065.0	0.0	99.54
25.1	230.8	2.520	0.414	1.61	7.06	0.347	6888.4	6888.4	0.0	99.54
28.8	229.3	4.517	0.643	2.00	16.05	0.587	5728.4	5728.4	0.0	99.54
29.1	229.1	4.542	0.654	2.04	16.06	0.586	5632.9	5632.9	0.0	99.54
30.1	228.5	4.349	0.681	2.16	16.07	0.584	5468.5	5468.5	0.0	99.54
33.5	227.0	3.903	0.722	2.50	16.07	0.577	5052.1	5052.1	0.0	99.54
39.7	226.1	3.476	0.745	3.00	16.07	0.561	4467.3	4467.3	0.0	99.54
43.1	226.2	3.315	0.749	3.25	16.07	0.552	4206.3	4206.3	0.0	99.54
44.1	226.2	3.666	0.752	3.32	16.07	0.583	4981.3	3925.1	0.0	99.20
46.5	226.5	3.571	0.754	3.50	16.07	0.579	4831.0	3770.9	0.0	99.19
53.6	227.7	3.340	0.757	4.00	16.07	0.568	4444.2	3374.7	0.0	99.16
55.2	228.1	3.232	0.758	4.10	16.07	0.557	1076.5	0.0	0.0	97.96
56.2	228.3	3.238	0.758	4.17	16.07	0.557	1074.8	0.0	0.0	97.96
62.2	231.0	3.182	0.760	4.55	16.07	0.558	1076.1	0.0	0.0	97.96
69.4	235.8	3.114	0.763	5.00	16.07	0.558	1077.7	0.0	0.0	97.96
78.2	243.2	3.023	0.766	5.54	16.07	0.558	1079.9	0.0	0.0	97.96
86.3	250.4	2.929	0.769	6.00	16.07	0.557	1082.3	0.0	0.0	97.96
97.2	258.7	2.785	0.773	6.60	16.07	0.554	1086.2	0.0	0.0	97.96
105.1	263.7	2.668	0.775	7.00	16.07	0.550	1089.6	0.0	0.0	97.96
117.2	270.1	2.462	0.778	7.58	16.07	0.537	1095.9	0.0	0.0	97.96
126.8	274.3	2.273	0.780	8.00	16.07	0.520	1101.7	0.0	0.0	97.96
141.2	279.4	1.924	0.781	8.56	16.07	0.471	1111.6	0.0	0.0	97.96
154.8	283.2	1.737	0.783	9.00	16.07	0.441	1115.6	0.0	0.0	97.96
173.2	287.4	1.643	0.789	9.56	16.07	0.441	1115.9	0.0	0.0	97.96
189.2	290.4	1.564	0.794	10.00	16.07	0.441	1116.2	0.0	0.0	97.96

TABLE 6.2.1-35

POST-REFLOOD MASS AND ENERGY RELEASES
DOUBLE-ENDED PUMP SUCTION - MINIMUM SAFETY INJECTION

Time	Break Path No. 1 Flow		Break Path No. 2 Flow	
(s)	(lbm/s)	Thousand (Btu/s)	(lbm/s)	Thousand (Btu/s)
225.9	238.4	298.4	281.8	134.5
260.9	237.1	296.8	283.0	133.4
295.9	234.4	293.4	285.8	132.6
300.9	234.8	293.9	285.4	132.3
310.9	233.6	292.4	286.6	132.2
315.9	233.8	292.7	286.4	131.9
325.9	233.2	291.9	287.0	131.6
330.9	232.4	290.9	287.8	131.7
340.9	232.3	290.8	287.9	131.2
360.9	230.3	288.2	289.9	130.9
365.9	230.6	288.6	289.6	130.6
375.9	230.0	287.9	290.2	130.3
290.9	228.4	285.9	291.8	130.1
395.9	228.6	286.1	291.6	129.8
425.9	226.0	282.9	294.2	129.2
430.9	226.0	282.9	294.2	129.0
465.9	223.6	279.9	296.5	128.1
615.9	223.4	279.6	296.8	128.0
616.0	98.7	122.6	421.5	160.9
645.9	97.9	121.6	422.3	159.6
795.9	93.7	116.3	426.5	157.6
800.9	93.5	116.2	426.6	157.3
945.9	90.3	112.1	429.9	153.5
950.9	90.2	112.0	430.0	153.1
160.9	86.5	107.4	433.7	152.6
1280.9	84.6	104.9	435.6	148.6
1305.9	84.2	104.4	436.0	148.0
1425.9	82.2	102.0	438.0	148.8
1570.9	82.1	101.9	438.1	147.0
1571.0	79.0	90.9	441.2	58.4
3600.0	64.0	73.7	456.2	61.1
3600.1	50.7	58.4	469.5	46.0
3952.2	50.7	58.4	469.5	46.0
3952.3	52.3	60.3	462.3	61.6
9999.9	52.3	60.3	462.3	61.6
10000.0	38.2	43.9	476.5	63.5
100000.0	20.4	23.5	494.3	65.9
1000000.0	8.7	10.1	505.9	67.4

TABLE 6.2.1-36

POST-REFLOOD MASS AND ENERGY RELEASES
DOUBLE-ENDED PUMP SUCTION - MAXIMUM SAFETY INJECTION

Time	Break Path No. 1 Flow		Break Path No. 2 Flow	
(s)	(lbm/s)	Thousand (Btu/s)	(lbm/s)	Thousand (Btu/s)
189.2	162.3	202.1	957.3	218.8
199.2	161.7	201.3	957.9	218.5
224.2	161.9	201.5	957.7	217.2
259.2	161.0	200.4	958.7	215.9
274.2	161.1	200.5	958.5	215.1
289.2	160.3	199.6	959.3	214.6
299.2	160.9	200.3	958.7	214.0
334.2	159.4	198.4	960.3	212.7
344.2	159.8	198.9	959.8	212.1
359.2	159.6	198.6	960.1	211.4
374.2	158.4	197.2	961.2	211.0
389.2	158.8	197.7	960.8	210.1
409.2	157.9	196.5	961.8	209.4
419.2	158.3	197.0	961.4	208.8
439.2	157.3	195.9	962.3	213.2
459.2	157.7	196.3	961.9	212.1
509.2	156.4	194.6	963.3	209.7
514.2	156.8	195.2	962.8	209.3
559.2	155.6	193.7	964.0	207.1
569.2	155.9	194.1	963.7	206.5
639.2	154.6	192.5	965.0	207.6
919.2	154.6	192.5	965.0	207.6
919.3	91.9	113.5	1027.7	217.7
924.2	91.8	113.3	1027.8	217.3
999.2	90.1	111.3	1029.5	214.3
1064.2	89.1	109.9	1030.6	214.5
1219.2	86.6	106.8	1033.1	214.5
1299.2	85.3	105.2	1034.3	212.9
1430.6	85.3	105.2	1034.3	212.9
1430.7	81.9	94.2	1037.7	115.2
2397.3	81.9	94.2	1037.7	115.2
2397.4	84.8	97.6	1016.2	170.8
3599.0	84.8	97.6	1016.2	170.8
3600.0	65.0	74.8	1036.0	174.1
3600.1	53.4	61.4	1047.7	157.5
10000.0	38.8	44.7	1062.2	159.6
100000.0	20.7	23.9	1080.3	162.4
1000000.0	8.9	10.2	1092.1	164.1

TABLE 6.2.1-37

MASS BALANCE
DOUBLE-ENDED PUMP SUCTION - MINIMUM SAFETY INJECTION

Time (s)	0.00	22.00	22.00	225.84	620.90	1570.94	3600.00
				Mass (Thousand lbm)			
Initial Mass (In RCS and ACC)	735.26	735.26	735.26	735.26	735.26	735.26	735.26
Added Mass (Pumped Injection)	0.00	0.00	0.00	92.43	297.90	791.36	1835.65
Total Available	735.26	735.26	735.26	827.68	1033.16	1526.62	2570.91
Distribution							
• Reactor Coolant	513.22	48.53	67.98	125.36	125.36	125.36	125.36
• Accumulator	222.04	176.45	156.99	0.00	0.00	0.00	0.00
Effluent							
• Break Flow	0.00	510.28	510.28	691.11	896.58	1390.04	2434.33
• ECCS Spill	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Accountable	735.26	735.25	735.25	816.46	1021.94	1515.40	2559.69

TABLE 6.2.1-38

MASS BALANCE
DOUBLE-ENDED PUMP SUCTION - MAXIMUM SAFETY INJECTION

	Time (s)	0.00	22.00	22.00	189.19	924.20	1430.60	3600.00
				Mass (Thousand lbm)				
Initial	In RCS and ACC	735.26	735.26	735.26	735.26	735.26	735.26	735.26
Added Mass	Pumped Injection	0.00	0.00	0.00	159.76	982.69	1547.31	3935.89
	Total Added	0.00	0.00	0.00	159.76	982.69	1547.31	3935.89
Total Available		735.26	735.26	735.26	895.02	1717.95	2282.57	4671.15
Distribution	Reactor Coolant	513.22	48.53	67.98	130.18	130.18	130.18	130.18
	Accumulator	222.04	176.45	156.99	0.00	0.00	0.00	0.00
	Total Contents	735.26	224.97	224.97	130.18	130.18	130.18	130.18
Effluent	Break Flow	0.00	510.28	510.28	753.61	1576.54	2141.16	4529.74
	ECCS Spill	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Total Effluent	0.00	510.28	510.28	753.61	1576.54	2141.16	4529.74
Total Accountable		735.26	735.25	735.25	883.80	1706.72	2271.35	4659.93

TABLE 6.2.1-39

MASS BALANCE
DOUBLE-ENDED HOT LEG GUILLOTINE

	Time (s)	0.00	25.00	25.00
			Mass (Thousand lbm)	
Initial	In RCS and ACC	735.26	735.26	735.26
Added Mass	Pumped Injection	0.00	0.00	0.00
	Total Added	0.00	0.00	0.00
Total Available		735.26	735.26	735.26
Distribution	Reactor Coolant	513.22	93.74	113.19
	Accumulator	222.04	144.38	124.93
	Total Contents	735.26	238.12	238.12
Effluent	Break Flow	0.00	497.13	497.13
	ECCS Spill	0.00	0.00	0.00
	Total Effluent	0.00	497.13	497.13
Total Accountable		735.26	735.25	735.25

TABLE 6.2.1-40

ENERGY BALANCE
DOUBLE-ENDED PUMP SUCTION - MINIMUM SAFETY INJECTION

	Time (s)	0.00	22.00	22.00	225.84	620.90	1570.94	3600.00
				Energy (Million Btu)				
Initial Energy	In RCS, ACC, S Gen	898.43	898.43	898.43	898.43	898.43	898.43	898.43
Added Energy	Pumped Injection	0.00	0.00	0.00	9.05	29.18	79.98	219.17
	Decay Heat	0.00	8.68	8.68	33.02	70.83	142.86	263.30
	Heat from Secondary	0.00	-1.78	-1.78	-1.78	0.92	6.51	6.51
	Total Added	0.00	6.90	6.90	40.89	100.94	229.35	488.98
Total Available		898.43	905.33	905.33	939.32	999.37	1127.78	1387.41
Distribution	Reactor Coolant	309.49	13.70	15.63	34.74	34.74	34.74	34.74
	Accumulator	22.10	17.56	15.63	0.00	0.00	0.00	0.00
	Core Stored	25.04	13.26	13.26	4.95	4.75	4.29	3.33
	Primary Metal	154.32	146.70	146.70	120.18	88.47	64.63	49.37
	Secondary Metal	114.85	114.42	114.42	105.35	85.36	57.36	43.51
	Steam Generator	272.63	277.93	277.93	251.64	200.58	135.68	103.99
	Total Contents	898.43	583.57	583.57	516.86	413.89	296.70	234.93
Effluent	Break Flow	0.00	321.76	321.76	410.78	573.79	819.39	1140.79
	ECCS Spill	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Total Effluent	0.00	321.76	321.76	410.78	573.79	819.39	1140.79
Total Accountable		898.43	905.33	905.33	927.63	987.68	1116.09	1375.72

TABLE 6.2.1-41

ENERGY BALANCE
DOUBLE-ENDED PUMP SUCTION - MAXIMUM SAFETY INJECTION

	Time (s)	0.00	22.00	22.00	189.19	924.20	1430.60	3600.00
				Energy (Million Btu)				
Initial Energy	In RCS, ACC, S Gen	898.43	898.43	898.43	898.43	898.43	898.43	898.43
Added Energy	Pumped Injection	0.00	0.00	0.00	15.65	96.26	158.86	517.84
	Decay Heat	0.00	8.68	8.68	29.71	95.70	133.17	263.31
	Heat From Secondary	0.00	-1.78	-1.78	-1.78	3.25	5.86	5.86
	Total Added	0.00	6.90	6.90	43.58	195.22	297.88	787.01
Total Available		898.43	905.33	905.33	942.00	1093.65	1196.31	1685.44
Distribution	Reactor Coolant	309.49	13.70	15.63	36.09	36.09	36.09	36.09
	Accumulator	22.10	17.56	15.63	0.00	0.00	0.00	0.00
	Core Stored	25.04	13.26	13.26	4.95	4.75	4.48	3.33
	Primary Metal	154.32	146.70	146.70	119.02	79.46	65.95	49.38
	Secondary Metal	114.85	114.42	114.42	103.30	73.72	57.94	43.50
	Steam Generator	272.63	277.93	277.93	245.74	171.66	136.25	103.29
	Total Contents	898.43	583.57	583.57	509.09	365.67	300.71	235.59
Effluent	Break Flow	0.00	321.76	321.76	421.23	716.30	883.92	1438.17
	ECCS Spill	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Total Effluent	0.00	321.76	321.76	421.23	716.30	883.92	1438.17
Total Accountable		898.43	905.33	905.33	930.32	1081.97	1184.63	1673.76

TABLE 6.2.1-42

ENERGY BALANCE
DOUBLE-ENDED HOT LEG GUILLOTINE

	Time (s)	0.00	25.00	25.00
		Energy (Million Btu)		
Initial Energy	In RCS, ACC, S Gen	898.43	898.43	898.43
Added Energy	Pumped Injection	0.00	0.00	0.00
	Decay Heat	0.00	9.63	9.63
	Heat From Secondary	0.00	-4.06	-4.06
	Total Added	0.00	5.57	5.57
Total Available		898.43	904.00	904.00
Distribution	Reactor Coolant	309.49	22.68	24.62
	Accumulator	22.10	14.37	12.44
	Core Stored	25.04	10.07	10.07
	Primary Metal	154.32	144.41	144.41
	Secondary Metal	114.85	112.08	112.08
	Steam Generator	272.63	270.80	270.80
	Total Contents	898.43	574.41	574.41
Effluent	Break Flow	0.00	329.59	329.59
	ECCS Spill	0.00	0.00	0.00
	Total Effluent	0.00	329.59	329.59
Total Accountable		898.43	904.00	904.00

TABLE 6.2.1-43

SYSTEM PARAMETERS

PARAMETER	VALUE
Core inlet temperature (includes +6.0°F allowance for instrument error and deadband)	562.8°F
Core outlet temperature (includes +6.0°F allowance for instrument error and deadband)	631.04°F
Initial steam generator steam pressure	969.6 psia
Assumed maximum containment back pressure	66.7 psia
Pumped injection, assumed for FROTH:	
Minimum (lb/s)	514.67
Maximum (lb/s)	1101.03

TABLE 6.2.1-44

THROUGH

TABLE 6.2.1-59

DELETED

TABLE 6.2.1-60

SPECTRUM OF SECONDARY SYSTEM PIPE RUPTURES ANALYZED[a]

Case	Size	Type Of Rupture	% Power[b]
1	Full	Double-ended	102
2	Full	Double-ended	70
3	Full	Double-ended	30
4	Full	Double-ended	0
5	0.60 ft^2	Double-ended with entrainment	102
6	0.53 ft^2	Double-ended with entrainment	70
7	0.36 ft^2	Double-ended with entrainment	30
8	0.20 ft^2	Double-ended with entrainment	0
9	0.33 ft^2	Double-ended without entrainment	102
10	0.32 ft^2	Double-ended without entrainment	70
11	0.22 ft^2	Double-ended without entrainment	30
12	0.10 ft^2	Double-ended without entrainment	0
13	0.86 ft^2	Split rupture	102
14	0.908 ft^2	Split rupture	70
15	0.944 ft^2	Split rupture	30
16	0.40 ft^2	Split rupture	0

a. Case 1 through case 16 have assumed the loss of a diesel.
b. % power of 3579 MWt.

TABLE 6.2.1-61 (SHEET 1 OF 9)

MASS AND ENERGY RELEASE DATA FOR CASE 16 - PEAK CALCULATED CONTAINMENT PRESSURE FOR MSLB

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
0.00	0.00	0.000	9.40	843.43	1.005	18.60	799.79	0.954
0.20	906.37	1.077	9.60	842.21	1.003	18.80	798.43	0.953
0.40	904.86	1.075	9.80	841.85	1.003	19.00	797.53	0.952
0.60	903.06	1.073	10.00	841.43	1.002	19.20	797.05	0.951
0.80	901.31	1.071	10.20	839.56	1.000	19.40	795.86	0.950
1.00	899.95	1.070	10.40	838.11	0.999	19.60	795.70	0.950
1.20	898.72	1.068	10.60	836.90	0.997	19.80	794.60	0.948
1.40	897.06	1.066	10.80	836.71	0.997	20.00	793.65	0.947
1.60	894.80	1.064	11.00	836.34	0.997	20.20	792.95	0.946
1.80	892.96	1.062	11.20	834.53	0.994	20.40	792.33	0.946
2.00	892.40	1.061	11.40	833.05	0.993	20.60	791.61	0.945
2.20	890.84	1.059	11.60	831.83	0.991	20.80	791.36	0.945
2.40	888.62	1.057	11.80	831.34	0.991	21.00	791.16	0.944
2.60	886.71	1.054	12.00	829.79	0.989	21.20	790.40	0.943
2.80	885.95	1.054	12.20	828.86	0.988	21.40	789.33	0.942
3.00	884.39	1.052	12.40	827.84	0.987	21.60	788.93	0.942
3.20	882.89	1.050	13.00	824.87	0.983	21.80	788.56	0.941
3.40	880.80	1.048	13.20	823.95	0.982	22.00	787.66	0.940
3.60	879.46	1.046	13.40	822.96	0.981	22.20	787.82	0.940
3.80	878.70	1.045	13.60	822.04	0.980	22.40	786.59	0.939
4.00	877.29	1.044	13.80	821.04	0.979	22.60	786.22	0.939
4.20	875.24	1.041	14.00	820.14	0.978	22.80	786.48	0.939
4.40	873.61	1.039	14.20	819.17	0.977	23.00	785.80	0.938
4.80	870.88	1.036	14.40	818.88	0.976	23.20	785.75	0.938
5.00	869.46	1.035	14.60	817.85	0.975	23.40	784.42	0.936
5.20	869.10	1.034	14.80	816.40	0.973	23.60	783.92	0.936
5.40	867.83	1.033	15.00	815.61	0.973	24.00	783.12	0.935
5.60	865.86	1.030	15.20	815.16	0.972	24.20	782.74	0.934
5.80	864.34	1.029	15.40	814.13	0.971	24.80	781.65	0.933
6.20	861.82	1.026	15.60	812.69	0.969	25.40	780.51	0.932
6.40	860.80	1.025	15.80	811.88	0.968	26.00	779.34	0.930
6.60	860.07	1.024	16.00	811.45	0.968	26.40	778.53	0.930
6.80	858.87	1.022	16.20	810.44	0.967	27.00	777.29	0.928
7.00	857.05	1.020	16.40	809.04	0.965	27.60	776.01	0.927
7.20	855.98	1.019	16.60	808.20	0.964	28.20	774.70	0.925
7.40	855.36	1.018	16.80	807.86	0.964	28.80	773.36	0.924
7.60	854.30	1.017	17.00	806.87	0.962	29.80	771.07	0.921
7.80	853.01	1.016	17.20	805.45	0.961	30.40	769.67	0.919
8.00	851.60	1.014	17.40	804.57	0.960	31.40	767.32	0.917
8.20	850.38	1.013	17.60	804.24	0.959	32.60	764.48	0.913

TABLE 6.2.1-61 (SHEET 2 OF 9)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
8.40	848.89	1.011	17.80	803.29	0.958	34.80	759.23	0.907
8.60	847.83	1.010	18.00	801.91	0.957	39.00	749.16	0.895
8.80	847.27	1.009	18.20	801.00	0.956	41.20	743.90	0.889
9.00	845.58	1.007	18.40	800.77	0.955	43.40	738.67	0.883
45.60	733.49	0.877	116.20	578.27	0.695	128.40	506.00	0.609
47.80	728.34	0.871	116.40	576.70	0.693	128.80	504.21	0.607
50.00	723.23	0.865	116.60	575.15	0.691	129.20	502.44	0.604
52.00	718.64	0.860	116.80	573.61	0.689	129.60	500.70	0.602
54.20	713.63	0.854	117.00	572.09	0.687	129.80	499.84	0.601
56.40	708.68	0.848	117.20	570.59	0.685	130.20	498.13	0.599
58.60	703.79	0.842	117.40	569.10	0.684	130.60	496.45	0.597
60.80	698.94	0.837	117.60	567.63	0.682	130.80	495.62	0.596
62.60	695.01	0.832	117.80	566.17	0.680	131.20	493.98	0.594
62.80	694.74	0.832	118.00	564.73	0.679	131.60	492.37	0.592
65.00	689.99	0.826	118.20	563.31	0.677	131.80	491.57	0.591
67.20	685.30	0.821	118.40	561.90	0.675	132.20	489.99	0.590
69.40	680.65	0.815	118.60	560.50	0.674	132.60	488.43	0.588
71.60	676.05	0.810	118.80	559.12	0.672	132.80	487.66	0.587
73.80	671.49	0.805	119.00	557.76	0.670	133.20	486.14	0.585
74.80	669.43	0.802	119.20	556.40	0.669	133.40	485.39	0.584
76.00	666.98	0.799	119.40	555.07	0.667	133.80	483.90	0.582
78.20	662.52	0.794	119.60	553.74	0.666	134.20	482.44	0.581
80.40	658.10	0.789	119.80	552.43	0.664	134.40	481.71	0.580
82.40	654.13	0.784	120.00	551.14	0.662	134.80	480.28	0.578
84.60	649.79	0.779	120.20	549.85	0.661	135.20	478.87	0.576
86.80	645.51	0.774	120.40	548.58	0.659	135.60	477.48	0.575
89.00	641.26	0.769	120.60	547.33	0.658	136.00	476.11	0.573
91.20	637.06	0.764	120.80	546.08	0.656	136.40	474.76	0.571
93.40	632.91	0.759	121.00	544.85	0.655	136.80	473.43	0.570
95.40	629.16	0.755	121.20	543.63	0.653	137.20	472.12	0.568
97.60	625.09	0.750	121.40	542.42	0.652	137.40	471.47	0.567
99.80	621.06	0.745	121.60	541.22	0.651	137.80	470.19	0.566
102.00	617.06	0.740	121.80	540.04	0.649	138.40	468.31	0.564
104.20	613.12	0.736	122.20	537.70	0.646	138.80	467.07	0.562
106.40	609.21	0.731	122.60	535.40	0.644	139.40	465.25	0.560
108.40	605.70	0.727	123.00	533.14	0.641	139.80	464.06	0.559
110.60	601.89	0.723	123.40	530.92	0.638	140.40	462.31	0.557
111.60	600.17	0.720	123.60	529.83	0.637	141.00	460.60	0.554
112.80	598.12	0.718	124.00	527.67	0.635	141.60	458.92	0.552
114.00	596.09	0.716	124.40	525.54	0.632	142.20	457.28	0.550
114.20	595.62	0.715	124.80	523.45	0.629	142.60	456.20	0.549
114.40	593.24	0.712	125.20	521.39	0.627	143.20	454.62	0.547
114.60	591.49	0.710	125.40	520.37	0.626	143.60	453.59	0.546

TABLE 6.2.1-61 (SHEET 3 OF 9)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
114.80	589.76	0.708	125.80	518.36	0.623	144.20	452.06	0.544
115.00	588.06	0.706	126.20	516.38	0.621	144.60	451.06	0.543
115.20	586.38	0.704	126.40	515.40	0.620	145.20	449.59	0.541
115.40	584.72	0.702	126.80	513.46	0.618	145.60	448.63	0.540
115.60	583.08	0.700	127.20	511.55	0.615	146.20	447.21	0.538
115.80	581.46	0.698	127.60	509.68	0.613	146.80	445.82	0.537
116.00	579.85	0.696	128.00	507.83	0.611			
147.40	444.46	0.535	172.20	407.41	0.491	218.80	387.01	0.466
147.80	443.57	0.534	172.80	406.86	0.490	221.00	386.69	0.466
148.40	442.26	0.532	173.40	406.32	0.489	223.20	386.41	0.465
149.00	440.97	0.531	174.40	405.44	0.488	225.40	386.15	0.465
149.60	439.71	0.529	175.40	404.60	0.487	227.40	385.93	0.465
150.20	438.48	0.528	176.40	403.79	0.486	229.60	385.72	0.465
150.80	437.28	0.527	177.00	403.32	0.486	231.80	385.52	0.464
151.40	436.10	0.525	177.60	402.85	0.485	234.00	385.34	0.464
151.80	435.33	0.524	178.20	402.40	0.485	236.20	385.19	0.464
152.40	434.19	0.523	178.80	401.96	0.484	238.40	385.04	0.464
152.80	433.45	0.522	179.80	401.25	0.483	240.40	384.93	0.464
153.40	432.35	0.521	180.80	400.56	0.482	242.60	384.81	0.463
154.00	431.27	0.519	181.40	400.16	0.482	244.80	384.71	0.463
154.60	430.22	0.518	182.00	399.77	0.481	247.00	384.62	0.463
155.20	429.19	0.517	182.60	399.38	0.481	249.20	384.54	0.463
155.60	428.52	0.516	183.20	399.01	0.481	251.40	384.47	0.463
156.20	427.52	0.515	184.20	398.40	0.480	255.60	384.36	0.463
156.60	426.87	0.514	185.20	397.82	0.479	260.00	384.28	0.463
157.20	425.91	0.513	186.40	397.15	0.478	264.20	384.22	0.463
157.60	425.28	0.512	187.40	396.62	0.478	268.60	384.18	0.463
158.20	424.35	0.511	188.40	396.10	0.477	273.00	384.16	0.463
158.60	423.75	0.510	189.40	395.61	0.476	277.60	384.15	0.463
159.20	422.85	0.509	190.60	395.03	0.476	283.80	384.16	0.463
159.80	421.98	0.508	191.80	394.49	0.475	290.00	384.19	0.463
160.40	421.12	0.507	192.80	394.05	0.475	296.00	384.23	0.463
160.80	420.55	0.506	193.80	393.64	0.474	302.20	384.29	0.463
161.40	419.72	0.505	195.00	393.15	0.474	314.60	384.41	0.463
162.00	418.90	0.504	196.20	392.69	0.473	376.40	385.10	0.464
162.60	418.11	0.504	197.20	392.33	0.473	401.00	385.36	0.464
163.20	417.32	0.503	198.60	391.84	0.472	425.80	385.61	0.464
163.80	416.56	0.502	198.80	391.65	0.472	592.20	387.24	0.466
164.40	415.81	0.501	200.40	391.13	0.471	611.40	387.41	0.467
164.80	415.32	0.500	201.40	390.82	0.471	623.80	387.51	0.467
165.40	414.60	0.499	202.40	390.52	0.470	636.00	387.59	0.467
165.80	414.13	0.499	203.60	390.18	0.470	648.40	387.66	0.467

TABLE 6.2.1-61 (SHEET 4 OF 9)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
166.40	413.43	0.498	204.80	389.85	0.470	654.60	387.69	0.467
167.00	412.75	0.497	205.80	389.59	0.469	667.00	387.72	0.467
167.60	412.08	0.496	206.80	389.34	0.469	686.80	387.73	0.467
168.20	411.42	0.495	208.00	389.05	0.469	692.40	387.71	0.467
168.60	410.99	0.495	209.20	388.78	0.468	698.00	387.69	0.467
169.20	410.36	0.494	210.20	388.56	0.468	703.80	387.66	0.467
169.60	409.95	0.494	211.20	388.35	0.468	716.80	387.56	0.467
170.20	409.34	0.493	212.40	388.11	0.467	720.80	387.52	0.467
170.60	408.94	0.493	213.40	387.92	0.467	732.20	387.40	0.467
171.20	408.36	0.492	215.40	387.56	0.467	743.60	387.25	0.466
171.60	407.97	0.491	216.60	387.35	0.467	755.00	387.07	0.466
766.20	386.88	0.466	868.00	370.26	0.446	878.80	333.01	0.401
777.60	386.66	0.466	868.40	369.25	0.445	879.00	332.11	0.400
789.00	386.42	0.465	868.80	368.22	0.444	879.20	331.20	0.399
800.20	386.18	0.465	869.00	367.70	0.443	879.40	330.28	0.398
811.60	385.91	0.465	869.40	366.62	0.442	879.60	329.36	0.397
823.00	385.63	0.464	869.80	365.53	0.440	879.80	328.43	0.396
834.40	385.34	0.464	870.00	364.97	0.440	880.00	327.50	0.394
858.80	384.69	0.463	870.40	363.83	0.438	880.20	326.55	0.393
859.00	384.68	0.463	870.80	362.67	0.437	880.40	325.60	0.392
859.20	384.62	0.463	871.20	361.47	0.435	880.80	323.68	0.390
859.40	384.52	0.463	871.40	360.87	0.435	881.20	321.74	0.387
859.60	384.41	0.463	871.80	359.63	0.433	881.60	319.77	0.385
859.80	384.27	0.463	872.00	359.00	0.432	882.00	317.77	0.383
860.00	384.11	0.463	872.20	358.36	0.432	882.20	316.77	0.381
860.20	383.95	0.462	872.40	357.72	0.431	882.60	314.74	0.379
860.40	383.77	0.462	872.60	357.07	0.430	883.00	312.69	0.377
860.60	383.57	0.462	872.80	356.41	0.429	883.20	311.66	0.375
860.80	383.37	0.462	873.00	355.74	0.428	883.60	309.59	0.373
861.00	383.16	0.461	873.20	355.07	0.428	884.00	307.50	0.370
861.20	382.94	0.461	873.40	354.39	0.427	884.40	305.39	0.368
861.40	382.71	0.461	873.60	353.70	0.426	884.80	303.26	0.365
861.60	382.46	0.461	873.80	353.00	0.425	885.00	302.20	0.364
861.80	382.21	0.460	874.00	352.30	0.424	885.40	300.06	0.361
862.00	381.95	0.460	874.20	351.58	0.423	886.00	296.83	0.357
862.20	381.68	0.460	874.40	350.86	0.423	886.40	294.66	0.355
862.40	381.40	0.459	874.60	350.14	0.422	886.80	292.49	0.352
862.60	381.11	0.459	874.80	349.40	0.421	887.40	289.23	0.348
862.80	380.81	0.459	875.00	348.65	0.420	889.00	280.51	0.338
863.00	380.51	0.458	875.20	347.90	0.419	889.60	277.25	0.334
863.20	380.19	0.458	875.40	347.14	0.418	890.20	274.01	0.330
863.40	379.87	0.458	875.60	346.38	0.417	890.60	271.86	0.327
863.60	379.53	0.457	875.80	345.60	0.416	891.00	269.72	0.325

TABLE 6.2.1-61 (SHEET 5 OF 9)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
863.80	379.19	0.457	876.00	344.82	0.415	891.40	267.59	0.322
864.00	378.84	0.456	876.20	344.02	0.414	891.80	265.47	0.319
864.20	378.48	0.456	876.40	343.22	0.413	892.00	264.42	0.318
864.40	378.11	0.455	876.60	342.42	0.412	892.40	262.33	0.316
864.60	377.74	0.455	876.80	341.60	0.411	892.80	260.25	0.313
864.80	377.36	0.455	877.00	340.78	0.410	893.20	258.19	0.311
865.20	376.57	0.454	877.20	339.94	0.409	893.60	256.15	0.308
865.60	375.75	0.453	877.40	339.10	0.408	893.80	255.13	0.307
866.00	374.90	0.452	877.60	338.26	0.407	894.00	254.12	0.306
866.20	374.47	0.451	877.80	337.40	0.406	894.20	253.12	0.305
866.60	373.58	0.450	878.00	336.54	0.405	894.40	252.12	0.303
867.00	372.66	0.449	878.20	335.67	0.404	894.60	251.13	0.302
867.40	371.72	0.448	878.40	334.79	0.403	894.80	250.14	0.301
867.80	370.75	0.447	878.60	333.90	0.402	895.00	249.16	0.300
895.20	248.18	0.299	904.40	210.60	0.253	913.80	185.77	0.223
895.40	247.22	0.297	904.60	209.93	0.252	914.00	185.37	0.222
895.60	246.25	0.296	904.80	209.28	0.251	914.20	184.97	0.222
895.80	245.30	0.295	905.00	208.63	0.251	914.40	184.57	0.221
896.00	244.35	0.294	905.20	207.99	0.250	914.60	184.18	0.221
896.20	243.41	0.293	905.40	207.35	0.249	914.80	183.79	0.221
896.40	242.47	0.292	905.60	206.72	0.248	915.00	183.41	0.220
896.60	241.54	0.291	905.80	206.10	0.248	915.20	183.03	0.220
896.80	240.62	0.289	906.00	205.48	0.247	915.40	182.66	0.219
897.00	239.71	0.288	906.20	204.87	0.246	915.60	182.29	0.219
897.20	238.80	0.287	906.40	204.27	0.245	915.80	181.92	0.218
897.40	237.90	0.286	906.60	203.67	0.245	916.00	181.56	0.218
897.60	237.00	0.285	906.80	203.08	0.244	916.20	181.20	0.217
897.80	236.11	0.284	907.00	202.49	0.243	916.40	180.85	0.217
898.00	235.23	0.283	907.20	201.91	0.242	916.60	180.50	0.217
898.20	234.36	0.282	907.40	201.34	0.242	916.80	180.16	0.216
898.40	233.49	0.281	907.60	200.77	0.241	917.00	179.81	0.216
898.60	232.63	0.280	907.80	200.20	0.240	917.20	179.48	0.215
898.80	231.78	0.279	908.20	199.09	0.239	917.40	179.14	0.215
899.00	230.93	0.278	908.40	198.54	0.238	917.60	178.81	0.214
899.20	230.09	0.277	908.60	198.00	0.238	917.80	178.49	0.214
899.40	229.26	0.276	908.80	197.46	0.237	918.00	178.16	0.214
899.60	228.43	0.275	909.00	196.93	0.236	918.20	177.85	0.213
899.80	227.62	0.274	909.20	196.40	0.236	918.40	177.53	0.213
900.00	226.80	0.273	909.40	195.88	0.235	918.60	177.22	0.213
900.20	226.00	0.272	909.60	195.37	0.235	918.80	176.91	0.212
900.40	225.20	0.271	909.80	194.86	0.234	919.20	176.31	0.211
900.60	224.41	0.270	910.00	194.35	0.233	919.40	176.01	0.211
900.80	223.62	0.269	910.20	193.85	0.233	919.80	175.43	0.210

TABLE 6.2.1-61 (SHEET 6 OF 9)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
901.00	222.84	0.268	910.40	193.36	0.232	920.00	175.14	0.210
901.20	222.07	0.267	910.60	192.87	0.232	920.40	174.57	0.209
901.40	221.30	0.266	910.80	192.39	0.231	920.80	174.02	0.209
901.60	220.54	0.265	911.00	191.91	0.230	921.00	173.75	0.208
901.80	219.79	0.264	911.20	191.44	0.230	921.40	173.22	0.208
902.00	219.04	0.263	911.40	190.97	0.229	921.80	172.70	0.207
902.20	218.30	0.262	911.60	190.51	0.229	922.00	172.45	0.207
902.40	217.57	0.261	911.80	190.05	0.228	922.20	172.19	0.206
902.60	216.84	0.261	912.00	189.60	0.228	922.60	171.70	0.206
902.80	216.12	0.260	912.20	189.15	0.227	923.00	171.21	0.205
903.00	215.41	0.259	912.40	188.71	0.226	923.20	170.98	0.205
903.20	214.70	0.258	912.60	188.27	0.226	923.60	170.51	0.204
903.40	214.00	0.257	912.80	187.84	0.225	924.00	170.05	0.204
903.60	213.31	0.256	913.00	187.41	0.225	924.20	169.83	0.204
903.80	212.62	0.255	913.20	187.02	0.224	924.60	169.39	0.203
904.00	211.94	0.255	913.40	186.60	0.224	925.00	168.96	0.203
904.20	211.26	0.254	913.60	186.18	0.223	925.40	168.54	0.202
925.80	168.13	0.202	945.00	156.66	0.188	1001.60	151.86	0.182
926.00	167.93	0.201	945.80	156.42	0.187	1004.40	151.83	0.182
926.40	167.53	0.201	946.40	156.24	0.187	1007.40	151.82	0.182
926.80	167.15	0.200	947.00	156.07	0.187	1010.00	151.80	0.182
927.20	166.77	0.200	947.80	155.86	0.187	1015.80	151.78	0.182
927.60	166.40	0.199	948.40	155.70	0.186	1021.40	151.76	0.182
928.00	166.04	0.199	949.20	155.51	0.186	1027.20	151.75	0.182
928.40	165.69	0.199	950.00	155.32	0.186	1038.40	151.74	0.182
928.80	165.35	0.198	950.60	155.18	0.186	1070.80	151.73	0.182
929.00	165.18	0.198	951.20	155.05	0.186	1800.00	151.74	0.182
929.40	164.85	0.198	952.00	154.89	0.185	1800.20	151.78	0.182
929.80	164.53	0.197	952.80	154.73	0.185	1800.40	152.30	0.182
930.00	164.37	0.197	953.40	154.62	0.185	1800.60	152.41	0.182
930.40	164.06	0.197	954.20	154.48	0.185	1800.80	152.36	0.182
930.80	163.76	0.196	955.00	154.34	0.185	1801.00	152.17	0.182
931.00	163.61	0.196	955.80	154.21	0.185	1801.20	151.84	0.182
931.40	163.32	0.196	956.40	154.12	0.185	1801.40	151.40	0.181
931.80	163.03	0.195	957.00	154.03	0.184	1801.60	150.84	0.181
932.20	162.75	0.195	957.80	153.91	0.184	1801.80	150.19	0.180
932.60	162.48	0.195	958.40	153.83	0.184	1802.00	149.44	0.179
933.00	162.22	0.194	959.20	153.72	0.184	1802.20	148.61	0.178
933.60	161.83	0.194	960.00	153.62	0.184	1802.40	147.70	0.177
934.20	161.46	0.193	961.20	153.48	0.184	1802.60	146.72	0.176
934.60	161.22	0.193	962.00	153.40	0.184	1802.80	145.69	0.174
935.00	160.98	0.193	962.80	153.31	0.184	1803.00	144.58	0.173
935.40	160.75	0.193	963.40	153.25	0.184	1803.20	143.43	0.172

TABLE 6.2.1-61 (SHEET 7 OF 9)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
935.80	160.53	0.192	964.80	153.12	0.183	1803.40	142.22	0.170
936.40	160.21	0.192	966.20	153.00	0.183	1803.60	140.93	0.169
937.00	159.89	0.192	967.60	152.89	0.183	1803.80	139.04	0.166
937.40	159.69	0.191	969.00	152.79	0.183	1804.00	137.19	0.164
937.80	159.49	0.191	970.40	152.70	0.183	1804.20	135.43	0.162
938.40	159.21	0.191	971.80	152.62	0.183	1804.40	133.68	0.160
938.80	159.02	0.190	973.20	152.54	0.183	1804.60	131.94	0.158
939.20	158.84	0.190	974.60	152.47	0.183	1804.80	130.22	0.156
939.60	158.66	0.190	976.00	152.40	0.182	1805.00	128.51	0.154
940.00	158.49	0.190	977.40	152.35	0.182	1805.20	126.82	0.151
940.60	158.24	0.190	979.00	152.29	0.182	1805.40	125.14	0.149
941.20	158.00	0.189	980.40	152.24	0.182	1805.60	123.48	0.147
941.60	157.84	0.189	981.80	152.19	0.182	1805.80	121.84	0.145
942.00	157.69	0.189	983.20	152.15	0.182	1806.00	120.22	0.143
942.40	157.54	0.189	984.60	152.12	0.182	1806.20	118.61	0.142
942.80	157.39	0.189	987.40	152.05	0.182	1806.40	117.03	0.140
943.40	157.18	0.188	990.20	152.00	0.182	1806.60	115.46	0.138
943.80	157.05	0.188	993.00	151.95	0.182	1806.80	113.92	0.136
944.20	156.92	0.188	996.00	151.91	0.182	1807.00	112.39	0.134
944.60	156.79	0.188	998.80	151.88	0.182	1807.20	110.89	0.132
1807.40	109.42	0.130	1816.60	60.30	0.071	1826.80	31.41	0.037
1807.60	107.97	0.129	1816.80	59.52	0.070	1827.00	30.99	0.036
1807.80	106.55	0.127	1817.00	58.75	0.069	1827.20	30.57	0.036
1808.00	105.14	0.125	1817.20	58.00	0.068	1827.40	30.15	0.035
1808.20	103.76	0.124	1817.40	57.25	0.068	1827.60	29.75	0.035
1808.40	102.39	0.122	1817.60	56.52	0.067	1827.80	29.35	0.034
1808.60	101.05	0.120	1817.80	55.80	0.066	1828.00	28.95	0.034
1808.80	99.73	0.119	1818.00	55.10	0.065	1828.20	28.57	0.033
1809.00	98.45	0.117	1818.20	54.40	0.064	1828.40	28.19	0.033
1809.20	97.17	0.116	1818.40	53.72	0.063	1828.60	27.83	0.032
1809.40	95.92	0.114	1818.60	53.04	0.062	1828.80	27.47	0.032
1809.60	94.69	0.113	1818.80	52.38	0.062	1829.00	27.11	0.032
1809.80	93.48	0.111	1819.00	51.73	0.061	1829.20	26.77	0.031
1810.00	92.28	0.110	1819.20	51.09	0.060	1829.40	26.44	0.031
1810.20	91.11	0.108	1819.40	50.46	0.059	1829.60	26.11	0.030
1810.40	89.96	0.107	1819.60	49.85	0.059	1829.80	25.79	0.030
1810.60	88.83	0.106	1819.80	49.24	0.058	1830.00	25.49	0.030
1810.80	87.72	0.104	1820.00	48.66	0.057	1830.20	25.19	0.029
1811.00	86.64	0.103	1820.20	48.08	0.057	1830.40	24.89	0.029
1811.20	85.57	0.102	1820.40	47.49	0.056	1830.60	24.59	0.029
1811.40	84.53	0.100	1821.20	45.15	0.053	1830.80	24.30	0.028
1811.60	83.46	0.099	1821.60	43.97	0.052	1831.00	24.00	0.028
1811.80	82.42	0.098	1821.80	43.39	0.051	1831.20	23.71	0.028

TABLE 6.2.1-61 (SHEET 8 OF 9)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
1812.00	81.38	0.097	1822.00	42.81	0.050	1831.40	23.41	0.027
1812.20	80.35	0.095	1822.20	42.23	0.050	1831.60	23.13	0.027
1812.40	79.34	0.094	1822.40	41.66	0.049	1831.80	22.84	0.027
1812.60	78.32	0.093	1822.60	41.11	0.048	1832.00	22.55	0.026
1812.80	77.32	0.092	1822.80	40.55	0.048	1832.20	22.30	0.026
1813.00	76.33	0.090	1823.00	40.00	0.047	1832.40	22.06	0.026
1813.20	75.34	0.089	1823.20	39.46	0.046	1832.60	21.81	0.025
1813.40	74.37	0.088	1823.40	38.92	0.046	1832.80	21.47	0.025
1813.60	73.40	0.087	1823.60	38.39	0.045	1833.00	21.17	0.025
1813.80	72.45	0.086	1823.80	37.87	0.044	1833.20	20.89	0.024
1814.00	71.51	0.085	1824.00	37.54	0.044	1833.40	20.62	0.024
1814.20	70.58	0.084	1824.20	37.13	0.043	1833.60	20.35	0.024
1814.40	69.66	0.082	1824.40	36.71	0.043	1833.80	20.08	0.023
1814.60	68.75	0.081	1824.60	36.28	0.042	1834.00	19.82	0.023
1814.80	67.85	0.080	1824.80	35.81	0.042	1834.20	19.56	0.023
1815.00	66.96	0.079	1825.00	35.37	0.041	1834.60	19.04	0.022
1815.20	66.09	0.078	1825.20	34.93	0.041	1834.80	18.79	0.022
1815.40	65.22	0.077	1825.40	34.49	0.040	1835.00	18.53	0.021
1815.60	64.37	0.076	1825.80	33.60	0.039	1835.20	18.26	0.021
1815.80	63.53	0.075	1826.00	33.16	0.039	1835.40	18.00	0.021
1816.00	62.71	0.074	1826.20	32.71	0.038	1835.60	17.73	0.021
1816.20	61.89	0.073	1826.40	32.28	0.038	1835.80	17.46	0.020
1816.40	61.09	0.072	1826.60	31.84	0.037	1836.00	17.19	0.020
1836.20	16.92	0.020						
1836.40	16.64	0.019						
1836.60	16.35	0.019						
1836.80	16.06	0.019						
1837.00	15.77	0.018						
1837.20	15.47	0.018						
1837.40	15.17	0.018						
1837.60	14.86	0.017						
1837.80	14.55	0.017						
1838.00	14.24	0.016						
1838.20	13.91	0.016						
1838.40	13.59	0.016						
1838.60	13.25	0.015						
1838.80	12.91	0.015						
1839.00	12.57	0.014						
1839.20	12.22	0.014						
1839.40	11.86	0.014						
1839.60	11.49	0.013						
1839.80	11.12	0.013						
1840.00	10.73	0.012						

TABLE 6.2.1-61 (SHEET 9 OF 9)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
1840.20	10.34	0.012						
1840.40	9.93	0.011						
1840.60	9.51	0.011						
1840.80	9.08	0.010						
1841.00	8.62	0.010						
1841.20	8.13	0.009						
1841.40	7.59	0.009						
1841.60	6.98	0.008						
1841.80	6.17	0.007						
1842.00	0.00	0.000						
1900.00	0.00	0.000						

TABLE 6.2.1-62 (SHEET 1 OF 8)

MASS AND ENERGY RELEASE DATA FOR CASE 13 - PEAK CALCULATED CONTAINMENT TEMPERATURE FOR MSLB

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
0.0000	0.0000	0.0000	10.60	1318.0	1.574	23.00	1850.0	2.201
0.2000	1485.0	1.770	10.80	1314.0	1.570	23.50	1844.0	2.194
0.4000	1485.0	1.770	11.00	1314.0	1.570	24.00	1841.0	2.190
0.6000	1485.0	1.770	11.20	1309.0	1.564	24.50	1837.0	2.186
0.8000	1485.0	1.770	11.40	1309.0	1.564	25.00	1833.0	2.182
1.000	1480.0	1.764	11.60	1305.0	1.559	25.50	1829.0	2.177
1.200	1475.0	1.758	11.80	1300.0	1.559	26.00	1826.0	2.173
1.400	1470.0	1.752	12.00	1596.0	1.554	26.50	1822.0	2.169
1.600	1465.0	1.746	12.20	1590.0	1.907	27.00	1818.0	2.164
1.800	1460.0	1.741	12.40	1604.0	1.900	27.50	1814.0	2.160
2.000	1455.0	1.735	12.60	1613.0	1.916	28.00	1810.0	2.155
2.200	1451.0	1.730	12.80	1622.0	1.927	28.50	1806.0	2.150
2.400	1446.0	1.725	13.00	1631.0	1.937	29.00	1801.0	2.145
2.600	1442.0	1.719	13.20	1640.0	1.947	29.50	1797.0	2.140
2.800	1437.0	1.714	13.40	1648.0	1.957	30.00	1793.0	2.135
3.000	1433.0	1.709	13.60	1657.0	1.968	30.50	1788.0	2.130
3.200	1429.0	1.704	13.80	1665.0	1.977	31.00	1784.0	2.125
3.400	1425.0	1.699	14.00	1674.0	1.987	31.50	1779.0	2.119
3.600	1421.0	1.695	14.20	1682.0	1.997	32.00	1774.0	2.113
3.800	1417.0	1.690	14.40	1690.0	2.007	32.50	1769.0	2.108
4.000	1413.0	1.685	14.60	1698.0	2.016	33.00	1764.0	2.103
4.200	1409.0	1.681	14.80	1706.0	2.025	33.50	1760.0	2.097
4.400	1405.0	1.676	15.00	1714.0	2.035	34.00	1755.0	2.092
4.600	1405.0	1.676	15.20	1722.0	2.044	34.50	1750.0	2.086
4.800	1400.0	1.671	15.40	1730.0	2.053	35.00	1745.0	2.080
5.000	1395.0	1.665	15.60	1737.0	2.062	35.50	1741.0	2.075
5.200	1391.0	1.661	15.80	1744.0	2.070	36.00	1736.0	2.069
5.400	1387.0	1.656	16.00	1751.0	2.079	36.50	1731.0	2.064
5.600	1387.0	1.656	16.20	1758.0	2.087	37.00	1726.0	2.058
5.800	1383.0	1.650	16.40	1765.0	2.095	37.50	1721.0	2.053
6.000	1378.0	1.645	16.60	1772.0	2.103	38.00	1717.0	2.047
6.200	1374.0	1.641	16.80	1778.0	2.110	38.50	1712.0	2.042
6.400	1374.0	1.641	17.00	1784.0	2.118	39.00	1707.0	2.037
6.600	1370.0	1.635	17.20	1795.0	2.125	39.50	1703.0	2.031
6.800	1366.0	1.631	17.40	1803.0	2.138	40.00	1698.0	2.026
7.000	1366.0	1.631	17.60	1809.0	2.147	40.50	1693.0	2.020
7.200	1361.0	1.625	17.80	1815.0	2.154	41.00	1689.0	2.015
7.400	1357.0	1.620	18.00	1821.0	2.161	41.50	1684.0	2.009
7.600	1357.0	1.620	18.20	1826.0	2.167	42.00	1679.0	2.004
7.800	1352.0	1.615	18.40	1830.0	2.173	42.50	1675.0	1.999
8.000	1348.0	1.610	18.60	1834.0	2.178	43.00	1670.0	1.993
8.200	1348.0	1.610	18.80	1837.0	2.182	43.50	1666.0	1.988
8.400	1344.0	1.605	19.00	1837.0	2.186	44.00	1661.0	1.983
8.600	1344.0	1.605	19.20	1840.0	2.186	44.50	1656.0	1.978
8.800	1339.0	1.600	19.40	1843.0	2.190	45.00	1652.0	1.972
9.000	1335.0	1.595	19.60	1845.0	2.193	45.50	1648.0	1.967
9.200	1335.0	1.595	19.80	1847.0	2.195	46.00	1643.0	1.962
9.400	1331.0	1.590	20.00	1851.0	2.198	46.50	1639.0	1.957
9.600	1331.0	1.590	20.50	1851.0	2.202	47.00	1634.0	1.952
9.800	1326.0	1.585	21.00	1852.0	2.202	47.50	1630.0	1.947
10.000	1326.0	1.585	21.50	1852.0	2.204	48.00	1626.0	1.942
10.20	1322.0	1.579	22.00	1852.0	2.204	48.50	1621.0	1.937
10.40	1322.0	1.580	22.50	1852.0	2.204	49.00	1617.0	1.932

TABLE 6.2.1-62 (SHEET 2 OF 8)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
49.50	1613.0	1.927	76.00	943.2	1.136	105.0	866.6	1.044
50.00	1608.0	1.922	76.50	939.6	1.131	106.0	866.0	1.043
50.50	1604.0	1.917	77.00	936.1	1.127	107.0	865.3	1.042
51.00	1621.0	1.938	77.50	932.8	1.123	108.0	864.7	1.041
51.50	1570.0	1.878	78.00	929.7	1.120	109.0	864.2	1.041
52.00	1538.0	1.840	78.50	926.7	1.116	110.0	863.6	1.040
52.50	1507.0	1.804	79.00	923.9	1.113	111.0	863.1	1.040
53.00	1478.0	1.770	79.50	921.1	1.109	112.0	862.6	1.039
53.50	1450.0	1.737	80.00	918.5	1.106	113.0	862.1	1.038
54.00	1423.0	1.706	80.50	916.0	1.103	114.0	861.7	1.038
54.50	1399.0	1.677	81.00	913.6	1.100	115.0	861.2	1.037
55.00	1375.0	1.649	81.50	911.3	1.097	116.0	860.8	1.037
55.50	1353.0	1.623	82.00	909.1	1.095	117.0	860.4	1.036
56.00	1332.0	1.598	82.50	907.1	1.092	118.0	860.0	1.036
56.50	1312.0	1.574	83.00	905.1	1.090	119.0	859.7	1.035
57.00	1293.0	1.552	83.50	903.2	1.088	120.0	859.3	1.035
57.50	1275.0	1.531	84.00	901.3	1.085	121.0	859.0	1.035
58.00	1258.0	1.511	84.50	899.6	1.083	122.0	858.6	1.034
58.50	1241.0	1.491	85.00	897.9	1.081	123.0	858.3	1.034
59.00	1226.0	1.473	85.50	896.3	1.079	124.0	858.0	1.033
59.50	1211.0	1.455	86.00	894.8	1.078	125.0	857.7	1.033
60.00	1197.0	1.438	86.50	893.4	1.076	126.0	857.4	1.033
60.50	1183.0	1.422	87.00	892.0	1.074	127.0	857.1	1.032
61.00	1170.0	1.406	87.50	890.7	1.073	128.0	856.8	1.032
61.50	1157.0	1.391	88.00	889.4	1.071	129.0	856.6	1.032
62.00	1145.0	1.377	88.50	888.2	1.070	130.0	856.3	1.031
62.50	1133.0	1.363	89.00	887.0	1.068	131.0	856.1	1.031
63.00	1122.0	1.350	89.50	885.9	1.067	132.0	855.8	1.031
63.50	1111.0	1.337	90.00	884.8	1.066	133.0	855.6	1.030
64.00	1101.0	1.324	90.50	883.8	1.064	134.0	855.3	1.030
64.50	1091.0	1.313	91.00	882.8	1.063	135.0	855.1	1.030
65.00	1081.0	1.301	91.50	881.9	1.062	136.0	854.9	1.030
65.50	1072.0	1.290	92.00	881.0	1.061	137.0	854.7	1.029
66.00	1063.0	1.279	92.50	880.1	1.060	138.0	854.5	1.029
66.50	1055.0	1.269	93.00	879.3	1.059	139.0	854.3	1.029
67.00	1046.0	1.259	93.50	878.5	1.058	140.0	854.1	1.029
67.50	1038.0	1.249	94.00	877.7	1.057	141.0	853.9	1.028
68.00	1030.0	1.240	94.50	877.0	1.056	142.0	853.7	1.028
68.50	1023.0	1.231	95.00	876.3	1.055	143.0	853.5	1.028
69.00	1016.0	1.223	95.50	875.6	1.055	144.0	853.3	1.028
69.50	1009.0	1.215	96.00	874.9	1.054	145.0	853.1	1.028
70.00	1003.0	1.207	96.50	874.3	1.053	146.0	853.0	1.027
70.50	996.3	1.199	97.00	873.7	1.052	147.0	852.8	1.027
71.00	990.3	1.192	97.50	873.1	1.052	148.0	852.6	1.027
71.50	984.6	1.185	98.00	872.6	1.051	149.0	852.5	1.027
72.00	979.1	1.179	98.50	872.0	1.050	150.0	852.3	1.027
72.50	973.9	1.172	99.00	871.5	1.050	151.0	852.2	1.026
73.00	968.9	1.166	99.50	871.0	1.049	152.0	852.0	1.026
73.50	964.1	1.161	100.0	870.5	1.048	153.0	851.9	1.026
74.00	959.5	1.155	101.0	869.8	1.048	154.0	851.7	1.026
74.50	955.2	1.150	102.0	868.9	1.046	155.0	851.6	1.026
75.00	951.0	1.145	103.0	868.1	1.046	156.0	851.4	1.025
75.50	947.0	1.140	104.0	867.3	1.045	157.0	851.3	1.025

TABLE 6.2.1-62 (SHEET 3 OF 8)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
158.0	851.2	1.025	220.0	489.7	0.5890	324.0	150.6	0.1783
159.0	851.1	1.025	222.0	525.8	0.6326	326.0	150.3	0.1779
160.0	850.9	1.025	224.0	573.5	0.6900	328.0	150.2	0.1778
161.0	850.8	1.025	226.0	545.0	0.6551	330.0	150.4	0.1780
162.0	850.7	1.025	228.0	448.8	0.5388	332.0	150.2	0.1778
163.0	850.6	1.024	230.0	404.4	0.4860	334.0	150.1	0.1776
164.0	850.5	1.024	232.0	492.3	0.5918	336.0	150.0	0.1776
165.0	850.4	1.024	234.0	488.1	0.5863	338.0	150.1	0.1777
166.0	850.3	1.024	236.0	419.5	0.5031	340.0	150.0	0.1776
167.0	850.2	1.024	238.0	360.1	0.4317	342.0	149.9	0.1775
168.0	850.1	1.024	240.0	366.8	0.4401	344.0	149.9	0.1774
169.0	850.0	1.024	242.0	401.8	0.4819	346.0	150.0	0.1775
170.0	849.9	1.024	244.0	365.6	0.4377	348.0	149.9	0.1774
171.0	849.8	1.024	246.0	302.6	0.3619	350.0	149.9	0.1774
172.0	849.7	1.023	248.0	291.9	0.3492	352.0	149.8	0.1774
173.0	849.6	1.023	250.0	315.4	0.3774	354.0	149.9	0.1774
174.0	849.6	1.023	252.0	307.5	0.3673	356.0	149.9	0.1774
175.0	849.5	1.023	254.0	251.5	0.2998	358.0	149.8	0.1773
176.0	849.4	1.023	256.0	235.1	0.2807	360.0	149.8	0.1773
177.0	849.3	1.023	258.0	271.8	0.3243	362.0	149.8	0.1773
178.0	849.3	1.023	260.0	248.6	0.2962	364.0	149.8	0.1773
179.0	849.2	1.023	262.0	213.2	0.2536	366.0	149.8	0.1773
180.0	849.1	1.023	264.0	198.2	0.2357	368.0	149.8	0.1773
181.0	849.1	1.023	266.0	206.3	0.2454	370.0	149.8	0.1773
182.0	849.0	1.023	268.0	206.7	0.2455	372.0	149.8	0.1773
183.0	843.9	1.016	270.0	184.5	0.2191	374.0	149.8	0.1773
184.0	824.8	0.9935	272.0	178.0	0.2115	376.0	149.8	0.1773
185.0	807.5	0.9726	274.0	188.2	0.2235	378.0	149.8	0.1773
186.0	785.5	0.9461	276.0	180.1	0.2137	380.0	149.8	0.1773
187.0	759.7	0.9150	278.0	168.8	0.2002	382.0	149.8	0.1773
188.0	730.0	0.8792	280.0	165.2	0.1960	384.0	149.8	0.1773
189.0	696.7	0.8389	282.0	170.6	0.2023	386.0	149.8	0.1773
190.0	660.2	0.7948	284.0	166.8	0.1977	388.0	149.8	0.1773
191.0	625.6	0.7530	286.O	159.7	0.1892	390.0	149.8	0.1773
192.0	622.4	0.7494	288.0	158.3	0.1876	392.0	149.8	0.1773
193.0	639.9	0.7704	290.0	162.0	0.1920	394.0	149.8	0.1773
194.0	659.3	0.7939	292.0	159.3	0.1887	396.0	149.8	0.1773
195.0	672.4	0.8097	294.0	155.4	0.1840	398.0	149.8	0.1773
196.0	678.8	0.8174	296.0	154.4	0.1829	400.0	149.8	0.1773
197.0	676.2	0.8143	298.0	156.3	0.1851	402.0	149.8	0.1773
198.0	663.9	0.7994	300.0	155.1	0.1836	404.0	149.8	0.1773
199.0	647.0	0.7790	302.0	152.9	0.1810	406.0	149.8	0.1773
200.0	635.1	0.7646	304.0	152.3	0.1803	408.0	149.8	0.1773
202.0	627.1	0.7550	306.0	153.3	0.1816	410.0	149.8	0.1773
204.0	631.5	0.7603	308.0	152.7	0.1808	412.0	149.8	0.1773
206.0	646.0	0.7778	310.0	151.5	0.1793	414.0	149.8	0.1773
208.0	626.7	0.7543	312.0	151.2	0.1790	416.0	149.8	0.1773
210.0	580.6	0.6988	314.0	151.7	0.1796	418.0	149.8	0.1773
212.0	573.1	0.6898	316.0	151.4	0.1792	420.0	149.8	0.1773
214.0	610.7	0.7351	318.0	150.7	0.1784	422.0	149.8	0.1773
216.0	605.4	0.7286	320.0	150.5	0.1782	424.0	149.8	0.1773
218.0	557.0	0.6699	322.0	150.9	0.1786	426.0	149.8	0.1773

TABLE 6.2.1-62 (SHEET 4 OF 8)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
428.0	149.8	0.1773
430.0	149.8	0.1773
432.0	149.8	0.1773
436.0	149.8	0.1773
438.0	149.8	0.1773
440.0	149.8	0.1773
442.0	149.8	0.1773
444.0	149.8	0.1773
446.0	149.8	0.1773
448.0	149.8	0.1773
450.0	149.8	0.1773
452.0	149.8	0.1774
454.0	149.8	0.1774
456.0	149.8	0.1774
458.0	149.8	0.1774
460.0	149.8	0.1774
462.0	149.9	0.1774
464.0	149.9	0.1774
466.0	149.9	0.1774
468.0	149.9	0.1774
470.0	149.9	0.1774
472.0	149.9	0.1774
474.0	149.9	0.1774
476.0	149.9	0.1774
478.0	149.9	0.1774
480.0	149.9	0.1774
482.0	149.9	0.1774
484.0	149.9	0.1774
486.0	149.9	0.1774
488.0	149.9	0.1774
490.0	149.9	0.1774
492.0	149.9	0.1774
494.0	149.9	0.1774
496.0	149.9	0.1774
498.0	149.9	0.1774
500.0	149.9	0.1774
502.0	149.9	0.1774
504.0	149.9	0.1774
506.0	149.9	0.1774
508.0	149.9	0.1774
510.0	149.9	0.1774
512.0	149.9	0.1774
514.0	149.9	0.1774
516.0	149.9	0.1774
518.0	149.9	0.1774
520.0	149.9	0.1774
522.0	149.9	0.1774
524.0	149.9	0.1774
526.0	149.9	0.1774
528.0	149.9	0.1774
530.0	149.9	0.1774
532.0	149.9	0.1774
536.0	149.9	0.1774
538.0	149.9	0.1774
540.0	149.9	0.1774
542.0	149.9	0.1774
544.0	149.9	0.1774
546.0	149.9	0.1774
548.0	149.9	0.1774
550.0	149.9	0.1774
552.0	149.9	0.1774
554.0	149.9	0.1774
556.0	149.9	0.1774
558.0	149.9	0.1774
560.0	149.9	0.1774
562.0	149.9	0.1774
564.0	149.8	0.1774
566.0	149.9	0.1774
568.0	149.8	0.1774
570.0	149.8	0.1774
572.0	149.8	0.1774
574.0	149.8	0.1774
576.0	149.8	0.1774
578.0	149.8	0.1774
580.0	149.8	0.1774
582.0	149.8	0.1774
584.0	149.8	0.1774
586.0	149.8	0.1774
588.0	149.8	0.1774
590.0	149.8	0.1774
592.0	149.8	0.1774
594.0	149.8	0.1774
596.0	149.8	0.1774
598.0	149.8	0.1774
600.0	149.8	0.1774
602.0	149.8	0.1774
604.0	149.8	0.1774
606.0	149.8	0.1774
608.0	149.8	0.1774
610.0	149.8	0.1774
612.0	149.8	0.1774
614.0	149.8	0.1774
616.0	149.8	0.1774
618.0	149.8	0.1774
620.0	149.8	0.1774
622.0	149.8	0.1774
624.0	149.8	0.1774
626.0	149.8	0.1774
628.0	149.8	0.1774
630.0	149.8	0.1774
632.0	149.8	0.1774
634.0	149.8	0.1774
636.0	149.8	0.1774
638.0	149.8	0.1774
640.0	149.8	0.1774
642.0	149.8	0.1774
644.0	149.8	0.1774
646.0	149.8	0.1774
648.0	149.8	0.1774
650.0	149.8	0.1774
652.0	149.8	0.1774
654.0	149.8	0.1774
656.0	149.8	0.1774
658.0	149.8	0.1774
660.0	149.8	0.1774
662.0	149.8	0.1774
664.0	149.8	0.1774
666.0	149.8	0.1774
668.0	149.8	0.1774
670.0	149.8	0.1774
672.0	149.8	0.1774
674.0	149.8	0.1774
676.0	149.8	0.1774
678.0	149.8	0.1774
680.0	149.8	0.1774
682.0	149.8	0.1774
684.0	149.8	0.1774
686.0	149.8	0.1774
688.0	149.8	0.1774
690.0	149.8	0.1774
692.0	149.8	0.1774
694.0	149.8	0.1774
696.0	149.8	0.1774
698.0	149.8	0.1774
700.0	149.8	0.1774
702.0	149.8	0.1774
704.0	149.8	0.1774
706.0	149.8	0.1774
708.0	149.8	0.1774
710.0	149.8	0.1774
712.0	149.8	0.1774
714.0	149.8	0.1774
716.0	149.8	0.1774
718.0	149.8	0.1774
720.0	149.8	0.1774
722.0	149.8	0.1774
724.0	149.8	0.1774
726.0	149.8	0.1774
728.0	149.8	0.1774
730.0	149.8	0.1774
732.0	149.8	0.1774
734.0	149.8	0.1774
736.0	149.8	0.1774
738.0	149.8	0.1774
740.0	149.8	0.1774

TABLE 6.2.1-62 (SHEET 5 OF 8)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
742.0	149.8	0.1774	848.0	149.8	0.1774	954.0	149.8	0.1774
744.0	149.8	0.1774	850.0	149.8	0.1774	956.0	149.8	0.1774
746.0	149.8	0.1774	852.0	149.8	0.1774	958.0	149.8	0.1774
748.0	149.8	0.1774	854.0	149.8	0.1774	960.0	149.8	0.1774
750.0	149.8	0.1774	856.0	149.8	0.1774	962.0	149.8	0.1774
752.0	149.8	0.1774	858.0	149.8	0.1774	964.0	149.8	0.1774
754.0	149.8	0.1774	860.0	149.8	0.1774	966.0	149.8	0.1774
756.0	149.8	0.1774	862.0	149.8	0.1774	968.0	149.8	0.1774
758.0	149.8	0.1774	864.0	149.8	0.1774	970.0	149.8	0.1774
760.0	149.8	0.1774	866.0	149.8	0.1774	972.0	149.8	0.1774
762.0	149.8	0.1774	868.0	149.8	0.1774	974.0	149.8	0.1774
764.0	149.8	0.1774	870.0	149.8	0.1774	976.0	149.8	0.1774
766.0	149.8	0.1774	872.0	149.8	0.1774	978.0	149.8	0.1774
768.0	149.8	0.1774	874.0	149.8	0.1774	980.0	149.8	0.1774
770.0	149.8	0.1774	876.0	149.8	0.1774	982.0	149.8	0.1774
772.0	149.8	0.1774	878.0	149.8	0.1774	984.0	149.8	0.1774
774.0	149.8	0.1774	880.0	149.8	0.1774	986.0	149.8	0.1774
776.0	149.8	0.1774	882.0	149.8	0.1774	988.0	149.8	0.1774
778.0	149.8	0.1774	884.0	149.8	0.1774	990.0	149.8	0.1774
780.0	149.8	0.1774	886.0	149.8	0.1774	992.0	149.8	0.1774
782.0	149.8	0.1774	888.0	149.8	0.1774	994.0	149.8	0.1774
784.0	149.8	0.1774	890.0	149.8	0.1774	996.0	149.8	0.1774
786.0	149.8	0.1774	892.0	149.8	0.1774	998.0	149.8	0.1774
788.0	149.8	0.1774	894.0	149.8	0.1774	1000.0	149.8	0.1774
790.0	149.8	0.1774	896.0	149.8	0.1774	1002.0	149.8	0.1774
792.0	149.8	0.1774	898.0	149.8	0.1774	1004.0	149.8	0.1774
794.0	149.8	0.1774	900.0	149.8	0.1774	1006.0	149.8	0.1774
796.0	149.8	0.1774	902.0	149.8	0.1774	1008.0	149.8	0.1774
798.0	149.8	0.1774	904.0	149.8	0.1774	1010.0	149.8	0.1774
800.0	149.8	0.1774	906.0	149.8	0.1774	1012.0	149.8	0.1774
802.0	149.8	0.1774	908.0	149.8	0.1774	1014.0	149.8	0.1774
804.0	149.8	0.1774	910.0	149.8	0.1774	1016.0	149.8	0.1774
806.0	149.8	0.1774	912.0	149.8	0.1774	1018.0	149.8	0.1774
808.0	149.8	0.1774	914.0	149.8	0.1774	1020.0	149.8	0.1774
810.0	149.8	0.1774	916.0	149.8	0.1774	1022.0	149.8	0.1774
812.0	149.8	0.1774	918.0	149.8	0.1774	1024.0	149.8	0.1774
814.0	149.8	0.1774	920.0	149.8	0.1774	1026.0	149.8	0.1774
816.0	149.8	0.1774	922.0	149.8	0.1774	1028.0	149.8	0.1774
818.0	149.8	0.1774	924.0	149.8	0.1774	1030.0	149.8	0.1774
820.0	149.8	0.1774	926.0	149.8	0.1774	1032.0	149.8	0.1774
822.0	149.8	0.1774	928.0	149.8	0.1774	1034.0	149.8	0.1774
824.0	149.8	0.1774	930.0	149.8	0.1774	1036.0	149.8	0.1774
826.0	149.8	0.1774	932.0	149.8	0.1774	1038.0	149.8	0.1774
828.0	149.8	0.1774	934.0	149.8	0.1774	1040.0	149.8	0.1774
830.0	149.8	0.1774	936.0	149.8	0.1774	1042.0	149.8	0.1774
832.0	149.8	0.1774	938.0	149.8	0.1774	1044.0	149.8	0.1774
834.0	149.8	0.1774	940.0	149.8	0.1774	1046.0	149.8	0.1774
836.0	149.8	0.1774	942.0	149.8	0.1774	1048.0	149.8	0.1774
838.0	149.8	0.1774	944.0	149.8	0.1774	1050.0	149.8	0.1774
840.0	149.8	0.1774	946.0	149.8	0.1774	1052.0	149.8	0.1774
842.0	149.8	0.1774	948.0	149.8	0.1774	1054.0	149.8	0.1774
844.0	149.8	0.1774	950.0	149.8	0.1774	1058.0	149.8	0.1774
846.0	149.8	0.1774	952.0	149.8	0.1774	1060.0	149.8	0.1774

TABLE 6.2.1-62 (SHEET 6 OF 8)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
1062.0	149.8	0.1774	1168.0	149.8	0.1774	1274.0	149.8	0.1774
1064.0	149.8	0.1774	1170.0	149.8	0.1774	1276.0	149.8	0.1774
1066.0	149.8	0.1774	1172.0	149.8	0.1774	1278.0	149.8	0.1774
1068.0	149.8	0.1774	1174.0	149.8	0.1774	1280.0	149.8	0.1774
1070.0	149.8	0.1774	1176.0	149.8	0.1774	1282.0	149.8	0.1774
1072.0	149.8	0.1774	1178.0	149.8	0.1774	1284.0	149.8	0.1774
1074.0	149.8	0.1774	1180.0	149.8	0.1774	1286.0	149.8	0.1774
1076.0	149.8	0.1774	1182.0	149.8	0.1774	1288.0	149.8	0.1774
1078.0	149.8	0.1774	1184.0	149.8	0.1774	1290.0	149.8	0.1774
1080.0	149.8	0.1774	1186.0	149.8	0.1774	1292.0	149.8	0.1774
1082.0	149.8	0.1774	1188.0	149.8	0.1774	1294.0	149.8	0.1774
1084.0	149.8	0.1774	1190.0	149.8	0.1774	1296.0	149.8	0.1774
1086.0	149.8	0.1774	1192.0	149.8	0.1774	1298.0	149.8	0.1774
1088.0	149.8	0.1774	1194.0	149.8	0.1774	1300.0	149.8	0.1774
1090.0	149.8	0.1774	1196.0	149.8	0.1774	1302.0	149.8	0.1774
1092.0	149.8	0.1774	1198.0	149.8	0.1774	1304.0	149.8	0.1774
1094.0	149.8	0.1774	1200.0	149.8	0.1774	1306.0	149.8	0.1774
1096.0	149.8	0.1774	1202.0	149.8	0.1774	1308.0	149.8	0.1774
1098.0	149.8	0.1774	1204.0	149.8	0.1774	1310.0	149.8	0.1774
1100.0	149.8	0.1774	1206.0	149.8	0.1774	1312.0	149.8	0.1774
1102.0	149.8	0.1774	1208.0	149.8	0.1774	1314.0	149.8	0.1774
1104.0	149.8	0.1774	1210.0	149.8	0.1774	1316.0	149.8	0.1774
1106.0	149.8	0.1774	1212.0	149.8	0.1774	1318.0	149.8	0.1774
1108.0	149.8	0.1774	1214.0	149.8	0.1774	1320.0	149.8	0.1774
1110.0	149.8	0.1774	1216.0	149.8	0.1774	1322.0	149.8	0.1774
1112.0	149.8	0.1774	1218.0	149.8	0.1774	1324.0	149.8	0.1774
1114.0	149.8	0.1774	1220.0	149.8	0.1774	1326.0	149.8	0.1774
1116.0	149.8	0.1774	1222.0	149.8	0.1774	1328.0	149.8	0.1774
1118.0	149.8	0.1774	1224.0	149.8	0.1774	1330.0	149.8	0.1774
1120.0	149.8	0.1774	1226.0	149.8	0.1774	1332.0	149.8	0.1774
1122.0	149.8	0.1774	1228.0	149.8	0.1774	1334.0	149.8	0.1774
1124.0	149.8	0.1774	1230.0	149.8	0.1774	1336.0	149.8	0.1774
1126.0	149.8	0.1774	1232.0	149.8	0.1774	1338.0	149.8	0.1774
1128.0	149.8	0.1774	1234.0	149.8	0.1774	1340.0	149.8	0.1774
1130.0	149.8	0.1774	1236.0	149.8	0.1774	1342.0	149.8	0.1774
1132.0	149.8	0.1774	1238.0	149.8	0.1774	1344.0	149.8	0.1774
1134.0	149.8	0.1774	1240.0	149.8	0.1774	1346.0	149.8	0.1774
1136.0	149.8	0.1774	1242.0	149.8	0.1774	1348.0	149.8	0.1774
1138.0	149.8	0.1774	1244.0	149.8	0.1774	1350.0	149.8	0.1774
1140.0	149.8	0.1774	1246.0	149.8	0.1774	1352.0	149.8	0.1774
1142.0	149.8	0.1774	1248.0	149.8	0.1774	1354.0	149.8	0.1774
1144.0	149.8	0.1774	1250.0	149.8	0.1774	1356.0	149.8	0.1774
1146.0	149.8	0.1774	1252.0	149.8	0.1774	1358.0	149.8	0.1774
1148.0	149.8	0.1774	1254.0	149.8	0.1774	1360.0	149.8	0.1774
1150.0	149.8	0.1774	1256.0	149.8	0.1774	1362.0	149.8	0.1774
1152.0	149.8	0.1774	1258.0	149.8	0.1774	1364.0	149.8	0.1774
1154.0	149.8	0.1774	1260.0	149.8	0.1774	1366.0	149.8	0.1774
1158.0	149.8	0.1774	1262.0	149.8	0.1774	1368.0	149.8	0.1774
1160.0	149.8	0.1774	1264.0	149.8	0.1774	1370.0	149.8	0.1774
1162.0	149.8	0.1774	1268.0	149.8	0.1774	1372.0	149.8	0.1774
1164.0	149.8	0.1774	1270.0	149.8	0.1774	1374.0	149.8	0.1774
1166.0	149.8	0.1774	1272.0	149.8	0.1774	1376.0	149.8	0.1774
						1378.0	149.8	0.1774
						1380.0	149.8	0.1774

TABLE 6.2.1-62 (SHEET 7 OF 8)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
1382.0	149.8	0.1774	1488.0	149.8	0.1774	1594.0	149.8	0.1774
1384.0	149.8	0.1774	1490.0	149.8	0.1774	1596.0	149.8	0.1774
1386.0	149.8	0.1774	1492.0	149.8	0.1774	1598.0	149.8	0.1774
1388.0	149.8	0.1774	1494.0	149.8	0.1774	1600.0	149.8	0.1774
1390.0	149.8	0.1774	1496.0	149.8	0.1774	1602.0	149.8	0.1774
1392.0	149.8	0.1774	1498.0	149.8	0.1774	1604.0	149.8	0.1774
1394.0	149.8	0.1774	1500.0	149.8	0.1774	1606.0	149.8	0.1774
1396.0	149.8	0.1774	1502.0	149.8	0.1774	1608.0	149.8	0.1774
1398.0	149.8	0.1774	1504.0	149.8	0.1774	1610.0	149.8	0.1774
1400.0	149.8	0.1774	1506.0	149.8	0.1774	1612.0	149.8	0.1774
1402.0	149.8	0.1774	1508.0	149.8	0.1774	1614.0	149.8	0.1774
1404.0	149.8	0.1774	1510.0	149.8	0.1774	1616.0	149.8	0.1774
1406.0	149.8	0.1774	1512.0	149.8	0.1774	1618.0	149.8	0.1774
1408.0	149.8	0.1774	1514.0	149.8	0.1774	1620.0	149.8	0.1774
1410.0	149.8	0.1774	1516.0	149.8	0.1774	1622.0	149.8	0.1774
1412.0	149.8	0.1774	1518.0	149.8	0.1774	1624.0	149.8	0.1774
1414.0	149.8	0.1774	1520.0	149.8	0.1774	1626.0	149.8	0.1774
1416.0	149.8	0.1774	1522.0	149.8	0.1774	1628.0	149.8	0.1774
1418.0	149.8	0.1774	1524.0	149.8	0.1774	1630.0	149.8	0.1774
1420.0	149.8	0.1774	1526.0	149.8	0.1774	1632.0	149.8	0.1774
1422.0	149.8	0.1774	1528.0	149.8	0.1774	1634.0	149.8	0.1774
1424.0	149.8	0.1774	1530.0	149.8	0.1774	1636.0	149.8	0.1774
1426.0	149.8	0.1774	1532.0	149.8	0.1774	1638.0	149.8	0.1774
1428.0	149.8	0.1774	1534.0	149.8	0.1774	1640.0	149.8	0.1774
1430.0	149.8	0.1774	1536.0	149.8	0.1774	1642.0	149.8	0.1774
1432.0	149.8	0.1774	1538.0	149.8	0.1774	1644.0	149.8	0.1774
1434.0	149.8	0.1774	1540.0	149.8	0.1774	1646.0	149.8	0.1774
1436.0	149.8	0.1774	1542.0	149.8	0.1774	1648.0	149.8	0.1774
1438.0	149.8	0.1774	1544.0	149.8	0.1774	1650.0	149.8	0.1774
1440.0	149.8	0.1774	1546.0	149.8	0.1774	1652.0	149.8	0.1774
1442.0	149.8	0.1774	1548.0	149.8	0.1774	1654.0	149.8	0.1774
1444.0	149.8	0.1774	1550.0	149.8	0.1774	1656.0	149.8	0.1774
1446.0	149.8	0.1774	1552.0	149.8	0.1774	1658.0	149.8	0.1774
1448.0	149.8	0.1774	1554.0	149.8	0.1774	1660.0	149.8	0.1774
1450.0	149.8	0.1774	1556.0	149.8	0.1774	1662.0	149.8	0.1774
1452.0	149.8	0.1774	1558.0	149.8	0.1774	1664.0	149.8	0.1774
1454.0	149.8	0.1774	1560.0	149.8	0.1774	1666.0	149.8	0.1774
1456.0	149.8	0.1774	1562.0	149.8	0.1774	1668.0	149.8	0.1774
1458.0	149.8	0.1774	1564.0	149.8	0.1774	1670.0	149.8	0.1774
1460.0	149.8	0.1774	1566.0	149.8	0.1774	1672.0	149.8	0.1774
1462.0	149.8	0.1774	1568.0	149.8	0.1774	1674.0	149.8	0.1774
1464.0	149.8	0.1774	1570.0	149.8	0.1774	1676.0	149.8	0.1774
1466.0	149.8	0.1774	1572.0	149.8	0.1774	1678.0	149.8	0.1774
1468.0	149.8	0.1774	1574.0	149.8	0.1774	1680.0	149.8	0.1774
1470.0	149.8	0.1774	1576.0	149.8	0.1774	1682.0	149.8	0.1774
1472.0	149.8	0.1774	1578.0	149.8	0.1774	1684.0	149.8	0.1774
1474.0	149.8	0.1774	1580.0	149.8	0.1774	1686.0	149.8	0.1774
1476.0	149.8	0.1774	1582.0	149.8	0.1774	1688.0	149.8	0.1774
1478.0	149.8	0.1774	1584.0	149.8	0.1774	1690.0	149.8	0.1774
1480.0	149.8	0.1774	1586.0	149.8	0.1774	1692.0	149.8	0.1774
1482.0	149.8	0.1774	1588.0	149.8	0.1774	1694.0	149.8	0.1774
1484.0	149.8	0.1774	1590.0	149.8	0.1774	1696.0	149.8	0.1774
1486.0	149.8	0.1774	1592.0	149.8	0.1774	1698.0	149.8	0.1774
						1700.0	149.8	0.1774

TABLE 6.2.1-62 (SHEET 8 OF 8)

Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)	Time (s)	Mass Flowrate lbm/s	Energy Flowrate Btu/s (E+06)
1702.	149.8	0.1774	1808.	102.4	0.1204			
1704.	149.8	0.1774	1810.	89.18	0.1045			
1706.	149.8	0.1774	1812.	79.05	0.0925			
1708.	149.8	0.1774	1814.	70.62	0.0824			
1710.	149.8	0.1774	1816.	52.93	0.0617			
1712.	149.8	0.1774	1818.	8.932	0.0104			
1714.	149.8	0.1774	1820.	1.206	0.0014			
1716.	149.8	0.1774	1822.	0.1610	0.0002			
1718.	149.8	0.1774	1824.	0.0214	0.0000			
1720.	149.8	0.1774	1826.	0.0028	0.0000			
1722.	149.8	0.1774	1828.	0.0004	0.0000			
1724.	149.8	0.1774	1830.	0.0001	0.0000			
1726.	149.8	0.1774	1832.	0.0000	0.0000			
1728.	149.8	0.1774	1834.	0.0000	0.0000			
1730.	149.8	0.1774	1836.	0.0000	0.0000			
1732.	149.8	0.1774	1838.	0.0000	0.0000			
1734.	149.8	0.1774	1840.	0.0000	0.0000			
1736.	149.8	0.1774	1842.	0.0000	0.0000			
1738.	149.8	0.1774	1844.	0.0000	0.0000			
1740.	149.8	0.1774	1846.	0.0000	0.0000			
1742.	149.8	0.1774	1848.	0.0000	0.0000			
1744.	149.8	0.1774	1850.	0.0000	0.0000			
1746.	149.8	0.1774	1852.	0.0000	0.0000			
1748.	149.8	0.1774	1854.	0.0000	0.0000			
1750.	149.8	0.1774	1856.	0.0000	0.0000			
1752.	149.8	0.1774	1858.	0.0000	0.0000			
1754.	149.8	0.1774	1860.	0.0000	0.0000			
1756.	149.8	0.1774	1862.	0.0000	0.0000			
1758.	149.8	0.1774	1864.	0.0000	0.0000			
1760.	149.8	0.1774	1866.	0.0000	0.0000			
1762.	149.8	0.1774	1868.	0.0000	0.0000			
1764.	149.8	0.1774	1870.	0.0000	0.0000			
1766.	149.8	0.1774	1872.	0.0000	0.0000			
1768.	149.8	0.1774	1874.	0.0000	0.0000			
1770.	149.8	0.1774	1876.	0.0000	0.0000			
1772.	149.8	0.1774	1878.	0.0000	0.0000			
1774.	149.8	0.1774	1880.	0.0000	0.0000			
1776.	149.8	0.1774	1882.	0.0000	0.0000			
1778.	149.8	0.1774	1884.	0.0000	0.0000			
1780.	149.8	0.1774	1886.	0.0000	0.0000			
1782.	149.8	0.1774	1888.	0.0000	0.0000			
1784.	149.8	0.1774	1890.	0.0000	0.0000			
1786.	149.8	0.1774	1892.	0.0000	0.0000			
1788.	149.8	0.1774	1894.	0.0000	0.0000			
1790.	149.8	0.1774	1896.	0.0000	0.0000			
1792.	149.8	0.1774	1898.	0.0000	0.0000			
1794.	149.8	0.1774	1900.	0.0000	0.0000			
1796.	149.8	0.1774						
1798.	149.8	0.1774						
1800.	149.8	0.1774						
1802.	150.4	0.1781						
1804.	154.4	0.1824						
1806.	118.5	0.1395						

TABLE 6.2.1-63

DELETED.

TABLE 6.2.1-64 (SHEET 1 OF 3)

SPECIFIC PLANT DESIGN INPUT FOR MSLB ANALYSIS

		1	2	3	4	5	6
Initial steam generator inventory (lbm)	Faulted Intact	115,000 107,700	129,000 123,500	154,700 142,700	186, 300 186, 300	114,500 107,500	129,000 123,500
Initial steam pressure (psia)		1003	1051	1112	1102	1003	1051
Mass added by feedwater pumping (lbm)		15,900	7,800	4,500	9,600	14,000	10,900
Mass added by feedwater flashing (lbm)		20,800	21,500	22,500	0	20,800	21,500
Unisolatable steam line volume (ft^3)		470	470	470	470	470	470
Auxiliary feedwater addition rate (lbm/h) (to faulted steam generator)		5.48E+05	5.48E+05	5.48E+05	5.48E+05	5.48E+05	5.48E+05
Main steam line isolation time (s)		11.2	11.0	10.9	10.9	11.4	11.8
Main feedwater line isolation time (s)		8.2	8.2	7.9	7.9	8.4	8.8
Termination of auxiliary feedwater addition (s)		1800	1800	1800	1800	1800	1800

TABLE 6.2.1-64 (SHEET 2 OF 3)

Case		7	8	9	10	11	12
Initial steam generator inventory (lbm)	Faulted	154,700	186,400	115,000	129,000	154,700	186,400
	Intact	142,800	154,500	107,700	123,500	142,800	154,500
Initial steam pressure (psia)		1112	1102	1003	1051	1112	1102
Mass added by feedwater pumping (lbm)		11,200	23,300	16,000	18,700	11,600	29,200
Mass added by feedwater flashing (lbm)		22,600	24,900	20,800	21,500	22,600	24,900
Unisolatable steamline volume (ft^3)		470	470	470	470	470	470
Auxiliary feedwater addition rate (lbm/h)		5.48E+05	5.48e+05	5.48E+05	5.48E+05	5.48E+05	5.48E+05
Main steam line isolation time (s)		61.6	280.9	14.2	63.1	92.1	188.1
Main feedwater line isolation signal reached (s)		16.4	112.7	11.2	18.3	22.2	185.1
Termination of auxiliary feedwater addition (s)		1800	1800	1800	1800	1800	1800

TABLE 6.2.1-64 (SHEET 3 OF 3)

Case		13	14	15	16
Initial steam generator inventory (lbm)	Faulted Intact	116,000 107,700	130,100 123,500	155,900 142,800	186,300 186,300
Initial steam pressure (psia)		1008	1051	1112	1102
Mass added by feedwater pumping (lbm)		25,100	17,600	10,700	4800
Mass added by feedwater flashing (lbm)		20,800	21,500	22,600	0
Unisolatable steam line volume (ft^3)		470	470	470	470
Auxiliary feedwater addition rate (lbm/h)		5.41E+05	5.41E+05	5.41E+05	5.47E+05
Main steam line isolation time (s)		50.1	48.0	46.5	114.0
Main feedwater line isolation time (s)		17.1	15.5	14.3	23.8
Termination of auxiliary feedwater addition (s)		1800	1800	1800	1800

TABLE 6.2.1-65

SUMMARY OF RESULTS FOR MSLB CONTAINMENT PRESSURE - TEMPERATURE ANALYSIS

Case No.	Power [a] Level (%)	Break Size (ft^2)	Break Type	Max. Press. at *Time (psig at Time)	Max. Vapor Temp *at Time (°F at s)	Dryout Time (s)	6.29 psig *at Time (s)	17.19 psig *at Time (s)	25.79 psig *at Time (s)
1	102	Full	Double ended	27.6 at 124	232 at 125	1822	2.9	28.6	94.5
2	70	Full	Double ended	27.5 at 1809	232 at 157	1822	2.4	27.5	107.6
3	30	Full	Double ended	30.4 at 197	238 at 197	1834	2.0	31.3	129.0
4	0	Full	Double ended	30.2 at 173	238 at 173	1808	1.6	10.5	109.0
5	102	0.60	Double ended	27.1 at 1808	271 at 107	1844	6.4	61.1	201.0
6	70	0.53	Double ended	28.3 at 1833	252 at 107	1864	6.8	80.3	409.2
7	30	0.36	Double ended	32.4 at 1820	260 at 95	1912	9.4	52.1	537.4
8	0	0.20	Double ended	22.9 at 6339	226 at 119	6283	17.6	133.1	-
9	102	0.33	Double ended	28.5 at 1831	249 at 111	1898	11.0	136.5	854.1
10	70	0.32	Double ended	30.8 at 1836	273 at 71	1920	11.2	53.6	442.9
11	30	0.22	Double ended	32.7 at 2177	262 at 97	2155	15.1	83.7	1085.7
12	0	0.10	Double ended	17.0 at 8603	220 at 135	8600	33.7	-	-
13	102	0.86	Split	31.9 at 190	303 at 109	1832	10.1	40.4	89.7
14	70	0.908	Split	31.3 at 226	301 at 109	1834	8.5	38.2	93.8
15	30	0.944	Split	31.8 at 327	299 at 107	1856	7.3	36.7	96.7
16	0	0.40	Split	36.5 at 898	260 at 117	1842	16.8	103.8	484.6

a. % Power Level of 3579 MWt.

TABLE 6.2.1-66

SEQUENCE OF EVENTS FOR CASE 16 -
PEAK CALCULATED CONTAINMENT PRESSURE FOR MSLB

Time (s)	Event
0.0	Break occurs, blowdown from all steam generators.
16.8	Containment pressure setpoint for isolation of main feedwater lines reached (6.29 psig).
23.8	Main feedwater line isolation valves closed.
103.8	Containment pressure setpoint for isolation of main steam lines reached (17.19 psig).
114	Main steam line isolation valves closed, blowdown from broken loop steam generator and unisolated steam piping only.
116.8	Air coolers start.
117.0	Peak containment vapor temperature of 260°F is reached.
484.6	Containment pressure setpoint for actuation of containment sprays reached (25.79 psig).
584.6	Containment sprays start.
897.6	Peak containment pressure of 36.5 psig is reached.
1800.0	Auxiliary feedwater addition is terminated.
1842	Dryout occurs, steam generator dry following auxiliary feedwater termination.

TABLE 6.2.1-67

SEQUENCE OF EVENTS FOR CASE 13 -
PEAK CALCULATED CONTAINMENT TEMPERATURE FOR MSLB

Time (s)	Event
0.0	Break occurs, blowdown from all four steam generators.
10.1	Containment pressure setpoint for isolation of main feedwater lines reached (6.29 psig).
22.0	Main feedwater line isolation valves closed.
110.2	Air coolers start.
40.4	Containment pressure setpoint for isolation of main steam lines reached (17.19 psig).
50.1	Main steam line isolation valves closed, blowdown from broken loop steam generator and unisolated steam piping only.
89.7	Containment pressure setpoint for actuation of containment sprays reached (25.79 psig).
109.0	Peak containment vapor temperature of (303.1°F) is reached.
190.3	Containment sprays start.
190.4	Peak containment pressure of 31.9 psig is reached.
1800.0	Auxiliary feedwater addition is terminated.
1832.0	Dryout occurs, steam generator dry following auxiliary feedwater termination.

TABLE 6.2.1-68

BLOWDOWN MASS/ENERGY RELEASES FOR LIMITING CASES
DECLG (C_D=0.6, Low T_{AVG})

Time (s)	Mass Flow Rate (lbm/s)	Energy Release Rate (Btu/s)
0	65184	34575122
1	65184	34575122
2	56193	30344429
3	41256	22657374
4	35492	19926677
5	28425	16498775
6	27812	16227833
7	25996	15413707
8	23030	14325843
9	19958	12961246
10	17589	11668789
11	15609	10428740
12	14522	9583417
13	13691	8944095
14	12766	8368203
15	11960	7867605
16	10654	7135285
17	8188	5941968
18	6797	5107798
19	6620	4742754
20	7356	4747262
21	7008	4062956
22	6965	3625279
23	6952	3234900
24	6121	2631796
25	4973	1945657
26	4443	1644792
27	3827	1333195
28	3141	1068429
29	2673	855329
30	2296	700974
31	1928	548382
32	1605	458978
33	1379	324043
33.39	878	186500

TABLE 6.2.1-69

REFLOOD MASS/ENERGY RELEASES FOR LIMITING CASE
DECLG (C_D=0.6, Low T_{AVG}, MIN SI)

Time (s)	Mass Flow Rate (lbm/s)	Energy Release Rate (Btu/s)
45.9	35	37993
55.9	72	82068
65.9	105	122137
75.9	122	144865
85.9	120	144282
95.9	119	143294
120.9	160	156786
145.9	324	195848
170.9	404	221510
195.9	406	221418
220.9	398	224504
245.9	389	230303
270.9	375	225886
295.9	378	226935

TABLE 6.2.1-70

BROKEN LOOP INJECTION SPILL DURING BLOWDOWN
(C_D=0.6, LOW T_{AVG})

Time (s)	Mass Flow Rate (lbm/s)
0	3138
1	2995
2	2798
3	2637
4	2503
5	2388
6	2288
7	2201
8	2122
9	2051
10	1988
11	1930
12	1877
13	1828
14	1783
15	1741
16	1702
17	1665
18	1631
19	1599
20	1569
21	1541
22	1514
23	1489
24	1466
25	1444
26	1423
27	1402
28	1383
29	1365
30	1347

TABLE 6.2.1-71

ACTIVE HEAT SINKS FOR MINIMUM
CONTAINMENT PRESSURE ANALYSIS

Containment Spray Parameters

Number of pumps operating	2
Maximum Spray Flow (gal/min)	6748
Fastest post-LOCA initiation of spray pumps, assuming offsite power loss and no diesel failure(s)	74.0

Containment Fan Coolers

Number of fan coolers	8
Maximum CCWS flow (gal/min)	8975
Maximum NSCW water flow (gal/min)	9625
Fastest post-LOCA initiation of fan coolers assuming offsite power loss and no diesel failure(s)	41.1

TABLE 6.2.1-72 (SHEET 1 OF 2)

PASSIVE HEAT SINKS[a]

Wall	Material	Thermal Conductivity (Btu/h-ft-°F)	Volumetric Heat Capacity (Btu/ft^3-°F)	Thickness (ft)	Area (ft^2)
1	Epoxy	3.5	20.0	0.00025	32340
	Zinc coating	1.5	20.0	0.00020833	
	Steel	27.0	58.8	0.0208333	
	Concrete	0.92	22.62	0.5	
	Concrete	0.92	22.62	5.5	
2	Zinc Coating	1.5	20.0	0.00020833	72043
	Steel	27.0	58.8	0.020833	
	Concrete	0.92	22.62	0.5	
	Concrete	0.92	22.62	5.5	
3	Steel	27.0	58.8	0.015833	11442
	Epoxy	3.5	20.0	0.001125	
	Concrete	0.92	22.62	0.5	
	Concrete	0.92	22.62	6.3333	
4	Epoxy	3.5	20.0	0.0015417	15086.5
	Concrete	0.92	22.62	0.5	
	Concrete	0.92	22.62	2.25	
	Steel	27.0	58.8	0.020833	
	Concrete	0.92	22.62	10.5	
5	Zinc Coating	1.5	20.0	0.00020833	4957
	Steel	27.0	58.8	0.020833	
	Concrete	0.92	22.62	0.5	
	Concrete	0.92	22.62	7.5	
6	Zinc Coating	1.5	20.0	0.00020833	595.7
	Steel	27.0	58.8	0.125	
7	Galvanization	65.0	41.0	0.00020833	316975
	Steel	27.0	58.8	0.004375	
8	Zinc Coating	1.5	20.0	0.00020833	159120.9
	Steel	27.0	58.8	0.055	
9	Epoxy	3.5	20.0	0.00025	30030
	Zinc Coating	1.5	20.0	0.00020833	
	Steel	27.0	58.8	0.020833	

TABLE 6.2.1-72 (SHEET 2 OF 2)

Wall	Material	Thermal Conductivity (Btu/h-ft-°F)	Volumetric Heat Capacity (Btu/ft^3-°F)	Thickness (ft)	Area (ft^2)
10	Zinc Coating Steel	1.5 27.0	20.0 58.8	0.00020833 0.020833	159134
11	Zinc coating Steel	1.5 27.0	20.0 58.8	0.00020833 0.041667	101266.4
12	Zinc Coating Steel	1.5 27.0	20.0 58.8	0.00020833 0.10417	30474.2
13	Steel	27.0	58.8	0.0083333	75041
14	Steel	27.0	58.8	0.09375	232
15	Steel	27.0	58.8	0.04425	1259.4
16	Epoxy Concrete Concrete	3.5 0.92 0.92	20.0 22.62 22.62	0.001125 0.5 6.8333333	44202
17	Epoxy Concrete Concrete	3.5 0.92 0.92	20.0 22.62 22.62	0.000875 0.5 3.5	19550.3
18	Zinc Coating Steel Concrete Concrete	1.5 27.0 0.92 0.92	20.0 58.8 22.62 22.62	0.00020833 0.083333 0.5 4.0	1546.8
19	Epoxy Concrete Concrete	3.5 0.92 0.92	20.0 22.62 22.62	0.001125 0.5 4.0	2215.4
20[(a)]	Steel	27.0	58.8	0.0083333	100000

a. Wall 20 is the additional 357,900 lbm of metalmass allowance in the containment which is documented in Westinghouse SECL-99-021, Rev. 2. The effects of this are included in the LB LOCA PCT in table 15.6.5-4.

TABLE 6.2.1-73

FORCES AND MOMENTS FOR BROKEN COLD LEG WITH SUPPORT

	Elevation/Nodes					
Loads	187.00' 1-16	190.56' 17-24	184.64' 25-32	179.06' 33-40	172.61' 41-48	Total 1-48
F_x(E-W)(lbf)	-6.76×10^5	-1.10×10^6	-1.09×10^6	-1.02×10^6	-4.96×10^5	-41.9×10^5
F_y(N-S)(lbf)	-2.59×10^5	-5.54×10^5	$-5x42x10^5$	-5.61×10^5	-2.43×10^5	-20.6×10^5
$F_{Lateral}$(lbf)	7.24×10^5	1.23×10^6	1.22×10^6	1.63×10^6	5.52×10^5	46.7×10^5
M_x(ft-lbf)	0	-1.94×10^6	1.27×10^6	4.49×10^6	3.50×10^6	77.3×10^5
M_y(ft-lbf)	0	3.84×10^6	-2.56×10^6	-8.15×10^6	-7.14×10^6	-149×10^5

TABLE 6.2.1-74

UPLIFT FORCES AND MOMENTS FOR BROKEN COLD LEG WITH SUPPORT

	Loads	
Nodes	F_Z (lbf)	M_Z (ft-lbf)
1-16	0	-1.27×10^5
17-24	8.45×10^5	-1.40×10^3
25-32	-4.76×10^4	-1.40×10^3
49	4.71×10^5	0
50	2.62×10^5	0
Total	7.40×10^5	-1.23×10^5

TABLE 6.2.2-1

CONTAINMENT FAN COOLER DESIGN DATA

Characteristic	Quantity
Number of units	8
Containment atmosphere design inlet conditions, post LOCA	
Design temperature, saturation (°F)	270
Design pressure (psia)	60
Steam partial pressure (psia)	42
Air partial pressure (psia)	18
Cooling water (nuclear service cooling water)	
Minimum flowrate (gal/min/unit)	700
nlet temperature post-LOCA (°F)	95
Peak outlet temperature (°F)	257
Design post-LOCA heat removal rate at 270°F containment temperature (Btu/h/unit)	56.3×10^6
Fan flowrate (actual ft^3/min)	43,500 (slow speed)
Static head (in. WG)	0.85
Motor horsepower	62.5

TABLE 6.2.2-2

CONTAINMENT FAN COOLING HEAT REMOVAL
CAPACITY (POST-LOCA MODES)

(NSCW Temperature 40°F)

Containment Temperature (°F)	Capacity (Btu/h)
120	16.39E+06
150	25.92E+06
180	39.65E+06
210	53.57E+06
240	66.09E+06
270	77.97E+06

(NSCW Temperature 95°F)

Containment Temperature (°F)	Capacity (Btu/h)
120	5.80E+06
145	12.96E+06
170	21.23E+06
195	29.92E+06
220	38.76E+06
245	47.54E+06
270	56.30E+06

TABLE 6.2.2-3 (SHEET 1 OF 5)

CONTAINMENT COOLING SYSTEM FAILURE MODE AND EFFECTS ANALYSIS

Item No.	Description of Component	Safety Function	Plant Operating Mode[a]	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability
1	Breakers for high-speed operation) on 1AB04, 480-V switchgear, 1E bus, train A, normally closed (NC) No. 04 breaker for A7-001 motor and fan. No. 08 breaker for A7-002 motor and fan. No. 12 breaker for A7-005 motor and fan. No. 16 breaker for A7-006 motor and fan.	Provide continuity and protection for item 3 motor and fan	A	Inadvertent open, one breaker	Switchgear alarm Motor indicating lights	None; loss of train A;[b] train B available.
2	Breakers (for low-speed operation) on 1AB04, 480-V switchgear, 1E bus, train A, normally open (NO) No. 05 breaker for A7-001 motor and fan. No. 09 breaker for A7-002 motor and fan. No. 13 breaker for A7-005 motor and fan. No. 17 breaker for A7-006 motor and fan.	Provide continuity and protection for item 3 motor and fan	A	Inadvertent open, one breaker	Switchgear alarm Motor indicating lights	None; loss of train A; train B available
			B	Fail to close, one breaker	Sequencer alarm Motor indicating lights	None; loss of train A; train B available
3	Containment building (CTB) cooling unit motor and fan, train A, normally energized (NE) 1-1501-A7-001-M01 1-1501-A7-002-M01 1-1501-A7-005-M01 1-1501-A7-006-M01	Provide motive power for circulating air in the CTB	A	Fail to operate, one motor and fan	(C)	None; loss of train A; train B available
			B	Fail to restart and operate, one motor and fan	(C)	None; loss of train A; train B available

TABLE 6.2.2-3 (SHEET 2 OF 5)

Item No.	Description of Component	Safety Function	Plant Operating Mode[a]	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability
4	Breakers, 480-V MCC, 1E bus, train A, NC No. 26 breaker on 1ABE for A7-001 damper (HV2582A) No. 27 breaker on 1ABE for A7-002 damper (HV2582B)	Provide continuity and protection for item 6, damper	A	Inadvertent open, one breaker	Motor control center (MCC) alarm Position indicating lights	None; no loss of train A. NO damper will remain open.
	No. 07 breaker on 1ABC for A7-005 damper (HV2584A) No. 08 breaker on 1ABC for A7-006 damper (HV2584B)		B	Inadvertent open, one breaker	MCC alarm Position indicating lights	None; no loss of train A. NO damper will remain open.
5	Motor starters for item 6, train A NO No. 26 motor starter on 1ABE for A7-001 damper No. 27 motor starter on 1ABE for A7-002 damper	Provide continuity to one damper, item 6	A	Inadvertent closed, one motor starter	Position indicating lights	None; loss of train A; train B available
	No. 07 motor starter on 1ABC for A7-005 damper No. 08 motor starter on 1ABC for A7-006 damper		B	Fail to close	Position indicating lights	None; no loss of train A. NO damper will remain open.
6	Motor-operated on-off dampers, train A, NO HV2582A damper on 1ABE for A7-001 HV2582B damper on 1ABE for A7-002	Allow flow of air to the containment building and prevent backflow	A	Inadvertent closed, one damper	Position indicating lights	None; loss of train A; train B available
	HV2584A damper on 1ABC for A7-005 HV2584B damper on 1ABC for A7-006		B	Inadvertent closed, one damper	Position indicating lights	None; loss of train A; train B available

TABLE 6.2.2-3 (SHEET 3 OF 5)

Item No.	Description of Component	Safety Function	Plant Operating Mode[a]	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability
7	001, 003 backflow damper, train A, NO	Allow flow of air to the containment building and prevent backflow	A	Inadvertent close, one damper		None; loss of train A; train B available.
			B	Fail to close, one damper	(C)	None; no loss of train A. NO damper will remain open.
				Fail to reopen, one damper		None; loss of train A; train B available.
8	Breakers, (for high speed operation) on 1BB06, 480-V switchgear, 1E bus, train B, NC No. 04 breaker for A7-003 motor and damper No. 08 breaker for A7-004 motor and damper No. 12 breaker for A7-007 motor and damper No. 16 breaker for A7-008 motor and damper	Provide continuity and protection for item 10 motor and fan	A	Inadvertent open, one breaker	Switchgear alarm Motor indicating lights	None; loss of train B; train A available.
9	Breakers, (for low-speed operation) on 1BB06, 480-V switchgear, 1E bus, train B NO No. 05 breaker for A7-003 motor and damper No. 09 breaker for A7-004 motor and damper No. 13 breaker for A7-007 motor and damper No. 17 breaker for A7-008 motor and damper	Provide continuity and protection for item 10	A	Inadvertent open, one breaker	Switchgear alarm Motor indicating lights	None; loss of train B; train A available.
			B	Fail to close, one breaker	Sequencer alarm Motor indicating lights	None; loss of train B; train A available.

TABLE 6.2.2-3 (SHEET 4 OF 5)

Item No.	Description of Component	Safety Function	Plant Operating Mode[a]	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability
10	CTB cooling unit motor and fan, train B, NE 1-1501-A7-003-M01 1-1501-A7-004-M01 1-1501-A7-007-M01 1-1501-A7-008-M01	Provide motive power for circulating air in the containment building	A	Fail to operate, one motor and fan	(C)	None; loss of train B; train A available
			B	Fail to restart and operate, one motor and fan	(C)	None; loss of train B; train A available
11	Breakers, 480-V MCC, 1E bus, train B, NC No. 26 breaker on 1BBE for A7-003 damper (HV2583A) No. 27 breaker on 1BBE for A7-004 damper (HV2583B) No. 07 breaker on 1BBC for A7-007 damper (HV2585A) No. 08 breaker on 1BBC for A7-008 damper (HV2585B)	Provide continuity and protection for item 13 damper	A	Inadvertent open, one breaker	MCC alarm Position indicating lights	None; no loss of train B. NO damper will remain open.
			B	Inadvertent open, one breaker	Switchgear alarm Position indicating lights	None, no loss of train B. NO damper will remain open.
12	Motor starter for item 13, train B NO No. 26 motor starter on 1BBE for A7-003 damper No. 27 motor starter on 1BBE for A7-004 damper No. 07 motor starter on 1BBC for A7-007 damper No. 08 motor starter on 1BBC for A7-008 damper	Provide continuity to one damper, item 13	A	Inadvertent close, one motor starter	Position indicating lights	None; loss of train B; train A available
			B	Fail to close	Position indicating lights	None; no loss of train B. NO damper will remain open.
13	Motor-operated on-off dampers, train B, NO HV2583A damper on 1BBE for A7-003 damper HV2583B damper on 1BBE for A7-004 damper HV2585A damper on 1BBC for A7-007 damper HV2585B damper on 1BBC for A7-008 damper	Allow flow of air to the containment building and prevent backflow	A	Inadvertent close, one damper	Position indicating lights	None; loss of train B; train A available
			B	Inadvertent close, one damper	Position indicating lights	None; loss of train B; train A available

TABLE 6.2.2-3 (SHEET 5 OF 5)

Item No.	Description of Component	Safety Function	Plant Operating Mode[a]	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability
14	002, 004 backflow damper train B, NO	Allow flow of air to the containment building and prevent backflow	A	Inadvertent close, one damper		None; loss of train B; train A available.
			B	Fail to close, one damper	(C)	None; no loss of train B; NO damper will remain open.
				Fail to reopen, one damper		None; loss of train B; train A available.
15	Fans and dampers for train A: 1-1501-A7-001-000 1-1501-A7-002-000 1-1501-A7-005-000 1-1501-A7-006-000	Provide circulation of air inside containment building	A	Mechanical failure	Motor indicating light	None, loss of train A; train B available
16	Fans and dampers for train B: 1-1501-A7-003-000 1-1501-A7-004-000 1-1501-A7-007-000 1-1501-A7-008-000	Provide circulation of air inside containment building	A	Mechanical failure	Motor indicating light	None; loss of train B; train A available
17	Cooling coils for train A: 1-1501-A7-001-000 1-1501-A7-002-000 1-1501-A7-005-000 1-1501-A7-006-000	Provide cooling and heat removal inside containment building	A	Leakage in cooling coil	Flow indication Temperature alarm high Leak detector level alarm	None; loss of train A; train B available
18	Cooling coils for train B: 1-1501-A7-003-000 1-1501-A7-004-000 1-1501-A7-007-000 1-1501-A7-008-000	Provide cooling and heat removal inside containment building	A	Leakage in cooling coil	Flow indication Temperature alarm high Leak detector level alarm	None; loss of train B; train A available

a. A - Normal, four fans in train A or four fans in train B operating; B - safety injection, same as normal except all fans are sequenced at low speed. Four fans required (load shed occurs only under loss of offsite power).
b. Loss of one breaker or fan degrades the performance of that train. The other train remains available.
c. Flow indication inferred by motor indicating lights and switchgear alarm.

TABLE 6.2.2-4

CONTAINMENT SPRAY SYSTEM COMPONENT DESIGN PARAMETERS

Containment Spray Pump

Type	Horizontal centrifugal
Quantity	2
Design pressure (psig)	300
Design temperature (°F)	250
Design flowrate (gal/min)	2600
Design head (ft)	450
Material	Stainless steel

Containment Spray Nozzle

Quantity	342[a]
Type	Spraco 1713A or equivalent
Flow per nozzle at 40 psi Δp (gal/min)	15.2
Material	Stainless steel

Refueling Water Storage Tank

Quantity	1
Nominal volume (gal)	715,000
Boric acid concentration (ppm)	2400-2600
Design pressure	Hydraulic head
Design temperature (°F)	150
Operating pressure	Hydraulic head
Material	Stainless steel-lined concrete

Eductors[b]

Quantity	2
Eductor inlet (spray water)	
Operating fluid	Borated water
Operating temperature	Ambient
Design flow (gal/min)	39.3
NaOH concentration (wt percent)	30-32
Design temperature (°F)	300
Design pressure (psig)	300
Material	Stainless steel

a. On Unit 1, one nozzle is currently obstructed to prevent flow through the nozzle.
b. The spray additive subsystem has been abandoned in place. The eductors currently serve only to maintain the spray system pressure boundary integrity.

TABLE 6.2.2-5 (SHEET 1 OF 2)

FAILURE MODE AND EFFECTS ANALYSIS - CONTAINMENT SPRAY SYSTEM - ACTIVE COMPONENTS

Component	Failure Mode	CS Operation Phase	Effect on System Operation[a]	Failure Detection Method[b]	Remarks
1. Motor-operated gate valve 1-9001A (1-9001B analogous)	Fails to open on demand	Containment spray injection and recirculation phases	Failure blocks flow of spray coolant to nozzles of spray header of train A of containment spray system, which reduces redundancy of spray system. No safety effect on system operation. Minimum containment spray requirements will be met by the flow of containment spray coolant from the operation of train B (train A).	Valve position indication (closed to open position change) at CB. Valve monitor light and alarm (closed position) for group monitoring of components at CB.	Valve is normally closed during power operations. Valve opens on actuation by a CS signal.
2. Containment spray pump 1 (pump 2 analogous)	Fails to deliver working fluid	Containment spray injection and recirculation phases	Failure reduces the redundancy of providing coolant spray to the containment. Fluid flow from CS pump 1 (pump 2) will be lost. Minimum flow requirements for containment spray will be met by CS pump 2(pump 1) delivering working fluid to spray header in train B (train A).	Open pump switchgear circuit breaker indication at CB. Circuit breaker close position monitor light and alarm for group monitoring of components at CB common breaker trip alarm at CB.	Pump circuit breaker is aligned to close on actuation by a CS signal.

TABLE 6.2.2-5 (SHEET 2 OF 2)

	Component	Failure Mode	CS Operation Phase	Effect on System Operation[a]	Failure Detection Method[b]	Remarks
3.	Motor-operated gate valve 1-9002A or 1-9003A (1-9002B or 1-9003B analogous)	Fails to open on demand	Containment spray recirculation phase	Failure blocks flow of coolant from the containment sump to the suction of CS pump 1 (pump 2). Coolant flow from CS pump 1 (pump 2) will be lost which reduces the redundancy of spray system. No safety effect on system operation. Minimum flow requirements for containment spray will be met by CS train B (train A).	Valve position indication (closed to open position change) at CB. Monitor light and alarm (valve open) for group monitoring of components at CB.	Valves are opened by operator during the switchover from the RWST to the containment sumps (paragraph 6.2.2.2).
4.	Motor-operated gate valve 1-9017A (1-9017B analogous)	Fails to close on demand	Containment spray recirculation phase	Failure reduces the redundancy of isolation provided to prevent coolant flow from containment sump to RWST. No safety effect on system operation. Check valve 001 (008) provides isolation.	Valve position indication) open to closed position change) at CB. Monitor light and alarm (valve closed) for group monitoring of components at CB.	Valves are closed by the operator during the switchover from the RWST to the containment sumps (paragraph 6.2.2.2).

5. Deleted.

a. List of abbreviations and acronyms:

CB - Control board
CI - Containment isolation
CS - Containment spray
RWST - Refueling water storage tank

b. As part of plant operation, periodic tests, surveillance inspections, and instrument calibrations are made to monitor equipment and performance. Failure may be detected during such monitoring of equipment, in addition to detection methods noted.

TABLE 6.2.4-1 (SHEET 1 OF 17)

CONTAINMENT PENETRATION/ISOLATION VALVE INFORMATION

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve			Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator	Essential or Nonessential	Primary	Secondary	Normal	Shutdown	Post-Accident					
1	57(i)	Main steam and steam to auxiliary feed-water pump driver	Secondary coolant	29.5	Yes	1X4DB159-3	11	HV-3006A	Out	A	20'-9"	Gate	E/H	N	Auto	Remote man.	O	C	C	FC	SLI	5	A	Out
				29.5		2X4DB159-3		HV-3006B	Out		28'-4"	Gate	E/H	N	Auto	Remote man.	O	C	C	FC	SLI	5	B	
				8				PV-3000	Out		45'-10"	Globe	E/H	E	Auto	Remote man.	C	C	C	FC	Process/Signal	NA	A	
				6				PSV-3001	Out		20'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3002	Out		20'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3003	Out		20'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3004	Out		20'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3005	Out		20'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				4				HV-3009	Out		10'-1"	Gate	Elec. motor	E	Remote man.	None	O	O	O	FAI	Remote man.	NA	B (dc)	
				2				176	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				4				088	Out		7'-10"	Gate	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				356	Out		29'-8"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				X-192	Out		28'-0"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				4				HV-13005B	Out		42'-4"	Globe	Air	N	Auto	Remote man.	O	C	C	FC	SLI	5	B	
				1				X-117	Out		23'-3"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-119	Out		30'-10"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				4				HV-13005A	Out		26'-6"	Globe	Air	N	Auto	Remote man.	O	C	C	FC	SLI	5	A	
				1				X-209(o)	Out		10'-0"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-211(o)	Out		45'-0"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-429(p)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-438(p)	Out		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
2	57(i)	Main steam and steam to auxiliary feed-water pump driver	Secondary coolant	29.5	Yes	1X4DB159-3	11	HV-3016A	Out	A	28'-4 9/16"	Gate	E/H	N	Auto	Remote man.	O	C	C	FC	SLI	5	A	Out
				29.5		2X4DB159-3		HV-3016B	Out		36'-11 9/16"	Gate	E/H	N	Auto	Remote man.	O	C	C	FC	SLI	5	B	
				8				PV-3010	Out		16'-3"	Globe	E/H	E	Auto	Remote man.	C	C	C	FC	Process/Signal	NA	B	
				6				PSV-3011	Out		27'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3012	Out		27'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3013	Out		27'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3014	Out		27'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3015	Out		27'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				4				HV-3019	Out		5'-9"	Gate	Elec. motor	E	Remote man.	None	O	O	O	FAI	Remote man.	NA	A (dc)	
				2				178	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				4				082	Out		4'-7"	Gate	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				X-196	Out		14'-1"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-207(o)	Out		27'-0"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-162(o)	Out		26'-11"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-121	Out		31'-11"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-123	Out		40'-6"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				358	Out		10'-8"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				4				HV-13007A	Out		51'-6"	Globe	Air	N	Auto	Remote man.	O	C	C	FC	SLI	5	A	
				4				HV-13007B	Out		57'-10"	Globe	Air	N	Auto	Remote man.	O	C	C	FC	SLI	5	B	
				1				X-215(p)	Out		46'-5"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-166(p)	Out		26'-0"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-430	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	

TABLE 6.2.4-1 (SHEET 2 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve		Essential or Nonessential	Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator		Primary	Secondary	Normal	Shutdown	Post-Accident					
3	57[i]	Main steam line	Secondary coolant	29.5	Yes	1X4DB159-1	11	HV-3026A	Out	A	27'-4 3/4"	Gate	E/H	N	Auto	Remote man.	O	C	C	FC	SLI	5	A	Out
				29.5		2X4DB159-1		HV-3026B	Out		35'-1 3/4"	Gate	E/H	N	Auto	Remote man.	O	C	C	FC	SLI	5	B	
				8				PV-3020	Out		15'-3"	Globe	E/H	E	Auto	Remote man.	C	C	C	FC	Process/Signal	NA	B	
				6				PSV-3021	Out		26'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3022	Out		26'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3023	Out		26'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3024	Out		26'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3025	Out		26'-9"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				2				218	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				4				093	Out		3'-7"	Gate	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				352	Out		13'-11"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				X-198	Out		26'-9"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-125	Out		30'-11"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-127	Out		39'-6"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				4				HV-13008A	Out		50'-6"	Globe	Air	N	Auto	Remote man.	O	C	C	FC	SLI	5	A	
				4				HV-13008B	Out		56'-10"	Globe	Air	N	Auto	Remote man.	O	C	C	FC	SLI	5	B	
				1				X-217	Out		57'-1"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-431	In		-	Globe	manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
4	57[i]	Main steam line	Secondary coolant	29.5	Yes	1X4DB159-1	11	HV-3036A	Out	A	20'-0"	Gate	E/H	N	Auto	Remote man.	O	C	C	FC	SLI	5	A	Out
				29.5		2X4DB159-1		HV-3036B	Out		27'-4"	Gate	E/H	N	Auto	Remote man.	O	C	C	FC	SLI	5	B	
				8				PV-3030	Out		49'-3"	Globe	E/H	E	Auto	Remote man.	C	C	C	FC	Process/Signal	NA	A	
				6				PSV-3031	Out		20'-9 3/4"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3032	Out		20'-9 3/4"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3033	Out		20'-9 3/4"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3034	Out		20'-9 3/4"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				6				PSV-3035	Out		20'-9 3/4"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				2				220	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				4				152	Out		6'-10"	Gate	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-213[o]	Out		28'-5"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				X-194	Out		27'-10"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				354	Out		27'-0"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-129	Out		22'-6"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-131	Out		29'-10"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				4				HV-13006A	Out		26'-6"	Globe	Air	N	Auto	Remote man.	O	C	C	FC	SLI	5	A	
				4				HV-13006B	Out		42'-4"	Globe	Air	N	Auto	Remote man.	O	C	C	FC	SLI	5	B	
				1				X-432[p]	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-440	Out		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
5	N/A	Eddy current/sludge lancing	N/A	0.75	NO	1X4DB159-1	56[k]	X-017	Out	B	-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	-
				10		2X4DB159-1			In		-	Flange	NA	N	NA	None	C	C	C	NA	NA	NA	NA	-
7	57[i]	Steam generator blowdown	Secondary coolant	3	No	1X4DB159-3	34	HV-7603A	Out	A	10'	Globe	Air	N	Auto	Remote man.	O	C	C	FC	AFS	15	A,B	Out
				1.5		2X4DB159-3		126	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				335	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				409	In		-	Globe	manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
8	57[i]	Steam generator blowdown	Secondary coolant	3	No	1X4DB159-3	34	HV-7603B	Out	A	1'-0"	Globe	Air	N	Auto	Remote man.	O	C	C	FC	AFS	15	A,B	Out
				1.5		2X4DB159-3		129	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				336	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-157	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-164	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				410	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	

TABLE 6.2.4-1 (SHEET 3 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve			Actuation Mode		Valve Position					Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator	Essential or Nonessential	Primary	Secondary	Normal	Shutdown	Post-Accident	Power Failure	Actuation Signal			
9	57[i]	Steam generator blowdown	Secondary coolant	3	No	1X4DB159-1	34	HV-7603C	Out	A	1'-0"	Globe	Air	N	Auto	Remote man.	O	C	C	FC	AFS	15	A,B	Out
				1.5		2X4DB159-1		132	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				337	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-155	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				407	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
10	57[i]	Steam generator blowdown	Secondary coolant	3	No	1X4DB159-1	34	HV-7603D	Out	A	1'-0"	Globe	Air	N	Auto	Remote man.	O	C	C	FC	AFS	15	A,B	Out
				1.5		2X4DB159-1		135	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				338	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				408	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
11A	54	Chemical addition	Secondary coolant chemicals	0.5	No	1X4DB159-1	3A	HV-5280	Out	C	2'-0"	Globe	Air	N	Remote man.	None	C	C	C	FC	Remote man.[h]	NA	NA	In
				0.5		2X4DB159-1		676	In		-	Globe	Manual	N	Manual	None	LC	LC	LC	NA	NA	NA	NA	
				0.5				081	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
11B	57[i]	Steam generator secondary side sample	Secondary coolant	0.5	No	1X4DB159-3	35	HV-9451	Out	A	2'-6"	Globe	Solenoid	N	Auto	Remote man.	O	C	C	FC	AFS	15	A	Out
				0.5		2X4DB159-3		HV-9553A	In		-	Globe	Solenoid	N	Remote man.	None	C	C	C	FC	Remote man.	NA	NA	
				0.5				HV-9553B	In		-	Globe	Solenoid	N	Remote man.	None	O	C	C	FC	Remote man.	NA	NA	
				0.5				047	Out		3'-3"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.375				043	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
11C	57[i]	Steam generator secondary side sample	Secondary coolant	0.5	No	1X4DB159-3	35	HV-9452	Out	A	2'-6"	Globe	Solenoid	N	Auto	Remote man.	O	C	C	FC	AFS	15	B	Out
				0.5		2X4DB159-3		HV-9554A	In		-	Globe	Solenoid	N	Remote man.	None	C	C	C	FC	Remote man.	NA	NA	
				0.5				HV-9554B	In		-	Globe	Solenoid	N	Remote man.	None	O	C	C	FC	Remote man.	NA	NA	
				0.5				048	Out		3'-3"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.375				044	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
12A	54	Chemical addition	Secondary coolant chemicals	0.5	No	1X4DB159-1	3A	HV-5281	Out	C	2'-0"	Globe	Air	N	Remote man.	None	C	C	C	FC	Remote man.[h]	NA	NA	In
				0.5		2X4DB159-1		677	In		-	Globe	Manual	N	Manual	None	LC	LC	LC	NA	NA	NA	NA	
				0.5				084	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
12B	57[i]	Steam generator secondary side sample	Secondary coolant	0.5	No	1X4DB159-1	35	HV-9453	Out	A	2'-6"	Globe	Solenoid	N	Auto	Remote man.	O	C	C	FC	AFS	15	B	Out
				0.5		2X4DB159-1		HV-9555A	In		-	Globe	Solenoid	N	Remote man.	None	C	C	C	FC	Remote man.	NA	NA	
				0.5				HV-9555B	In		-	Globe	Solenoid	N	Remote man.	None	O	C	C	FC	Remote man.	NA	NA	
				0.5				049	Out		3'-3"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.375				045	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-185	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-423[o]	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
12C	57[i]	Steam generator secondary side sample	Secondary coolant	0.5		1X4DB159-1	35	HV-9454	Out	A	2'-6"	Globe	Solenoid	N	Auto	Remote man.	O	C	C	FC	AFS	15	A	Out
				0.5		2X4DB159-1		HV-9556A	In		-	Globe	Solenoid	N	Remote man.	None	C	C	C	FC	Remote man.	NA	NA	
				0.5				HV-9556B	In		-	Globe	Solenoid	N	Remote man.	None	O	C	C	FC	Remote man.	NA	NA	
				0.5				050	Out		3'-3"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.375				046	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	

TABLE 6.2.4-1 (SHEET 4 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve			Actuation Mode		Valve Position					Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator	Essential or Nonessential	Primary	Secondary	Normal	Shutdown	Post-Accident	Power Failure	Actuation Signal			
13A	56	Containment air radioactivity monitor inlet	Containment atmosphere	1 1 1	No	1X4DB213-2 2X4DB213-2	36	HV-12975 HV-12976 X-001	In Out In	C C	- 3'-0" -	Gate Gate Globe	Solenoid Solenoid Manual	N N N	Auto Auto Manual	Remote man. Remote man. None	O O C	C C C	C C C	FC FC NA	CVI CVI NA	15 15 NA	A B NA	Out
13B	56	Containment air radioactivity monitor outlet	Containment atmosphere	1 1 1	No	1X4DB213-2 2X4DB213-2	45	HV-12977 HV-12978 X-003	Out In In	C C	2'-0" - -	Globe Globe Globe	Solenoid Solenoid Manual	N N N	Auto Auto Manual	Remote man. Remote man. None	O O C	C C C	C C C	FC FC NA	CVI CVI NA	15 15 NA	B A NA	In
13C	1.141	Containment pressure detector	DC 702 silicone oil	Tubing	Yes	1X4DB131 2X4DB131	48	None	-	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
14A	1.141	Reactor vessel water level instrumentation	Water	Tubing	Yes	1X4DB113 2X4DB113	54	None	-	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
14B	1.141	Reactor vessel water level instrumentation	Water	Tubing	Yes	1X4DB113 2X4DB113	54	None	-	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
14C	1.141	Reactor vessel water level instrumentation	Water	Tubing	Yes	1X4DB113 2X4DB113	54	None	-	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
15	54	Purification water supply to refueling cavity	Borated water	3 3	No	1X4DB130 2X4DB130	37	050 051	Out In	C	1'-6" -	Dia Dia	Manual Manual	N N	Manual Manual	None None	LC LC	LC[c] LC[c]	LC LC	NA NA	NA NA	NA NA	NA NA	In
18	57[i]	Feedwater	Secondary coolant	16 1	No	1X4DB168-3 2X4DB168-3	12	HV-5229 X-031	Out Out	A	11'-0" 4'-1"	Gate Globe	E/H Manual	N N	Auto Manual	Remote man. NA	O C	C C	C C	FC NA	FI NA	5 NA	A,B	In
19	57[i]	Feedwater	Secondary coolant	16 1 1	No	1X4DB168-3 2X4DB168-3	12	HV-5228 X-036[b] X-037[o]	Out Out Out	A	11'-0" 4'-5" -	Gate Globe Globe	E/H Manual Manual	N N N	Auto Manual Manual	Remote man. NA NA	O C C	C C C	C C C	FC NA NA	FI NA NA	5 NA NA	A,B NA NA	In
20	57[i]	Feedwater	Secondary coolant	16 1	No	1X4DB168-3 2X4DB168-3	12	HV-5230 X-073	Out Out	A	8'-0" 3'-0"	Gate Globe	E/H Manual	N N	Auto Manual	Remote man. NA	O C	C C	C C	FC NA	FI NA	5 NA	A,B NA	In
21	57[i]	Feedwater	Secondary coolant	16 1 1	No	1X4DB168-3 2X4DB168-3	12	HV-5227 X-075[b] X-076[p]	Out Out Out	A	8'-9" 3'-0" -	Gate Globe Globe	E/H Manual Manaul	N N N	Auto Manual Manual	Remote man. NA NA	O C C	C C C	C C C	FC NA NA	FI NA NA	5 NA NA	A,B NA NA	In
22	54	Demineralized water supply	Demin. water	2 2 1.5 1.0 1.0	No	AX4DB190-2	38	005 038 PSV-17589 X-065 X-950	Out In In In In	C	2'-2" - - - -	Globe Check Relief Globe Globe	Manual Self Self Manual Manual	N N N N N	Manual Auto Auto Manual Manual	None None None None None	LC - C LC LC	LC - C LC LC	LC - C LC LC	NA NA NA NA NA	NA NA NA NA NA	NA NA NA NA NA	NA NA NA NA NA	In

TABLE 6.2.4-1 (SHEET 5 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve		Essential or Nonessential	Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator		Primary	Secondary	Normal	Shutdown	Post-Accident					
23	54	Breathing air supply	Compressed air	1.5 1.5 0.75	No	1X4DB186-1 2X4DB186-1	53	211 184 226	Out In In	C	2'-9" - -	Gate Check Globe	Manual Self Manual	N N N	Manual Auto Manual	None None None	LC - C	LC - C	LC - C	NA NA NA	NA NA NA	NA NA NA	NA NA NA	In
24	55	Hot leg sample line	Primary coolant	0.5 0.5	No	1X4DB140 2X4DB140	52	HV-3502 HV-3548	Out In	C	8" -	Globe Globe	Air Elec. motor	N N	Auto Auto	Remote man. Remote man.	O O	O O	C C	FC FAI	CIA CIA	15 20	A B	Out
28	54	ACCW supply	Water with corrosion inhibitors	10 10	No	1X4DB138-2 1X4DB138-1	40	HV-1978 HV-1979	In Out	C	- 1'-6"	B-fly B-fly	Elec. motor Elec. motor	E[g] E	Remote man. Remote man.	Manual Manual	O O	O O	C[g] C[g]	FAI FAI	Remote man. Remote man.	NA NA	B A	In
				0.75	No	1X4DB138-2 2X4DB138-2	40	PSV-1978	In		1'-6"	Relief	Self	N	Auto	None	C	C	C	N/A	N/A	N/A	N/A	In
29	54	ACCW return	Water with corrosion inhibitors	10 10 0.75	No	1X4DB138-2 1X4DB138-1	24	HV-1974 HV-1975 113	In Out In	C	- 1'-6"	B-fly B-fly Check	Elec. motor Elec. motor Self	E[g] E N	Remote man. Remote man. Auto	Manual Manual None	O O -	O O -	C[g] C[g] -	FAI FAI NA	Remote man. Remote man. NA	NA NA NA	B A NA	Out
30	55	Safety injection to cold leg	Borated water	4 2 2 2 2 0.75 0.75 0.75 0.75 0.75 0.75 0.75 0.75 0.75 0.75 0.75 1 1 1 1 1	Yes	1X4DB121 2X4DB121	19	HV-8835 143 144 145 146 113 HV-8823 X-119 X-120 X-121 X-122 X-123 X-124 X-125 X-126 X-181 X-239 X-241 X-346 X-348	Out In In In In In In In In In In In In In In In In In In In In	A	2'-0" -	Gate Check Check Check Check Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe	Elec. motor Self Self Self Self Manual Air Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual	E E E E E N N N N N N N N N N N N N N N	Remote man. Auto Auto Auto Auto Manual Auto Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual	Manual None None None None None Remote None None None None None None None None None None None None None	O - - - - C C C C C C C C C C C C C C C	O - - - - C C C C C C C C C C C C C C C	O - - - - C C C C C C C C C C C C C C C	FAI NA NA NA NA NA FC NA NA NA NA NA NA NA NA NA NA NA NA NA	Remote man. NA NA NA NA NA CIA NA NA NA NA NA NA NA NA NA NA NA NA NA	NA NA NA NA NA NA 15 NA NA NA NA NA NA NA NA NA NA NA NA NA	A NA NA NA NA NA B NA NA NA NA NA NA NA NA NA NA NA NA NA	In
31	55	Safety injection to hot leg	Borated water	4 0.75 2 2 0.75 0.75 0.75 0.75 0.75 1 1	Yes	1X4DB121 2X4DB121	20	HV-8802B[1/] HV-8824 122 123 063 X-288 X-289 X-290 X-291 X-286 X-310	Out In In In In In In In In In In	A	2'-6" - - - - - - - - - -	Gate Globe Check Check Globe Globe Globe Globe Globe Globe Globe	Elec. motor Air Self Self Manual Manual Manual Manual Manual Manual Manual	E N E E N N N N N N N	Remote man. Auto Auto Auto Manual Manual Manual Manual Manual Manual Manual	Manual Remote man. None None None None None None None None None	C C - - C C C C C C C	C C - - C C C C C C C	O[b] C - - C C C C C C C	FAI FC NA NA NA NA NA NA NA NA NA	Remote man. CIA NA NA NA NA NA NA NA NA NA	NA 15 NA NA NA NA NA NA NA NA NA	B B NA NA NA NA NA NA NA NA NA	In

TABLE 6.2.4-1 (SHEET 6 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/Outside	Type Tests	Length of Pipe (ft-in.)	Valve			Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator	Essential or Nonessential	Primary	Secondary	Normal	Shutdown	Post-Accident					
32	55	Boron injection line to cold leg	Borated water	4	Yes	1X4DB119	25	HV-8801A	Out	A	16'-5"	Gate	Elec. motor	E	Auto	Remote man.	C	C	O	FAI	SI	NA	A	In
				4		2X4DB119		HV-8801B	Out		16'-3"	Gate	Elec. motor	E	Auto	Remote man.	C	C	O	FAI	SI	NA	B	
				3				013	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				0.75				HV-8843	In		-	Globe	Air	N	Auto	Remote man.	C	C	C	FC	CI	NA	NA	
				0.75				173	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
33	55	Safety injection to hot leg	Borated water	4	Yes	1X4DB121	20	HV-8802A[v/]	Out	A	1'-8"	Gate	Elec. motor	E	Remote man.	Manual	C	C	O[(b)]	FAI	Remote man.	NA	A	
				0.75		2X4DB121		HV-8881	In		-	Globe	Air	N	Auto	Remote man.	C	C	C	FC	CIA	15	B	
				2				120	In		-	Check	Self	E	Auto	None	-	-		NA	NA	NA	NA	
				2				121	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				0.75				290	In		-	Globe	Manual	N	Manual	None	C	C		NA	NA	NA	NA	
				0.75				X-292	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				X-293	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				X-294	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				X-295	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
34	56	Containment spray supply	Borated water	8	Yes	1X4DB131	26	HV-9001B	Out	C	2'-0"	Gate	Elec. motor	E	Auto	Remote man.	C	C	O	FAI	CS	NA	B	In
				8		2X4DB131		016	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				2				014	Out		2'-0"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				046	Out		2'-0"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
35	56	Containment spray supply	Borated water	8	Yes	1X4DB131	26	HV-9001A	Out	C	7'-5"	Gate	Elec. motor	E	Auto	Remote man.	C	C	O	FAI	CS	NA	A	In
				8		2X4DB131		015	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				2				013	Out		2'-1"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				047	Out		2'-1"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
36	56	RHR emergency sump suction	Borated water	14	Yes	1X4DB122 2X4DB122	27	HV-8811B[(i)]	Out	A	6'-6"	Gate	Elec. motor	E	Auto	Remote man.	C	C	O[(b)]	FAI	SI[(e)]	NA	B	Out
37	56	RHR emergency sump suction	Borated water	14	Yes	1X4DB122 2X4DB122	27	HV-8811A[(i)]	Out	A	6'-6"	Gate	Elec. motor	E	Auto	Remote man.	C	C	O[(b)]	FAI	SI[(e)]	NA	A	Out
38	56	Containment spray emergency sump suction	Borated water	12	Yes	1X4DB131 2X4DB131	23	HV-9002B[(i)]	Out	A	6'-9"	Gate	Elec. motor	E	Remote man.	Manual	C	C	O[(b)]	FAI	Remote man.	NA	B	Out

TABLE 6.2.4-1 (SHEET 7 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve		Essential or Nonessential	Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator		Primary	Secondary	Normal	Shutdown	Post-Accident					
39	56	Containment spray emergency sump suction	Borated water	12	Yes	1X4DB131 2X4DB131	23	HV-9002A[l]	Out	A	6'-9"	Gate	Elec. motor	E	Remote man.	Manual	C	C	O[b]	FAI	Remote man.	NA	A	Out
40	54	Fire protection water	Well water	4	No	1X4DB174-4	41	HV-27901	Out	C	2'-6"	Gate	Air	N	Auto	Remote man.	C	O	C	FC	CIA	20	A,B	In
				6		2X4DB174-4		036	In		-	Check	Self	N	Auto	None	-	-	-	NA	NA	NA	NA	
				1				018	In		-	Gate	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
41	54	Accumulator test and drain line	Borated water	0.75	No	1X4DB121	1	HV-8871	In	C	3'-7"	Globe	Air	N	Auto	Remote man.	C	C	C	FC	CIA	15	A	Out
				0.75		2X4DB121		HV-8964	Out		2'-7"	Globe	Air	N	Auto	Remote man.	C	C	C	FC	CIA	15	B	
				0.75				HV-8888	Out		2'-9"	Globe	Air	N	Auto	Remote man.	C	C	C	FC	CIA	15	B	
				0.75				016	Out		2'-9"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				X-165[o]	Out		2'-6"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				X-444[o]	Out		6'-8"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				X-835[o]	Out		1'-6"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				293[p]	Out		1'-0"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				324[p]	Out		5'-6"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				PSV-8871	In		3'-7"	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	Out
42	54	Nitrogen supply to accumulator	N_2	1	No	1X4DB120	3	HV-8880	Out	C	2'-6"	Globe	Air	N	Auto	Remote man.	C	C	C	FC	CIA	15	A	In
				1		2X4DB120		017	In		-	Check	Self	N	Auto	None	-	-	-	NA	NA	NA	NA	
				0.75				013	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
43	57	NSCW supply to reactor cavity coolers	Treated well water	8	No	1X4DB135-1	17	HV-2134	Out	A	3'-0"	B-fly	Elec. motor	N	Auto	Remote man.	O	O	C	FAI	SI	40	A	In
				0.75		2X4DB135-1		PSV-11673	In		-	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				1				225	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-658	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-185	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-186	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-181[p]	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-182[p]	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				309	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				315	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				317	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				323	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				325	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				346	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				348	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-206[o]	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	

TABLE 6.2.4-1 (SHEET 8 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/Outside	Type Tests	Length of Pipe (ft-in.)	Valve		Essential or Nonessential	Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator		Primary	Secondary	Normal	Shutdown	Post-Accident					
44	57	NSCW return from reactor cavity coolers	Treated well water	8	No	1X4DB135-1	46	HV-2138	Out	A	2'-0"	B-fly	Elec. motor	N	Auto	Remote man.	O	O	C	FAI	SI	40	A	Out
				1		2X4DB135-1		X-183(p)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-184(p)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-187	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-188	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				198	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				PSV-2136	In		-	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				1				313	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				311	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				321	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				319	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				327	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				350	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				352	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-222(o)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
45	57	NSCW supply to reactor cavity coolers	Treated well water	8	No	1X4DB135-2	17	HV-2135	Out	A	1'-8"	B-fly	Elec. motor	N	Auto	Remote man.	O	O	C	FAI	SI	40	B	In
				1		2X4DB135-2		X-062(o)	Out		2'-0"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				159	Out		2'-0"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				PSV-2137	In		-	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				1				022	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-654	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-565	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-389	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-490	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-388	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.1				X-405	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.1				X-431	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-391	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-392	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				427	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				443	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				447	In		-	Globe	Manual	N	Manual	None	C	O	C	NA	NA	NA	NA	
				1				455	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				461	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				429	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
46	57	NSCW return from reactor cavity coolers	Treated well water	8	No	1X4DB135-2	46	HV-2139	Out	A	1'-8"	B-fly	Elec. motor	N	Auto	Remote man.	O	O	C	FAI	SI	40	B	Out
				1		2X4DB135-2		X-040	Out		1'-6"	Globe	Manual	N	Manual	Manual	C	C	C	NA	NA	NA	NA	
				1				164(o)	In		-	Globe	Manual	N	Manual	Manual	C	C	C	NA	NA	NA	NA	
				1				X-491(o)	In		-	Globe	Manual	N	Manual	Manual	C	C	C	NA	NA	NA	NA	
				1				X-568(o)	In		-	Globe	Manual	N	Manual	Manual	C	C	C	NA	NA	NA	NA	
				1				X-390(o)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				X-387	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				PSV-11772	In		-	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				1				431	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				433	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				445	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				449	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				453	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				457(p)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				461(o)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-816(p)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-241(p)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-676(p)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				451(o)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
								459(o)																

TABLE 6.2.4-1 (SHEET 9 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/Outside	Type Tests	Length of Pipe (ft-in.)	Valve			Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator	Essential or Nonessential	Primary	Secondary	Normal	Shutdown	Post-Accident					
48	54	Normal letdown line	Primary coolant	3	No	1X4DB114	7	HV-8160	In	C	-	Globe	Air	N	Auto	Remote man.	O	O	C	FC	CIA	15	A	Out
				3		2X4DB114		HV-8152	Out		4'-5"	Globe	Air	N	Auto	Remote man.	O	O	C	FC	CIA	15	B	
				0.75				502[f][o]	Out		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
49	54	Excess letdown and seal water leakoff	Primary coolant	2	No	1X4DB114	31	HV-8100	Out	C	2'-4"	Globe	Elec. motor	N	Auto	Remote man.	O	O	C	FAI	CIA	15	B	Out
				2		2X4DB114		HV-8112	In		-	Globe	Elec. motor	N	Auto	Remote man.	O	O	C	FAI	CIA	15	A	
				0.75				021	In		-	Check	Self	N	Auto	None	-	-	-	NA	NA	NA	NA	
50	54	Normal charging line	Primary coolant	3	No	1X4DB114 2X4DB114	33	HV-8105	Out	C	3'-0"	Gate	Elec. motor	N	Auto	Remote man.	O	O	C	FAI	SI	17[m]	B	In
				3				032	In		-	Check	Self	N	Auto	None	-	-	-	NA	NA	NA	NA	
				0.75				465	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
51	54	Reactor coolant pump seal water supply (pump loop No. 4)	Primary coolant	1.5	Yes	1X4DB114	32	HV-8103D	Out	A	1'-3"	Globe	Elec. motor	E	Remote man.	Manual	O	O	O	FAI	Remote man.	NA	B	In
				1.5		2X4DB114		355	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				0.75				452	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
52	54	Reactor coolant pump seal water supply (pump loop No. 3)	Primary coolant	1.5	Yes	1X4DB114	32	HV-8103C	Out	A	1'-3"	Globe	Elec. motor	E	Remote man.	Manual	O	O	O	FAI	Remote man.	NA	B	In
				1.5		2X4DB114		354	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				0.75				451	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
53	54	Reactor coolant pump seal water supply (pump loop No. 2)	Primary coolant	1.5	Yes	1X4DB114	32	HV-8103B	Out	A	1'-4"	Globe	Elec. motor	E	Remote man.	Manual	O	O	O	FAI	Remote man.	NA	B	In
				1.5		2X4DB114		353	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				0.75				450	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
54	54	Reactor coolant pump seal water supply (pump loop No. 1)	Primary coolant	1.5	Yes	1X4DB114	32	HV-8103A	Out	A	1'-4"	Globe	Elec. motor	E	Remote man.	Manual	O	O	O	FAI	Remote man.	NA	B	In
				1.5		2X4DB114		004	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				0.75				449	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
55	NA	Eddy current/sludge lancing	NA	0.75	No	1X4DB159-1	56[k]	X-018	Out	B	-	Globe[q]	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	-
				10		2X4DB159-1			In		-	Flange	NA	N		None	C	C	C	NA	NA	NA		NA
56	55	RHR pump discharge to hot leg	Borated water	12	Yes	1X4DB121	21	HV-8840(l)	Out	A	3'-0"	Gate	Elec. motor	E	Remote man.	Manual	C	C	O[b]	FAI	Remote man.	NA	B	In
				8		2X4DB121		128	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				8				129	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				0.75				HV-8825	In		-	Globe	Air	N	Auto	Remote man.	C	C	-	FC	CIA	15	B	
				0.75				112[o]	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-435	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				296[p]	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
57	55	RHR loop into cold leg	Borated water	8	Yes	1X4DB121	30	HV-8809A	Out	A	2'-0"	Gate	Elec. Motor	E	Remote man.	Manual	O	O	O[b]	FAI	Remote man.	NA	A	In
				6		2X4DB121		147	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				6				148	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				0.75				111	In		-	Globe	Manual	N	Manual	None	C	C	-	NA	NA	NA	NA	
				0.75				HV-8890A	In		-	Globe	Air	N	Auto	Remote man.	C	C	C	FC	CIA	15	B	
58	55	RHR loop into cold leg	Borated water	8	Yes	1X4DB121	30	HV-8809B	Out	A	1'-10"	Gate	Elec. motor	E	Remote man.	Manual	O	O	O[b]	FAI	Remote man.	NA	B	In
				6		2X4DB121		149	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				6				150	In		-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	
				0.75				110	In		-	Globe	Manual	N	Manual	None	C	C	-	NA	NA	NA	NA	
				0.75				HV-8890B	In		-	Globe	Air	N	Auto	Remote man.	C	C	C	FC	CIA	15	B	
				1				X-410	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	

TABLE 6.2.4-1 (SHEET 10 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve			Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator	Essential or Nonessential	Primary	Secondary	Normal	Shutdown	Post-Accident					
59	55	RHR suction from hot leg	Primary coolant	12 3	Yes	1X4DB122 2X4DB122	4	HV-8701A PSV-8708A	In In	A	- -	Gate Relief	Elec. motor Self	E N	Remote man. Auto	Manual None	C C	O C	C C	FAI NA	Remote man. NA	NA NA	A NA	Out
60	55	RHR suction from hot leg	Primary coolant	12 3	Yes	1X4DB122 2X4DB122	4	HV-8702A PSV-8708B	In In	A	- -	Gate Relief	Elec. motor Self	E N	Remote man. Auto	Manual None	C	O C	C C	FAI NA	Remote man. NA	NA NA	B NA	Out
62	54	Pressurizer relief tank sample to waste gas compressor suction and N supply	Waste gas/N	1 1	No	1X4DB112 2X4DB112	2	HV-8033 HV-8047	Out In	C	1'-6" -	Dia Dia	Air Air	N N	Auto Auto	Remote man. Remote man.	C C	C C	C C	FC FC	CIA CIA	15 15	B A	Out/In
63	54	Pressurizer relief tank makeup water supply	Demin. water	3 3 1	No	1X4DB112 2X4DB112	39	HV-8028 112 020	Out In In	C	1'-6" - -	Dia Check Globe	Air Self Manual	N N N	Auto Auto Manual	Remote man. None None	O - LC	O - LC	C - LC	FC NA NA	CIA NA NA	15 NA NA	B NA NA	In
64A	NA	Flow verification and pressure sensing piping	Containment atmosphere	0.5 1	No	1X4DB132 2X4DB132	51	119 NA	In In	B	-	Globe Flange	Manual NA	N N	Manual NA	None None	C NA	C NA	C NA	NA NA	NA NA	NA NA	NA NA	
64B	NA	Flow verification and pressure sensing piping	Containment atmosphere	0.5 1	No	1X4DB132 2X4DB132	51	120 NA	In In	B	-	Globe Flange	Manual NA	N N	Manual NA	None None	C NA	C NA	C NA	NA NA	NA NA	NA NA	NA NA	
67A	55	Pressurizer steam sample line	Primary coolant	0.5 0.5	No	1X4DB140 2X4DB140	7	HV-3514 HV-3513	Out In	C	4'-0" -	Globe Globe	Air Air	N N	Auto Auto	Remote man. Remote man.	C C	C C	C C	FC FC	CIA CIA	15 15	A B	Out
67B	55	Pressurizer liquid sample line	Primary coolant	0.5 0.5	No	1X4DB140 2X4DB140	7	HV-3507 HV-3508	In Out	C	- 3'-0"	Globe Globe	Air Air	N N	Auto Auto	Remote man. Remote man.	C C	C C	C C	FC FC	CIA CIA	15 15	B A	Out
67C	1.141	Containment pressure detector	DC 702 Silicone oil	Tubing	Yes	1X4DB131 2X4DB131	48		-	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
68	NA	Containment leak rate test	Containment atmosphere	0.75 8 0.75	No	1X4DB132 2X4DB132	18	018(c) NA(p) 019(p)	In In In	B	- - -	Globe Flange Globe	Manual NA Manual	N N N	Manual NA Manual	None None None	C NA C	C NA C	C NA C	NA NA NA	NA NA NA	NA NA NA	NA NA NA	
69A	54	Chemical addition	Secondary coolant chemicals	0.5 0.5 0.5	No	1X4DB159-3 2X4DB159-3	3B	HV-5278 678 087	Out In In	C	1'-3" - -	Globe Globe Globe	Air Manual Manual	N N N	Remote Manual Manual	None None None	C LC C	C LC C	C LC C	FC NA NA	Remote man.(h) NA NA	NA NA NA	NA NA NA	In
69B	54	Chemical addition	Secondary coolant chemicals	0.5 0.5 0.5	No	1X4DB159-3 2X4DB159-3	3B	HV-5279 679 090	Out In In	C	1'-6" - -	Globe Globe Globe	Air Manual Manual	N N N	Remote Manual Manual	None None None	C LC C	C LC C	C LC C	FC NA NA	Remote man.(h) NA NA	NA NA NA	NA NA NA	In

TABLE 6.2.4-1 (SHEET 11 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve		Essential or Nonessential	Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator		Primary	Secondary	Normal	Shutdown	Post-Accident					
69C	1.141	Containment pressure detector	DC 702 silicone oil	Tubing	Yes	1X4DB131 2X4DB131	48	None	-	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
70A	56	Containment H_2 monitor suction	Containment atmosphere	0.75 0.75 0.75	Yes	1X4DB213-2 2X4DB213-2	14	HV-2791A HV-2790A HV-2790B	Out In In	C	2'-9" - -	Globe Globe Globe	Solenoid Solenoid Solenoid	E E E	Remote man. Remote man. Remote man.	Remote man.[n] None None	C C C	C C C	O O O	FC FC FC	Remote man. Remote man. Remote man.	NA NA NA	A B B	Out
70B	56	Containment H_2 monitor discharge	Containment atmosphere	0.75 0.75 0.75	Yes	1X4DB213-2 2X4DB213-2	8	HV-2793A 001 039	Out In In	C	1'-2" - -	Globe Check Globe	Solenoid Self Manual	E E N	Remote man. Auto Manual	Remote man.[n] None None	C - C	C - C	O - C	FC NA NA	Remote man. NA NA	NA NA NA	A NA NA	In
70C	1.141	Containment pressure detector	DC 702 silicone oil	Tubing	Yes	1X4DB131 2X4DB131	48	None	-	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
71A	56	Containment H_2 monitor suction	Containment atmosphere	0.75 0.75 0.75	Yes	1X4DB213-2 2X4DB213-2	14	HV-2792B HV-2791B HV-2792A	In Out In	C	- 1'-9" -	Globe Globe Globe	Solenoid Solenoid Solenoid	E E E	Remote man. Remote man. Remote man.	None Remote man.[n] None	C C C	C C C	O O O	FC FC FC	Remote man. Remote man. Remote man.	NA NA NA	A B A	Out
71B	56	Containment H_2 monitor discharge	Containment atmosphere	0.75 0.75 0.75	Yes	1X4DB213-2 2X4DB213-2	8	HV-2793B 002 040	Out In In	C	2'-3" - -	Globe Check Globe	Solenoid Self Manual	E E N	Remote man. Auto Manual	Remote man.[n] None Manual	C - C	C - C	O - C	FC NA NA	Remote man. NA NA	NA NA NA	B NA NA	In
71C	1.141	Containment pressure detector	DC 702 silicone oil	Tubing	Yes	1X4DB131 2X4DB131	48	None	-	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
72A	54	Accumulator sample line	Borated water	0.75 0.75 0.75	No	1X4DB120 2X4DB120	28	HV-10950 159 178	In Out In	C	- 2'-9" -	Globe Globe Globe	Solenoid Manual Manual	N N N	Auto Manual Manual	Remote man. None None	C LC C	C LC C	C LC C	FC NA NA	CIA NA NA	15 NA NA	A NA NA	Out
72B	54	Accumulator sample line	Borated water	0.75 0.75 0.75	No	1X4DB120 2X4DB120	28	HV-10952 161 183	In Out In	C	- 2'-4" -	Globe Globe Globe	Solenoid Manual Manual	N N N	Auto Manual Manual	Remote man. None None	C LC C	C LC C	C LC C	FC NA NA	CIA NA NA	15 NA NA	A NA NA	Out
73A	54	Accumulator sample line	Borated water	0.75 0.75 0.75	No	1X4DB120 2X4DB120	28	HV-10951 160 181	In Out In	C	- 2'-9" -	Globe Globe Globe	Solenoid Manual Manual	N N N	Auto Manual Manual	Remote man. None None	C LC C	C LC C	C LC C	FC NA NA	CIA NA NA	15 NA NA	B NA NA	Out
73B	54	Accumulator sample line	Borated water	0.75 0.75 0.75	No	1X4DB120 2X4DB120	28	HV-10953 162 185	In Out In	C	- 2'-4" -	Globe Globe Globe	Solenoid Manual Manual	N N N	Auto Manual Manual	Remote man. None None	C LC C	C LC C	C LC C	FC NA NA	CIA NA NA	15 NA NA	B NA NA	Out
77	54	Reactor coolant drain tank pump discharge	Primary coolant	3 3 1 1 1 1 1 1 1 1 1 0.75	No	1X4DB127 2X4DB127	29	HV-7699 HV-7136 X-186(o) X-189(o) X-218(o) X-220(p) X-153(p) X-154(p) X-173(p) X-229 PSV-7699	In Out In In In In In In In In In	C	- 1'-0" - - - - - - - - 1'-0"	Dia Dia Globe Globe Globe Globe Globe Globe Globe Globe Relief	Air Air Manual Manual Manual Manual Manual Manual Manual Manual Self	N N N N N N N N N N N	Auto Auto Manual Manual Manual Manual Manual Manual Manual Manual Auto	Remote man. Remote man. None None None None None None None None None	O O C C C C C C C C C	O O C C C C C C C C C	C C C C C C C C C C C	FC FC NA NA NA NA NA NA NA NA N/A	CIA CIA NA NA NA NA NA NA NA NA N/A	15 15 NA NA NA NA NA NA NA NA N/A	A B NA NA NA NA NA NA NA NA N/A	Out

TABLE 6.2.4-1 (SHEET 12 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve			Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator	Essential or Nonessential	Primary	Secondary	Normal	Shutdown	Post-Accident					
78	54	Normal containment sump pumps discharge	Drains	3 3 0.75	No	1X4DB143 2X4DB143	10	HV-780 HV-781 PSV-0780	In Out In	C	- 3'-9" 3'-9"	Gate Gate Relief	Air Air Self	N N N	Auto Auto Auto	Remote man. Remote man. None	O O C	C C C	C C C	FC FC N/A	CIA CIA N/A	15 15 N/A	A B N/A	Out Out
79	54	Reactor coolant drain tank vent and H_2 supply	Gas	0.75 0.75	No	1X4DB127 2X4DB127	9	HV-7126 HV-7150	In Out	C	- 1'-9"	Dia Dia	Air Air	N N	Auto Auto	Remote man. Remote man.	O O	C C	C C	FC FC	CIA CIA	15 15	A B	Out
80	56	Service air and post-LOCA purge air supply	Compressed air	4 4 0.75 0.75	No	1X4DB186-1 2X4DB186-1	6	HV-9385 034 228 229	Out In Out In	C	2'-0" - 1'-3" -	Gate Check Globe Globe	Air Self Manual Manual	N N N N	Auto Auto Manual Manual	Remote man. None None None	C - C C	C - C C	C - C C	FC NA NA NA	CIA NA NA NA	20 NA NA NA	A,B NA NA NA	In
81	54	Instrument air	Compressed air	2 2 0.75	No	1X4DB186-4 2X4DB186-4	5	HV-9378 049 256	Out In In	C	1'-6" - -	Globe Check Globe	Air Self Manual	N N N	Auto Auto Manual	Remote man. None None	O - C	O - C	C - C	FC NA NA	CIA NA NA	15 NA NA	A,B NA NA	In
83	56	Normal containment purge supply and equalizing	Containment atmosphere	24 24 14 14 0.75	No	1X4DB213-1 2X4DB213-1	15	HV-2626A HV-2627A HV-2626B HV-2627B 001	In Out In Out Out	C	- 7'-0" - 6'-0" 2'-3"	B-fly B-fly B-fly B-fly Gate	Elec. motor Elec. motor Air Air Manual	N N N N N	Auto Auto Auto Auto Manual	Remote man. Remote man. Remote man. Remote man. None	LC LC C C LC	O O C C LC	C C C C LC	FAI FAI FC FC NA	CVI CVI CVI CVI NA	10 10 5 5 NA	A B A B NA	In
84	56	Normal containment purge exhaust and equalizing	Containment atmosphere	24 24 14 14 0.75	No	1X4DB213-1 2X4DB213-1	47	HV-2628A HV-2629A HV-2628B HV-2629B 001	In Out In Out Out	C	- 13'-0" - 7'-0" 2'-3"	B-fly B-fly B-fly B-fly Gate	Elec. motor Elec. motor Air Air Manual	N N N N N	Auto Auto Auto Auto Manual	Remote man. Remote man. Remote man. Remote man. None	LC LC C C LC	O O C C LC	C C C C LC	FAI FAI FC FC NA	CVI CVI CVI CVI NA	10 10 5 5 NA	A B A B NA	Out
85C	1.141	Containment pressure detector	DC 702 silicone oil	Tubing	Yes	1X4DB131 2X4DB131	48	None	-	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
86A	56	Post-accident sampling	Containment atmosphere	1.0 1.0 0.5	No	1X4DB110 2X4DB110	45	HV-8211 HV-8212 X-002	In Out In	C	- 8'-6" -	Globe Globe Globe	Solenoid Solenoid Manual	N N N	Auto Auto Manual	Remote man. Remote man. None	C C C	C C C	C C C	FC FC NA	CIA CIA NA	15 15 NA	B A NA	In
87	NA	Containment leak rate test	Containment atmosphere	0.75 8.0 0.75	No	1X4DB132 2X4DB132	18	019(c) NA(p) 018	In In In	B	- - -	Globe Flange Globe	Manual NA Manual	N N N	Manual NA Manual	None None None	C NA C	C NA C	C NA C	NA NA NA	NA NA NA	NA NA NA	NA NA NA	-
88A	1.141	Reactor vessel water level instrumentation	Water	Tubing	Yes	1X4DB113 2X4DB113	54	None	-	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
88B	1.141	Reactor vessel water level instrumentation	Water	Tubing	Yes	1X4DB113 2X4DB113	54	None	-	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-

TABLE 6.2.4-1 (SHEET 13 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve			Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator	Essential or Nonessential	Primary	Secondary	Normal	Shutdown	Post-Accident					
88C	1.141	Reactor vessel water level instrumentation	Water	Tubing	Yes	1X4DB113 2X4DB113	54	None	-	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
89	NA	Transfer tube	Refueling pool water	20 0.50	No	1X4DB130 2X4DB130	13	NA 090	In In	B	-	Flange Globe	NA Manual	N N	NA Manual	None None	C C	C[c] C	C C	NA NA	NA NA	NA NA	NA NA	-
90	NA	Eddy current/sludge lancing	NA	0.75 10	No	1X4DB159-1 2X4DB159-1	56[k]	X-019	Out In	B	- -	Globe Flange	Manual	N N	Manual	None None	C NA	C NA	C NA	NA NA	NA NA	NA NA	NA NA	-
91	57	NSCW supply to containment coolers	Treated well water	8 1 0.75 1 1 1 0.5 0.5 1 0.5 0.5 1 1 0.75 1 1 1 1 1 1 1 1	Yes	1X4DB135-2 2X4DB135-2	55	HV-1809 019 PSV-1817 X-005 X-652 X-486 X-394 X-395 X-393 X-397 X-398 X-396 X-487 PSV-11774 365 369 375 377 387 381 439 441	Out In In Out In In In In In In In In In In In In In In In In In In	A	2'-6" - - 1'-6" - - - - - - - - - - - - - - - - -	B-fly Globe Relief Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Relief Globe Globe Globe Globe Globe Globe Globe Globe	Elec. motor Manual Self Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Self Manual Manual Manual Manual Manual Manual Manual Manual	E N N N N N N N N N N N N N N N N N N N N N	Auto Manual Auto Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Auto Manual Manual Manual Manual Manual Manual Manual Manual	Remote man. None None None None None None None None None None None None None None None None None None None None None	O C C C C C C C C C C C C C C C C C C C C C C	O C C C C C C C C C C C C C C C C C C C C C C	O C C C C C C C C C C C C C C C C C C C C C C	FAI NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	SI NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	B NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	In
92	57	NSCW supply to containment coolers	Treated well water	8 3 1 0.75 1 1 0.5 0.5 1 1 0.5 0.5 1 1 1 1 1 1 1 1 1	Yes	1X4DB135-2 2X4DB135-2	44	HV-1807 001 018 PSV-1815 X-077 X-656 X-400 X-401[o] X-500 X-245 X-403 X-404 393 397 403 405 409 415 435 437 X-399[p]	Out In In In Out In In In In In In In In In In In In In In In In In	A	2'-6" - - - 1'-6" - - - - - - - - - - - - - - - - -	B-fly Gate Globe Relief Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe Globe	Elec. motor Manual Manual Self Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual	E N N N N N N N N N N N N N N N N N N N N N	Auto Manual Manual Auto Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual Manual	Remote man. None None None None None None None None None None None None None None None None None None None None None	O C C C C C C C C C C C C C C C C C C C C C C	O C C C C C C C C C C C C C C C C C C C C C C	O C C C C C C C C C C C C C C C C C C C C C C	FAI NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	SI NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	B NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	In

TABLE 6.2.4-1 (SHEET 14 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve			Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator	Essential or Nonessential	Primary	Secondary	Normal	Shutdown	Post-Accident					
93	57	NSCW supply to containment coolers	Treated well water	8	Yes	1X4DB135-1	44	HV-1806	Out	A	2'-0"	B-fly	Elec. motor	E	Auto	Remote man.	O	O	O	FAI	SI	NA	A	In
				3		2X4DB135-1		002	In		-	Gate	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				PSV-1814	In		-	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				1				X-127	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-196	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-197	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-199	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-120	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-121(o)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-212(o)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				PSV-11672	In		-	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				1				285	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				291	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				293	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				297	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				303	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				305	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				338	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				340	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				482	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-198(p)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
94	57	NSCW supply to containment coolers	Treated well water	8	Yes	1X4DB135-1	55	HV-1808	Out	A	3'-0"	B-fly	Elec. motor	E	Auto	Remote man.	O	O	O	FAI	SI	NA	A	In
				1		2X4DB135-1		015	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				PSV-1816	In		-	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				1				X-259	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-193	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-194	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-192	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-189	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-190	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				X-191	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				PSV-11671	In		-	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				1				261	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				267	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				269	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				273	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				279	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				281	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				330	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				332	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	

TABLE 6.2.4-1 (SHEET 15 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve			Actuation Mode		Valve Position					Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator	Essential or Nonessential	Primary	Secondary	Normal	Shutdown	Post-Accident	Power Failure	Actuation Signal			
95	57	NSCW return from containment coolers	Treated well water	8	Yes	1X4DB135-2	16	HV-1831	Out	A	2'-0"	B-fly	Elec. motor	E	Auto	Remote man.	O	O	O	FAI	SI	NA	B	Out
				1				X-046	Out		1'-6"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				034	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-488	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-489	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-383	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-384	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				367	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				373	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				371	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				379	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				385	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				383	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				389	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				391	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
96	57	NSCW return from containment coolers	Treated well water	8	Yes	1X4DB135-2	16	HV-1823	Out	A	2'-6"	B-fly	Elec. motor	E	Auto	Remote man.	O	O	O	FAI	SI	NA	B	Out
				1		2X4DB135-2		X-052	Out		1'-6"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				032	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-501 (o)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-386	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.75				PSV-11773 (o)	In		-	Relief	Self	N	Auto	None	C	C	C	NA	NA	NA	NA	
				1				395	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				401	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				399	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				407	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				413	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				411 (o)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				417	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				419 (p)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-385 (p)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				396 (p)	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
								X-813																
97	57	NSCW return from containment coolers	Treated well water	8	Yes	1X4DB135-1	16	HV-1830	Out	A	2'-0"	B-fly	Elec. motor	E	Auto	Remote man.	O	O	O	FAI	SI	NA	A	Out
				1		2X4DB135-1		017	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-177	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-178	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				263	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				265	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				271	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				277	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				275	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				283	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				334	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				336	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	

TABLE 6.2.4-1 (SHEET 16 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve			Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator	Essential or Nonessential	Primary	Secondary	Normal	Shutdown	Post-Accident					
98	57	NSCW return from containment coolers	Treated well water	8	Yes	1X4DB135-1	16	HV-1822	Out	A	3'-0"	B-fly	Elec. motor	E	Auto	Remote man.	O	O	O	FAI	SI	NA	A	Out
				1		2X4DB135-1		016	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-179	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				289	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				287	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				295	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				301	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				299	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				307	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				342	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				344	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-180	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
100	56	Post-accident air exhaust	Containment atmosphere	4	No	1X4DB213-1	22	HV-2624A	In	C	-	B-fly	Elec. motor	N	Auto	Remote man.	C	C	C	FAI	CVI	NA[(j)]	A	Out
				4		2X4DB213-1		HV-2624B	In		-	B-fly	Elec. motor	N	Auto	Remote man.	C	C	C	FAI	CVI		B	
				4				012	Out		6'-3"	Gate	Manual	N	Manual	None	LC	LC	LC	NA	NA	NA[(j)]	NA	
				0.75				001	Out		6'-6"	Gate	Manual	N	Manual	None	LC	LC	LC	NA	NA	NA	NA	
101	57[(i)]	Auxiliary feedwater	Secondary coolant	6	Yes	1X4DB168-3	49	128	In	A	-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	In
				6		2X4DB168-3		120	Out		5'-8"	Check	Self	N	Auto	None	-	-	-	NA	NA	NA	NA	
				4				115	Out		3'-3"	Stop	Self	E	Auto	None	LO	LO	LO	NA	NA	NA	NA	
				6				HV-15198	Out		8'-3"	Gate	Air	N	Auto	Remote man.	O	C	C	FC	FI	NA	A,B	
				1				X-194[(o)]	Out		1'-8"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				136	Out		2'-2"	Check	Self	N	Auto	None	-	-	-	NA	NA	NA	NA	
				0.5				HV-5196	Out		5'-2"	Globe	Air	N	Remote man.	None	C	C	C	FC	Remote man.[(h)]	NA	NA	
				1				X-186	Out		18'-9"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-191[(p)]	Out		31'-4"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-193[(p)]	Out		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-241	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
102	57[(i)]	Auxiliary feedwater	Secondary coolant	6	Yes	1X4DB168-3	49	126	In	A	-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	In
				6		2X4DB168-3		118	Out		6'-4"	Check	Self	N	Auto	None	-	-	-	NA	NA	NA	NA	
				4				114	Out		6'-6"	Stop	Self	E	Auto	None	LO	LO	LO	NA	NA	NA	NA	
				6				HV-15197	Out		9'-0"	Gate	Air	N	Auto	Remote man.	O	C	C	FC	FI	NA	A,B	
				1				X-195	Out		4'-7"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				134	Out		2'-9"	Check	Self	N	Auto	None	-	-	-	NA	NA	NA	NA	
				0.5				HV-5195	Out		8'-0"	Globe	Air	N	Remote man.	None	C	C	C	FC	Remote Man.[(h)]	NA	NA	
				1.0				X-188	Out		20'-0"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-197	Out		30'-7"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				1				X-237	In		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	

TABLE 6.2.4-1 (SHEET 17 OF 17)

Penetration Number	GDC or RG	System Name	Fluid	Line Size (in.)	ESF or Support Systems	Drawing Number	Valve Arrangement Fig. 6.2.4-1	Valve Number	Location Relative to Containment Inside/ Outside	Type Tests	Length of Pipe (ft-in.)	Valve		Essential or Nonessential	Actuation Mode		Valve Position			Power Failure	Actuation Signal	Valve Closure Time (s)	Power Source 1E Bus A or B	Normal Direction of Flow
												Type	Operator		Primary	Secondary	Normal	Shutdown	Post-Accident					
103	57[i]	Auxiliary feedwater	Secondary coolant	6	Yes	1X4DB168-3	49	127	In	A	-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	In
				6		2X4DB168-3		119	Out		13'-0"	Check	Self	N	Auto	None	-	-	-	NA	NA	NA	NA	
				4				116	Out		10'-9"	Stop	Self	E	Auto	None	LO	LO	LO	NA	NA	NA	NA	
				6				HV-15199	Out		15'-10"	Gate	Air	N	Auto	Remote man.	O	C	C	FC	FI	NA	A,B	
				1				X-180[b]	Out		5'-2"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				135	Out		7'-1"	Check	Self	N	Auto	None	-	-	-	NA	NA	NA	NA	
				0.5				HV-5197	Out		7'-10"	Globe	Air	N	Remote man.	None	C	C	C	FC	Remote Man.[h]	NA	NA	
				1				X-215	In		-	Globe	Manual	N	Manual	None	C	C	C	NA		NA	NA	
				1				X-181[o]	Out		-	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
																					NA			
104	57[i]	Auxiliary feedwater	Secondary coolant	6	Yes	1X4DB168-3	49	125	In	A	-	Check	Self	E	Auto	None	-	-	-	NA	NA	NA	NA	In
				6		2X4DB168-3		117	Out		3'-9"	Check	Self	N	Auto	None	-	-	-	NA	NA	NA	NA	
				4				113	Out		9'-1"	Stop	Self	E	Auto	None	LO	LO	LO	NA	NA	NA	NA	
				6				HV-15196	Out		17'-2"	Gate	Air	N	Auto	Remote man.	O	C	C	FC	FI	NA	A,B	
				1				X-178	Out		5'-2"	Globe	Manual	N	Manual	None	C	C	C	NA	NA	NA	NA	
				0.5				133	Out		7'-1"	Check	Self	N	Auto	None	-	-	-	NA	NA	NA	NA	
				0.5				HV-5194	Out		7'-10"	Globe	Air	N	Remote man.	None	C	C	C	FC	Remote Man.[h]	NA	NA	
				1				X-233	In		-	Globe	Manual	N	Manual	None	C	C	C	NA		NA	NA	
																					NA			
	56	Equipment hatch	Containment atmosphere	NA	No	NA	42	NA	NA	B	NA	None	None	N	Manual	Manual	C	C	C	NA	NA	NA	NA	
	56	Personnel locks	Containment atmosphere	NA	No	NA	43	NA	NA	B	NA	None	None	N	Manual	Manual	C	C	C	NA	NA	NA	NA	
	56	Emergency doors	Containment	NA	No	NA	43	NA	NA	B	NA	None	None	N	Manual	Manual	C	C	C	NA	NA	NA	NA	

a. The following is a list of abbreviations:

GDC	-	General Design Criteria	SLI	-	steam line isolation
RG	-	NRC Regulatory Guides	FI	-	feedwater isolation
Dia	-	diaphragm	CVI	-	containment ventilation isolation
B-fly	-	butterfly	CIA	-	containment isolation phase A
O	-	open	CIB	-	containment isolation phase B
C	-	closed	SI	-	safety injection signal
LC	-	locked closed	CS	-	containment spray signal
FC	-	failed closed	E/H	-	electrohydraulic
Self	-	actuated by the fluid pressure	FAI	-	fail as is
NA	-	not applicable	AFS	-	auxiliary feedwater automatic start signal
Auto	-	automatic			

b. These valves are required to function for long term cooldown and recirculation.

c. These valves and/or flanges are opened for the refueling operation.

d. Valves identified in this table are shown with an asterisk (*) in figure 6.2.4-1.

e. Opens coincident on SI and RWST low-low.

f. Valve closure time is defined in SRP 6.2.4 paragraph II.6.N.

g. The ACCW penetrations are not part of an engineered safety features system nor are they required for safe shutdown; however, the penetrations are classified as essential due to the importance of maintaining cooling water to the reactor coolant pump. These valves remain open post-accident and are only closed for certain accident conditions.

h. Power to the solenoid shall be removed during all modes of operation, except during wet layup operation and periodic testing of the valves, by moving the slide links of the states terminal blocks, located in the terminal box, to the open position.

i. These penetrations are associated with the secondary side of the steam generators. The present design isolation provisions for secondary system lines meet the intent of GDC 57 (i.e., one isolation valve capable of remote manual operation is provided), but GDC 57 was not the design basis. The steam generator and associated secondary system piping form the primary barrier to the outside environment, much the same as the containment liner Plate. The valves associated with these penetrations do not receive a containment isolation signal and are not credited with affecting containment isolation in the safety analysis. The verification of the integrity of this barrier is accomplished in the type A test.

j. The stroke time of these values is 20 s. This penetration is verified closed by the Technical Specifications.

k. These penetrations are associated with the steam generator eddy current and sludge lancing operation. During modes 5 and 6, the blind flanges may be removed and replaced with a fixture that allows cables and hoses to pass through. The cables are sealed, and manually operated valves are provided on the hoses both inside and outside containment to provide the capability for containment closure during fuel movement as required by Technical Specifications.

l. Valve disk is provided with bonnet vent on the containment side of disk.

m. See table 6.3.2-3 for other stroke time information.

n. For LOCA conditions with a loss of one train of dc power, the secondary actuation mode for valves HV-2791A/B and HV-2793A/B is accomplished by providing an alternate train power connection in the QPCP panel per plant procedures.

o. Unit 1 only.

p. Unit 2 only.

q. Unit 2 is a gate valve.

TABLE 6.2.4-2 (SHEET 1 OF 15)

CONTAINMENT ISOLATION VALVES[a]

Valve Number	Function	Valve Closure Time(s)
1. Containment Isolation Phase "A"		
1HV-3502	Hot leg sample line	≤ 15
1HV-3548	Hot leg sample line	≤ 20
2HV-3502	Hot leg sample line and gross failed fuel detector	≤ 15
2HV-3548	Hot leg sample line and gross failed fuel detector	≤ 20
HV-8823	Safety injection pump discharge to cold leg	≤ 15
HV-8824	Safety injection pump discharge to hot leg	≤ 15
HV-8843	Boron injection line to cold leg	≤ 15
HV-8881	Safety injection pump discharge to hot leg	≤ 15
HV-27901	Fire protection water	≤ 20
HV-8871	Accumulator test and drain line	≤ 15
HV-8964	Accumulator test and drain line	≤ 15
HV-8888	Accumulator test and fill line	≤ 15
HV-8880	Nitrogen supply to accumulator	≤ 15
HV-8160	Normal letdown line	≤ 15

TABLE 6.2.4-2 (SHEET 2 OF 15)

Valve Number	Function	Valve Closure Time(s)
1. Containment Isolation Phase "A" (continued)		
HV-8152	Normal letdown line	≤ 15
HV-8100	Excess letdown and seal water leakoff	≤ 15
HV-8112	Excess letdown and seal water leakoff	≤ 15
HV-8825	RHR pump discharge to hot leg	≤ 15
HV-8890A	RHR pump discharge to cold leg	≤ 15
HV-8890B	RHR pump discharge to cold leg	≤ 15
HV-8033	Pressurizer relief tank sample to waste gas compressor suction	≤ 15
HV-8047	Pressurizer relief tank sample to waste gas compressor suction	≤ 15
HV-8028	Pressurizer relief tank makeup water supply	≤ 15
HV-3514	Pressurizer steam sample line	≤ 15
HV-3513	Pressurizer steam sample line	≤ 15
HV-3507	Pressurizer liquid sample line	≤ 15
HV-3508	Pressurizer liquid sample line	≤ 15
HV-10950	Accumulator sample line	≤ 15
HV-10952	Accumulator sample line	≤ 15
HV-10951	Accumulator sample line	≤ 15
HV-10953	Accumulator sample line	≤ 15
HV-7699	Reactor coolant drain tank pump discharge	≤ 15

TABLE 6.2.4-2 (SHEET 3 OF 15)

	Valve Number	Function	Valve Closure Time(s)
1.	Containment Isolation Phase "A" (continued)		
	HV-7136	Reactor coolant drain tank pump discharge	≤ 15
	HV-0780	Normal containment sump pumps discharge	≤ 15
	HV-0781	Normal containment sump pumps discharge	≤ 15
	HV-7126	Reactor coolant drain tank vent and H_2 supply	≤ 15
	HV-7150	Reactor coolant drain tank vent and H_2 supply	≤ 15
	HV-9385	Service air and post-LOCA purge air supply	≤ 20
	HV-9378	Instrument air	≤ 15
	HV-8211	Post-accident sampling	≤ 15
	HV-8212	Post-accident sampling	≤ 15
2.	Containment Ventilation Isolation		
	HV-12975	Containment air radioacitvity monitor inlet	≤ 15
	HV-12976	Containment air radioactivity monitor inlet	≤ 15
	HV-12977	Containment air radioactivity monitor outlet	≤ 15
	HV-12978	Containment air radioactivity monitor outlet	≤ 15
	HV-2626A	Containment pre-access purge supply and equalizing	≤ 10
	HV-2627A	Containment pre-access purge supply and equalizing	≤ 10

TABLE 6.2.4-2 (SHEET 4 OF 15)

Valve Number	Function	Valve Closure Time(s)
2. Containment Ventilation Isolation (continued)		
HV-2626B	Containment mini-purge supply and equalizing	≤ 5
HV-2627B	Containment mini-purge supply and equalizing	≤ 5
HV-2628A	Containment pre-access purge exhaust and equalizing	≤ 10
HV-2629A	Containment pre-access purge exhaust and equalizing	≤ 10
HV-2628B	Containment mini-purge exhaust and equalizing	≤ 5
HV-2629B	Containment mini-purge exhaust and equalizing	≤ 5
HV-2624A	Post-accident air exhaust	N/A
HV-2624B	Post-accident air exhaust	N/A
3. Safety Injection		
HV-8811B[(d)]	RHR emergency sump suction	N/A
HV-8811A[(d)]	RHR emergency sump suction	N/A
HV-2134[(e)]	NSCW supply to reactor cavity coolers	≤ 40
HV-2138[(e)]	NSCW return from reactor cavity coolers	≤ 40
HV-2135[(e)]	NSCW supply to reactor cavity coolers	≤ 40
HV-2139[(e)]	NSCW return from reactor cavity coolers	≤ 40
HV-8105	Normal charging line	≤ 17[(j)]
HV-1809[(e)]	NSCW supply to containment coolers	N/A

TABLE 6.2.4-2 (SHEET 5 OF 15)

	Valve Number	Function	Valve Closure Time(s)
3.	Safety Injection (continued)		
	HV-1807[e]	NSCW supply to containment coolers	N/A
	HV-1806[e]	NSCW supply to containment coolers	N/A
	HV-1808[e]	NSCW supply to containment coolers	N/A
	HV-1831[e]	NSCW return from containment coolers	N/A
	HV-1823[e]	NSCW return from containment coolers	N/A
	HV-1830[e]	NSCW return from containment coolers	N/A
	HV-1822[e]	NSCW return from containment coolers	N/A
	HV-8801A[d]	Boron injection line to cold leg	N/A
	HV-8801B[d]	Boron injection line to cold leg	N/A
4.	Check Valves		
	1418-U4-038	Demineralized water supply	N/A
	2401-U4-184	Breathing air supply	N/A
	1217-U4-113	ACCW return	N/A
	1204-U4-143	Safety injection to cold leg	N/A
	1204-U4-144	Safety injection to cold leg	N/A
	1204-U4-145	Safety injection to cold leg	N/A
	1204-U4-146	Safety injection to cold leg	N/A
	1204-U4-122	Safety injection to hot leg	N/A
	1204-U4-123	Safety injection to hot leg	N/A
	1204-U6-013	Boron injection to cold leg	N/A

TABLE 6.2.4-2 (SHEET 6 OF 15)

Valve Number	Function	Valve Closure Time(s)
4. Check Valves (continued)		
1204-U4-120	Safety injection to hot leg	N/A
1204-U4-121	Safety injection to hot leg	N/A
1206-U6-016	Containment spray supply	N/A
1206-U6-015	Containment spray supply	N/A
2301-U4-036	Fire protection water	N/A
2402-U4-017	Nitrogen supply to accumulator	N/A
1208-U4-021	Excess letdown and seal water leakoff	N/A
1208-U6-032	Normal charging line	N/A
1208-U4-355	Reactor coolant pump seal water supply	N/A
1208-U4-354	Reactor coolant pump seal water supply	N/A
1208-U4-353	Reactor coolant pump seal water supply	N/A
1208-U4-004	Reactor coolant pump seal water supply	N/A
1204-U6-128	RHR pump discharge to hot leg	N/A
1204-U6-129	RHR pump discharge to hot leg	N/A
1204-U6-147	RHR loop into cold leg	N/A
1204-U6-148	RHR loop into cold leg	N/A
1204-U6-149	RHR loop into cold leg	N/A
1204-U6-150	RHR loop into cold leg	N/A
1201-U6-112	Pressurizer relief tank makeup water supply	N/A
1513-U4-001	Containment H_2 monitor discharge	N/A

TABLE 6.2.4-2 (SHEET 7 OF 15)

	Valve Number	Function	Valve Closure Time(s)
4.	Check Valves (continued)		
	1513-U4-002	Containment H_2 monitor discharge	N/A
	2401-U4-034	Service air and post-LOCA purge air supply	N/A
	2420-U4-049	Instrument air	N/A
	1302-U4-126[(k)]	Auxiliary feedwater	N/A
	1302-U4-118[(k)]	Auxiliary feedwater	N/A
	1302-U4-114[(k)]	Auxiliary feedwater	N/A
	1302-U4-134[(k)]	Auxiliary feedwater	N/A
	1302-U4-128[(k)]	Auxiliary feedwater	N/A
	1302-U4-120[(k)]	Auxiliary feedwater	N/A
	1302-U4-115[(k)]	Auxiliary feedwater	N/A
	1302-U4-136[(k)]	Auxiliary feedwater	N/A
	1302-U4-127[(k)]	Auxiliary feedwater	N/A
	1302-U4-119[(k)]	Auxiliary feedwater	N/A
	1302-U4-116[(k)]	Auxiliary feedwater	N/A
	1302-U4-135[(k)]	Auxiliary feedwater	N/A
	1302-U4-125[(k)]	Auxiliary feedwater	N/A
	1302-U4-117[(k)]	Auxiliary feedwater	N/A
	1302-U4-113[(k)]	Auxiliary feedwater	N/A
	1302-U4-133[(k)]	Auxiliary feedwater	N/A

TABLE 6.2.4-2 (SHEET 8 OF 15)

	Valve Number	Function	Valve Closure Time(s)
5.	Remote Manual		
	HV-5280[b]	Chemical Addition	N/A
	HV-5281[b]	Chemical Addition	N/A
	HV-1978	ACCW Supply	N/A
	HV-1979	ACCW Supply	N/A
	HV-1974	ACCW Return	N/A
	HV-1975	ACCW Return	N/A
	HV-8835	Safety injection to cold leg	N/A
	HV-8802B	Safety injection to hot leg	N/A
	HV-8802A	Safety injection to hot leg	N/A
	HV-9002B[f]	Containment spray emergency sump suction	N/A
	HV-9002A[f]	Containment spray emergency sump suction	N/A
	HV-8103D	Reactor coolant pump seal water supply	N/A
	HV-8103B	Reactor coolant pump seal water supply	N/A
	HV-8103C	Reactor coolant pump seal water supply	N/A
	HV-8103A	Reactor coolant pump seal water supply	N/A
	HV-8840	RHR pump discharge to hot leg	N/A
	HV-8809A	RHR loop into cold leg	N/A
	HV-8809B	RHR loop into cold leg	N/A
	HV-8701A	RHR suction from hot leg	N/A
	HV-8702A	RHR suction from hot leg	N/A

TABLE 6.2.4-2 (SHEET 9 OF 15)

Valve Number	Function	Valve Closure Time(s)
5. Remote Manual (continued)		
HV-5278[b]	Chemical addition	N/A
HV-5279[b]	Chemical addition	N/A
HV-2791A[i]	Containment H_2 monitor suction	N/A
HV-2790A[i]	Containment H_2 monitor suction	N/A
HV-2790B[i]	Containment H_2 monitor suction	N/A
HV-2793A[i]	Containment H_2 monitor discharge	N/A
HV-2792B[i]	Containment H_2 monitor suction	N/A
HV-2791B[i]	Containment H_2 monitor suction	N/A
HV-2792A[i]	Containment H_2 monitor suction	N/A
HV-2793B[i]	Containment H_2 monitor discharge	N/A
HV-5194[b][k]	Auxiliary feedwater	N/A
HV-5197[b][k]	Auxiliary feedwater	N/A
HV-5195[b][k]	Auxiliary feedwater	N/A
HV-5196[b][k]	Auxiliary feedwater	N/A
HV-9556A[k]	Steam generator secondary side sample	N/A
HV-9556B[k]	Steam generator secondary side sample	N/A
HV-9555A[k]	Steam generator secondary side sample	N/A
HV-9555B[k]	Steam generator secondary side sample	N/A
HV-9554A[k]	Steam generator secondary side sample	N/A
HV-9554B[k]	Steam generator secondary side sample	N/A

TABLE 6.2.4-2 (SHEET 10 OF 15)

Valve Number	Function	Valve Closure Time(s)
5. Remote Manual (continued)		
HV-9553A[k]	Steam generator secondary side sample	N/A
HV-9553B[k]	Steam generator secondary side sample	N/A
HV-3009[h]	Main steam to auxiliary feedwater pump driver	N/A
HV-3019[h]	Main steam to auxiliary feedwater pump driver	N/A
6. Manual		
1213-U6-050[c]	Purification water supply to refueling cavity	N/A
1213-U6-051[c]	Purification water supply to refueling cavity	N/A
1418-U4-005[c]	Demineralized water supply	N/A
2401-U4-211[c]	Breathing air supply	N/A
1411-U4-676[c]	Chemical addition	N/A
1411-U4-677[c]	Chemical addition	N/A
1411-U4-678[c]	Chemical addition	N/A
1411-U4-679[c]	Chemical addition	N/A
1204-U4-159[c]	Accumulator sample line	N/A
1204-U4-161[c]	Accumulator sample line	N/A
1204-U4-160[c]	Accumulator sample line	N/A
1204-U4-162[c]	Accumulator sample line	N/A
1202-U4-001	NSCW supply to containment fire protection	N/A
1202-U4-002	NSCW supply to containment fire protection	N/A
1508-U4-012[c]	Post-accident air exhaust	N/A

TABLE 6.2.4-2 (SHEET 11 OF 15)

	Valve Number	Function	Valve Closure Time(s)
7.	Containment Spray		
	HV-9001A	Containment spray supply	N/A
	HV-9001B	Containment spray supply	N/A
8.	Pressure Relief Valves		
	PSV-17589	Plant demineralized water to containment	N/A
	PSV-11673	NSCW supply to reactor cavity coolers	N/A
	PSV-2136	NSCW from reactor cavity coolers	N/A
	PSV-2137	NSCW supply to reactor cavity coolers	N/A
	PSV-11772	NSCW from reactor cavity coolers	N/A
	PSV-8708A	RHR suction from hot leg	N/A
	PSV-8708B	RHR suction from hot leg	N/A
	PSV-1817	NSCW supply to containment coolers	N/A
	PSV-11774	NSCW supply to containment coolers	N/A
	PSV-1815	NSCW supply to containment coolers	N/A
	PSV-1814	NSCW supply to containment coolers	N/A
	PSV-11672	NSCW supply to containment coolers	N/A
	PSV-1816	NSCW supply to containment coolers	N/A
	PSV-11671	NSCW supply to containment coolers	N/A
	PSV-11773	NSCW supply to containment coolers	N/A
	PSV-3001	Main steam	N/A
	PSV-1978	ACCW Supply	N/A

TABLE 6.2.4-2 (SHEET 12 OF 15)

Valve Number	Function	Valve Closure Time(s)
8. Pressure Relief Valves (continued)		
PSV-8871	Accumulator Test and Drain	N/A
PSV-7699	RCDT Pump Discharge	N/A
PSV-0780	Normal Containment Sump Pump	N/A
PSV-3002	Main steam	N/A
PSV-3003	Main steam	N/A
PSV-3004	Main steam	N/A
PSV-3005	Main steam	N/A
PSV-3011	Main steam	N/A
PSV-3012	Main steam	N/A
PSV-3013	Main steam	N/A
PSV-3014	Main steam	N/A
PSV-3015	Main steam	N/A
PSV-3021	Main steam	N/A
PSV-3022	Main steam	N/A
PSV-3023	Main steam	N/A
PSV-3024	Main steam	N/A
PSV-3025	Main steam	N/A
PSV-3031	Main steam	N/A
PSV-3032	Main steam	N/A
PSV-3033	Main steam	N/A

TABLE 6.2.4-2 (SHEET 13 OF 15)

	Valve Number	Function	Valve Closure Time(s)
8.	Pressure Relief Valves (continued)		
	PSV-3034	Main steam	N/A
	PSV-3035	Main steam	N/A
9.	Other Automatic Valves		
	HV-3006A[(g)]	Main steam	≤5
	HV-3006B[(g)]	Main steam	≤5
	HV-3016A[(g)]	Main steam	≤5
	HV-3016B[(g)]	Main steam	≤5
	HV-13005A[(g)]	Main steam	≤5
	HV-13005B[(g)]	Main steam	≤5
	HV-13007A[(g)]	Main steam	≤5
	HV-13007B[(g)]	Main steam	≤5
	HV-3026A[(g)]	Main steam	≤5
	HV-3026B[(g)]	Main steam	≤5
	HV-13008A[(g)]	Main steam	≤5
	HV-13008B[(g)]	Main steam	≤5
	HV-3036A[(g)]	Main steam	≤5
	HV-3036B[(g)]	Main steam	≤5
	HV-13006A[(g)]	Main steam	≤5
	HV-13006B[(g)]	Main steam	≤5

TABLE 6.2.4-2 (SHEET 14 OF 15)

Valve Number	Function	Valve Closure Time(s)
9. Other Automatic Valves (continued)		
HV-7603A[h]	Steam generator blowdown	≤15
HV-7603B[h]	Steam generator blowdown	≤15
HV-7603C[h]	Steam generator blowdown	≤15
HV-7603D[h]	Steam generator blowdown	≤15
HV-9451[h]	Steam generator secondary side sample	≤15
HV-9452[h]	Steam generator secondary side sample	≤15
HV-9453[h]	Steam generator secondary side sample	≤15
HV-9454[h]	Steam generator secondary side sample	≤15
HV-5229[g]	Feedwater	≤5
HV-5228[g]	Feedwater	≤5
HV-5230[g]	Feedwater	≤5
HV-5227[g]	Feedwater	≤5
HV-15198[g]	Feedwater	≤5
HV-15197[g]	Feedwater	≤5
HV-15196[g]	Feedwater	≤5
HV-15199[g]	Feedwater	≤5
PV-3000	Main steam	N/A
PV-3010	Main steam	N/A
PV-3020	Main steam	N/A
PV-3030	Main steam	N/A

TABLE 6.2.4-2 (SHEET 15 OF 15)

a. See FSAR subsection 6.2.4 for discussion of the containment isolation system.

b. The containment isolation valves will be maintained closed by administratively controlling the air supply valve links in an open position.

c. Locked closed.

d. These valves are included for table completeness. The requirements of Technical Specification 3.6.3 do not apply; instead the requirements of Technical Specification 3.5.2 apply. Valve stroke times where specified will be tested pursuant to the IST program.

e. These valves are included for table completeness. The requirements of Technical Specification 3.6.3 do not apply; instead the requirements of Technical Specification 3.7.8 apply. Valve stroke times where specified will be tested pursuant to the IST program.

f. These valves are included for table completeness. The requirements of Technical Specification 3.6.3 do not apply; instead the requirements of Technical Specification 3.6.6 apply. Valve stroke times where specified will be tested pursuant to the IST program.

g. These valves are included for table completeness. The requirements of Technical Specification 3.6.3 do not apply; instead the requirements of Technical Specification 3.7.2 apply to the main steam isolation and bypass valves and Technical Specifications 3.3.2 and 3.7.3 apply to the main feedwater isolation valves.

h. These valves are included for table completeness. The requirements of Technical Specification 3.6.3 do not apply; instead the requirements of Technical Specification 3.7.5 apply. Valve stroke times where specified will be tested pursuant to the IST program.

i. These valves may be opened on an intermittent basis under administrative control.

j. See table 6.3.2-3 for other stroke time information.

k. These valves are included for table completeness. The requirements of Technical Specification 3.6.3 do not apply; instead verification of containment isolation is accomplished in the type A leakage test which credits the steam generator tubes and associated piping as the primary barrier to the outside environment. These valves are associated with the secondary side of the steam generators and are not subject to GDC 57. These valves do not receive a containment isolation signal and are not credited with effecting containment isolation in the safety analyses. Reference UFSAR paragraph 6.2.4.2.1 and table 6.2.4-1, penetrations 102, 103, and 104 and reference note "I."

NOTE: The valve table as shown reflects the inservice test program as modified by the second 10-year interval update. The updated IST program will be fully implemented by May 31, 1998.

TABLE 6.2.5-1

DESIGN DATA FOR HYDROGEN RECOMBINERS

Quantity	2 per unit
Power (each) (maximum/minimum) (kW)	75/50
Capacity (each) (minimum) (sf^3/min)	100
Heaters (per recombiner)	
Number	4 banks
Maximum heat flux (Btu/h-ft^2)	2850
Maximum sheath temperature (°F)	1550
Gas temperatures	
Inlet (°F)	80-155
Outlet of heater section (°F)	1150 to 1400
Exhaust (°F)	Approximately 50 above ambient
Materials	
Outer structure	Type 300 series SS
Inner structure	Incoloy 800
Heater element sheath	Incoloy 800
Base skid	Type 300 series SS
Weight (lb)	4500 lb
Codes and standards	American Society of Mechanical Engineers Section IX, Underwriters Laboratory, National Electric Manufacturers Association, National Fire Protection Association, Institute of Electrical and Electronic Engineers 279, 308, 323, 344, and 383, Safety Class 0 (See table 3.2.2-1.)

TABLE 6.2.5-2 (SHEET 1 OF 2)

CONTAINMENT BUILDING - FAILURE MODES AND EFFECTS ANALYSIS, POST-LOCA CAVITY PURGE SYSTEM

Item No.	Description of Component	Safety Function	Plant Operating Mode[(a)]	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability
1	No. 29 breaker on 1ABE, 480-V, 1E, MCC, train A normally closed (NC)	Provide continuity and protection to fan motor, item 3	A	Inadvertent open	Motor control center (MCC) alarm Fan motor lights Flow alarm low	None; loss of train A; train B available
2	No. 29 motor starter, for item 3, train A normally open (NO)	Provide continuity to fan motor, item 3	A	Fail to close	Fan motor lights Flow alarm low	None; loss of train A; train B available
3	1-1516-B7-001-M01, Containment building (CTB) post-LOCA cavity purge unit motor and fan train A, normally deenergized (ND)	Provide motive power by supplying air to the reactor cavity	A	Fail to start and operate	Flow alarm low	None; loss of train A; train B available
4	003 backflow damper, train A, NC	Allow flow of air to the reactor cavity and prevent backflow	A	Fail to open	Flow alarm low	None; loss of train A; train B available
5	No. 29 breaker on 1BBE, 480-V, 1E, MCC, train B, NC	Provide continuity and protection to fan motor, item 7	A	Inadvertent open	MCC alarm Fan motor lights	None; loss of train B; train A available Flow alarm low
6	No. 29 motor starter, item 7 train B, NO	Provide continuity to fan motor, item 7	A	Fail to close	Fan motor lights Flow alarm low	None; loss of train B; train A available
7	1-1516-B7-002-M01 CTB post-LOCA cavity purge unit motor and fan, train B, NO	Provide motive power by supplying air to the reactor cavity	A	Fail to start and operate	Flow alarm low	None; loss of train B; train A available

TABLE 6.2.5-2 (SHEET 2 OF 2)

Item No.	Description of Component	Safety Function	Plant Operating Mode[a]	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability
8	004 backflow damper, train B, NC	Allow flow of air to the reactor cavity and prevent backflow	A	Fail to open	Flow alarm low	None; loss of train B; train A available
9	Fan, fan shaft, bearing, etc., 1-1516-B7-001-000	Provide circulation of air	A	Mechanical failure	Flow alarm, low Motor indicating light	None; loss of train A; train B available
10	Fan, fan shaft, bearing, etc., 1-1516-B7-002-000	Provide circulation of air	A	Mechanical failure	Flow alarm low Motor indicating light	None; loss of train B; train A available

a. A - Safety injection for trains A and B start automatically and operate on safety injection only.

TABLE 6.2.5-3

DESIGN DATA FOR PRINCIPAL COMPONENTS OF POST-LOCA CONTAINMENT HYDROGEN PURGE SYSTEM

Containment Building Post-LOCA Purge Filter Exhaust Unit	
Quantity	1
Capacity (ft^3/min)	500
System Components	
Charcoal Adsorber	
Efficiency (%)	89.0[(a)]
Face velocity (ft/min)	40
Residence time (s/2-in. thickness)	0.25
Nominal size (Tyler mesh)	8 x 16
HEPA Filters	
Filter element	Pleated fiberglass
Size (in.)	24 x 24 x 12
Efficiency (%)	99.97% for 0.3-μm and larger particulates
Moisture Eliminator	
Separator element	Fiberglass or galvanized steel
Efficiency (%)	99% for 5 to 10 μm droplets
Electric Heating Coil	
Heater element	80% Ni, 20% Cr
Heating capacity (kW)	2.5

(a) Four-inch filter tested per Table 2 of Regulatory Guide 1.140. With bypass leakage of 1%, the charcoal adsorber section efficiency is 89%.

TABLE 6.2.5-4

DESIGN DATA FOR PRINCIPAL COMPONENTS
OF POST-LOCA CAVITY PURGE SYSTEM

Fan	
Quantity	2
Type	Centrifugal
Capacity (ft^3/min)	300
Static pressure (in. WG)	30
Motor (hp)	5

TABLE 6.2.5-5

DESIGN DATA FOR PRINCIPAL COMPONENTS OF
THE CONTAINMENT HYDROGEN MONITORING SYSTEM

Hydrogen Analyzer	
Quantity	2 per unit
Type	Thermal conductivity
Range	0 to 1 and 0 to 10 volume percent
Accuracy	±5 percent of full scale
Valves (isolation)	
Quantity	10
Type	8 solenoid-operated globe valves and 2 check valves
Tubing Material	Stainless steel (Class 2)

TABLE 6.2.5-6 (SHEET 1 OF 4)

PLANT PARAMETERS USED TO CALCULATE POST-ACCIDENT HYDROGEN PRODUCTION

Core thermal power (MWt) [(a)]	3565
Containment free volume (ft^3)	2.75×10^6
Normal containment temperature (°F)	120
Weight of zirconium clad incore (lb)	45,914
Percent zirconium-water reaction (%)	1.5
Hydrogen recombiner flowrate (sf^3/min)	100
Hydrogen recombiner efficiency (%)	95

Baseline Aluminum Inventory in Containment

Component	Weight (lb)	Surface (ft^2)
Flux map drive system	183	48
Nuclear instrumentation system	244	57
Digital rod position indicators	199	241
Control rod drive mechanism (CRDM) connectors	129	68
Miscellaneous valves (nuclear steam supply system) (NSSS)	230	86
Radiation monitoring system	4	4
Containment fan cooler return bend assemblies	765	3000
Communication equipment	65	10
Miscellaneous valves (balance of plant (BOP))	11	2
Containment lighting fixture plugs	10	6
Contingency (NSSS)	250	85
Baseline Total	2,090	3,607

TABLE 6.2.5-6 (SHEET 2 OF 4)

Baseline Zinc Inventory in Containment

	Type	Weight (lb)	Surface (ft^2)
Snubbers	Zn[b]	11	555
Integrated reactor vessel (RV) head/CRDM shroud	ZBP[c]	17,734	180,100
Platform grating	GS[d]	12,453	99,625
Pressurizer grating	GS	696	5,566
Steam generator grating	GS	813	6,500
Cables and related items	Zn	4,988	56,127
Cable tray supports	Zn	1,710	13,680
Inorganic zinc-based paint	ZBP	23,476	500,691
Miscellaneous BOP items	ZBP	36	31
Baseline Total		61,917	862,875

TABLE 6.2.5-6 (SHEET 3 OF 4)

Regulatory Guide 1.7 Hydrogen Production
Calculational Assumptions

Core Cooling Solution Radiolysis

Sources:

Percent of total halogens retained in the core	50
Percent of total noble gases retained in the core	0
Percent of other fission products retained in the core	99

Energy absorption by core cooling solution:

Percent of gamma energy absorbed by solution	10
Percent of beta energy absorbed by solution	0

Hydrogen production:

Molecules hydrogen produced per 100 eV energy absorbed by solution	0.50

TABLE 6.2.5-6 (SHEET 4 OF 4)

Sump Solution Radiolysis	
Sources:	
Percent of total halogens released to sump solution	50
Percent of noble gases released to sump solution	0
Percent of other fission products released to sump solution	1
Energy absorption by sump solution:	
Percent of total energy (beta and gamma) which is absorbed by the sump solution	100
Hydrogen production:	
Molecules of hydrogen produced per 100 eV of energy absorbed by the sump solution	0.5
Long-Term Aluminum Corrosion Rate	
Mils per year	200

a. Hydrogen generation analysis for core thermal power of 3565 MWt is bounding for MUR power uprate of 3625.6 MWt (see figure 6.2.5-7)

b. Zn - zinc metal.

c. ZBP - zinc-based paint.

d. GS - galvanized steel.

TABLE 6.2.5-7

CORE FISSION PRODUCT ENERGY AFTER 650 FULL-POWER DAYS

DELETED

TABLE 6.2.5-8

FISSION PRODUCT DECAY DEPOSITION IN SUMP SOLUTION

DELETED

TABLE 6.2.5-9

POST-ACCIDENT CONTAINMENT TEMPERATURE TRANSIENT
FOR HYDROGEN GENERATION ANALYSIS

Time Interval (s)	Temperature (°F)
0 - 1.0	rapid increase from 120 to 259
1.0 - 4 E3	259
4 E3 - 6 E3	Ramp down from 250 to 240
6 E3 - 1 E4	Ramp down from 240 to 225
1 E4 - 2 E4	Ramp down from 225 to 205
2 E4 - 4 E4	Ramp down from 205 to 190
4 E4 - 6 E4	Ramp down from 190 to 175
6 E4 - 1 E5	Ramp down from 175 to 160
1 E5 - 1.00 E6	Ramp down from 160 to 131[(a)]
> 1.00 E6	131

a. The long-term aluminum corrosion rate of 200 mils per year begins when the containment temperature drops below 154°F.

TABLE 6.2.5-10

CORROSION RATES USED IN THE POST-ACCIDENT
CONTAINMENT HYDROGEN GENERATION ANALYSIS

Temperature (°F)	Aluminum Corrosion Rate ($lb\text{-}moles_{al}/ft^2\text{-}h$)	Zinc Corrosion Rate[a] ($lb\text{-}moles_{zn}/ft^2\text{-}h$)
100	1.19 E-5	7.53 E-9
140	-	7.16 E-8
154	1.19 E-5	-
160	1.49 E-5	2.21 E-7
180	3.15 E-5	6.80 E-7
200	6.65 E-5	2.10 E-6
220	1.40 E-4	6.47 E-6
240	2.97 E-4	1.99 E-5
260	6.27 E-4	6.15 E-5
280	1.32 E-3	1.90 E-4
300	2.80 E-3	5.84 E-4

a. Zinc corrosion rate also applies to zinc-based paint.

TABLE 6.2.6-1 (SHEET 1 OF 2)

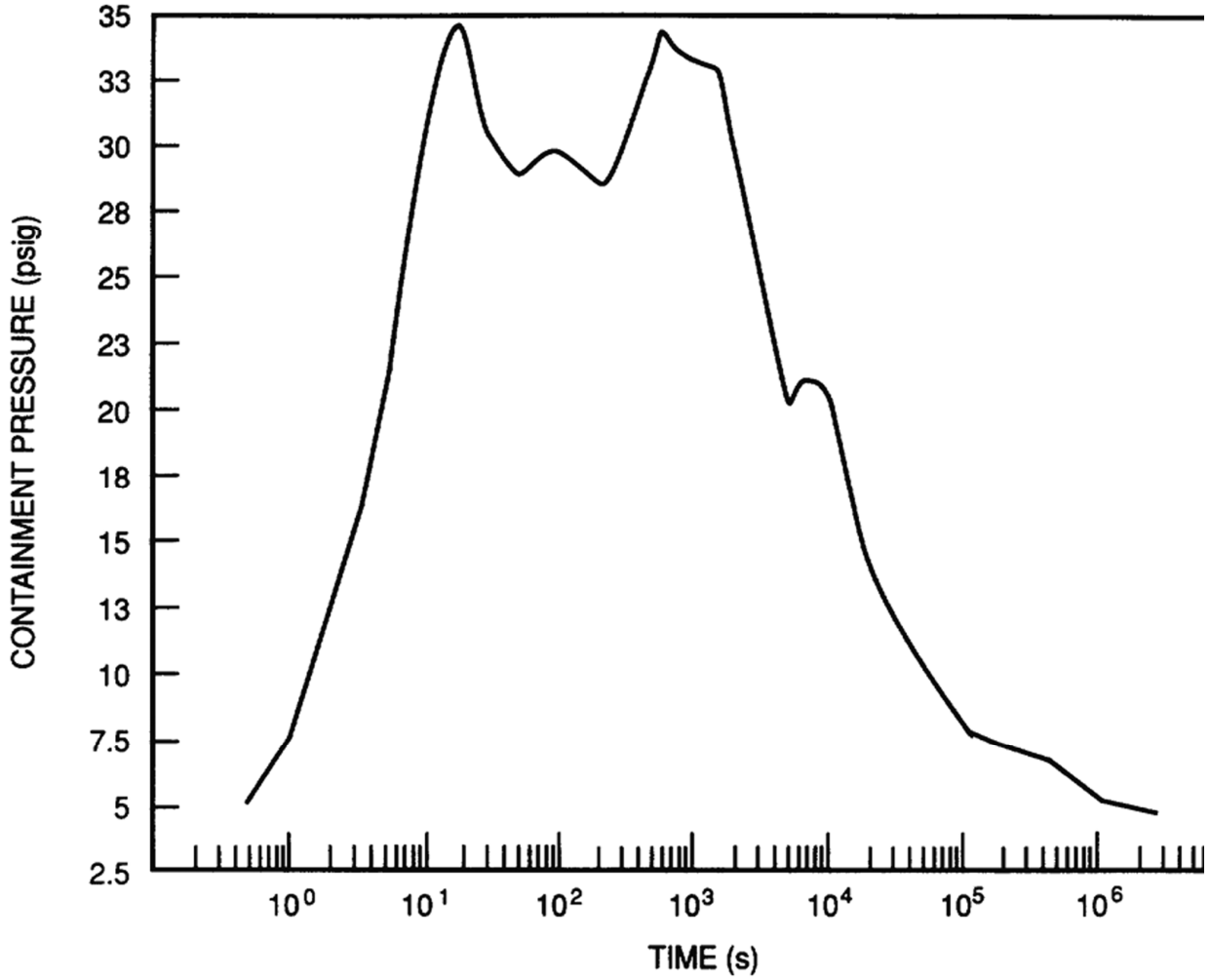
PENETRATIONS NOT VENTED TO CONTAINMENT OR DRAINED DURING TYPE A TESTING

Penetration Number[a]	Description[a]	Justification
1-4, 7-10, 11B, 11C, 12B, 12C, 18-21, 101-104	Steam generator secondary side (main steam blowdown, sampling, feedwater, and auxiliary feedwater)	The steam generator secondary side is considered an extension of the containment. The system is not part of the reactor coolant pressure boundary and does not open directly to the containment atmosphere during normal and post-accident conditions. However, since the secondary side of the steam generator is a viable leak path, the main steam lines are vented outside of containment.
30, 31, 33	Safety injection line	The system is normally filled with water from refueling water storage tank and operating under post-accident conditions.
32	Boron injection line (high head safety injection)	The system is normally filled with water from the charging pump discharge and operating under post-accident conditions.
36-39	Residual heat removal and containment spray pump suction from containment emergency sump	The system is normally filled with water from refueling water storage tank and operating under post-accident conditions. After an accident the static head between emergency sump and valve provides water seal to prevent leakage of containment atmosphere.
56, 57, 58	Residual heat removal discharge to reactor coolant system	The system is normally filled with water from the refueling water storage tank and operating under post-accident conditions.
59, 60	Residual heat removal suction from hot leg	The system is closed outside containment and constructed to ASME III, Class 2, and Seismic Category 1 standards.

TABLE 6.2.6-1 (SHEET 2 OF 2)

Penetration Number	Description	Justification
43-46, 91-98	Nuclear service cooling water supply and return	The system is a closed system inside containment per GDC 57 and thus not a potential atmospheric leak path. The system is normally filled with water and operating post-accident. The system may be operated to cool the containment atmosphere during the Type A tests.
51-54	RCP seal water supply	The system is closed outside containment and constructed to ASME III, Class 2, and Seismic Category 1 standards. The system is filled with water during all modes of plant operation (normal and post-accident) by the charging pumps.
13C, 67C, 69C, 70C, 71C, 85C	Containment pressure detectors	The system is filled with liquid and designed to satisfy the requirements of Regulatory Guide 1.141. Also the system is closed both inside and outside the containment.
14A, 14B, 14C, 88A, 88B, 88C	Reactor vessel water level instrumentation	The system is filled with liquid and designed to satisfy the requirements of Regulatory Guide 1.141. Also the system is closed both inside and outside.

a. See table 6.2.4-1 for further description.



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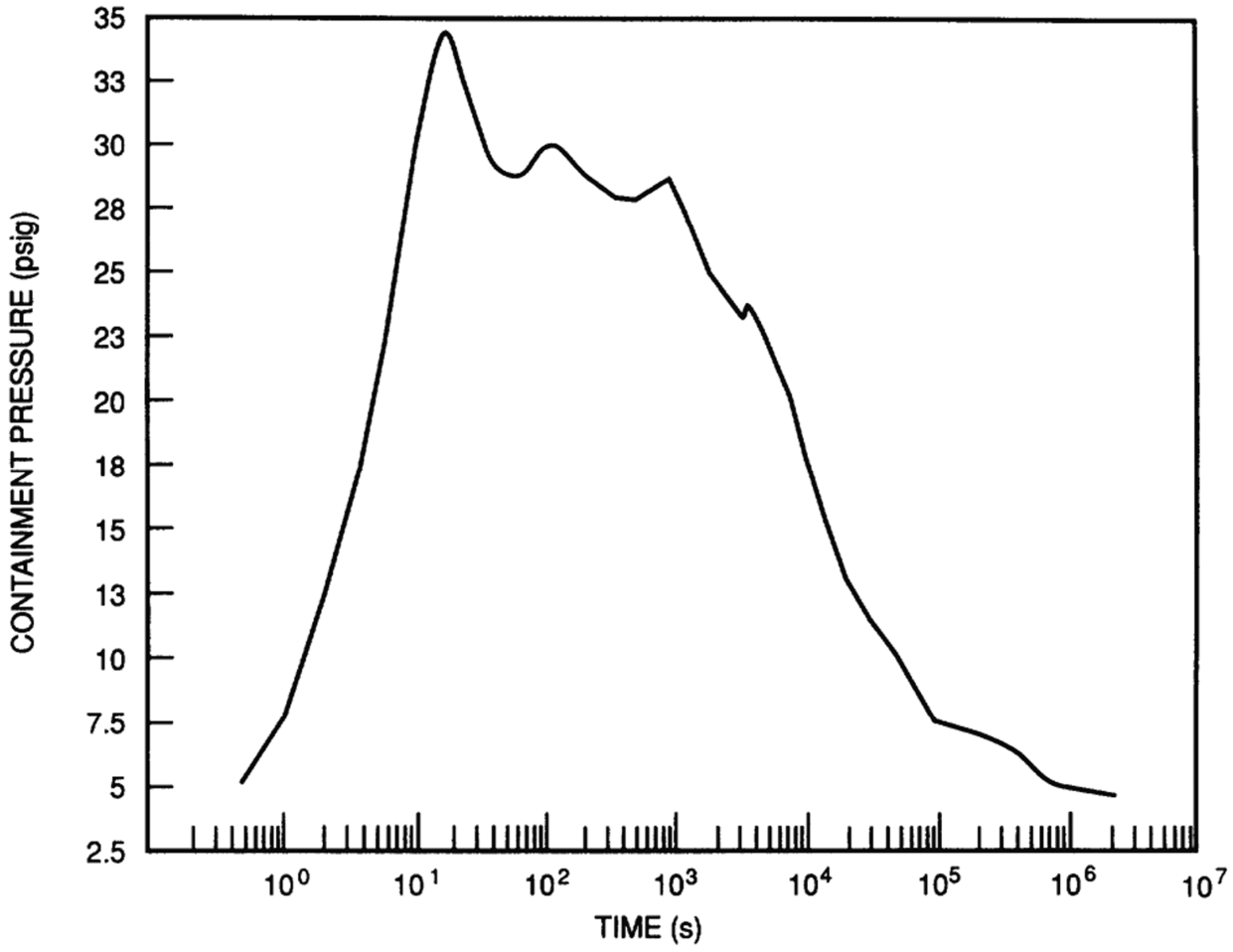
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CONTAINMENT PRESSURE TRANSIENT
LOCA–DOUBLE–ENDED PUMP SUCTION–
MINIMUM SAFETY INJECTION

FIGURE 6.2.1–1



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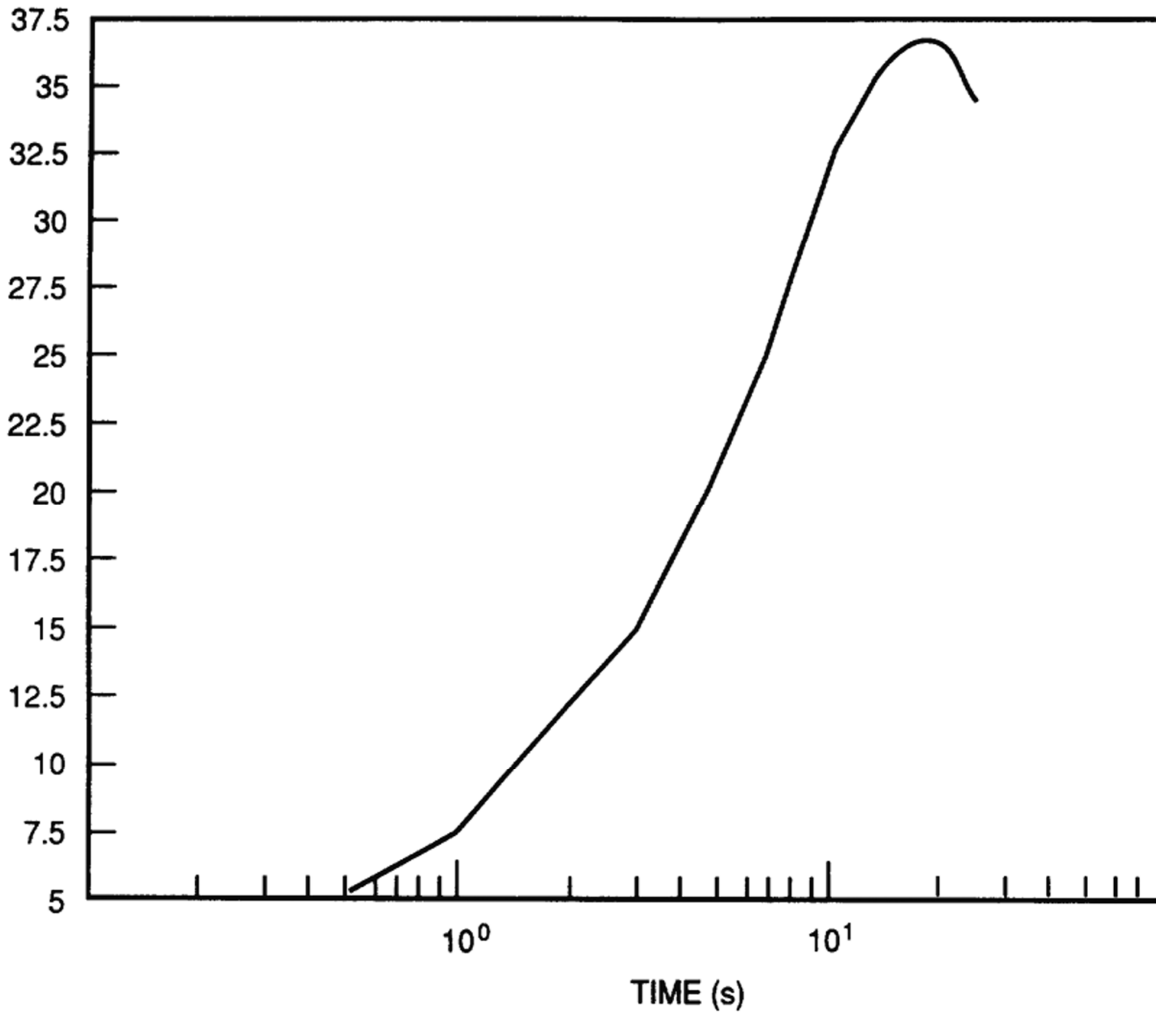
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	CONTAINMENT PRESSURE TRANSIENT LOCA–DOUBLE–ENDED PUMP SUCTION, MAXIMUM SAFETY INJECTION
	FIGURE 6.2.1–2



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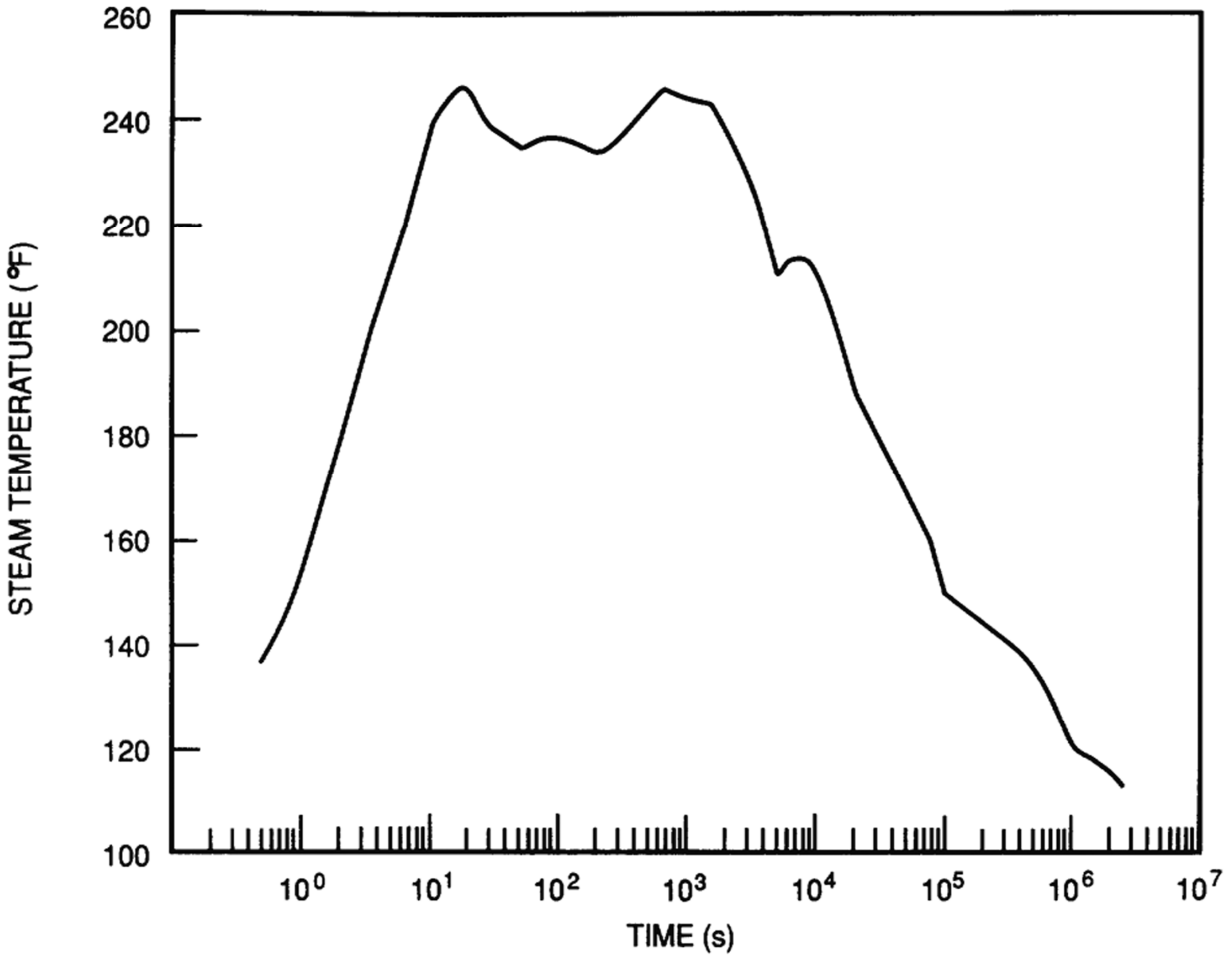
UNIT 1 AND UNIT 2

CONTAINMENT PRESSURE TRANSIENT
LOCA–DOUBLE–ENDED HOT LEG
GUILLOTINE

FIGURE 6.2.1–3



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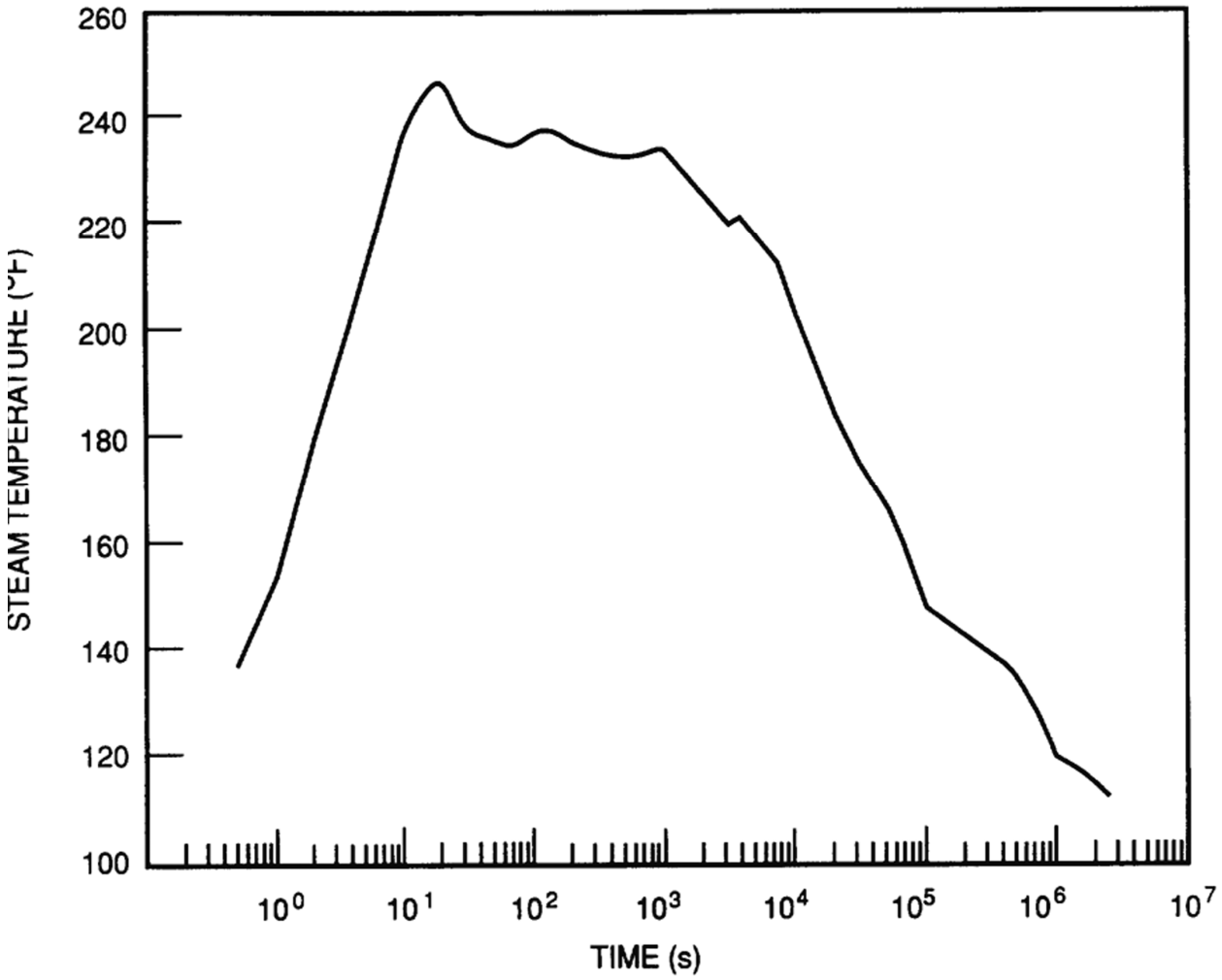
UNIT 1 AND UNIT 2

CONTAINMENT TEMPERATURE TRANSIENT
LOCA–DOUBLE–ENDED PUMP SUCTION–
MINIMUM SAFETY INJECTION

FIGURE 6.2.1–4



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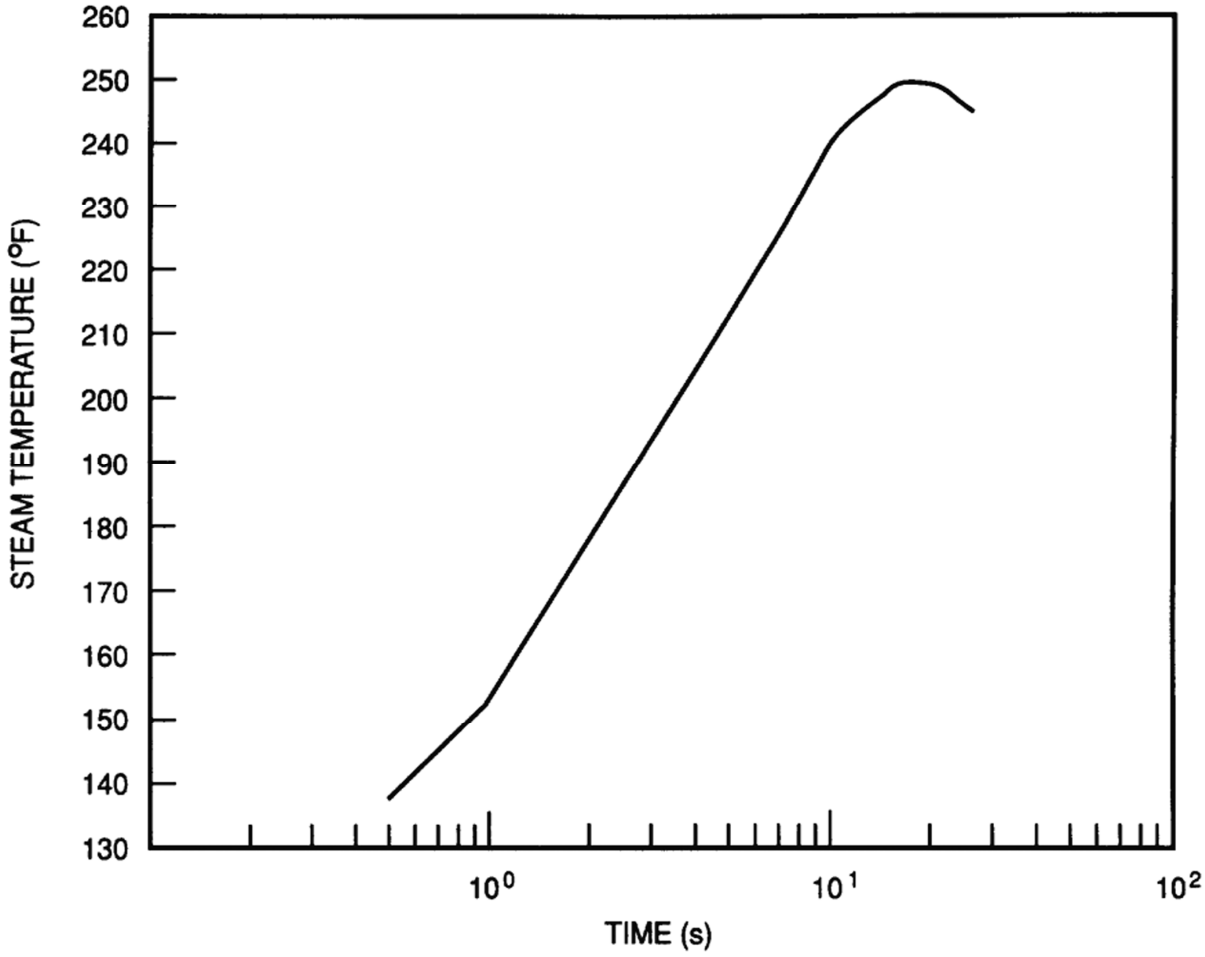
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	CONTAINMENT TEMPERATURE TRANSIENT LOCA–DOUBLE–ENDED PUMP SUCTION– MAXIMUM SAFETY INJECTION
	FIGURE 6.2.1–5



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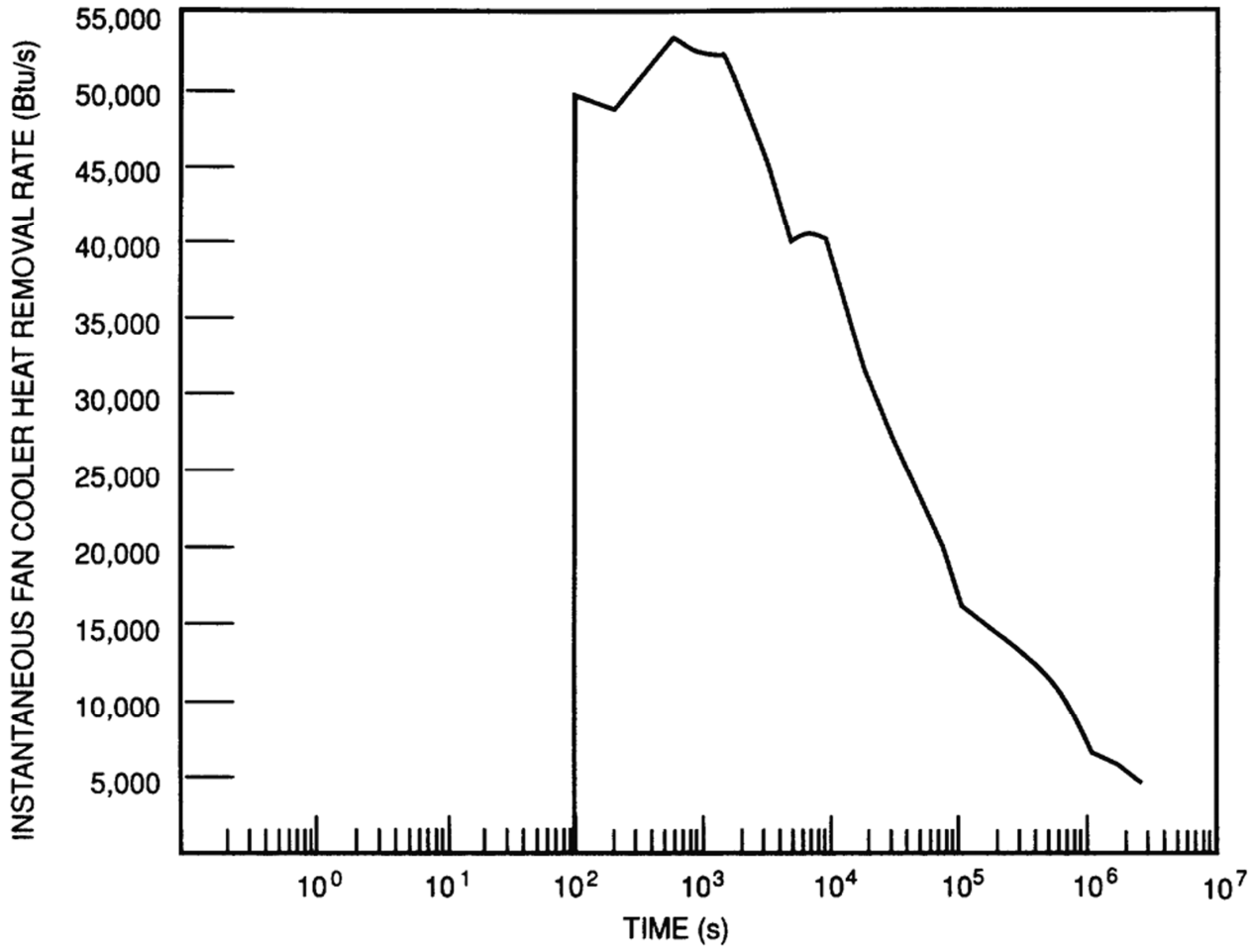
UNIT 1 AND UNIT 2

CONTAINMENT TEMPERATURE TRANSIENT
LOCA–DOUBLE–ENDED HOT LEG GUILLOTINE

FIGURE 6.2.1–6



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UNIT 1 AND UNIT 2

CONTAINMENT FAN COOLER PERFORMANCE
LOCA–DOUBLE–ENDED PUMP SUCTION–
MAXIMUM SAFETY INJECTION

FIGURE 6.2.1–7

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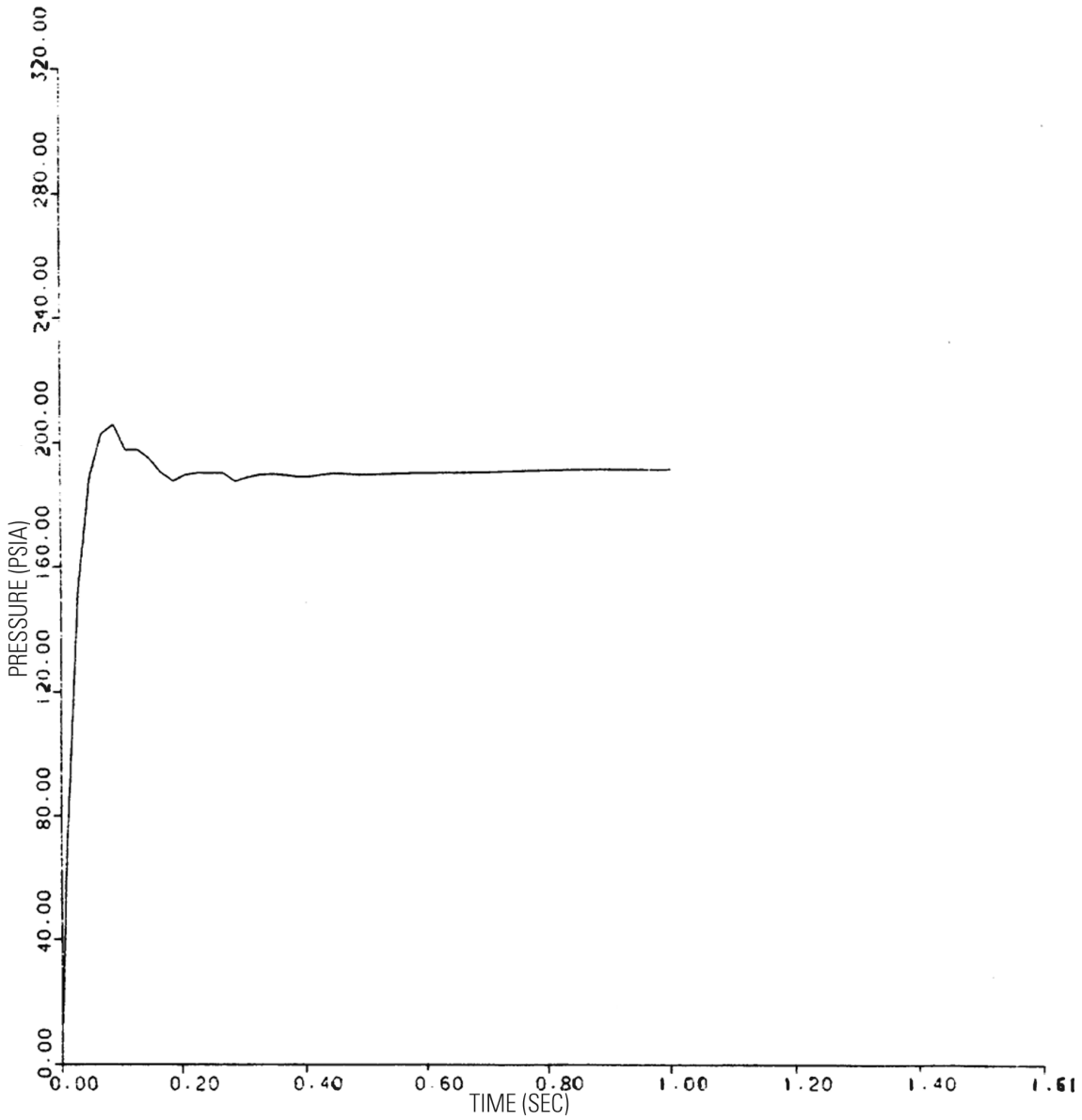

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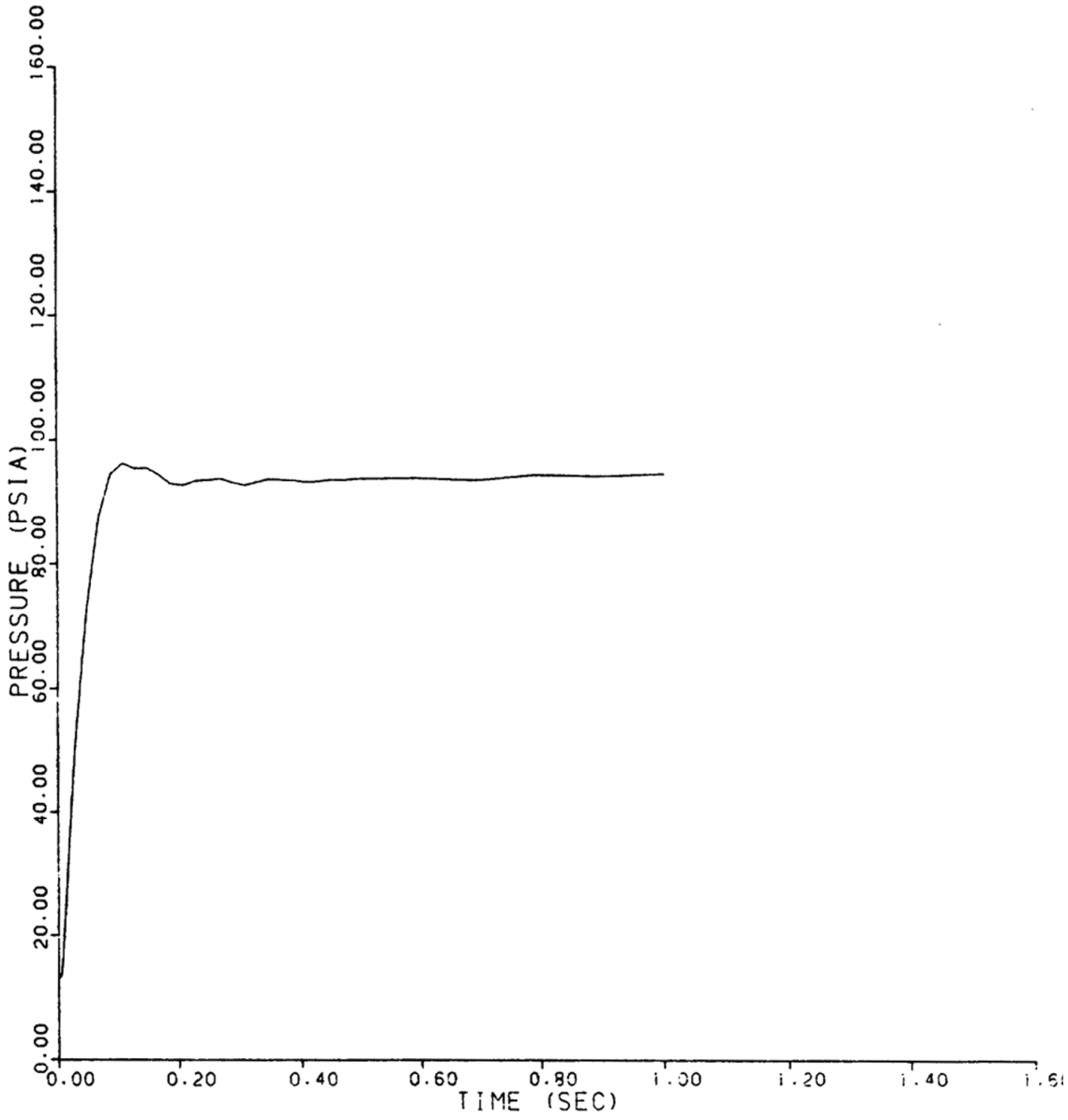
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E1
	FIGURE 6.2.1–15 (SHEET 1 OF 65)



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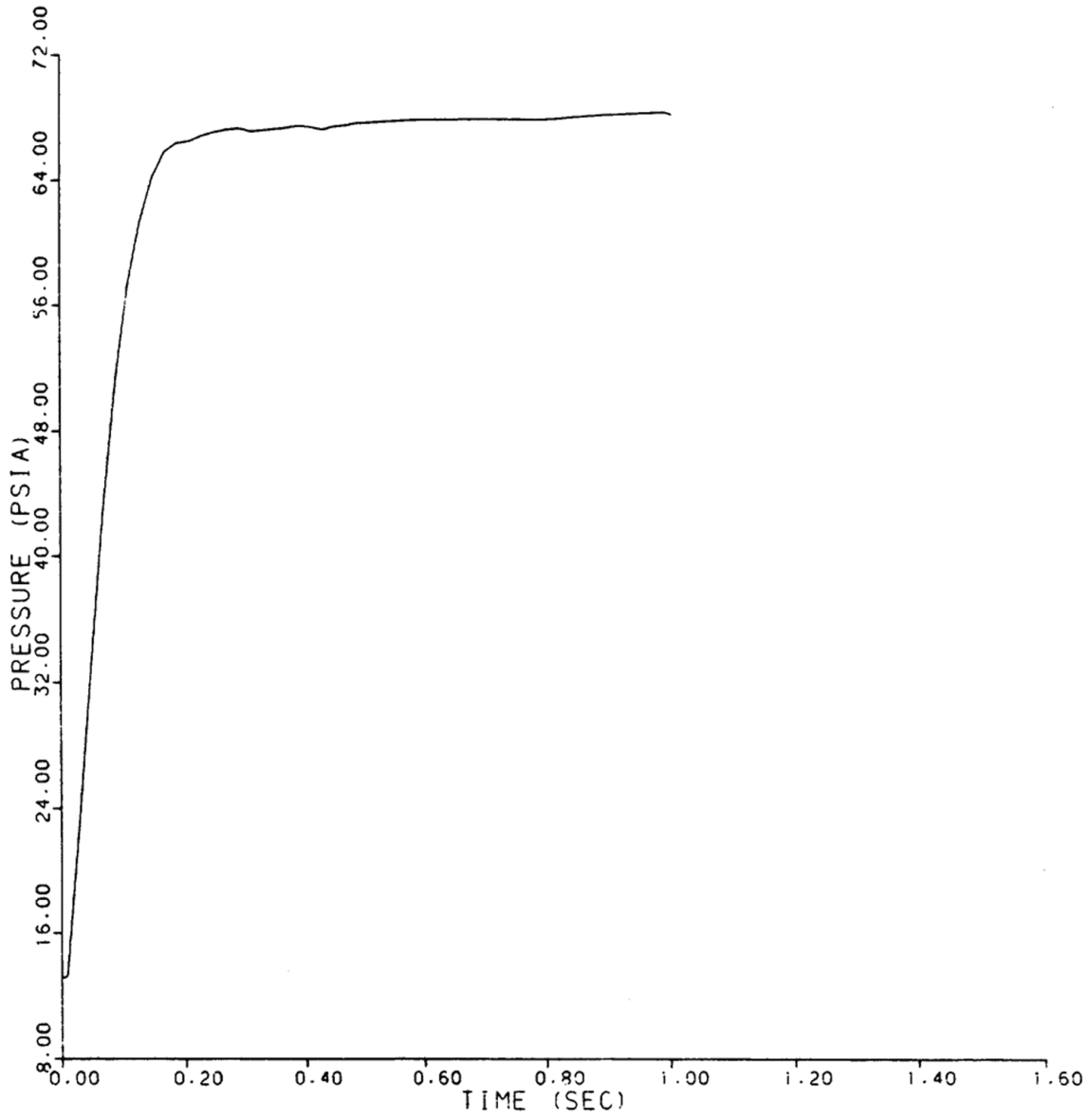
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E2
	FIGURE 6.2.1–15 (SHEET 2 OF 65)



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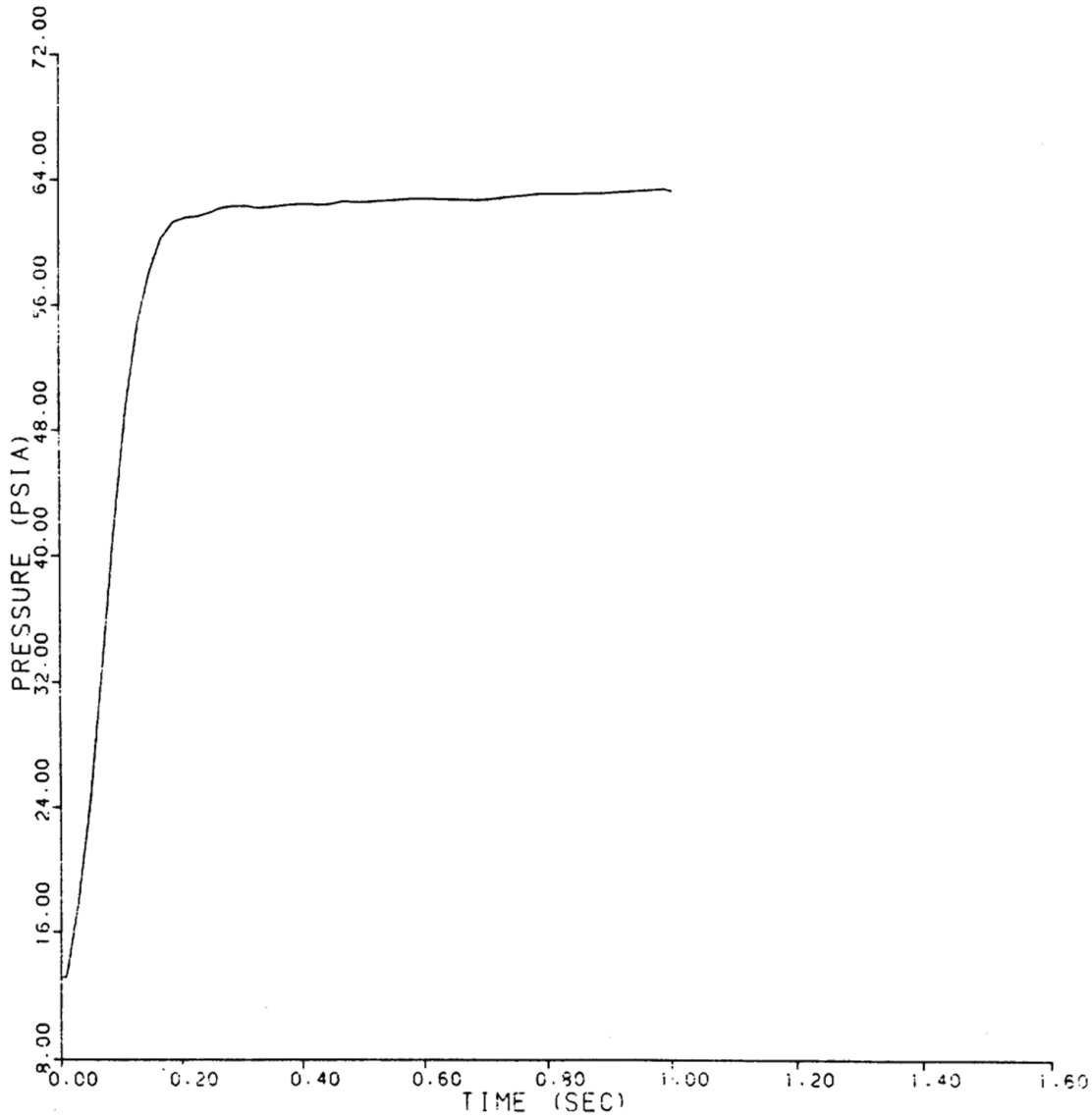
SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E3
		FIGURE 6.2.1–15 (SHEET 3 OF 65)



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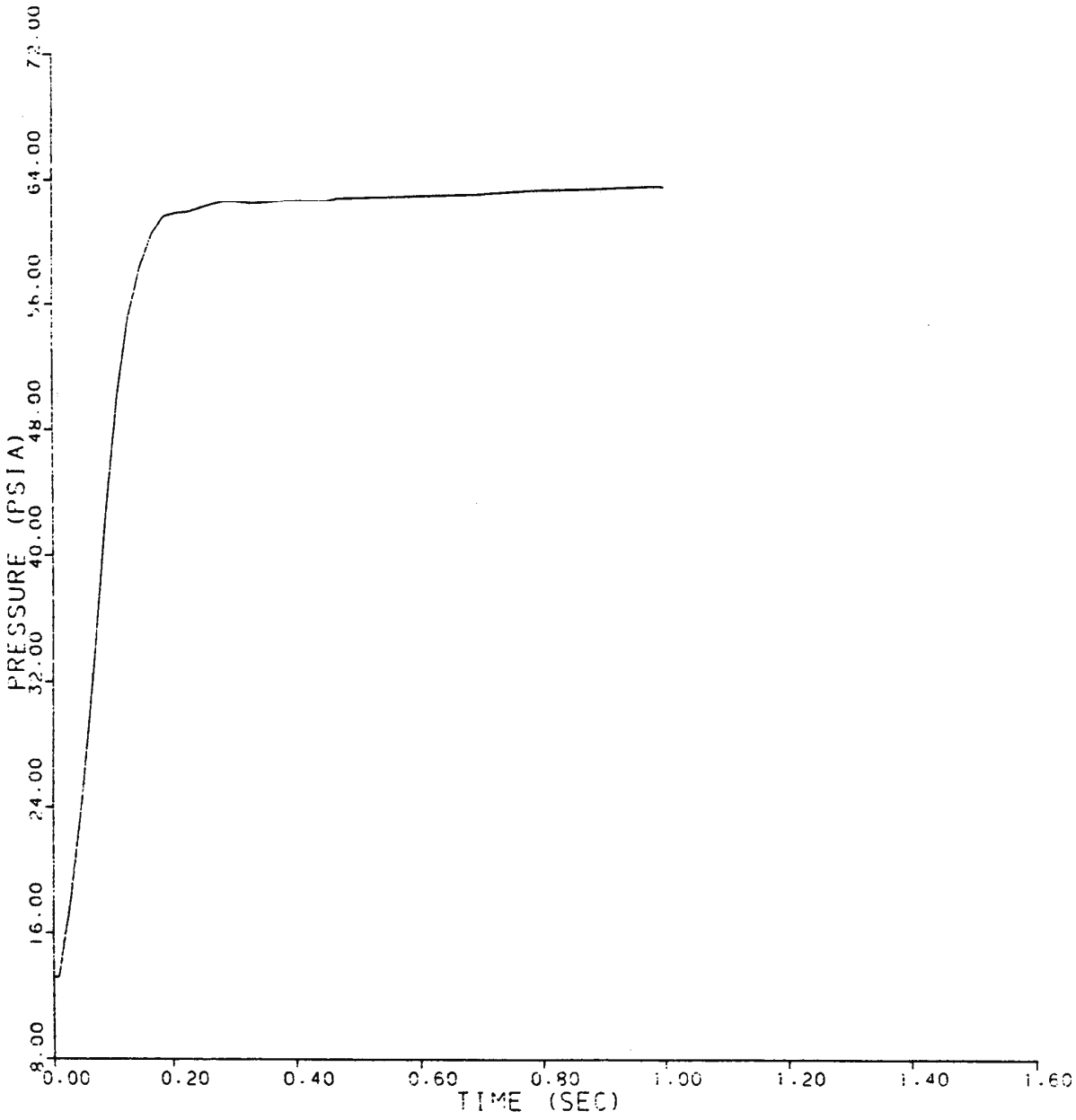
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E4
	FIGURE 6.2.1–15 (SHEET 4 OF 65)



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SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E5
	FIGURE 6.2.1–15 (SHEET 5 OF 65)

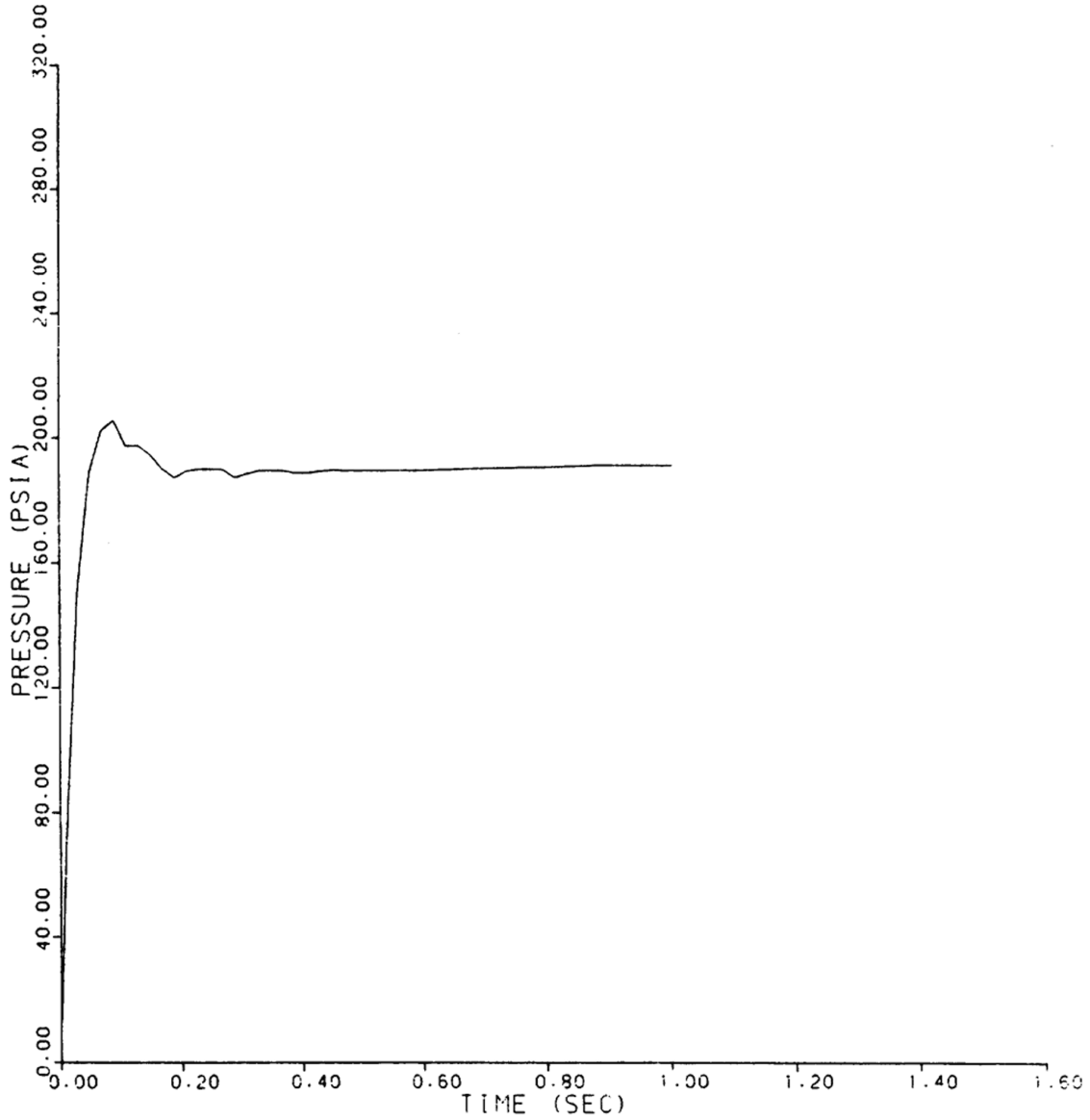

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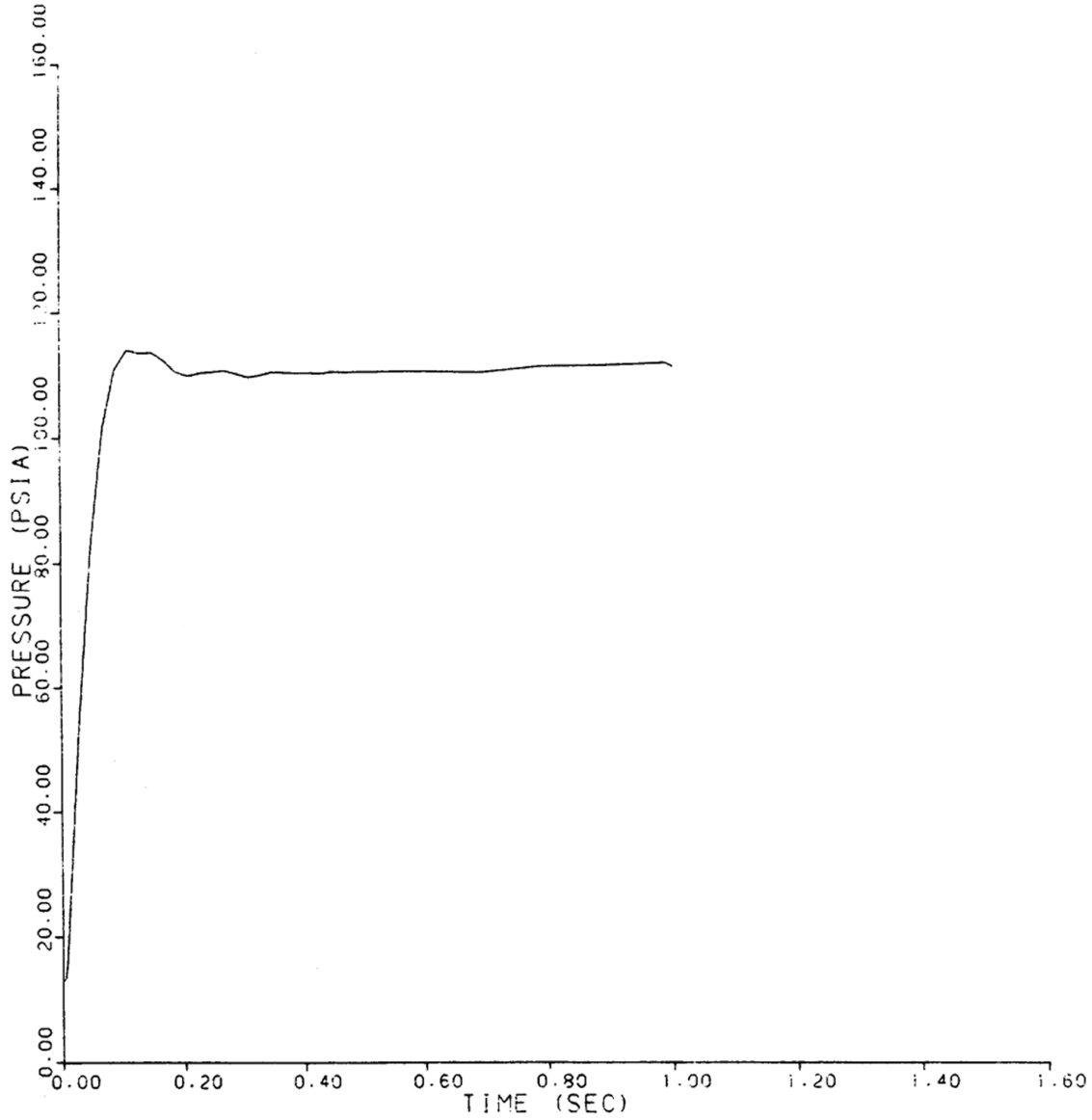
	FIGURE 6.2.1–15 (SHEET 6 OF 65)



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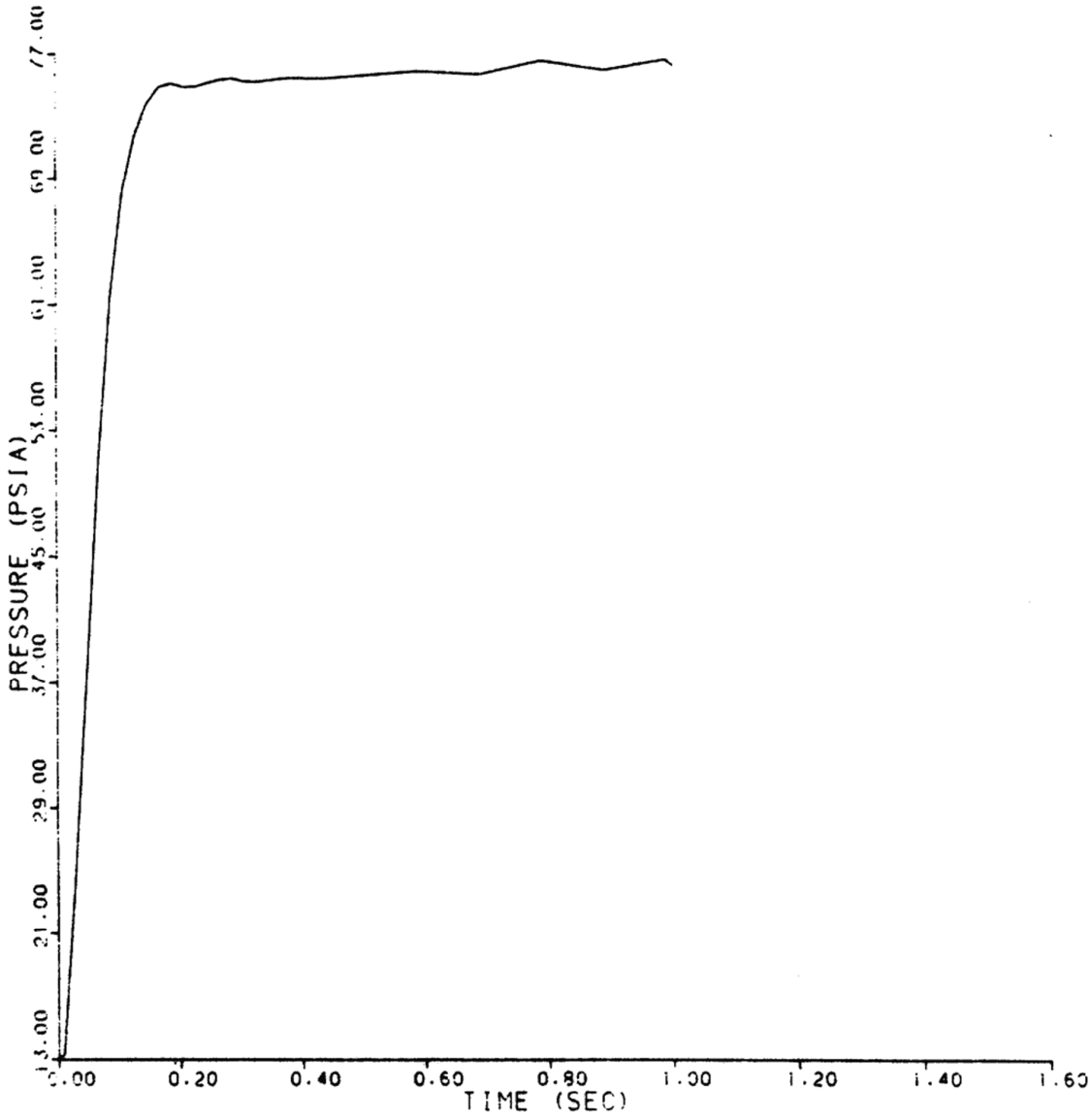
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR CAVITY PRESSURE RESPONSE – NODE E7

FIGURE 6.2.1–15 (SHEET 7 OF 65)



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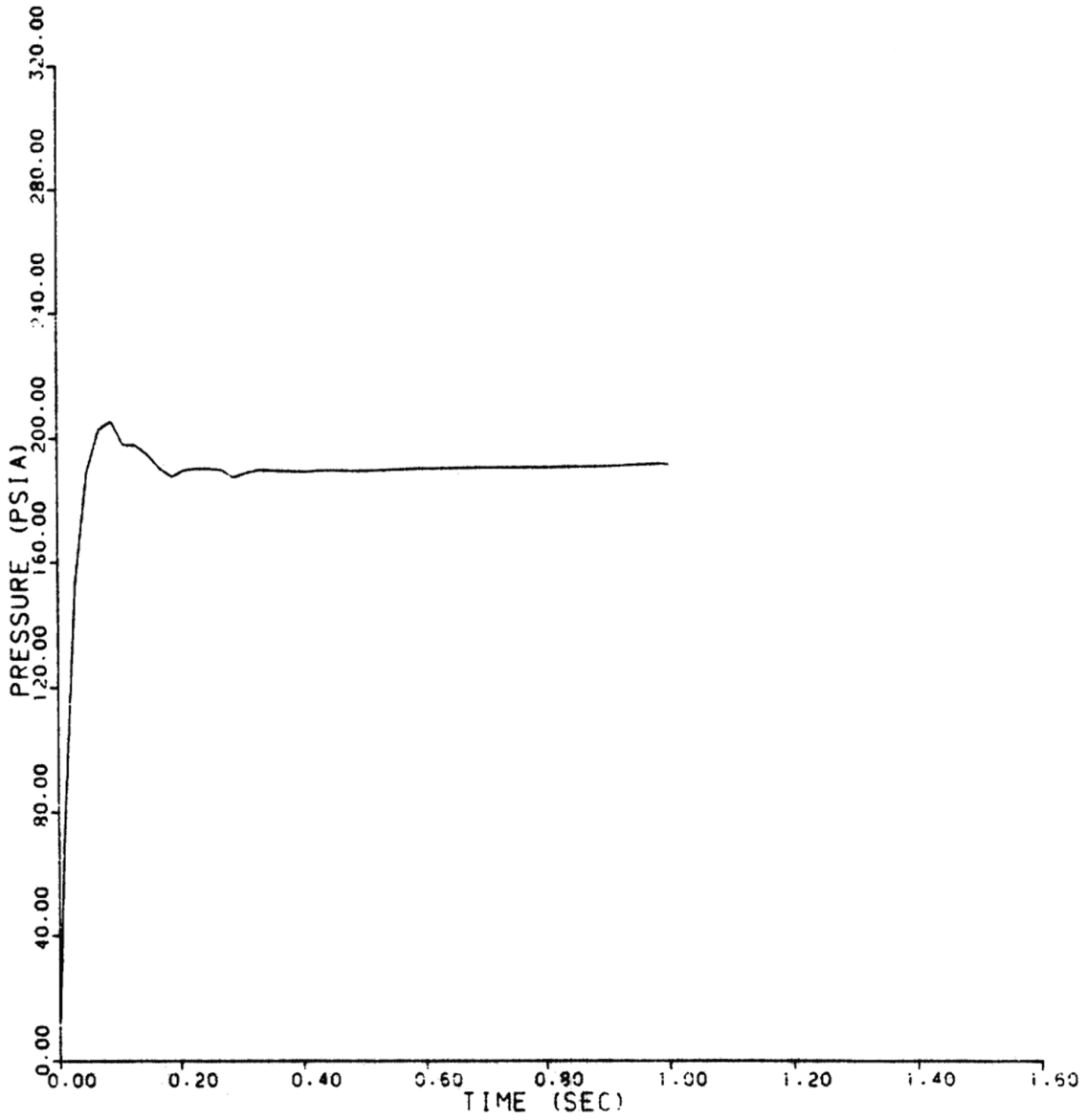
SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E8
		FIGURE 6.2.1–15 (SHEET 8 OF 65)



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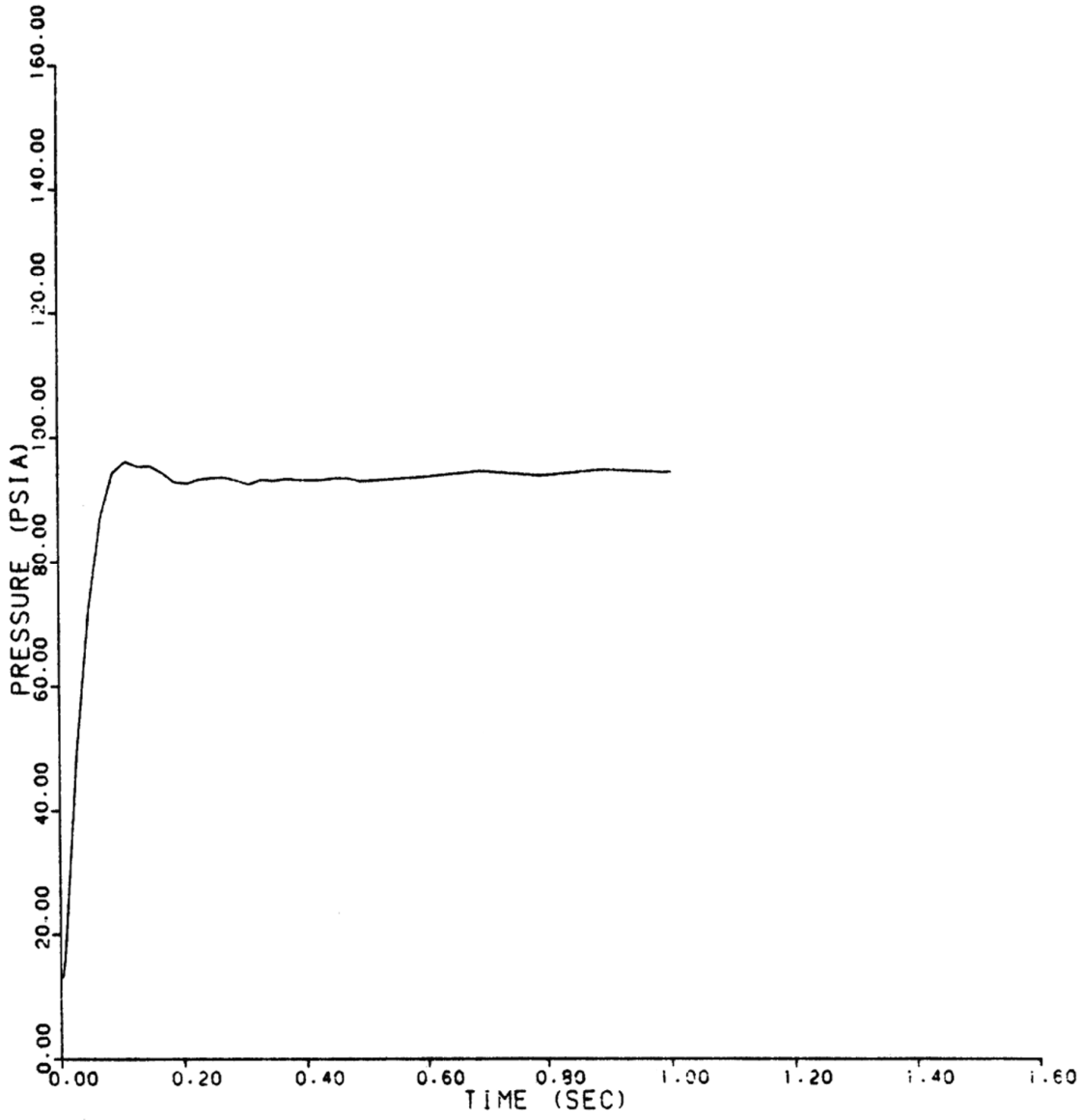
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E9
	FIGURE 6.2.1–15 (SHEET 9 OF 65)



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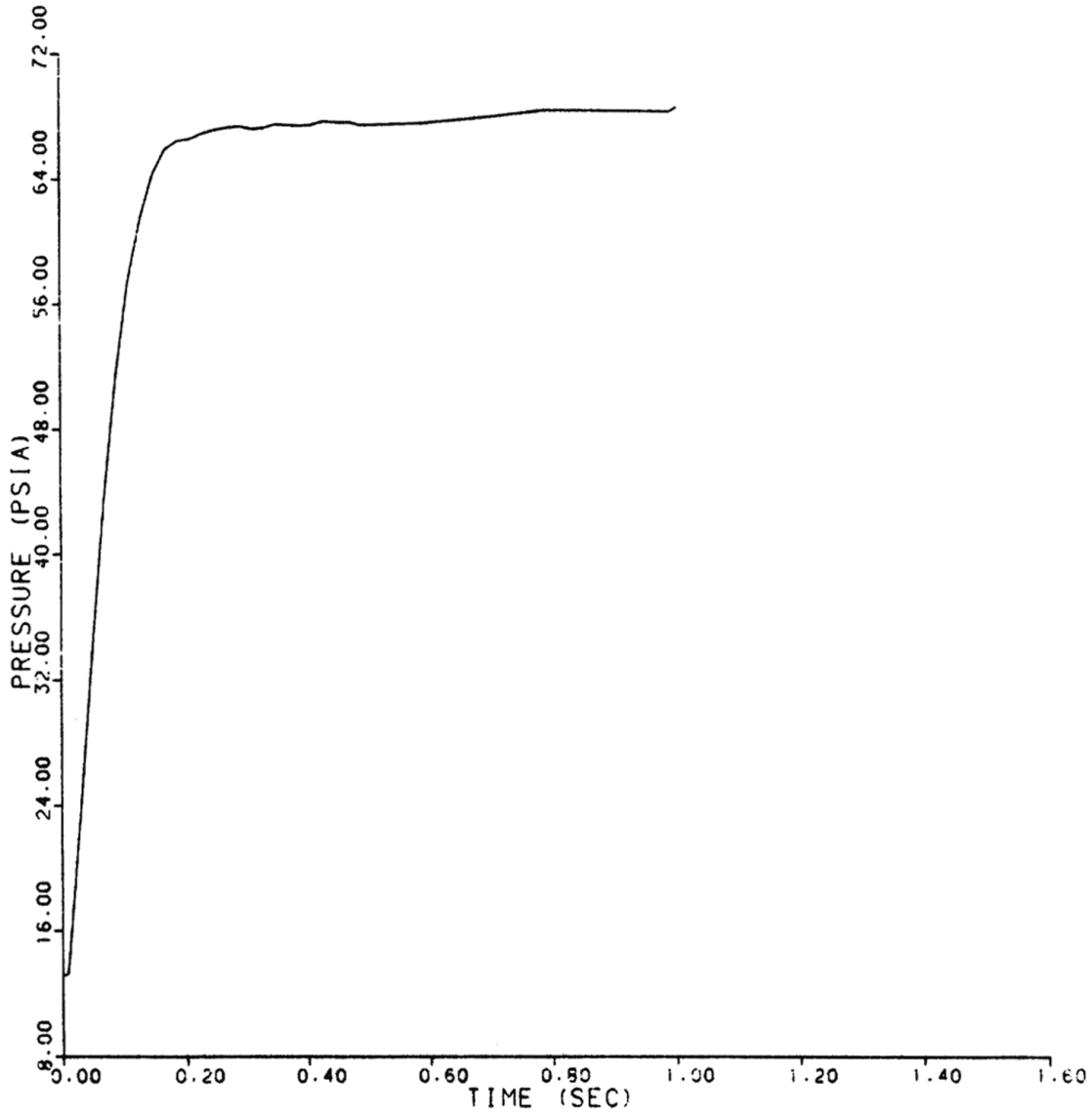
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E10
	FIGURE 6.2.1–15 (SHEET 10 OF 65)



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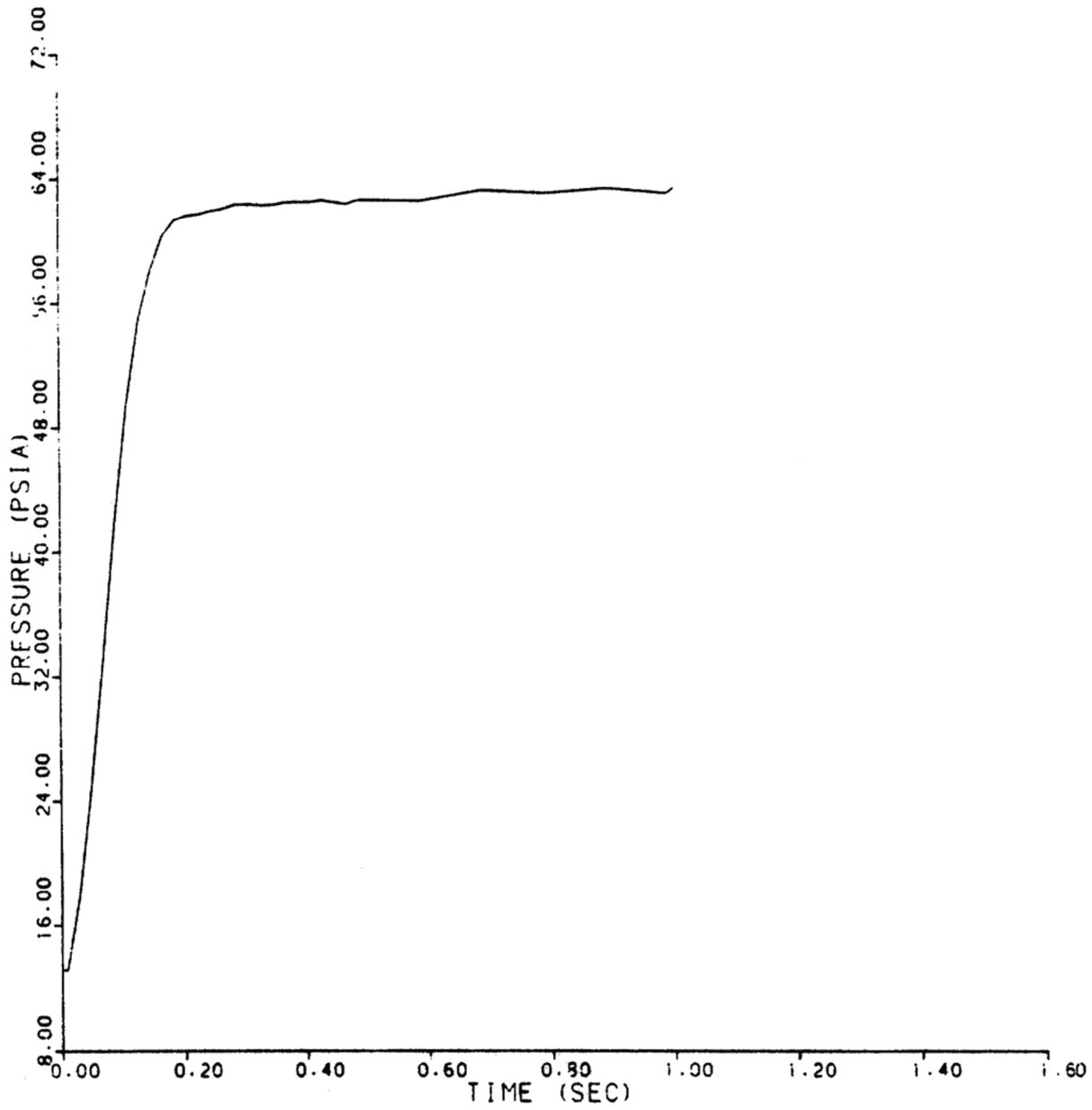
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E11
	FIGURE 6.2.1–15 (SHEET 11 OF 65)



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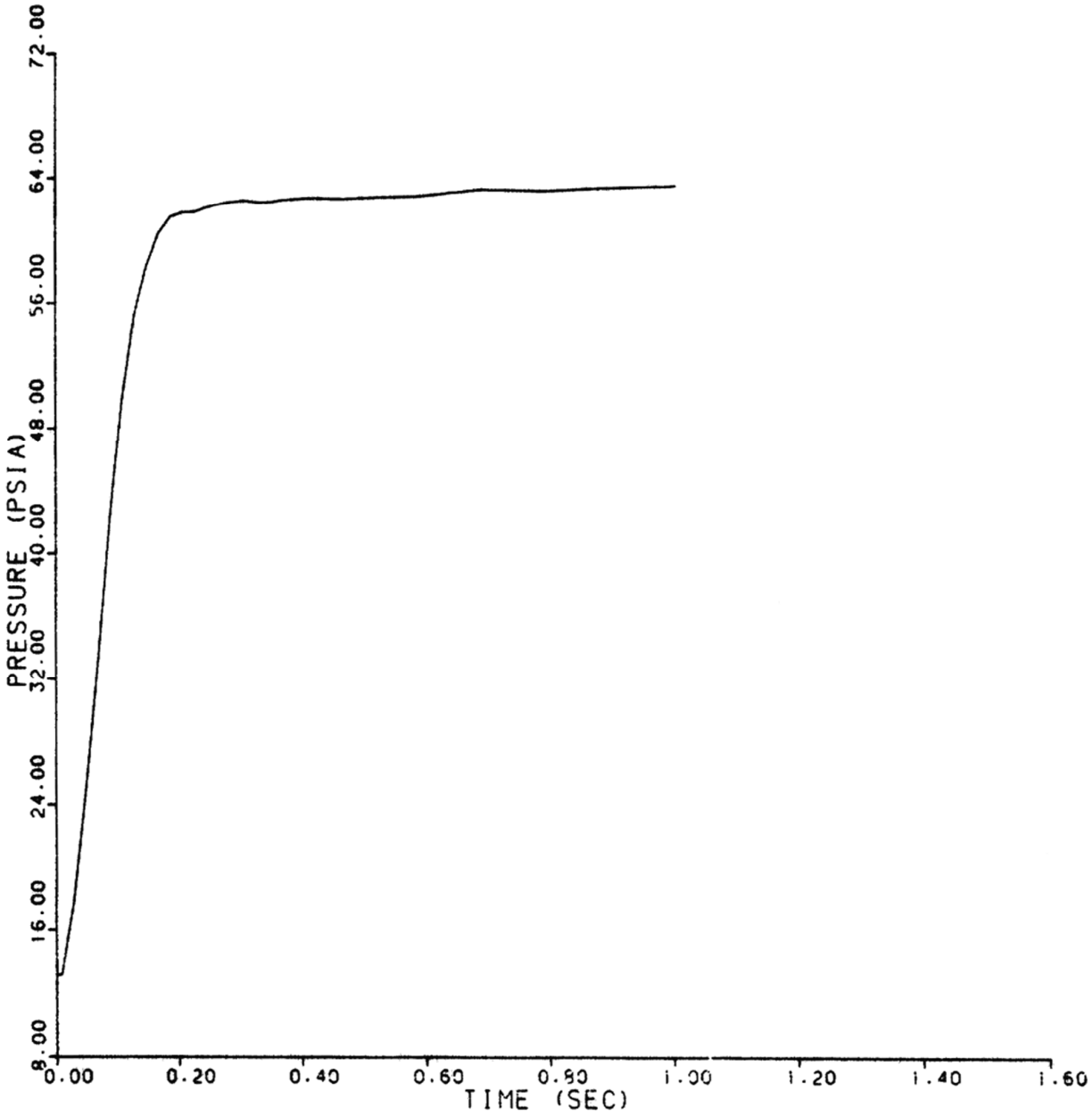
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E12
	FIGURE 6.2.1–15 (SHEET 12 OF 65)



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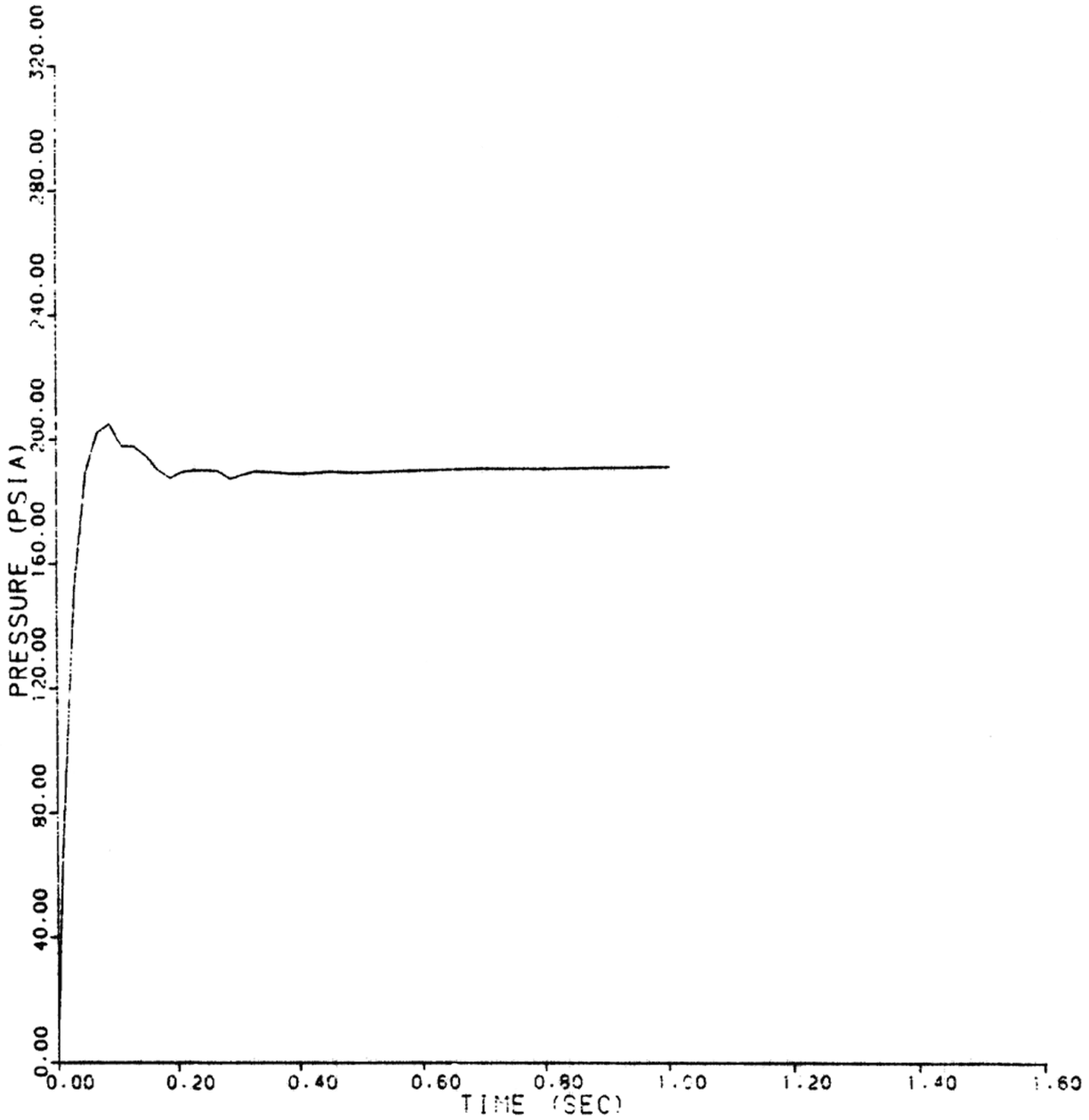
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E13
	FIGURE 6.2.1–15 (SHEET 13 OF 65)



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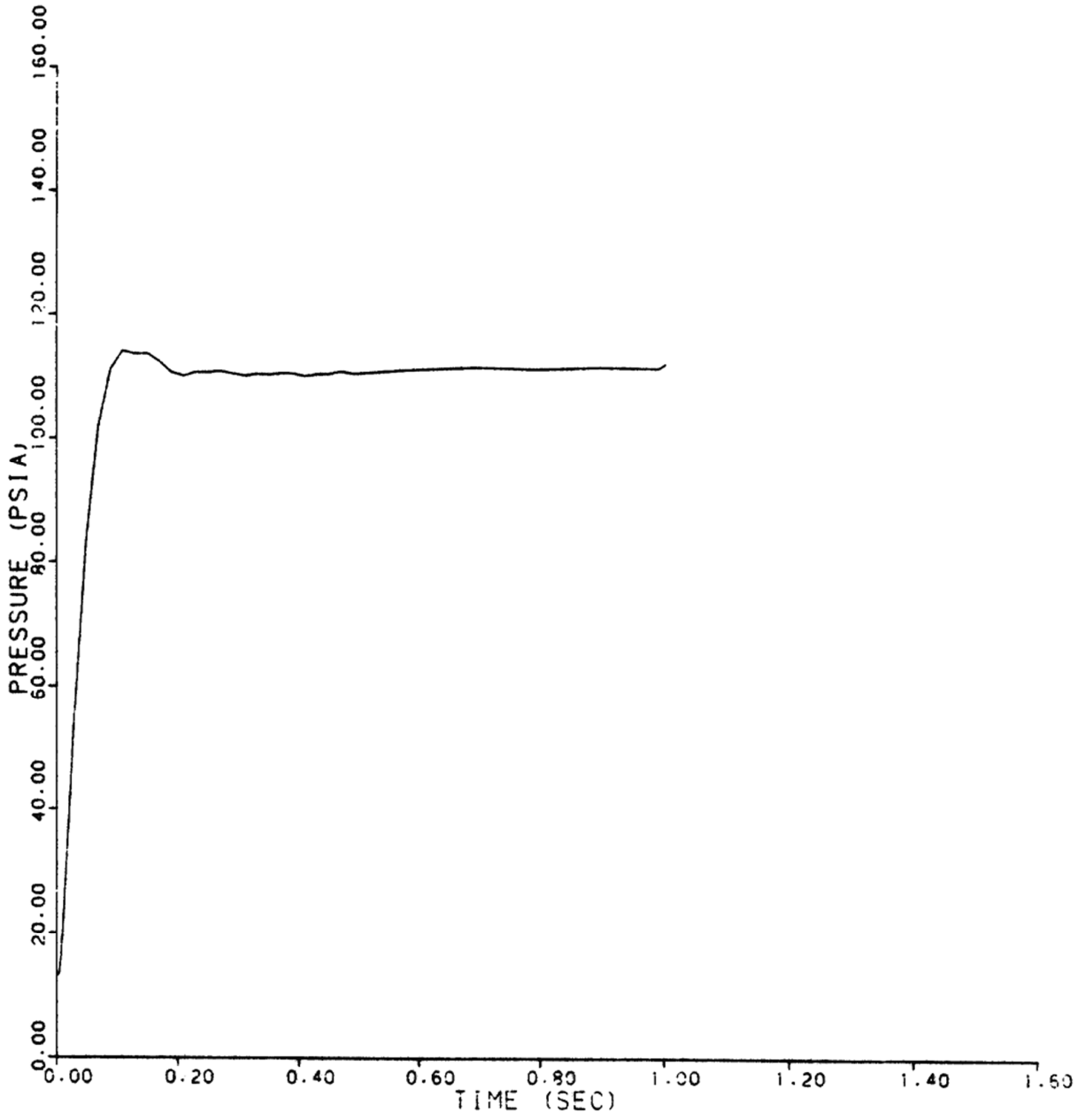
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E14
	FIGURE 6.2.1–15 (SHEET 14 OF 65)



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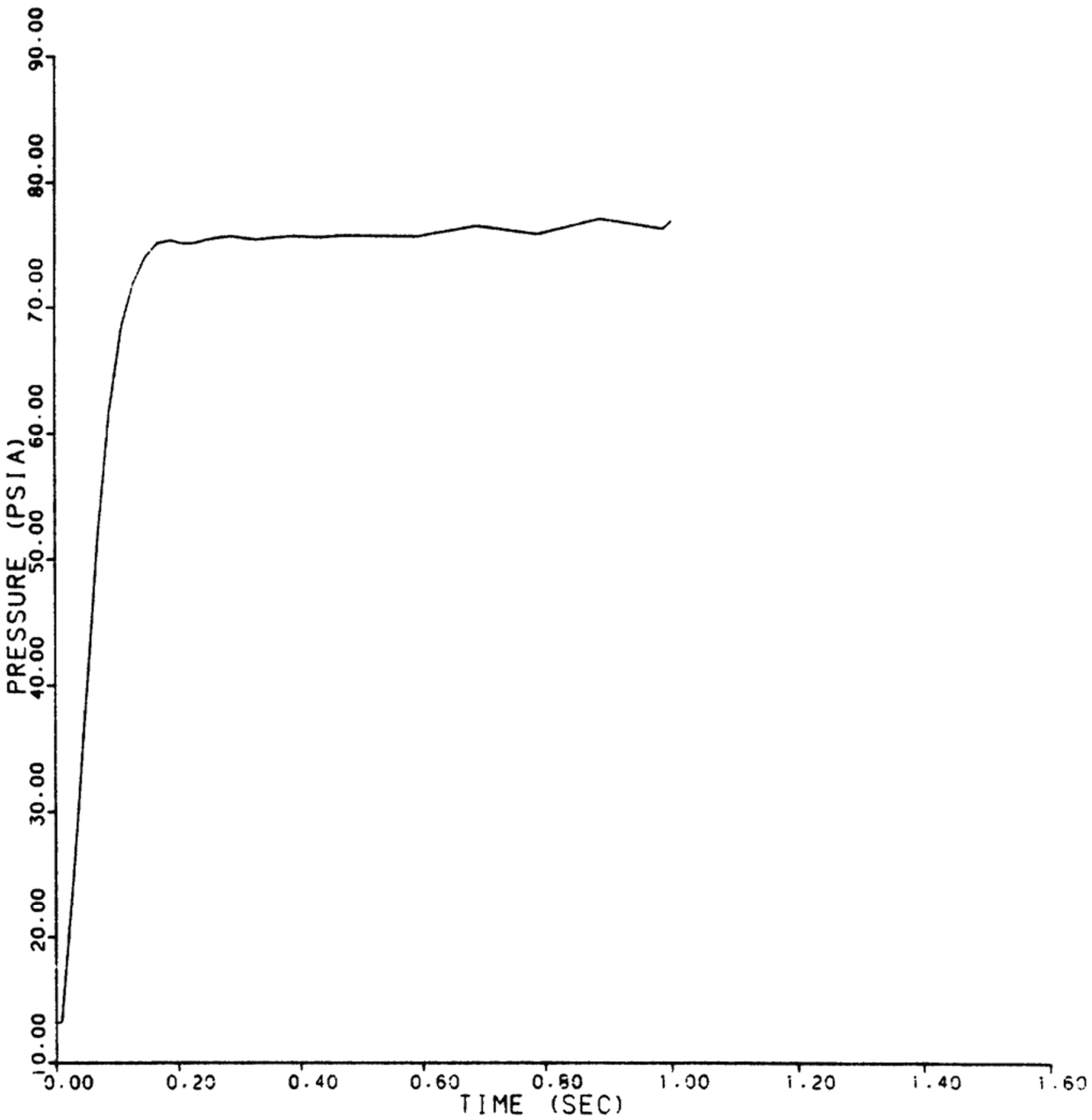
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E15
	FIGURE 6.2.1–15 (SHEET 15 OF 65)



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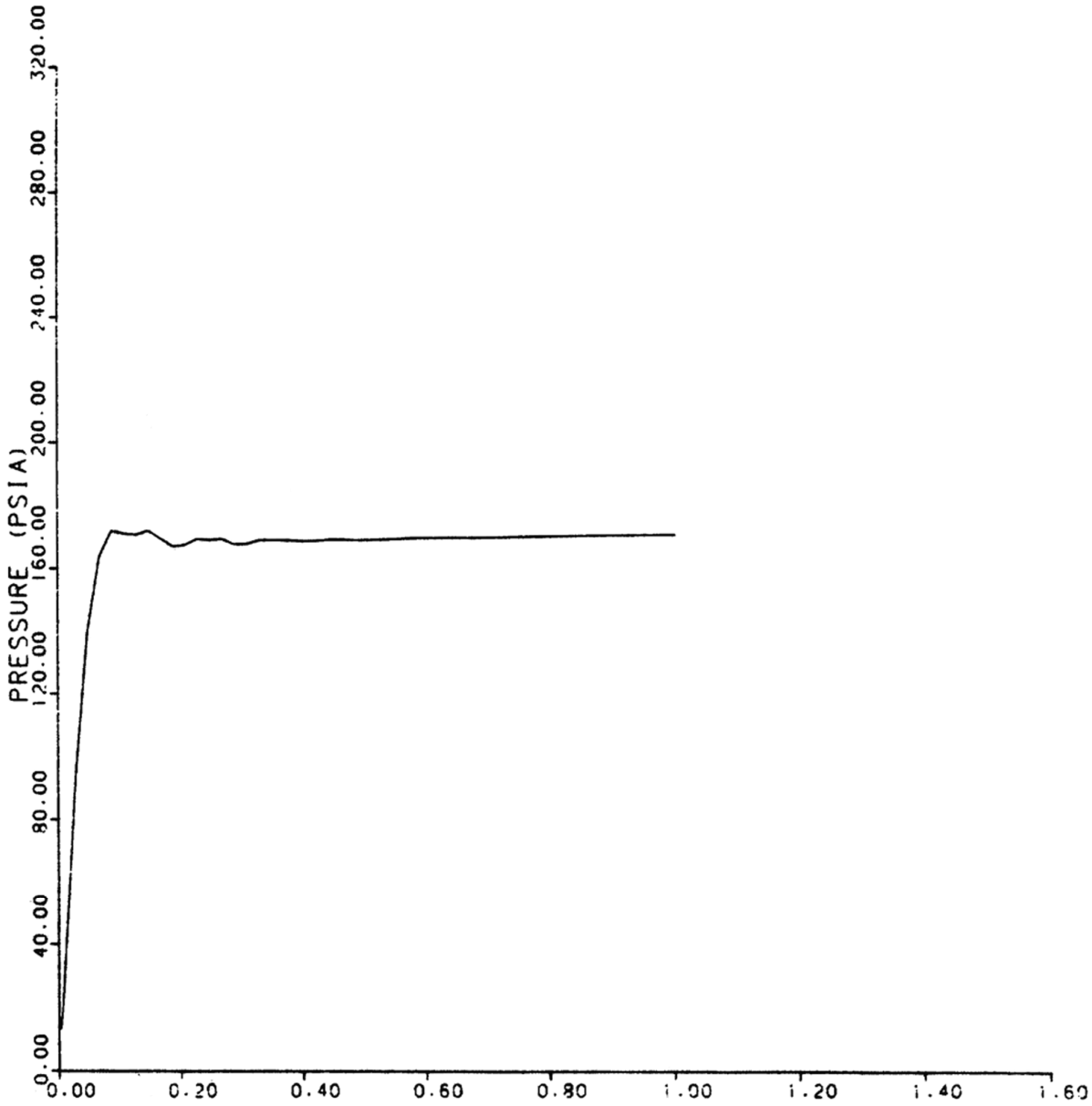
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E16
	FIGURE 6.2.1–15 (SHEET 16 OF 65)



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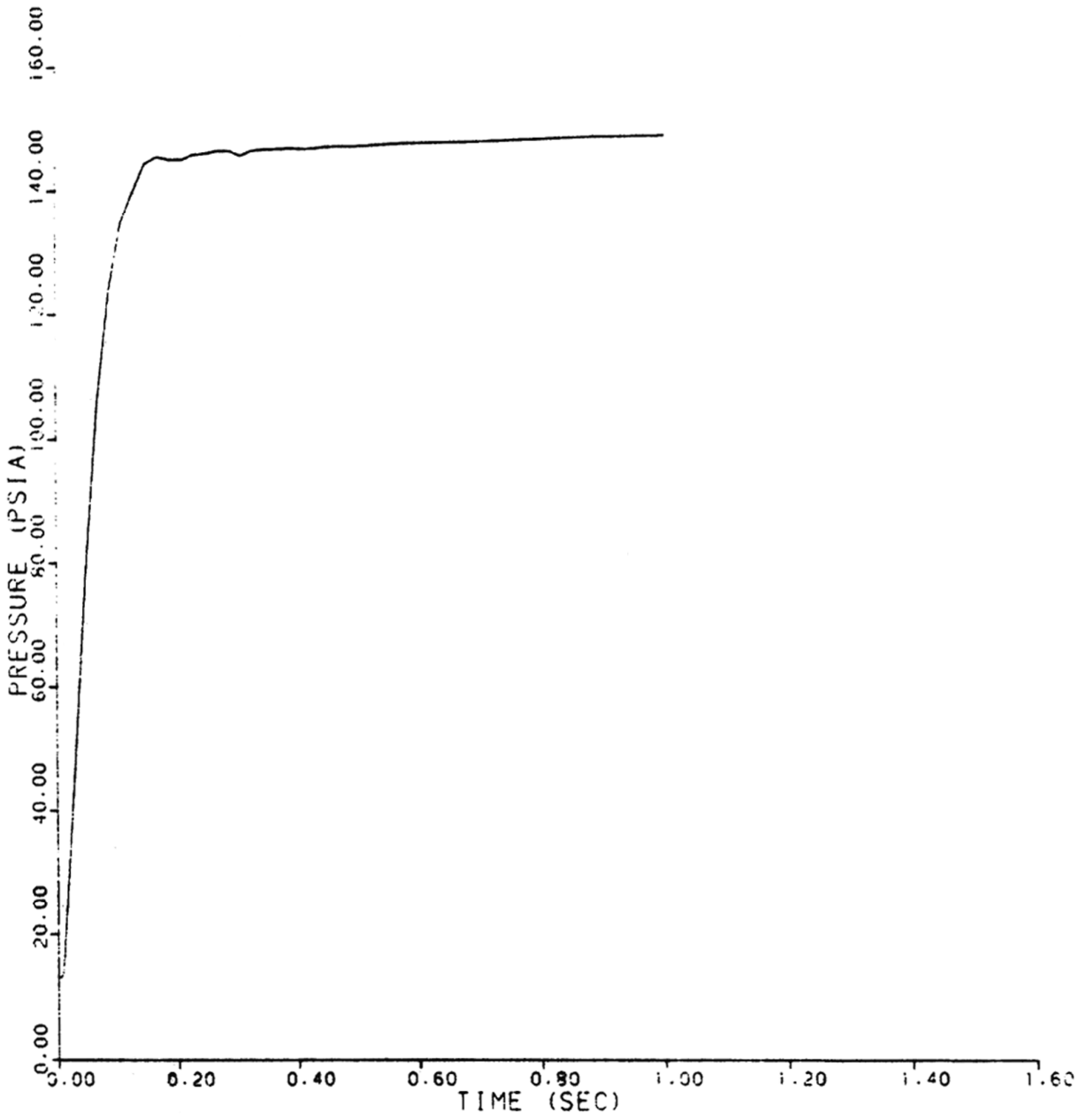
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E17
	FIGURE 6.2.1–15 (SHEET 17 OF 65)



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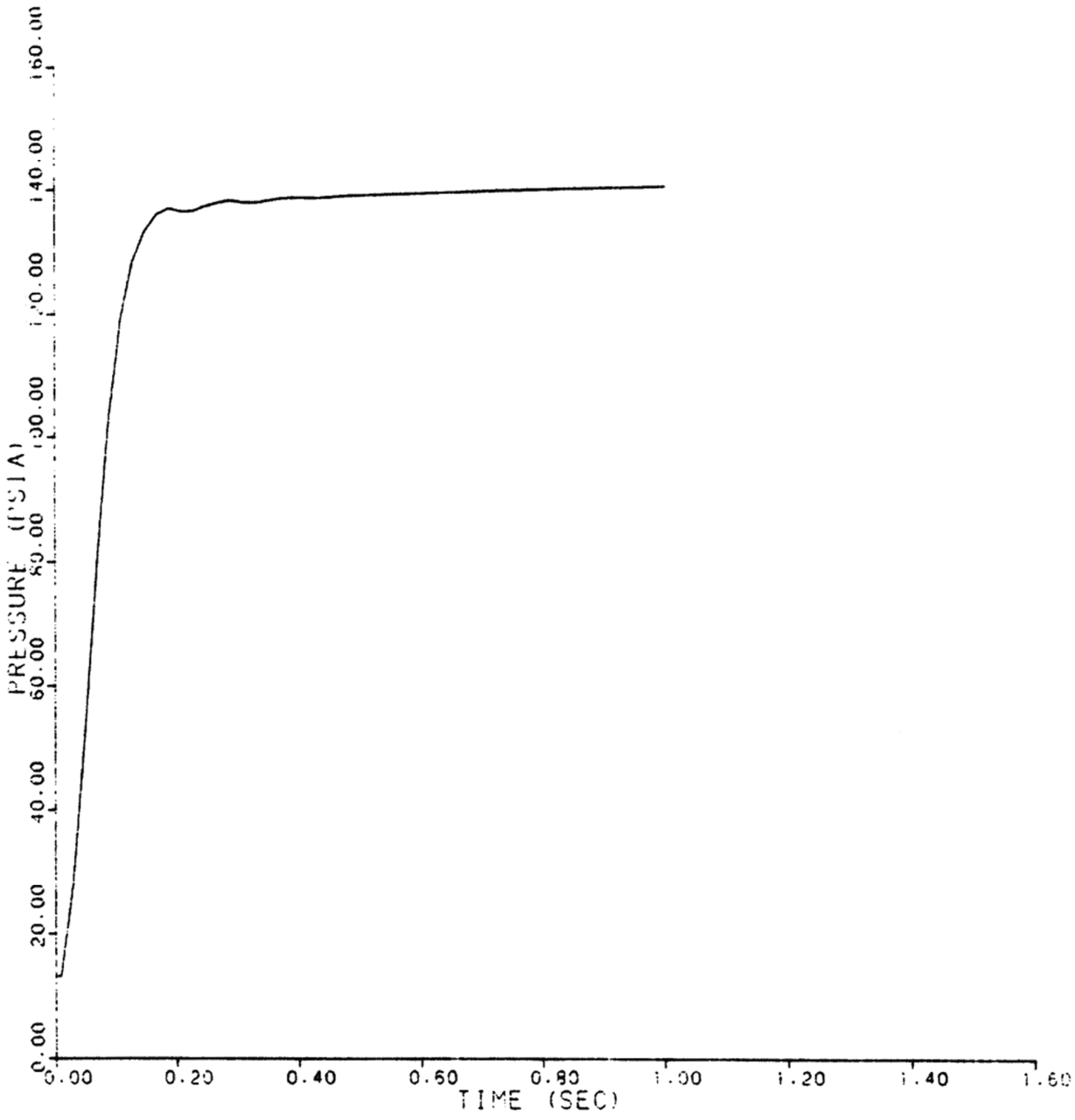
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E18
	FIGURE 6.2.1–15 (SHEET 18 OF 65)



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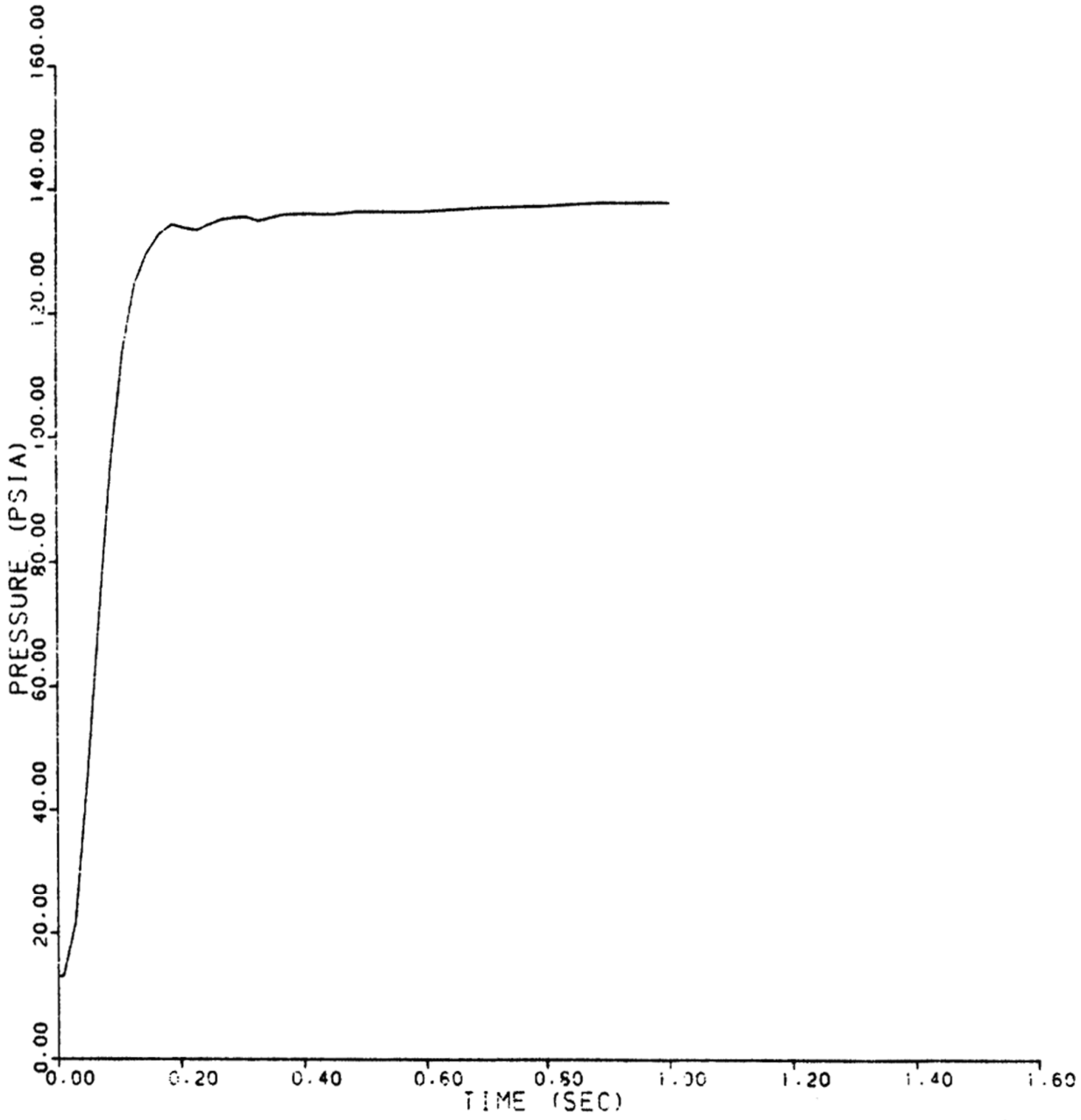
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E19
	FIGURE 6.2.1–15 (SHEET 19 OF 65)



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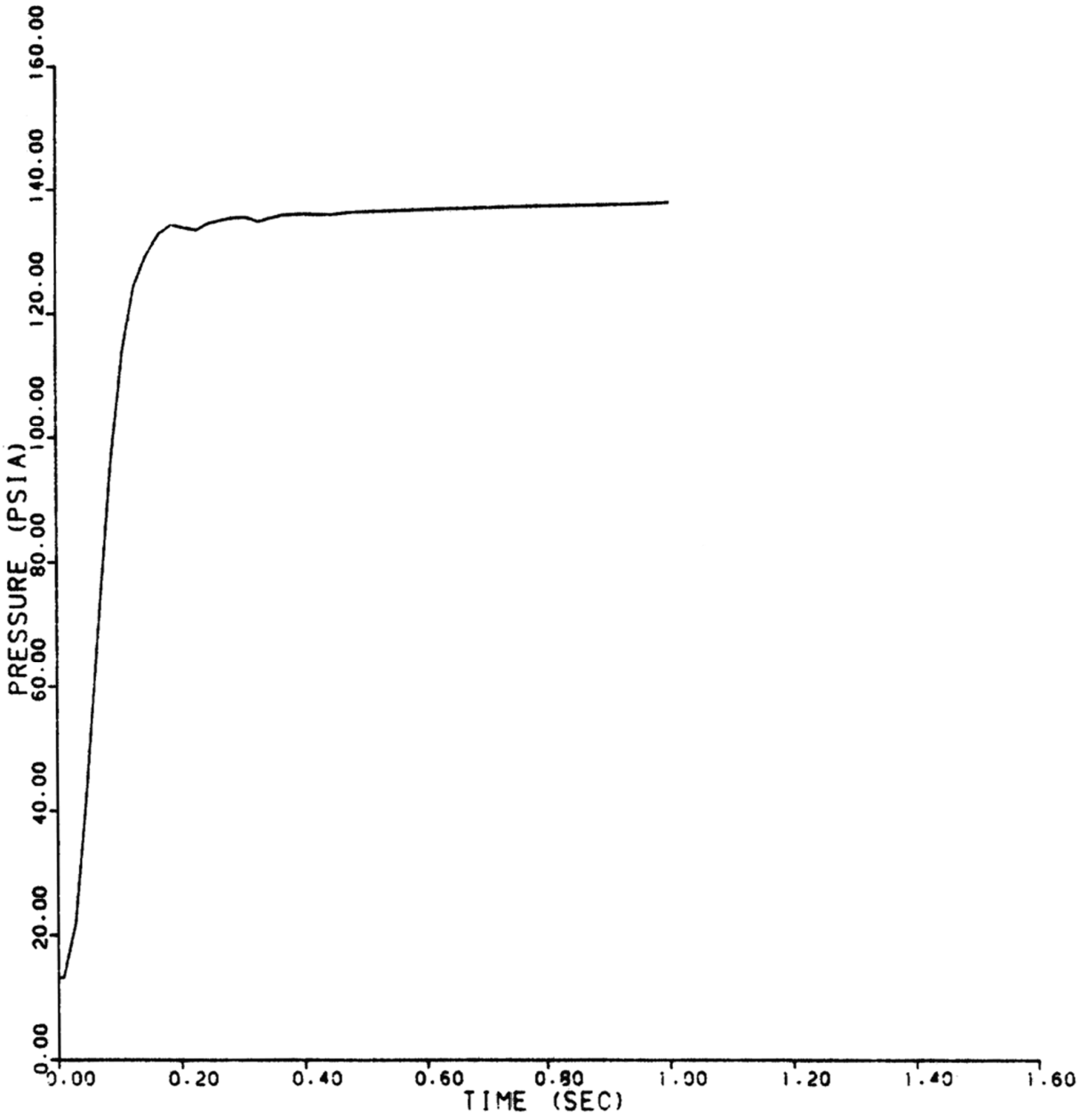
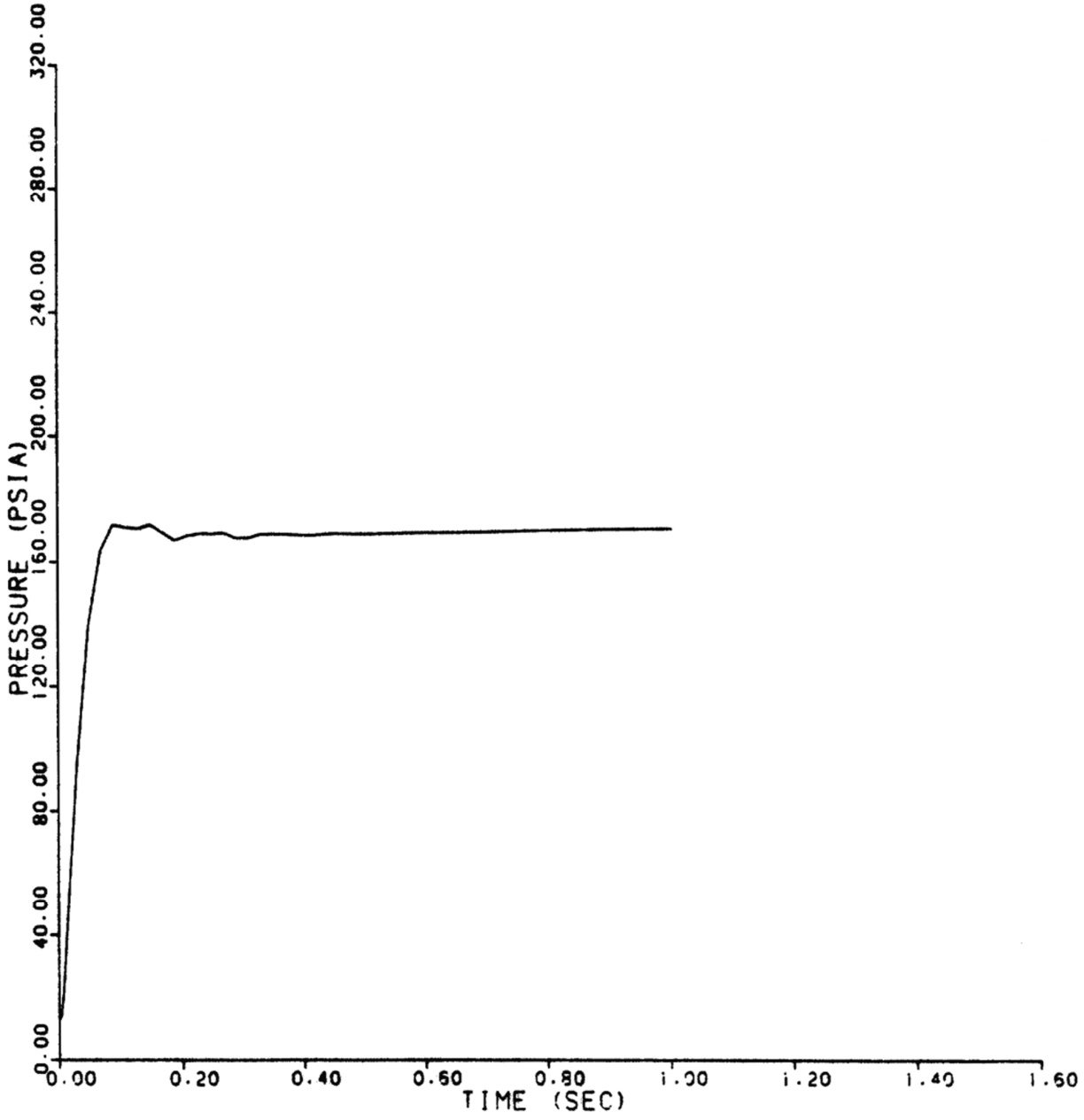
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E20
	FIGURE 6.2.1–15 (SHEET 20 OF 65)



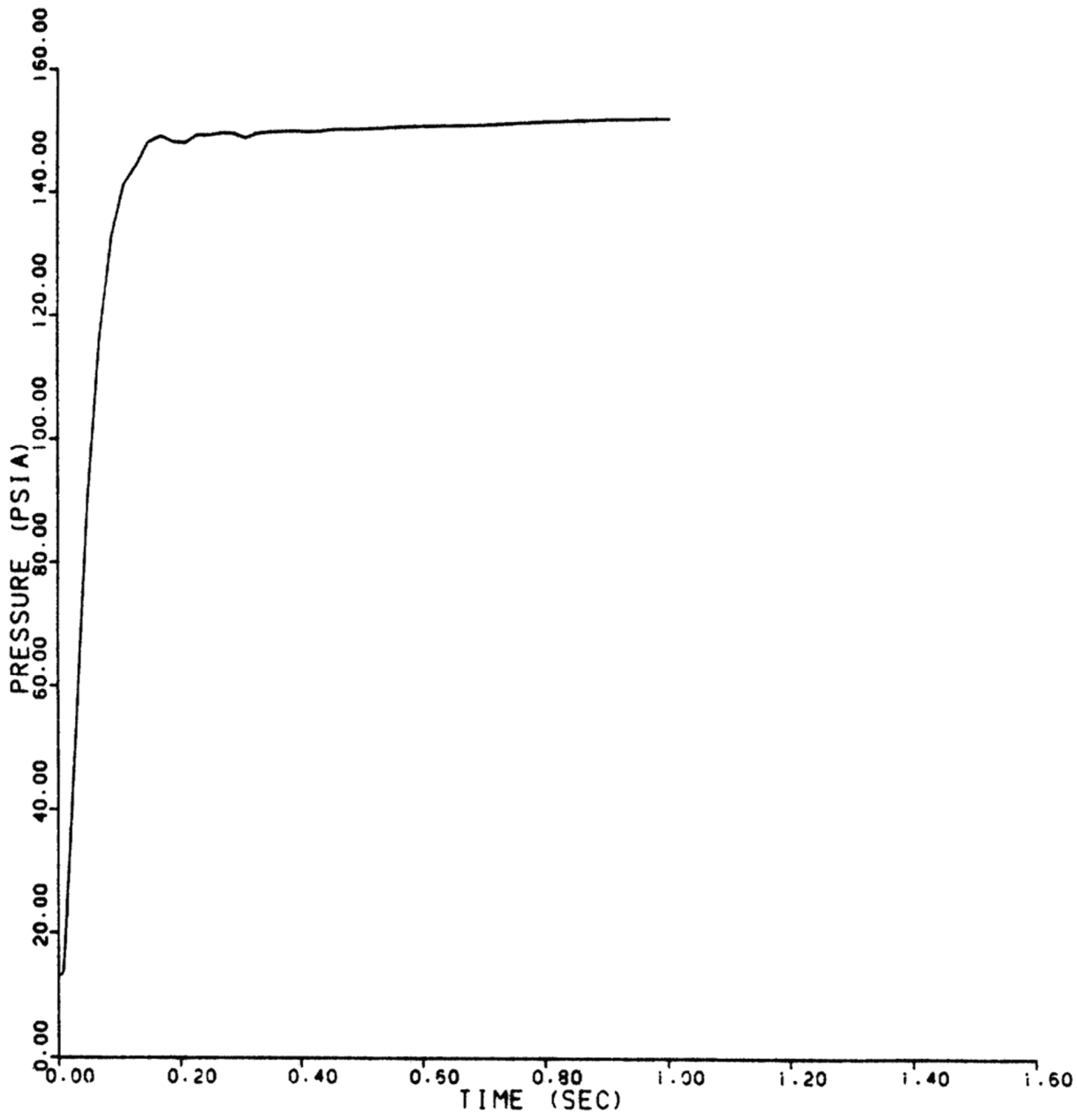
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E21
	FIGURE 6.2.1–15 (SHEET 21 OF 65)



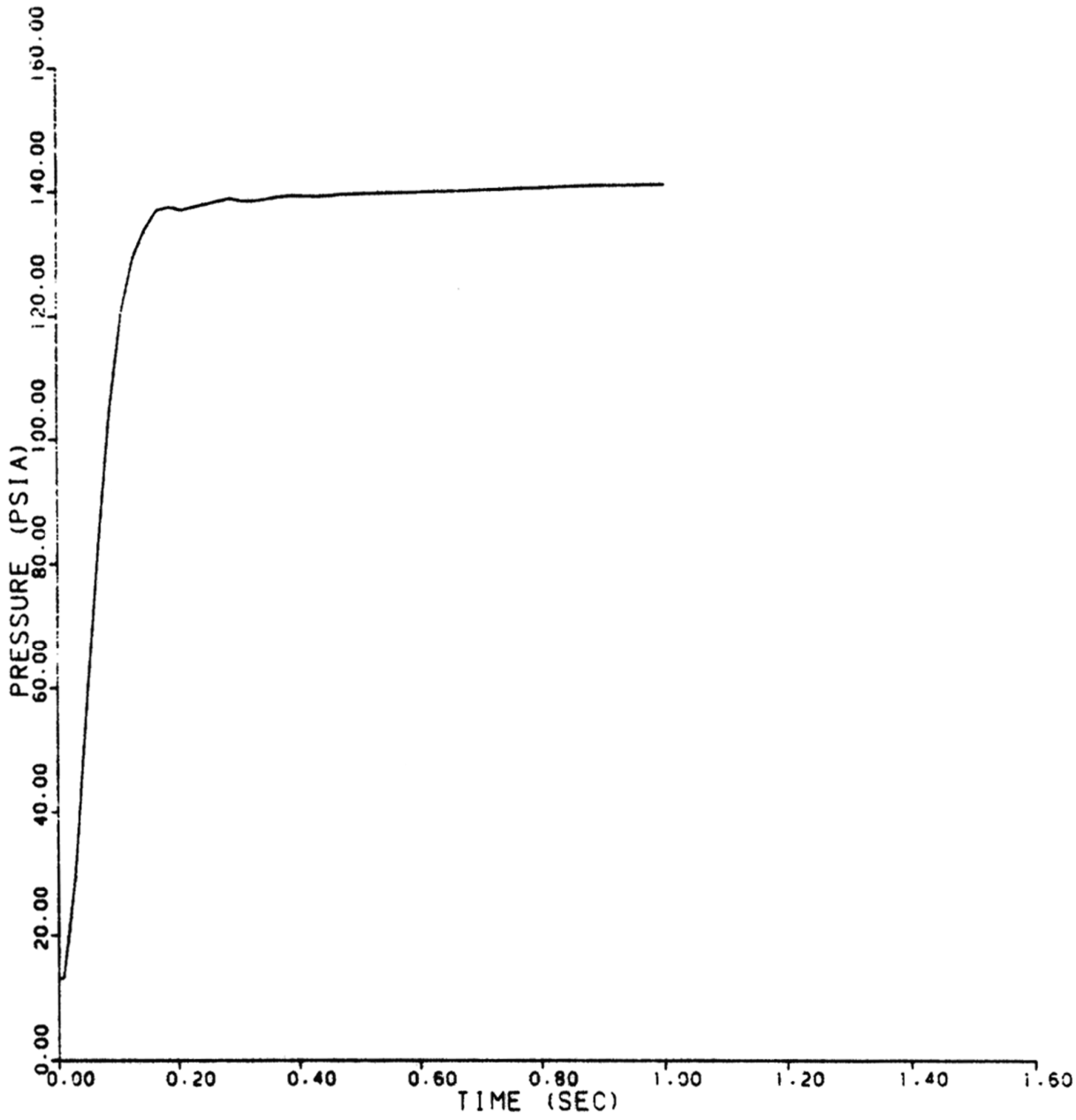
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E22
	FIGURE 6.2.1–15 (SHEET 22 OF 65)



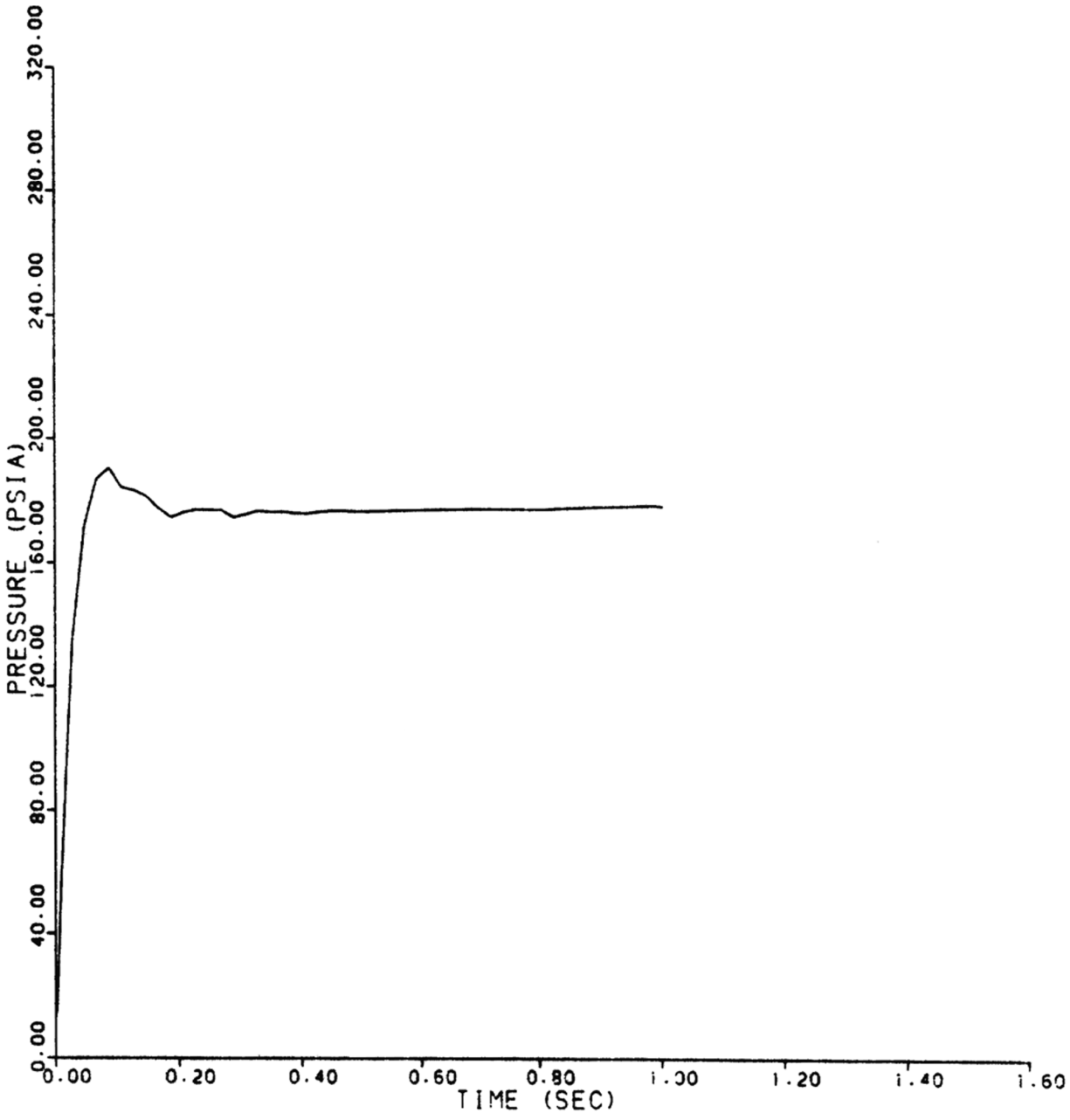
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E23
	FIGURE 6.2.1–15 (SHEET 23 OF 65)



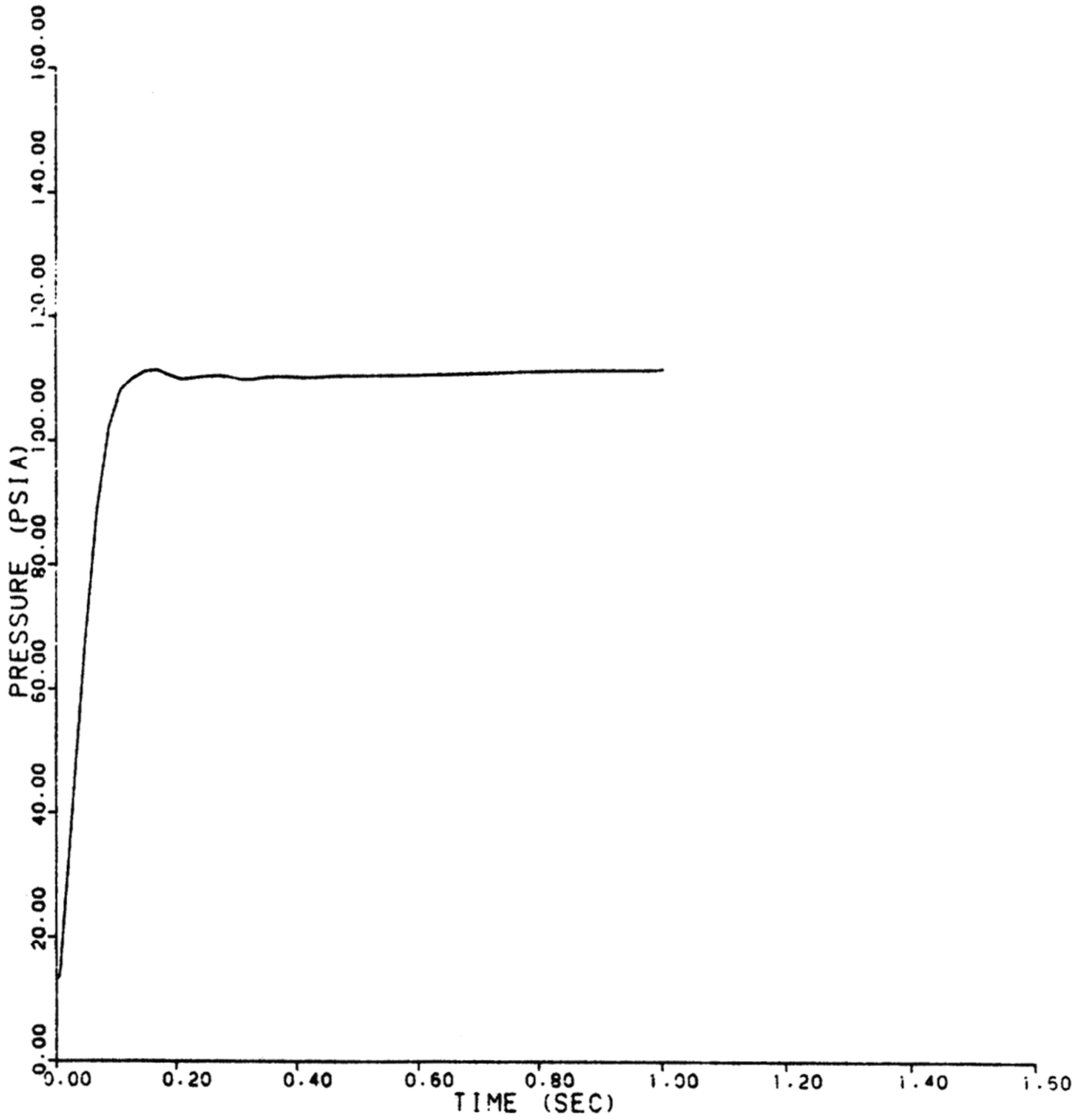
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E24
	FIGURE 6.2.1–15 (SHEET 24 OF 65)



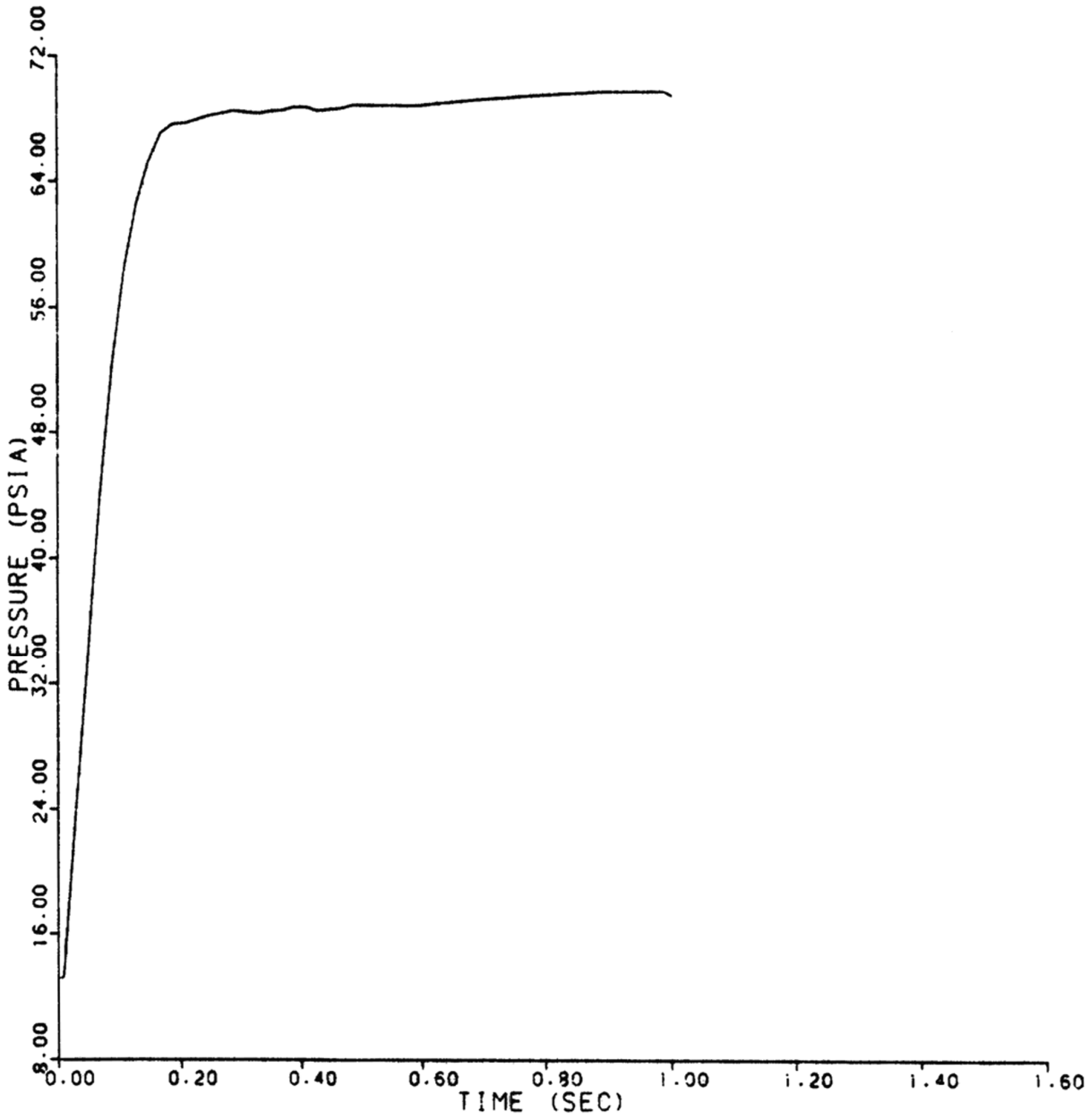
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E25
	FIGURE 6.2.1–15 (SHEET 25 OF 65)



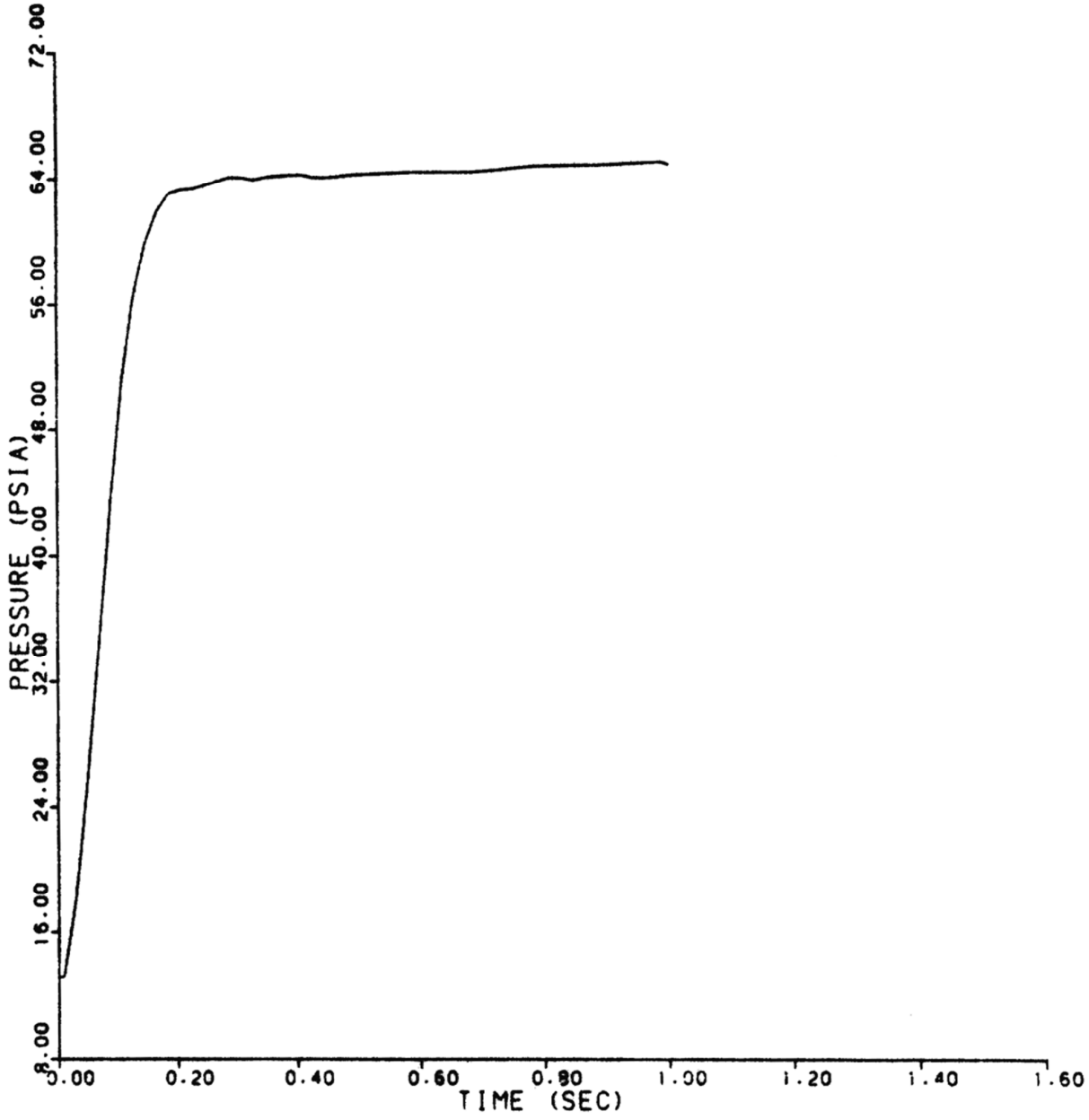
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E26
	FIGURE 6.2.1–15 (SHEET 26 OF 65)



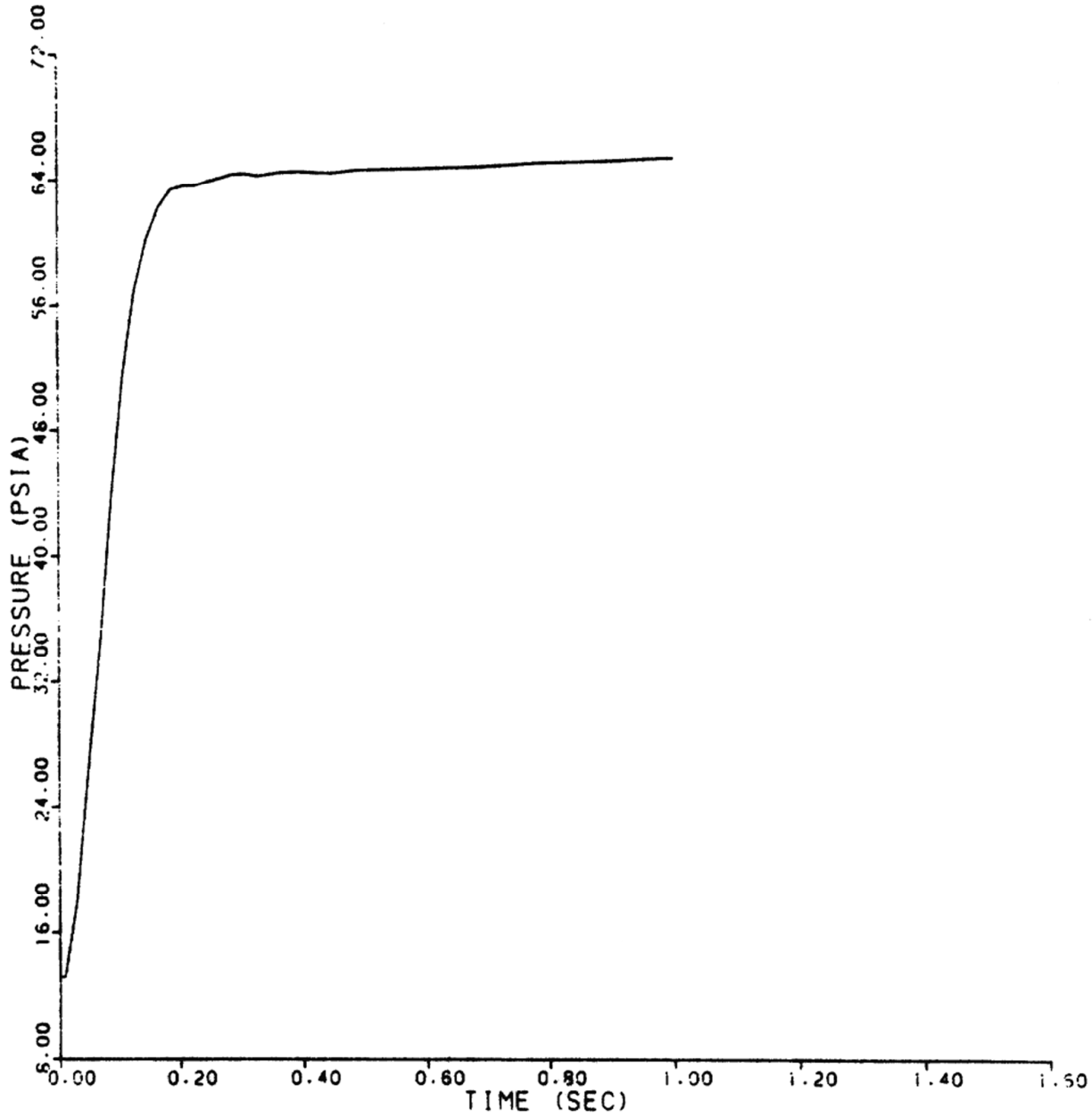
REV 13 4/06

SOUTHERN COMPANY Energy to Serve Your World® VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	REACTOR CAVITY PRESSURE RESPONSE – NODE E27
	FIGURE 6.2.1–15 (SHEET 27 OF 65)



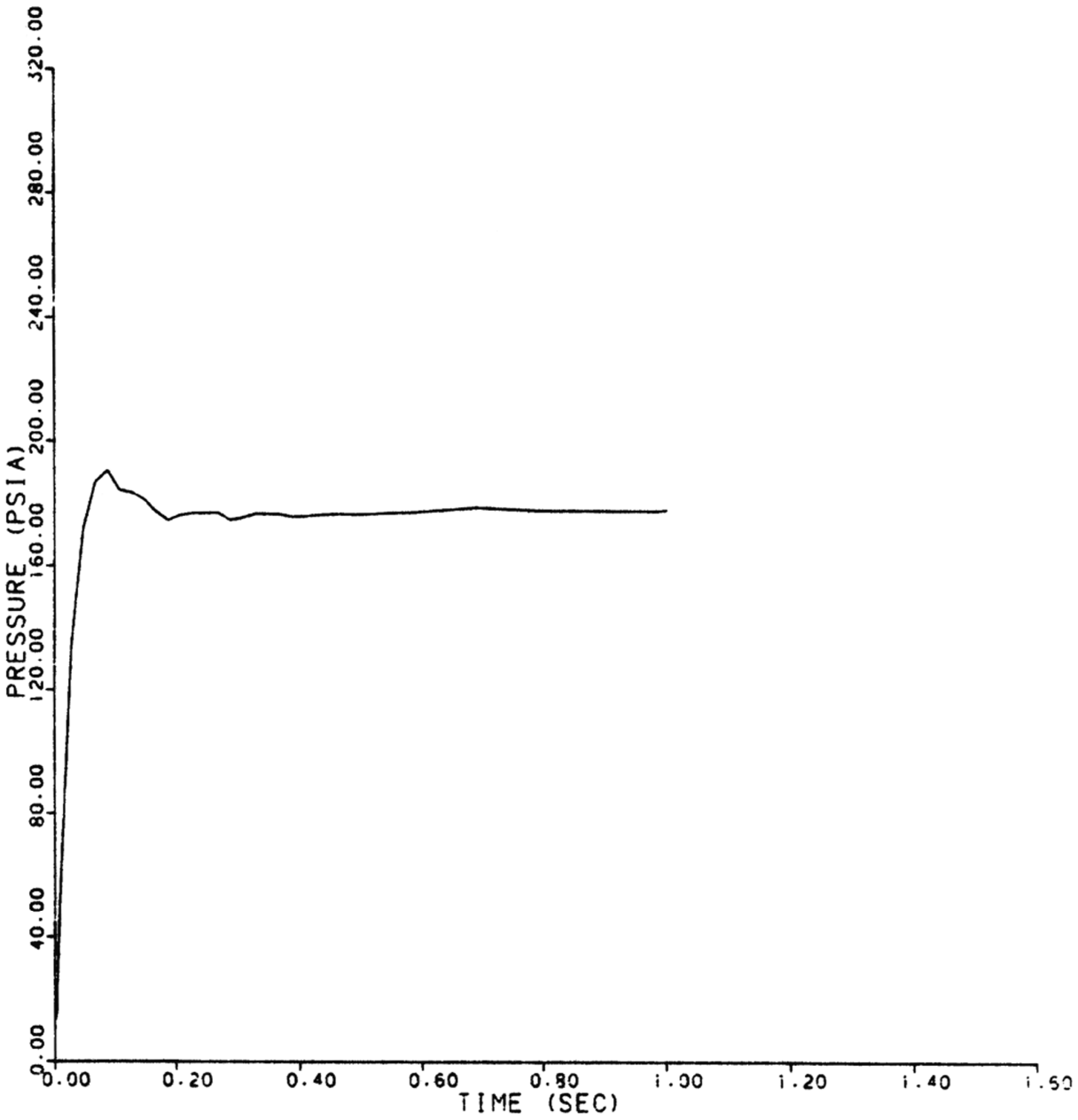
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	REACTOR CAVITY PRESSURE RESPONSE – NODE E28
	FIGURE 6.2.1–15 (SHEET 28 OF 65)



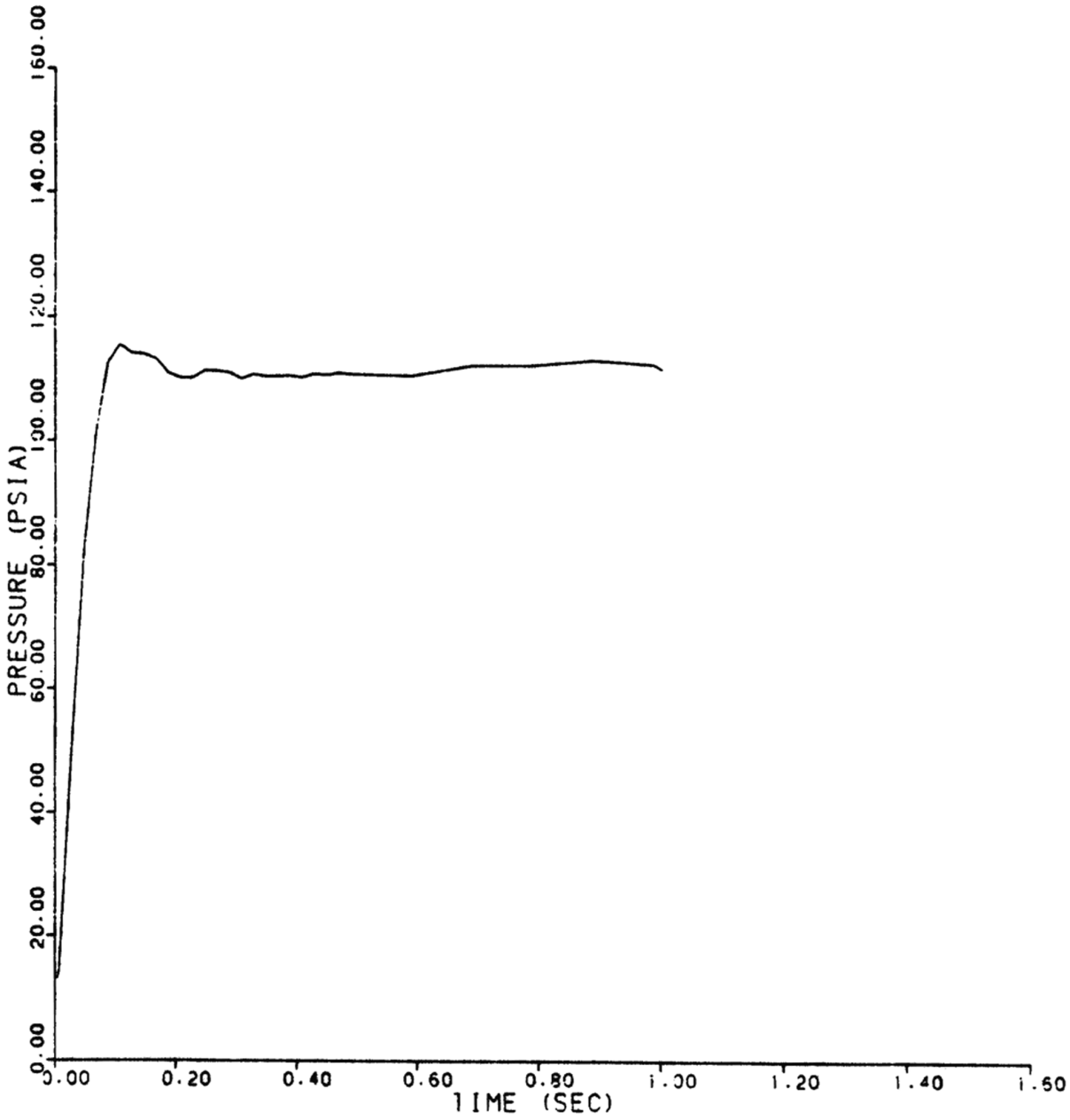
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	REACTOR CAVITY PRESSURE RESPONSE – NODE E29
	FIGURE 6.2.1–15 (SHEET 29 OF 65)



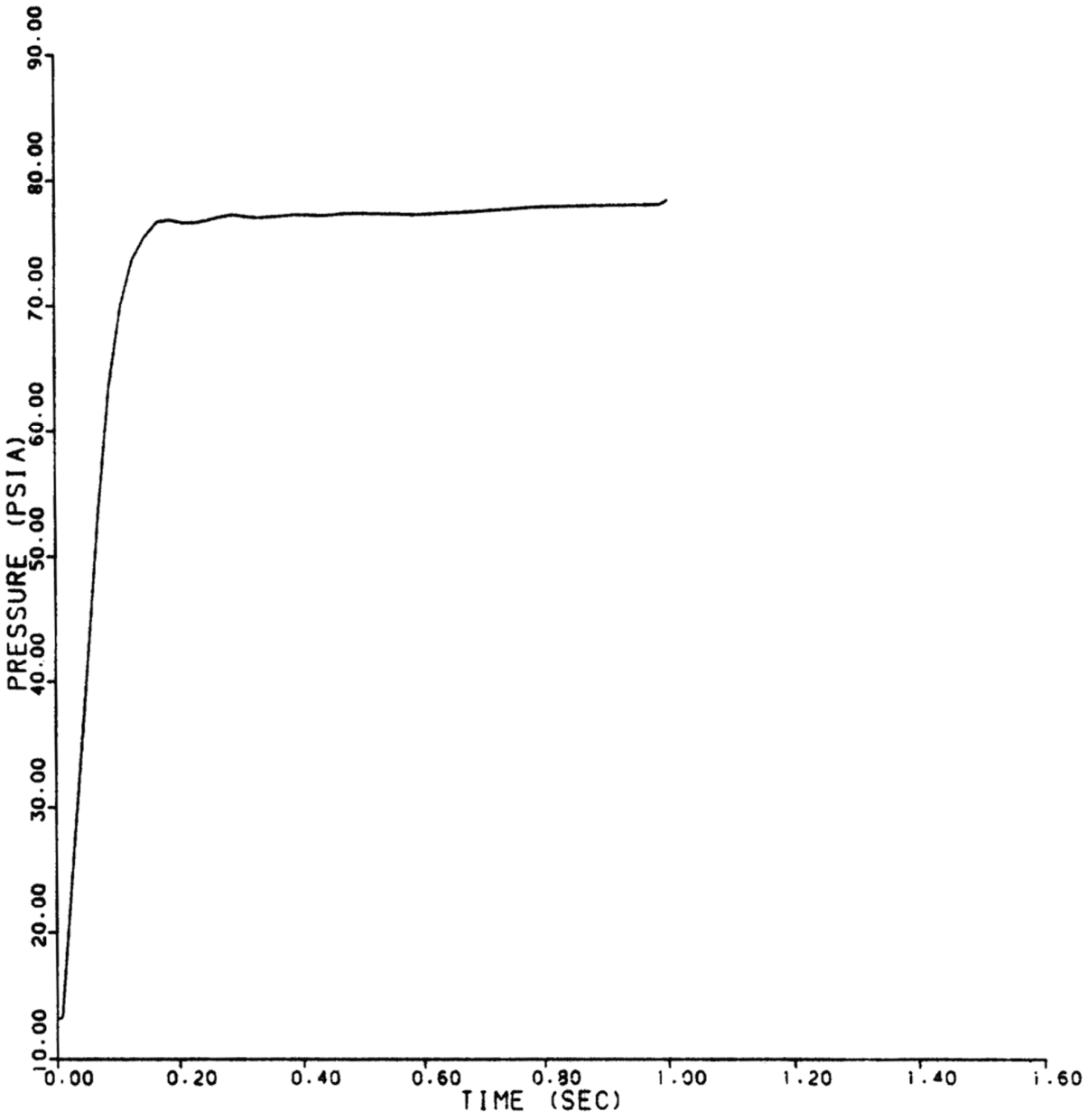
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E30 FIGURE 6.2.1–15 (SHEET 30 OF 65)



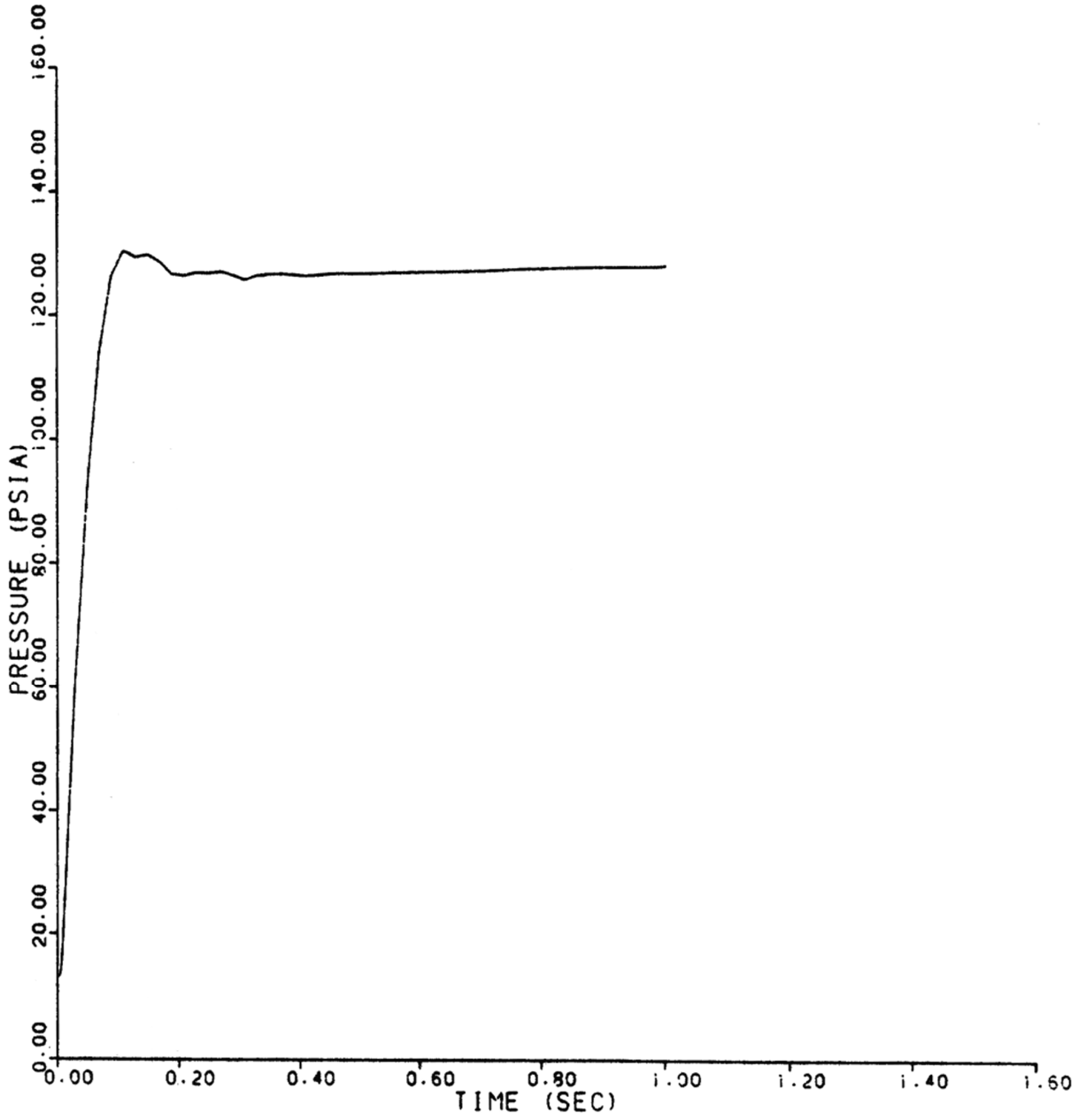
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E31
	FIGURE 6.2.1–15 (SHEET 31 OF 65)



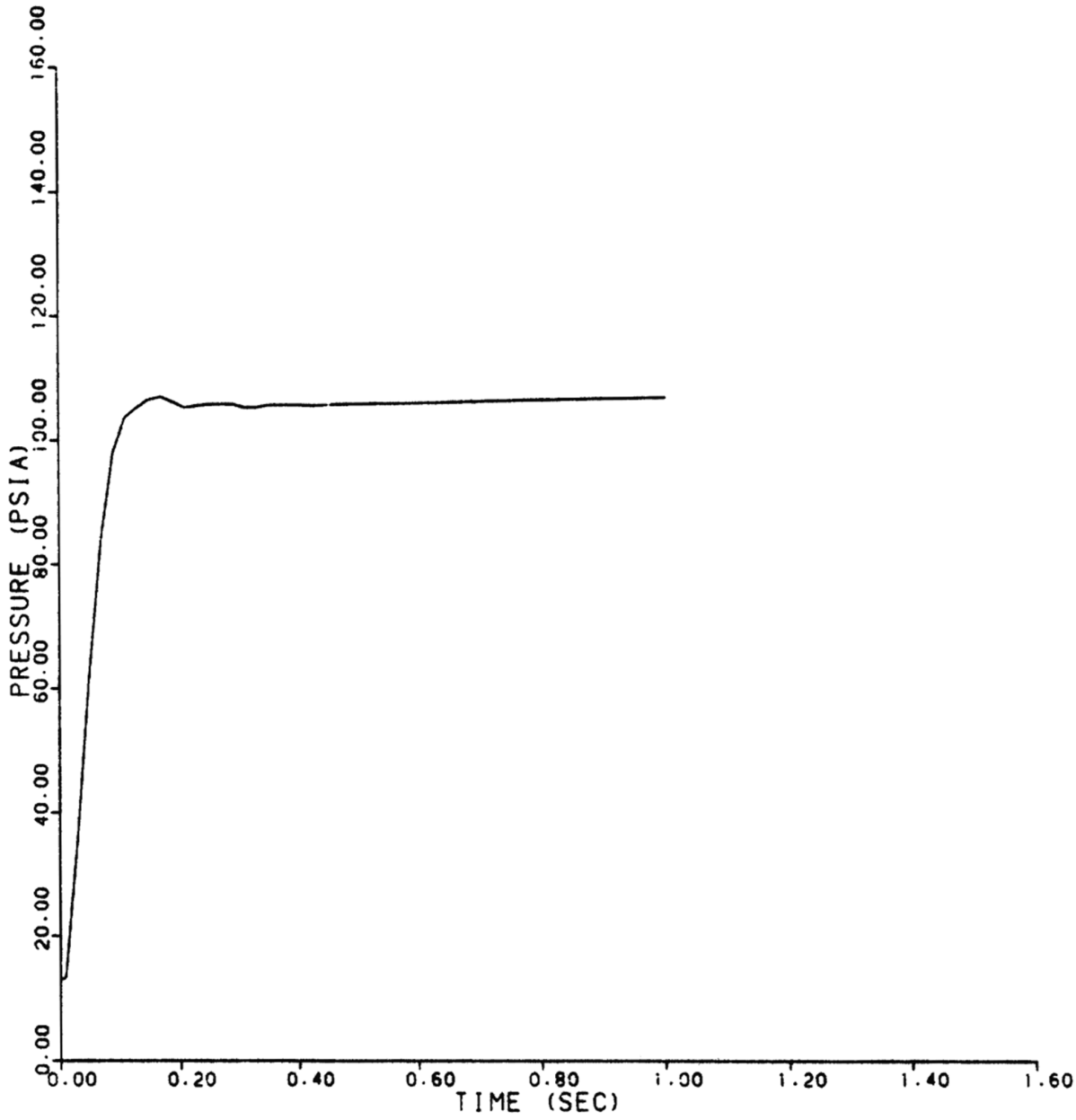
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	REACTOR CAVITY PRESSURE RESPONSE – NODE E32
	FIGURE 6.2.1–15 (SHEET 32 OF 65)



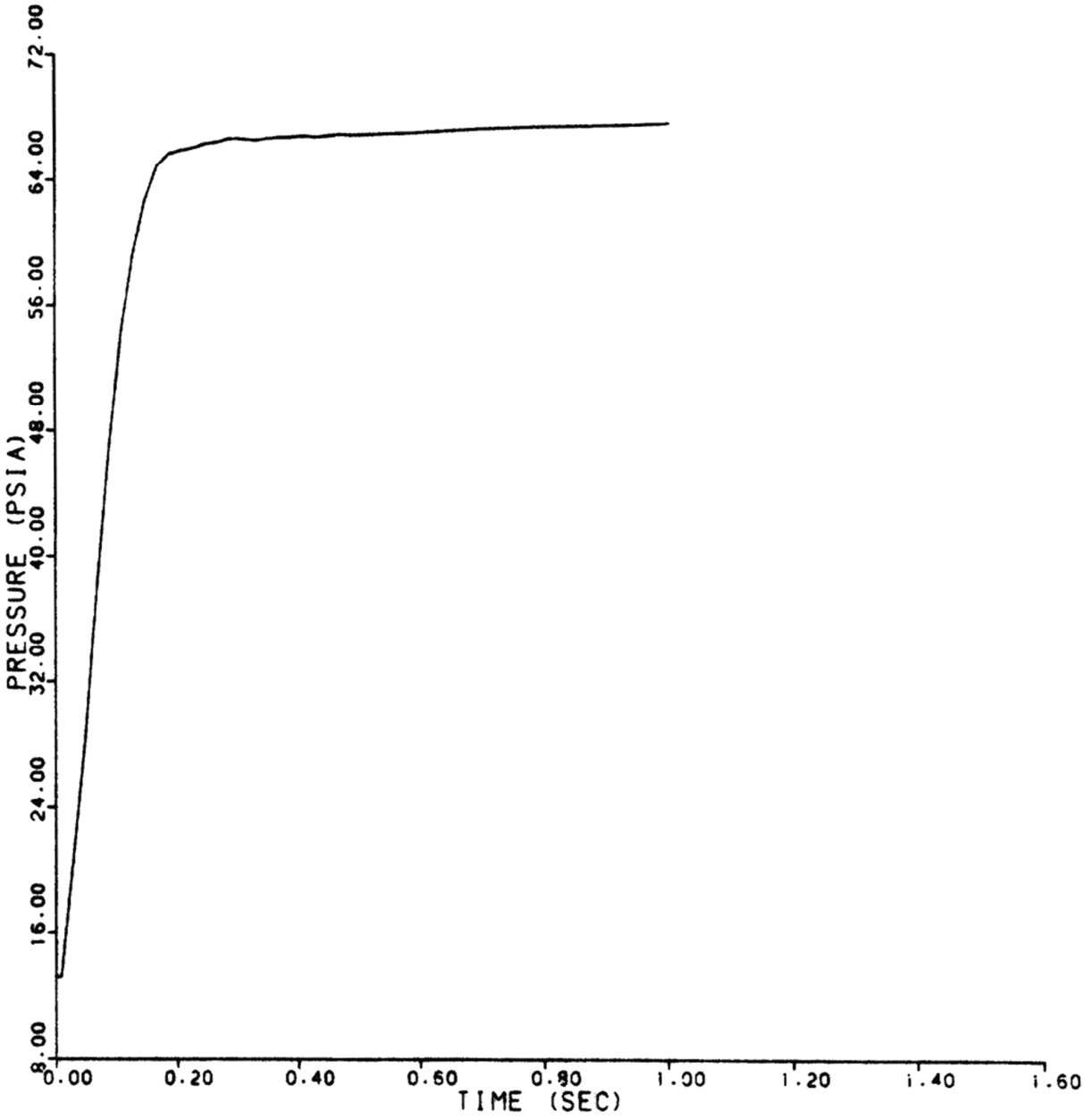
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E33
	FIGURE 6.2.1–15 (SHEET 33 OF 65)



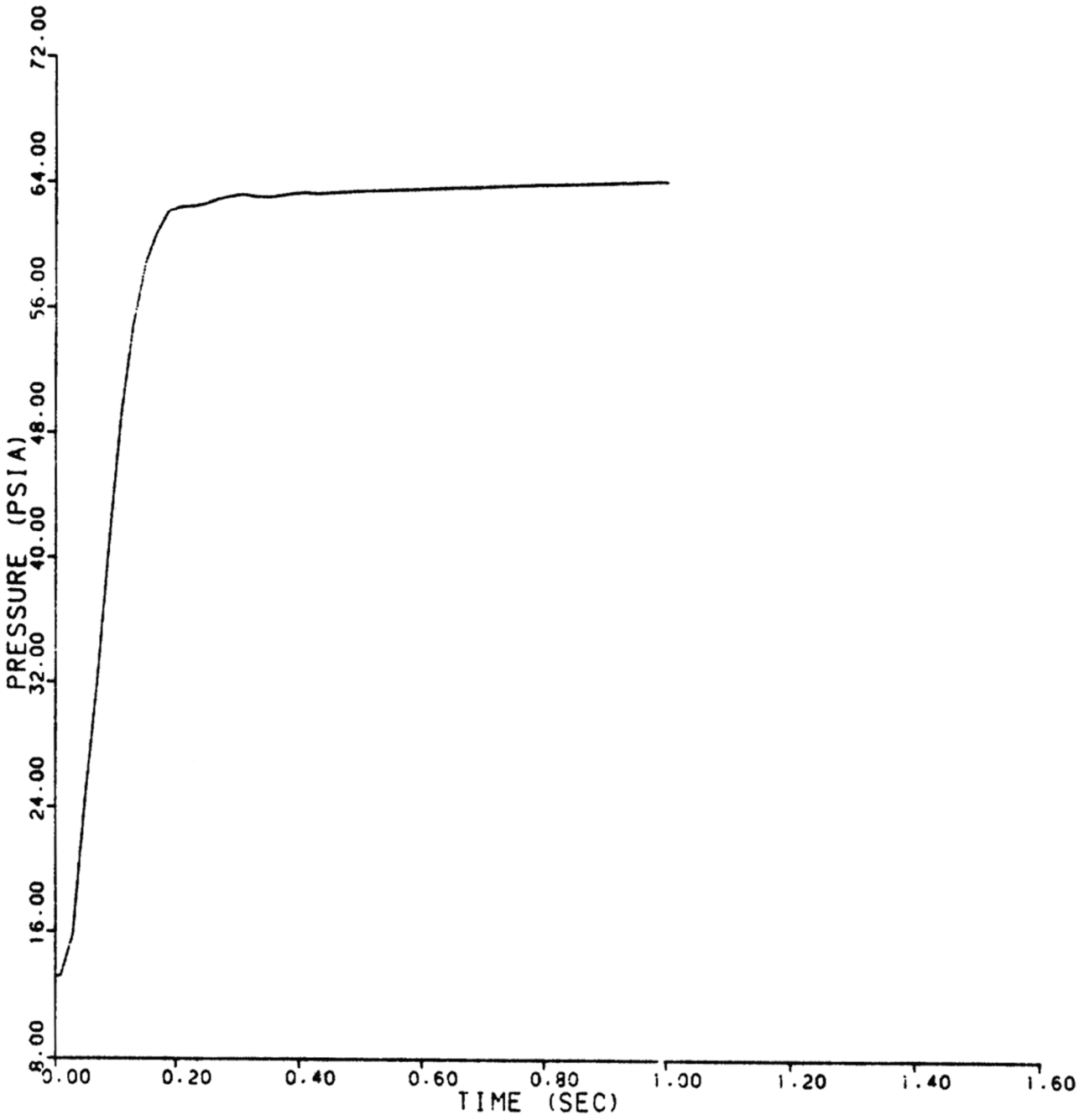
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	REACTOR CAVITY PRESSURE RESPONSE – NODE E34
	FIGURE 6.2.1–15 (SHEET 34 OF 65)



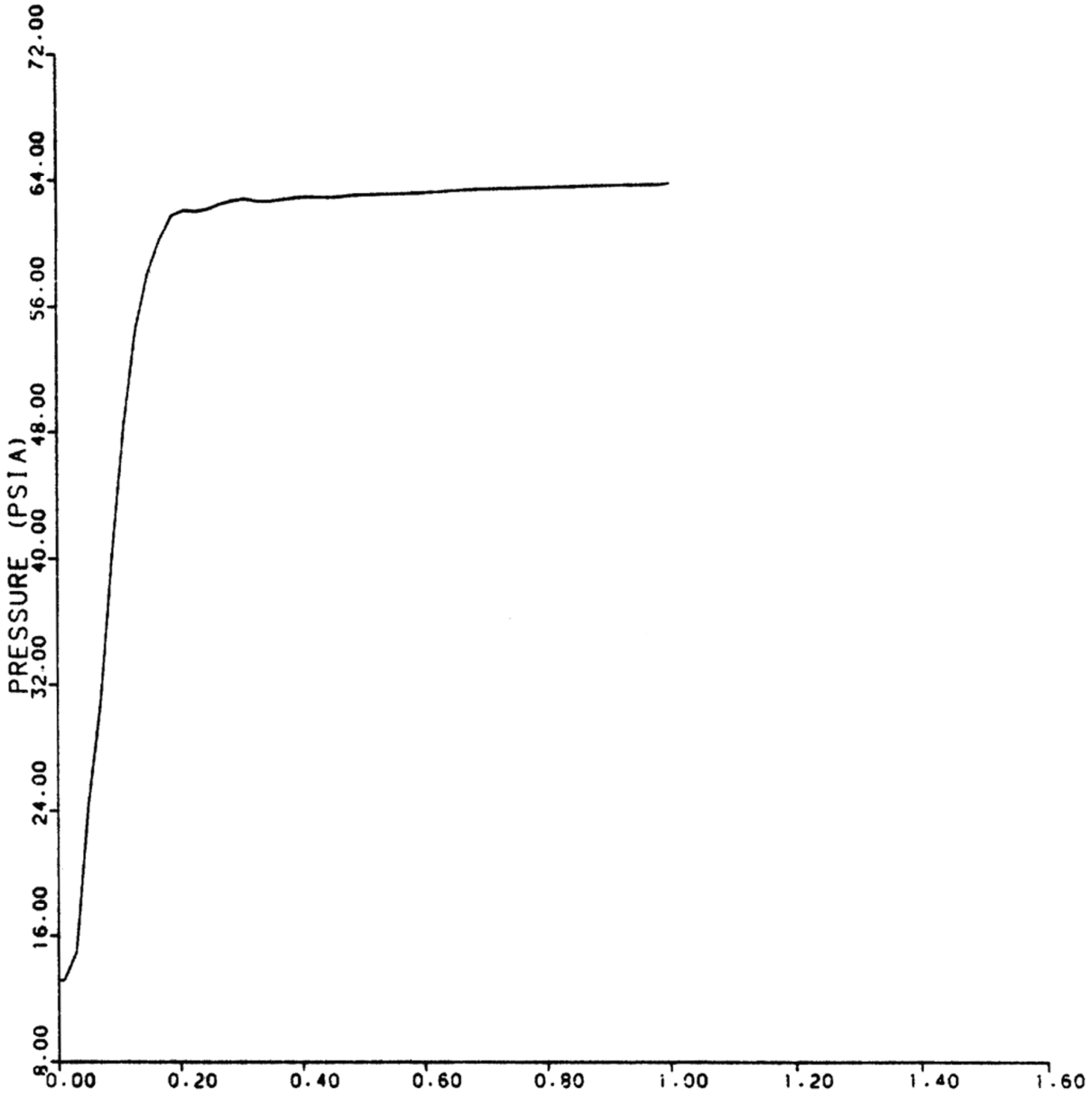
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	REACTOR CAVITY PRESSURE RESPONSE – NODE E35
	FIGURE 6.2.1–15 (SHEET 35 OF 65)



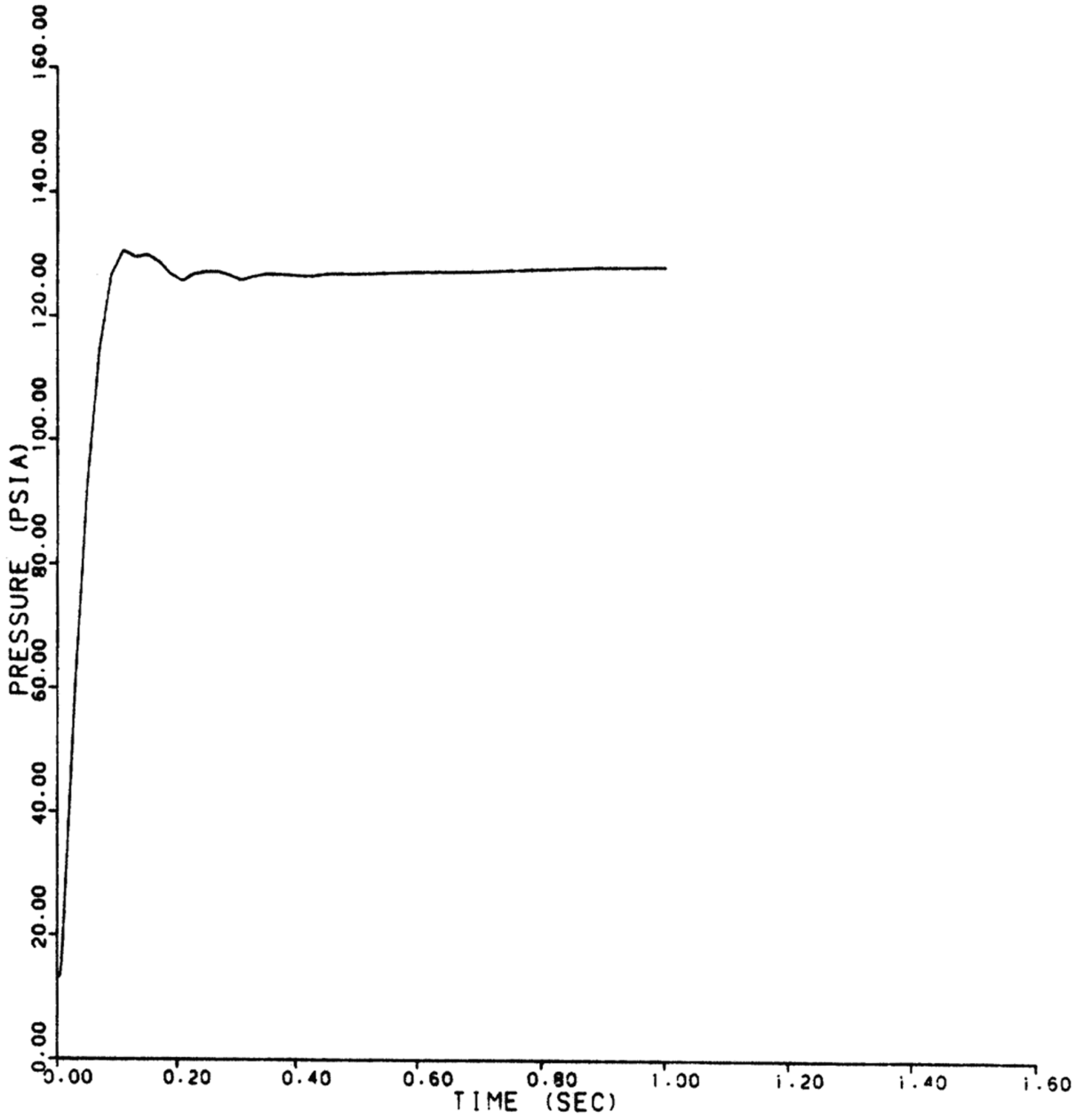
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E36
	FIGURE 6.2.1–15 (SHEET 36 OF 65)



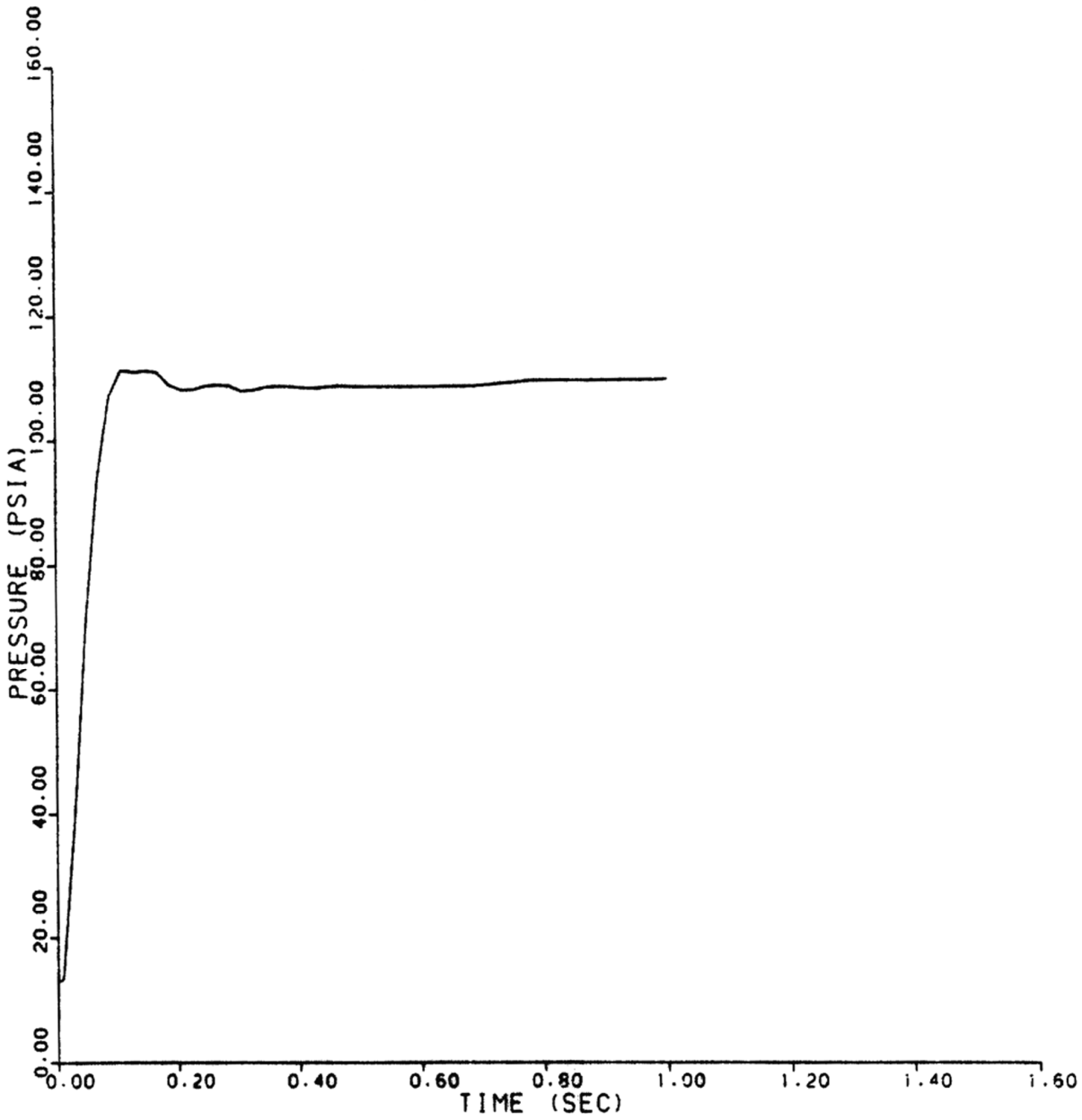
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E37
	FIGURE 6.2.1–15 (SHEET 37 OF 65)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E38
	FIGURE 6.2.1–15 (SHEET 38 OF 65)



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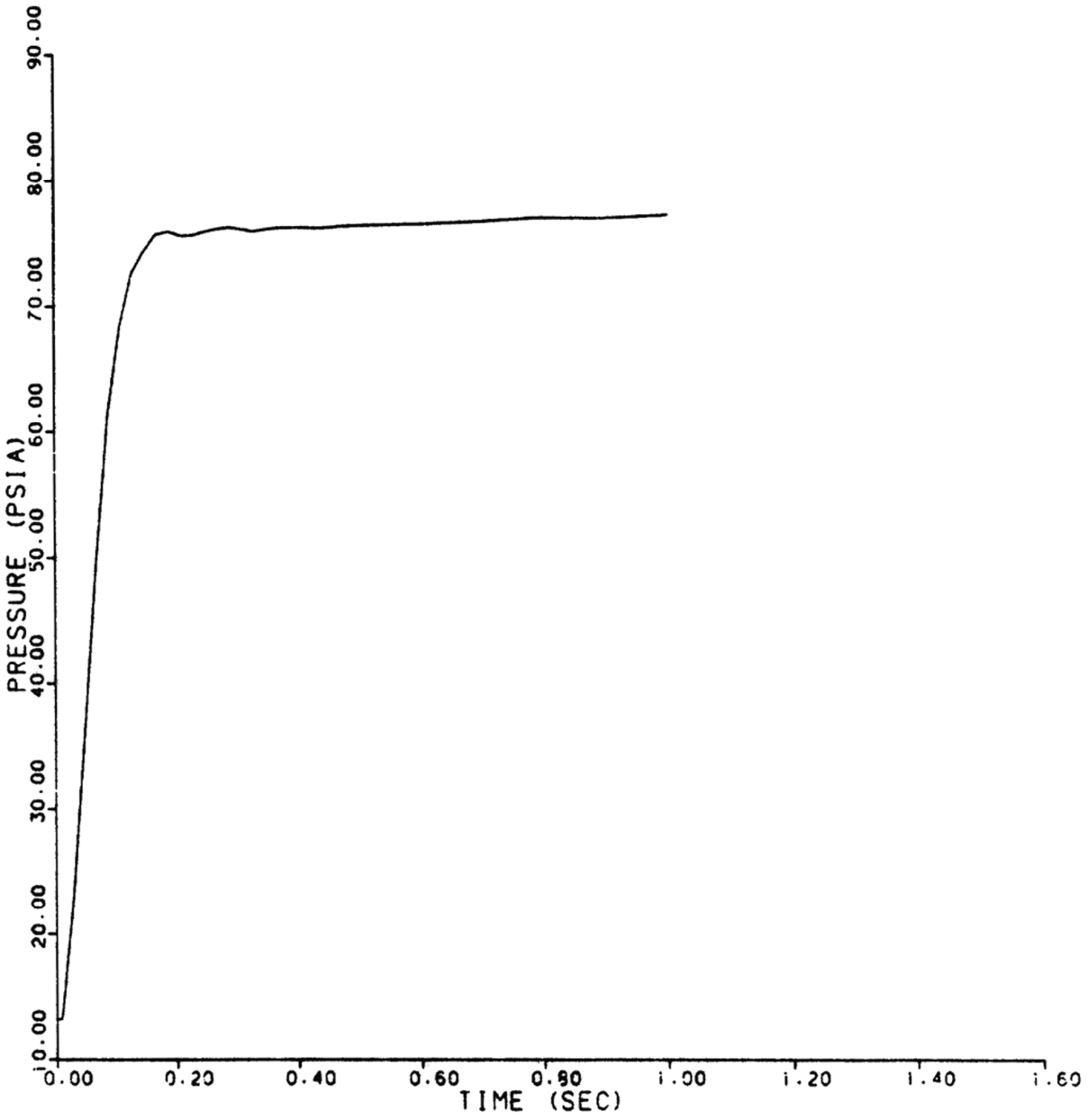
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

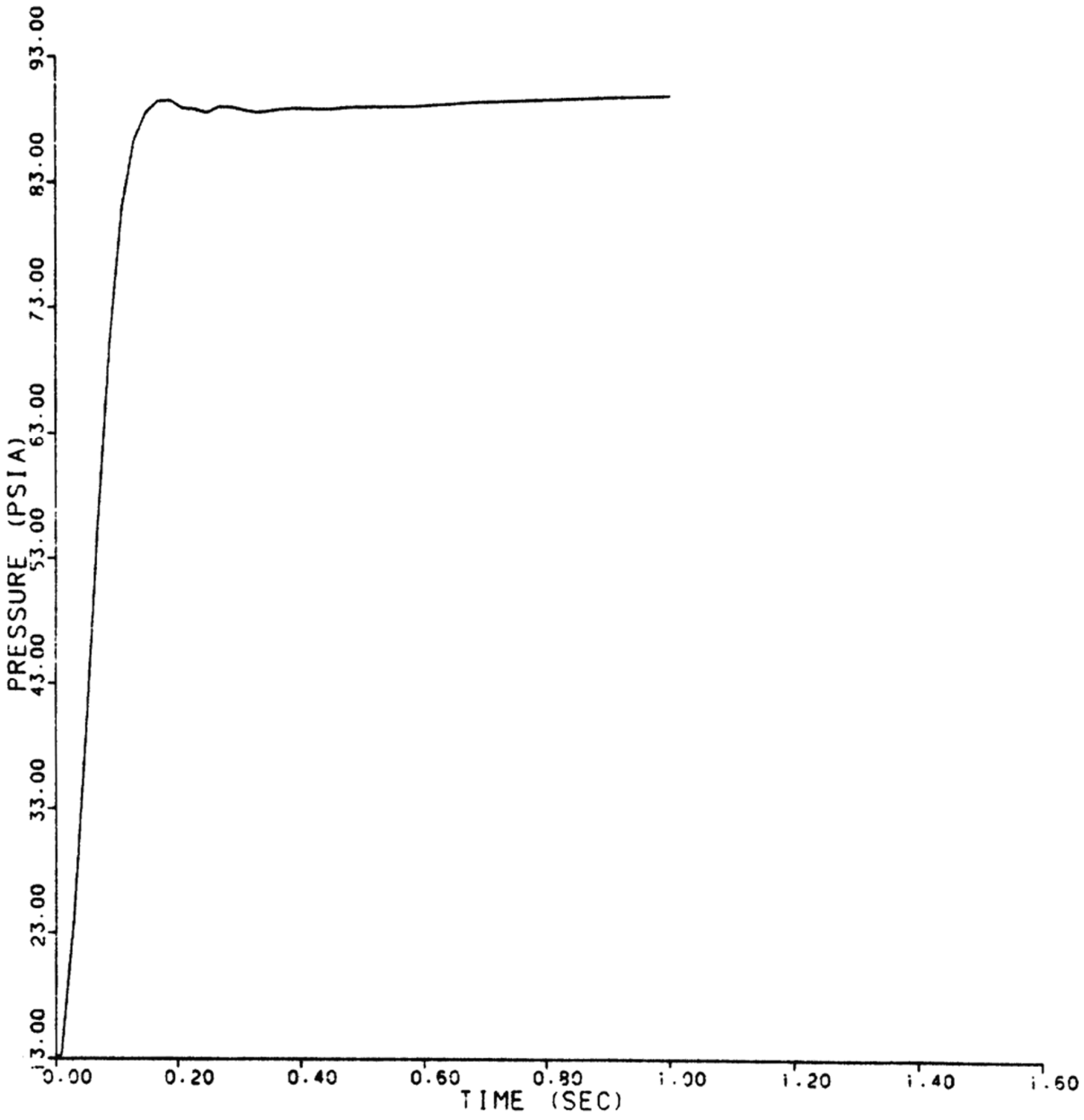
REACTOR CAVITY PRESSURE RESPONSE – NODE E39

FIGURE 6.2.1–15 (SHEET 39 OF 65)



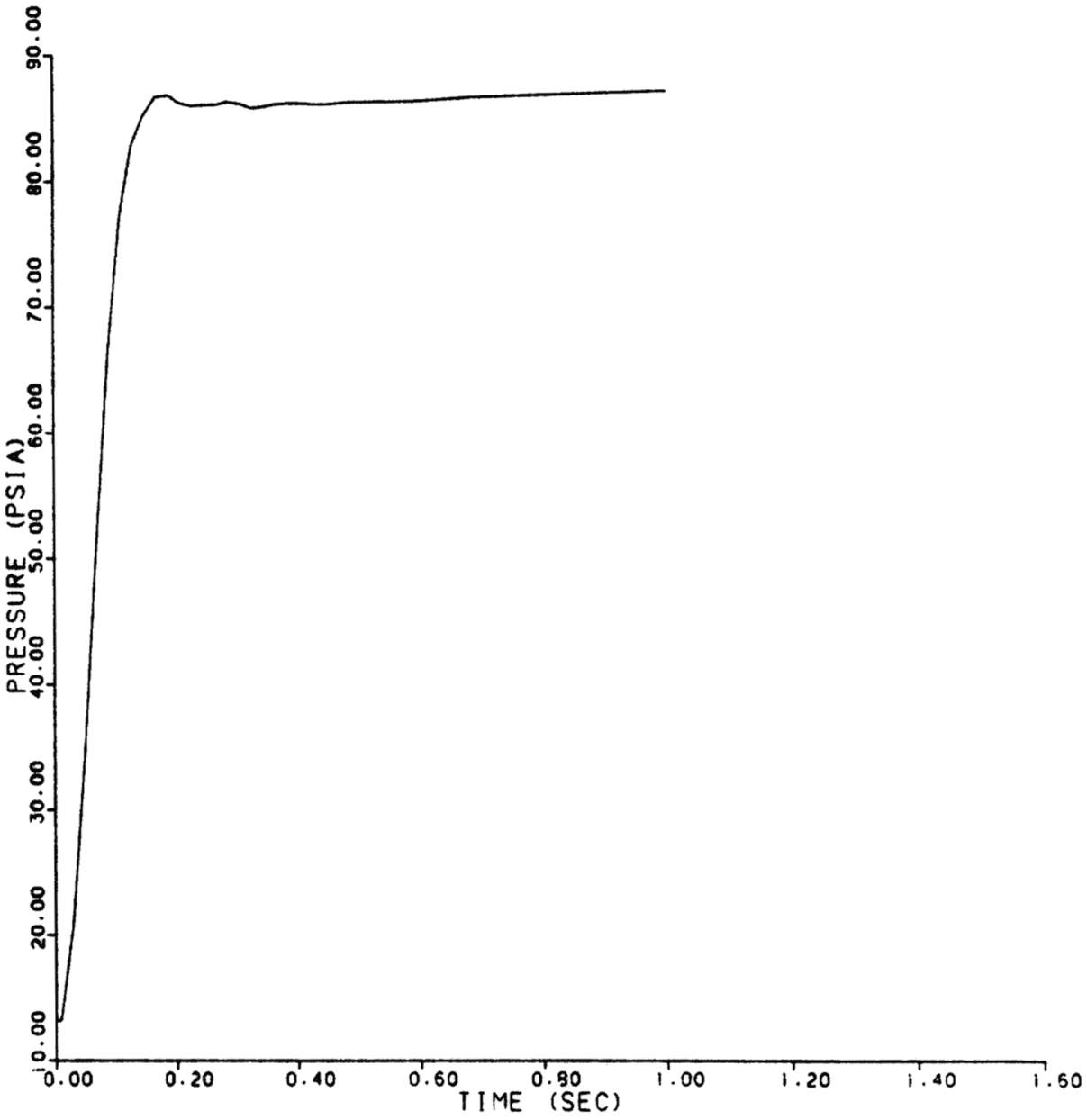
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E40
	FIGURE 6.2.1–15 (SHEET 40 OF 65)



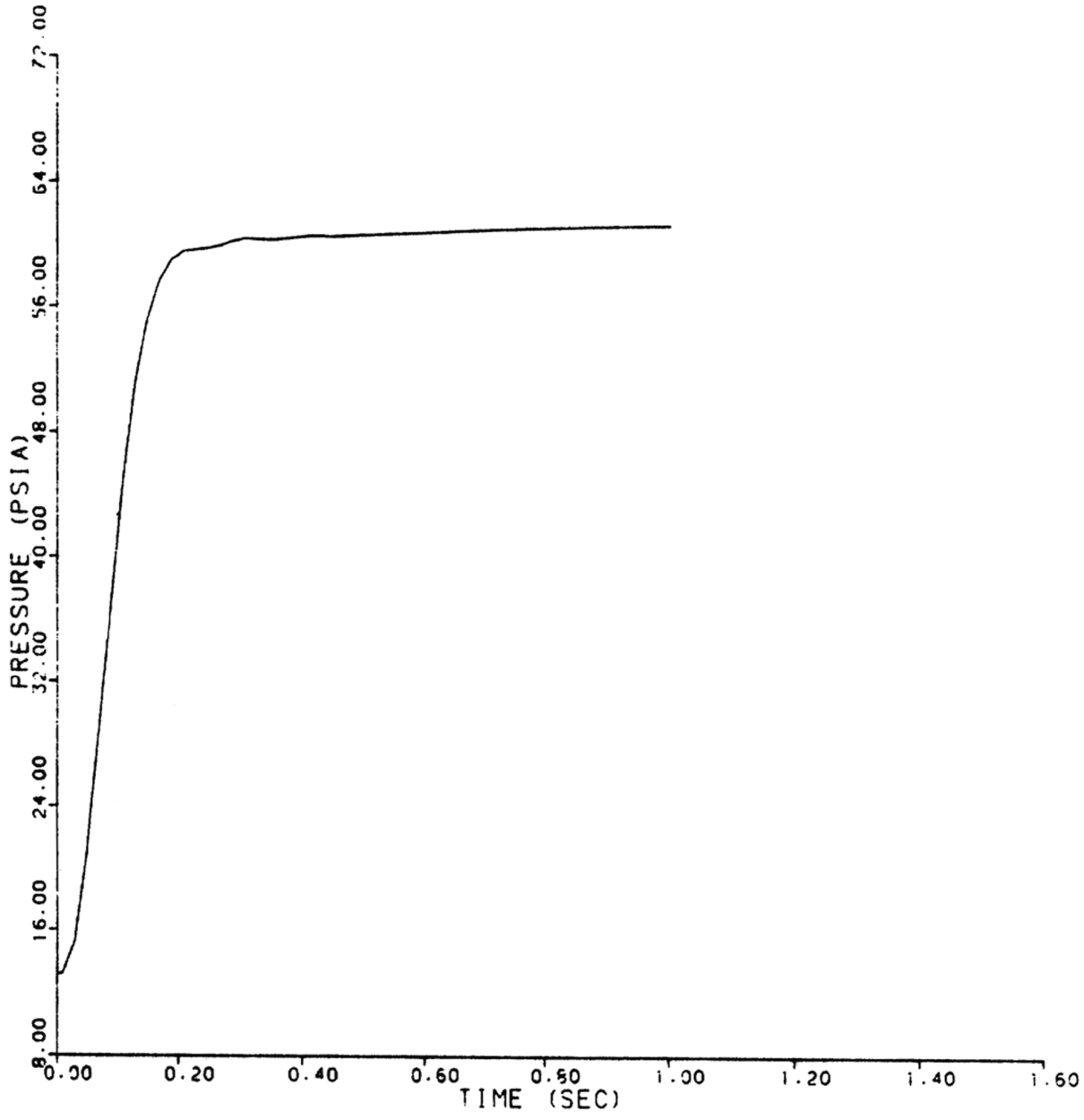
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E41
	FIGURE 6.2.1–15 (SHEET 41 OF 65)



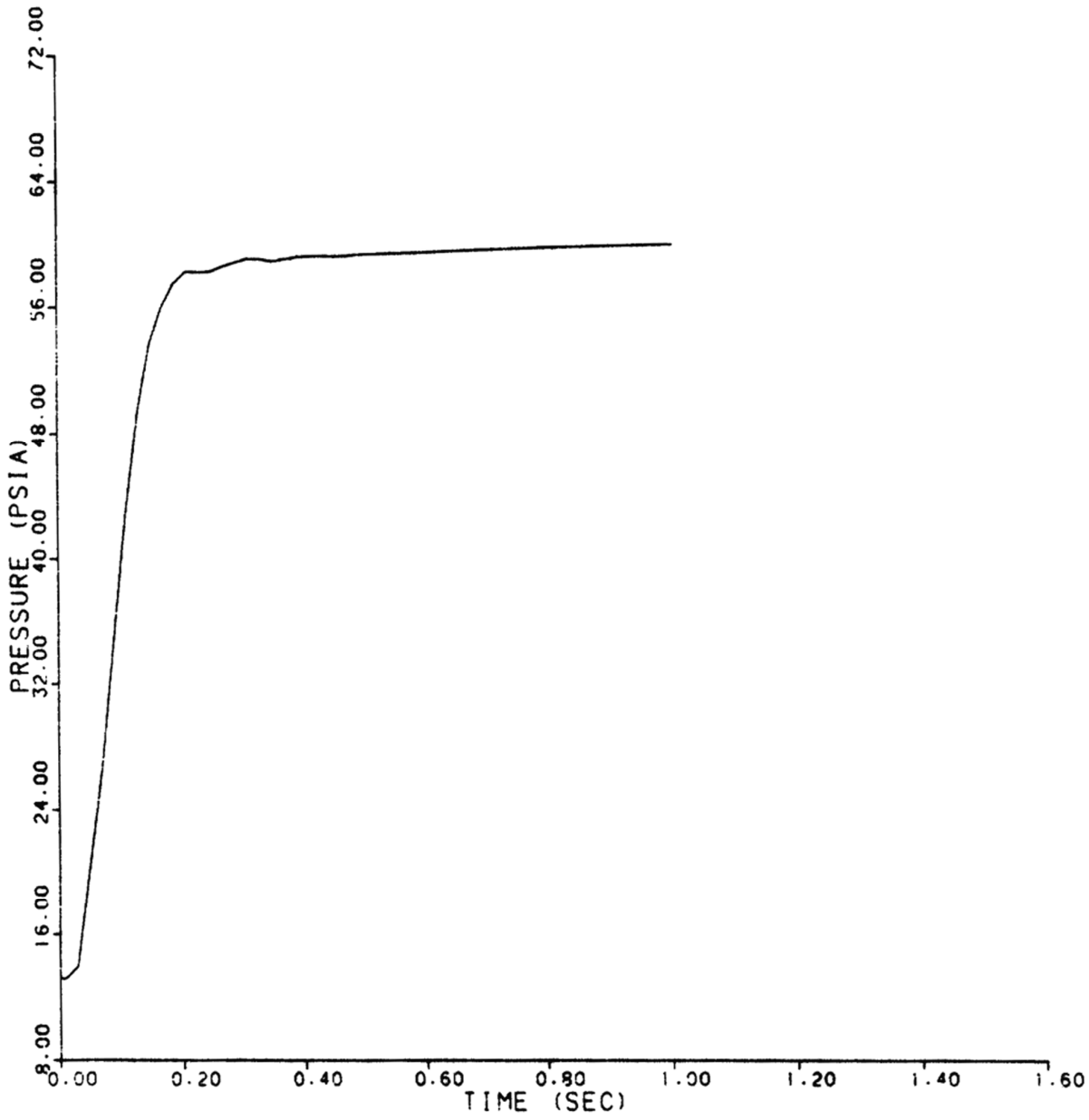
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E42
	FIGURE 6.2.1–15 (SHEET 42 OF 65)



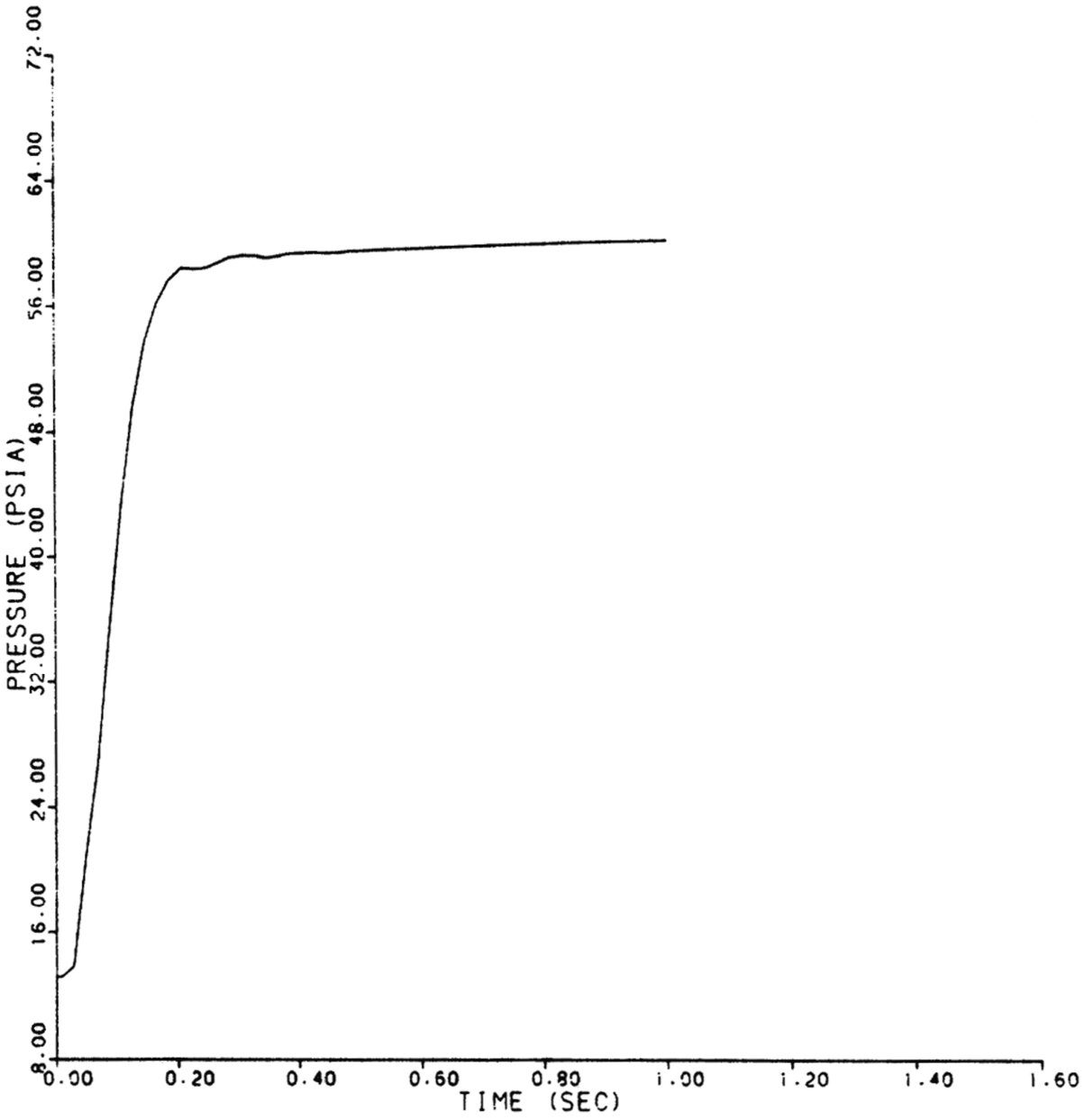
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E43
	FIGURE 6.2.1–15 (SHEET 43 OF 65)



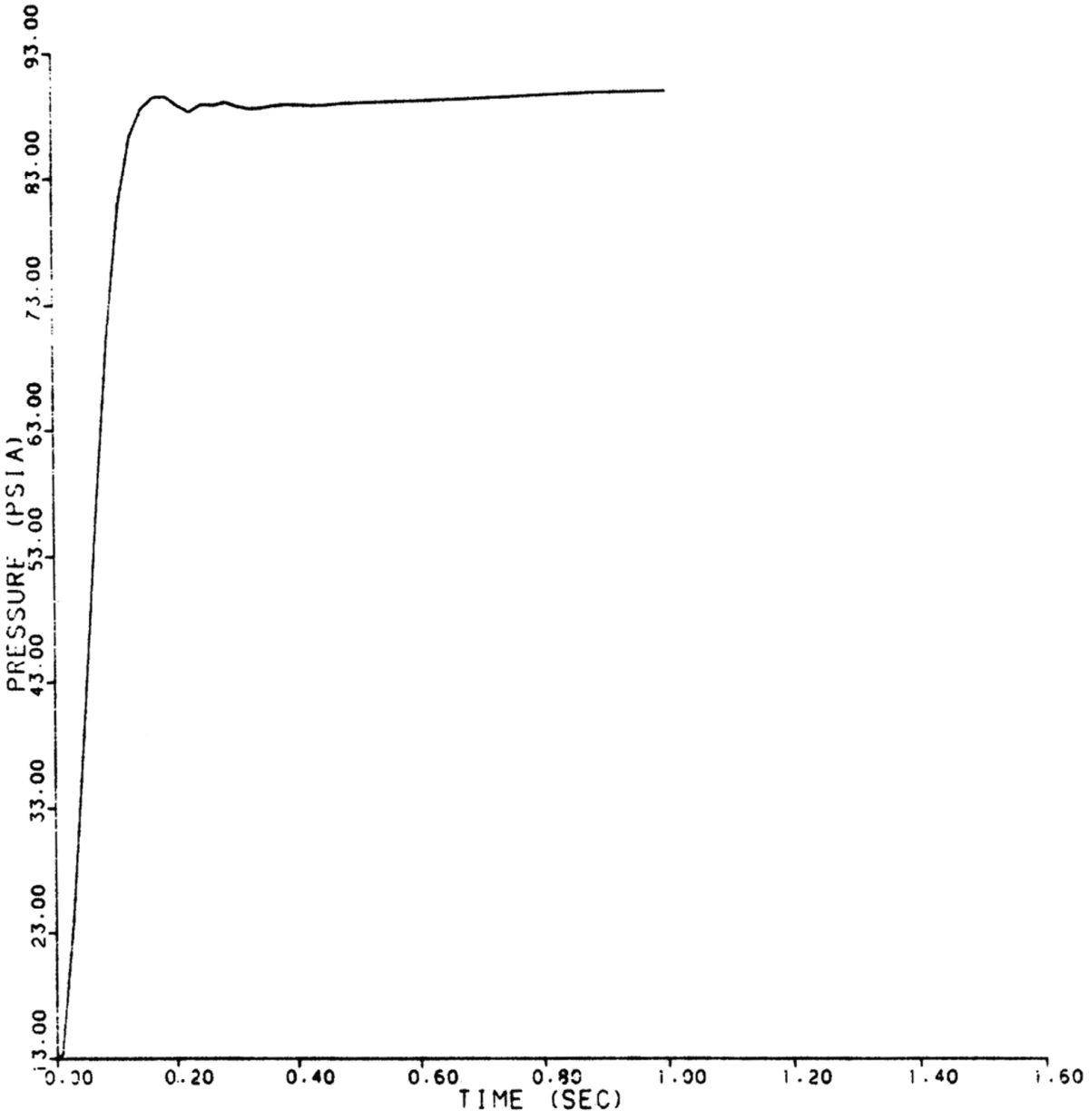
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E44
	FIGURE 6.2.1–15 (SHEET 44 OF 65)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E45
	FIGURE 6.2.1–15 (SHEET 45 OF 65)



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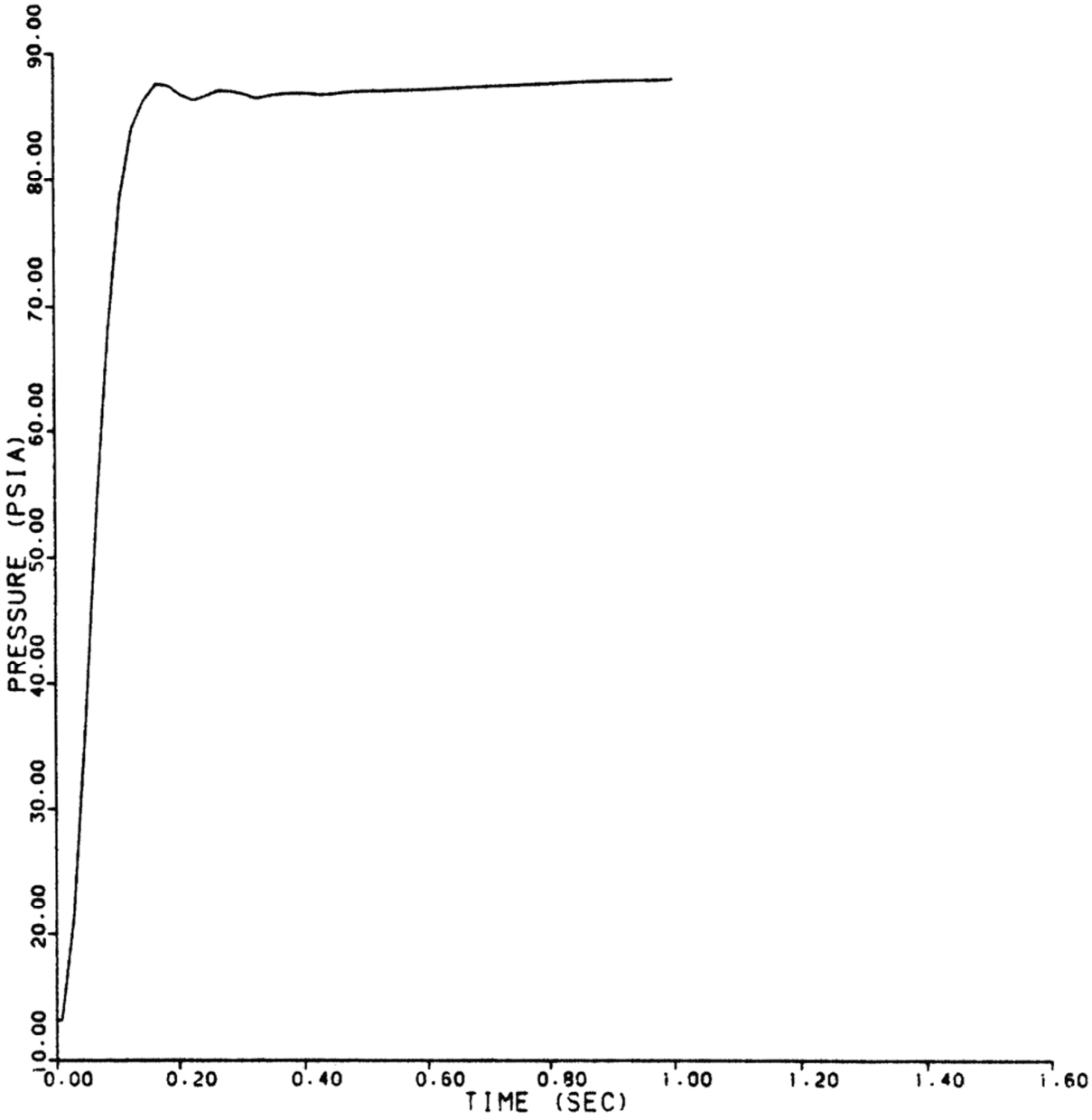
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E46
	FIGURE 6.2.1–15 (SHEET 46 OF 65)



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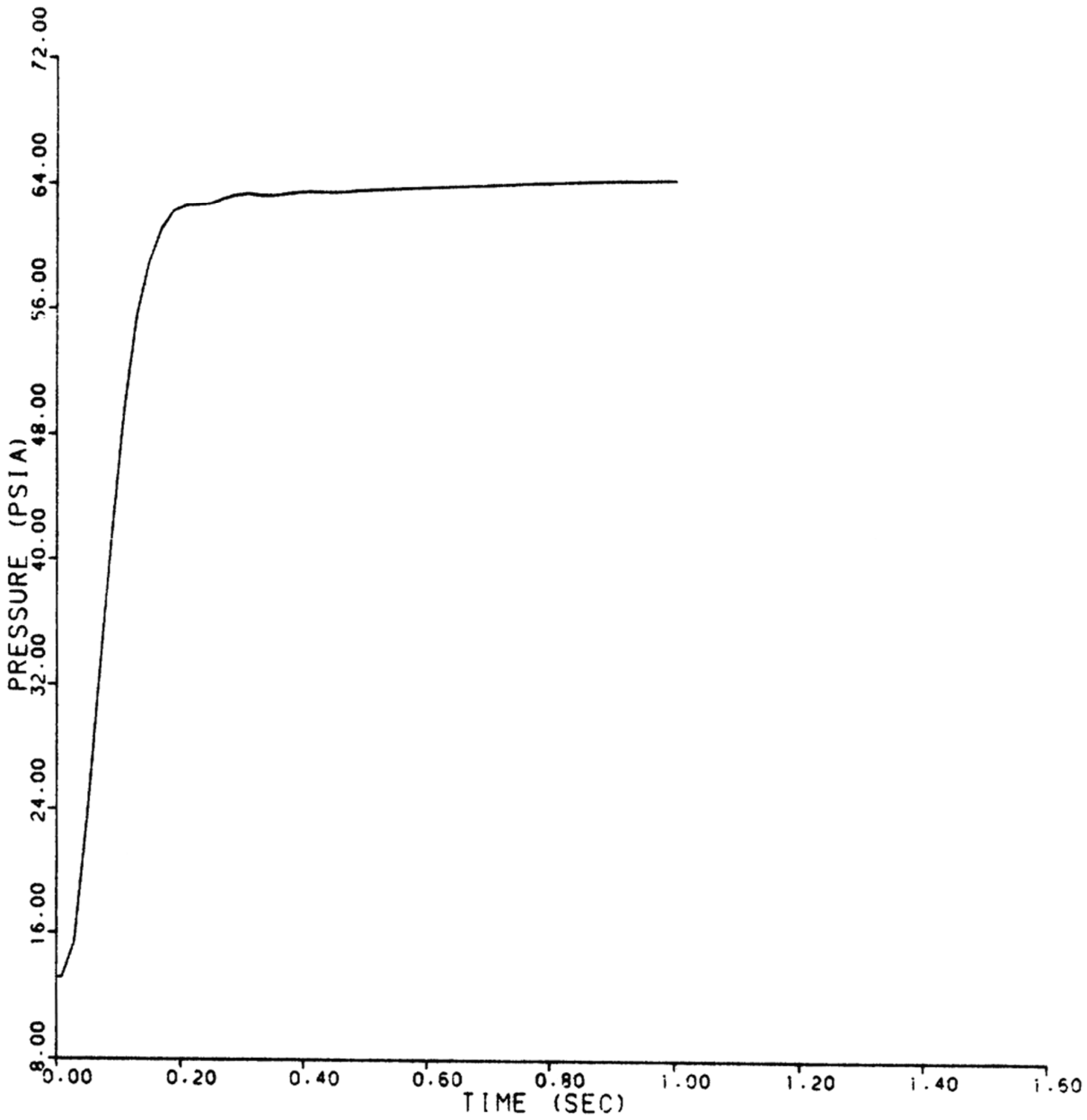
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E47
	FIGURE 6.2.1–15 (SHEET 47 OF 65)



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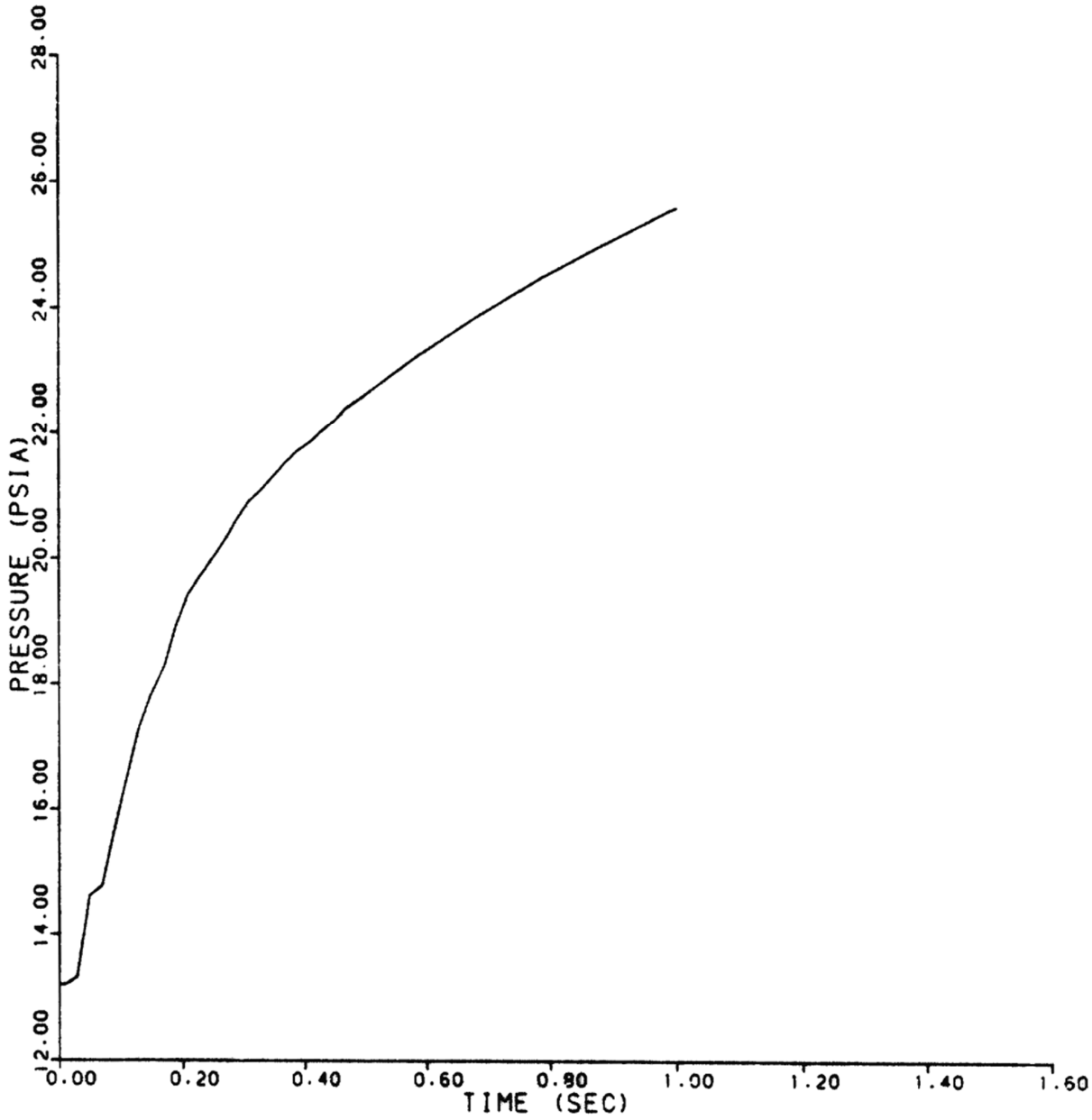
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E48
	FIGURE 6.2.1–15 (SHEET 48 OF 65)



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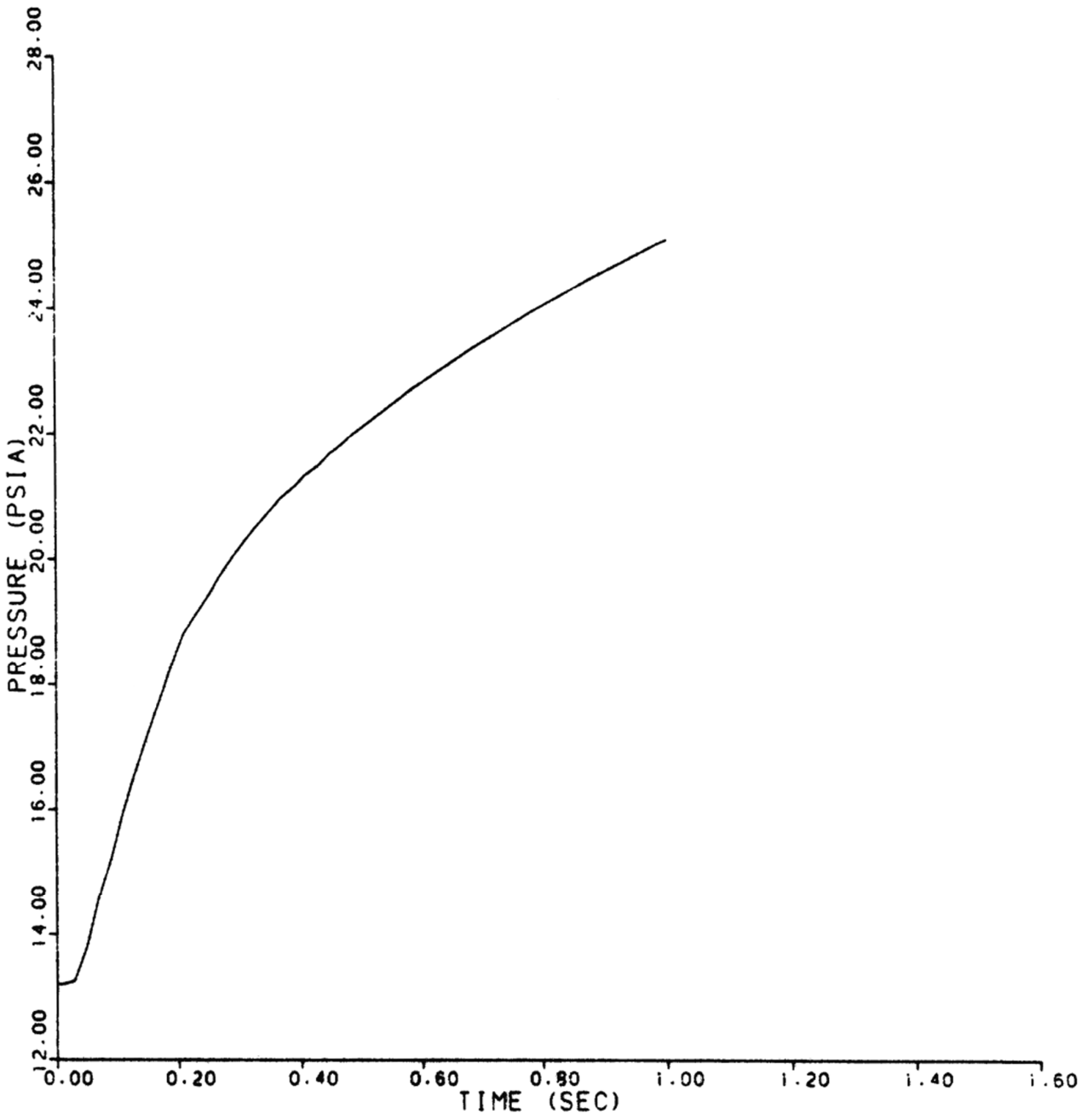
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E49
	FIGURE 6.2.1–15 (SHEET 49 OF 65)



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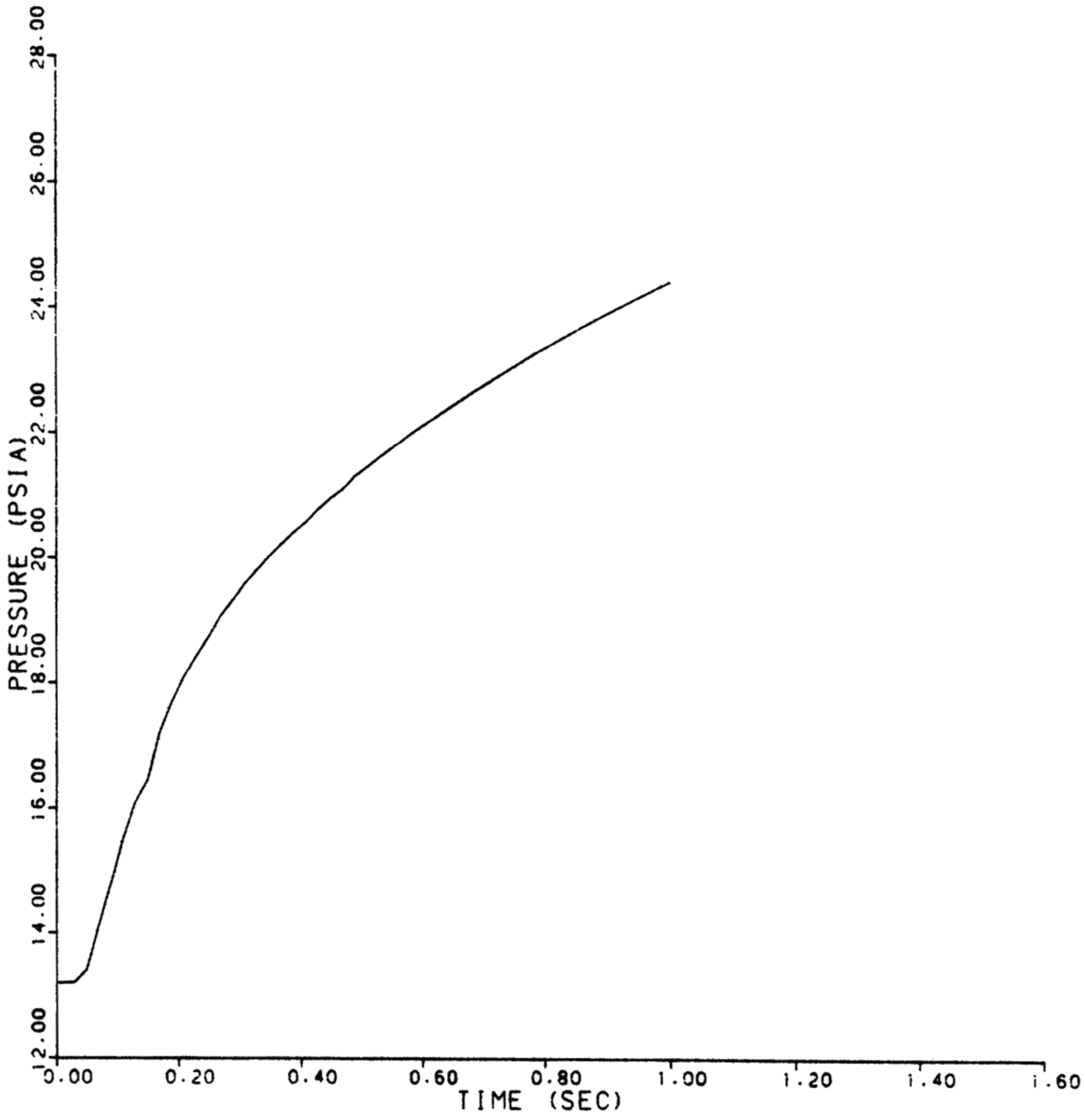
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E50
	FIGURE 6.2.1–15 (SHEET 50 OF 65)



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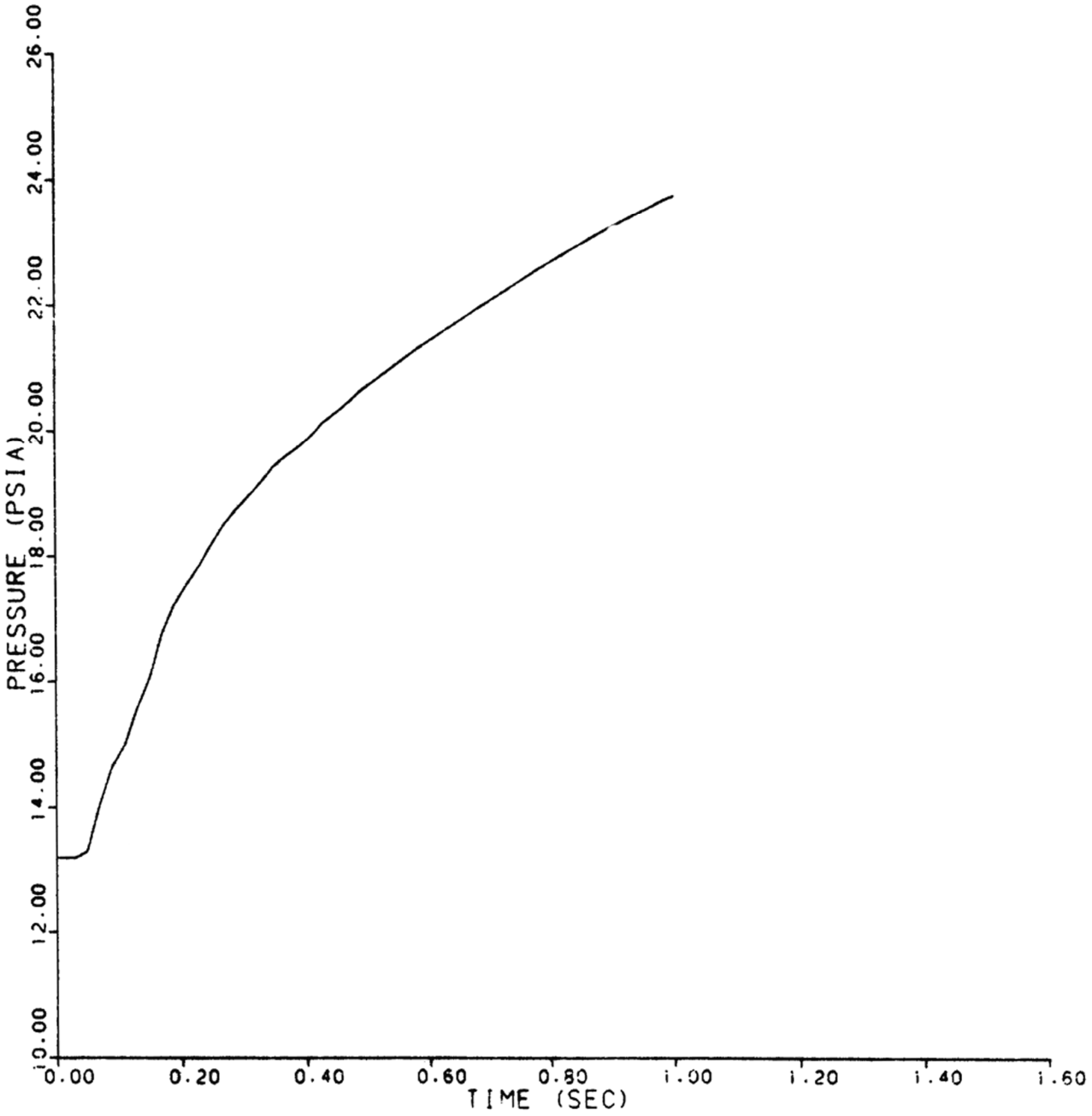
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E51
	FIGURE 6.2.1–15 (SHEET 51 OF 65)



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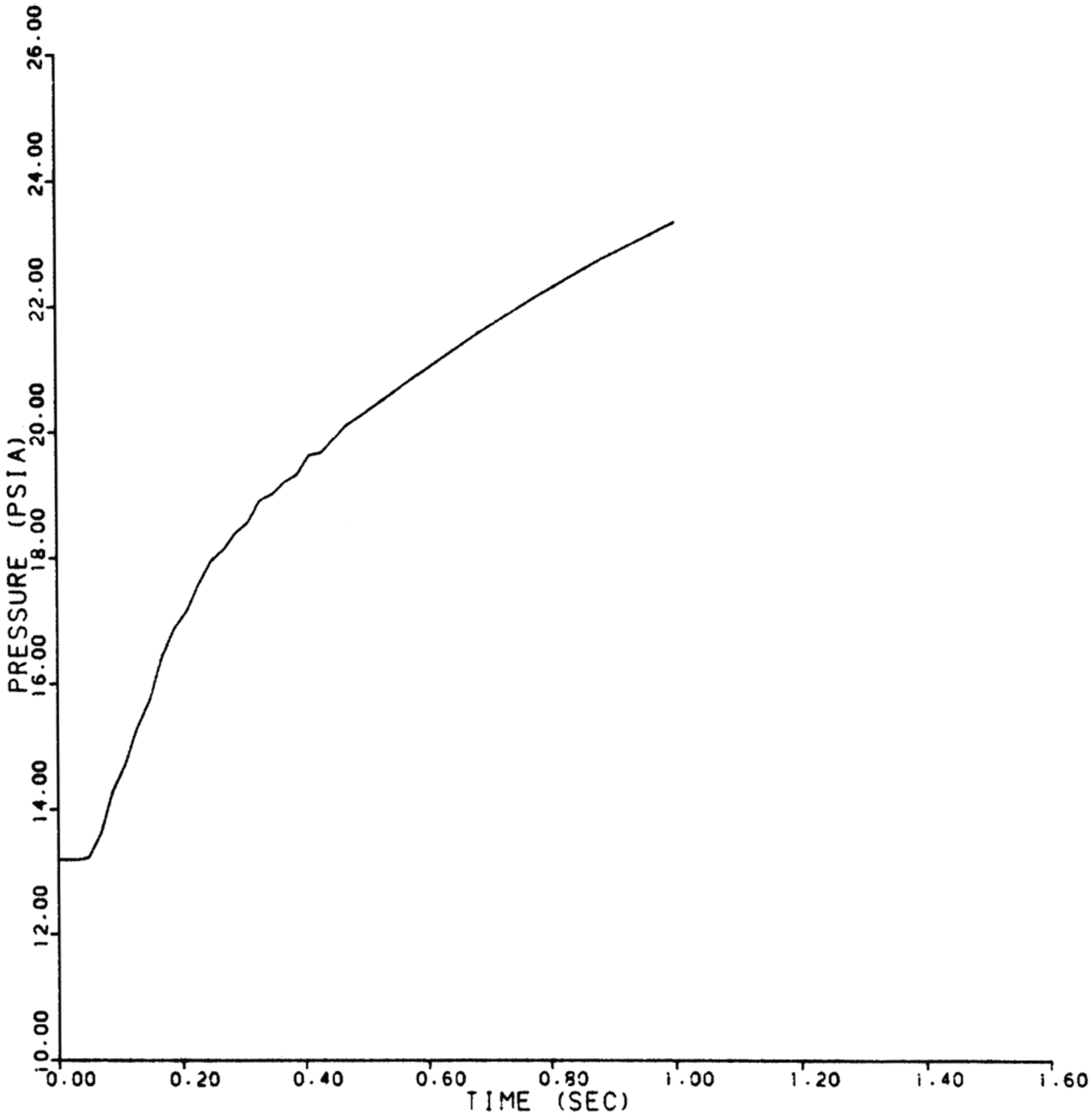
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E52
	FIGURE 6.2.1–15 (SHEET 52 OF 65)



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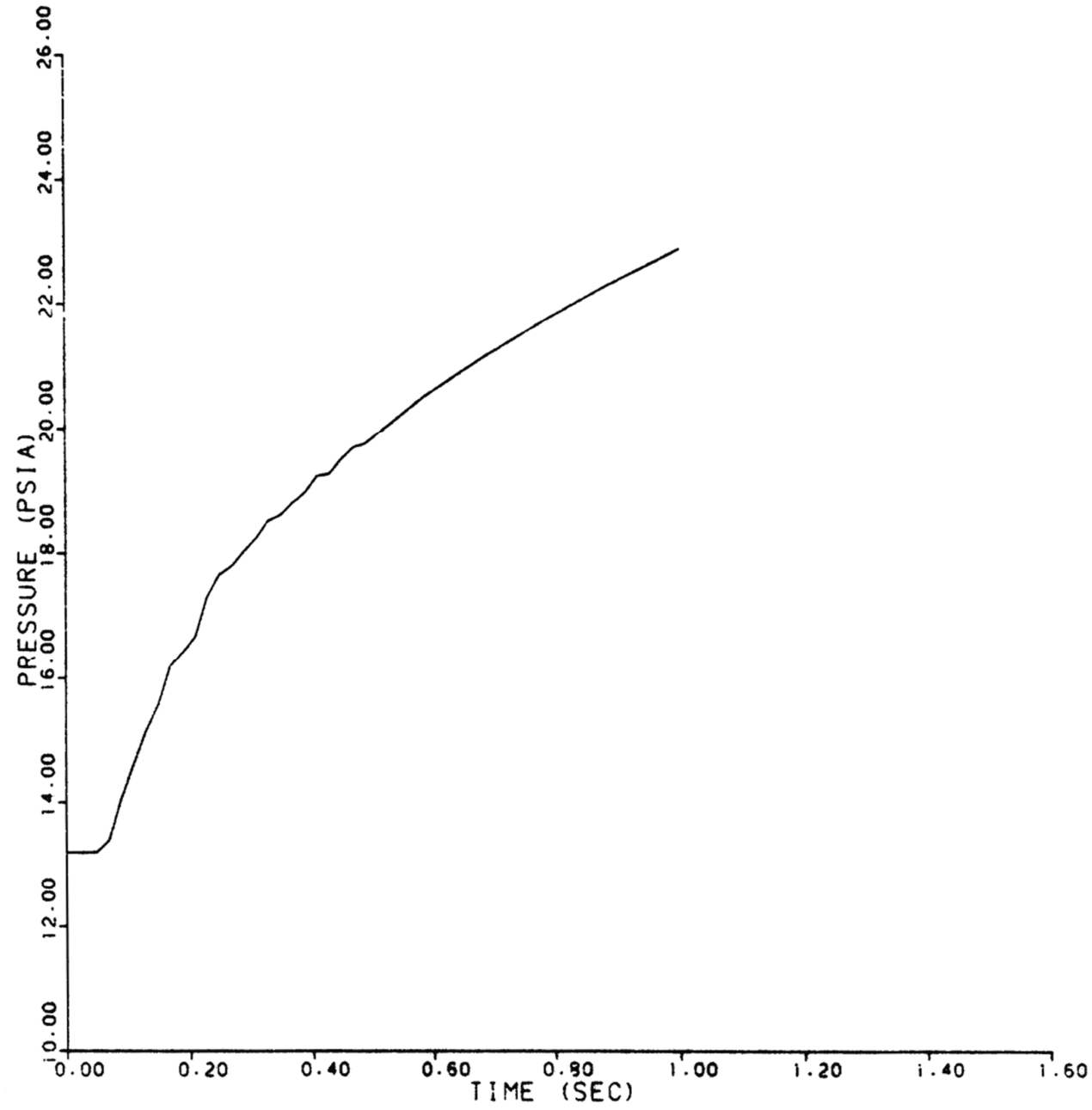
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – ODE E53
	FIGURE 6.2.1–15 (SHEET 53 OF 65)



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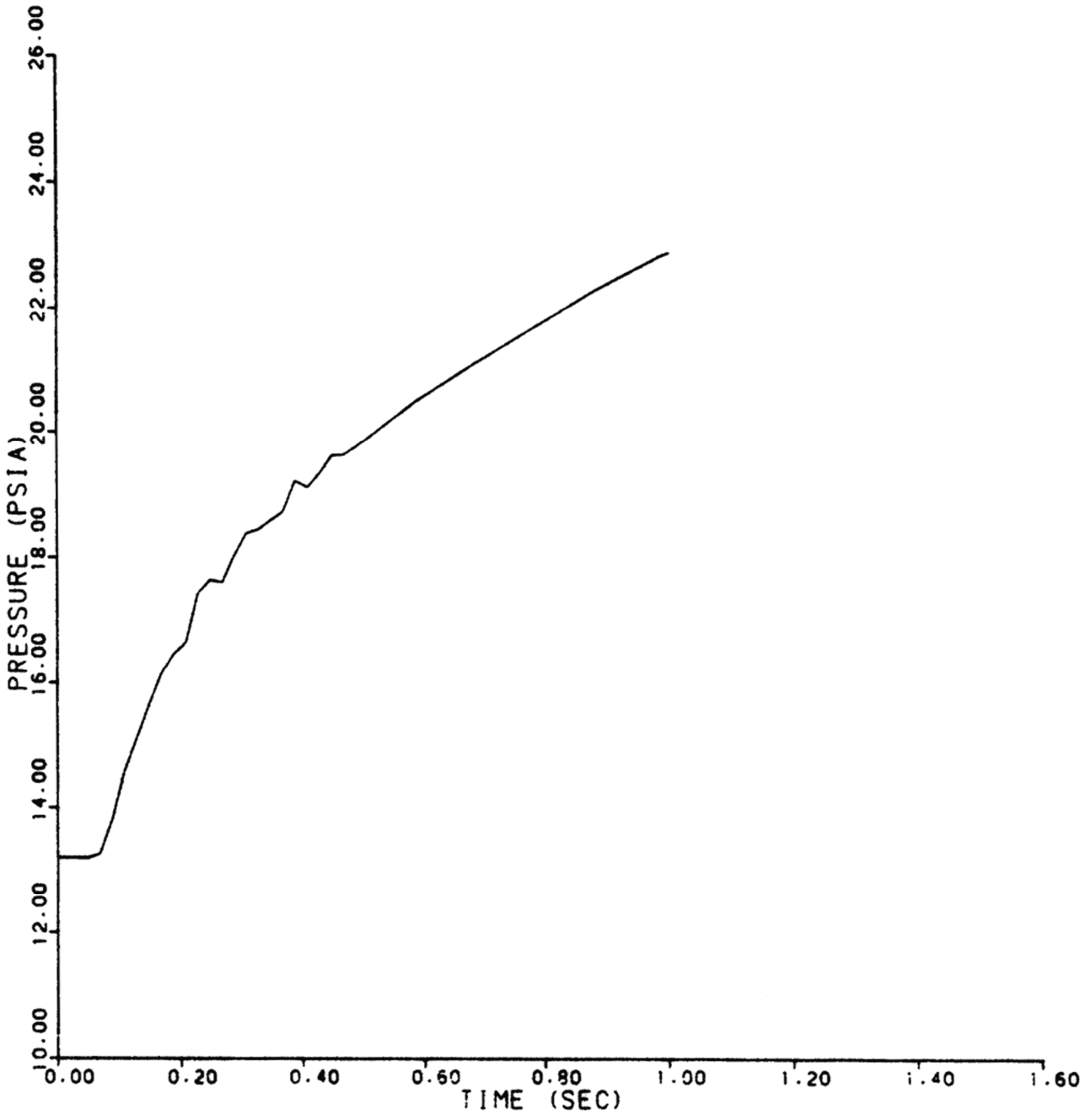
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E54
	FIGURE 6.2.1–15 (SHEET 54 OF 65)



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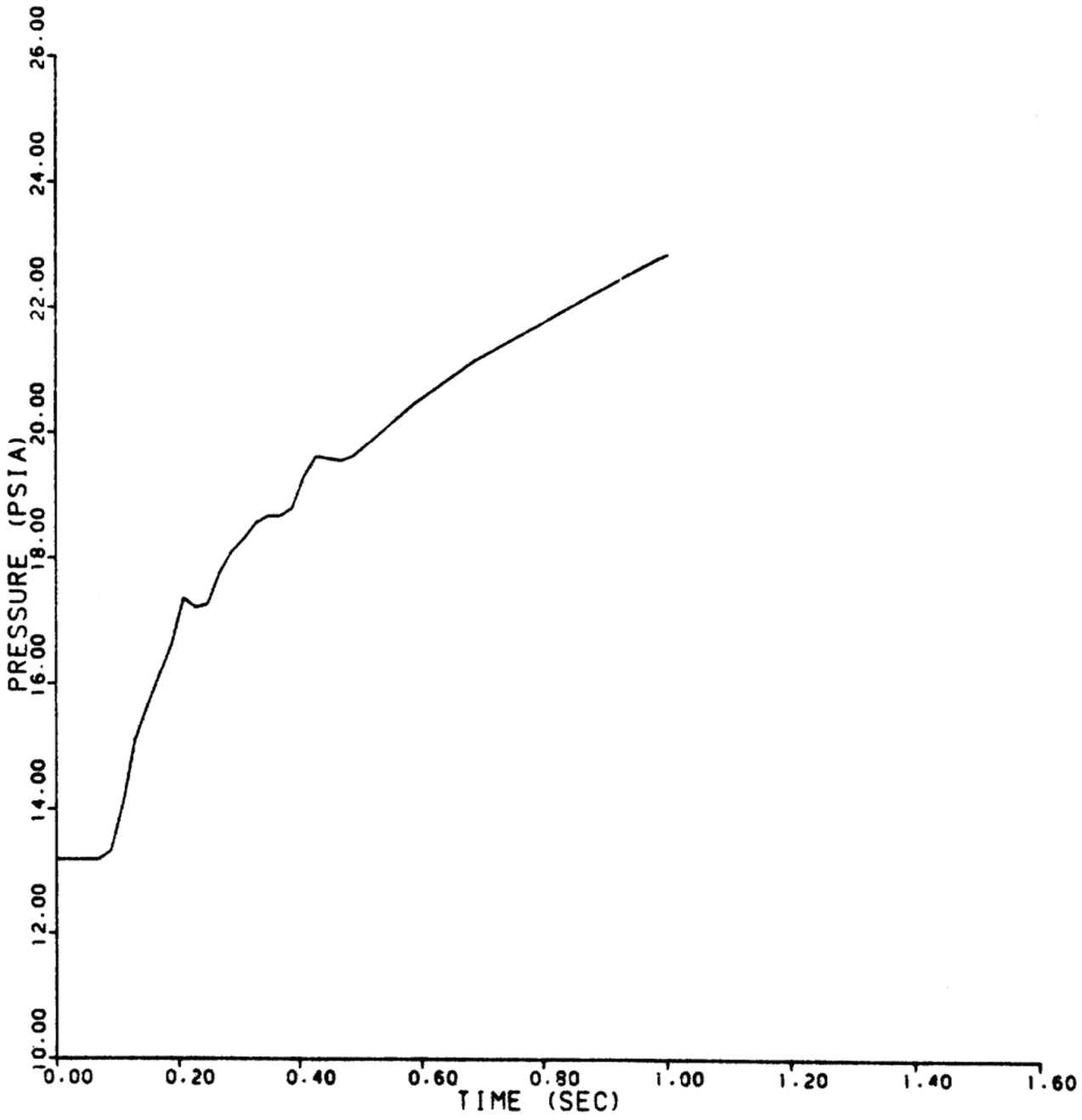
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E55
	FIGURE 6.2.1–15 (SHEET 55 OF 65)



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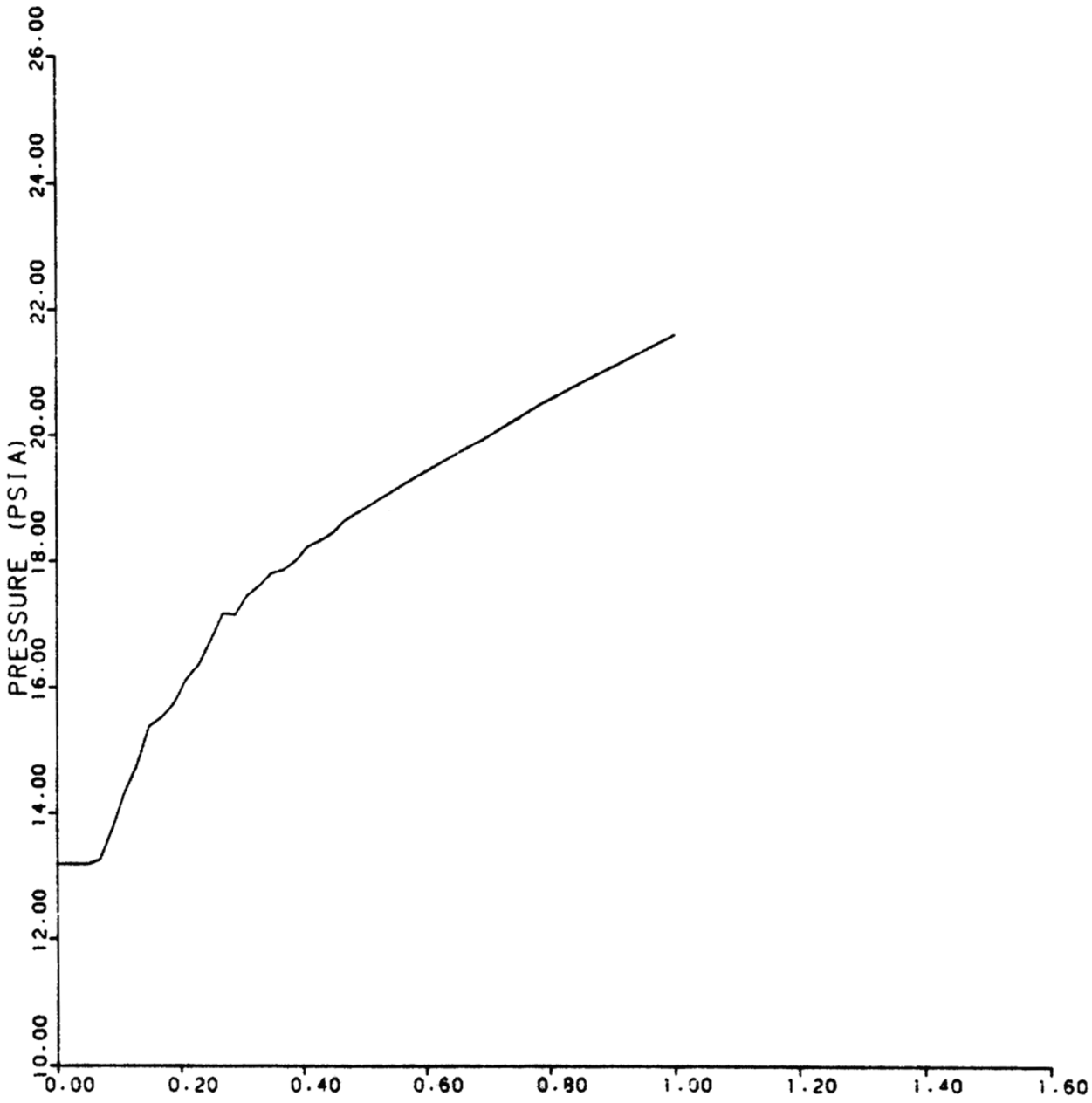
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E56
	FIGURE 6.2.1–15 (SHEET 56 OF 65)



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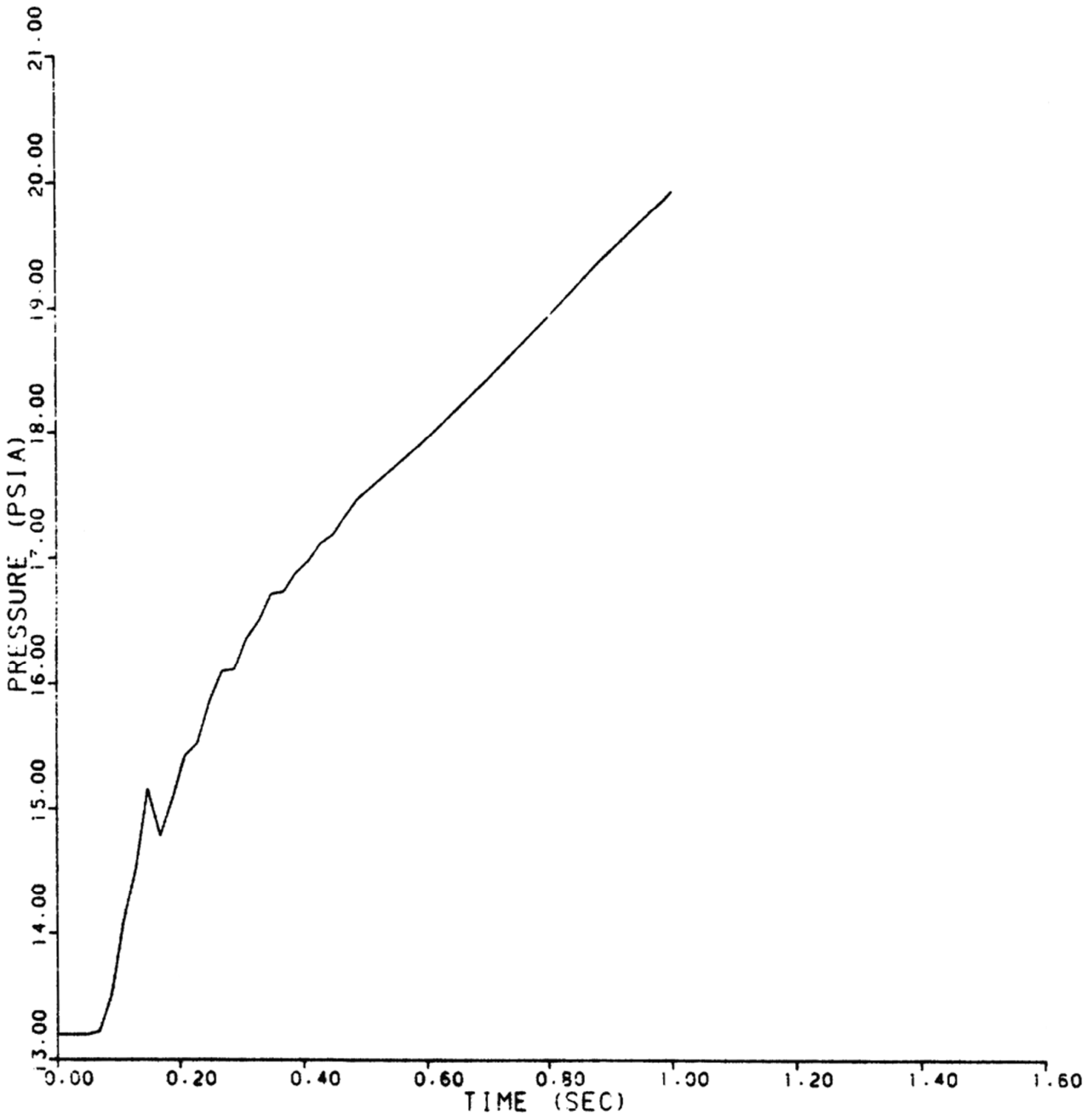
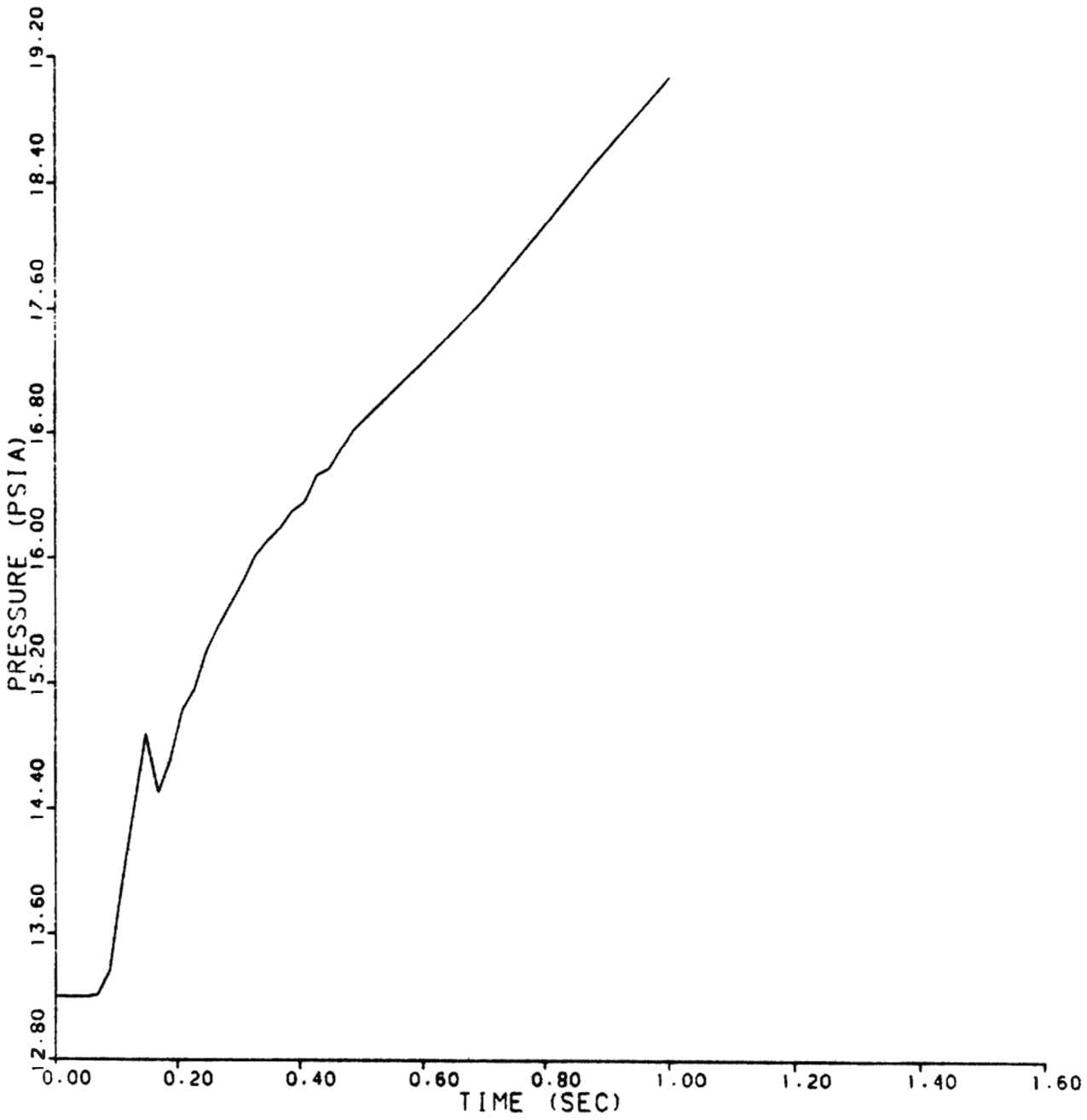
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E57
	FIGURE 6.2.1–15 (SHEET 57 OF 65)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E58
	FIGURE 6.2.1–15 (SHEET 58 OF 65)



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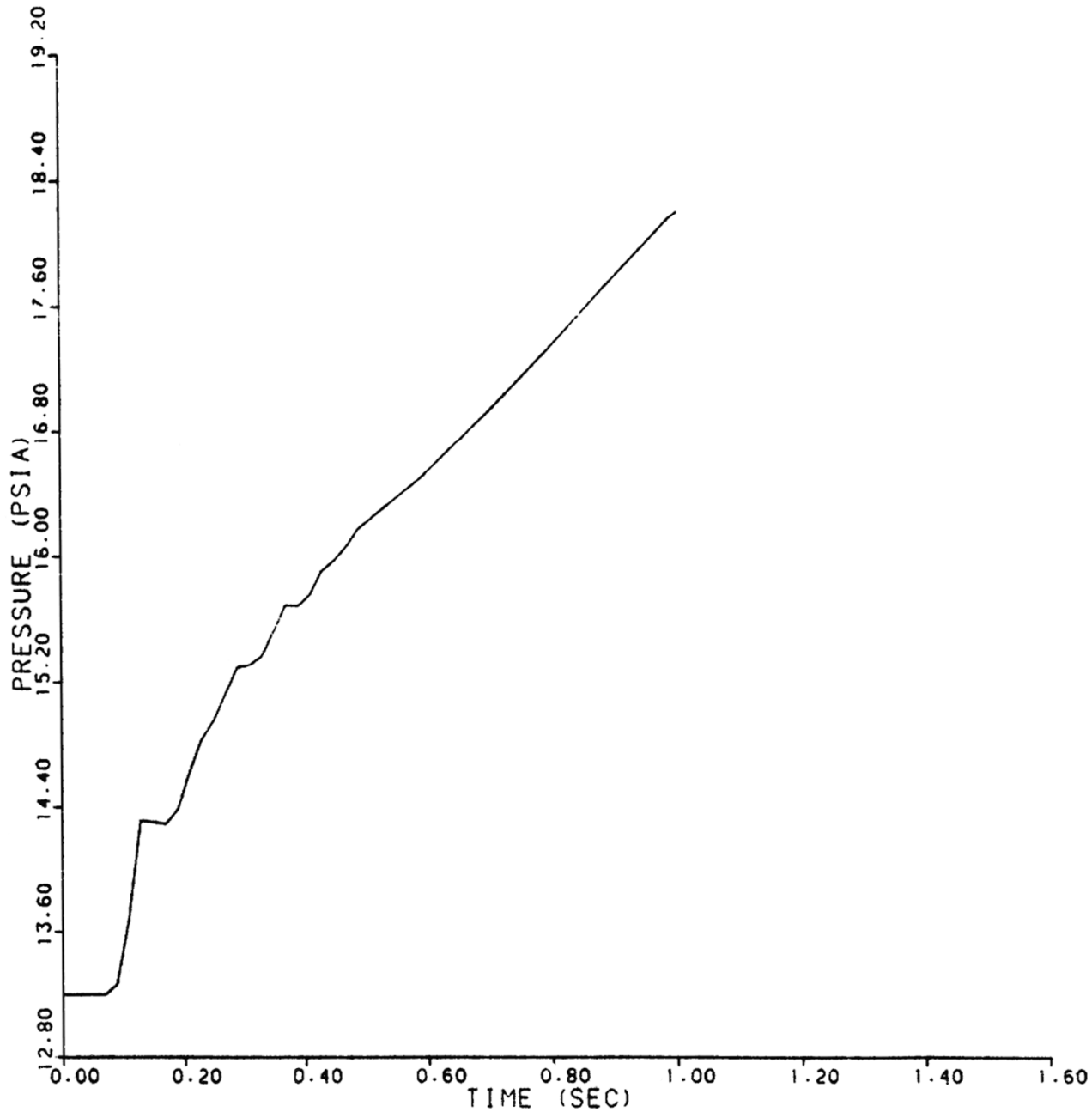
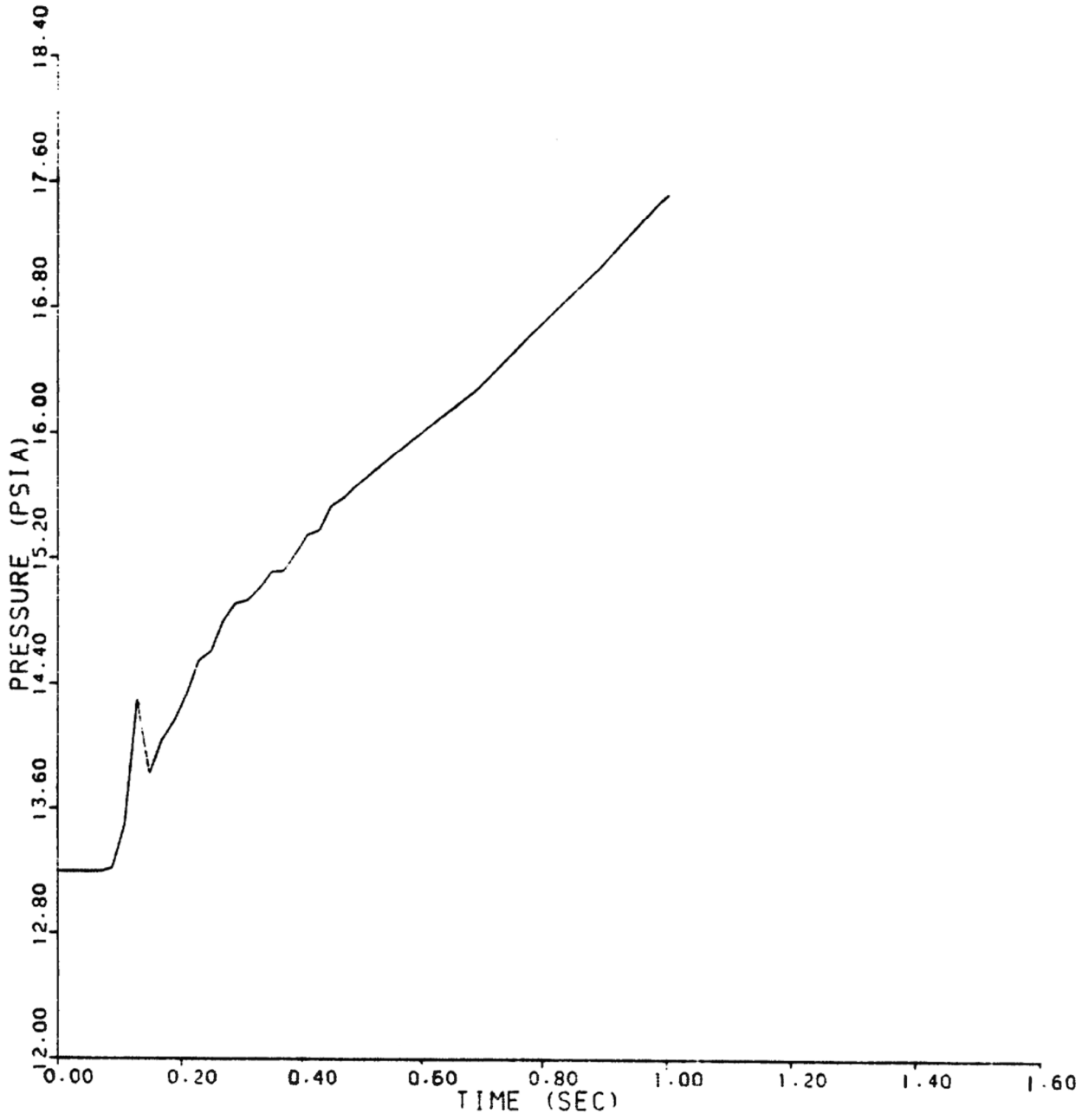
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E59
	FIGURE 6.2.1–15 (SHEET 59 OF 65)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E60 FIGURE 6.2.1–15 (SHEET 60 OF 65)



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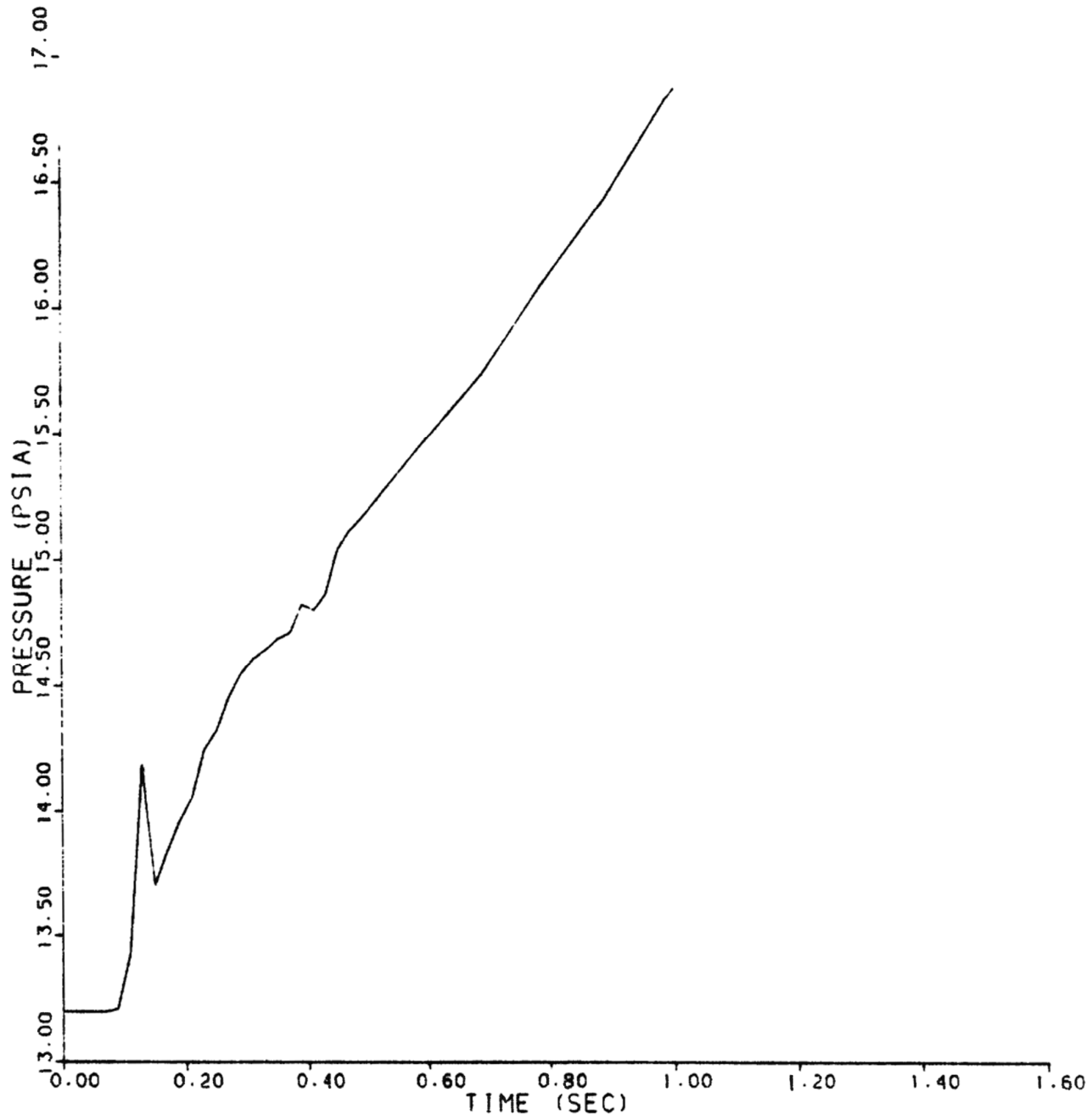
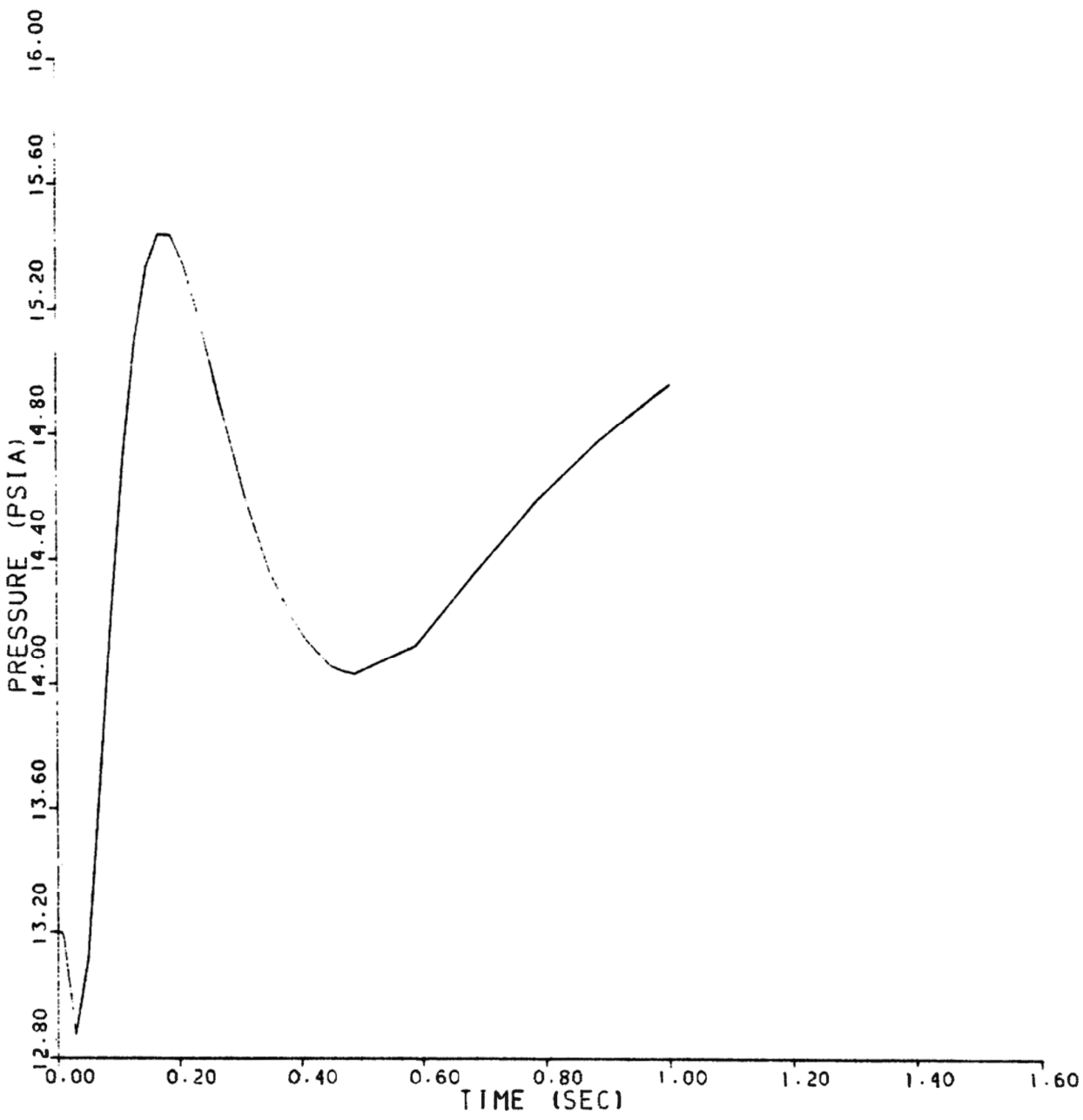
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E61
	FIGURE 6.2.1–15 (SHEET 61 OF 65)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E62
	FIGURE 6.2.1–15 (SHEET 62 OF 65)



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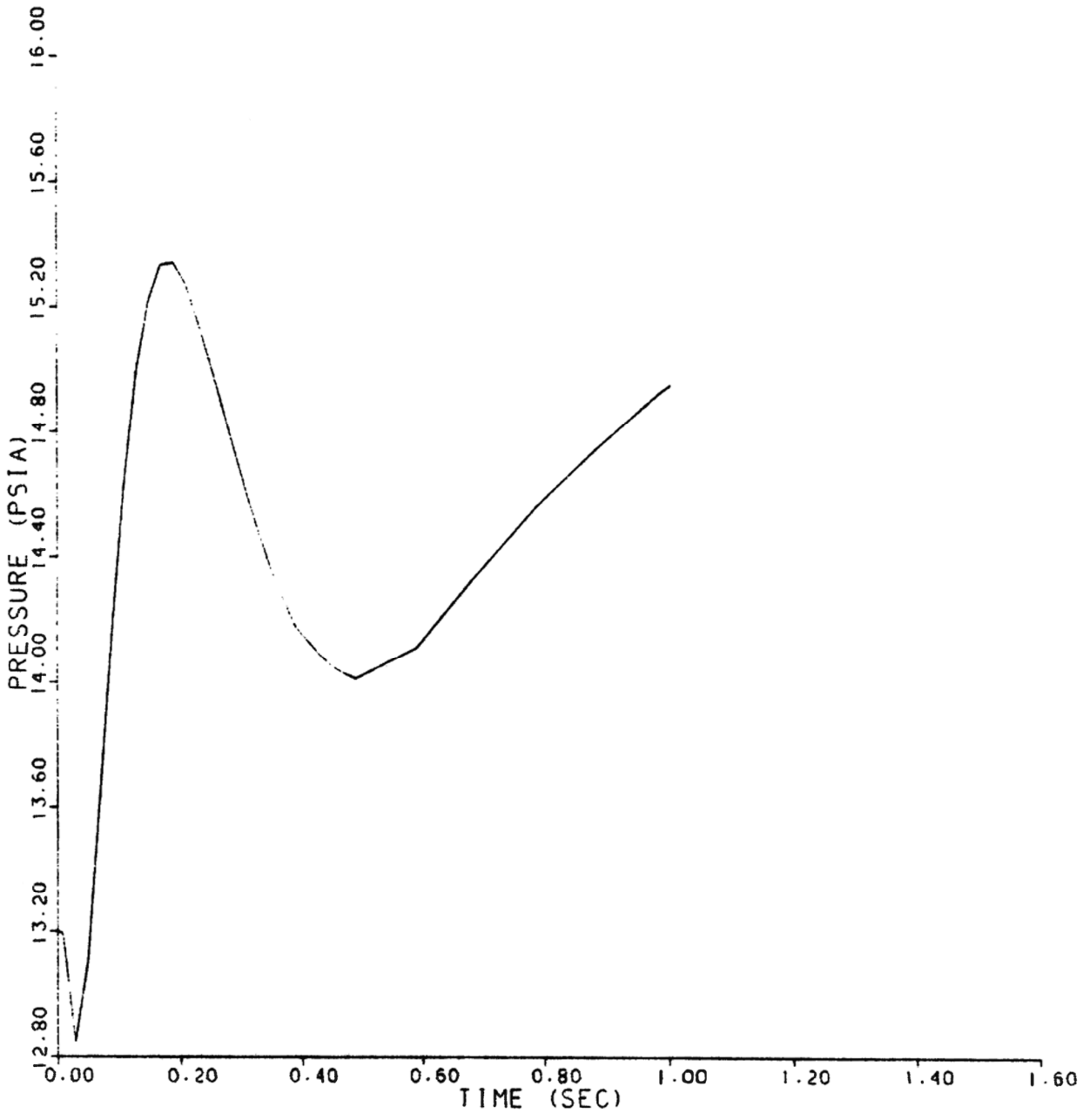
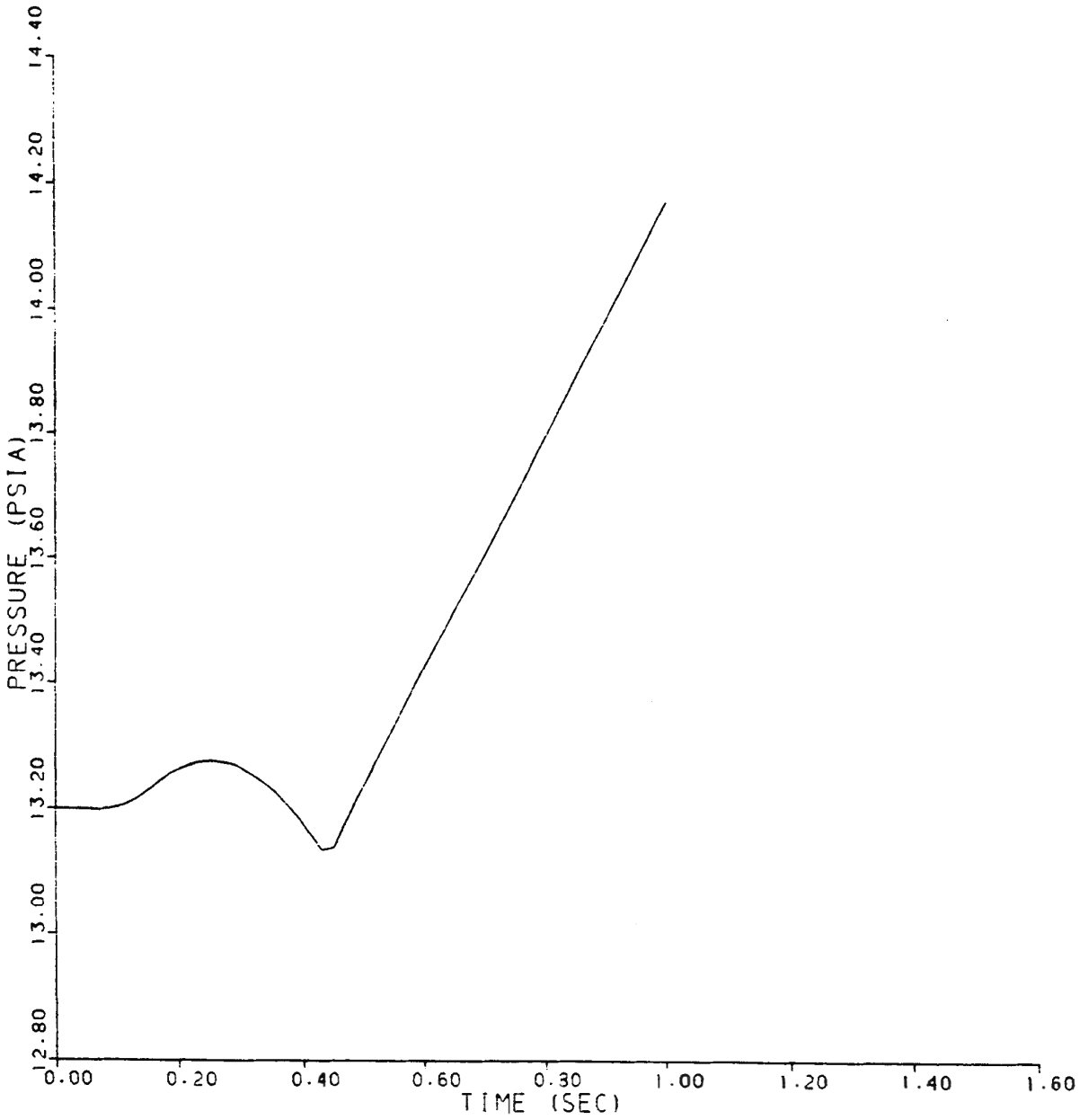
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E63
	FIGURE 6.2.1–15 (SHEET 63 OF 65)



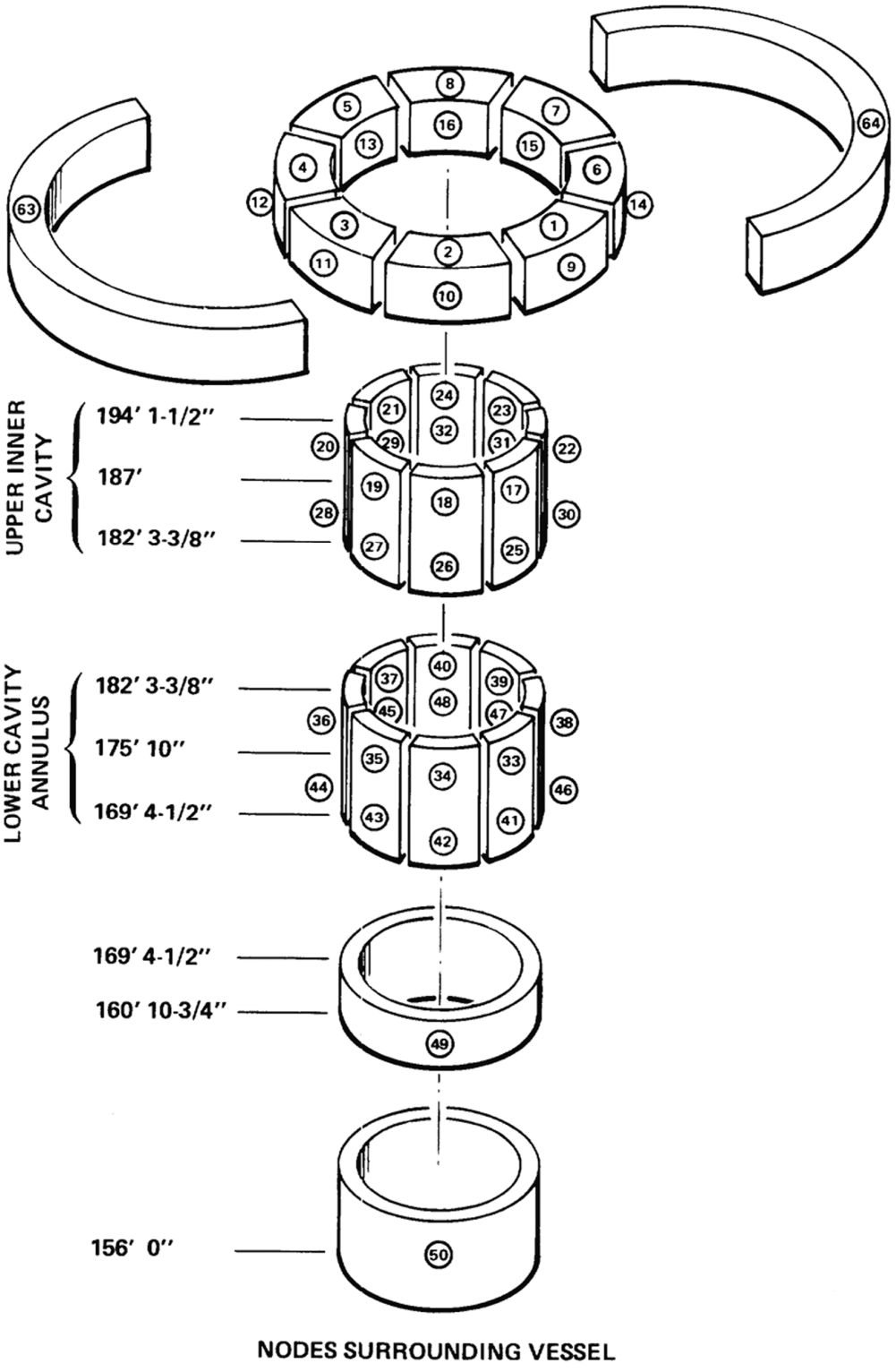
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E64
	FIGURE 6.2.1–15 (SHEET 64 OF 65)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR CAVITY PRESSURE RESPONSE – NODE E65
	FIGURE 6.2.1–15 (SHEET 65 OF 65)



NODES SURROUNDING VESSEL

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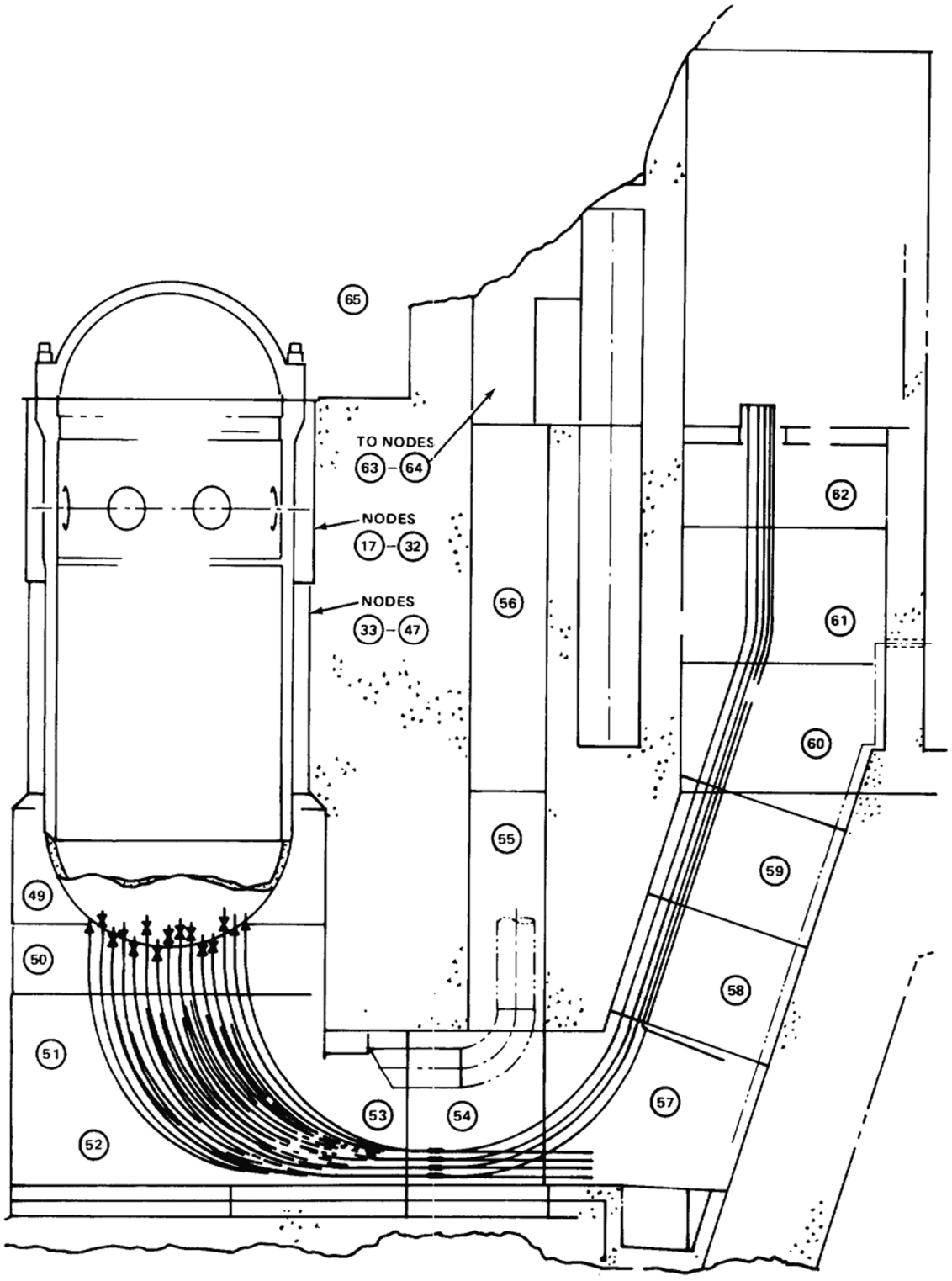
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY MODEL
	FIGURE 6.2.1–16 (SHEET 1 OF 3)



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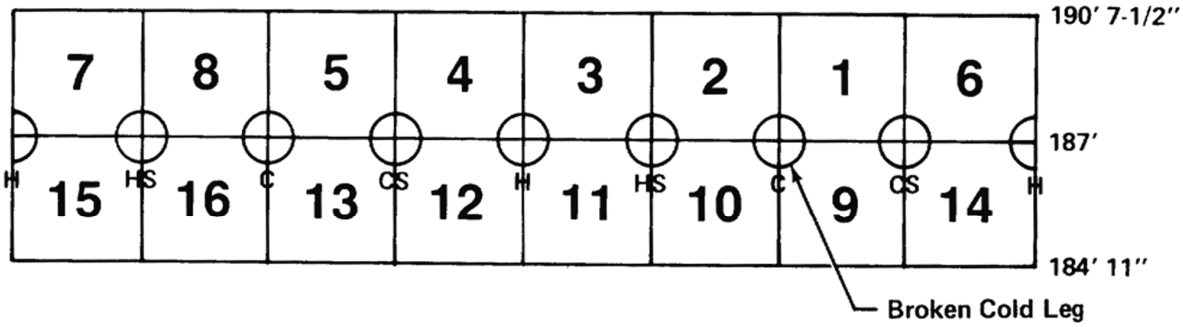
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

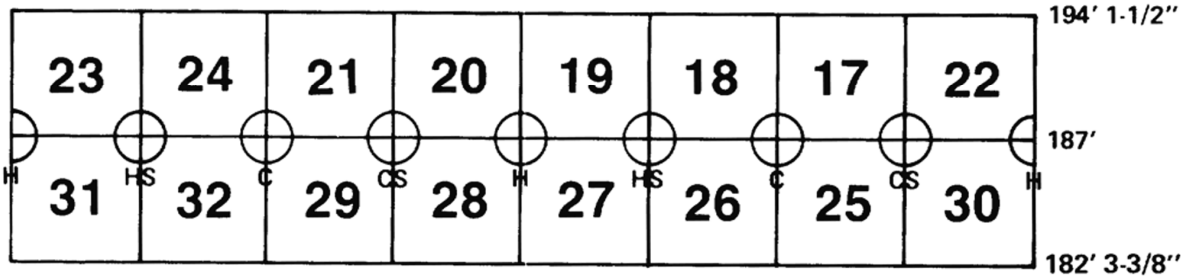
REACTOR CAVITY MODEL

FIGURE 6.2.1–16 (SHEET 2 OF 3)

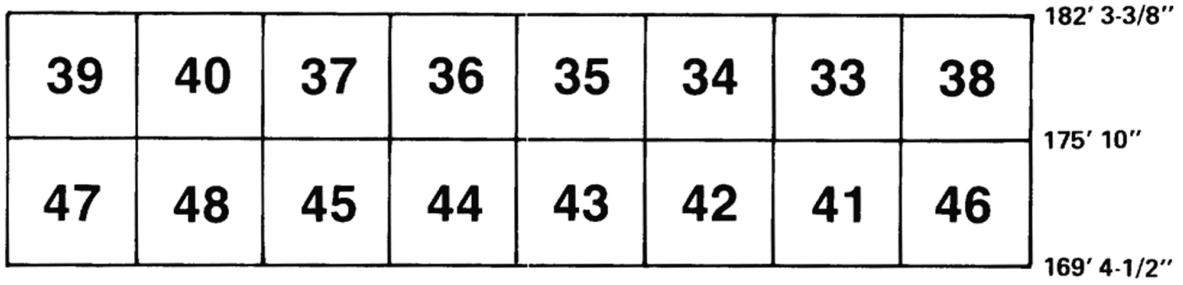
(I) INSPECTION ANNULUS (360°)



(II) UPPER INNER CAVITY



(III) LOWER CAVITY ANNULUS



* C = COLD LEG
H = HOT LEG
CS = COLD LEG WITH SUPPORT
HS = HOT LEG WITH SUPPORT

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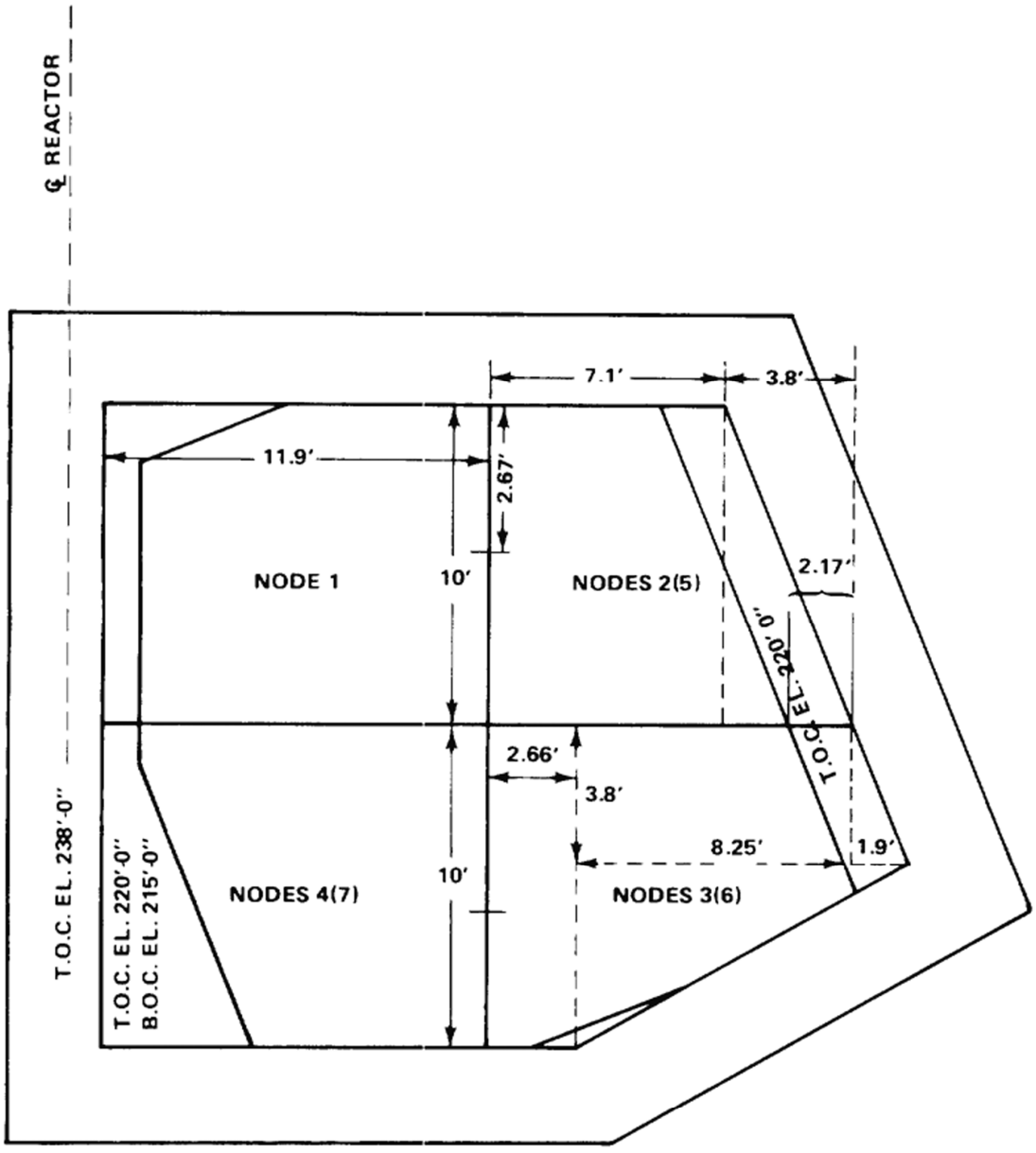
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR CAVITY MODEL
	FIGURE 6.2.1–16 (SHEET 3 OF 3)

STEAM GENERATOR DOGHOUSE (TOP VIEW)
NODALIZATION



NODE NUMBERS IN PARENTHESES ARE THE NODES BELOW ANOTHER NODE.

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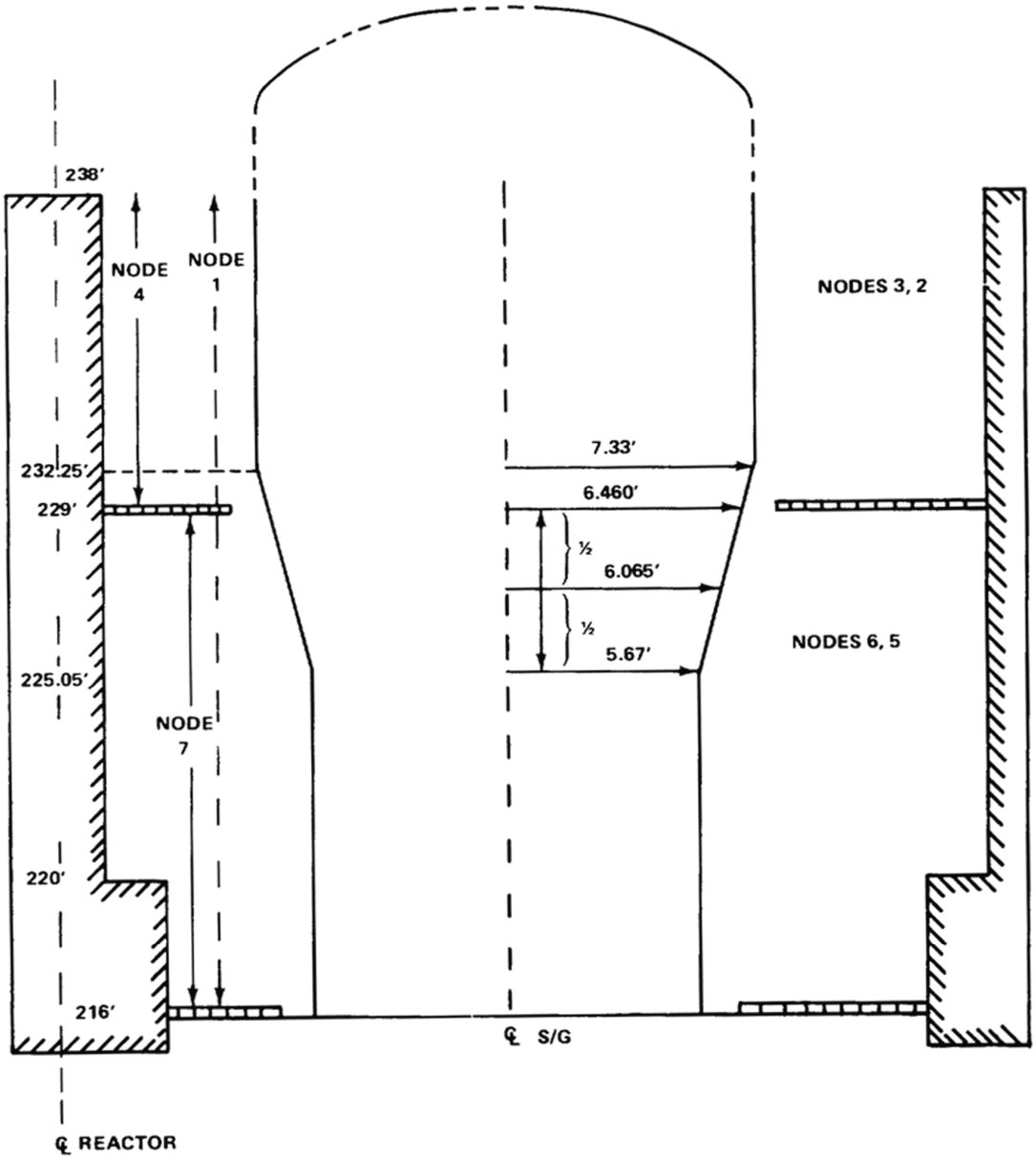
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK ABOVE el. 220 ft.) FIGURE 6.2.1–16A (SHEET 1 OF 3)



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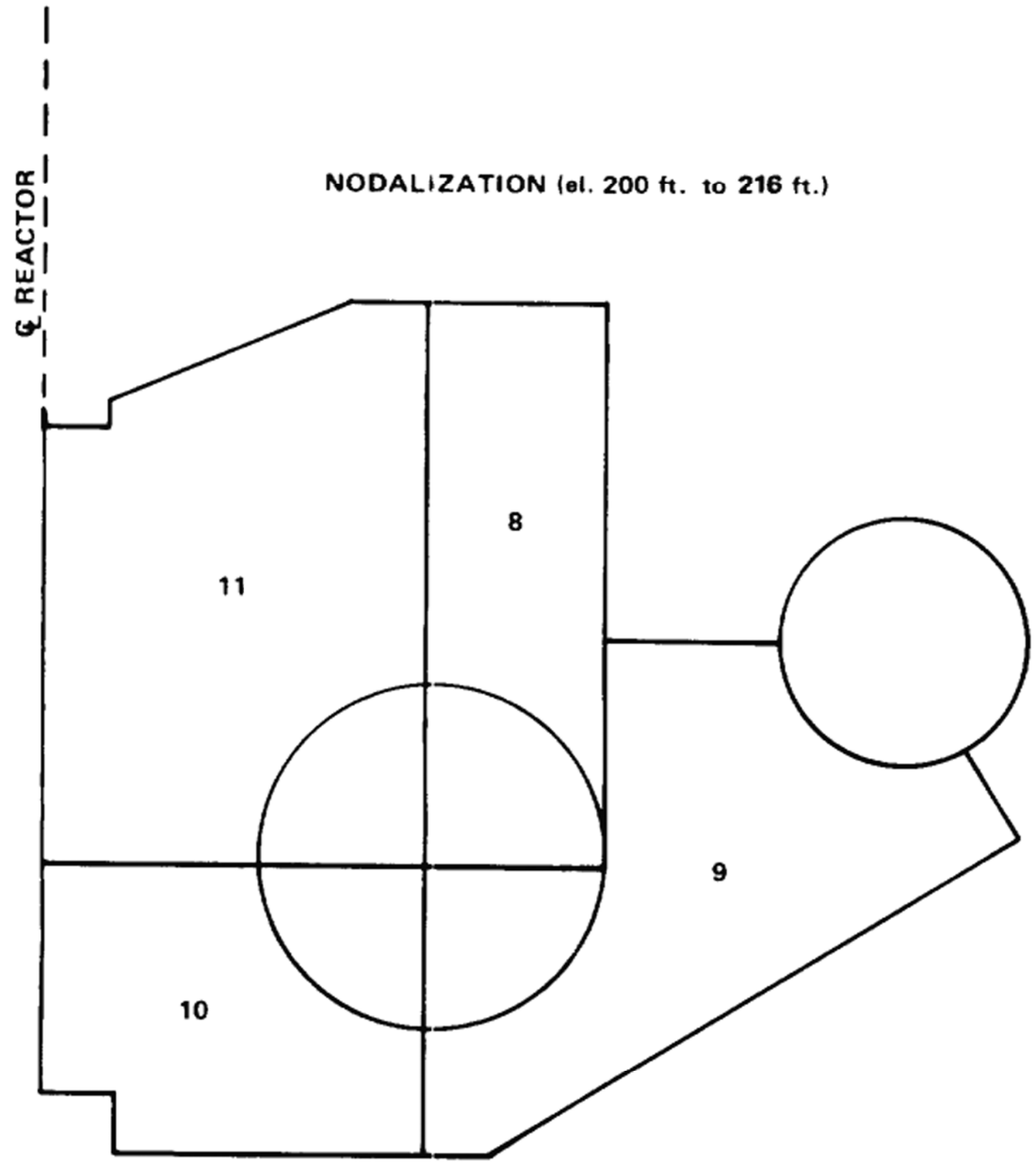
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK ABOVE el. 220 ft.)
	FIGURE 6.2.1–16A (SHEET 2 OF 3)



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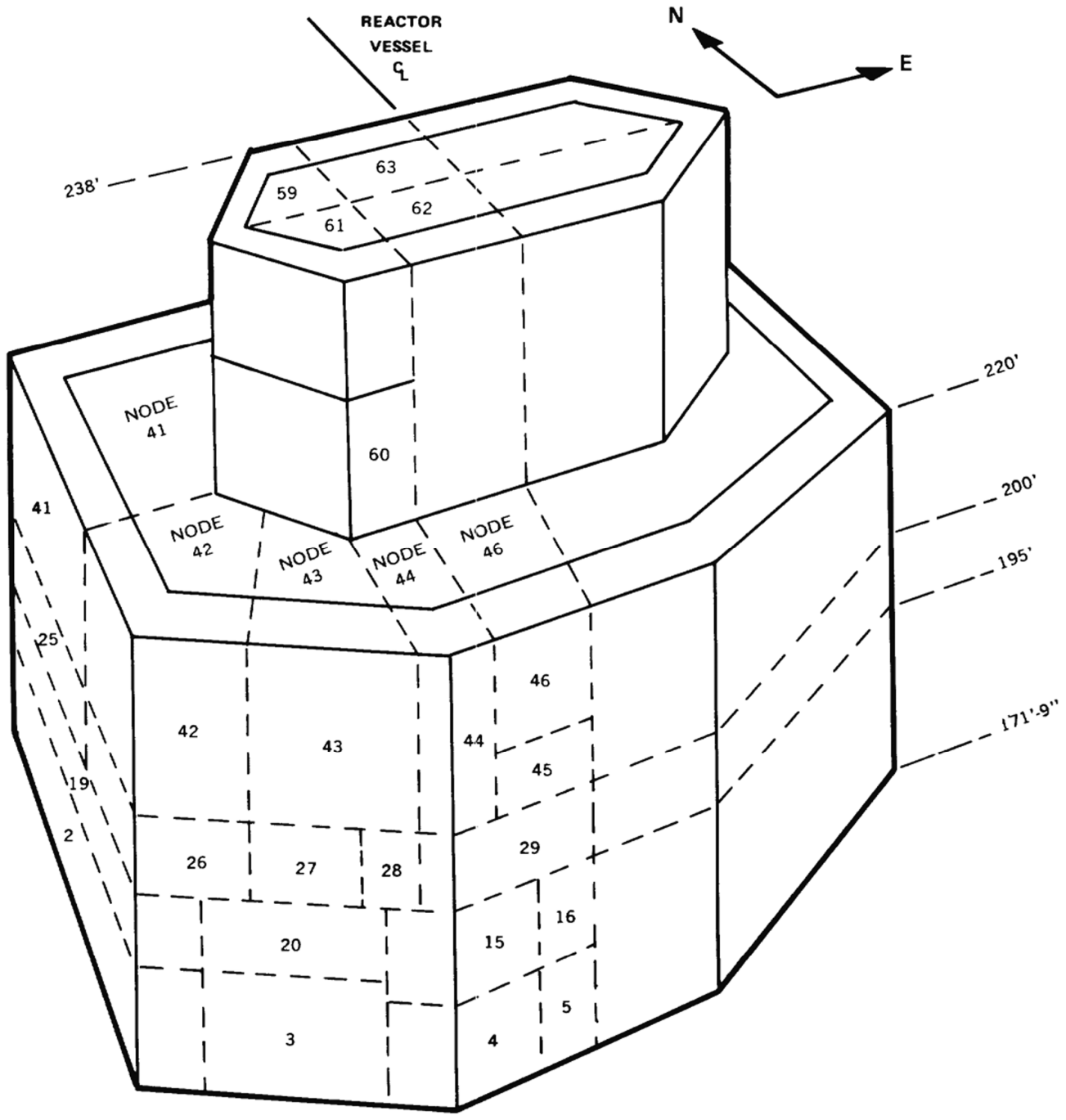
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT MODEL
(PIPE BREAK ABOVE el. 220 ft.)

FIGURE 6.2.1–16A (SHEET 3 OF 3)



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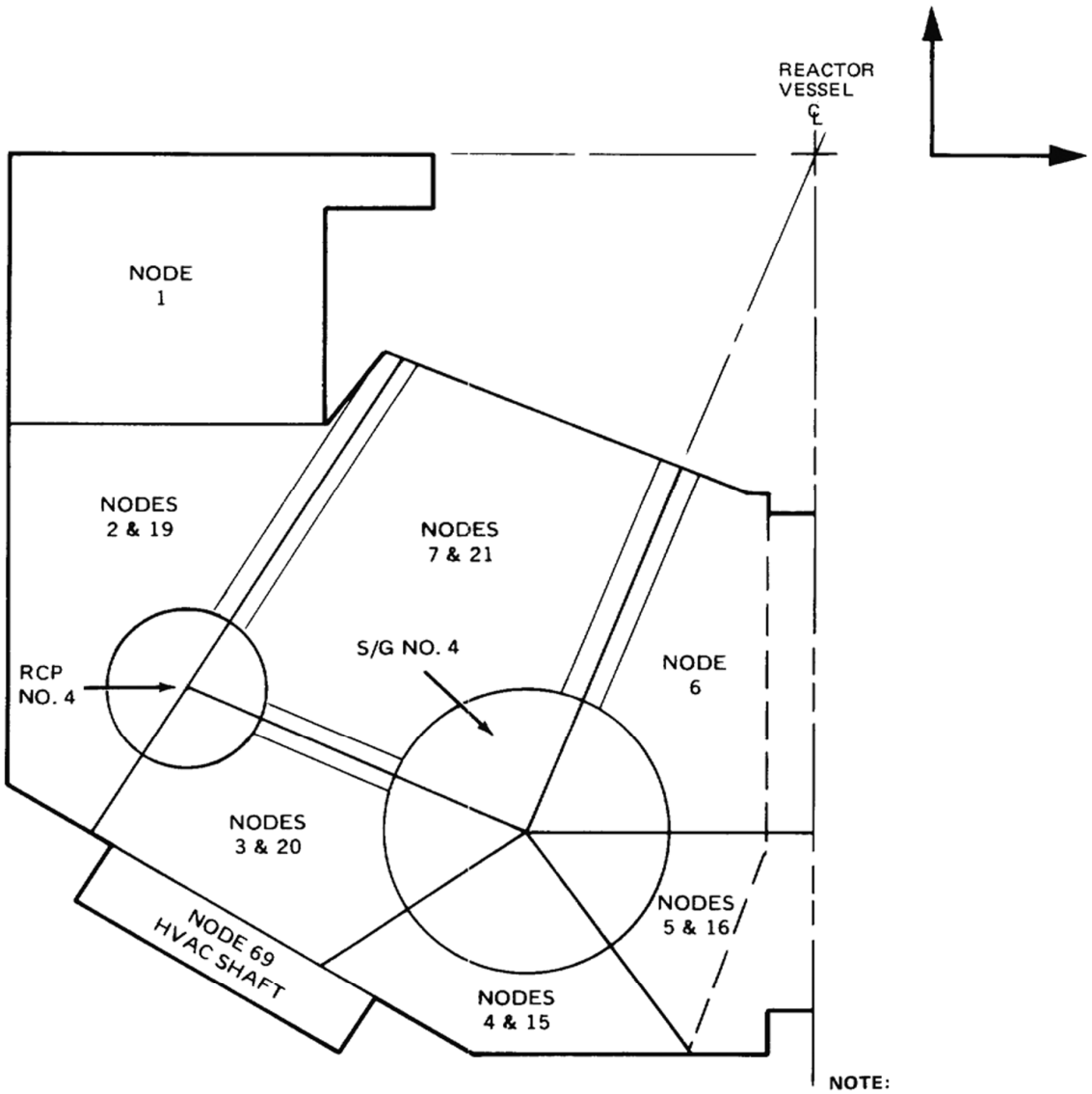
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT MODEL
(PIPE BREAK BELOW el.220 ft.)

FIGURE 6.2.1–17 (SHEET 1 OF 6)



NODES	BOTTOM ELEVATION	TOP ELEVATION
1	171' 9"	178' 3 3/4"
4, 5	171' 9"	183' 0"
2, 3, 7	171' 9"	187' 0"
15, 16	183' 0"	195' 0"
19, 20, 21	187' 0"	195' 0"
6	171' 9"	195' 0"

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SOUTHERN COMPANY
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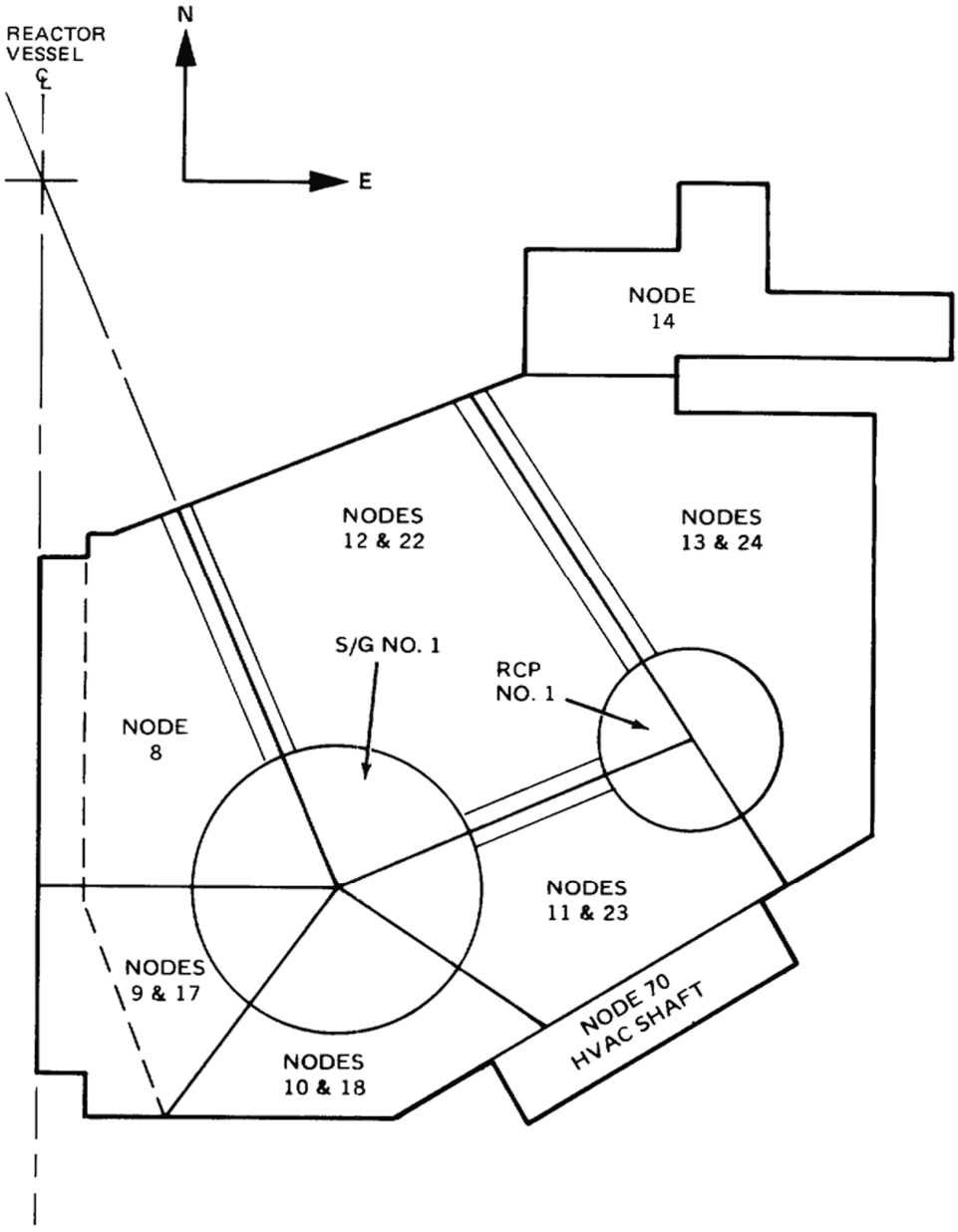
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT MODEL
(PIPE BREAK BELOW el. 220 ft.) LEVEL 1
(LEFT HALF TOP VIEW)

FIGURE 6.2.1–17 (SHEET 2 OF 6)



NODES (SYMMETRIC)	BOTTOM ELEVATION	TOP ELEVATION
9 (5), 10 (4)	171' 9"	183' 0"
11 (3), 13 (2), 12 (7)	171' 9"	187' 0"
17 (16), 18 (15)	183' 0"	195' 0"
22 (21), 23 (20), 24 (19)	187' 0"	195' 0"
14 (1 APPROX.)	171' 9"	192' 0"
8 (6)	171' 9"	195' 0"

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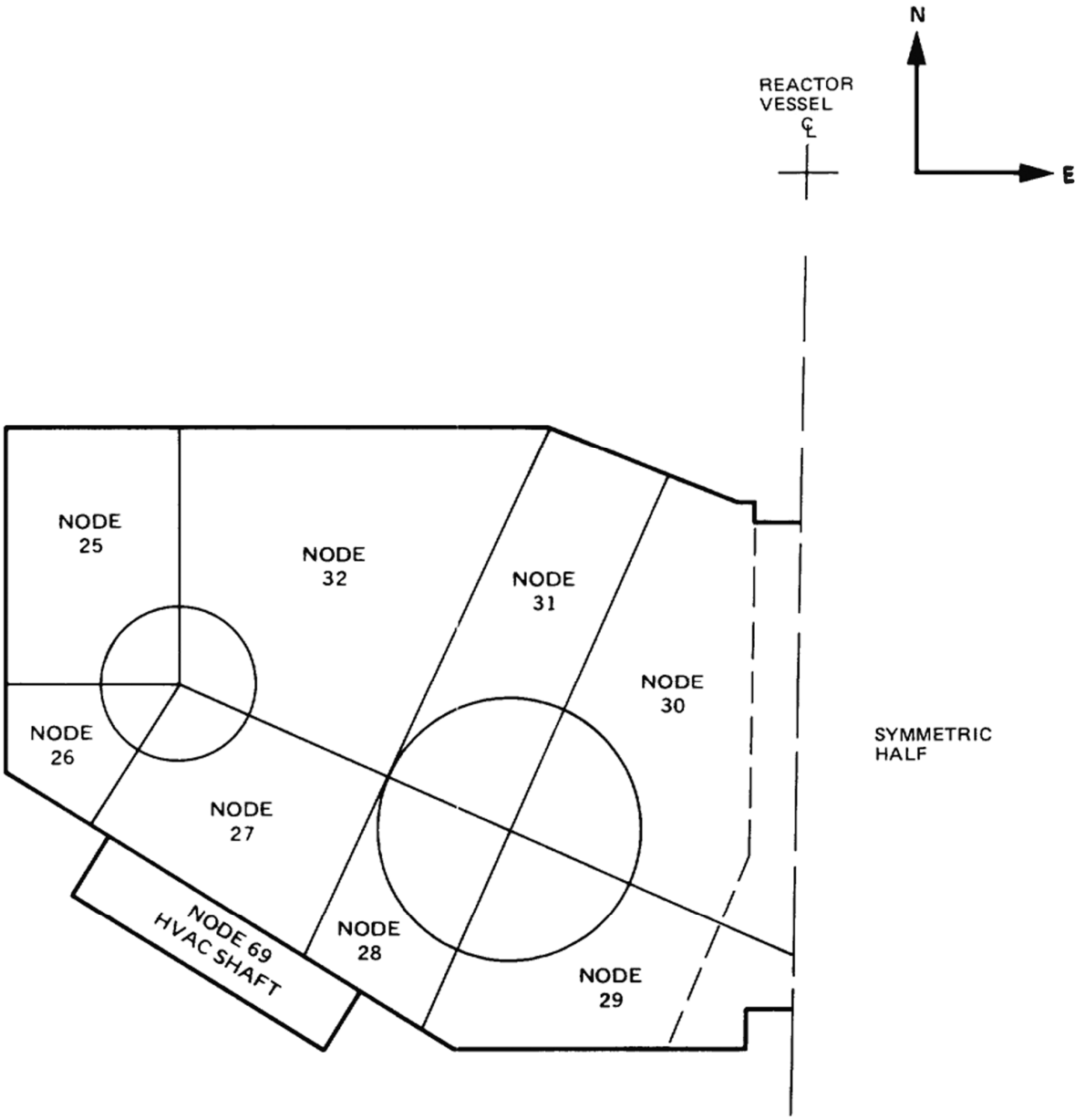
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT MODEL
(PIPE BREAK BELOW el. 220 ft.) LEVEL 1
(RIGHT HALF TOP VIEW)

FIGURE 6.2.1–17 (SHEET 3 of 6)



SYMMETRIC NODES:

25 — 38	29 — 34
26 — 37	30 — 33
27 — 36	31 — 40
28 — 35	32 — 39

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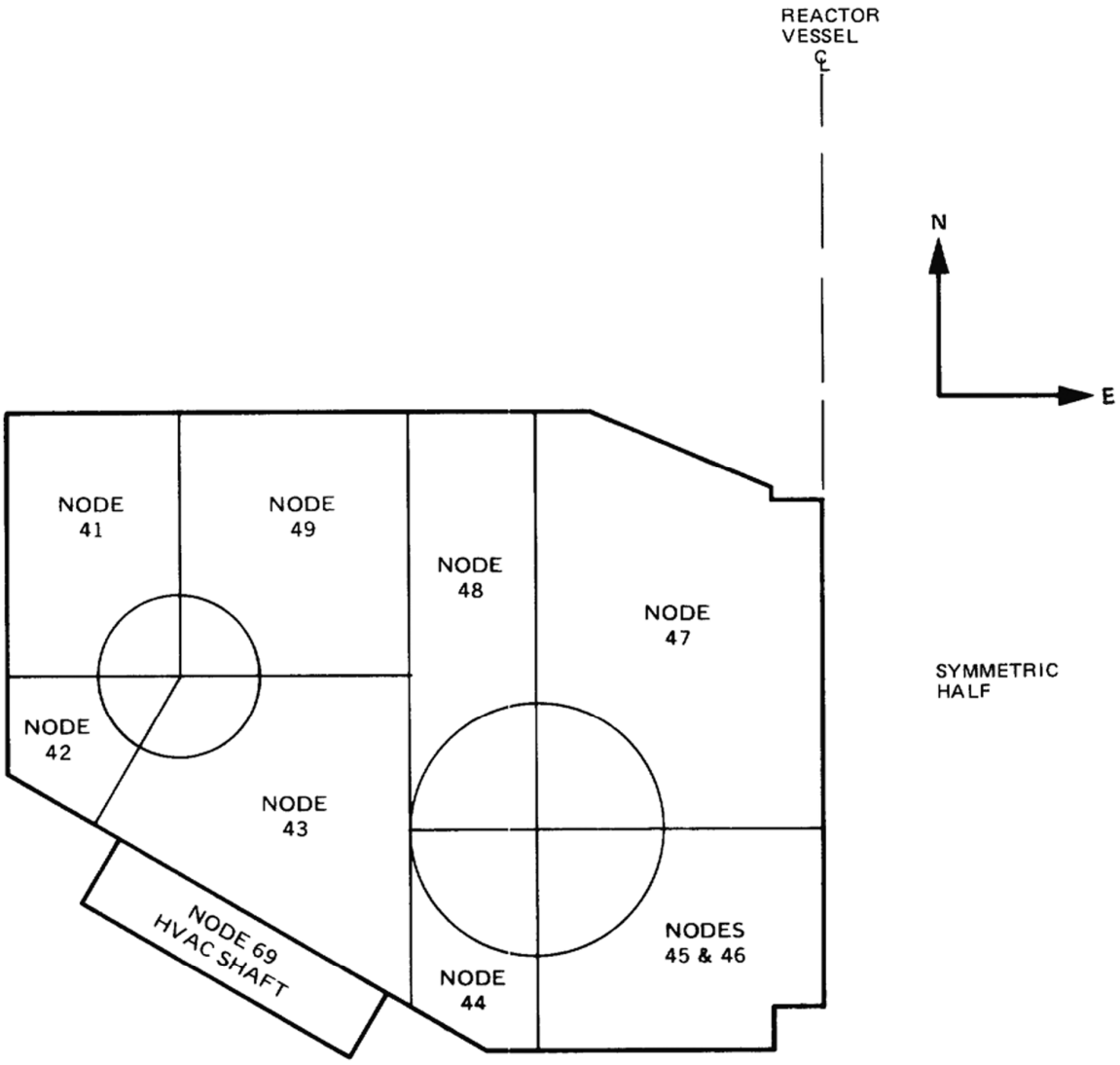
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT MODEL (PIPE BREAK BELOW el. 220445 ft.) LEVEL 2, EL.195 TO 200 ft. (TOP VIEW)

FIGURE 6.2.1–17 (SHEET 4 OF 6)



NOTE:

1. SKETCH HAS NO SCALE.
   200' TO 207' (NODE 45)
   207' TO 215' (NODE 46)
   200' TO 215' (NODES 44, 47, 48)
   200' TO 220' (NODES 41, 42, 43, 49)

SYMMETRIC NODES

41 — 56	45 — 51
42 — 55	46 — 52
43 — 54	47 — 50
44 — 53	48 — 58
	49 — 57

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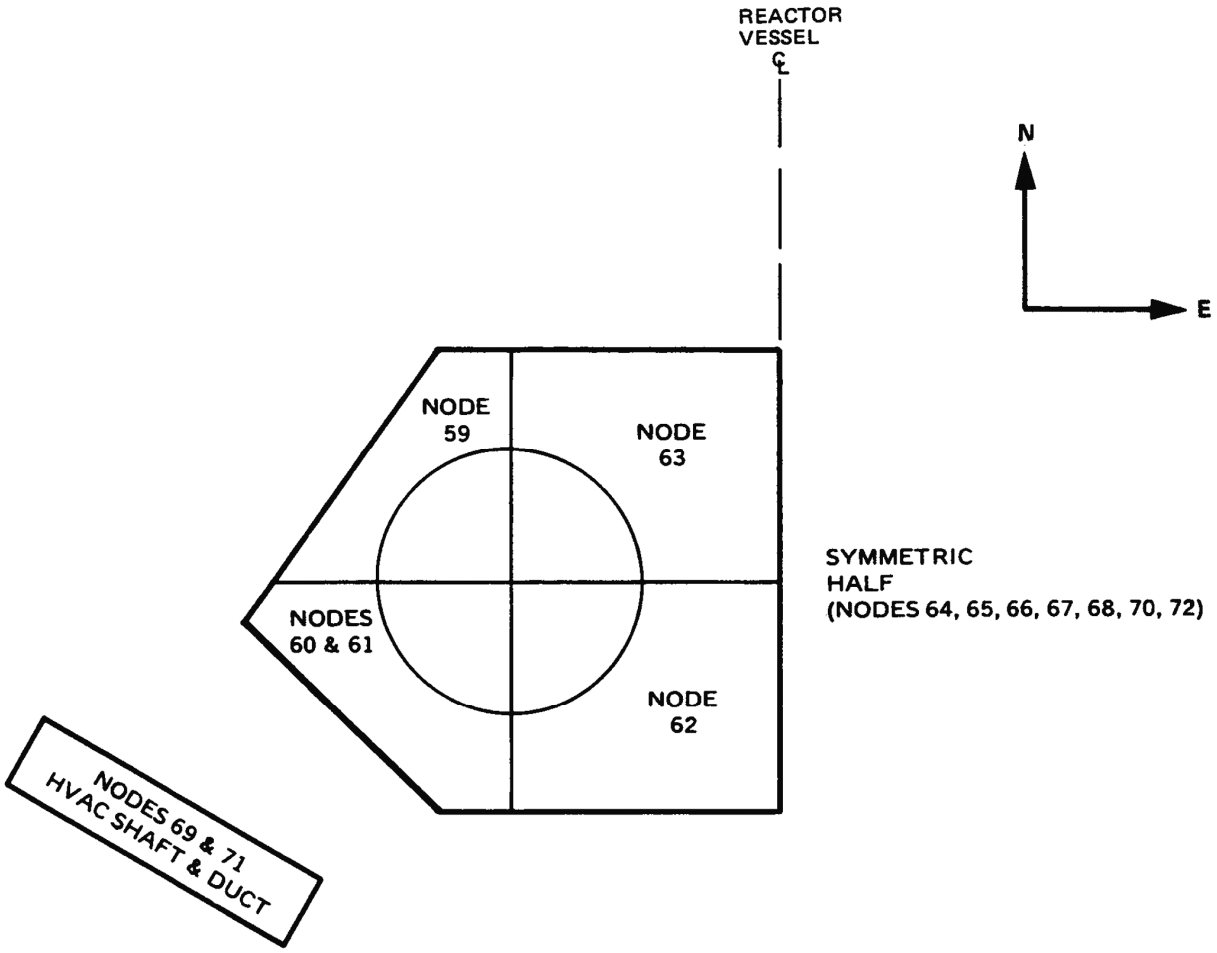
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT MODEL
(PIPE BREAK BELOW el. 220 ft.)
LEVEL 3 (TOP VIEW)

FIGURE 6.2.1–17 (SHEET 5 OF 6)



NOTES:

1. SKETCH HAS NO SCALE.

215' TO 229' (NODE 60)
229' TO 238' (NODE 61)
215' TO 238' (NODES 59, 62, 63)

SYMMETRIC NODES

59 — 68
60 — 66
61 — 67
62 — 65
63 — 64
69 — 70
71 — 72

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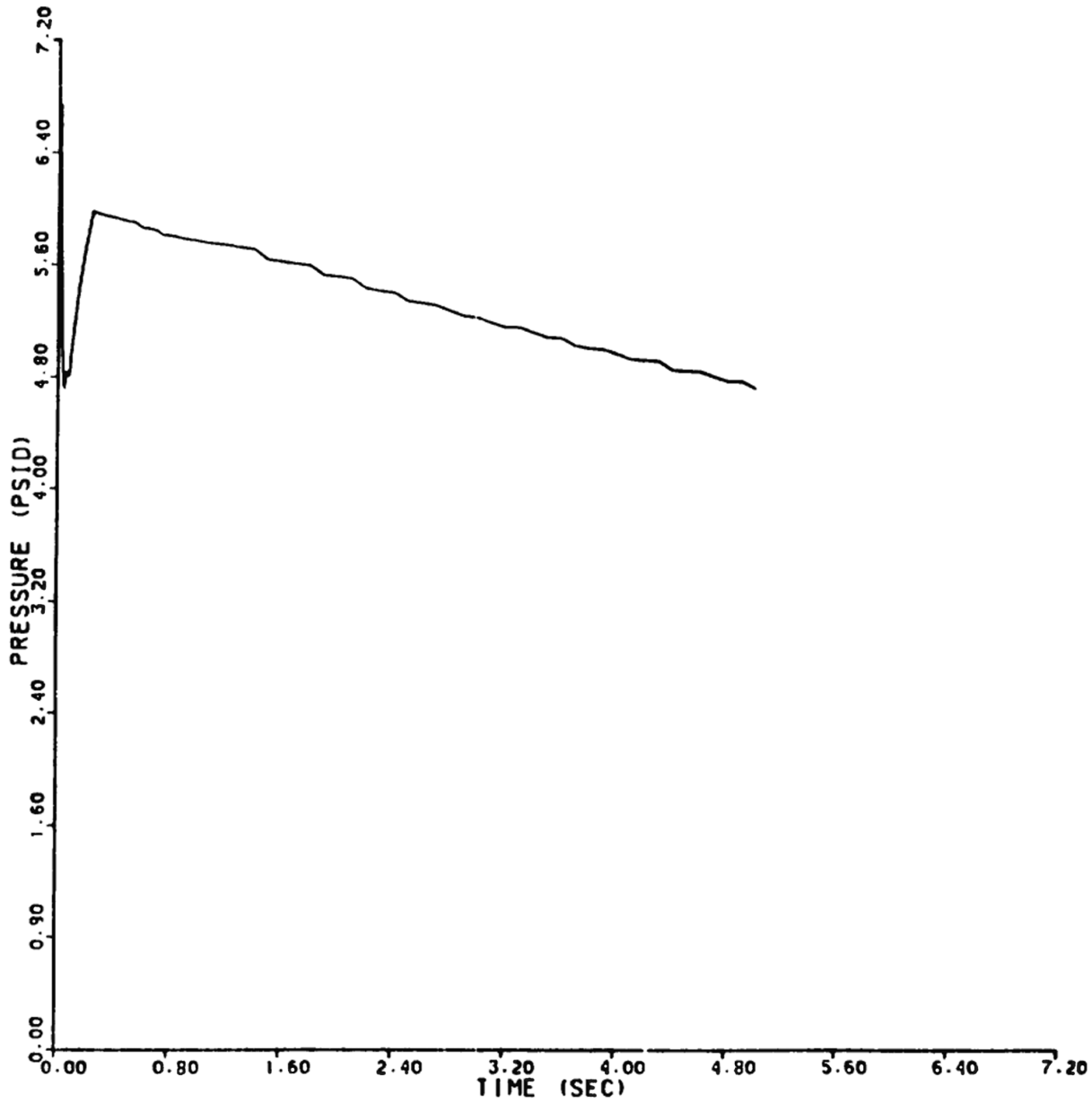
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT MODEL
(PIPE BREAK BELOW el. 220 ft.)
LEVEL 4 (TOP VIEW)

FIGURE 6.2.1–17 (SHEET 6 OF 6)



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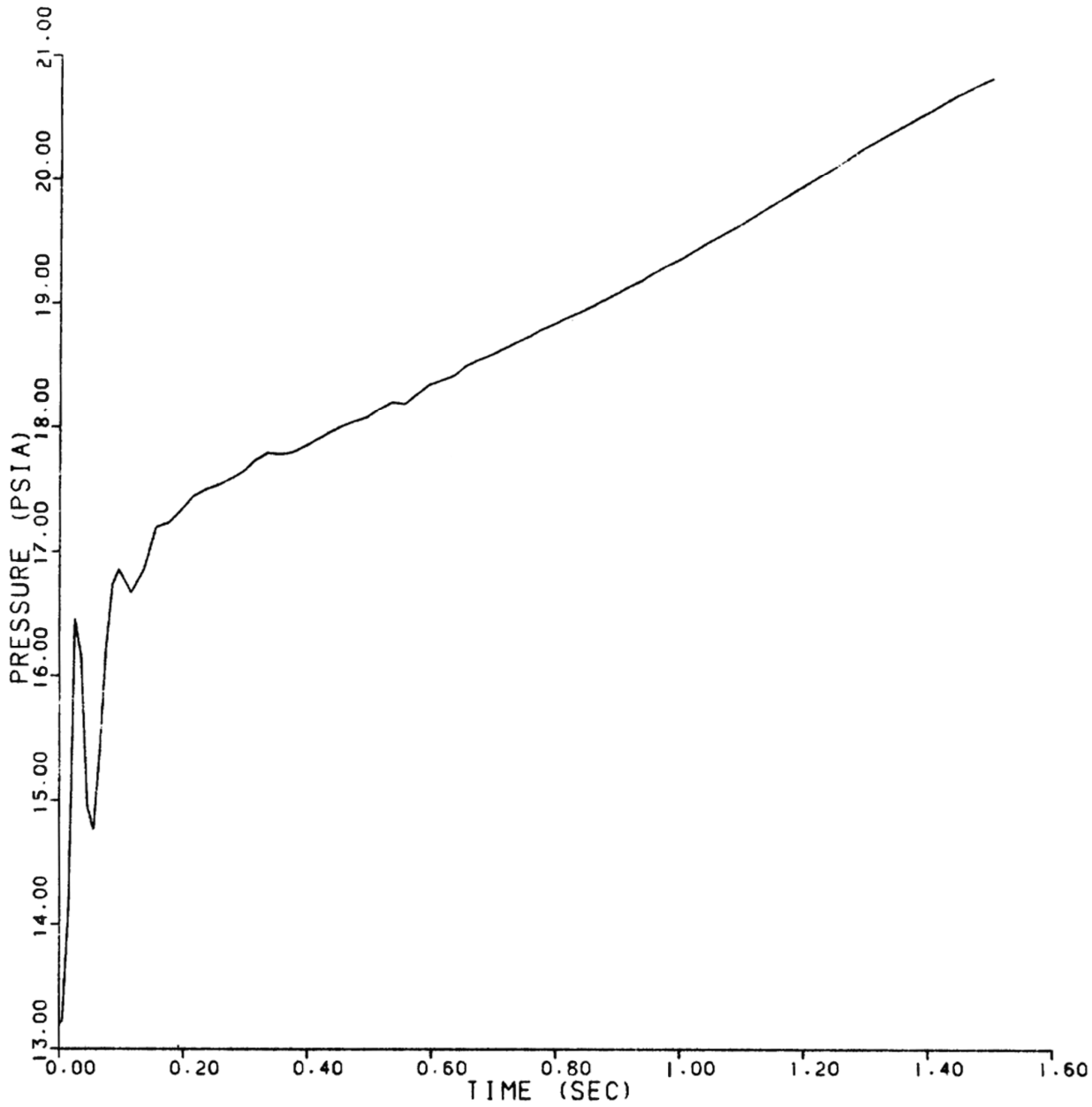
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENTS DIFFERENTIAL PRESSURE RESPONSE ACROSS DOGHOUSE WALL (144.5 in^2 MAIN FEEDWATER LINE BREAK)

FIGURE 6.2.1–17A



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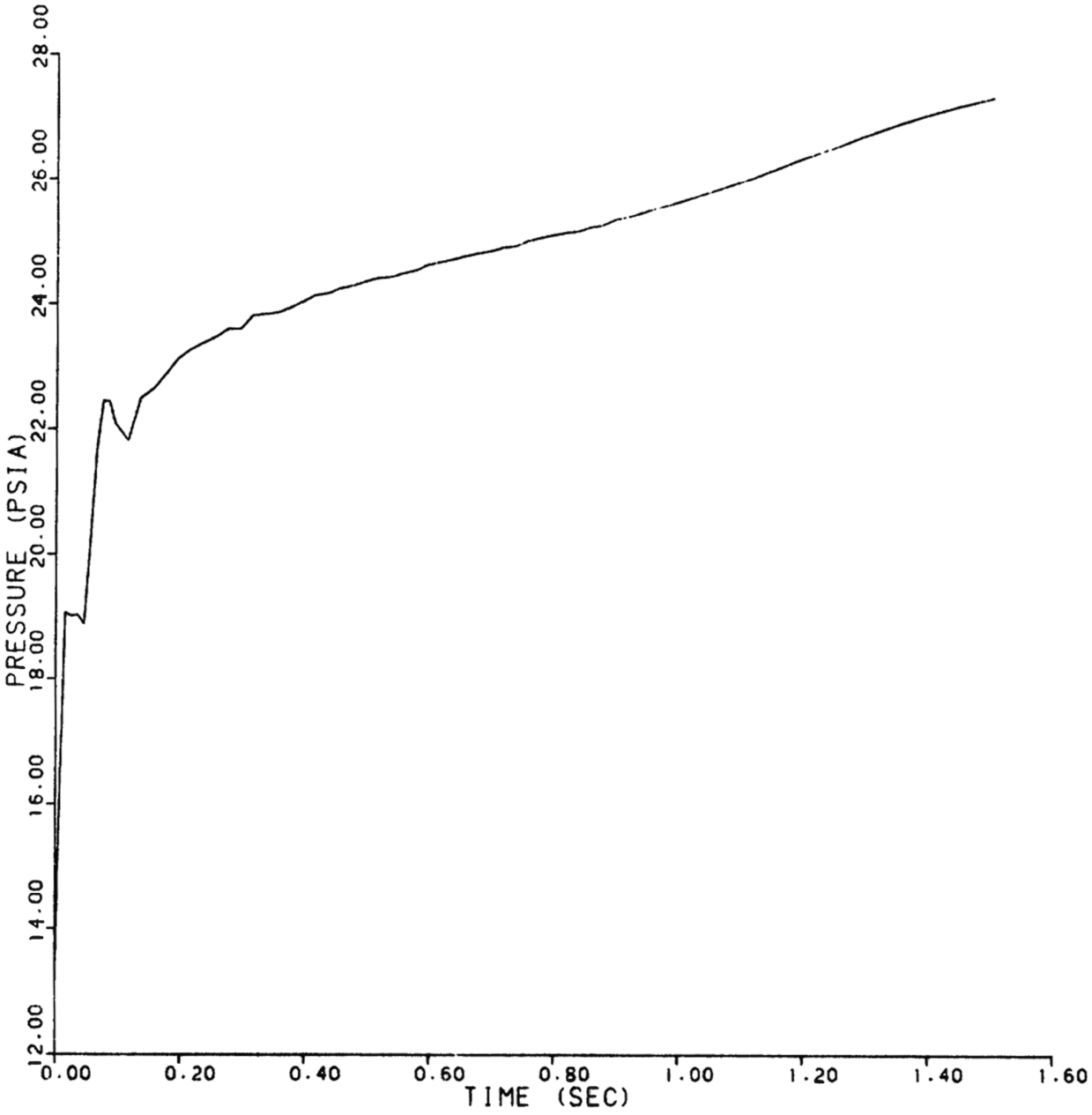
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E1
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 1 OF 74)



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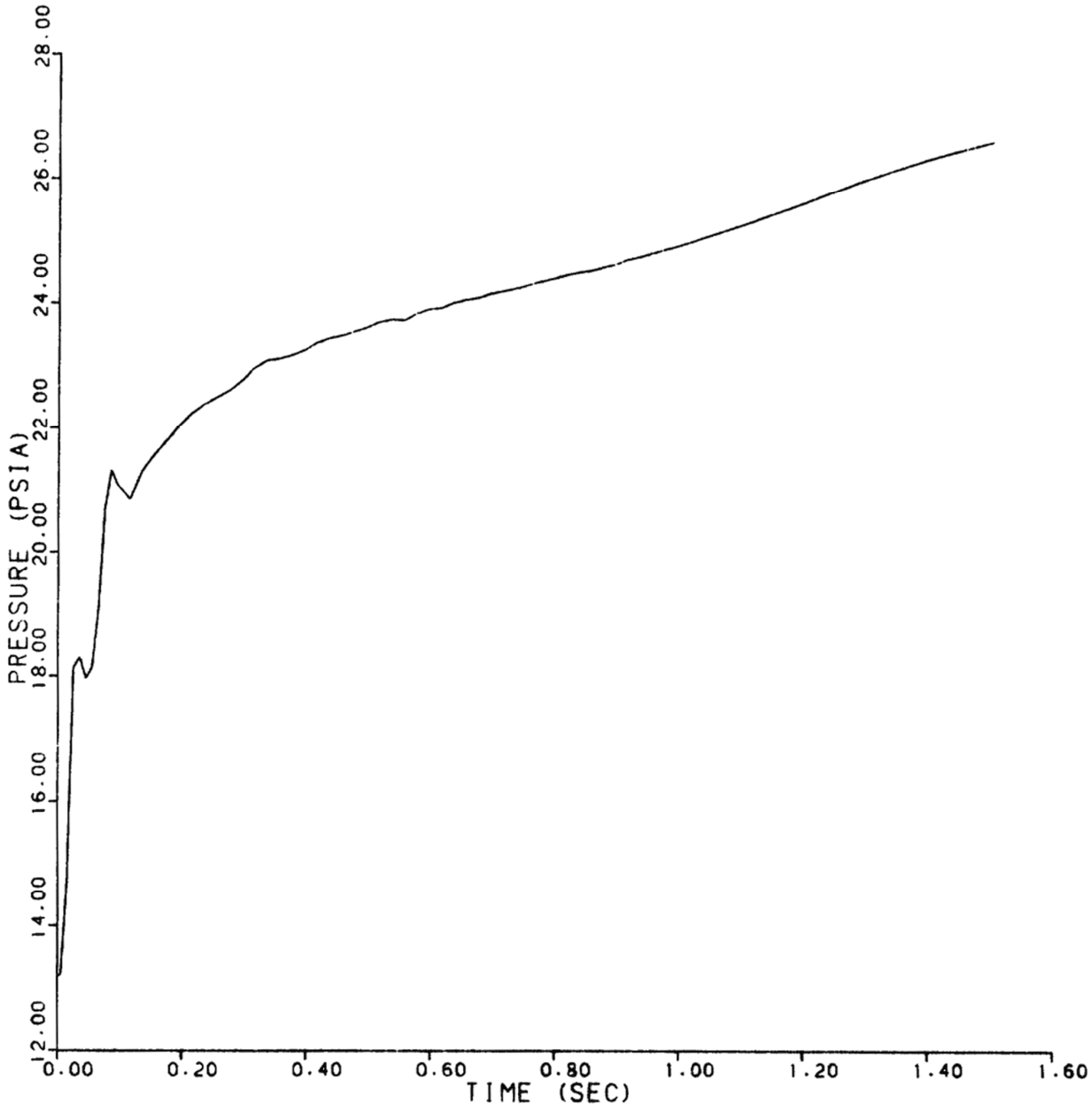
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR COMPARTMENT PRESSURE RESPONSE – NODE E2 (236–in.^{2} BREAK AREA)
	FIGURE 6.2.1–18 (SHEET 2 OF 74)



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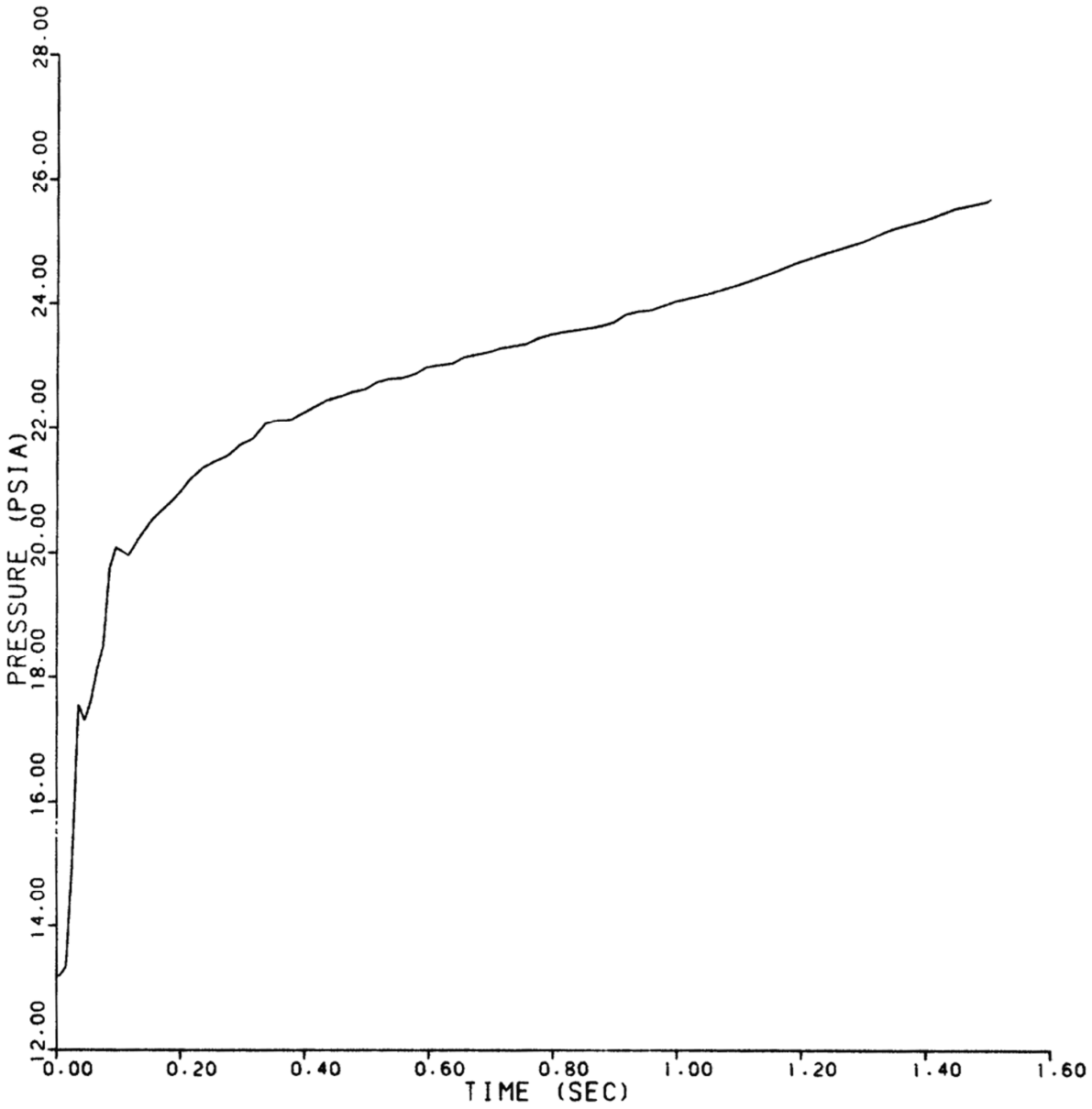
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E3
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 3 OF 74)



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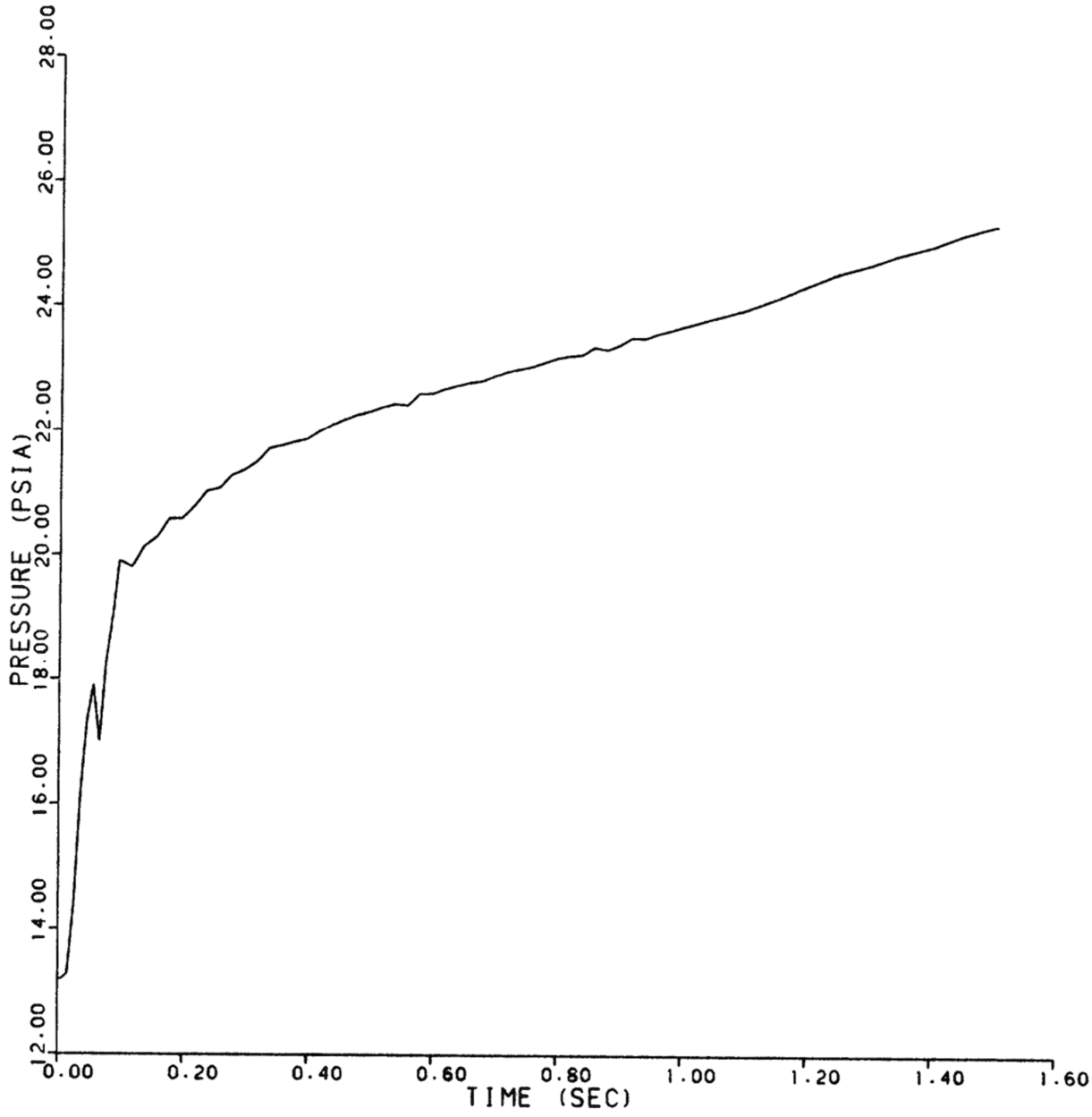
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E4
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 4 OF 74)



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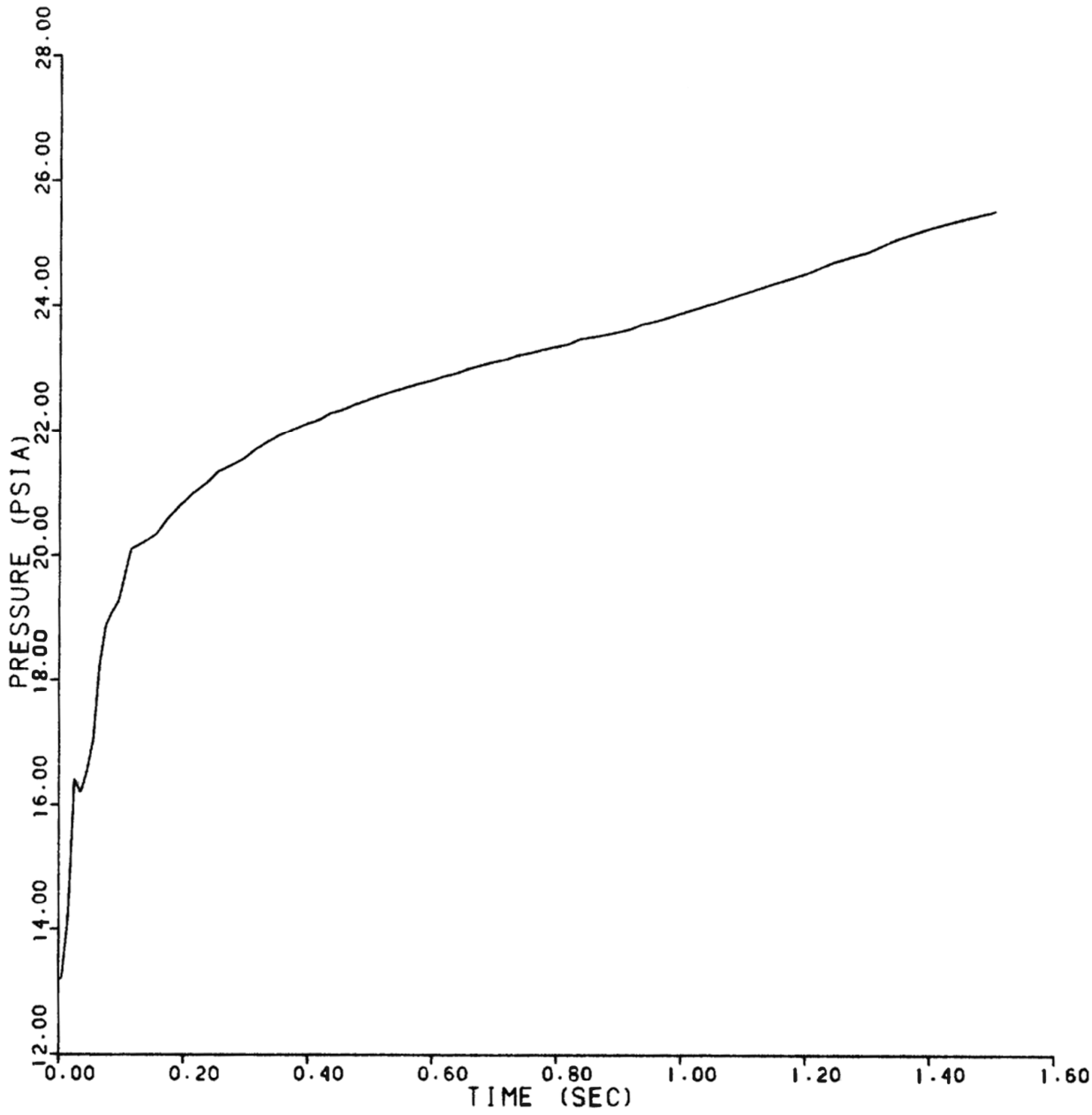
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E5
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 5 OF 74)



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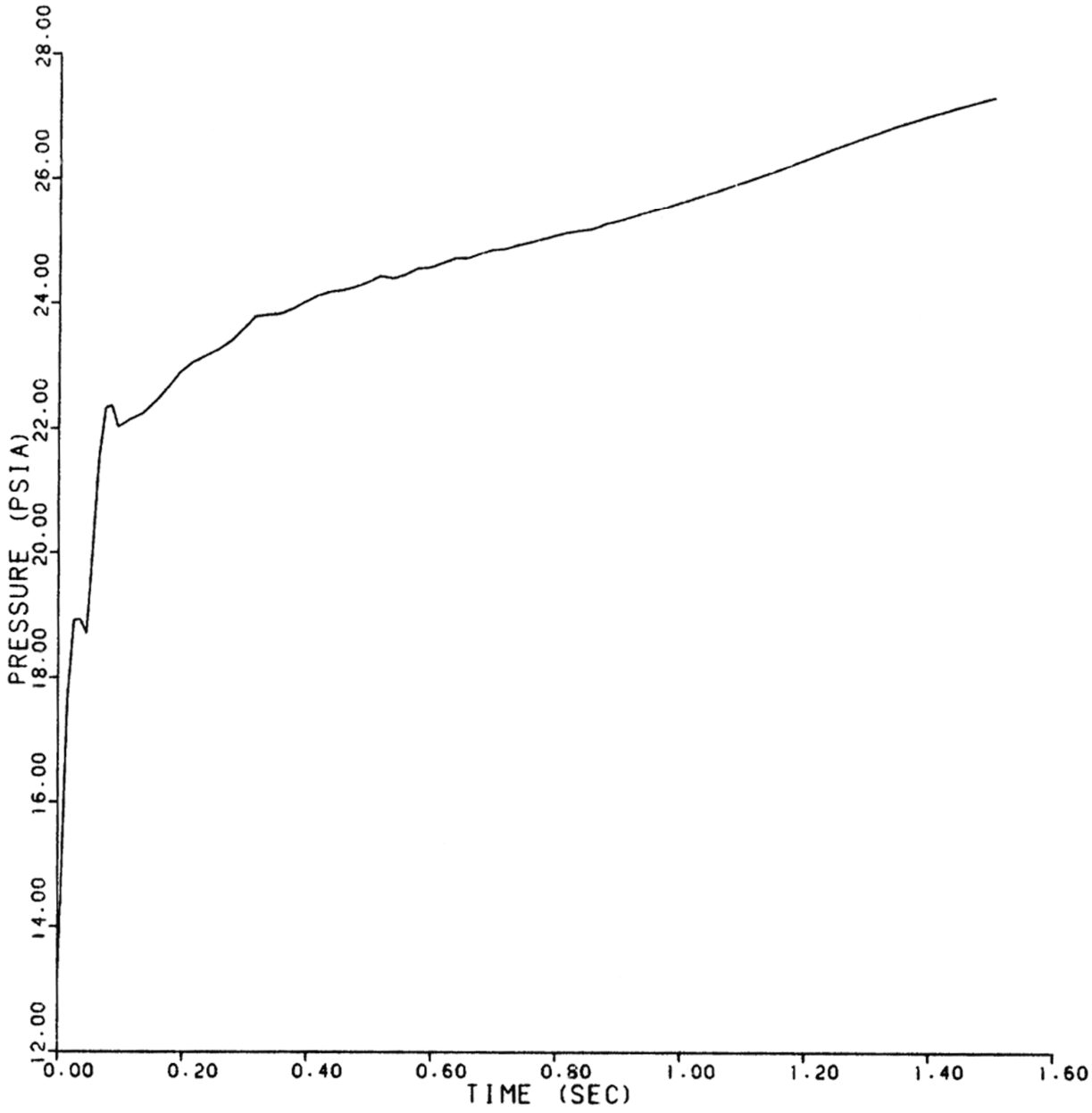
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E6
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 6 OF 74)



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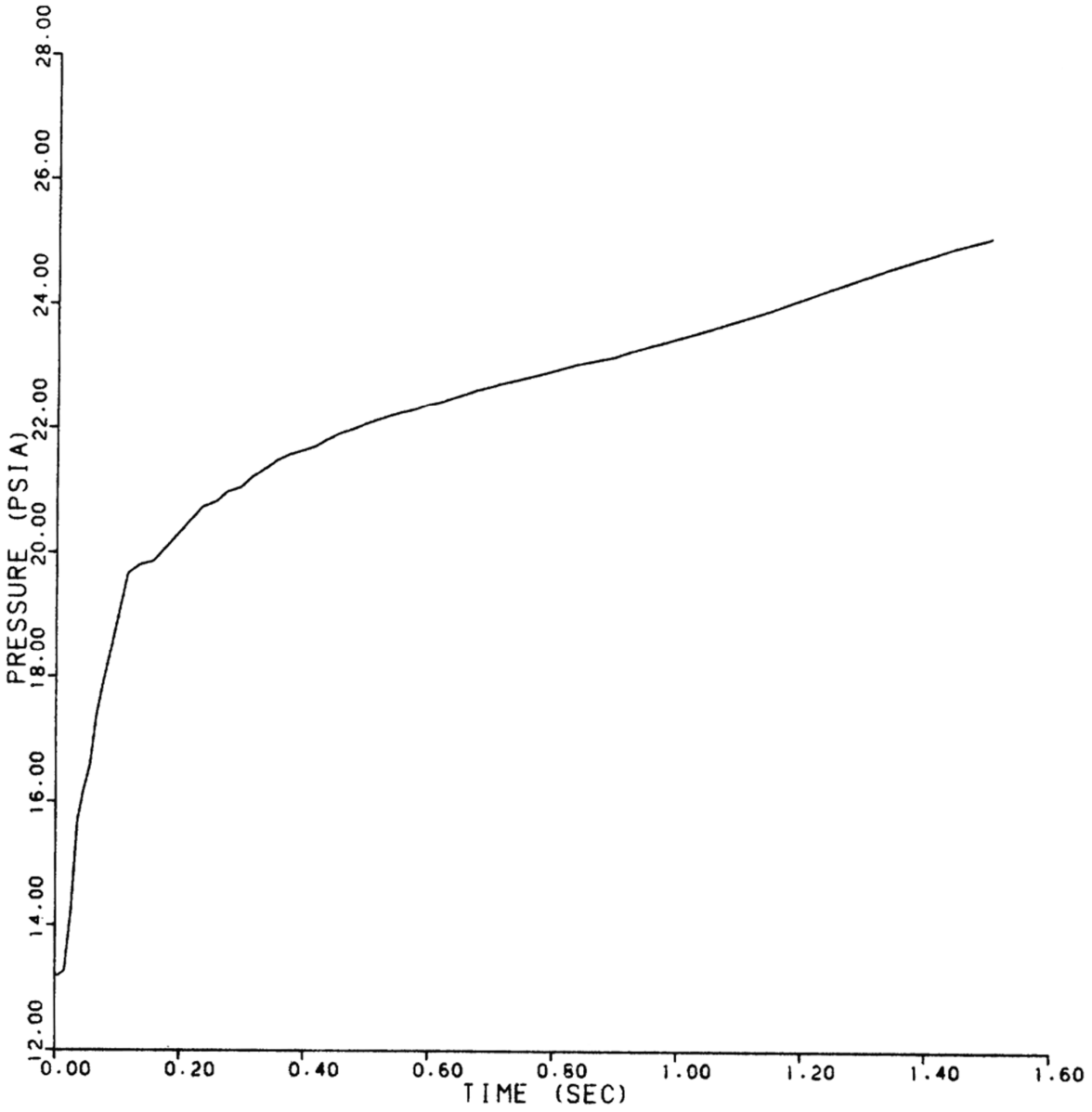
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E7
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 7 OF 74)



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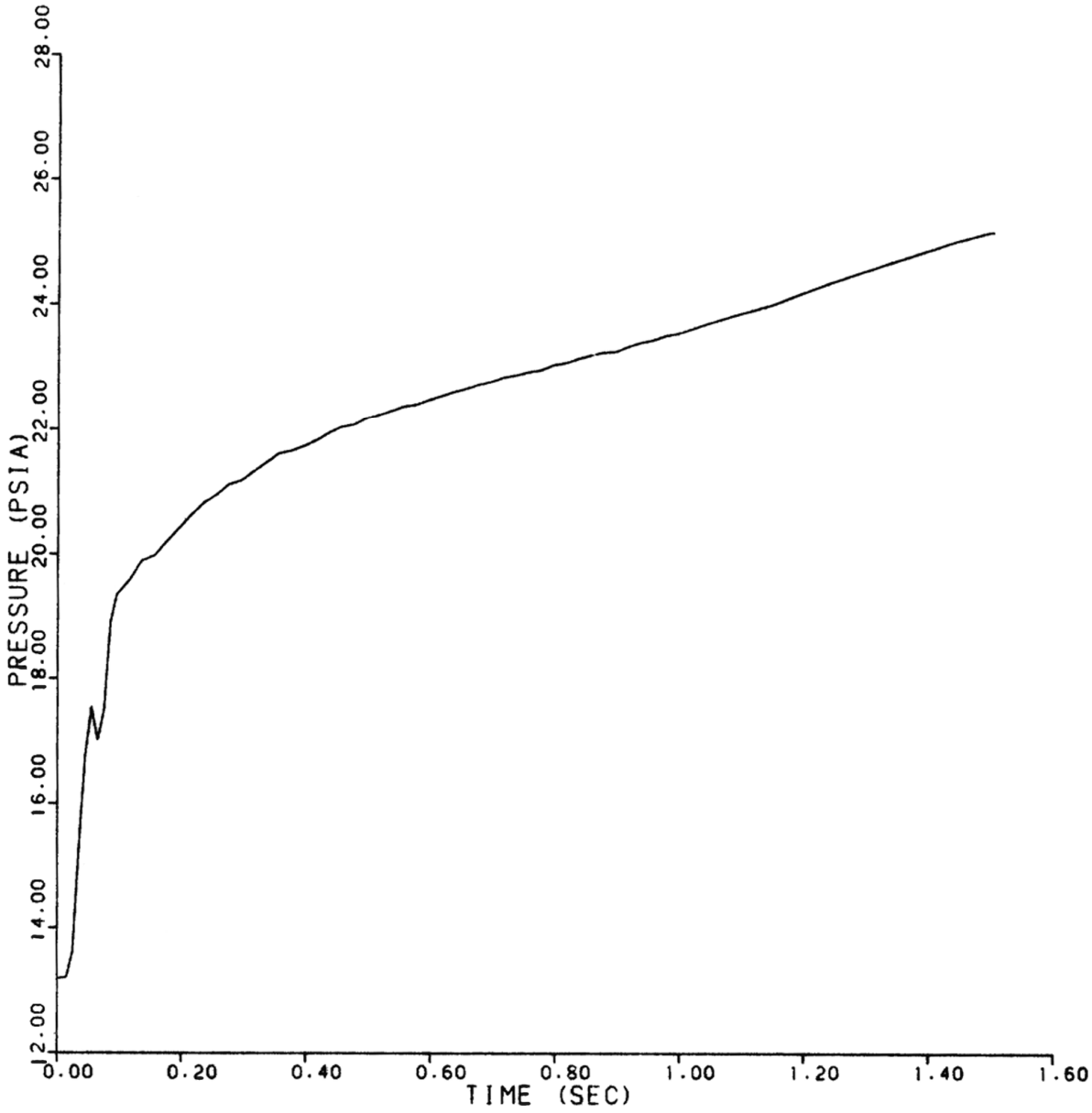
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E8
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 8 OF 74)



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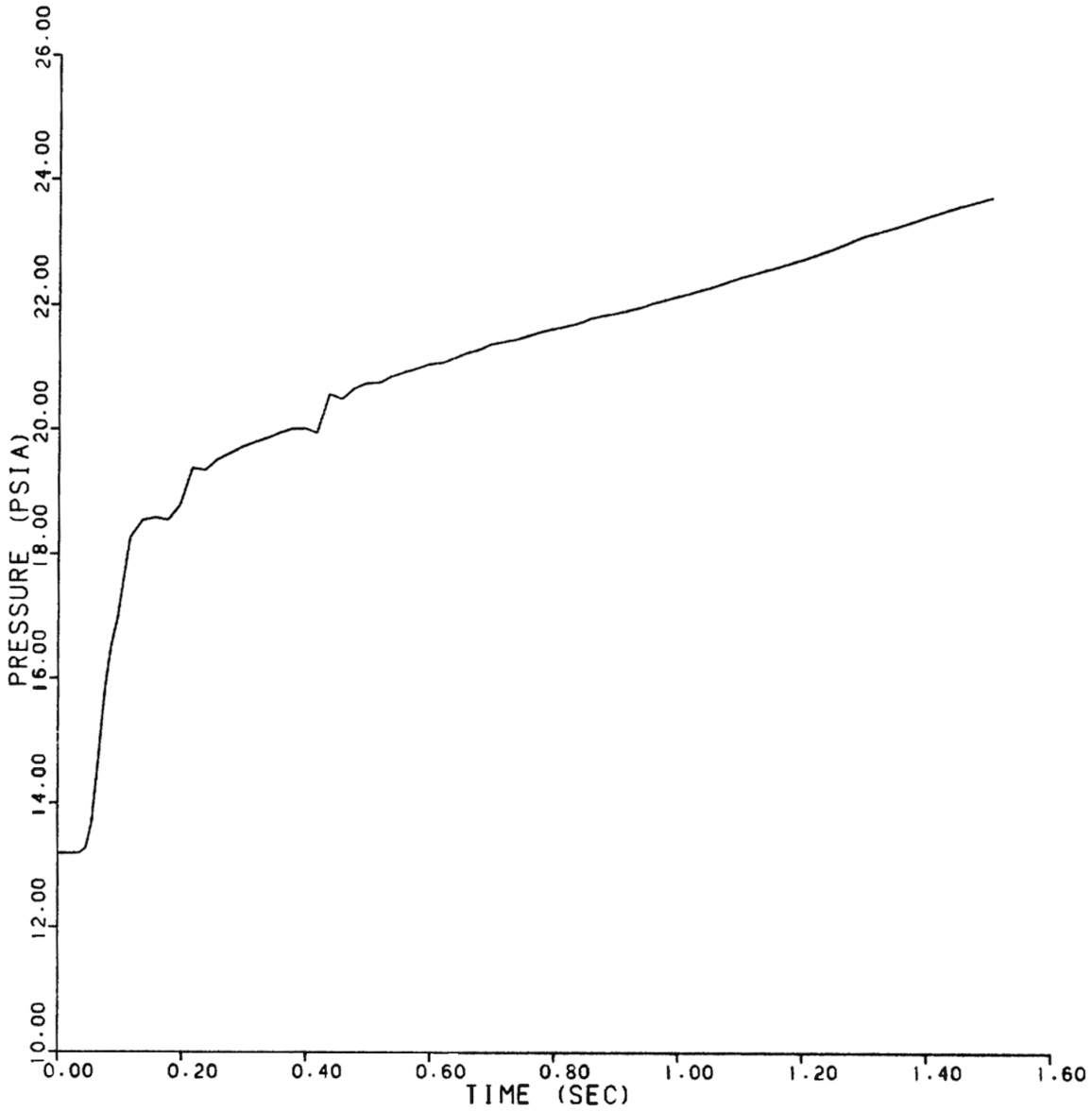
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E9
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 9 OF 74)



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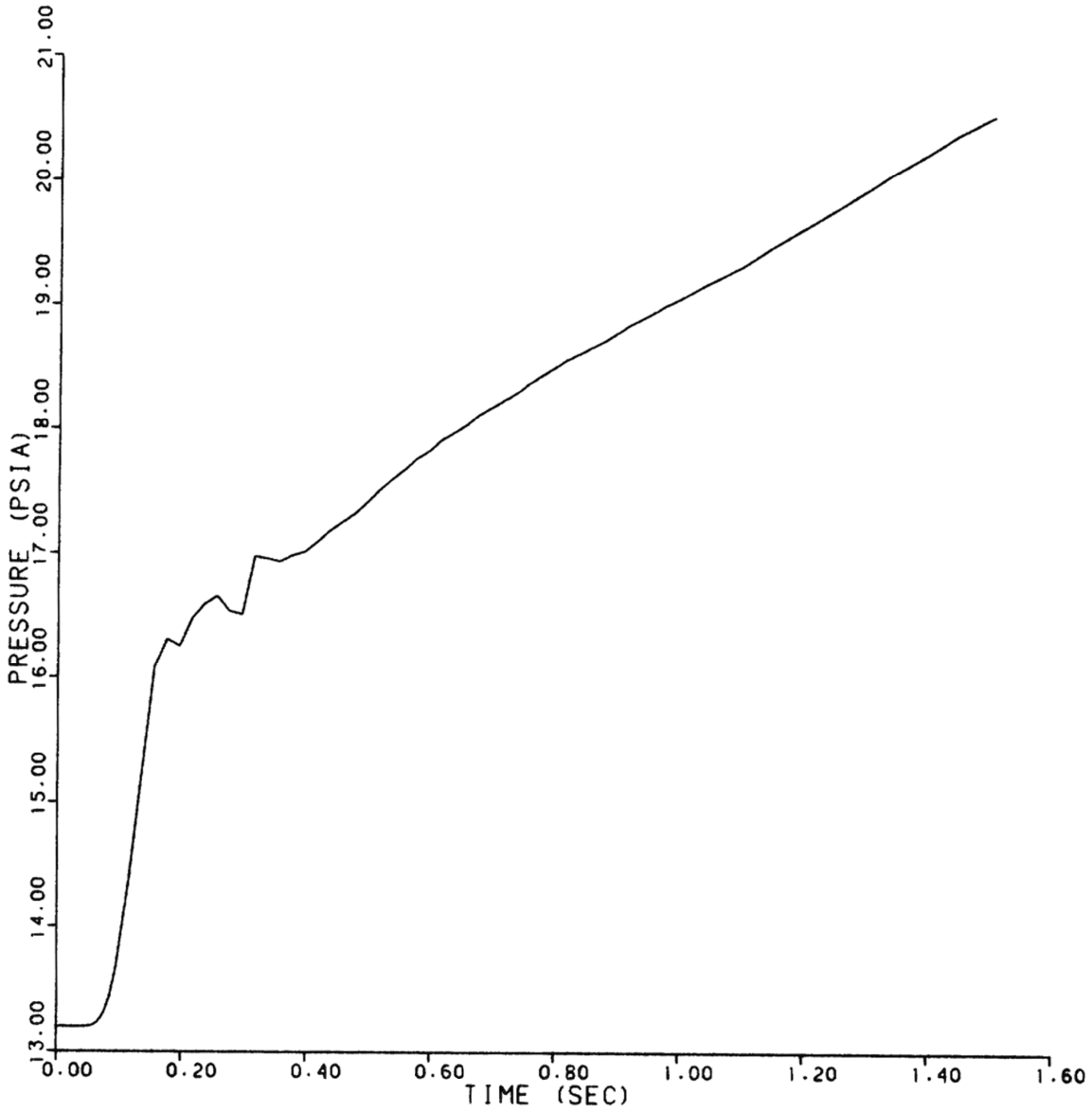
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT PRESSURE RESPONSE – NODE E10 (236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 10 OF 74)



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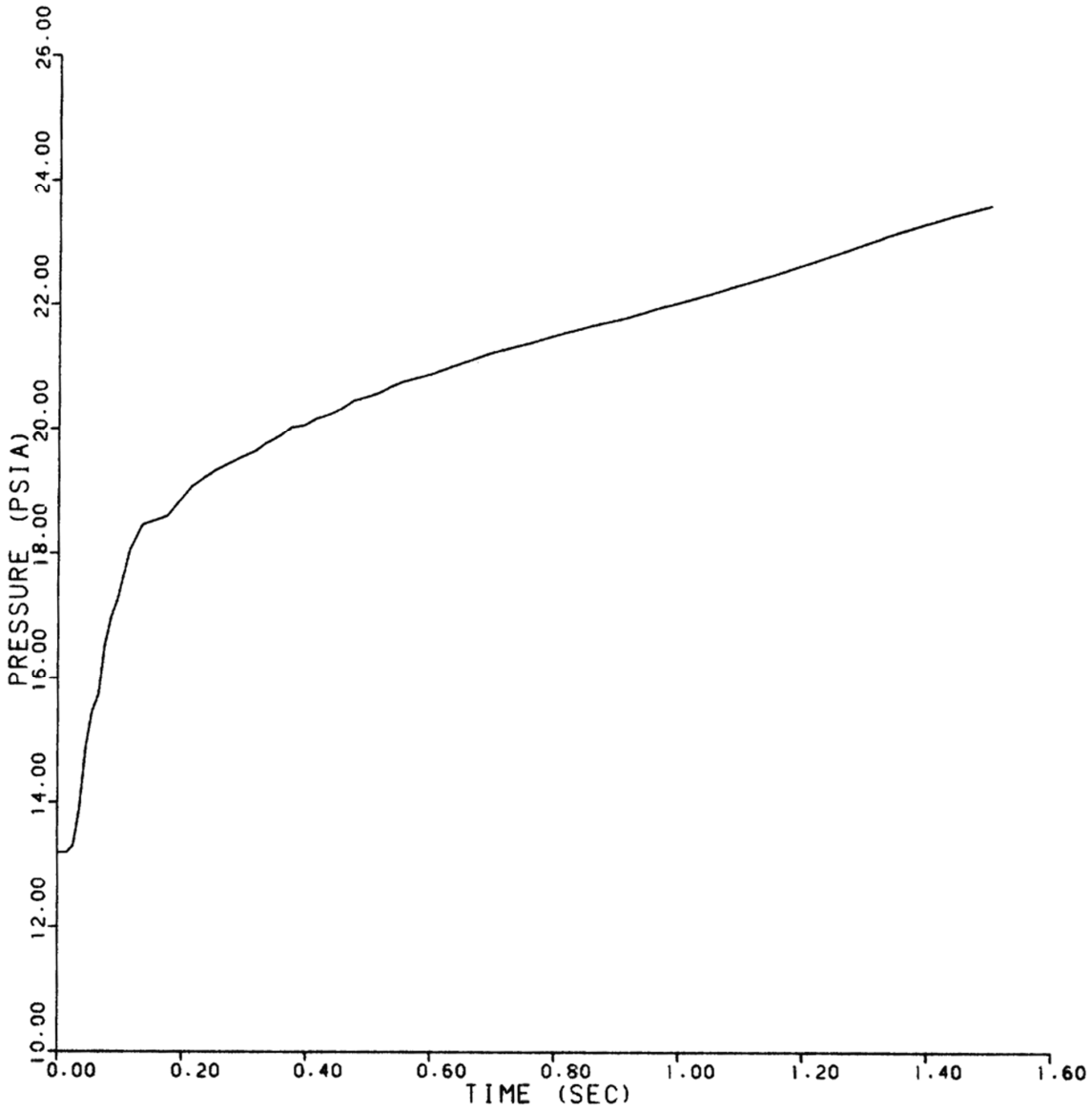
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT PRESSURE RESPONSE – NODE E11 (236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 11 OF 74)



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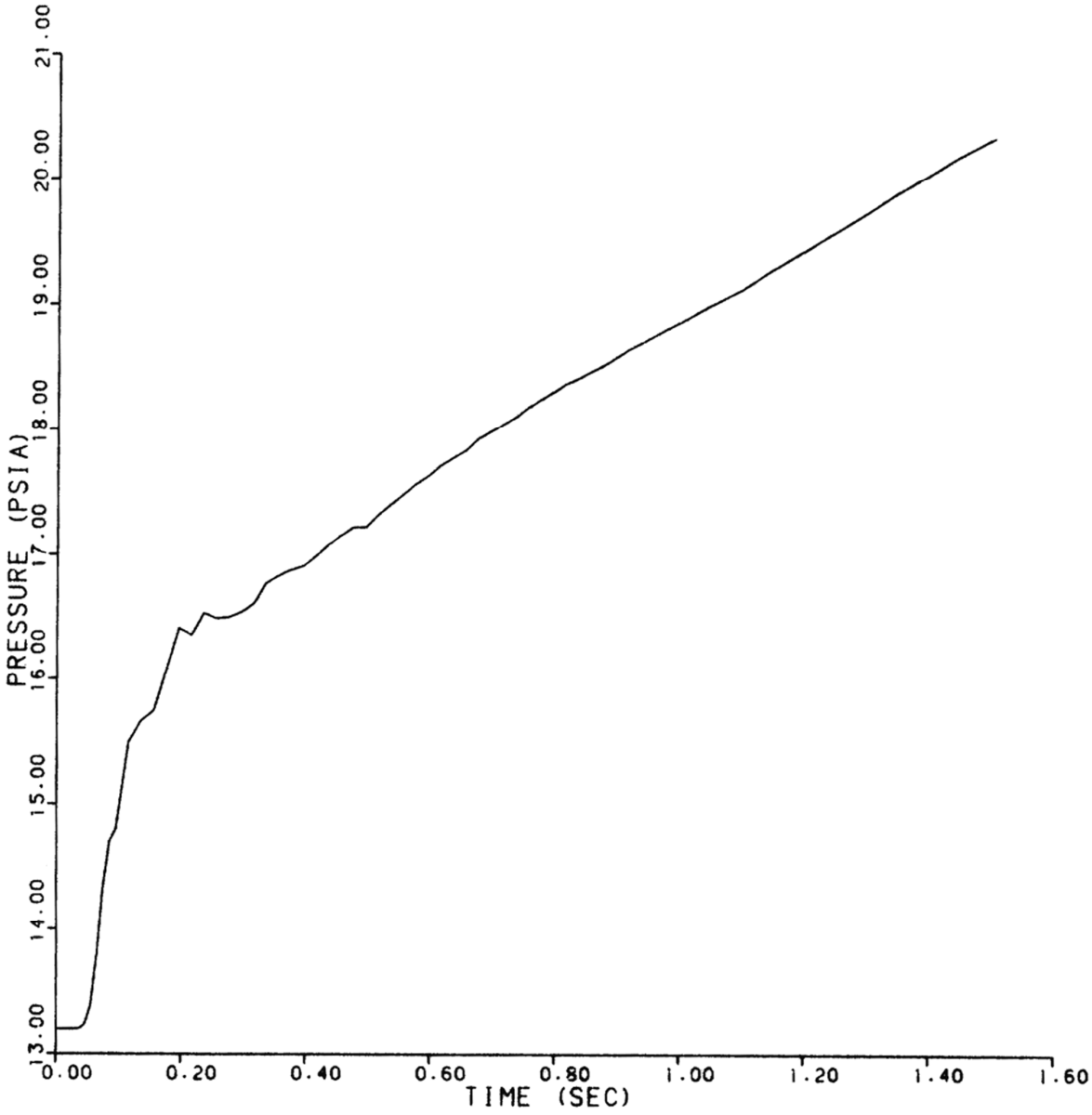
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT PRESSURE RESPONSE – NODE E12 (236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 12 OF 74)



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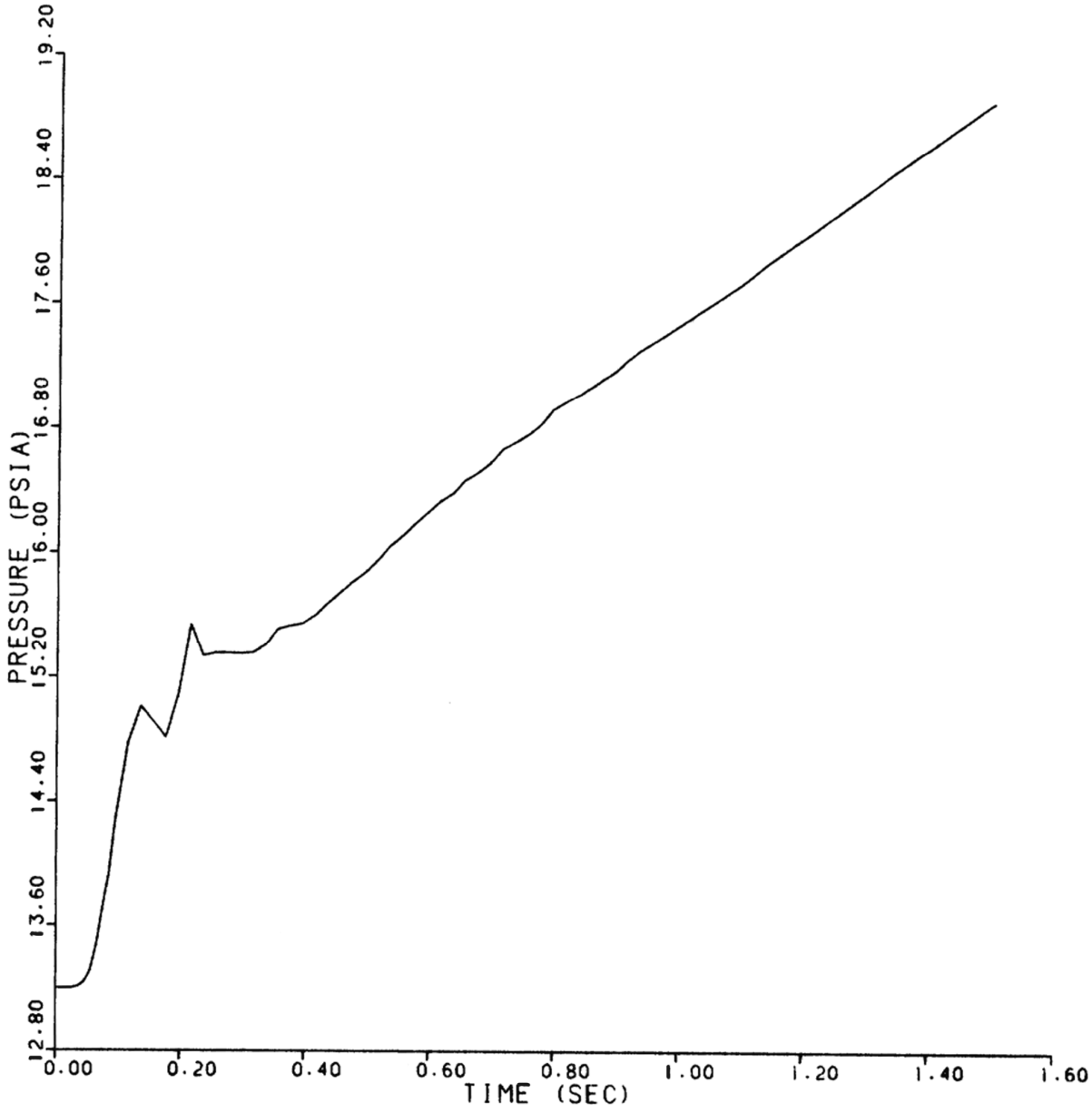
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E13
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 13 OF 74)



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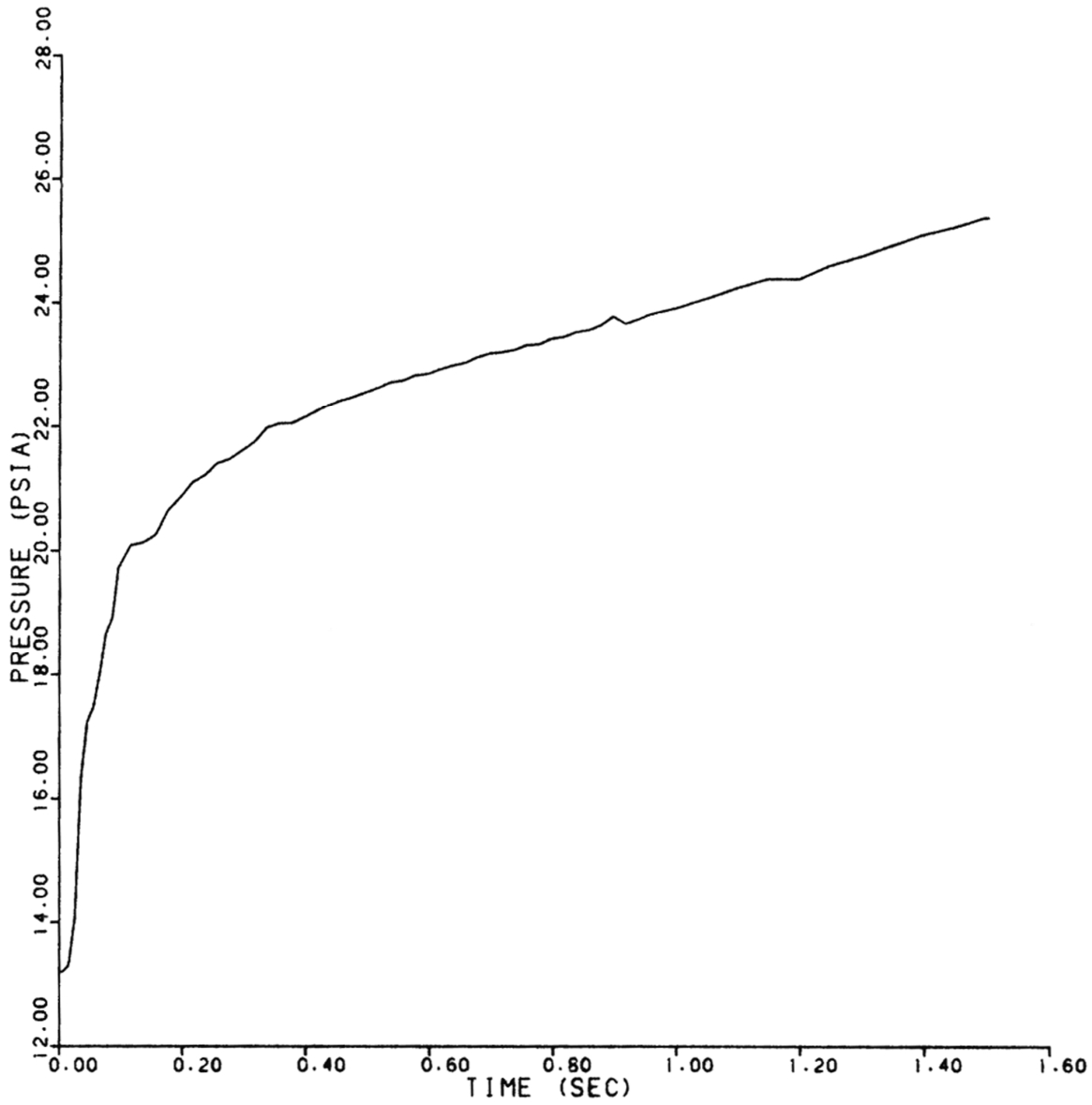
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E14
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 14 OF 74)



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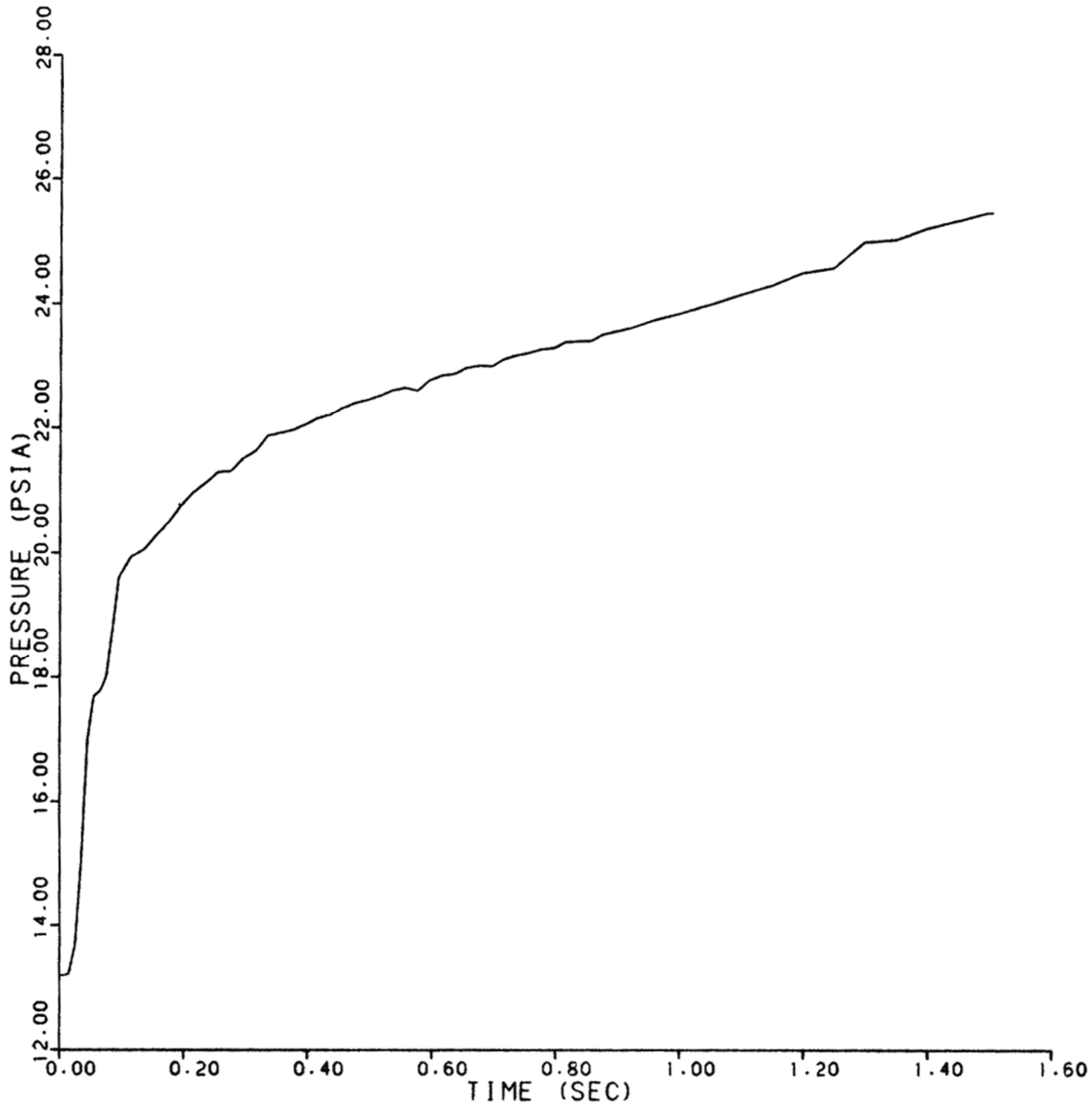
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E15
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 15 OF 74)



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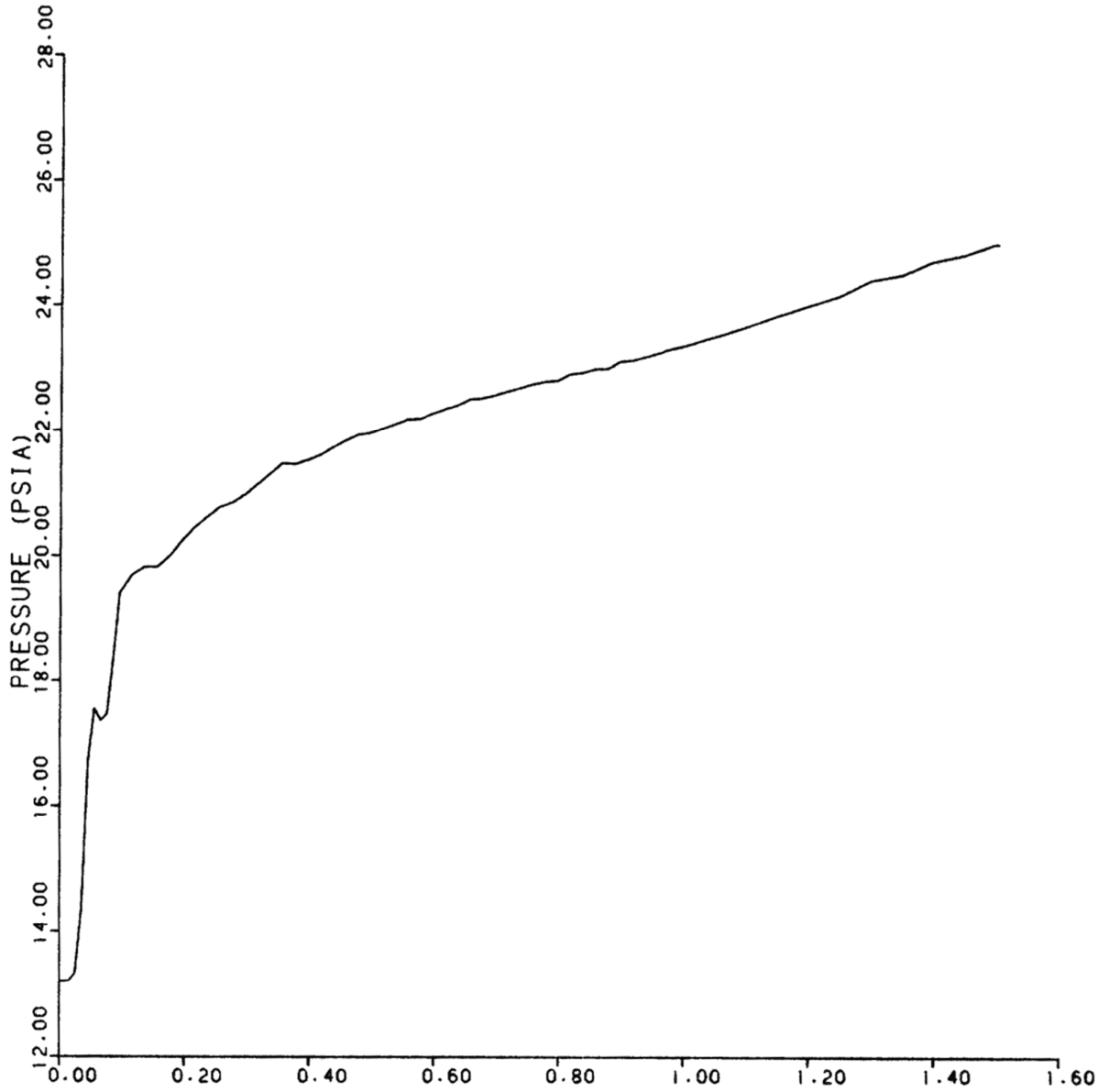
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E16
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 16 OF 74)



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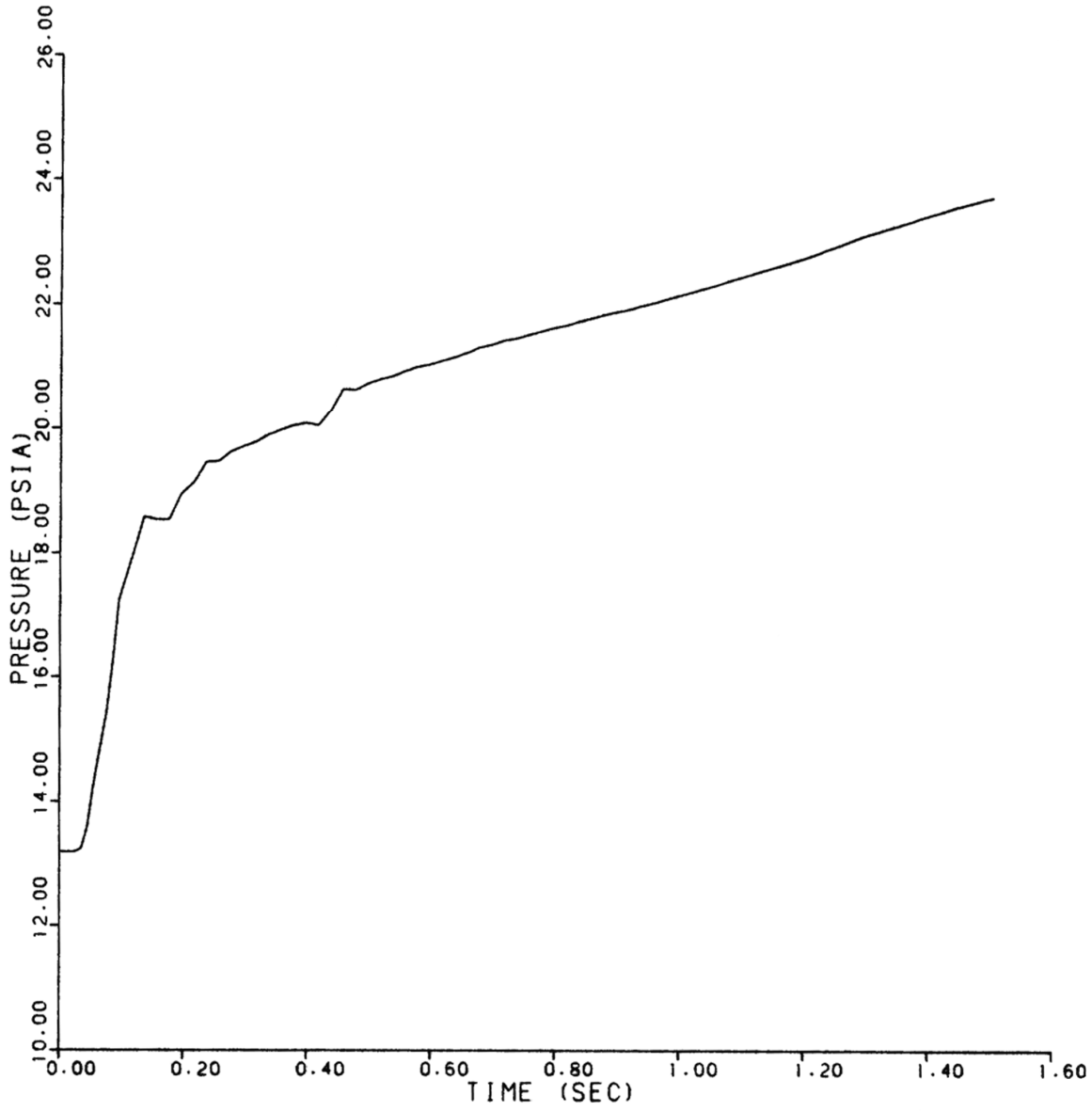
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E17
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 17 OF 74)



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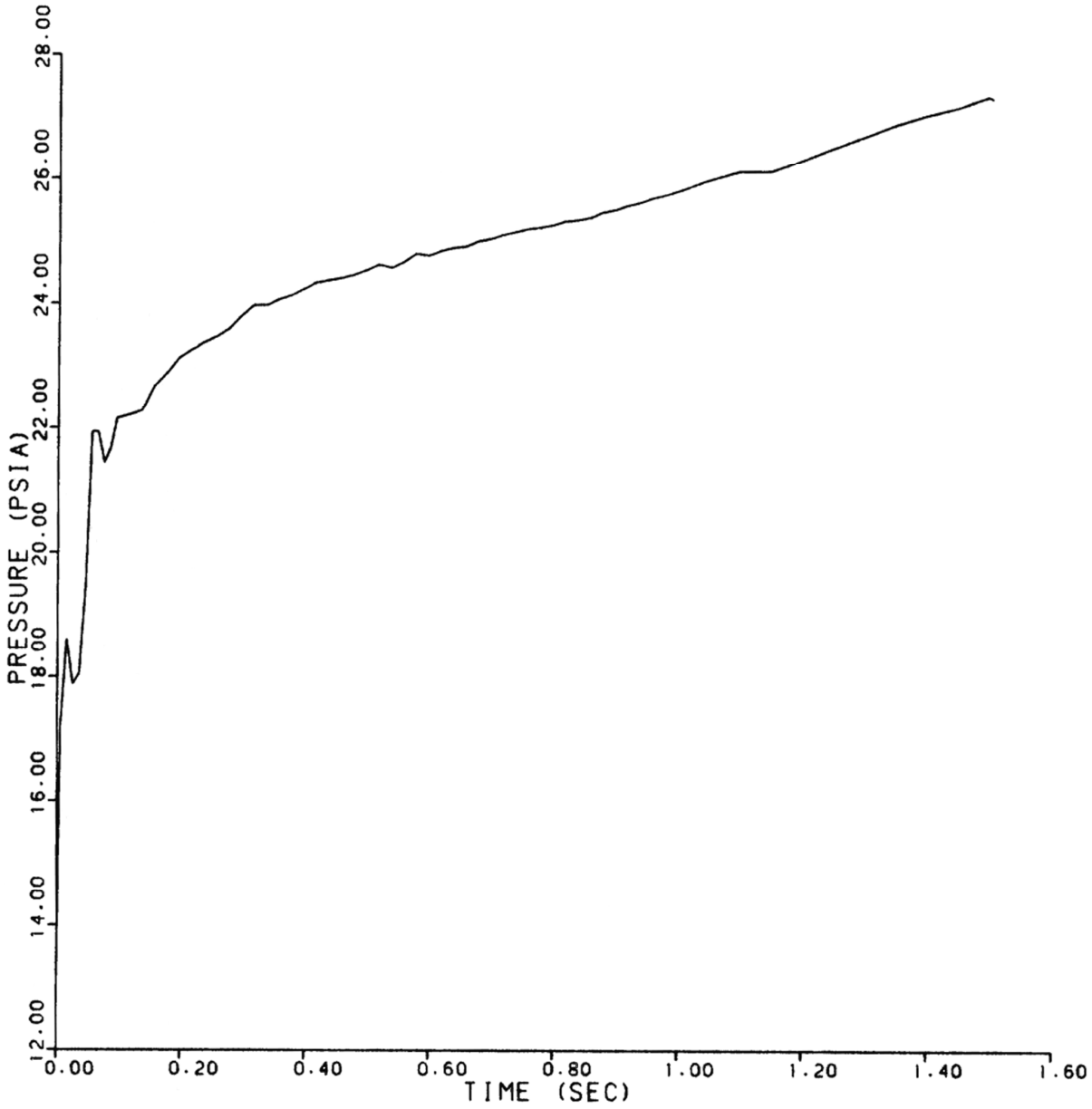
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E18
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 18 OF 74)



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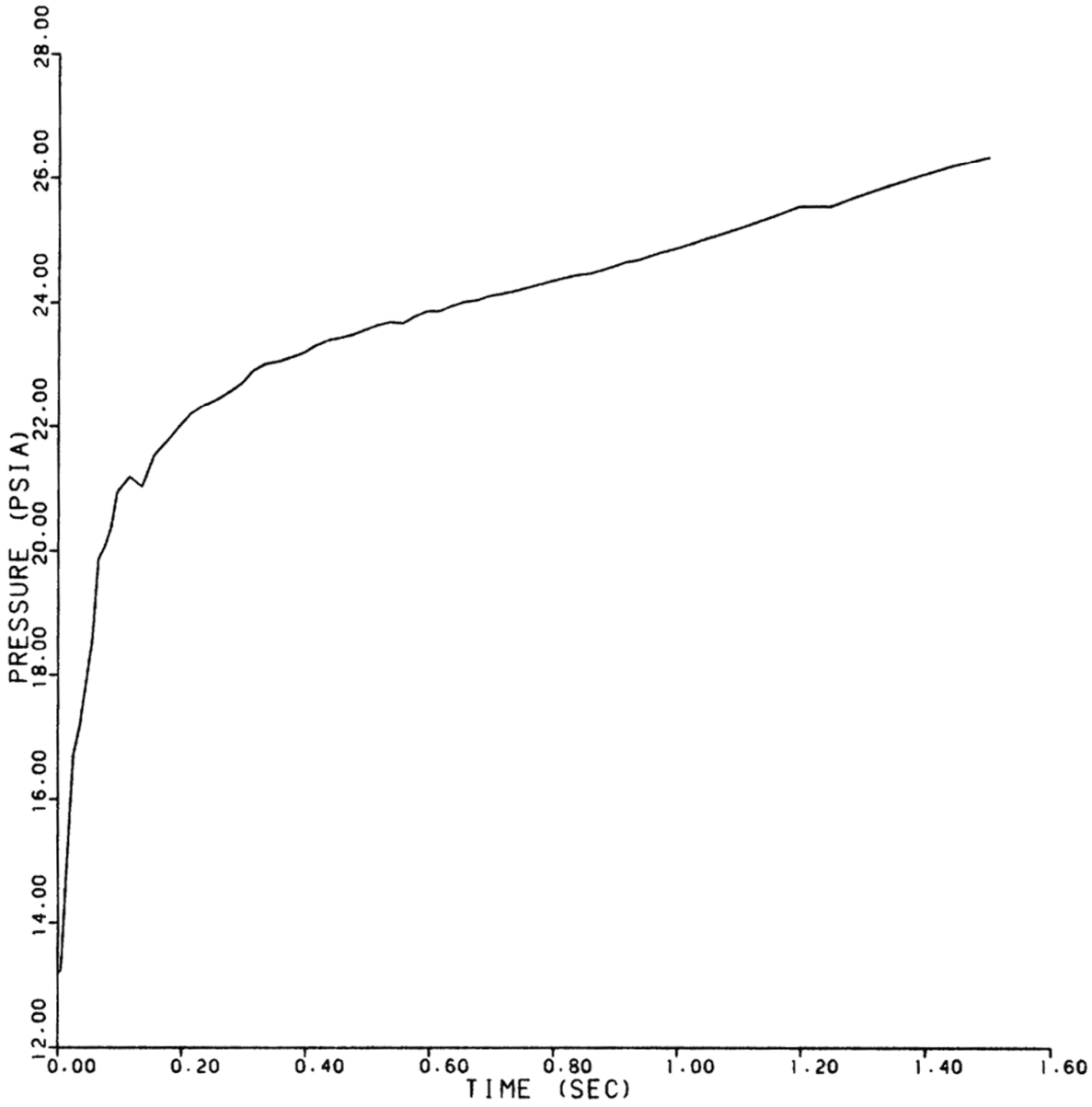
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR COMPARTMENT PRESSURE RESPONSE – NODE E19 (236–in.^{2} BREAK AREA)
	FIGURE 6.2.1–18 (SHEET 19 OF 74)



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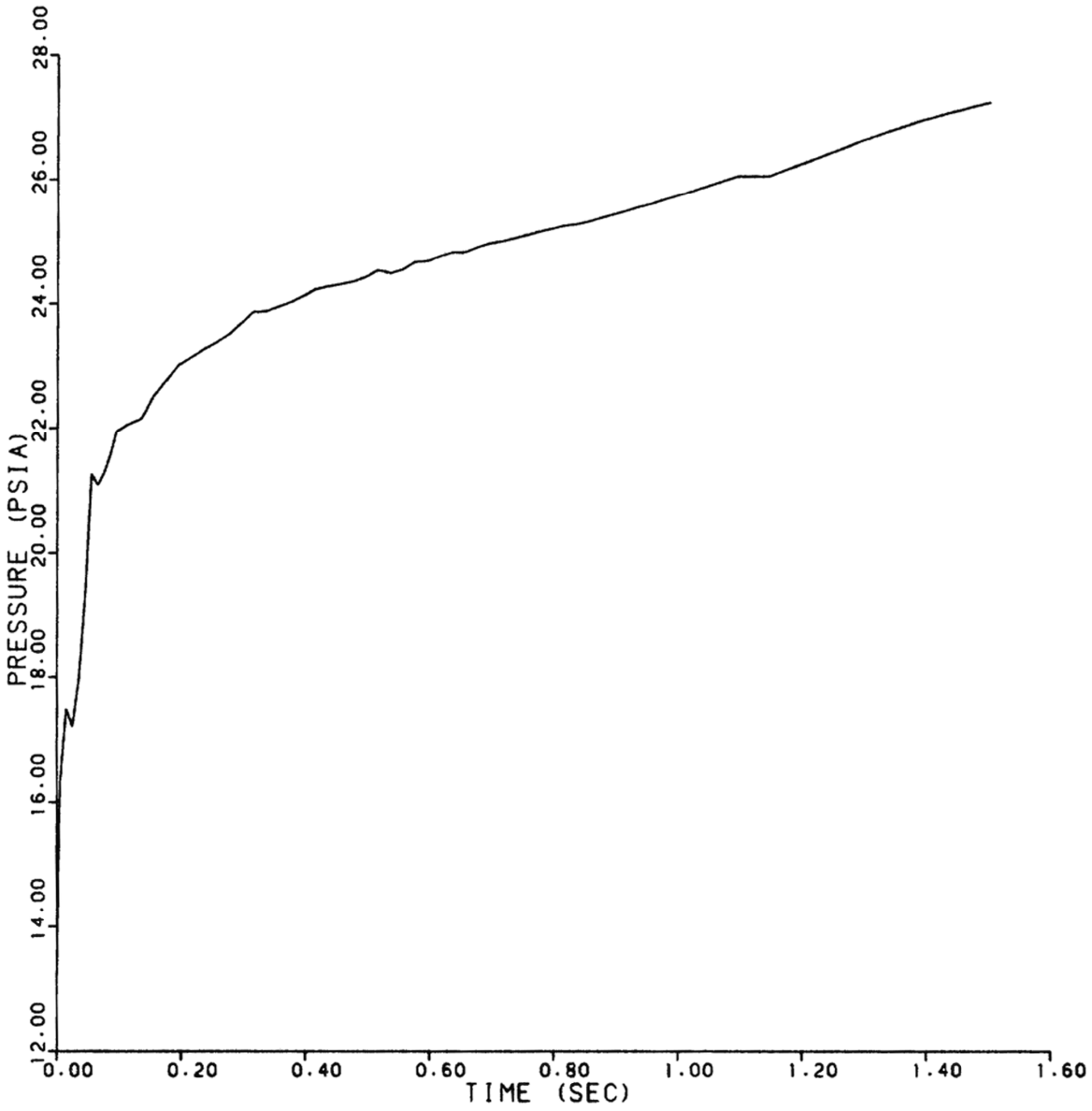
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT PRESSURE RESPONSE – NODE E20 (236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 20 OF 74)



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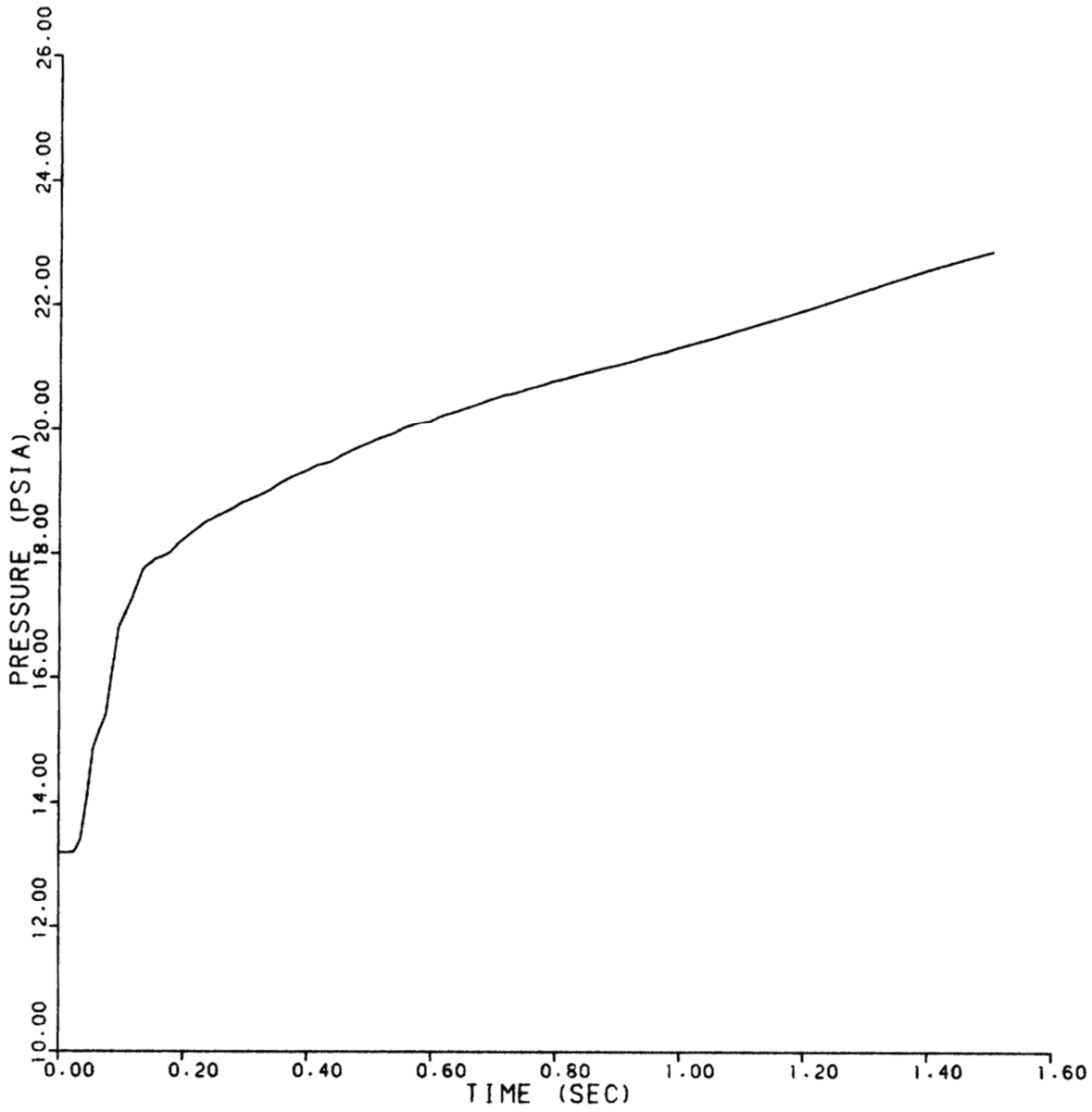
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E21
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 21 OF 74)



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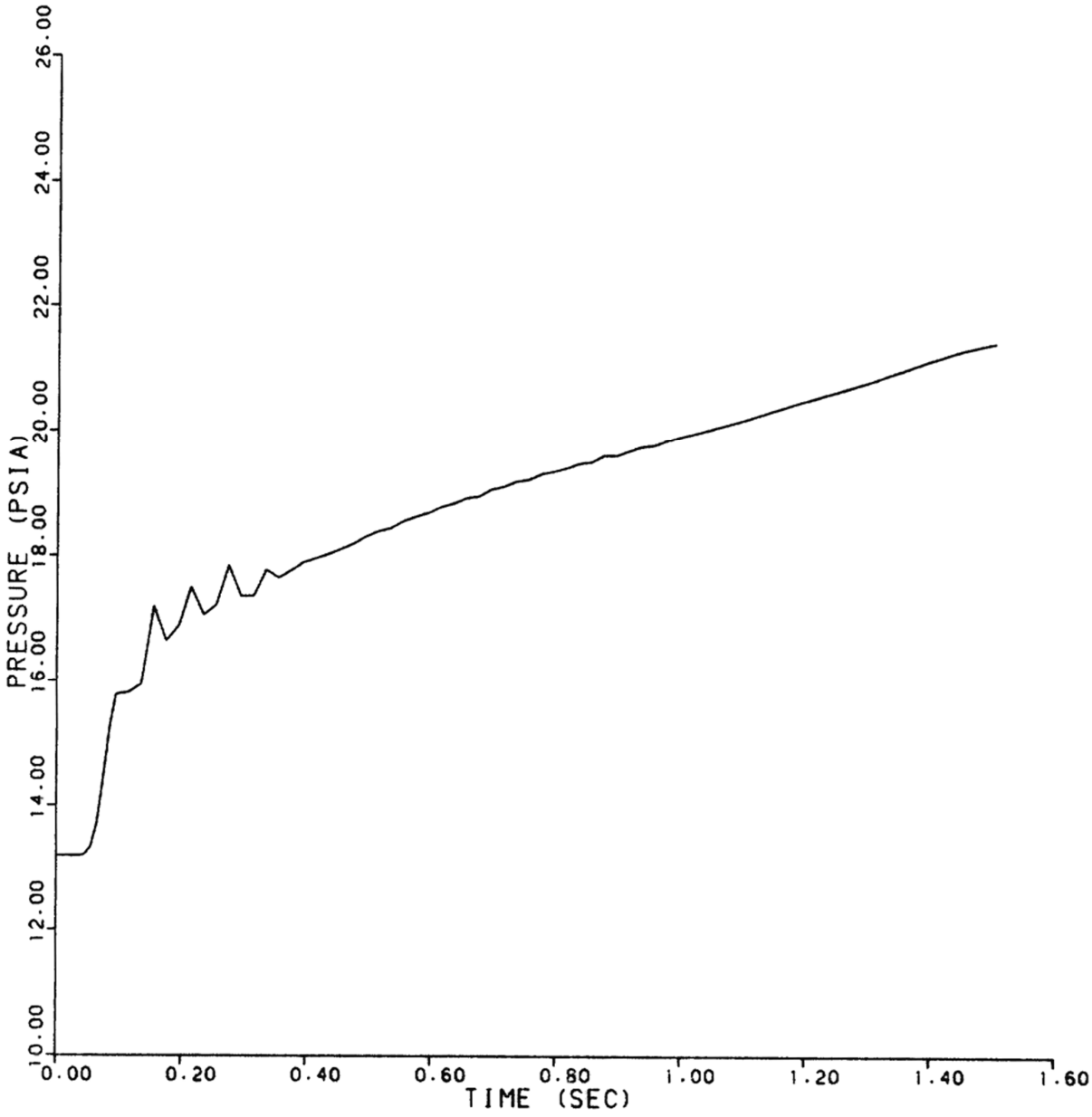
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT PRESSURE RESPONSE – NODE E22 (236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 22 OF 74)



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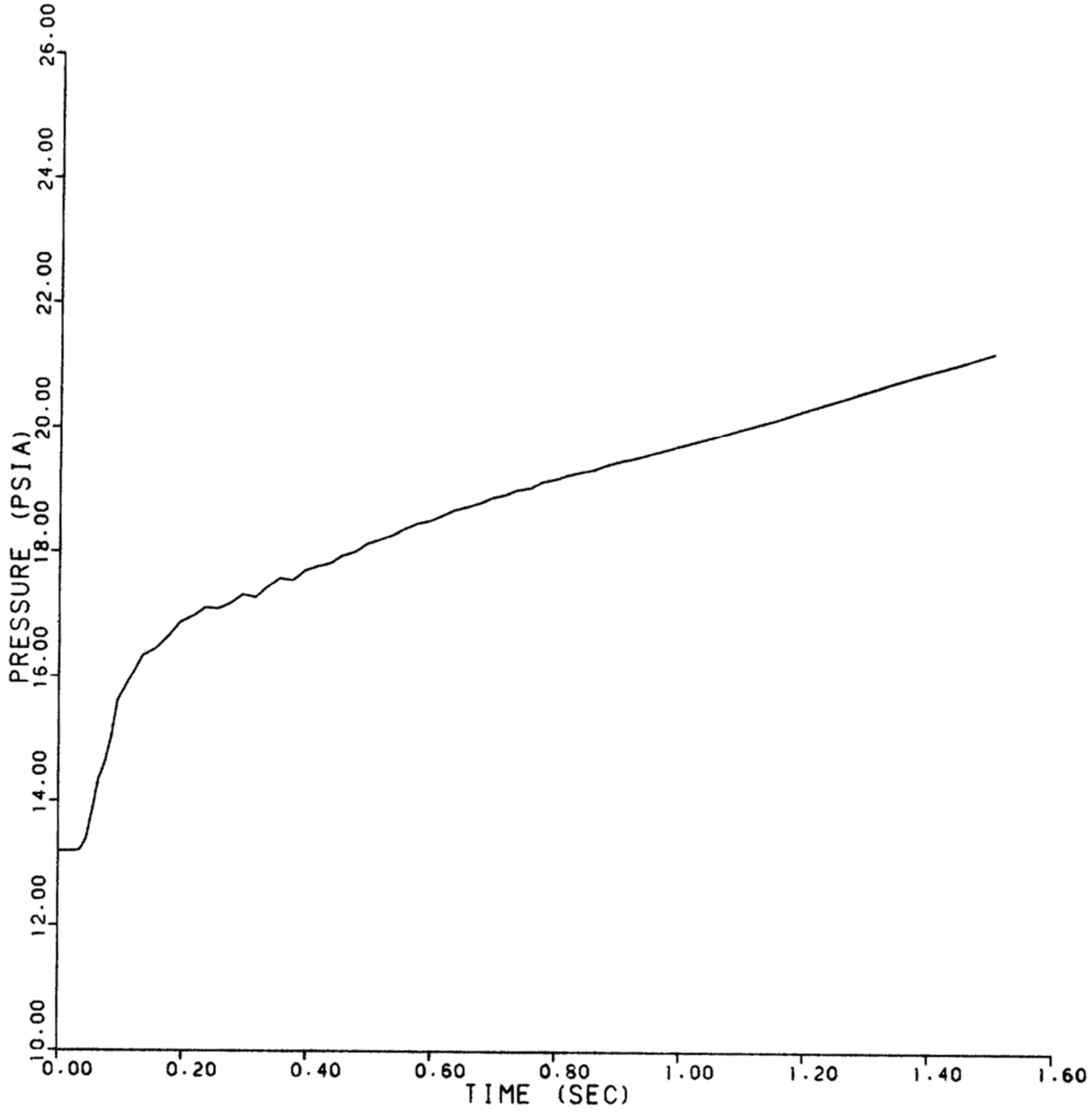
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E23
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 23 OF 74)



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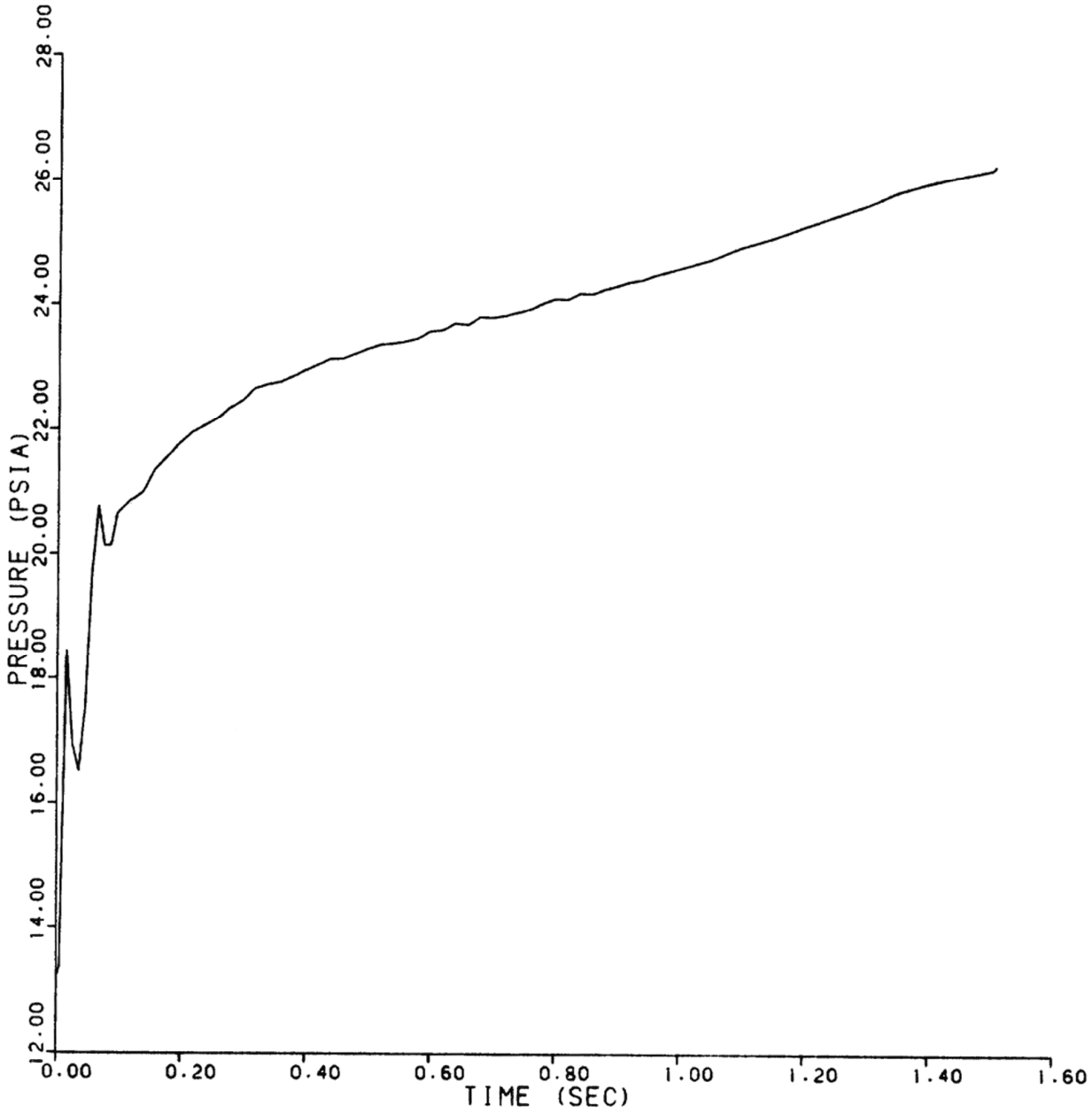
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E24
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 24 OF 74)



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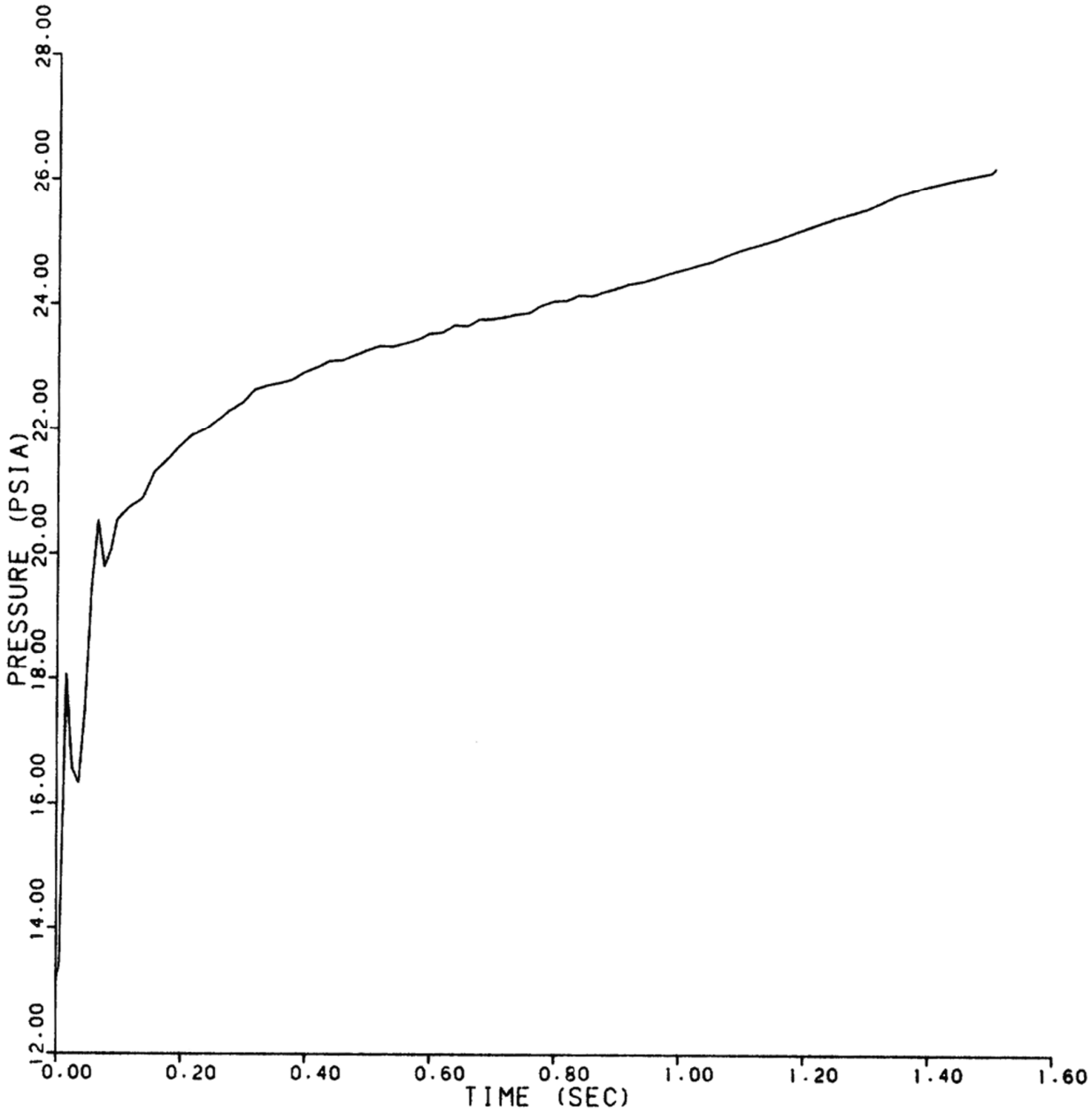
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E25
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 25 OF 74)



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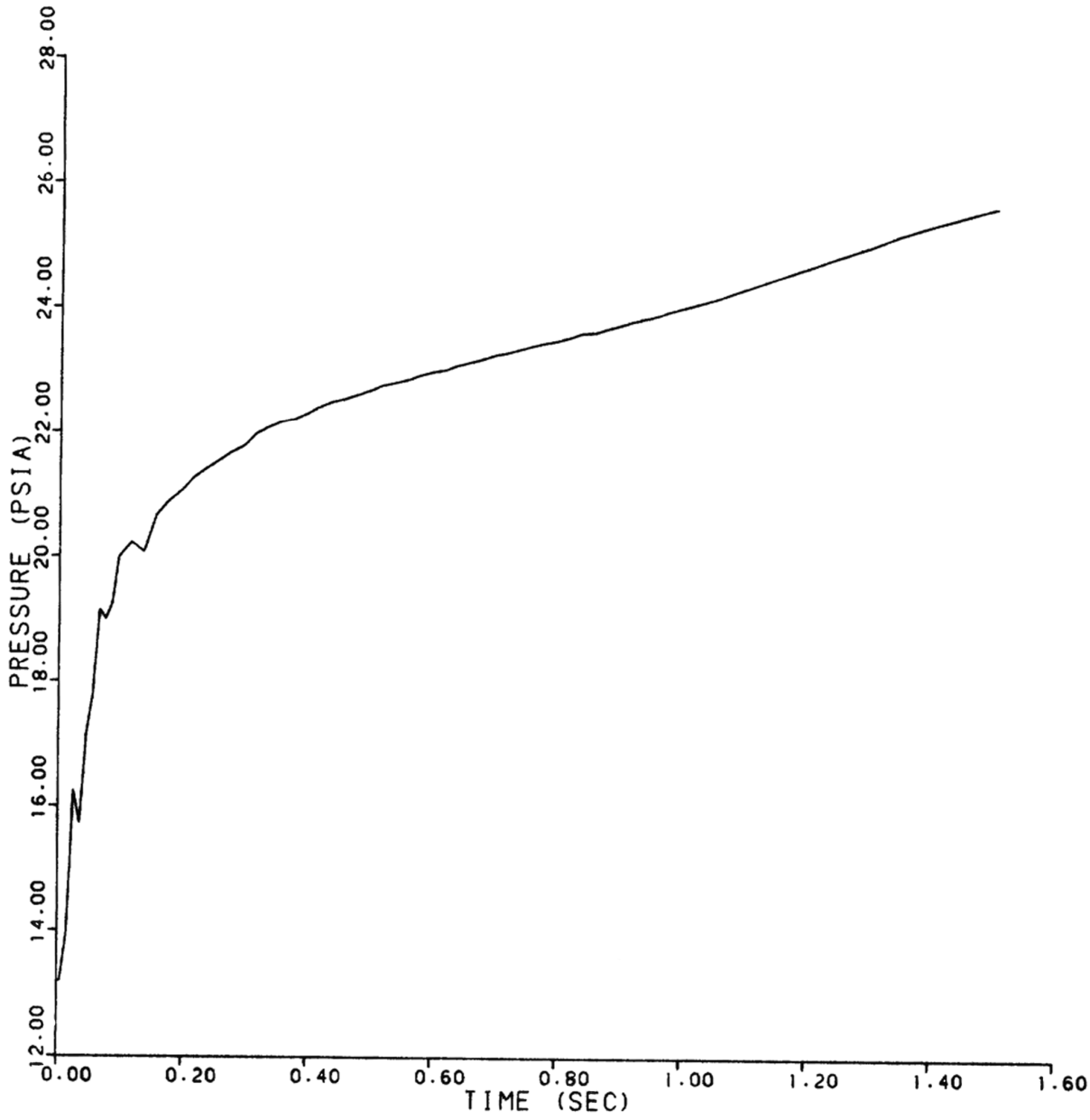
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E26
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 26 OF 74)



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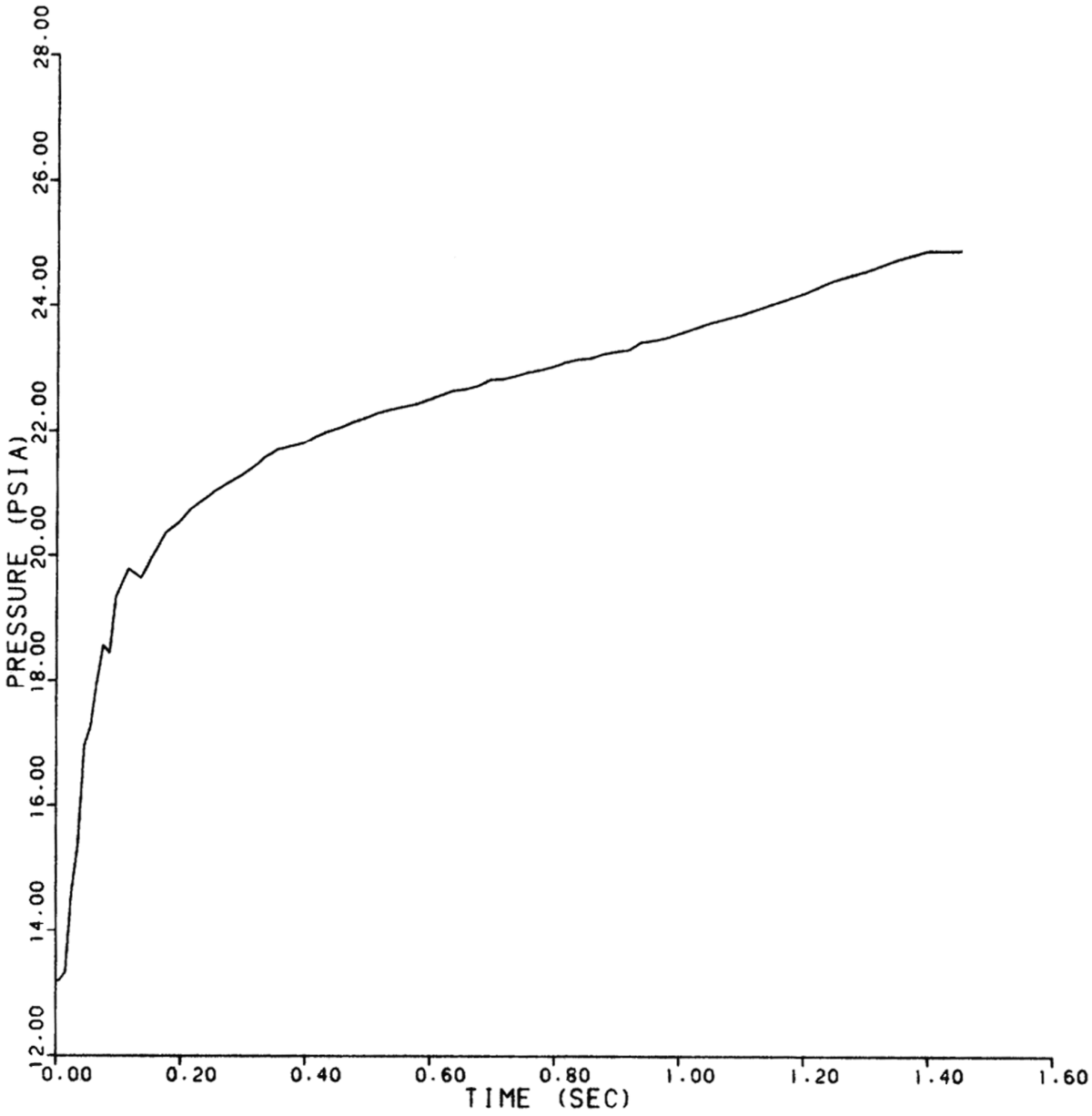
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT PRESSURE RESPONSE – NODE E27 (236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 27 OF 74)



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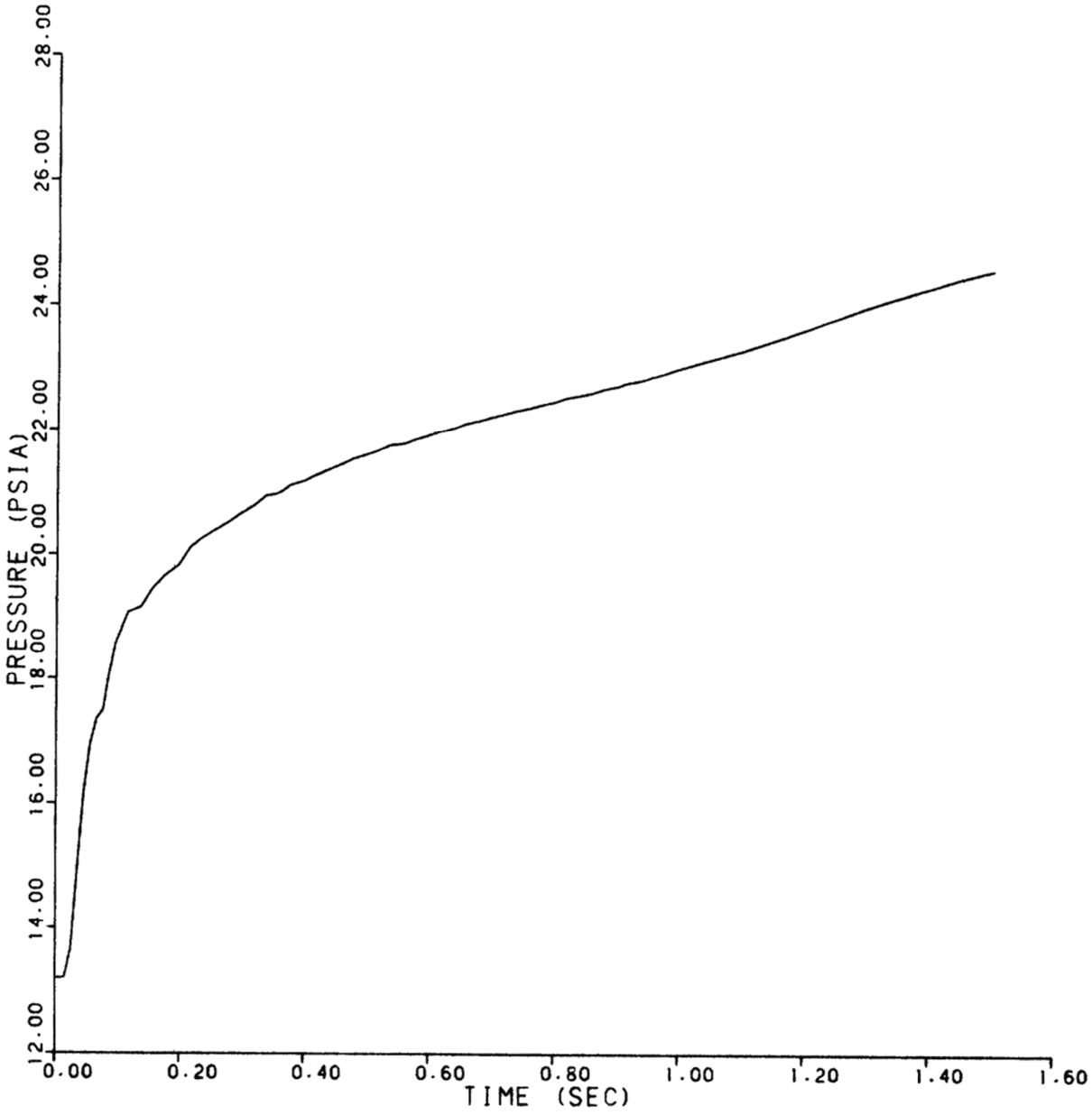
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E28
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 28 OF 74)



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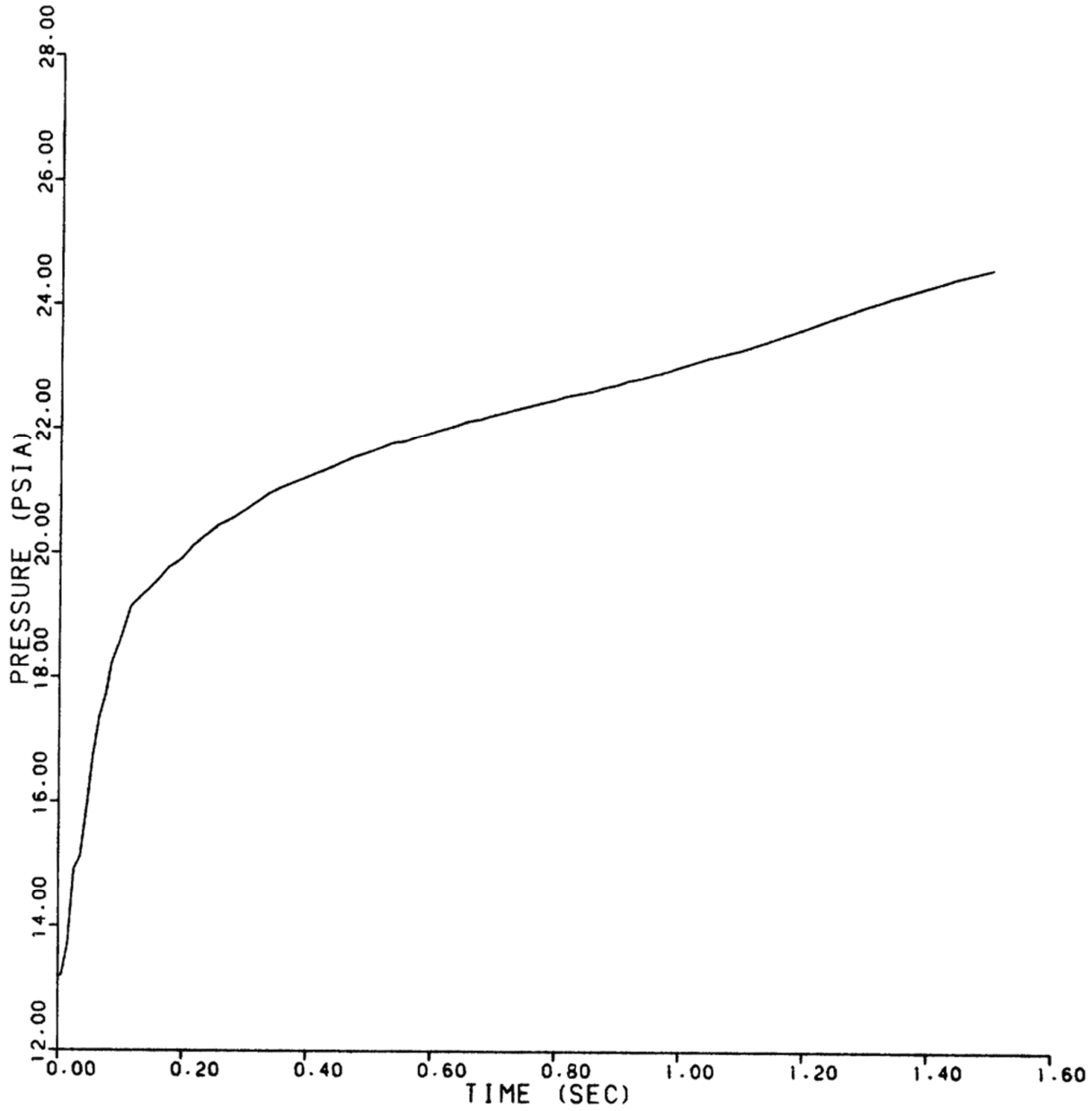
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E29
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 29 OF 74)



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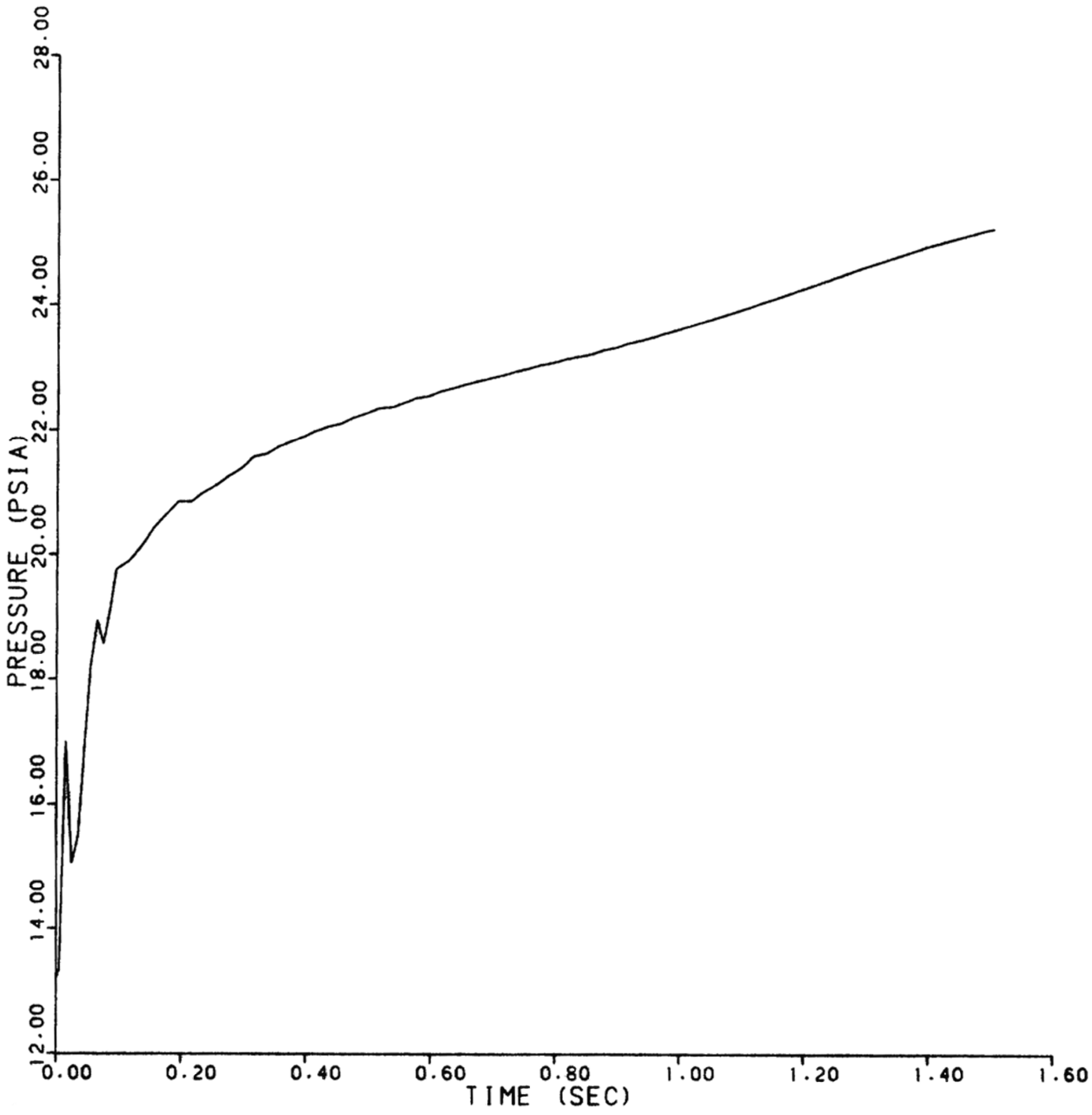
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E30
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 30 OF 74)



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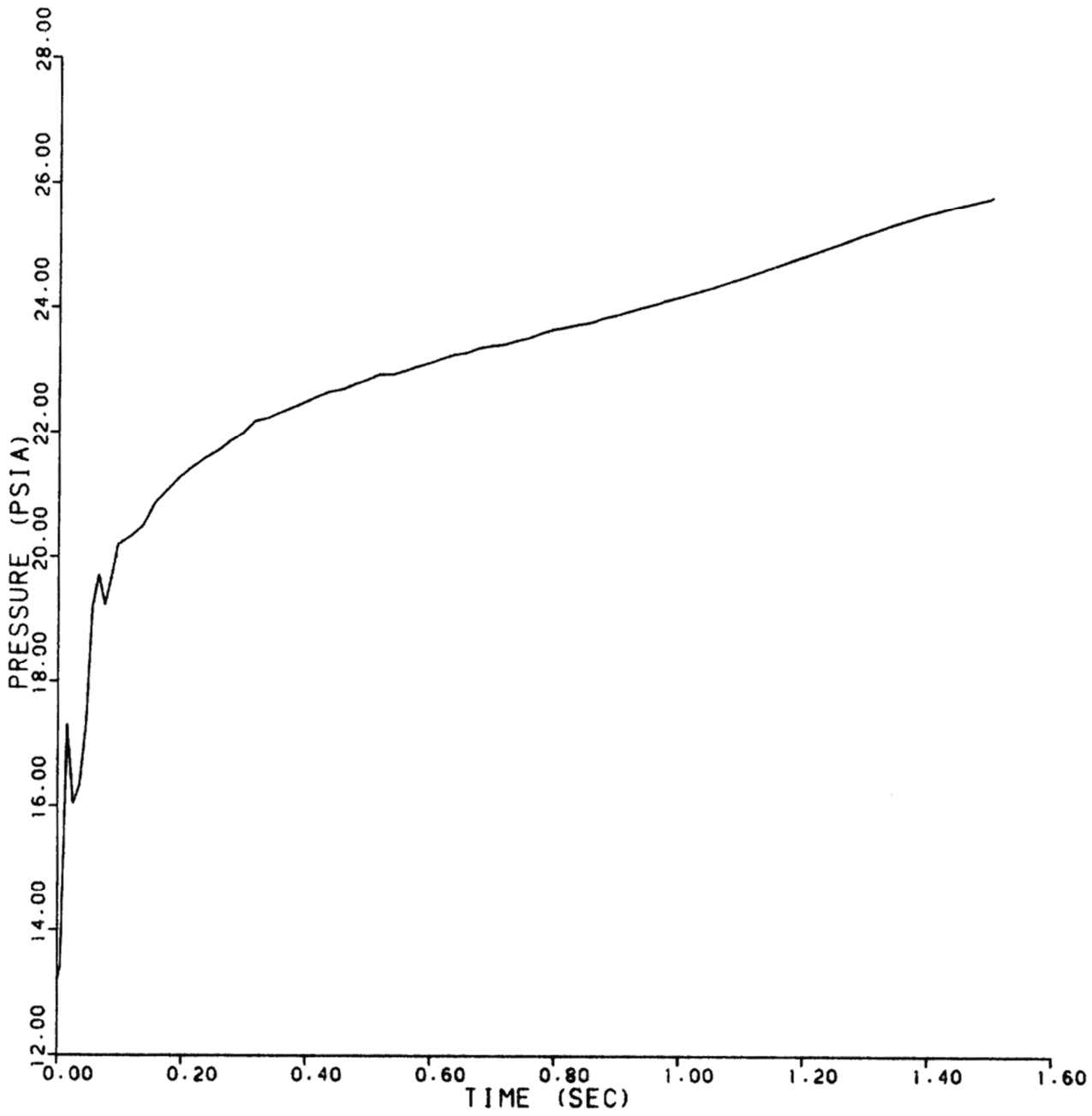
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E31
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 31 OF 74)



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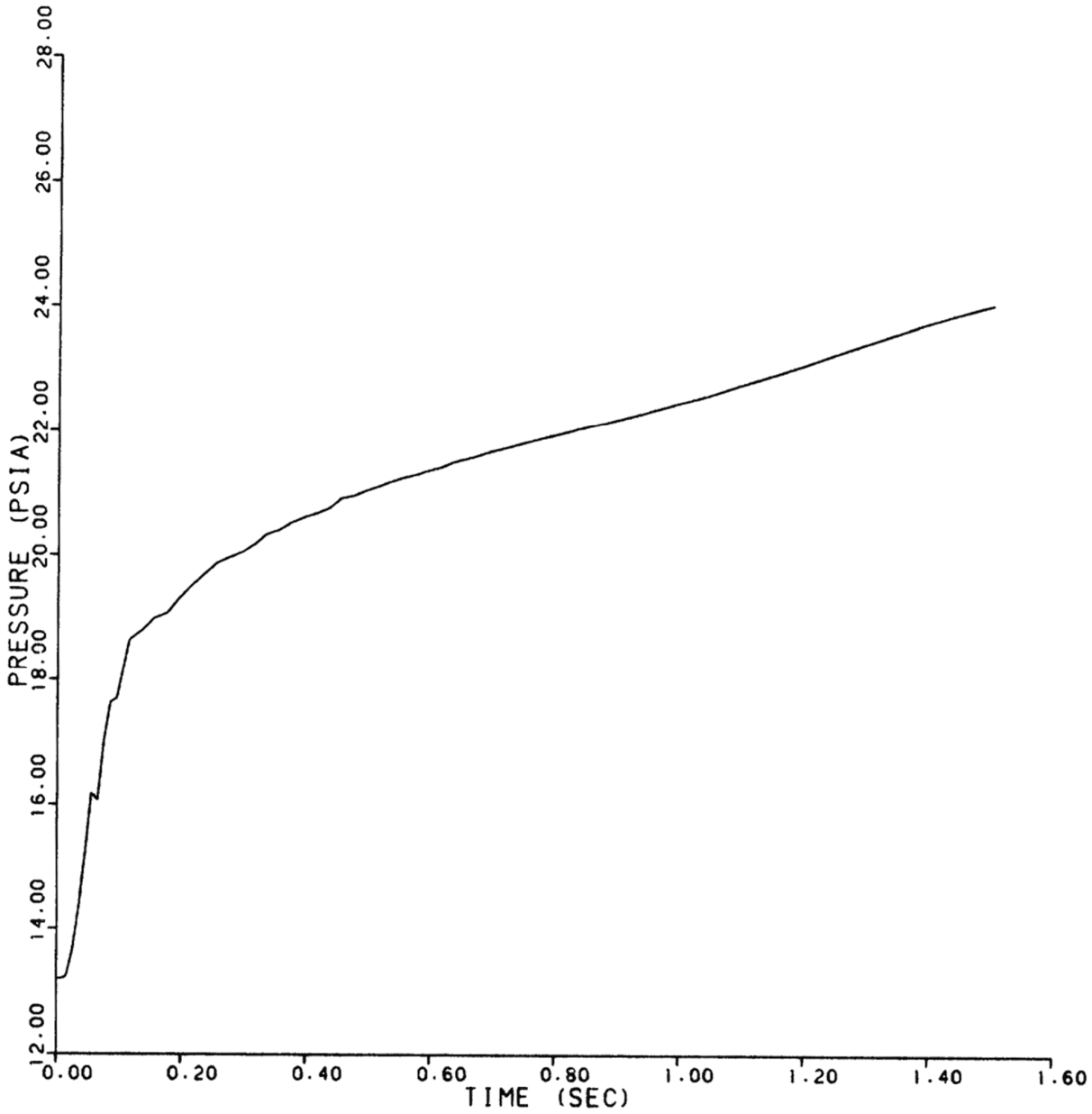
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E32
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 32 OF 74)



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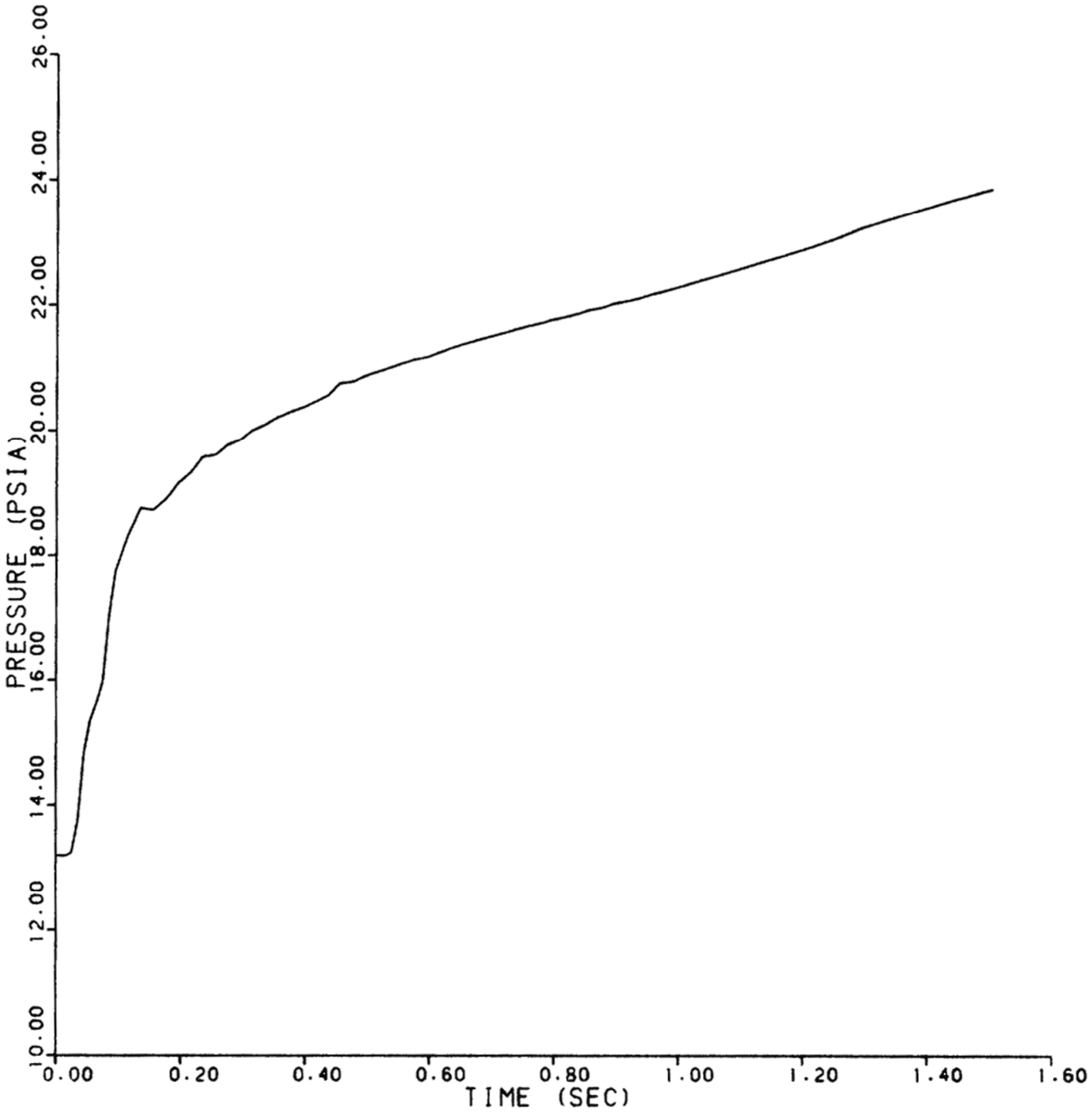
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E33
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 33 OF 74)



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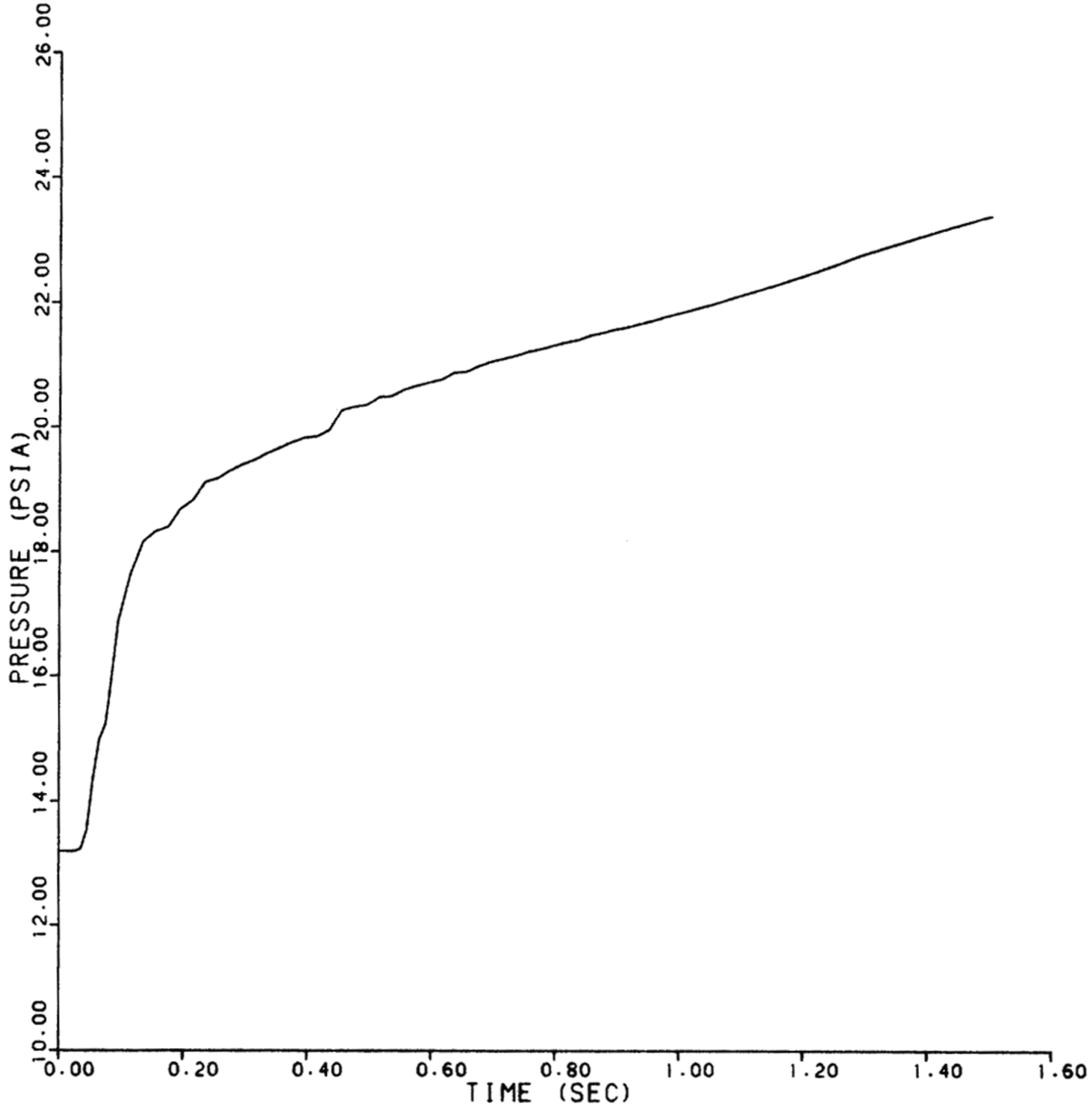
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E34
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 34 OF 74)



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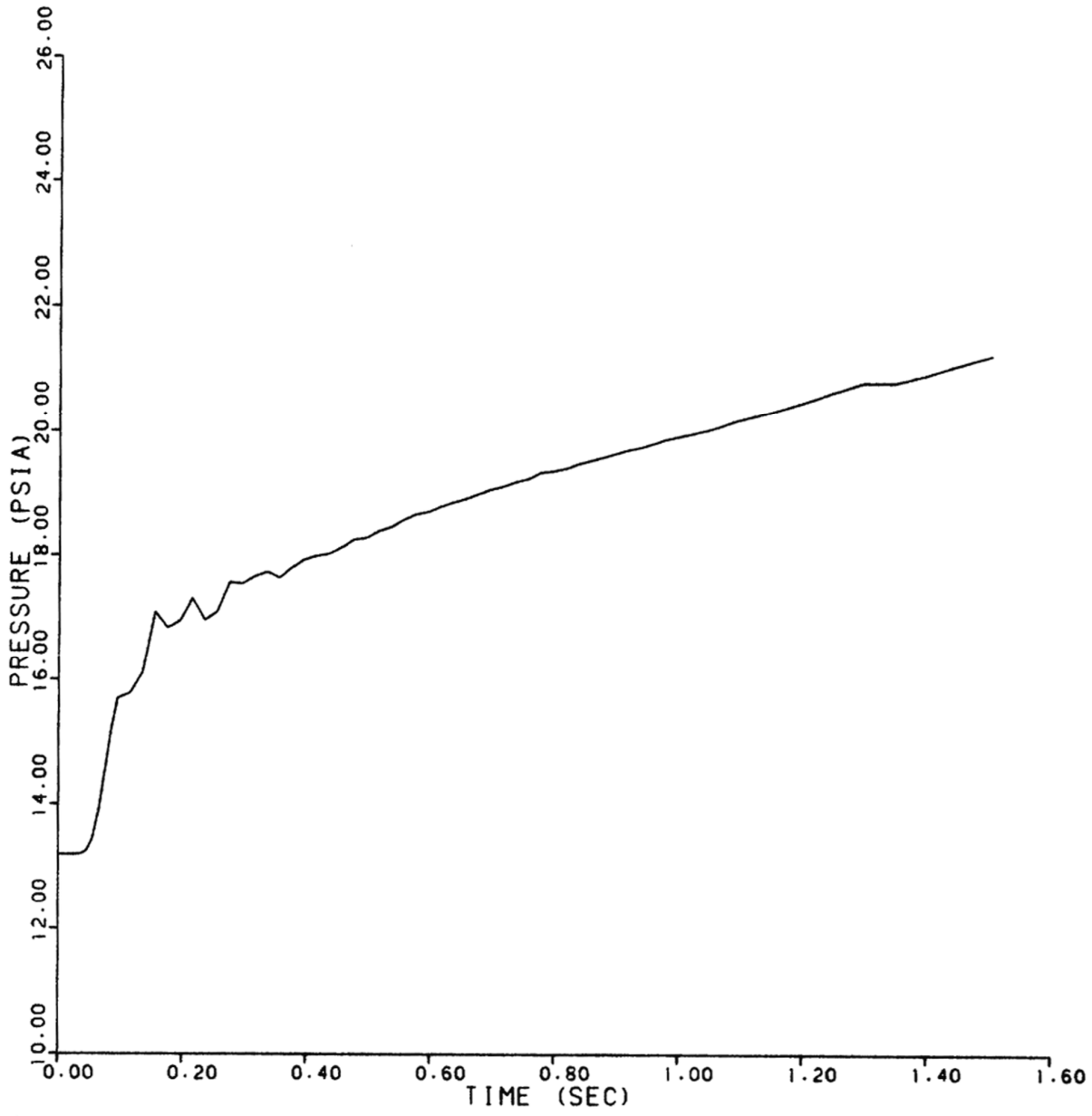
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E35
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 35 OF 74)



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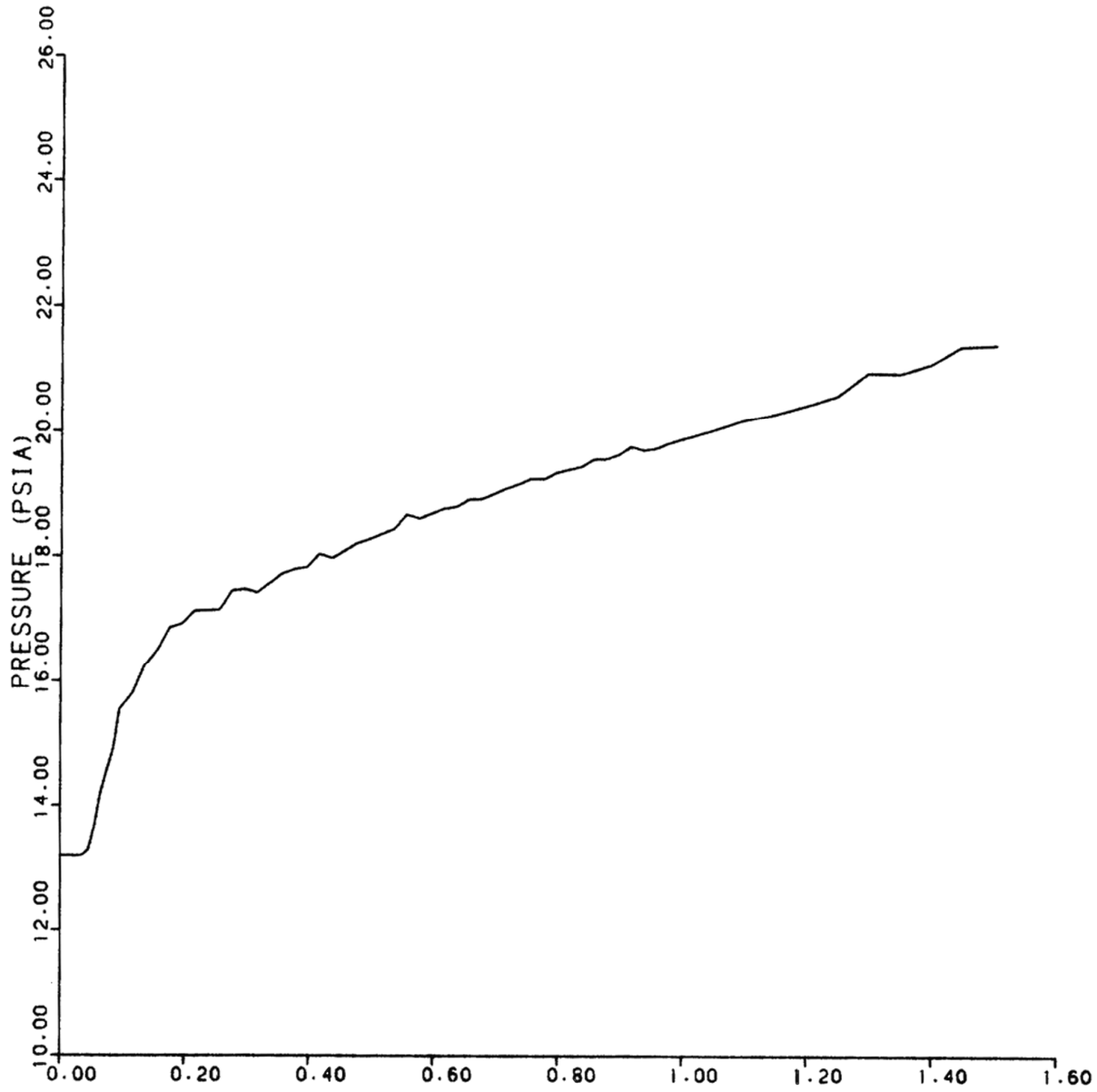
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E36
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 36 OF 74)



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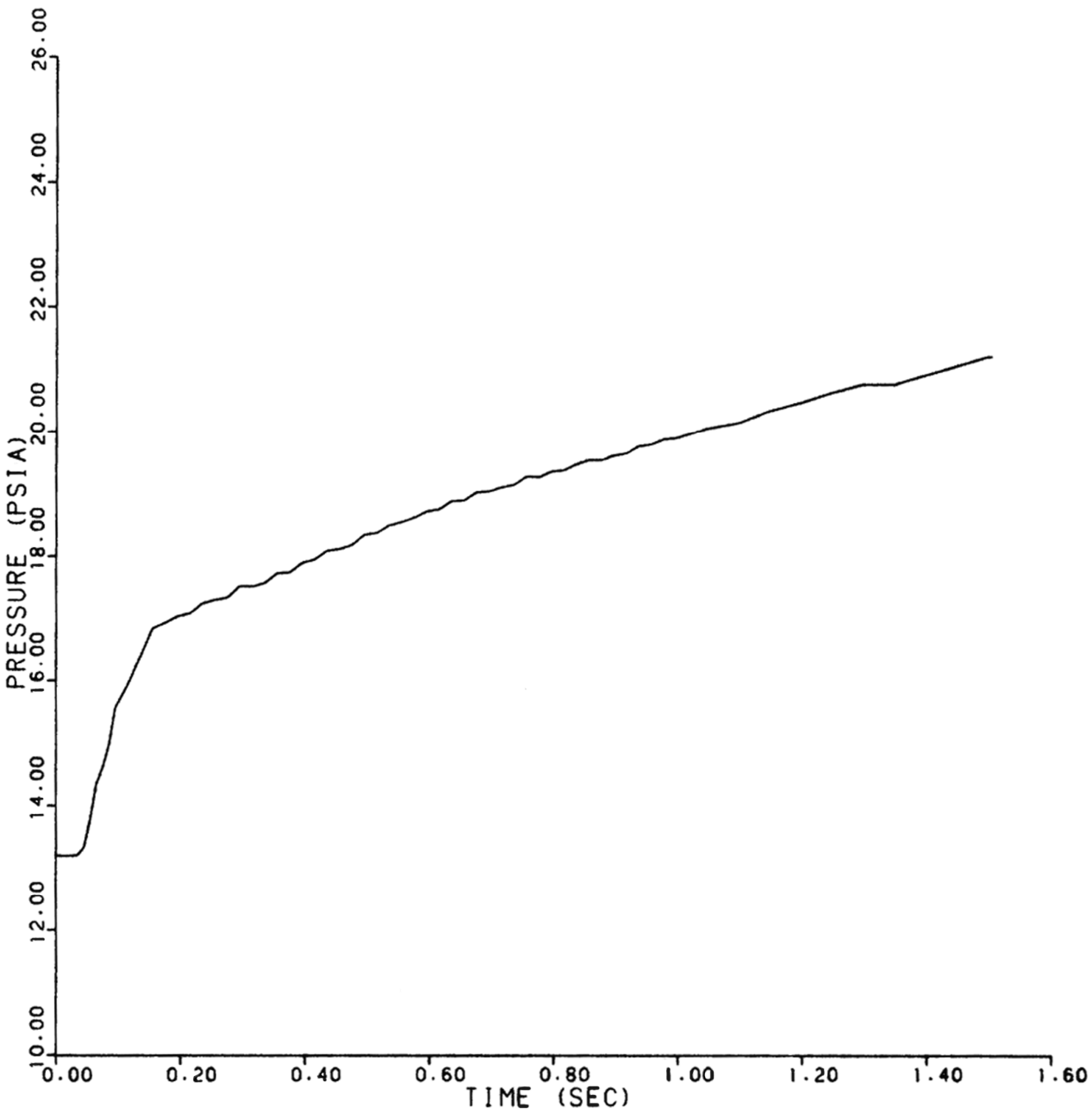
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E37
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 37 OF 74)



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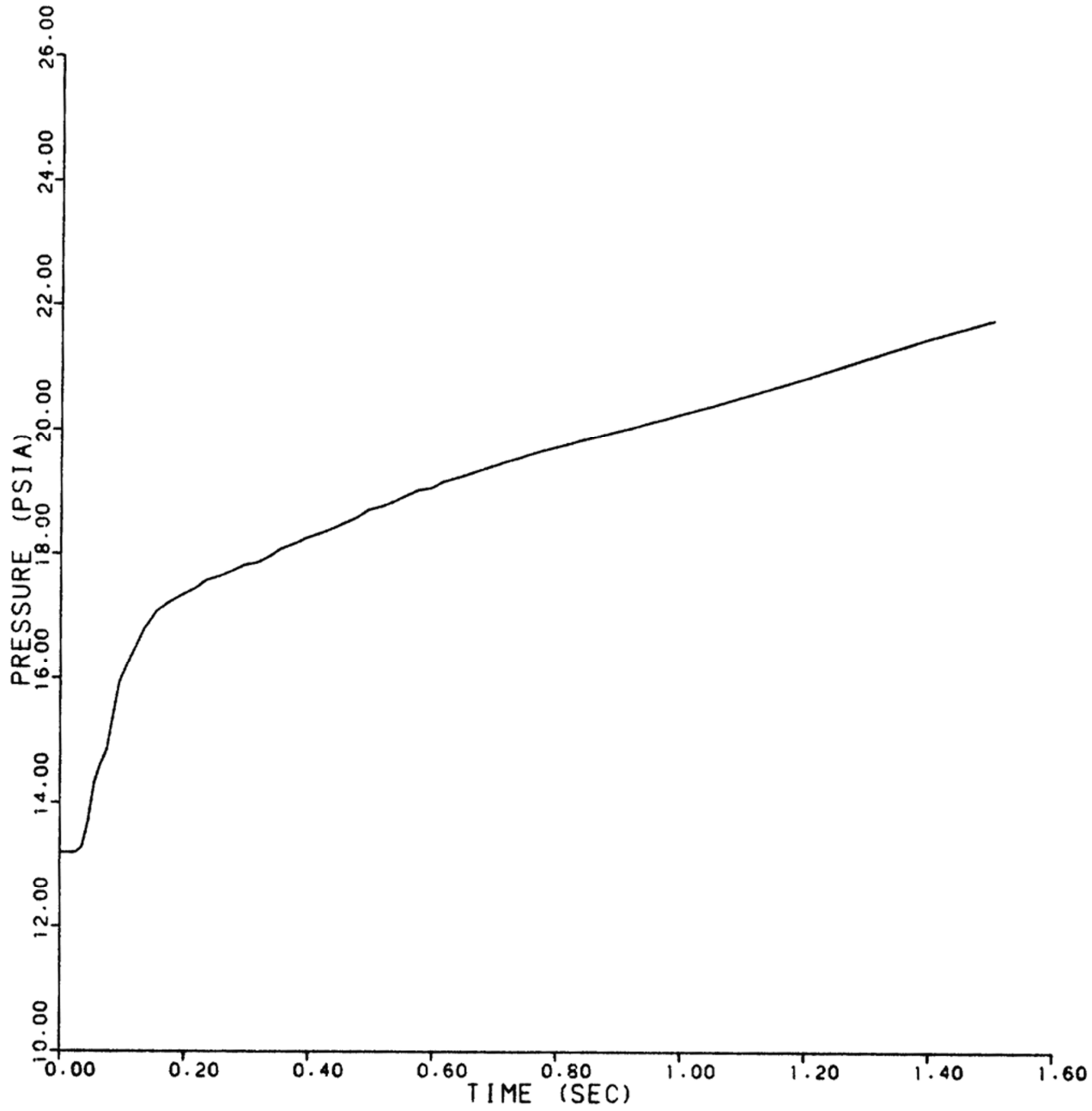
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E38
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 38 OF 74)



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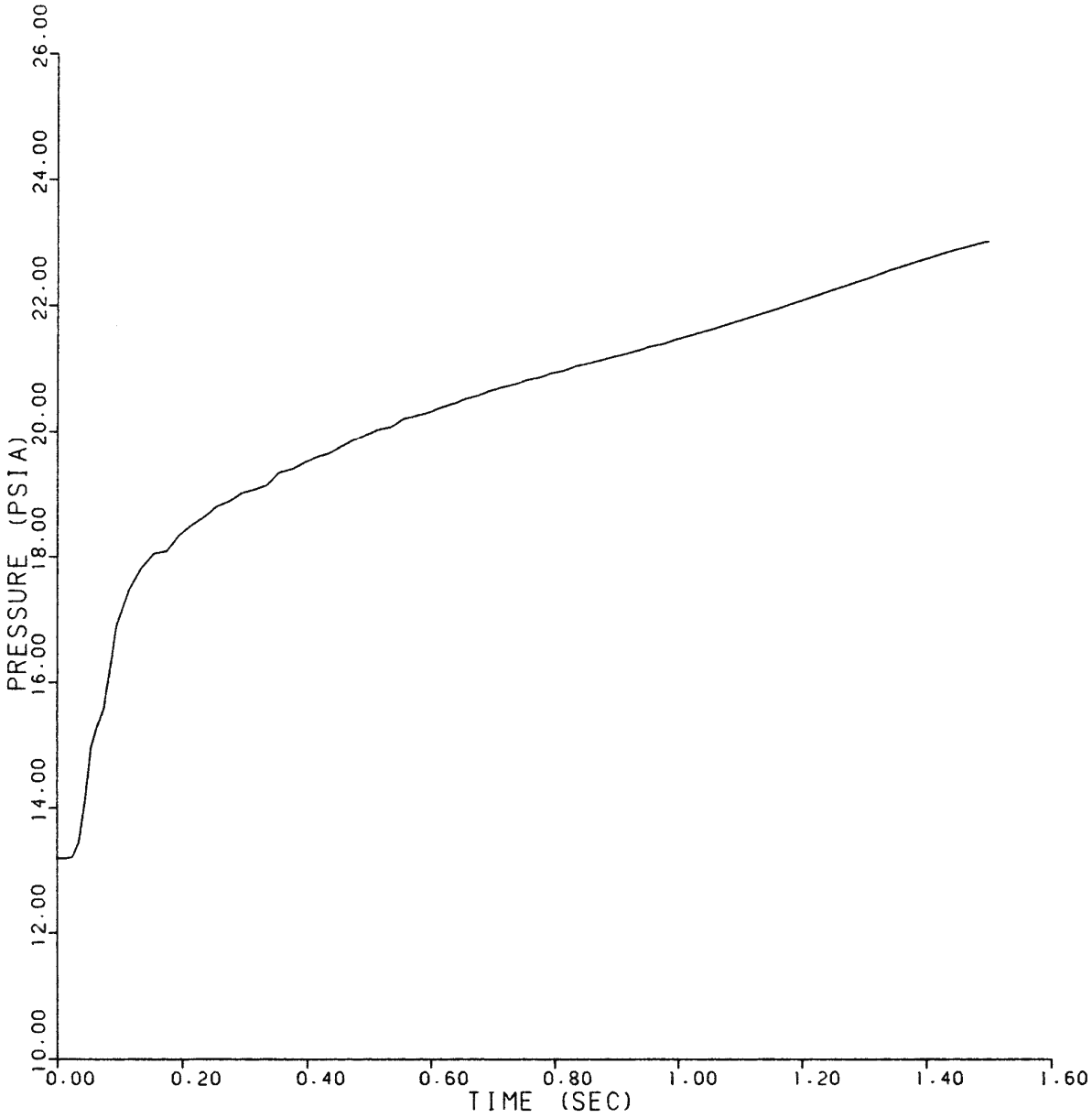
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E39
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 39 OF 74)



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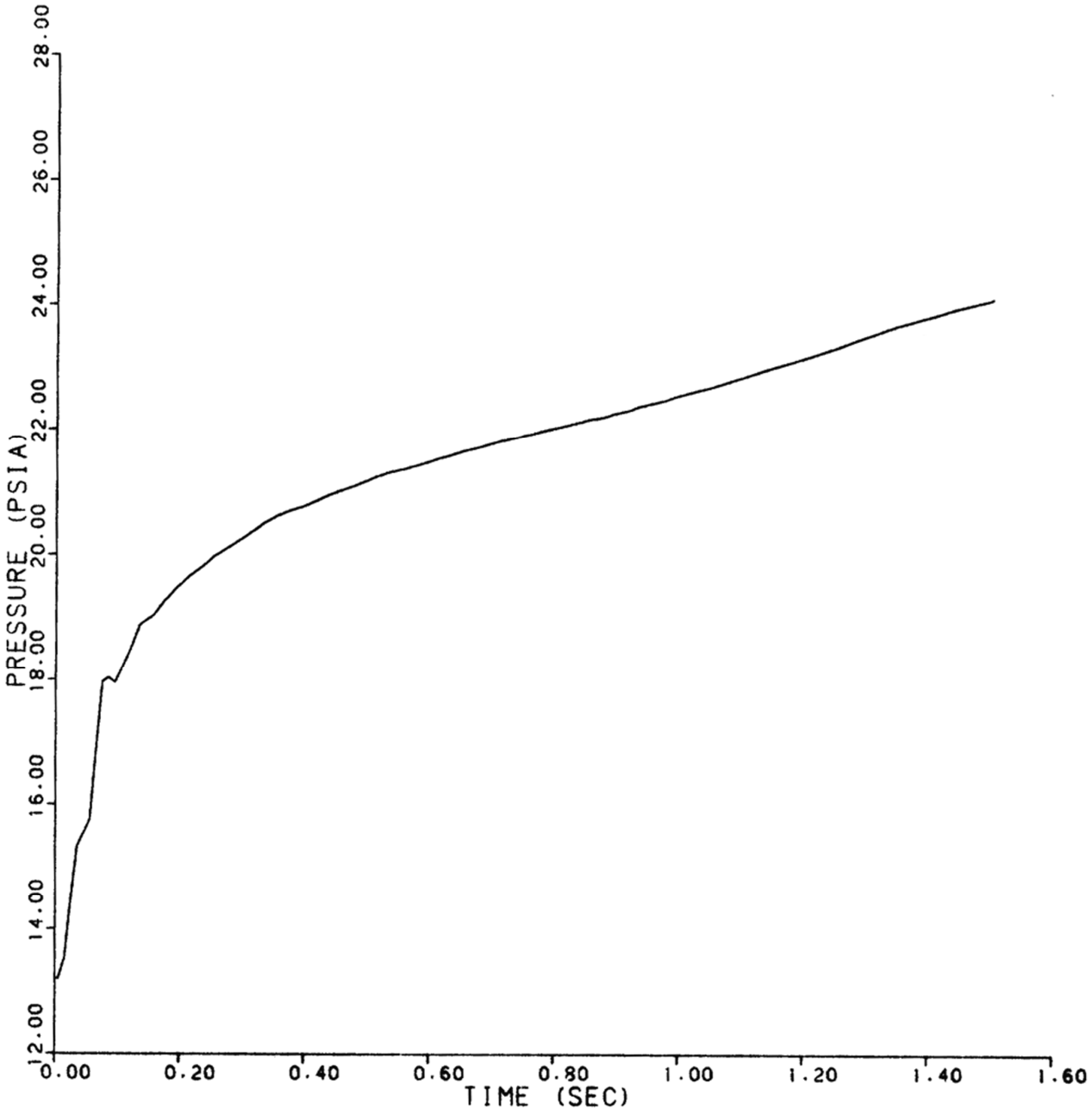
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E40
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 40 OF 74)



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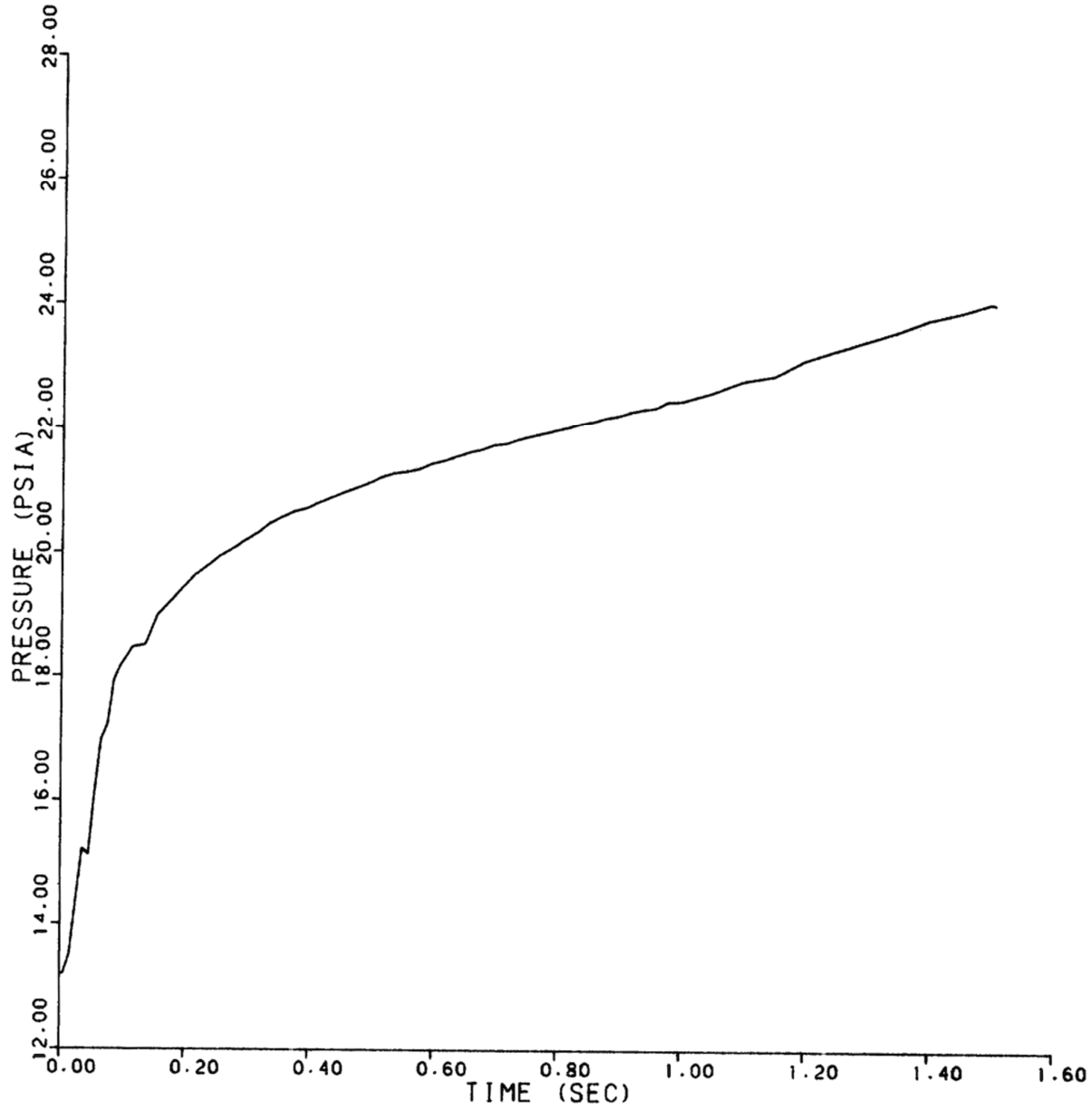
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E41
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 41 OF 74)



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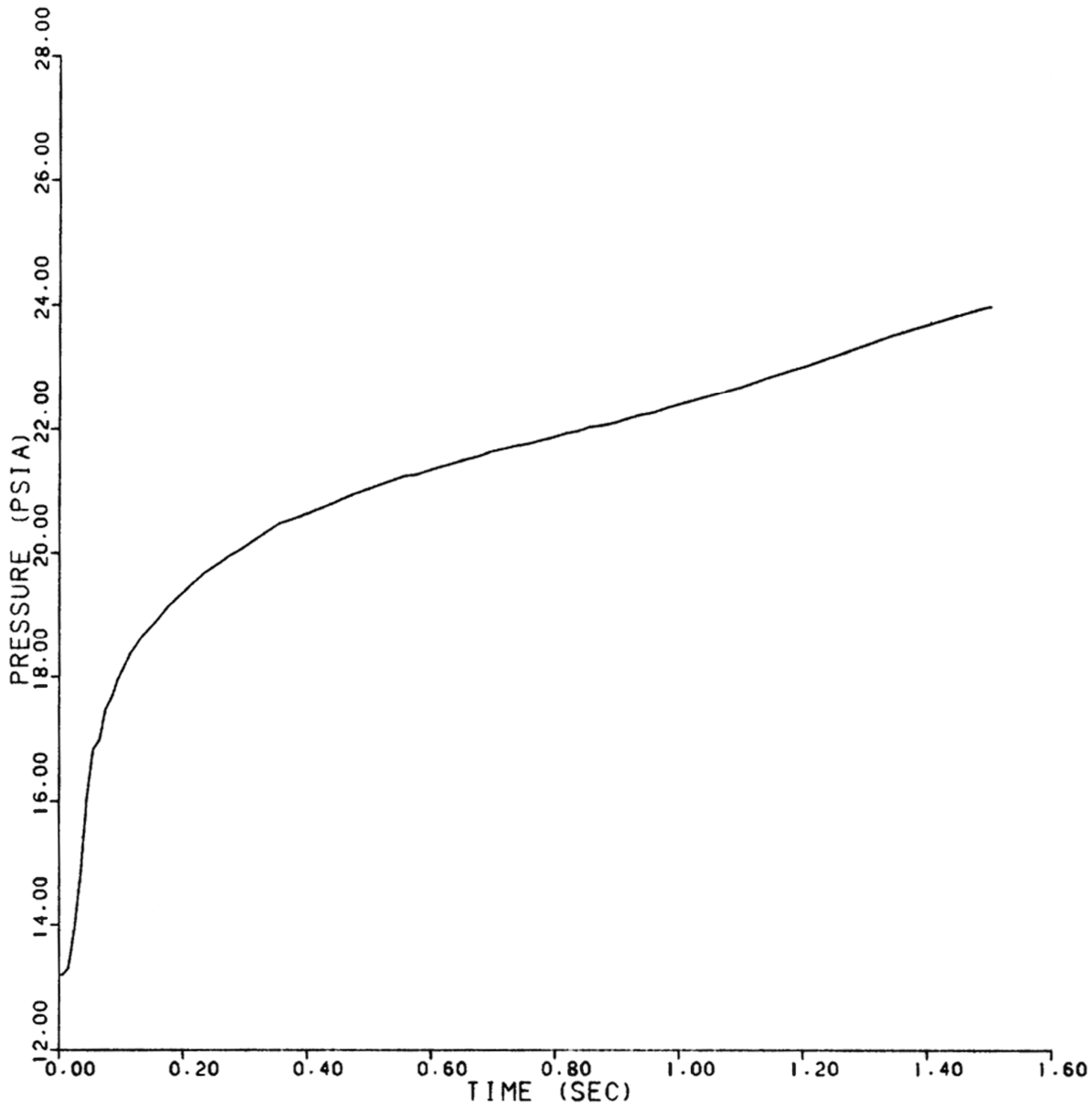
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E42
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 42 OF 74)



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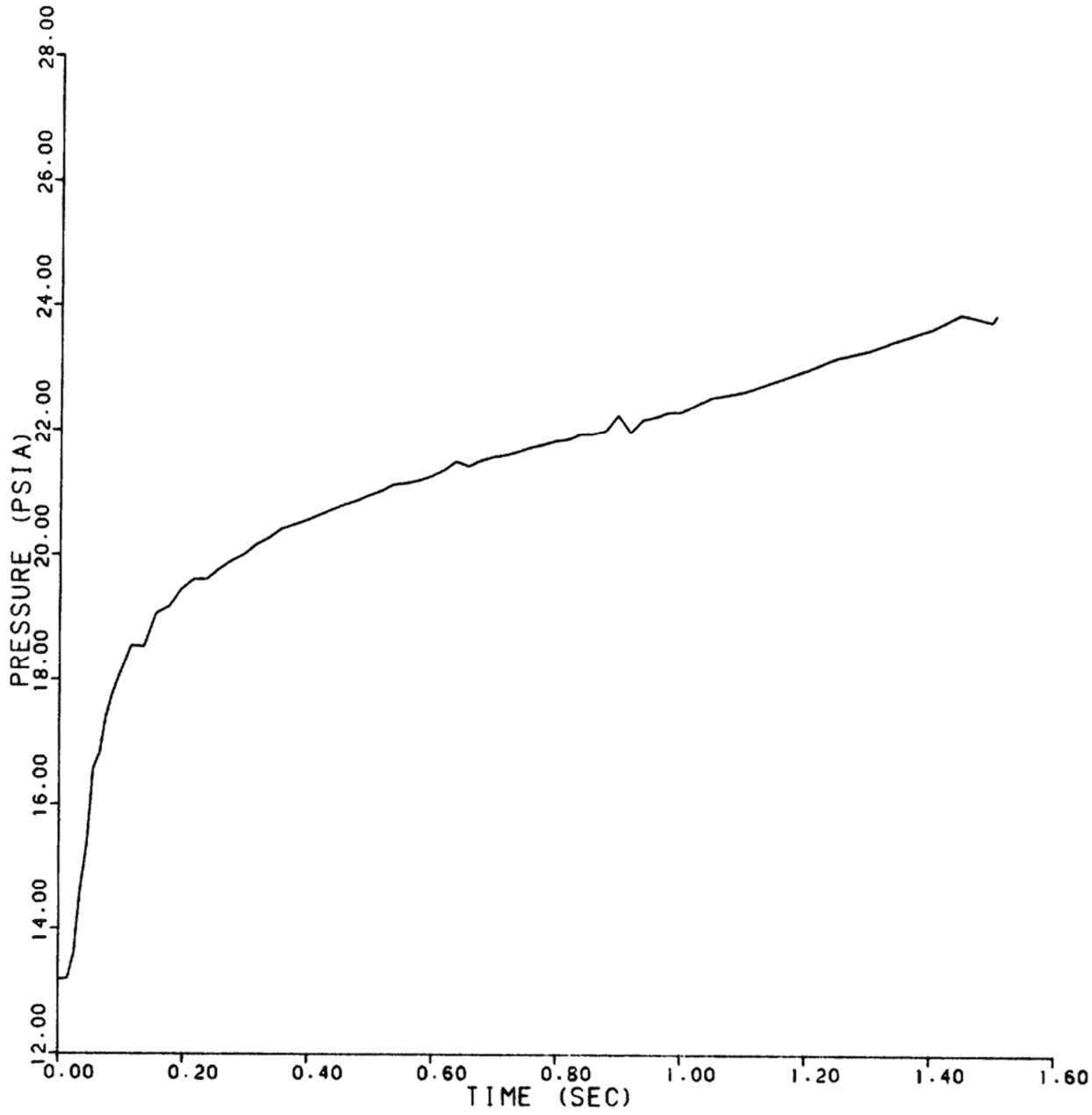
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E43
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 43 OF 74)



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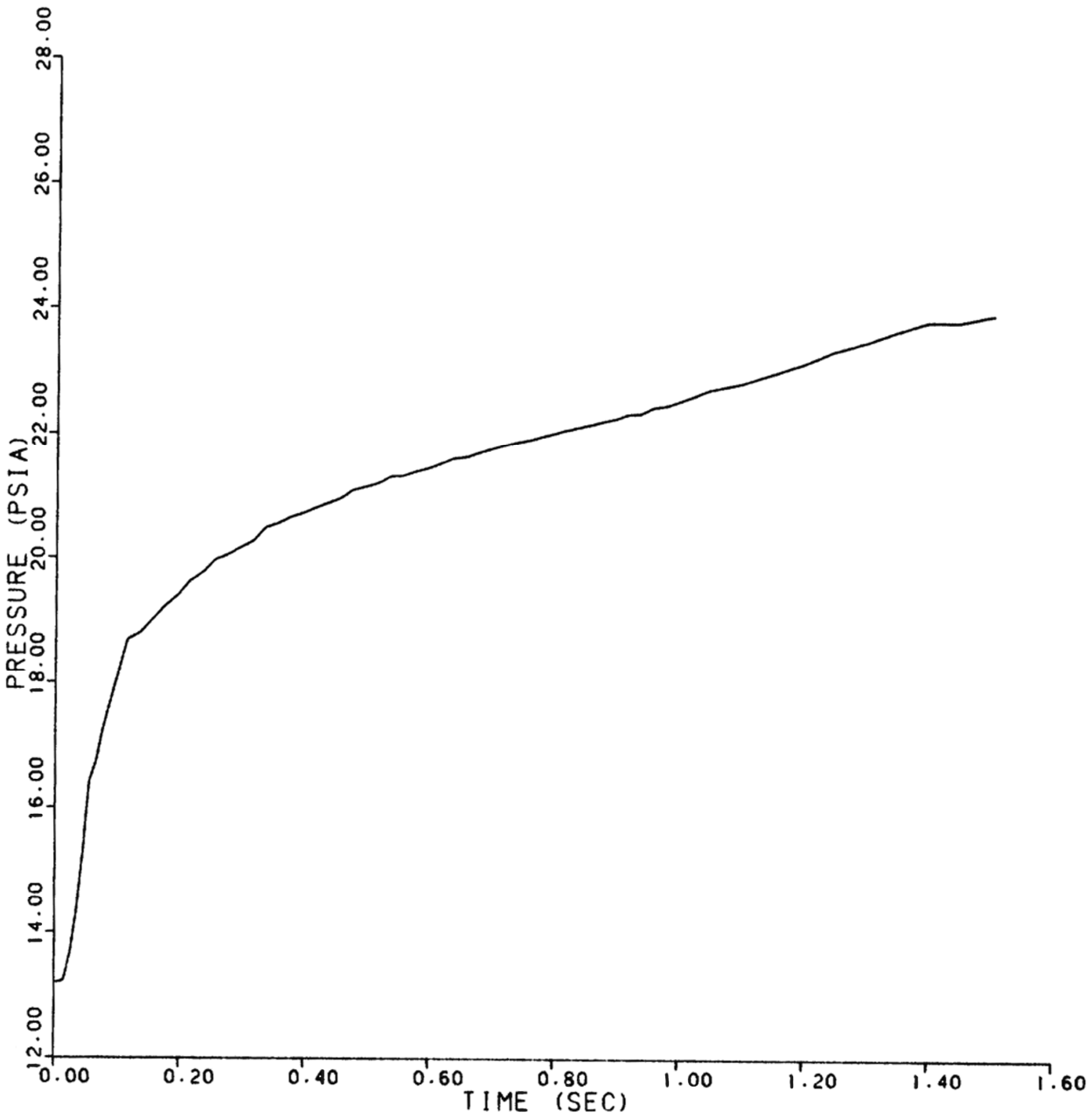
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E44
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 44 OF 74)



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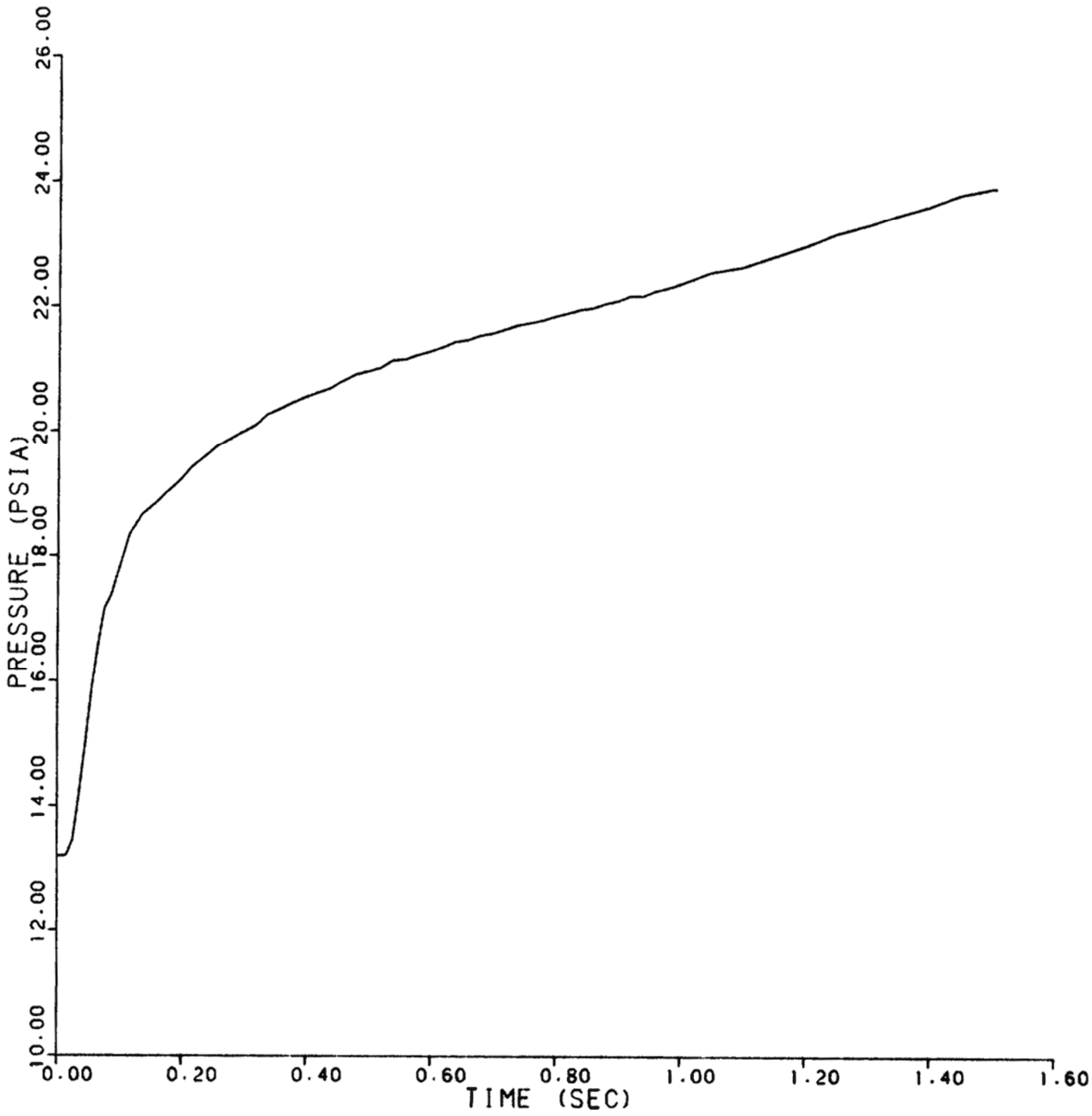
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E45
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 45 OF 74)



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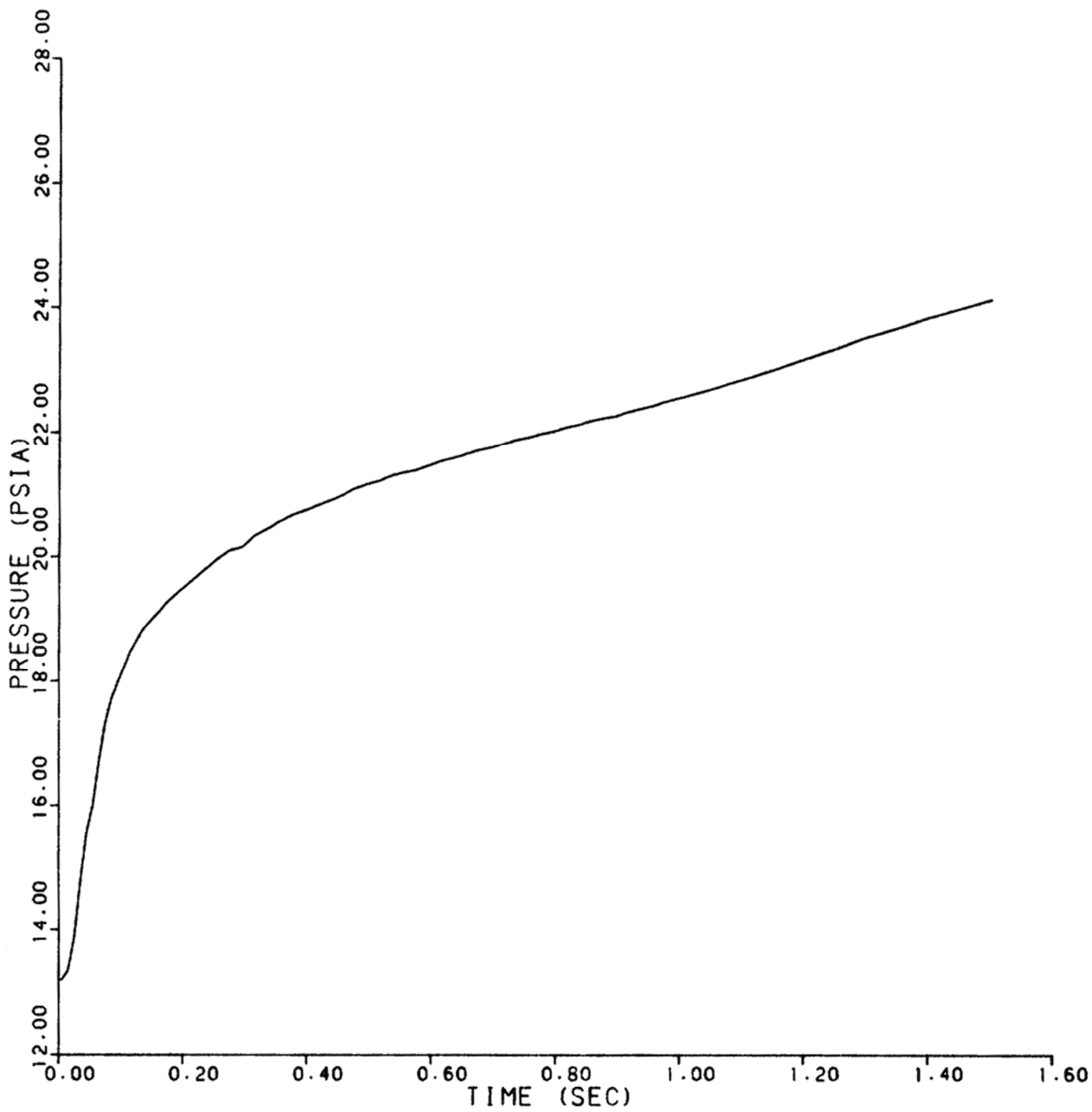
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E46
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 46 OF 74)



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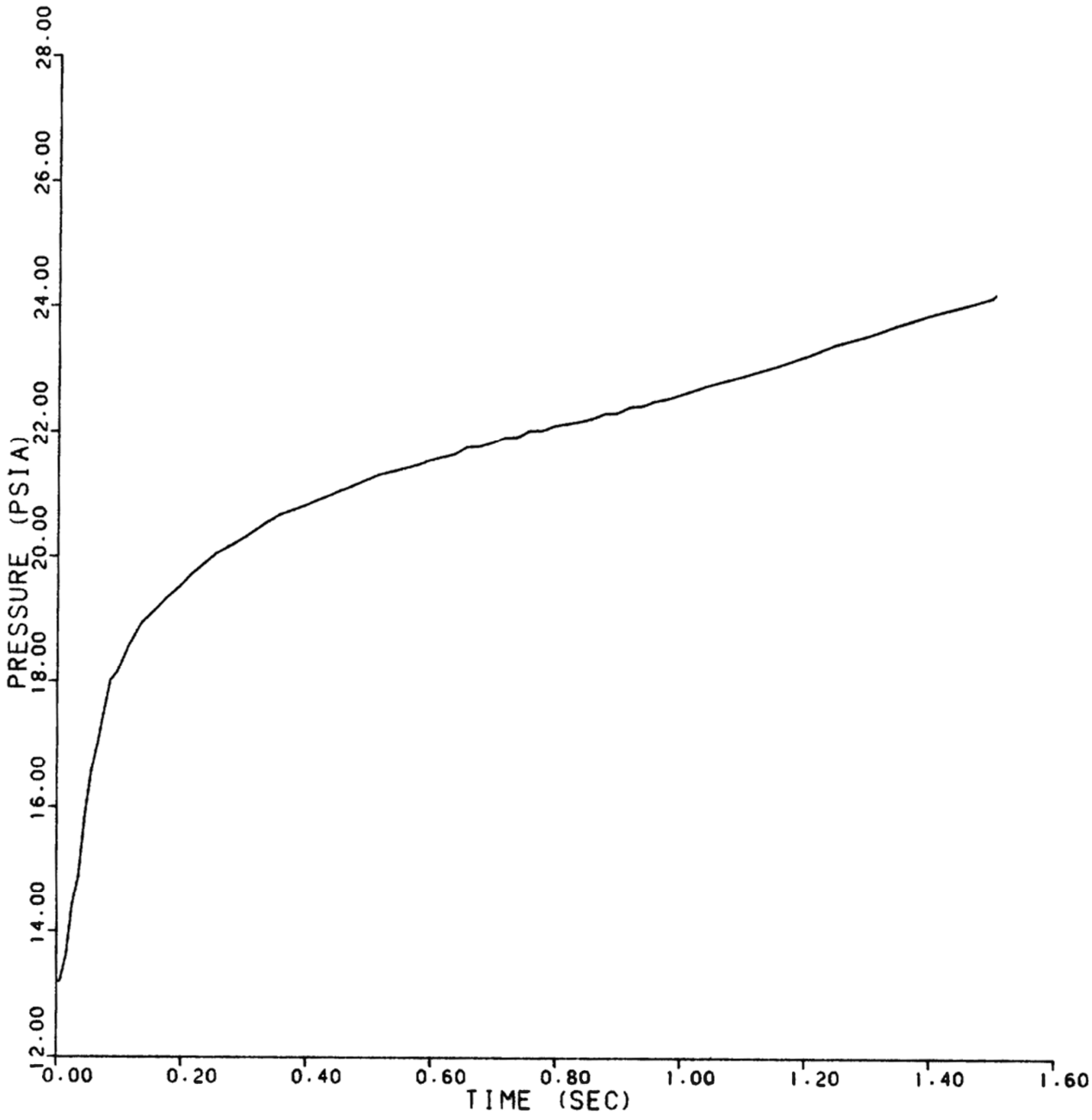
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E47
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 47 OF 74)



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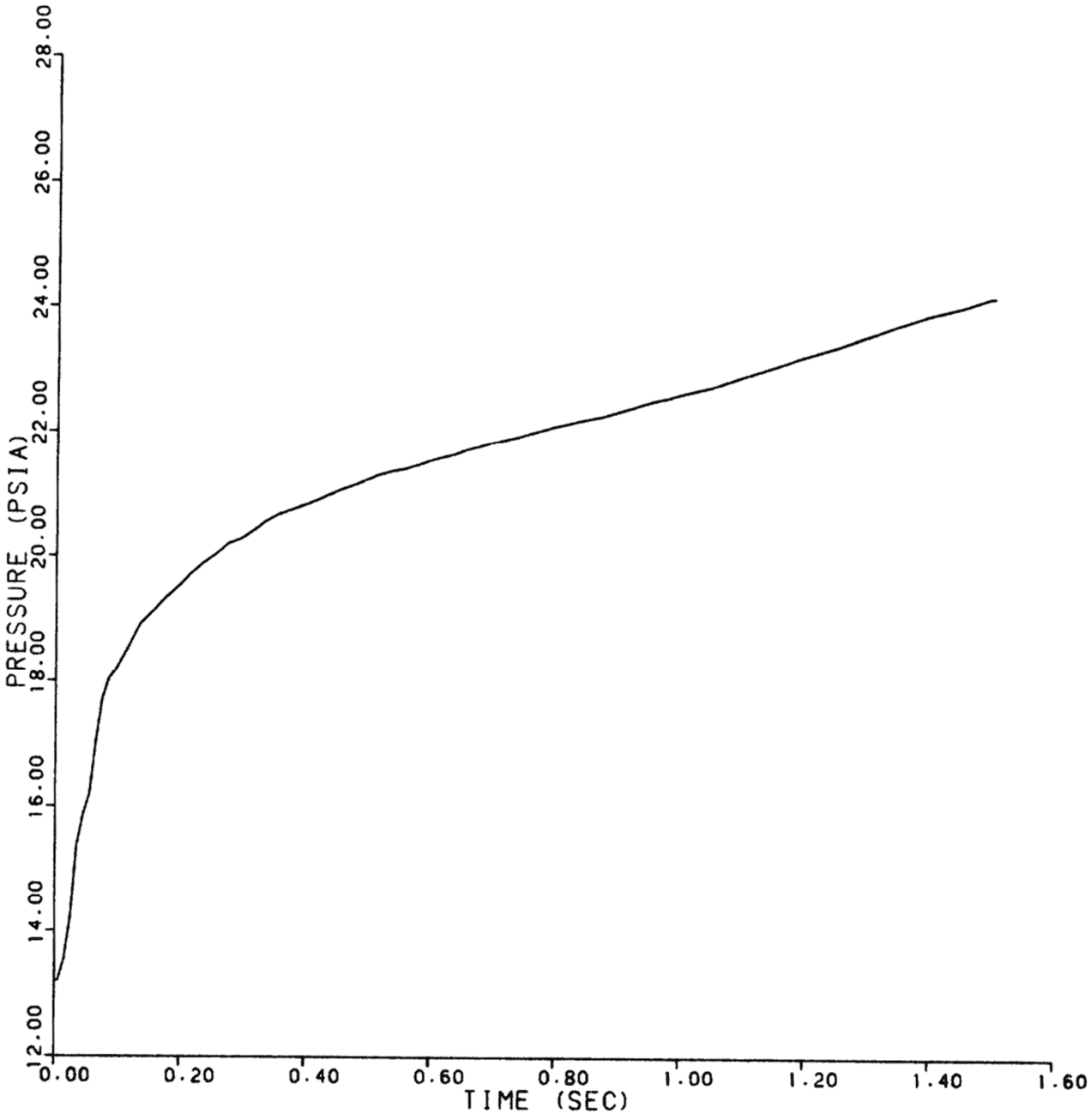
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E48
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 48 OF 74)



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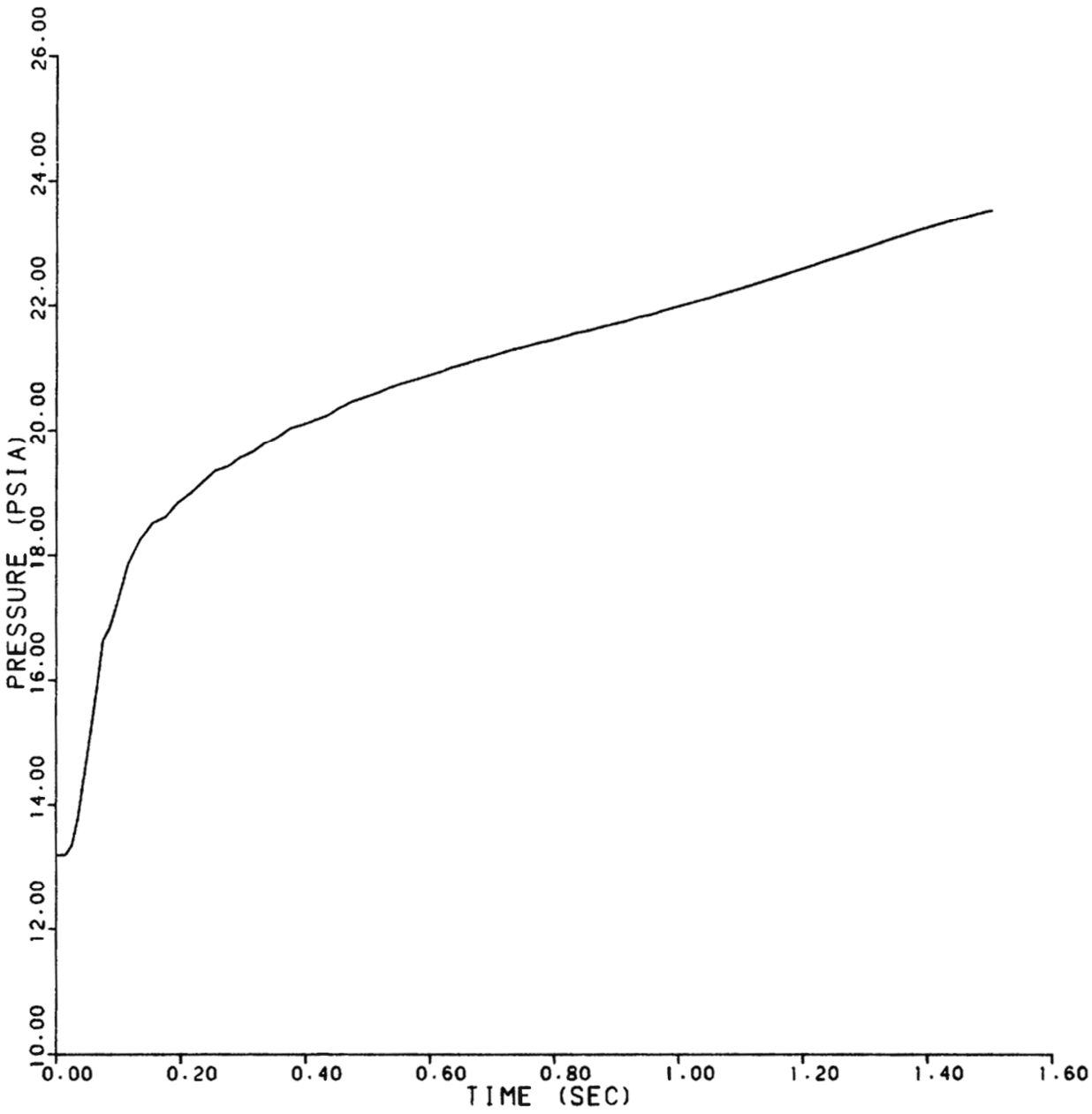
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E49
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 49 OF 74)



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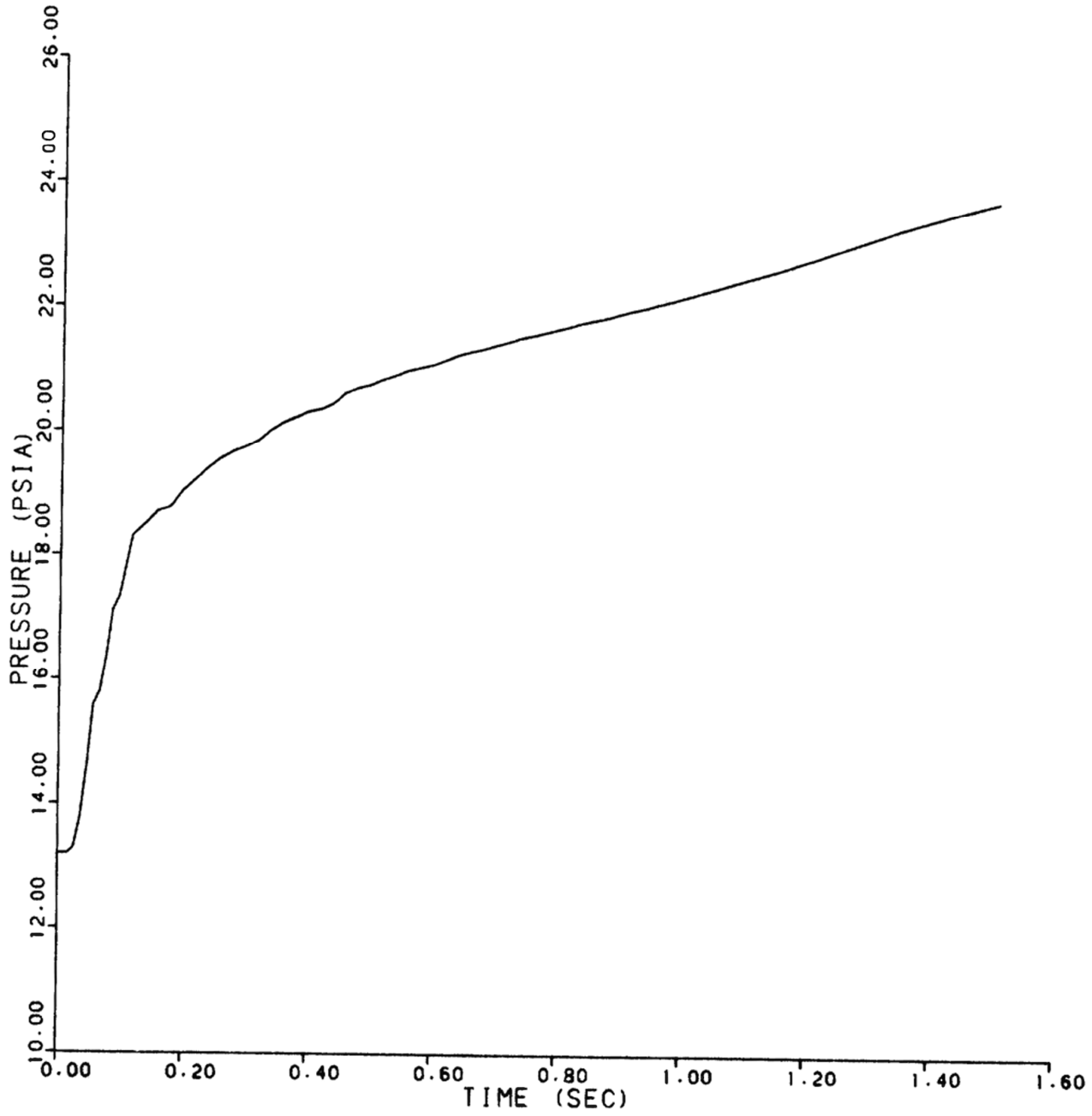
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E50
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 50 OF 74)



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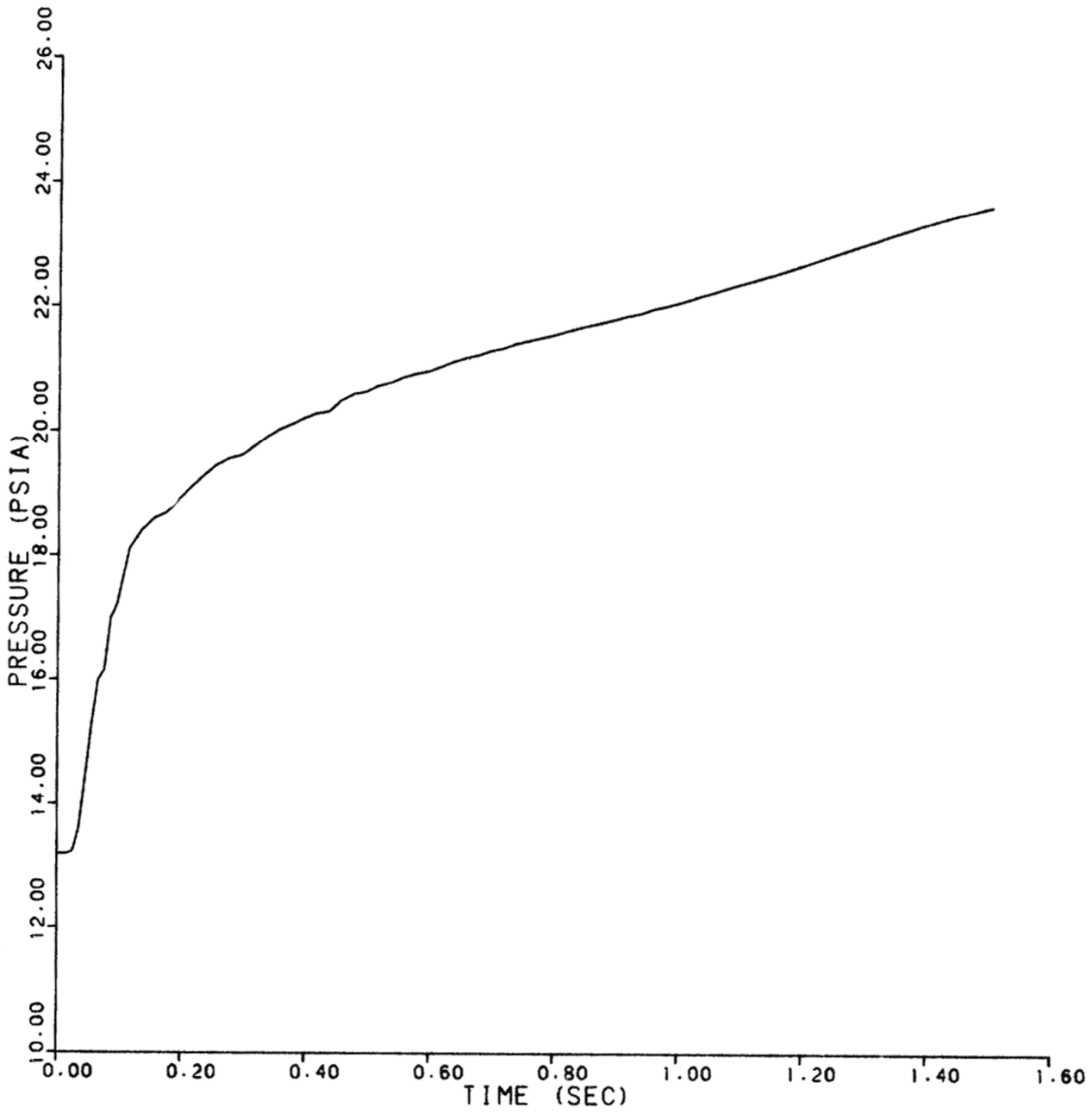
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E51
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 51 OF 74)



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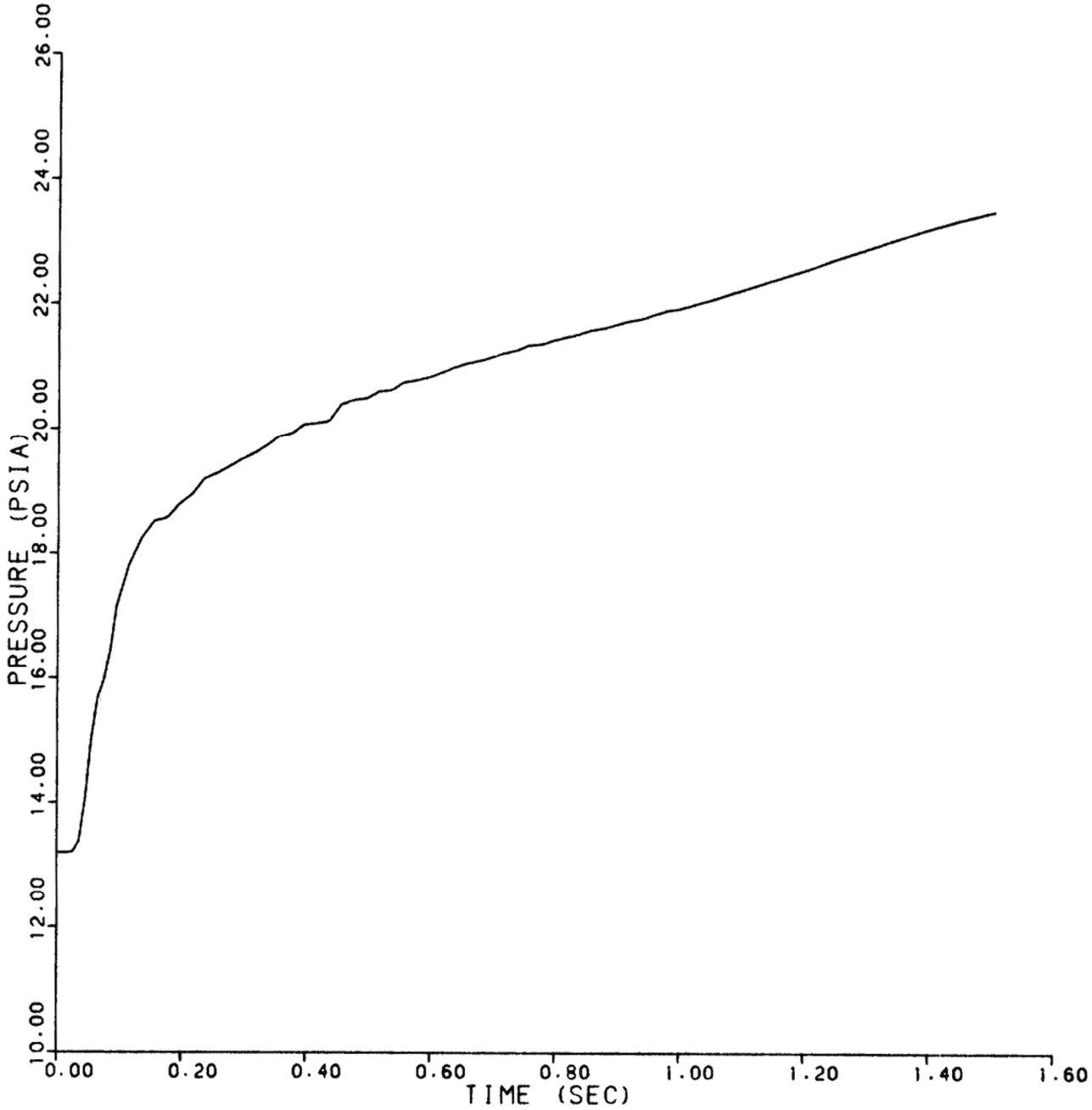
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E52
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 52 OF 74)



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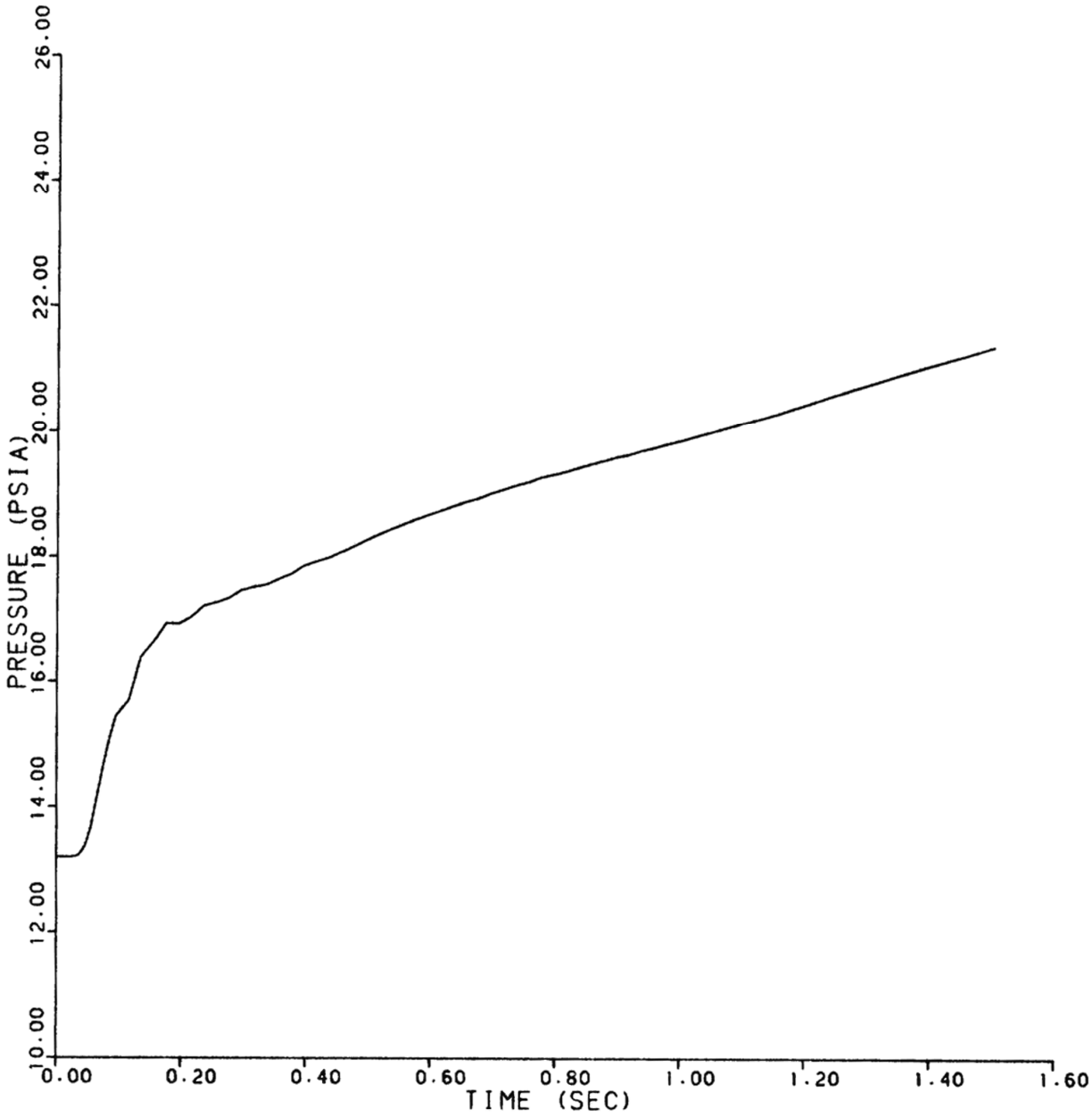
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E53
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 53 OF 74)



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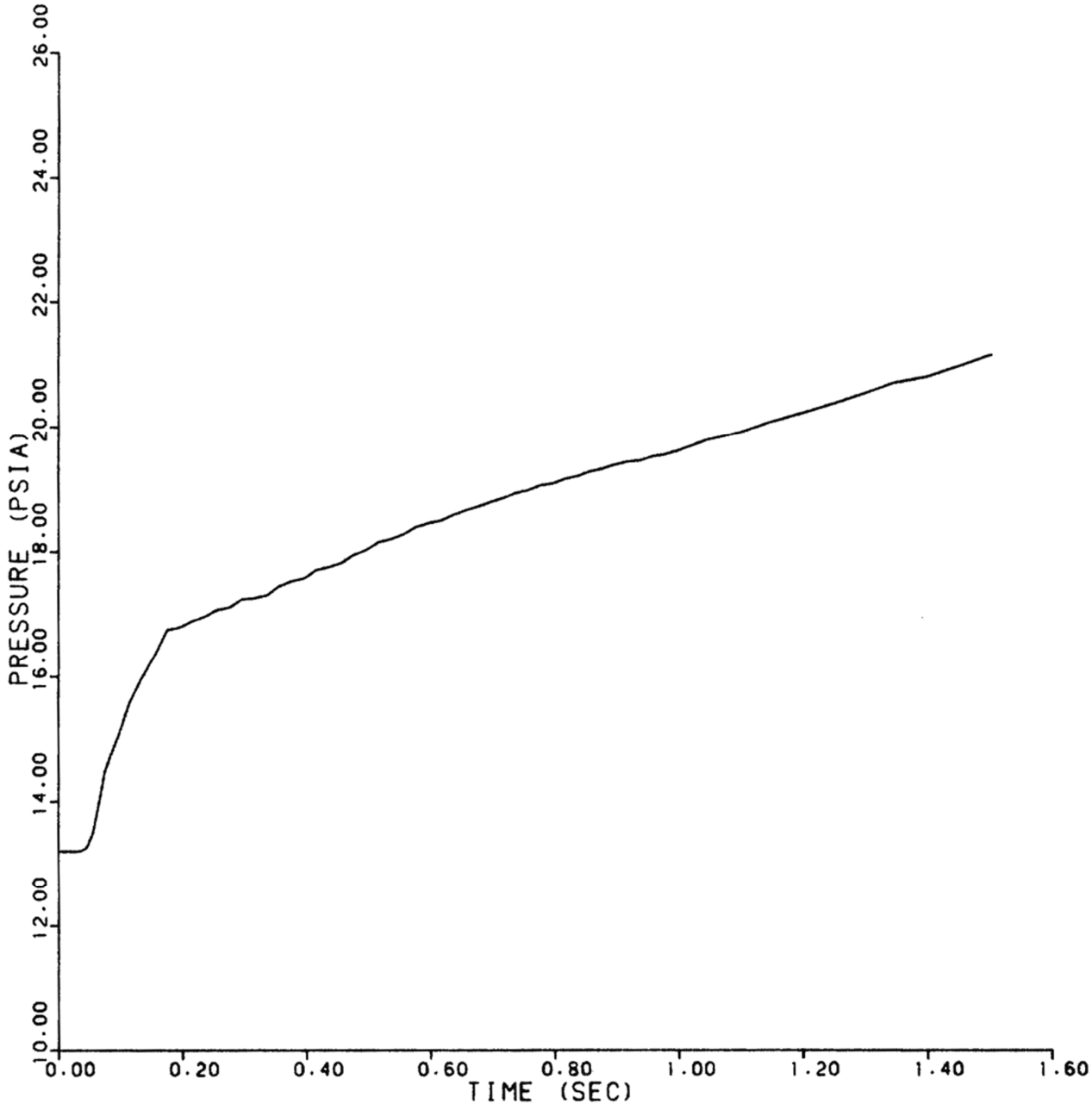
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E54
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 54 OF 74)



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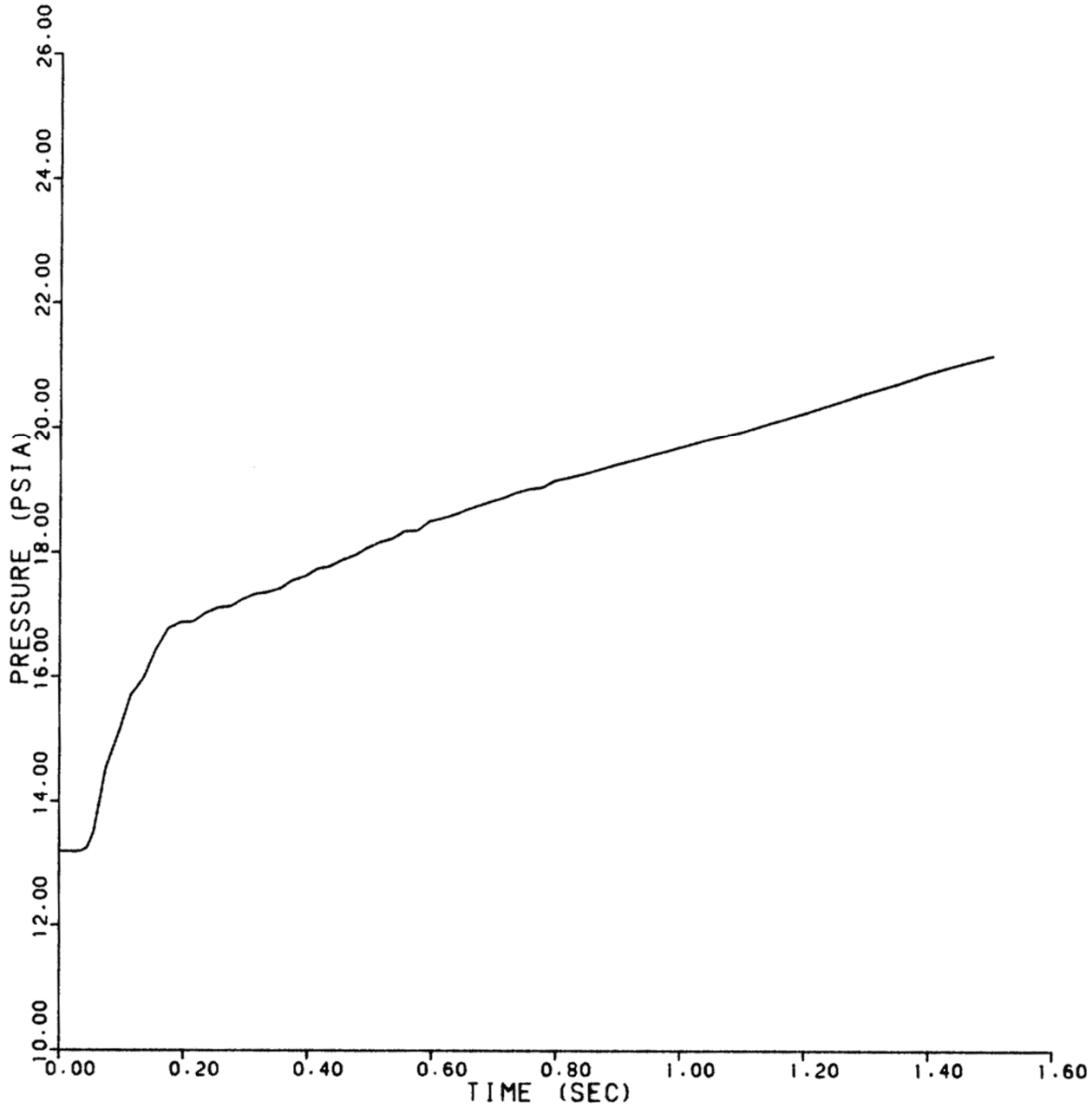
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E55
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 55 OF 74)



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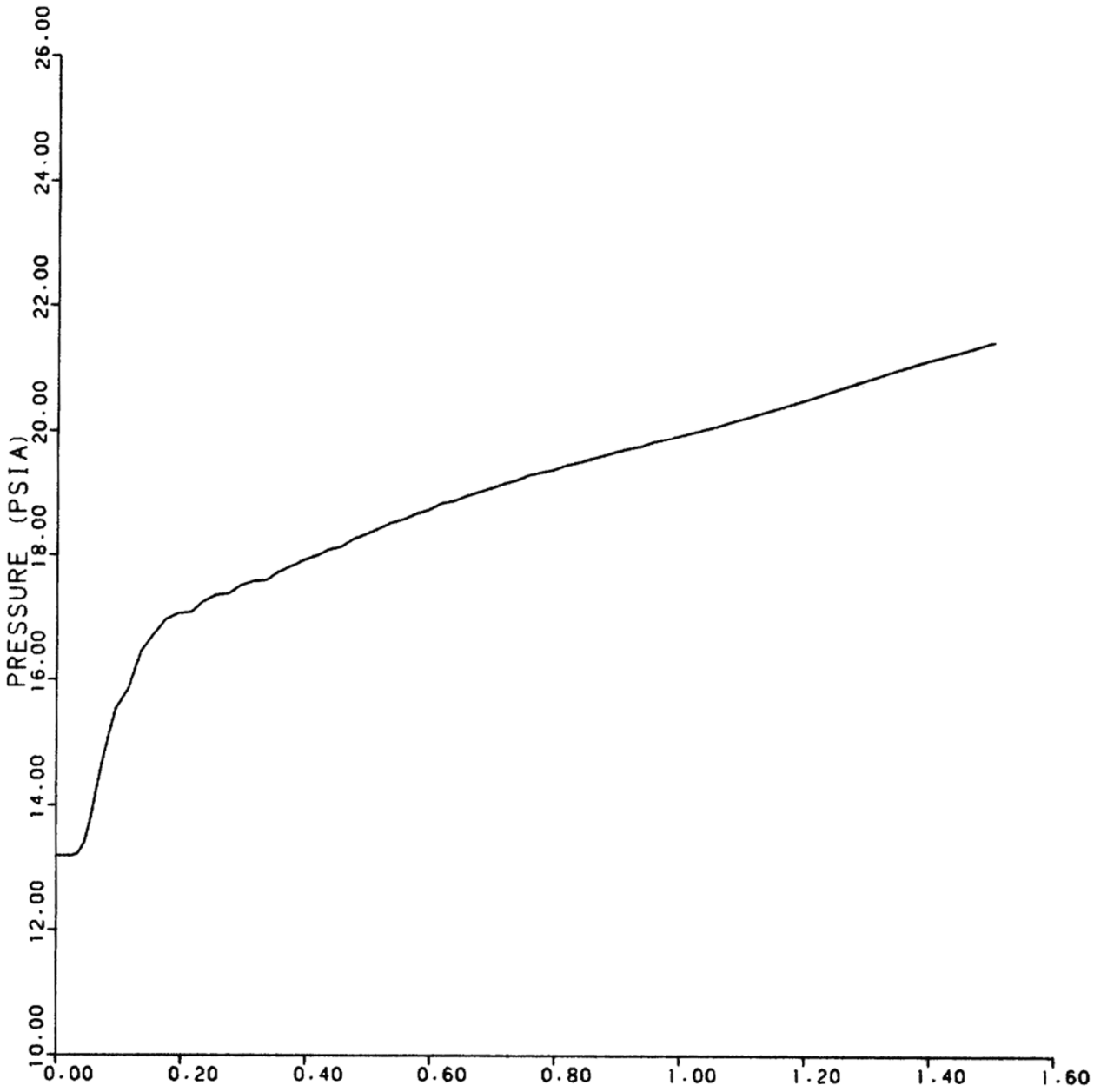
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E56
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 56 OF 74)



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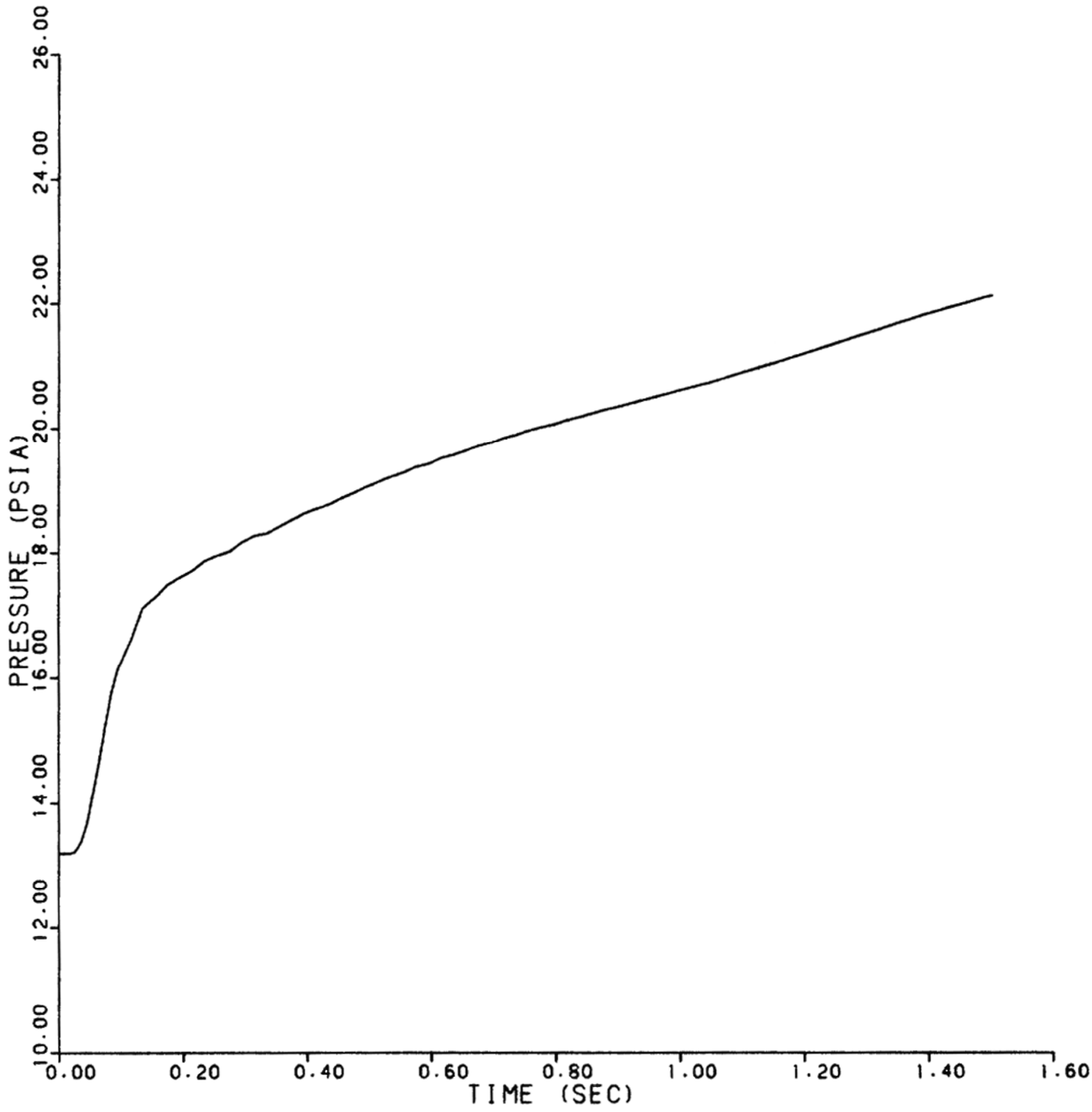
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E57
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 57 OF 74)



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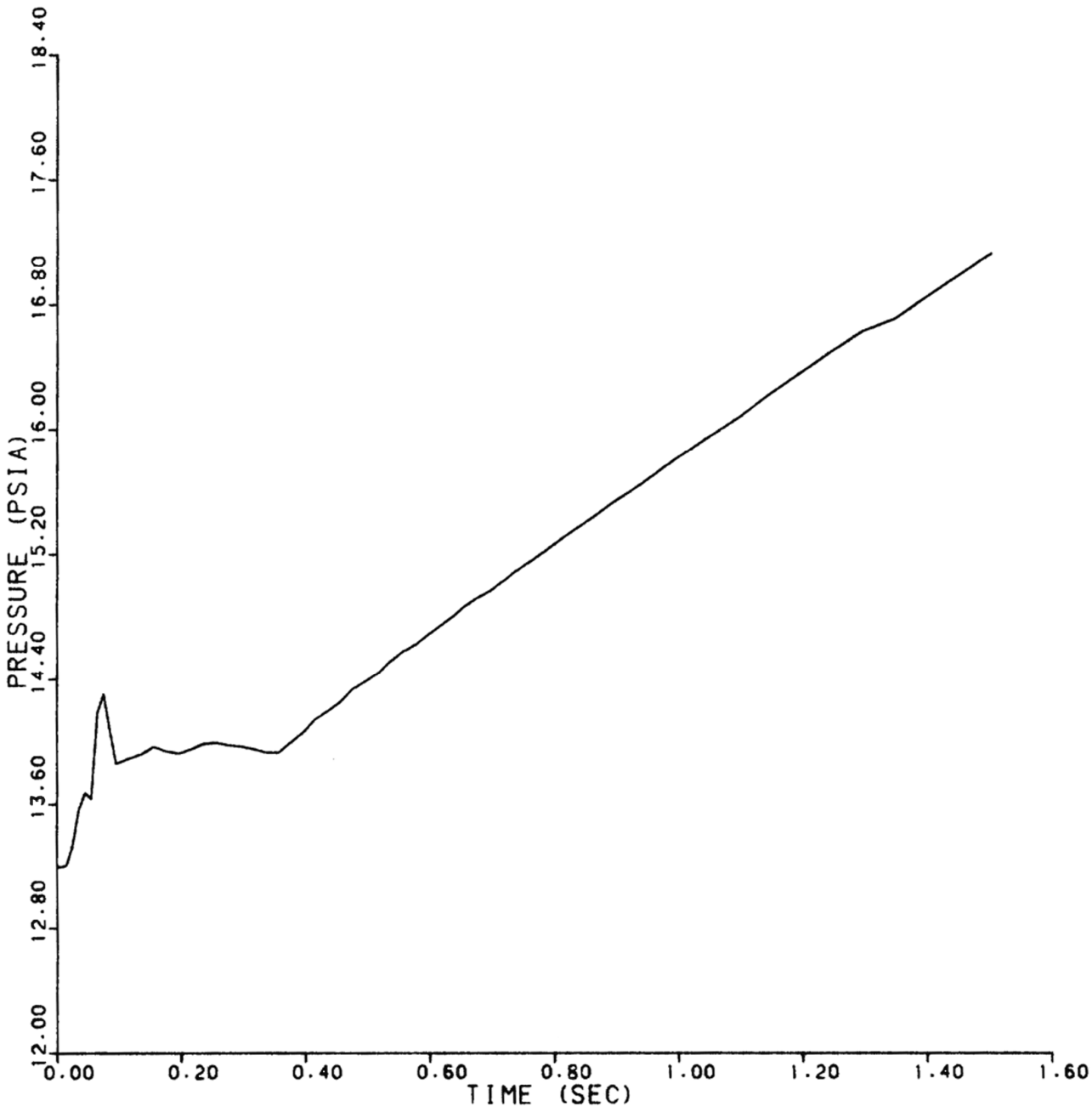
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E58
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 58 OF 74)



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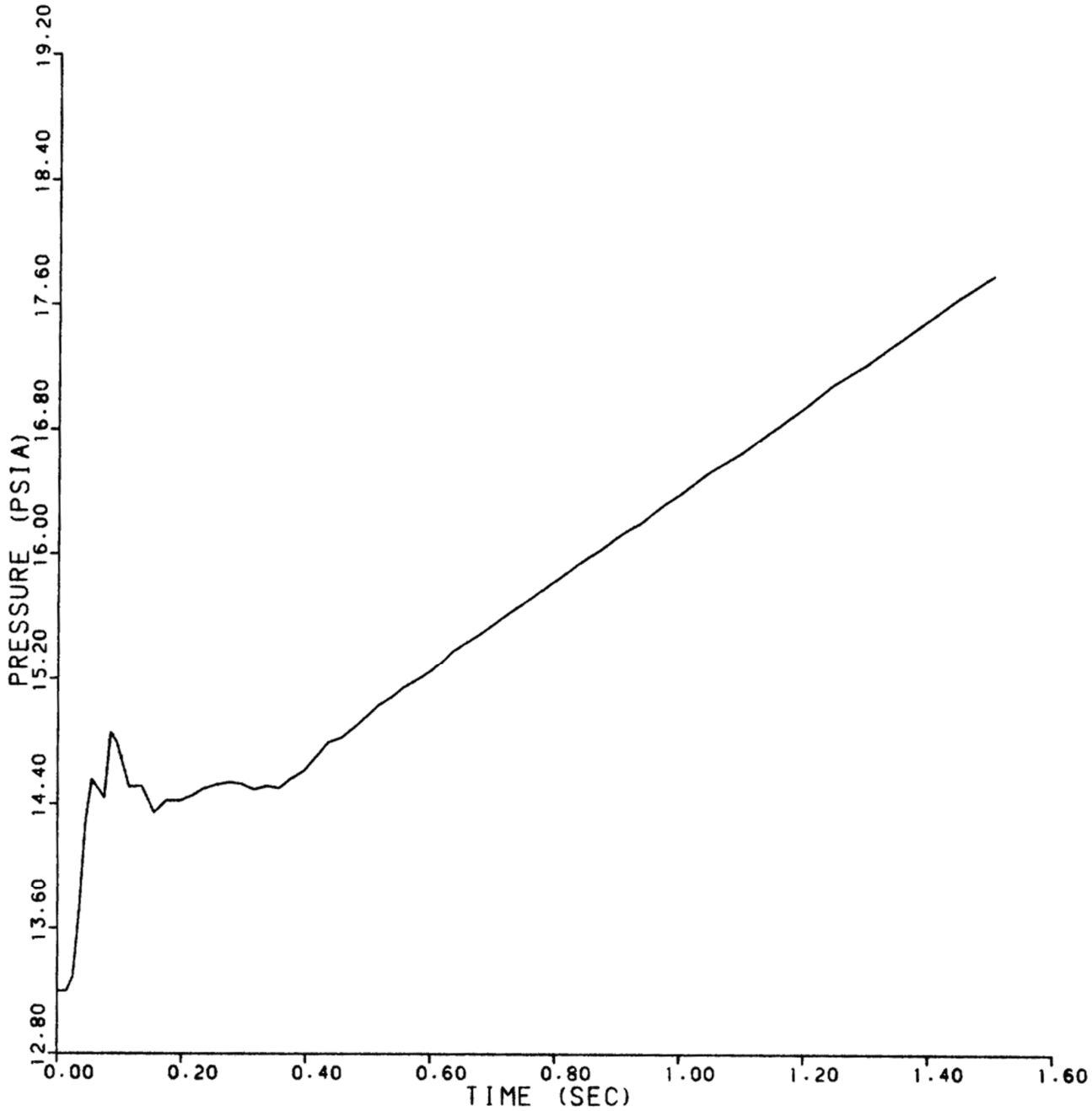
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E59
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 59 OF 74)



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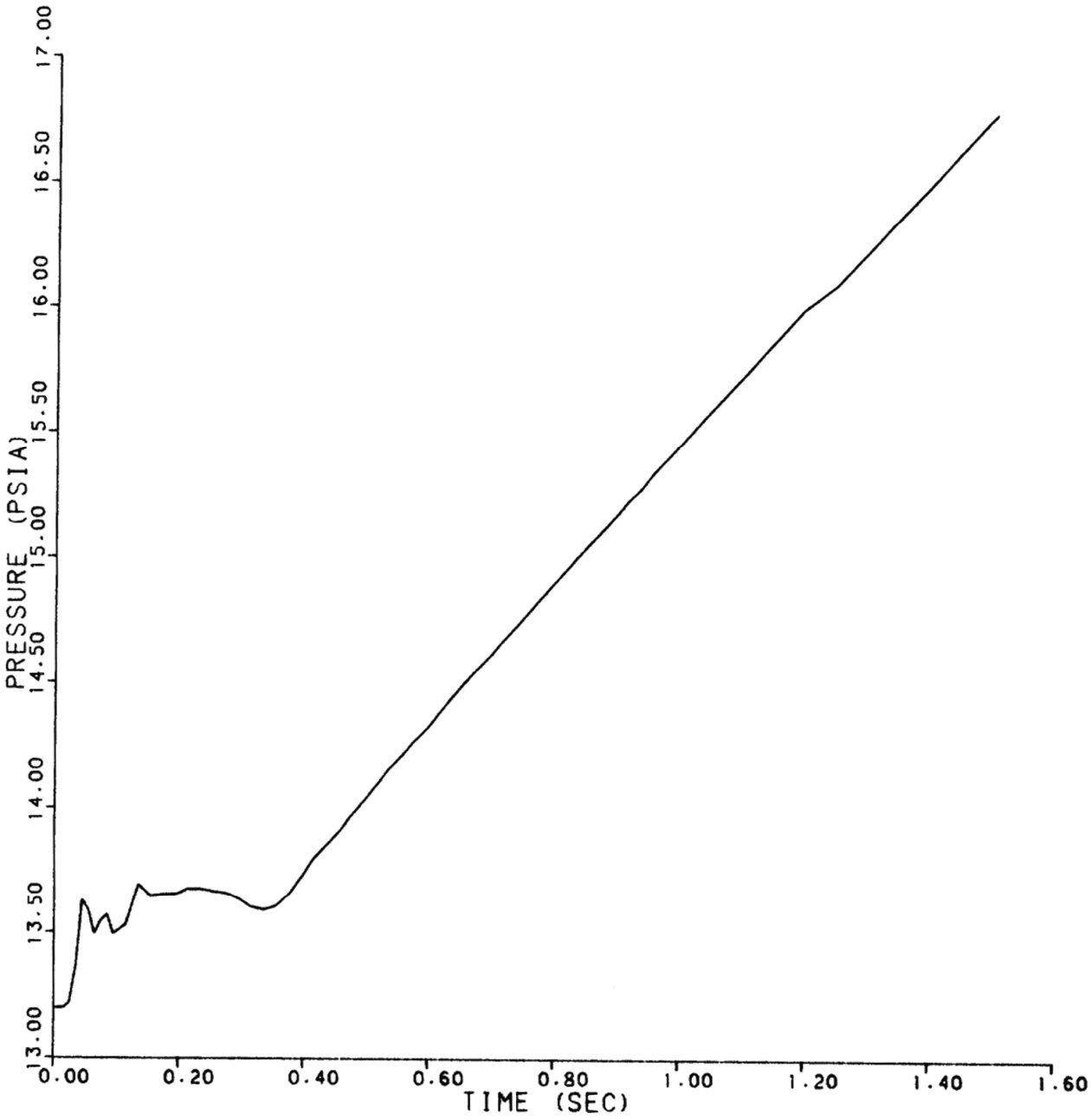
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E4
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 60 OF 74)



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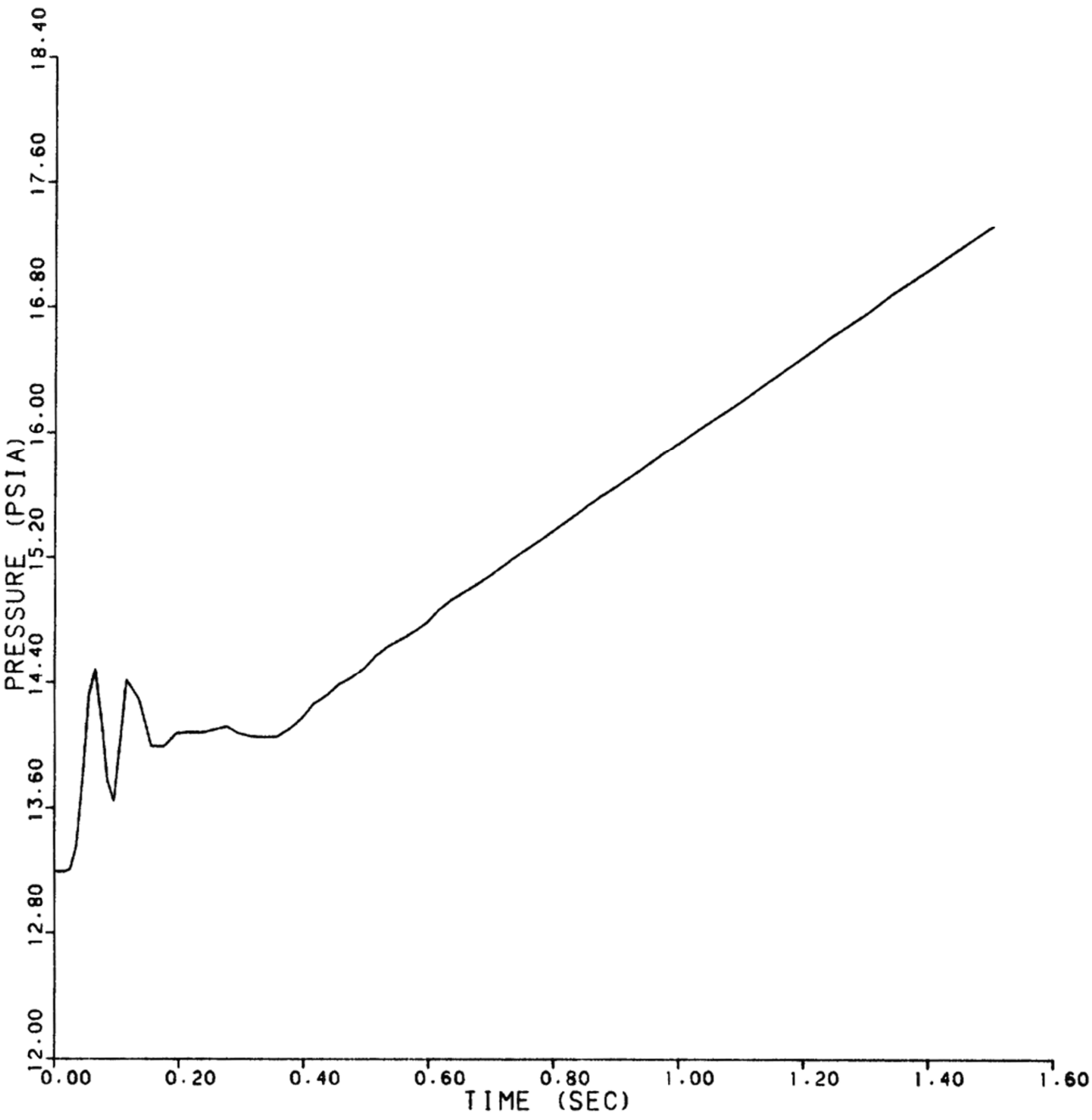
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E61
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 61 OF 74)



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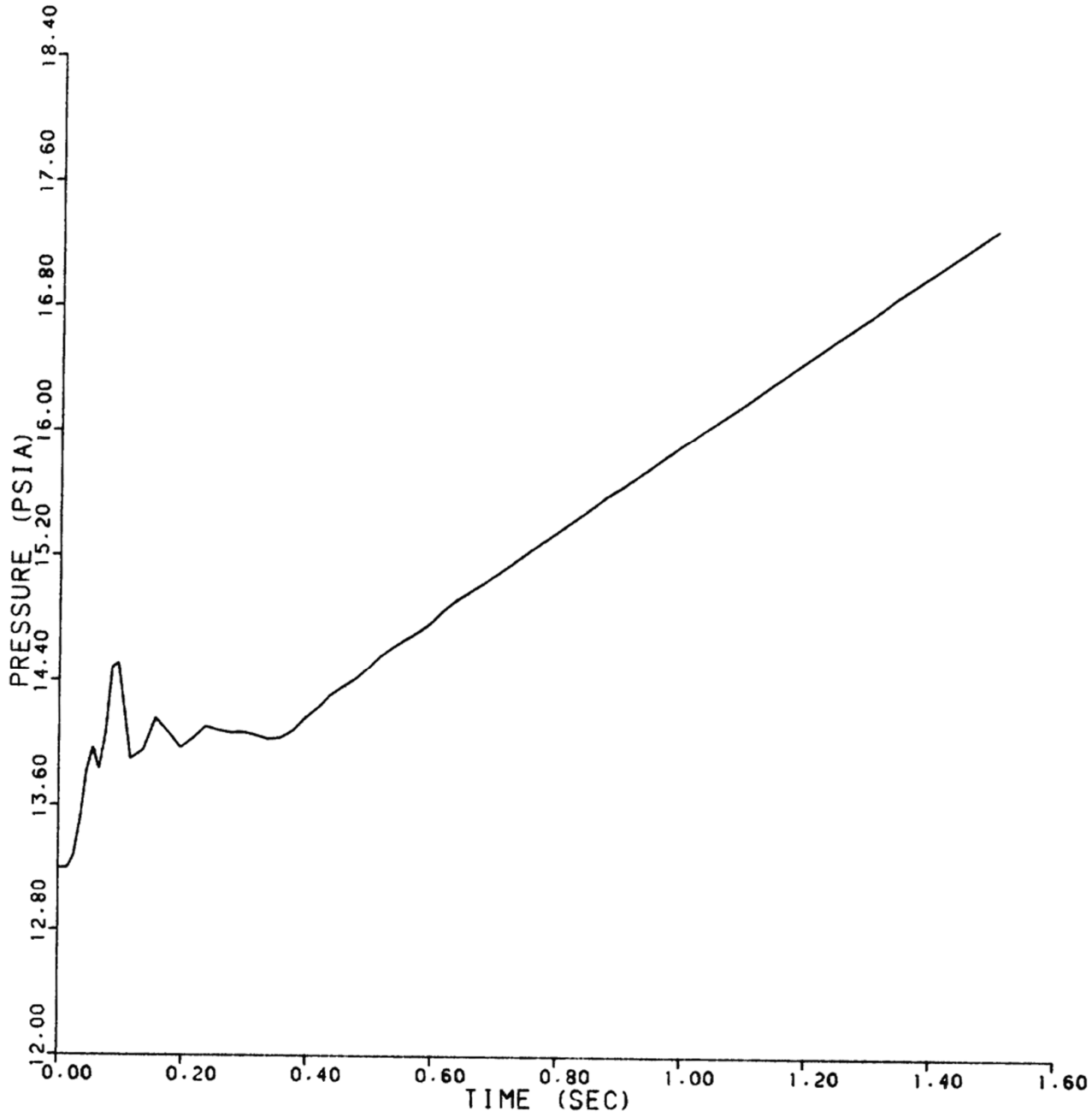
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E62
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 62 OF 74)



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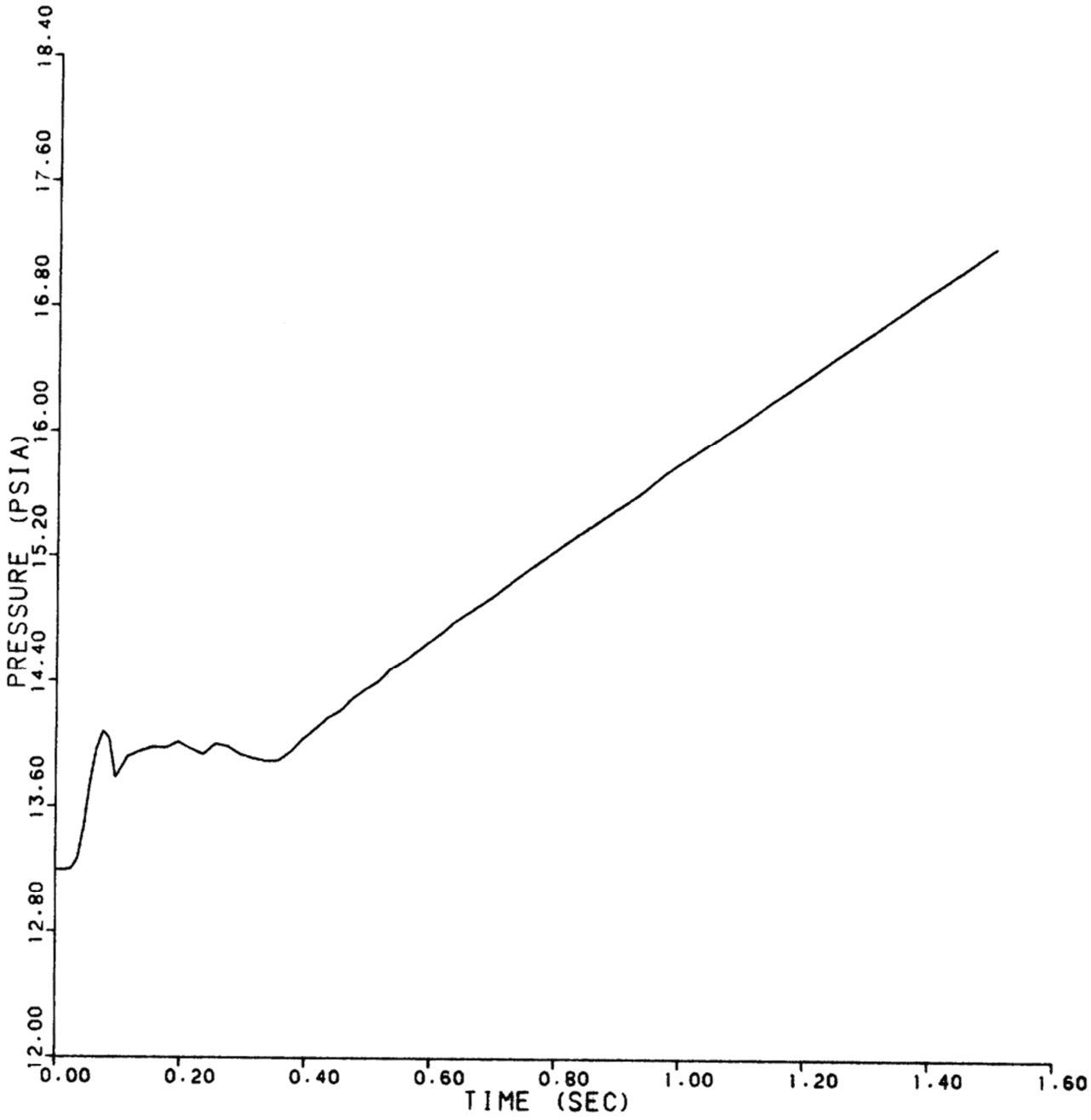
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E63
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 63 OF 74)



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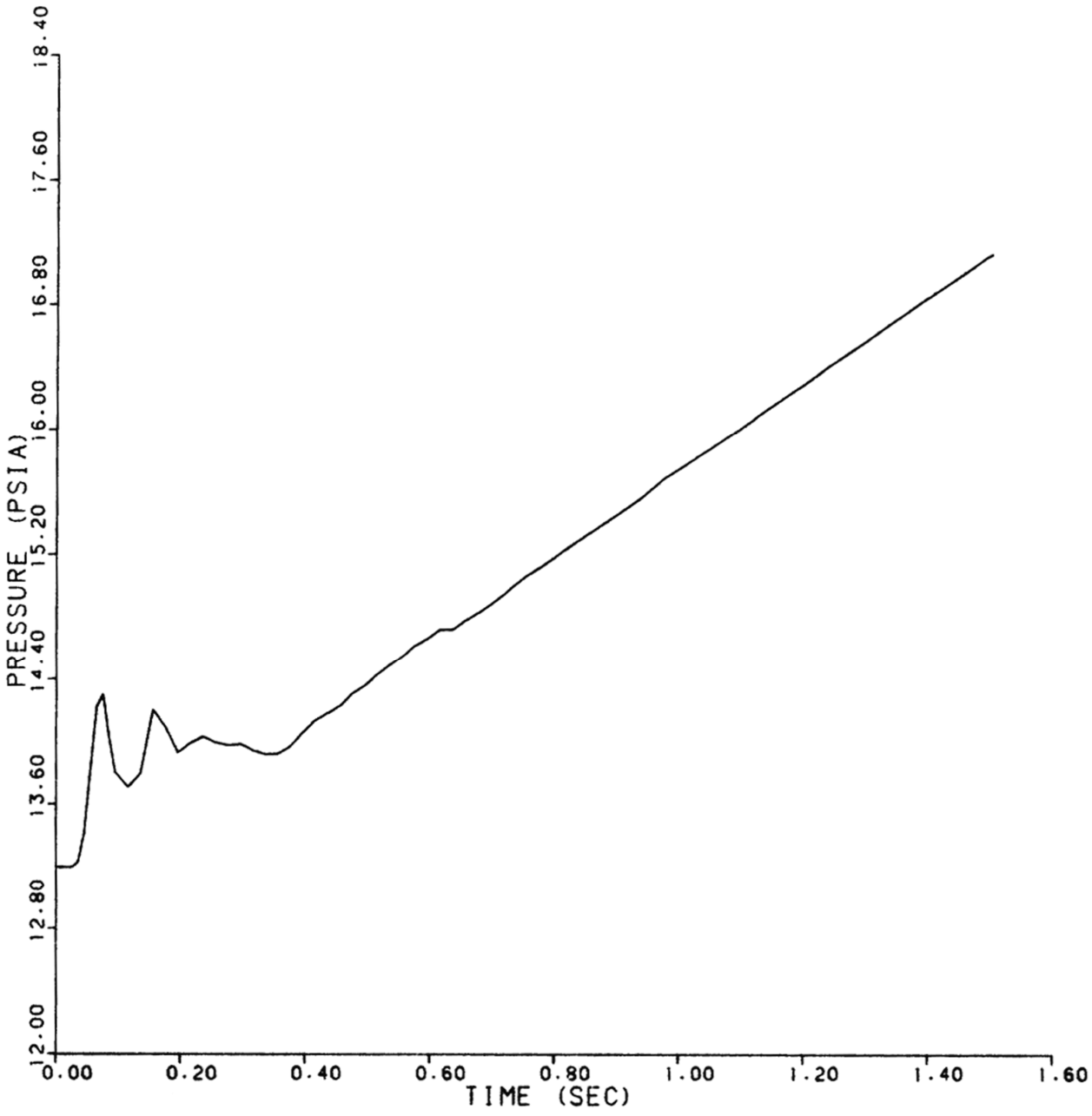
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E64
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 64 OF 74)



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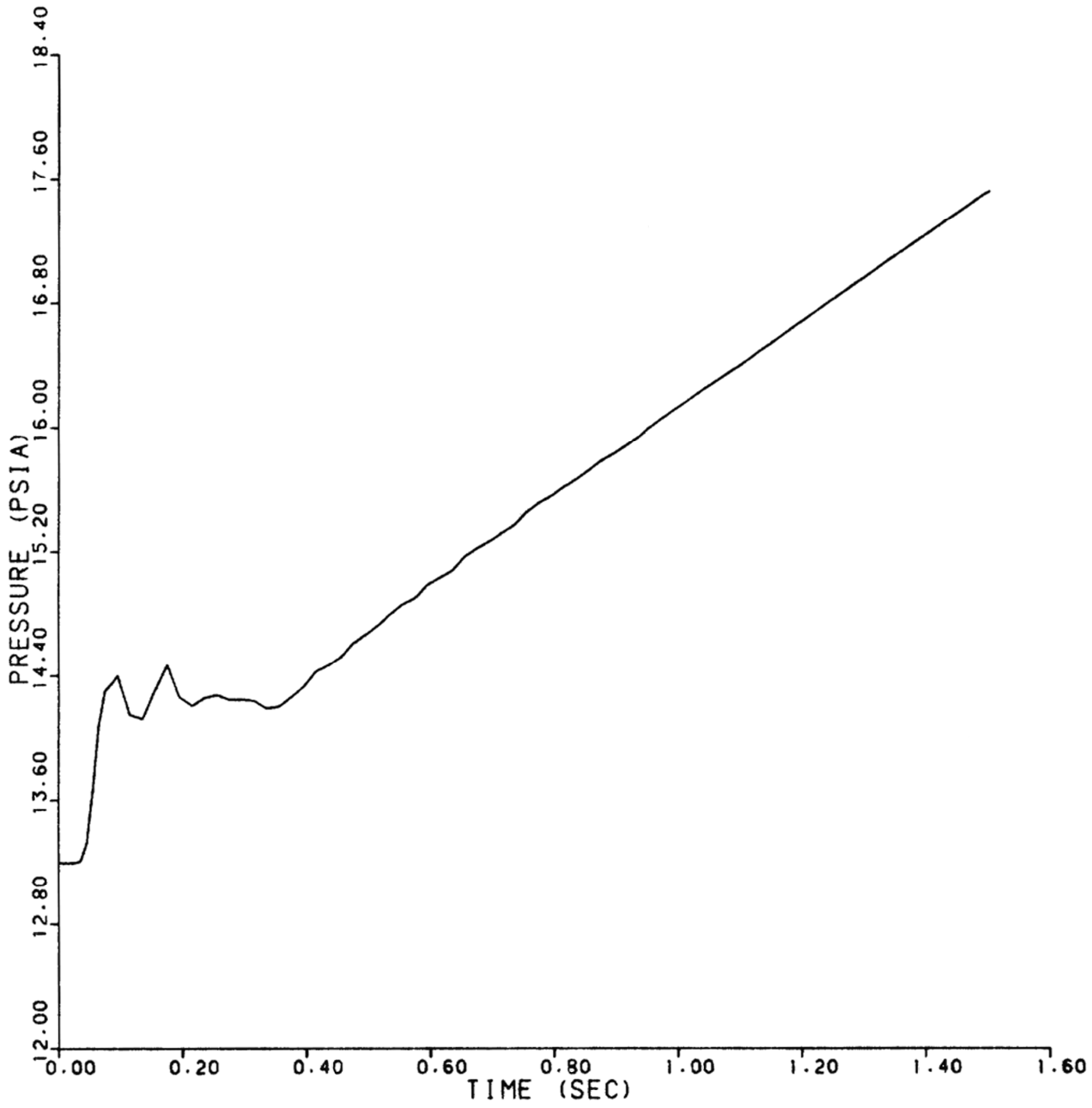
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E65
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 65 OF 74)



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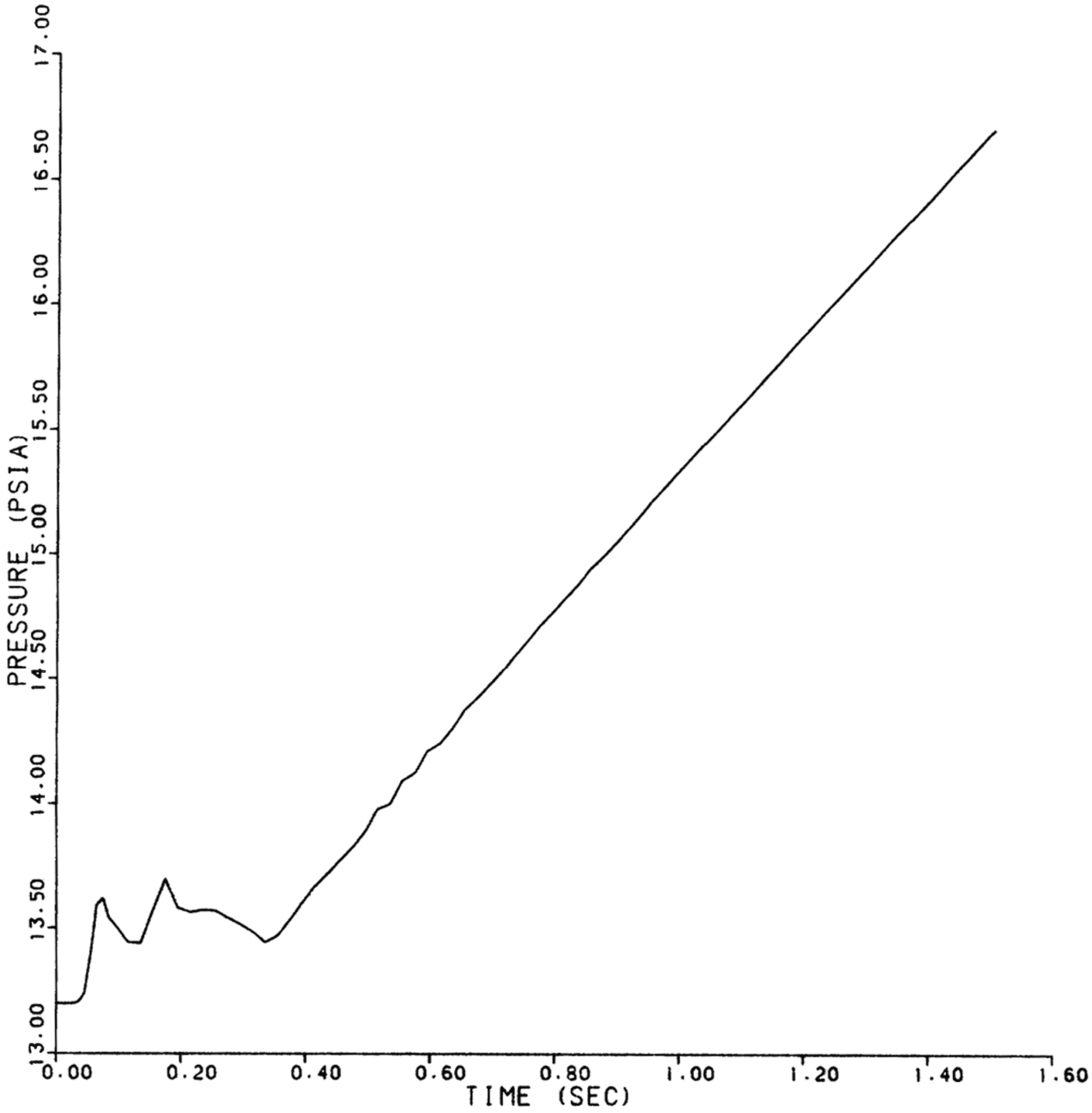
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E66
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 66 OF 74)



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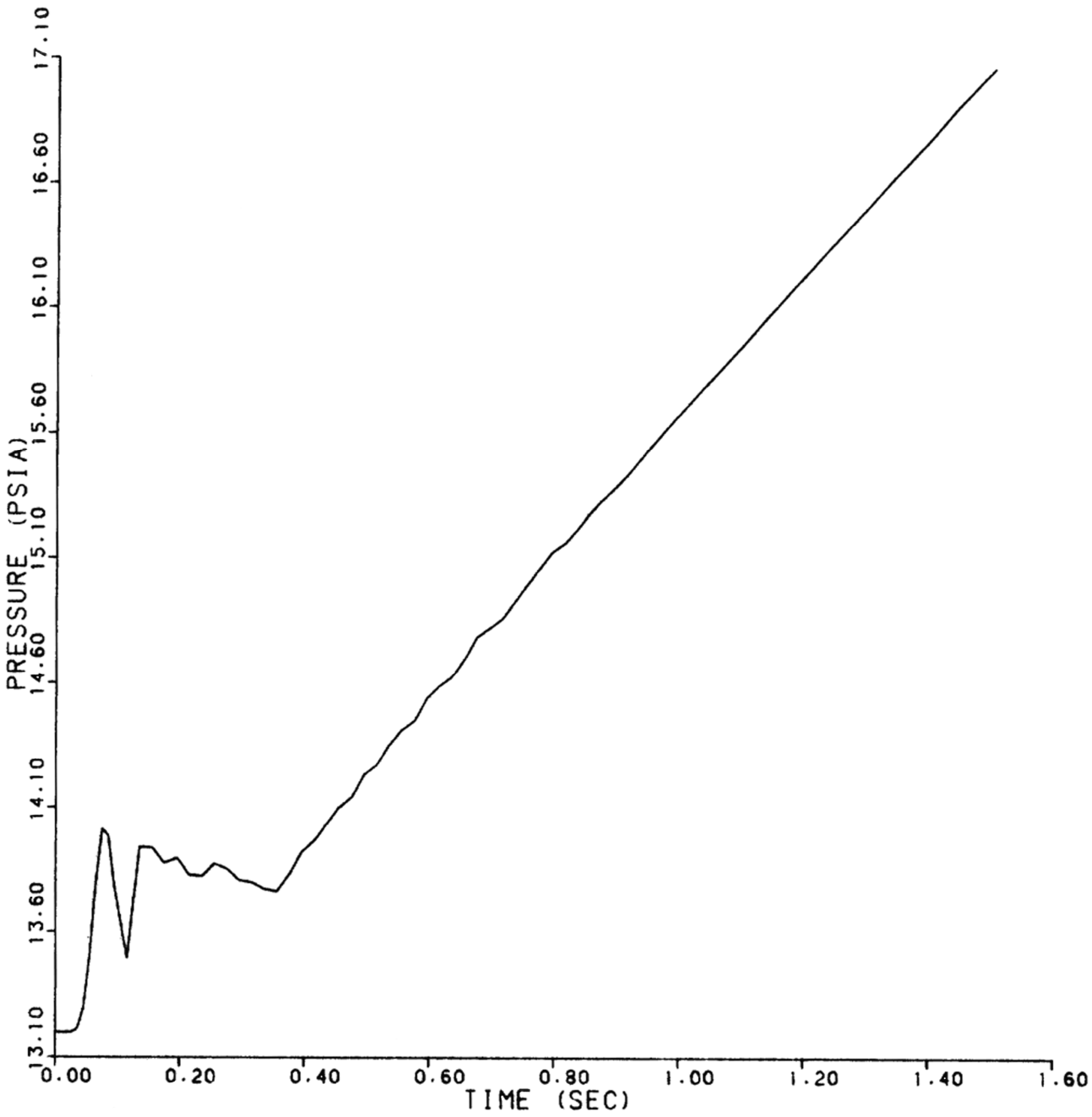
STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E67
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 67 OF 74)



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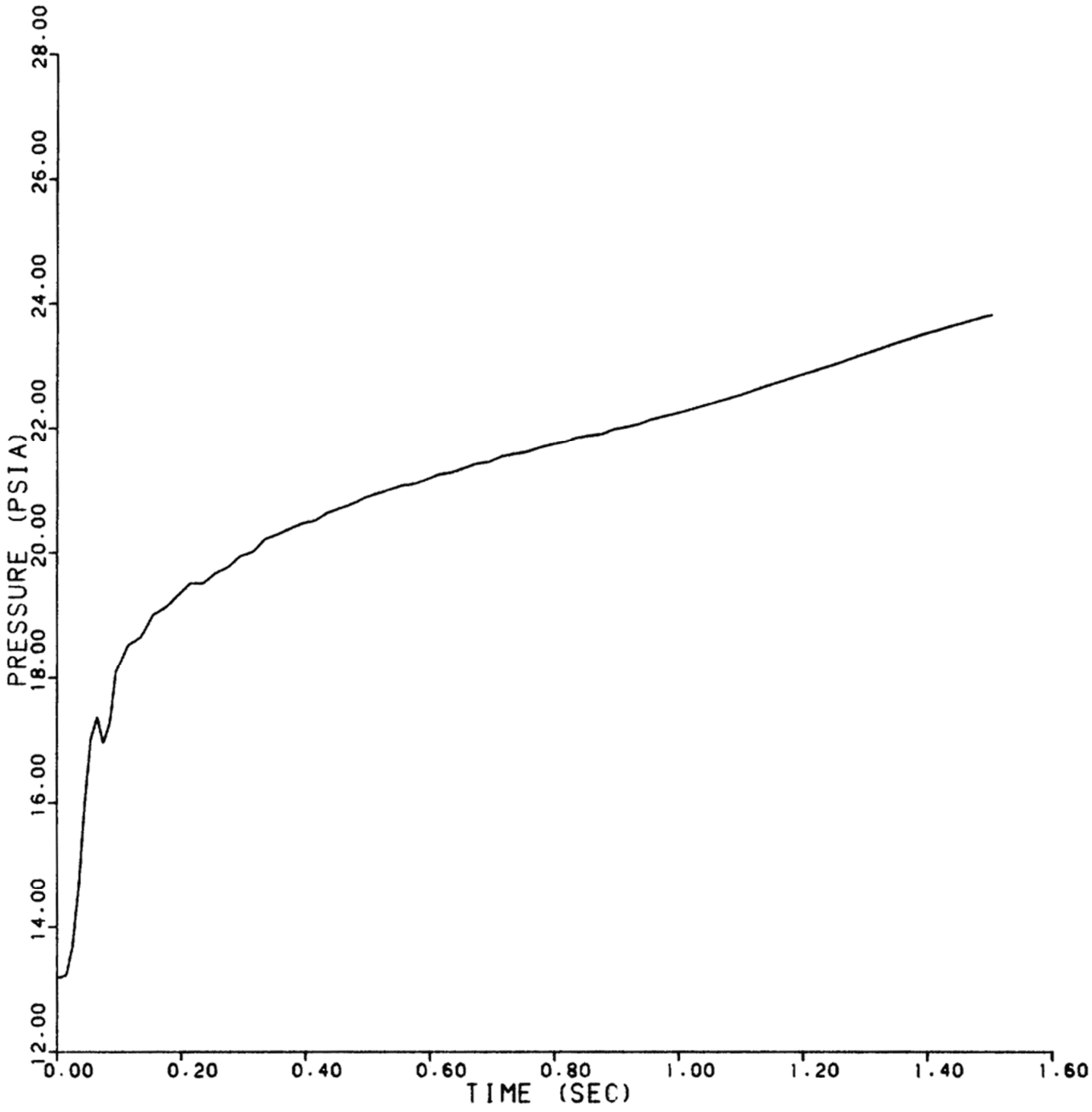
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E68
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 68 OF 74)



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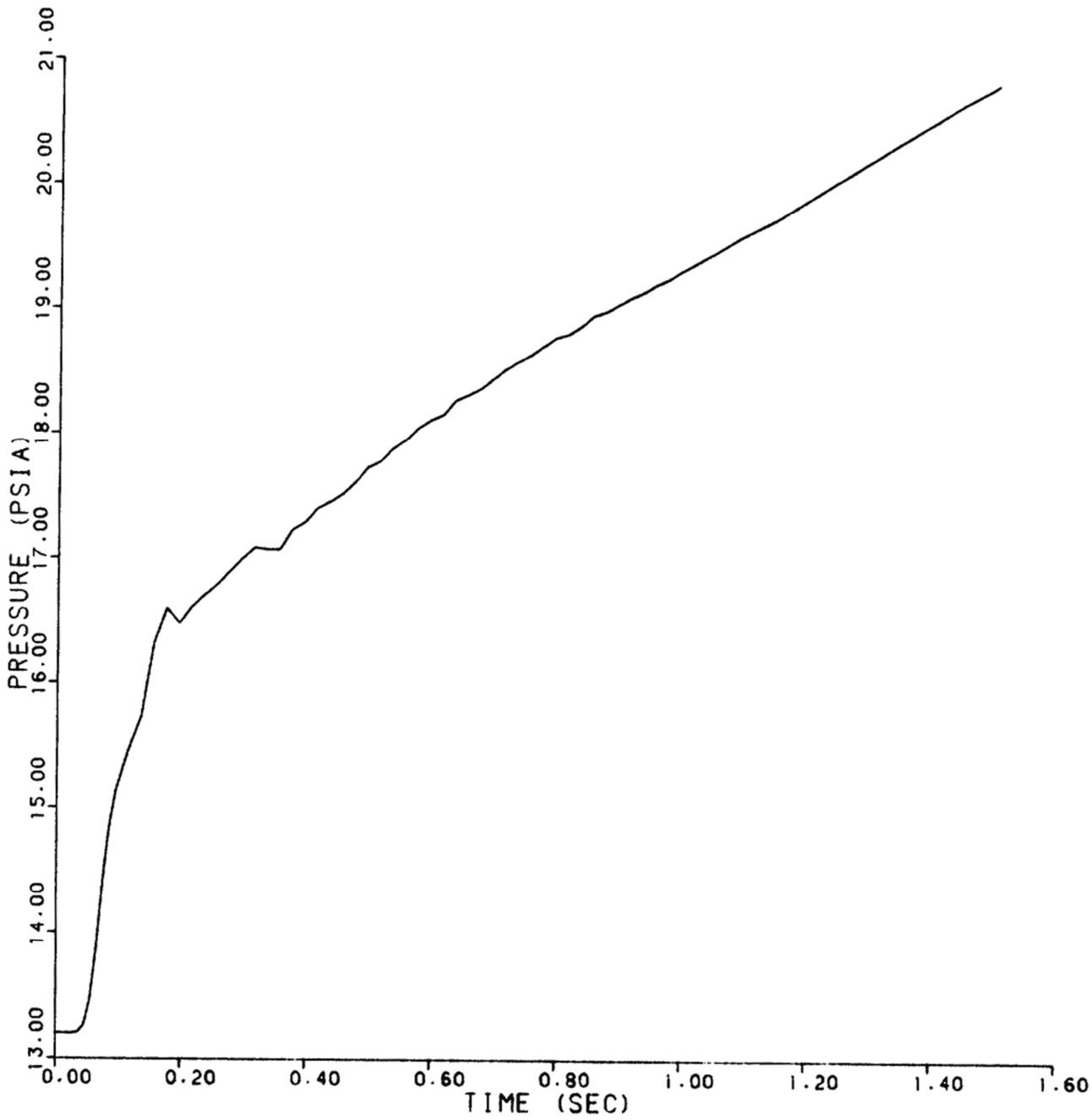
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E69
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 69 OF 74)



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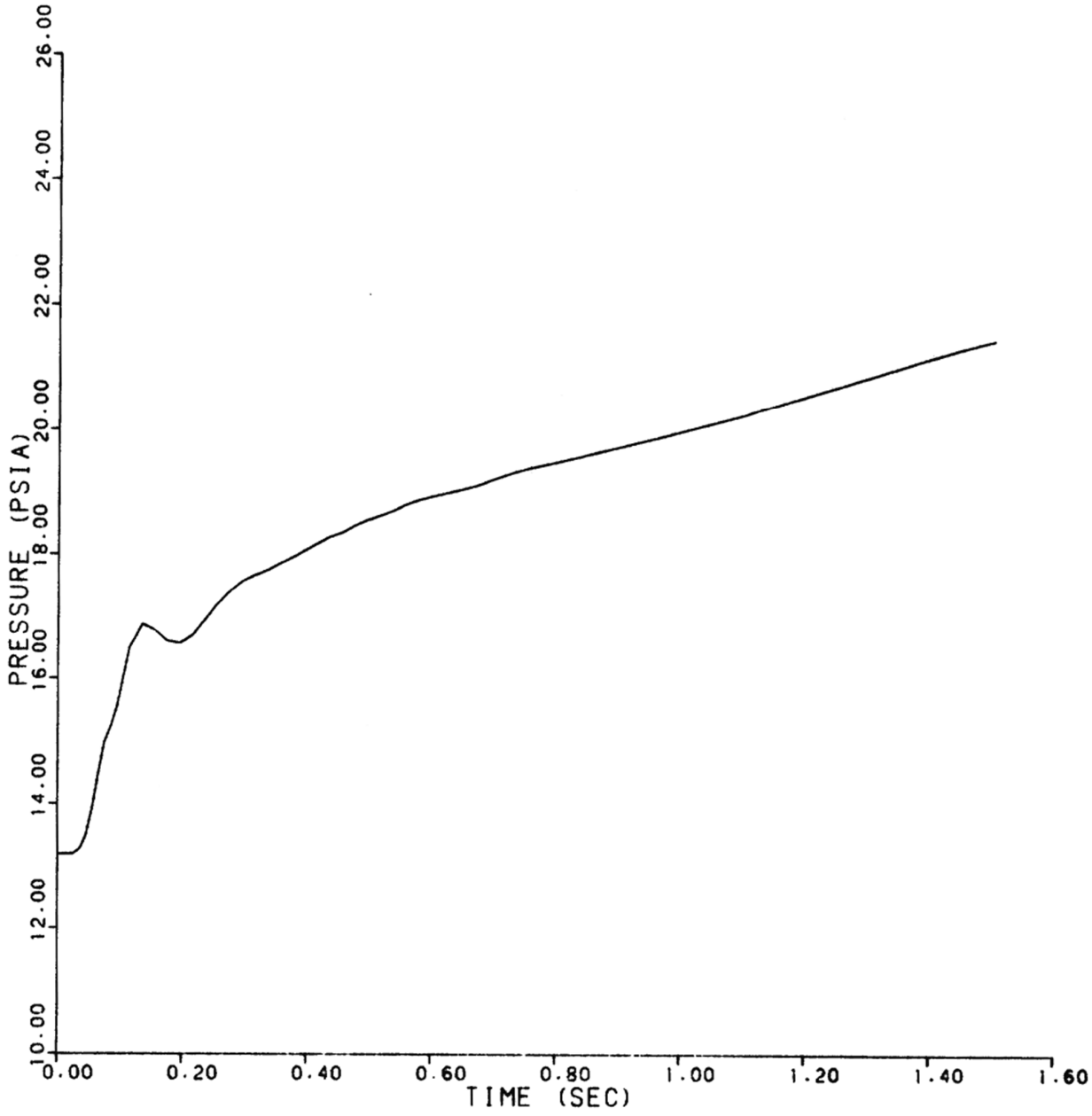
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E70
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 70 OF 74)



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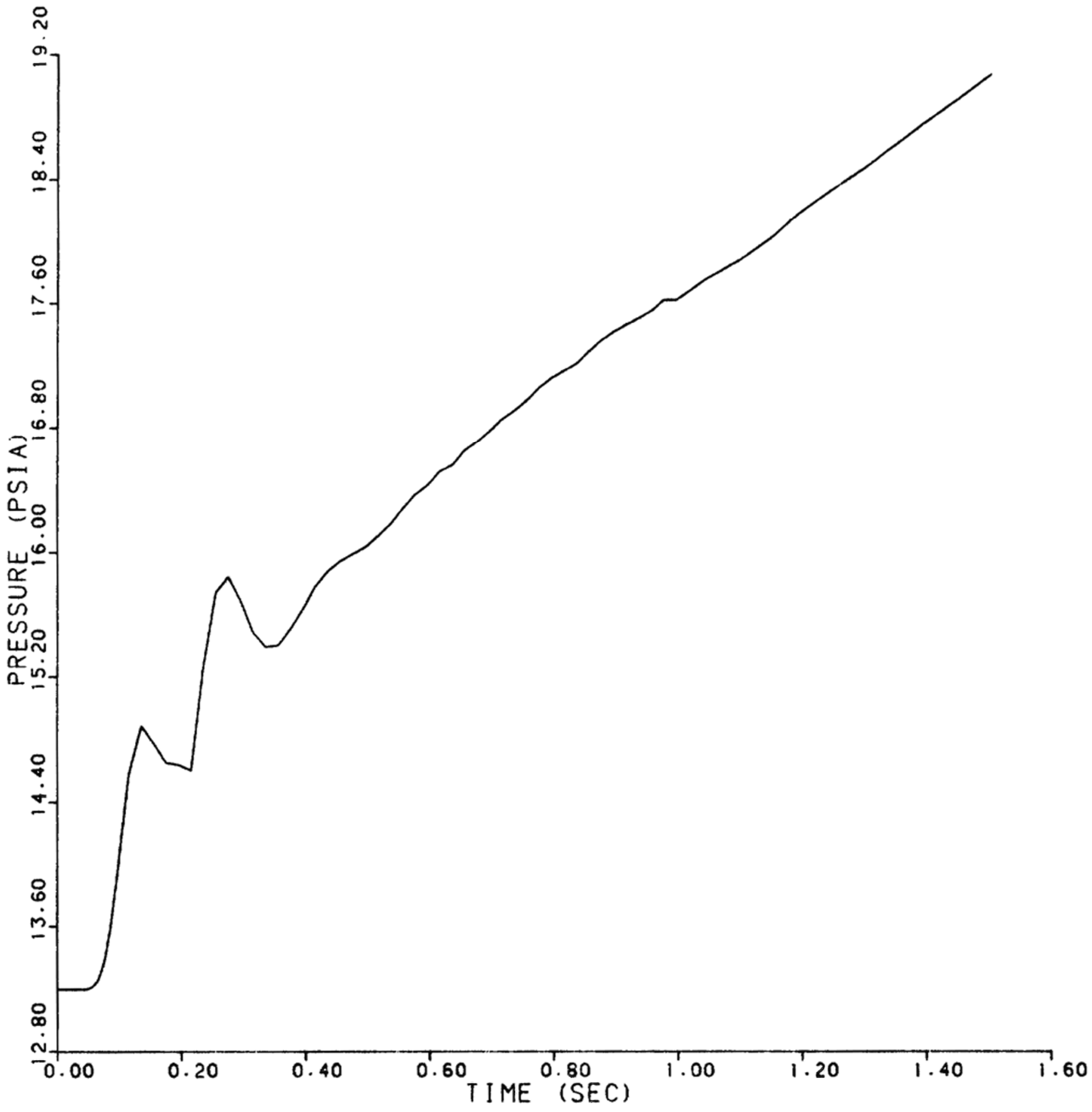
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E71
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 71 OF 74)



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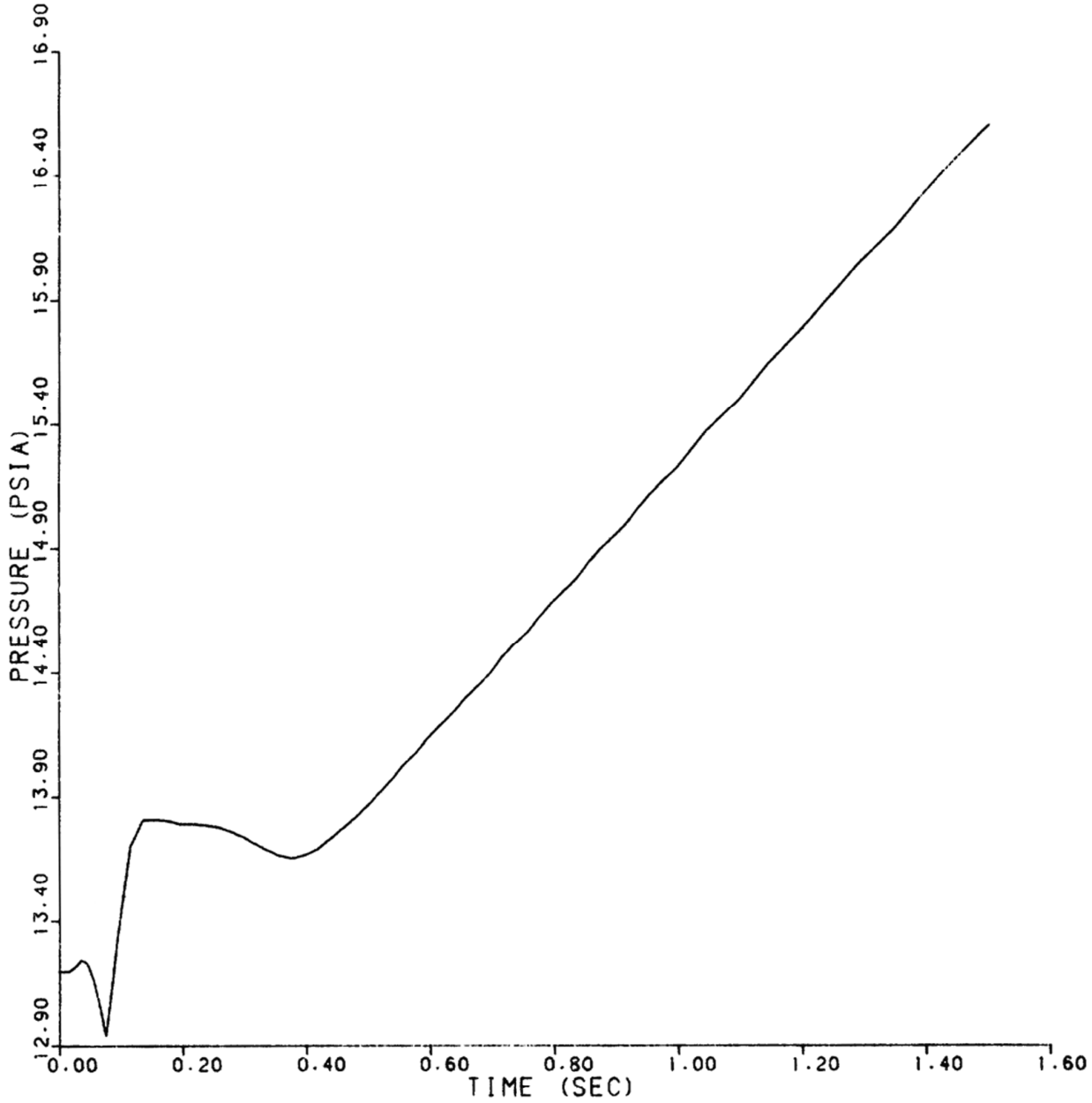
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E72
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 72 OF 74)



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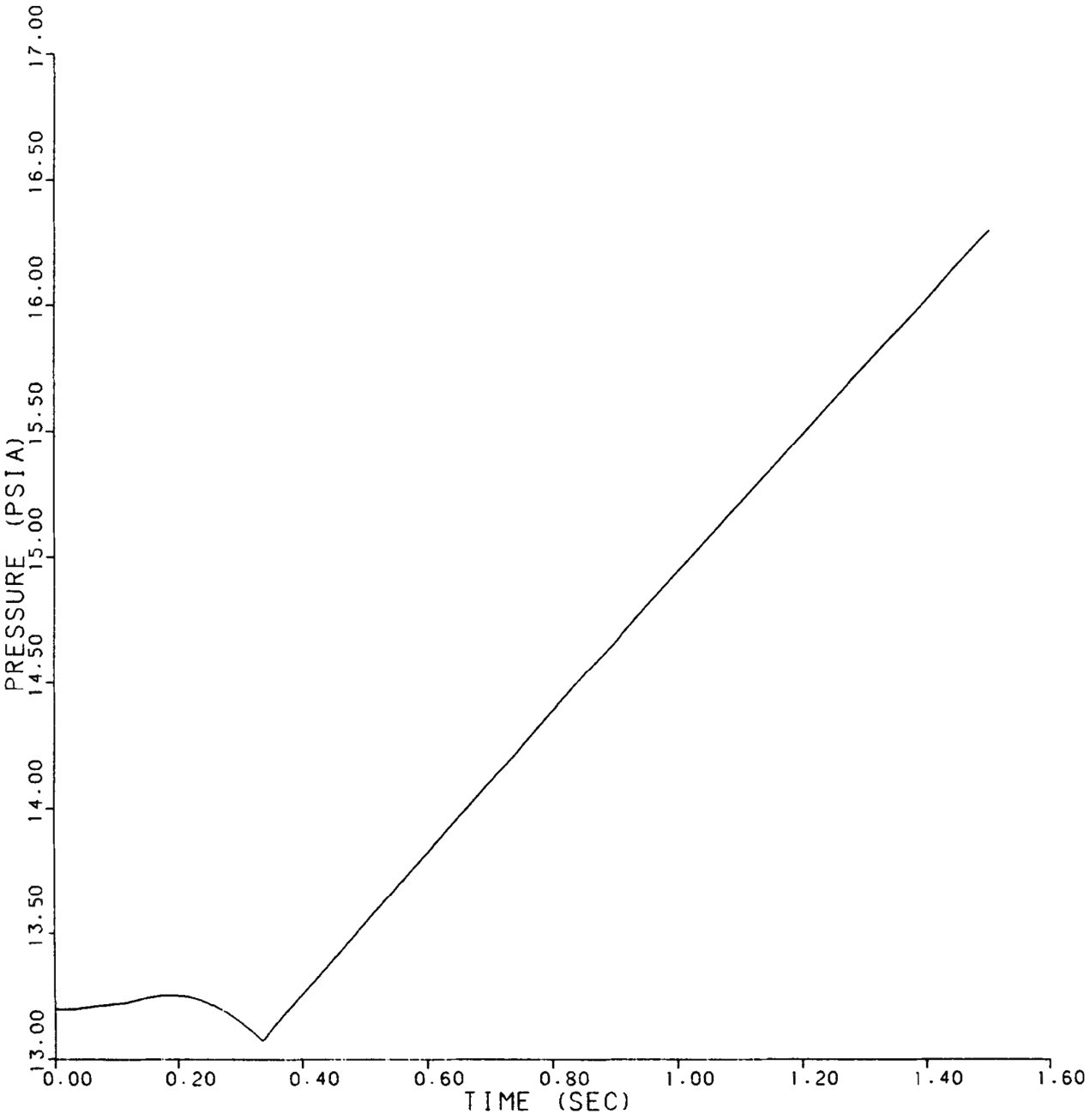
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E73
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 73 OF 74)



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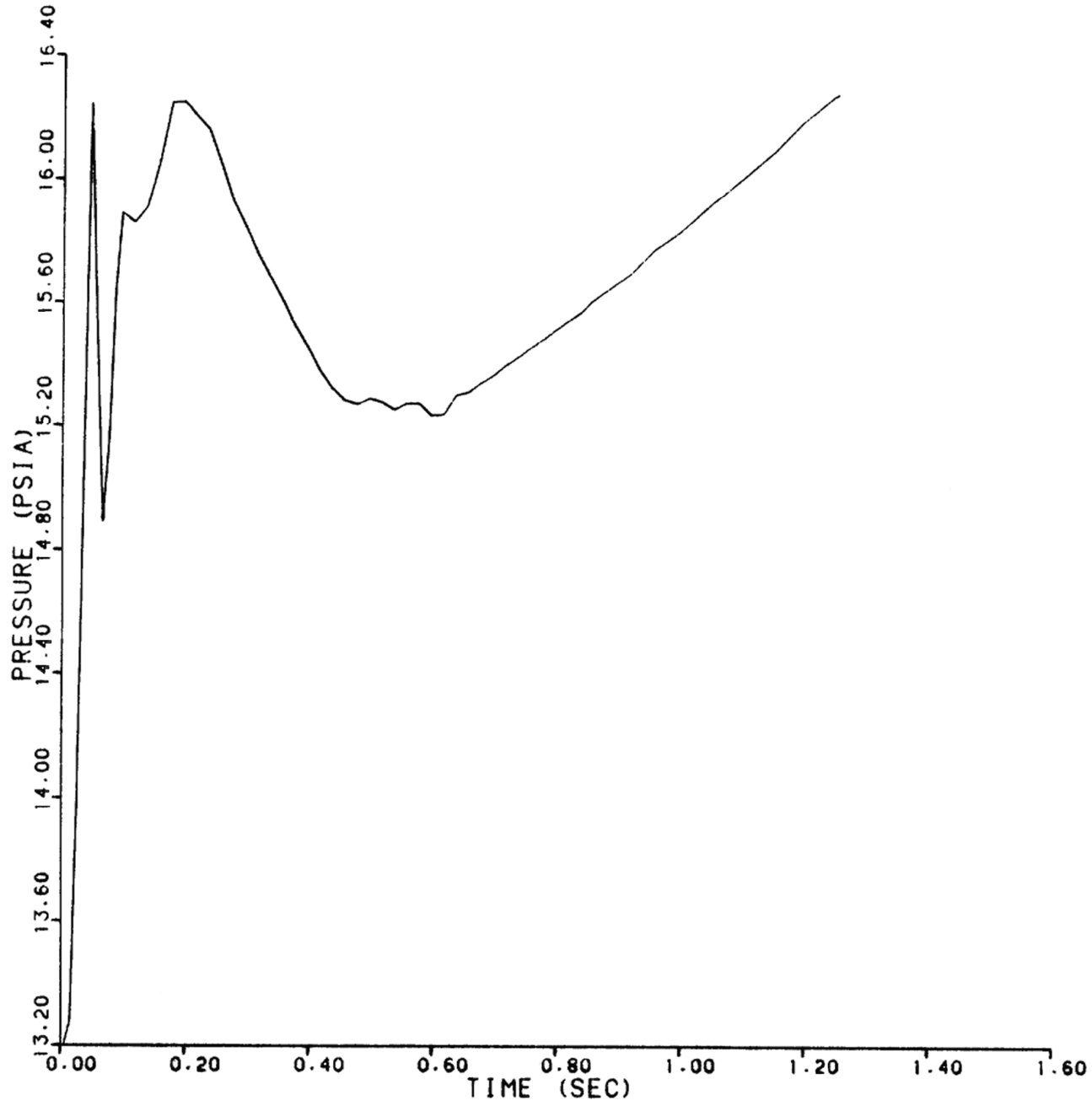
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR COMPARTMENT
PRESSURE RESPONSE – NODE E74
(236–in.^{2} BREAK AREA)

FIGURE 6.2.1–18 (SHEET 74 OF 74)



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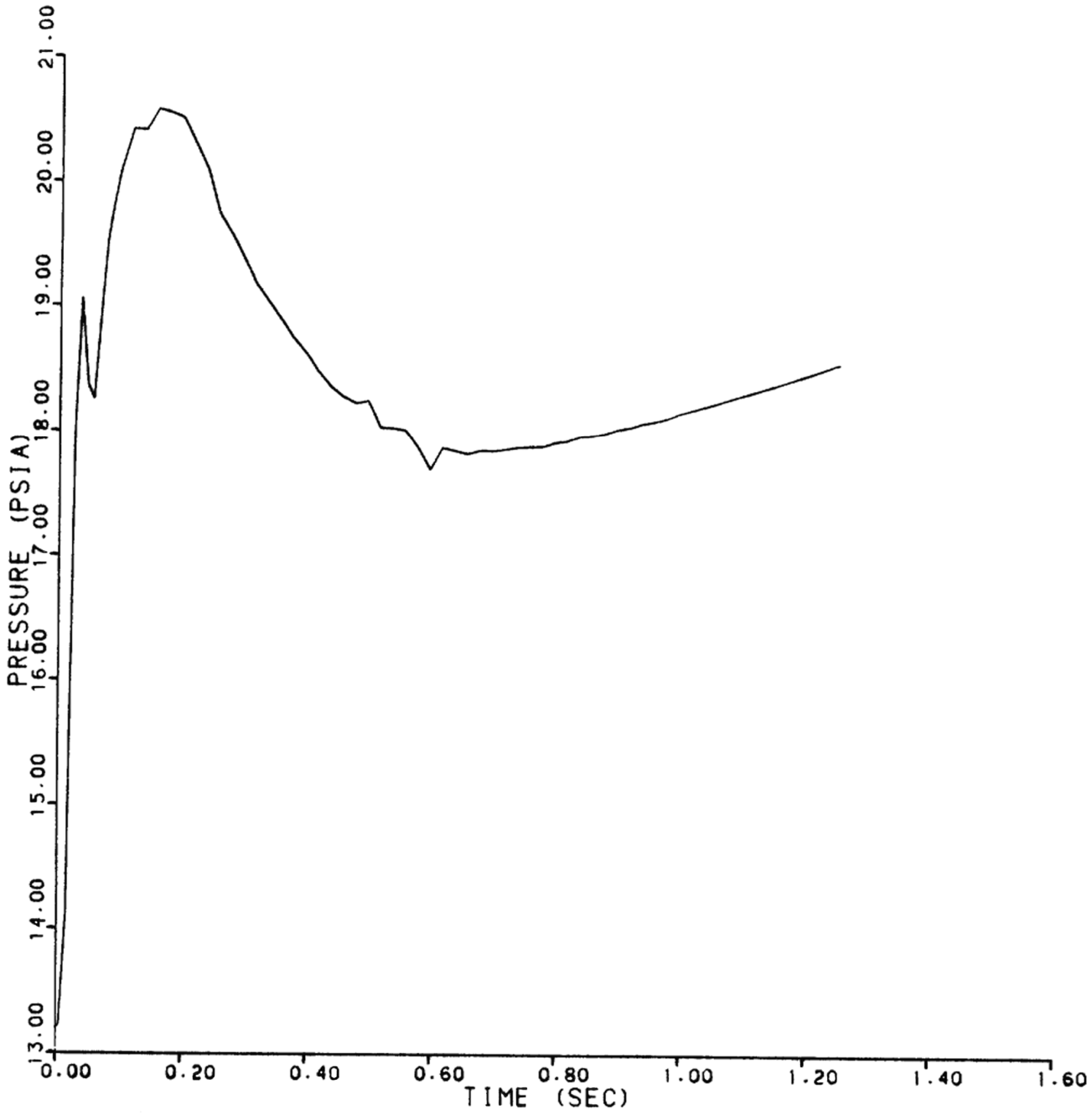
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E1
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 1 OF 74)



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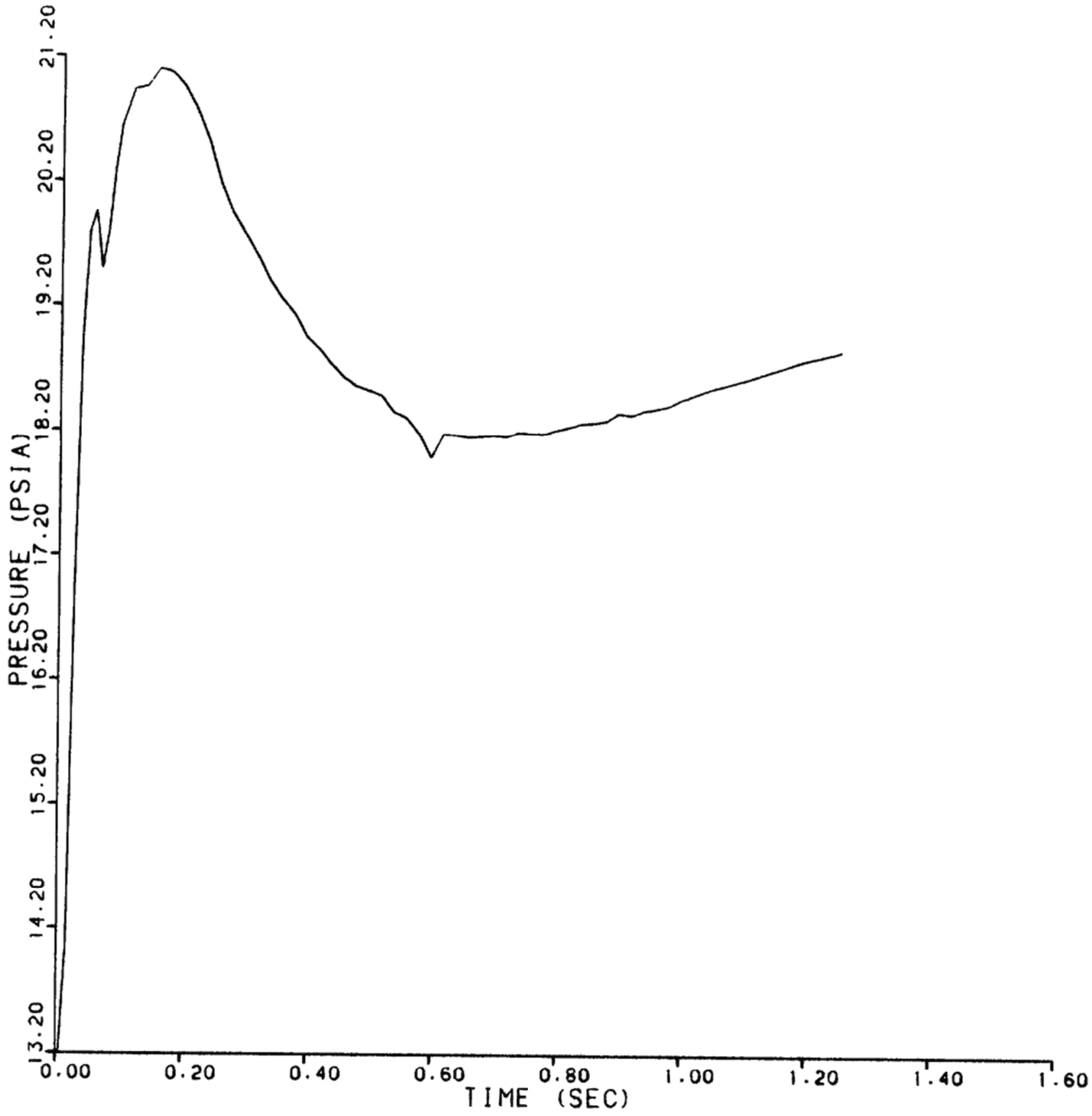
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E2
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 2 OF 74)



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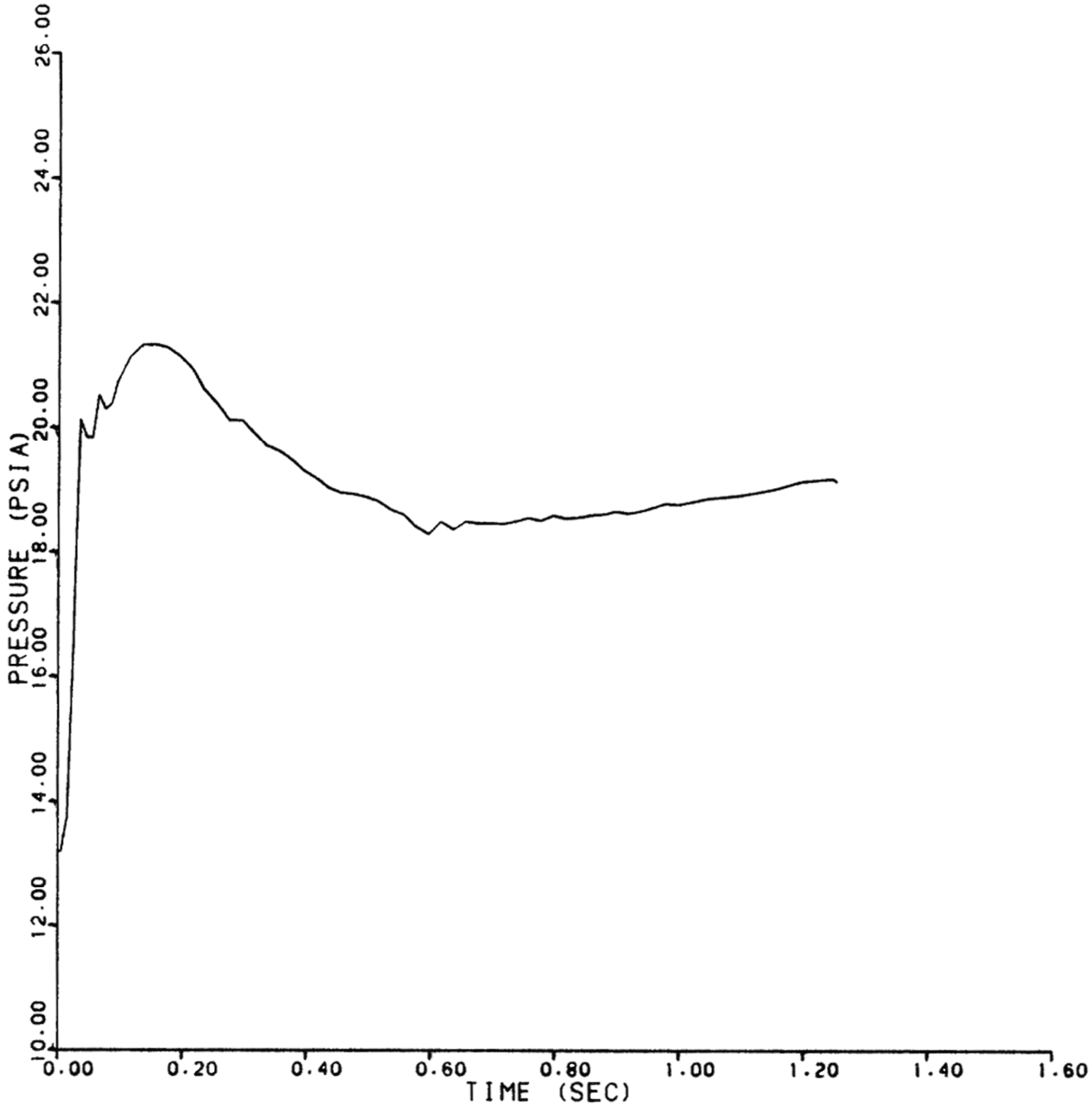
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E3
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 3 OF 74)



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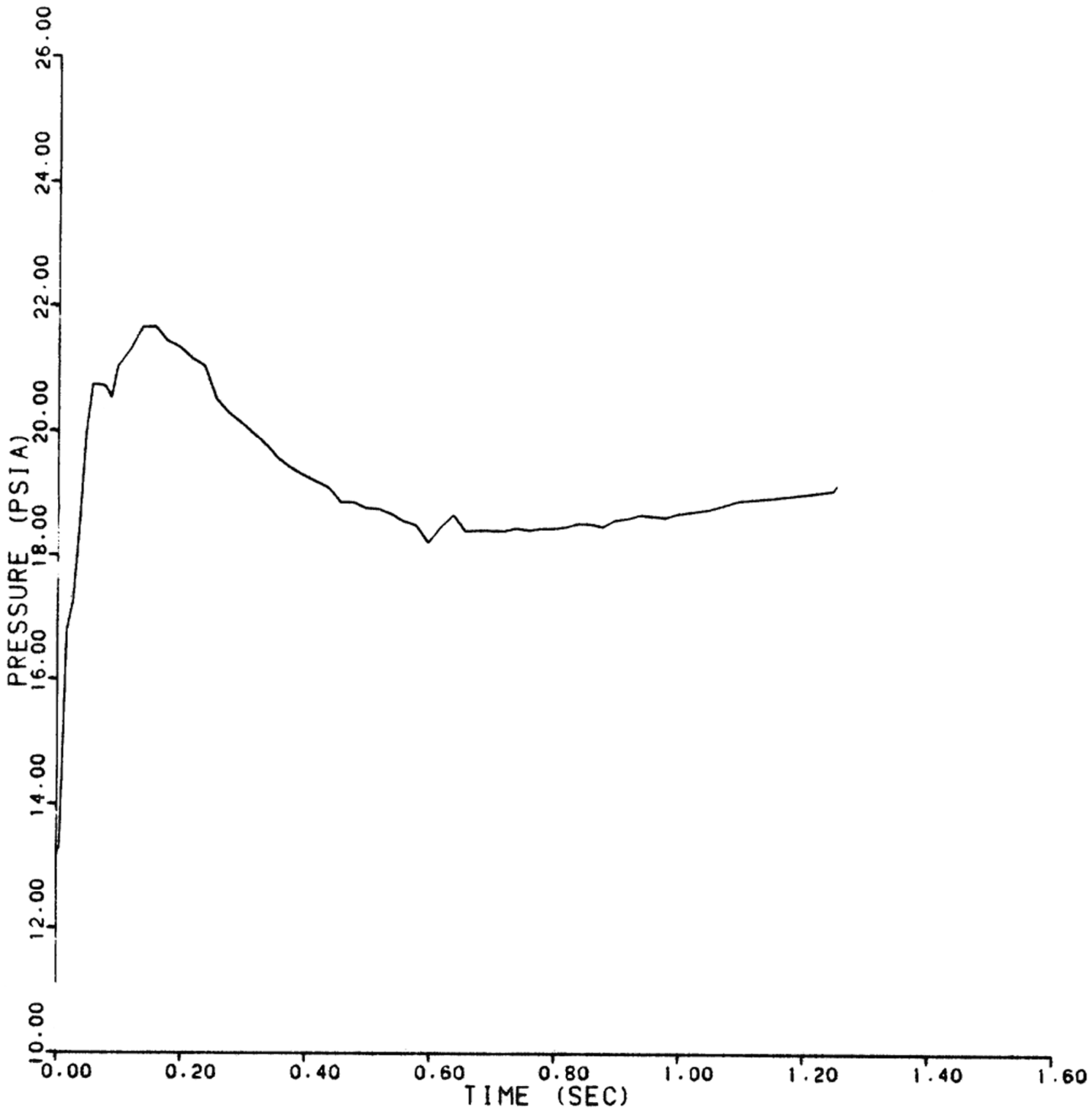
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E4 (306–in.^{2} BREAK AREA) FIGURE 6.2.1–19 (SHEET 4 OF 74)



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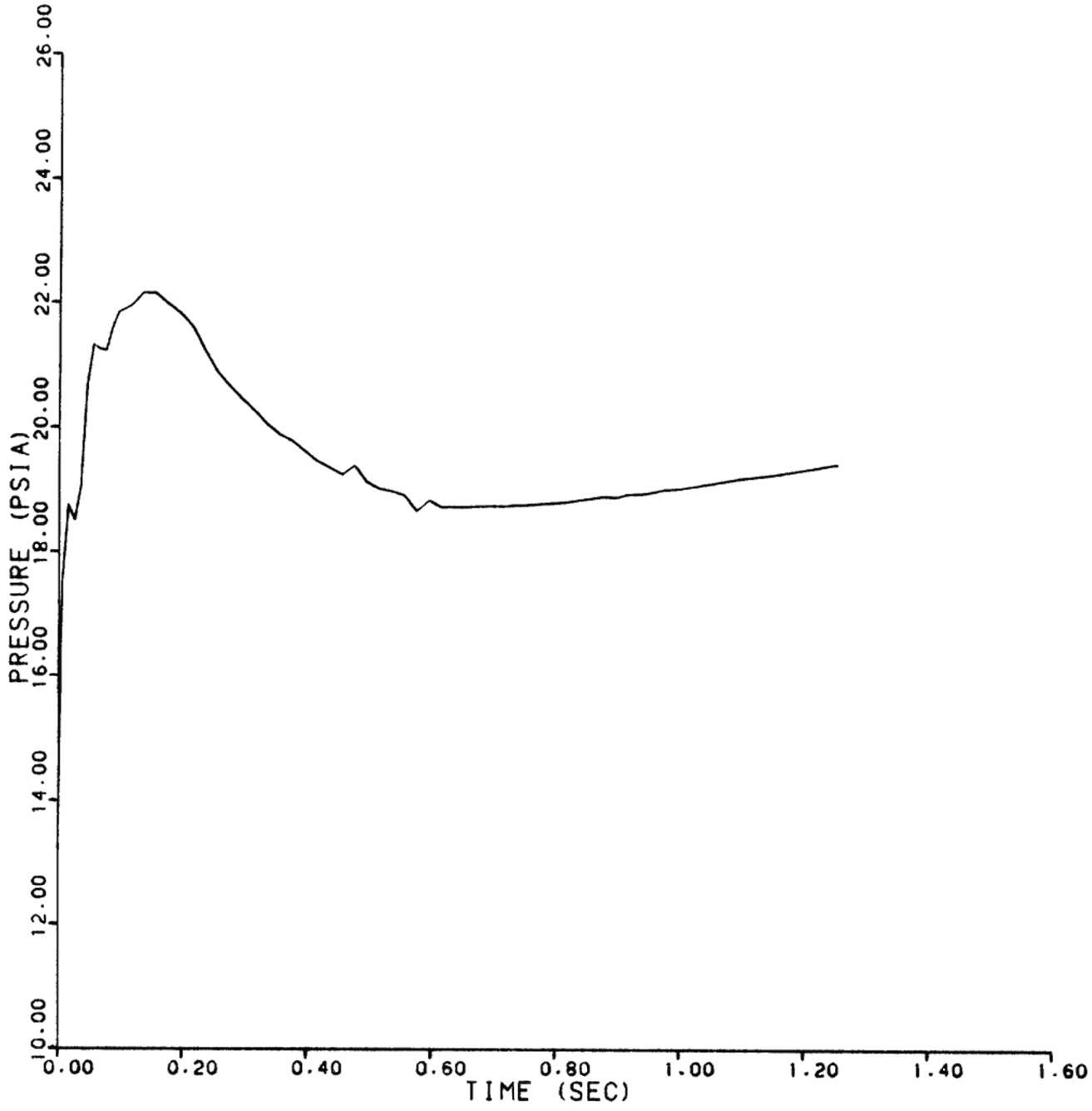
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E5 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 5 OF 74)



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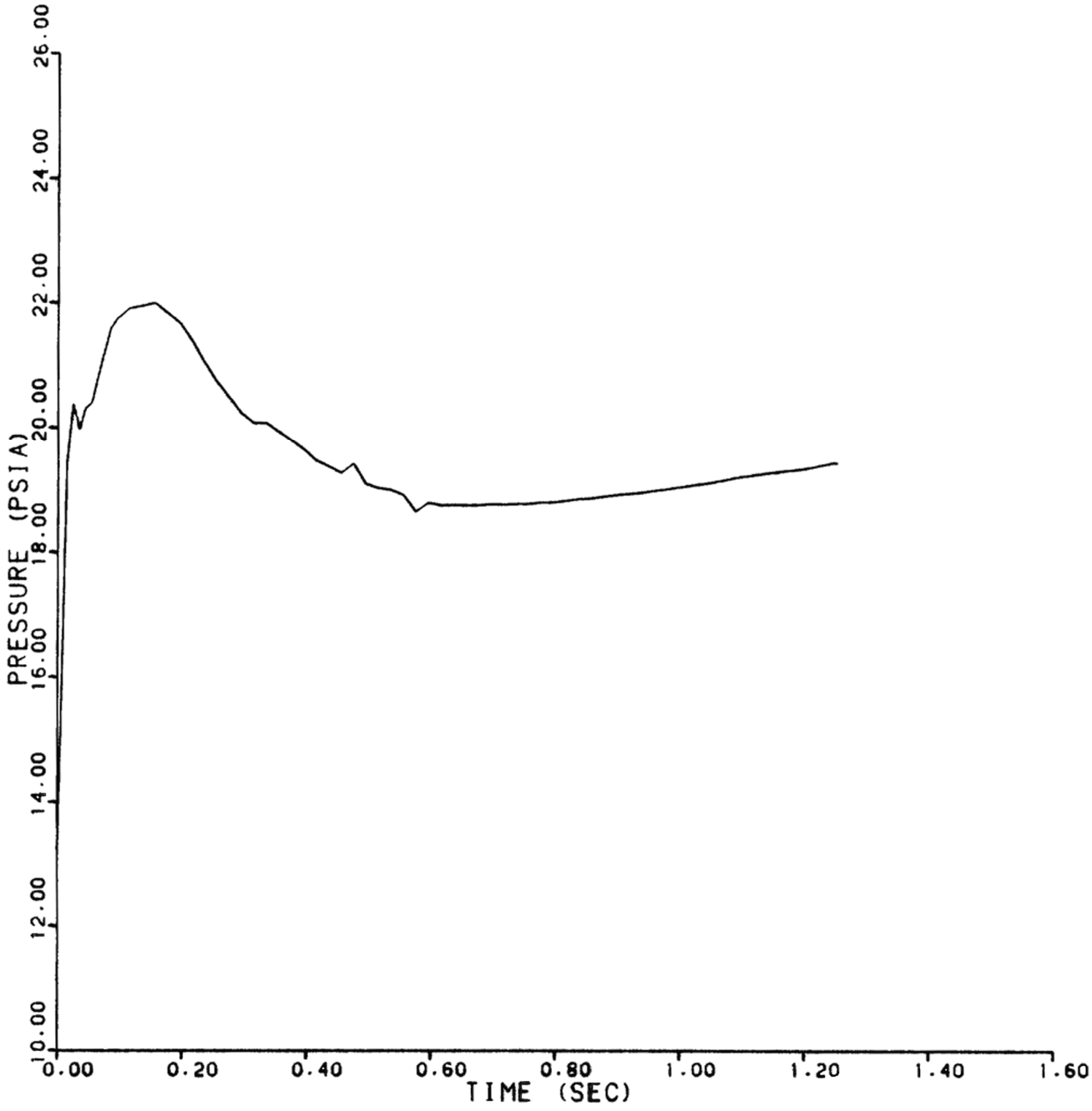
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E6
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 6 OF 74)



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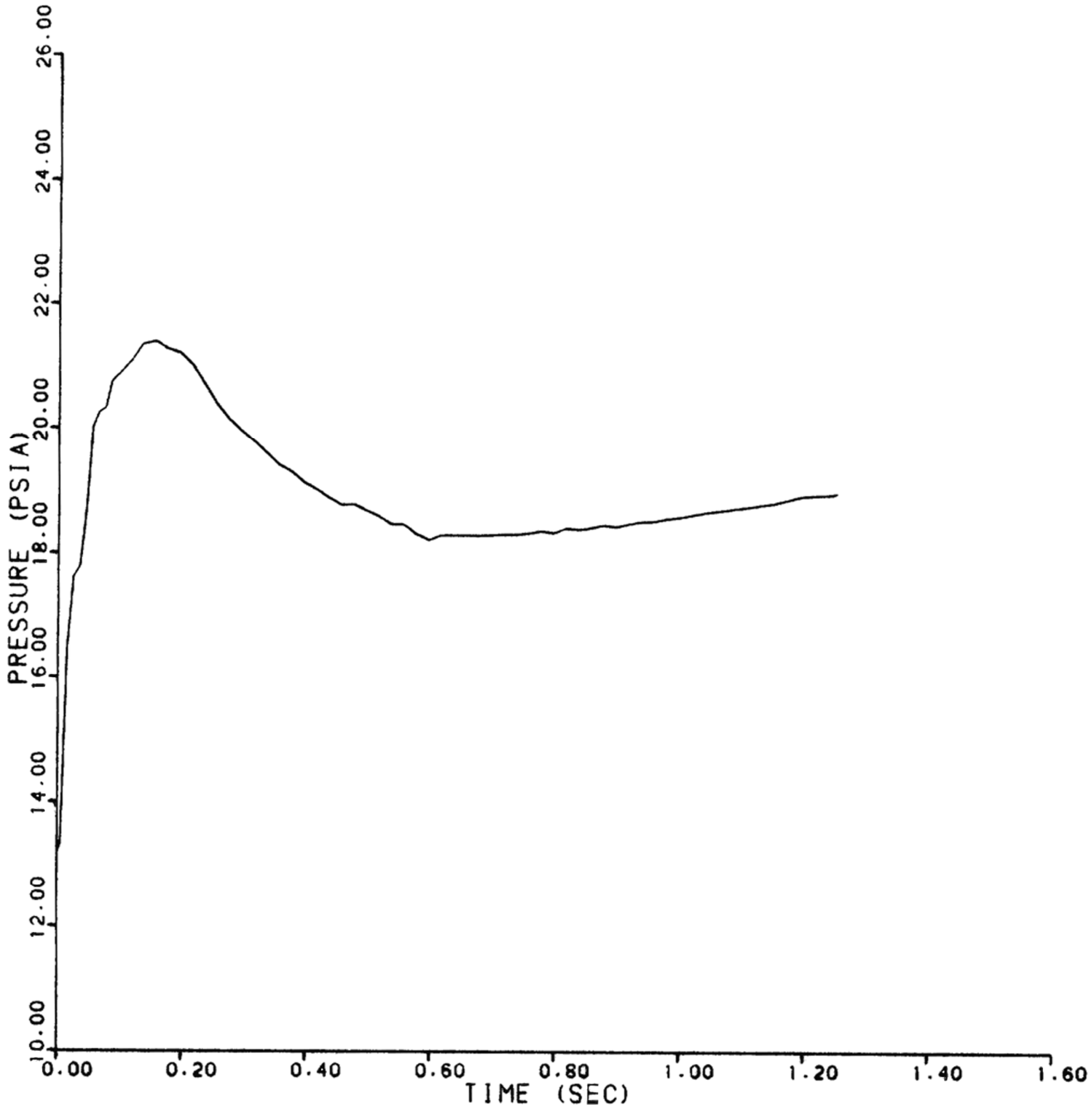
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E7
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 7 OF 74)



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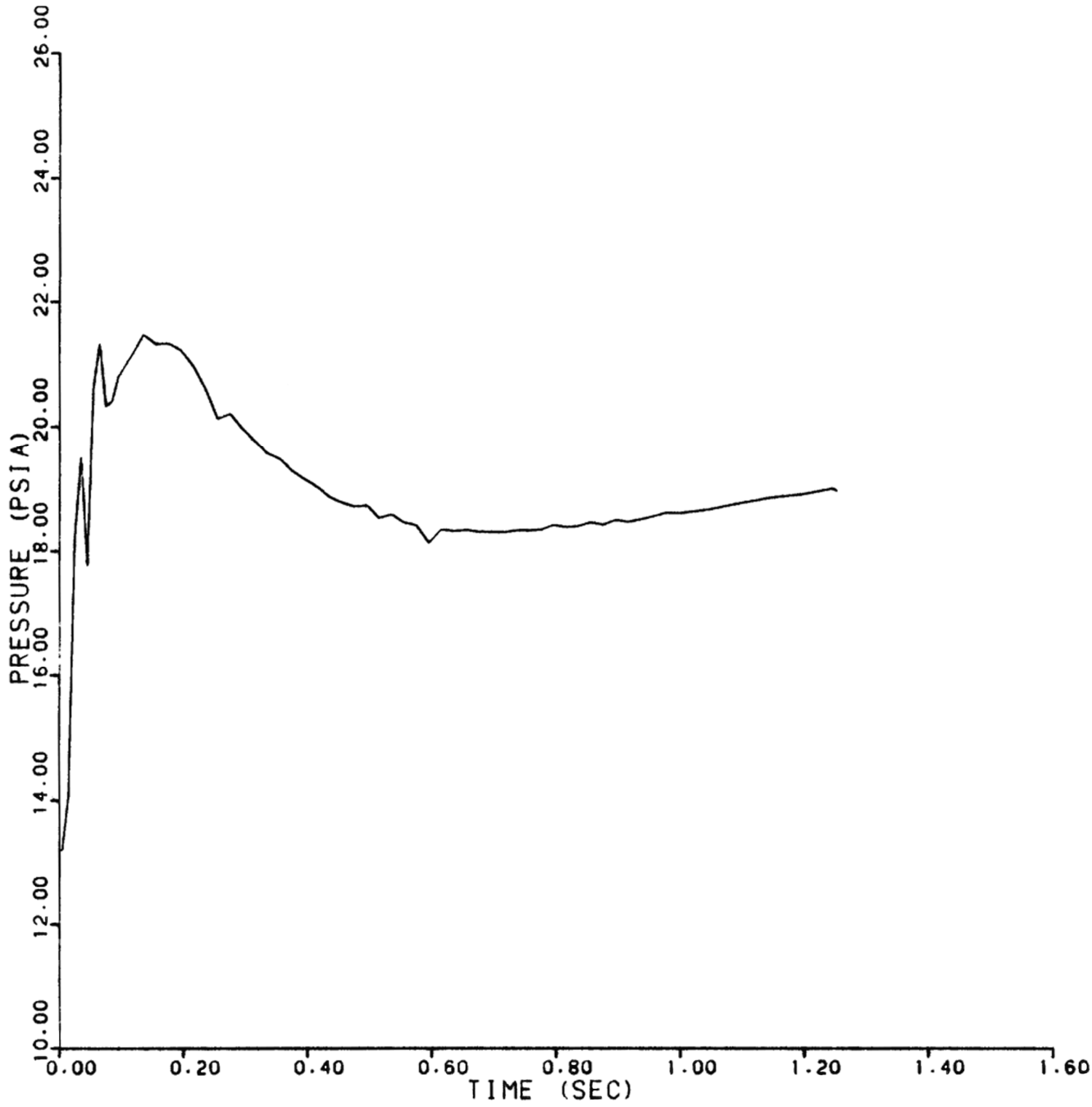
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E8
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 8 OF 74)



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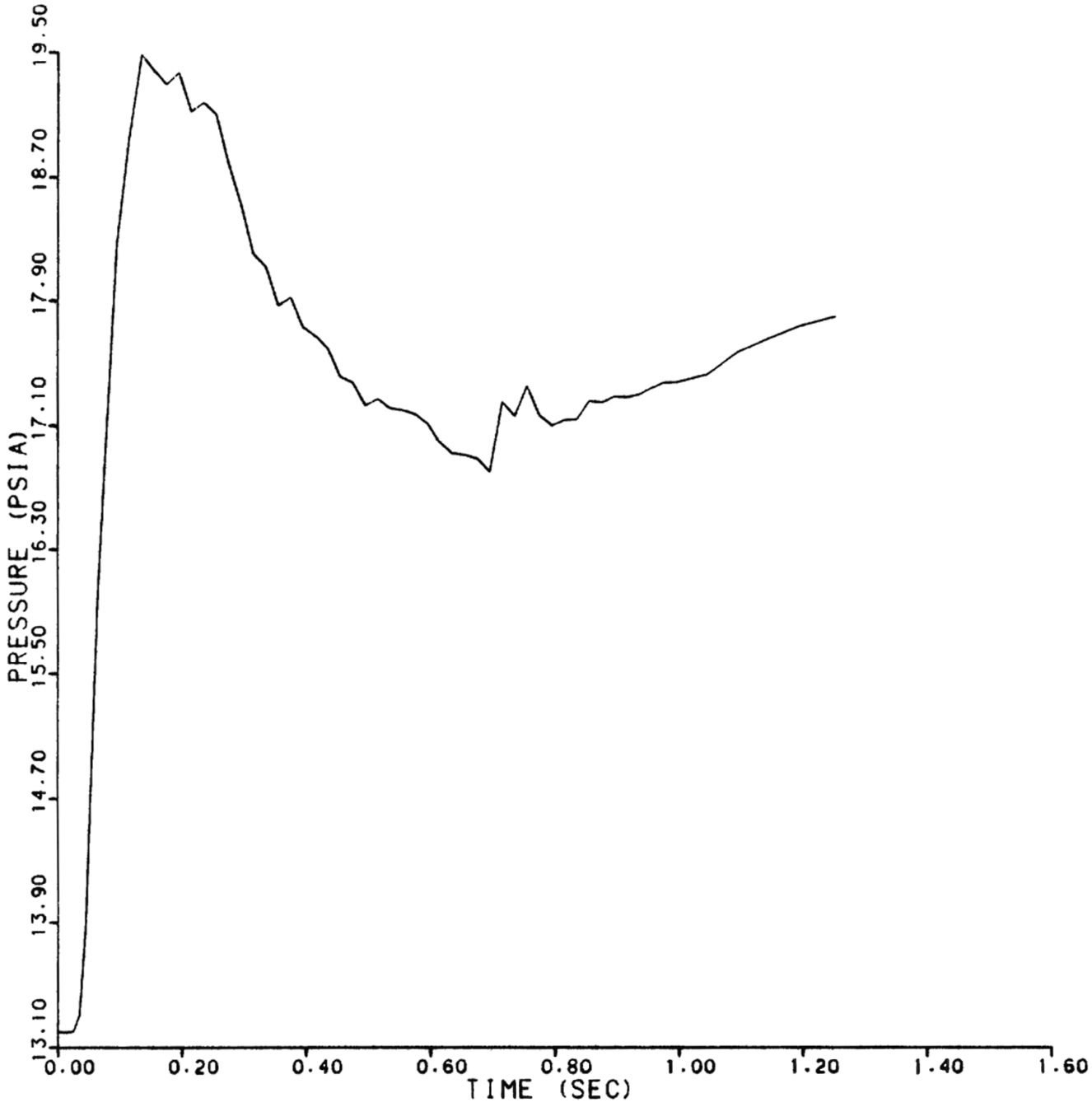
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E9
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 9 OF 74)



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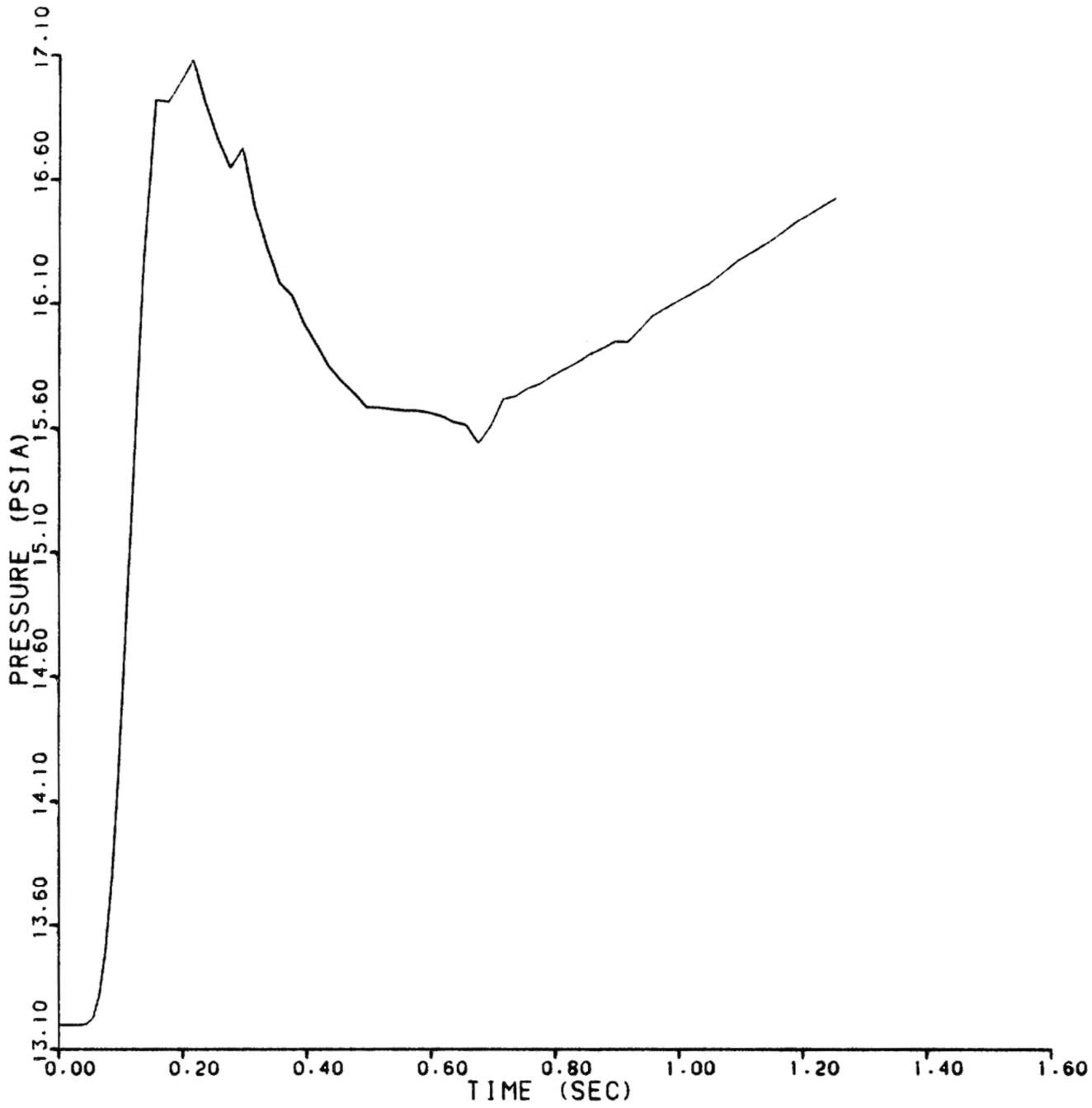
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E10
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 10 OF 74)



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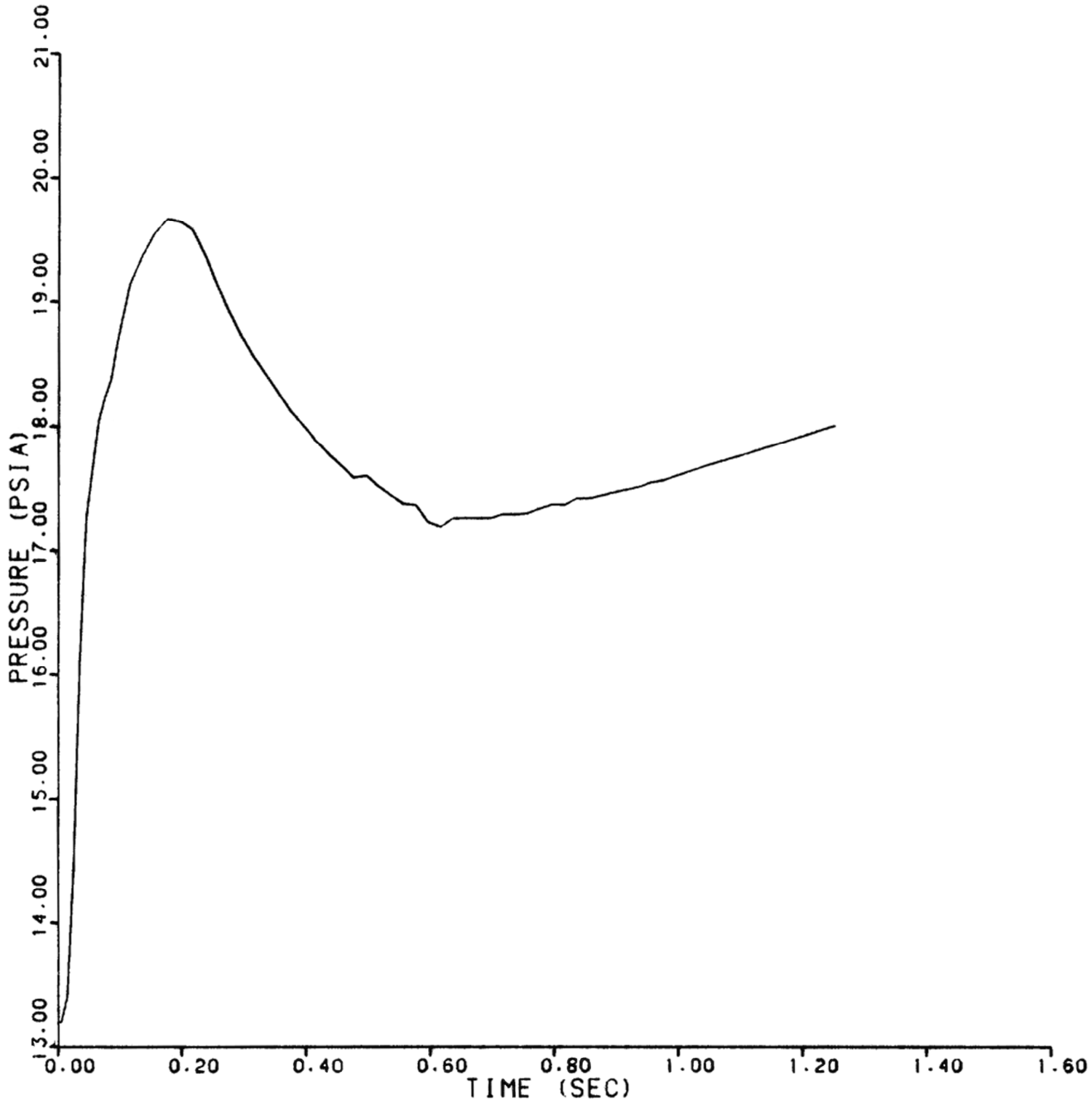
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E11
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 11 OF 74)



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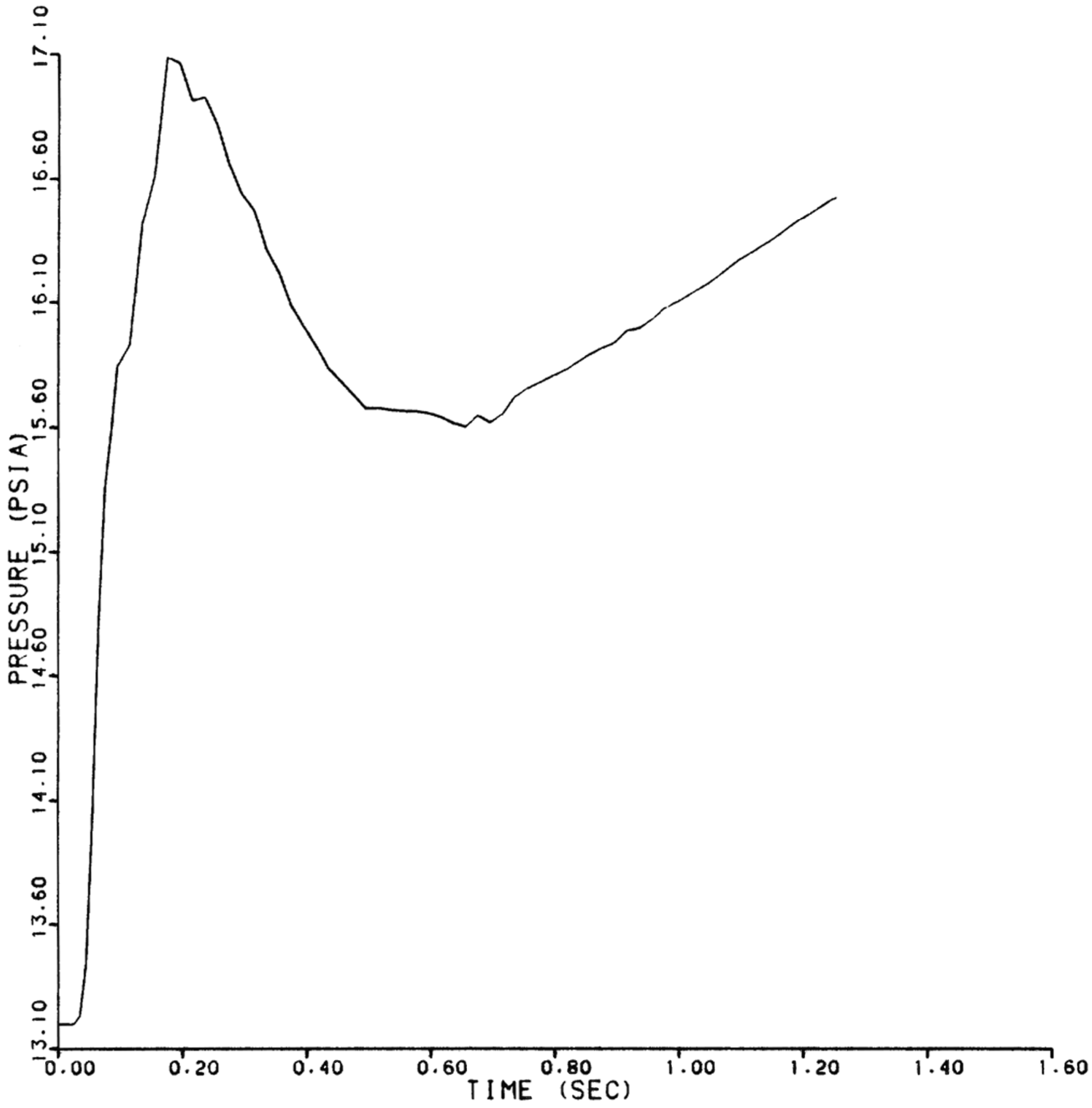
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E12
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 12 OF 74)



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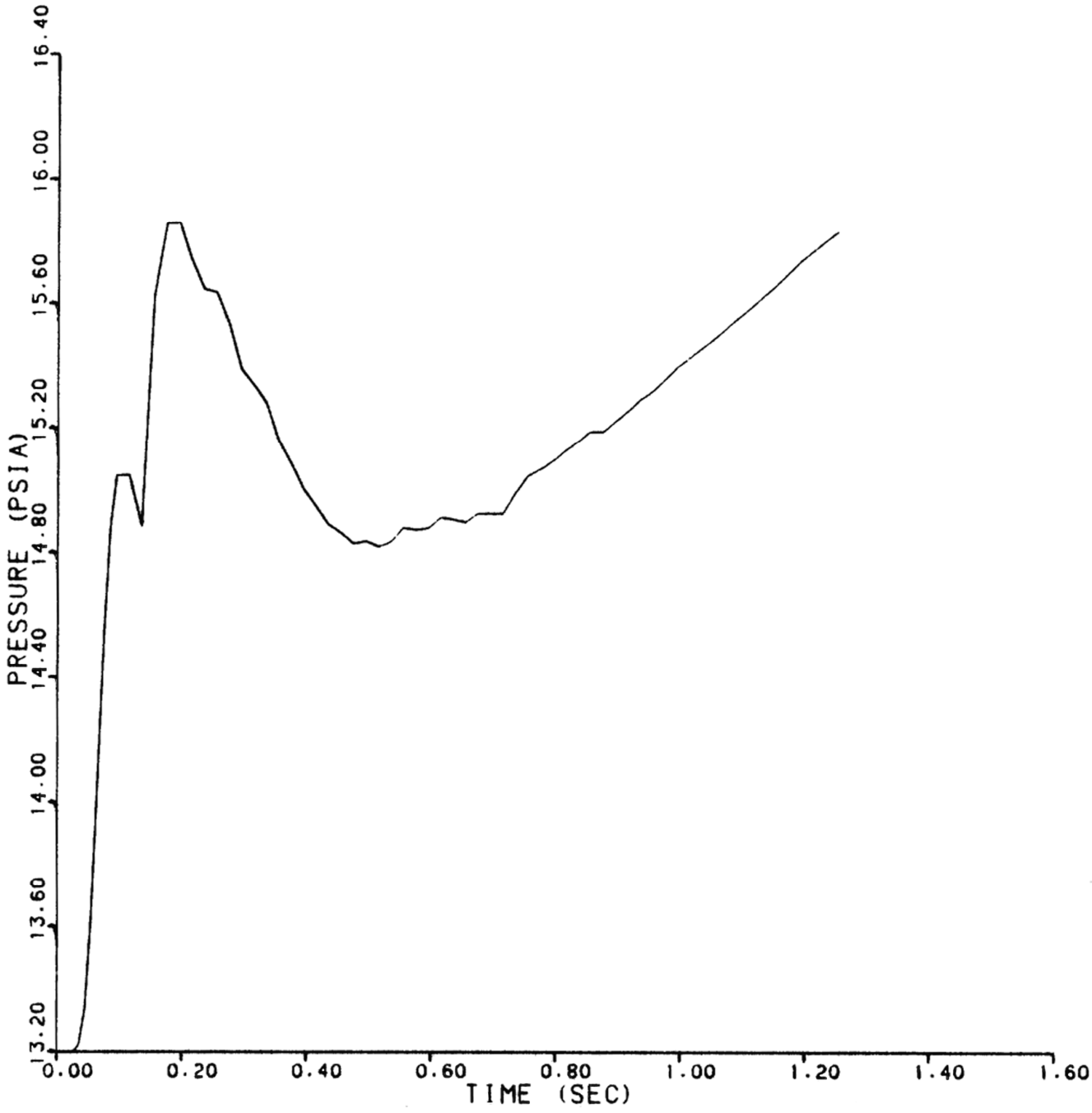
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E13
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 13 OF 74)



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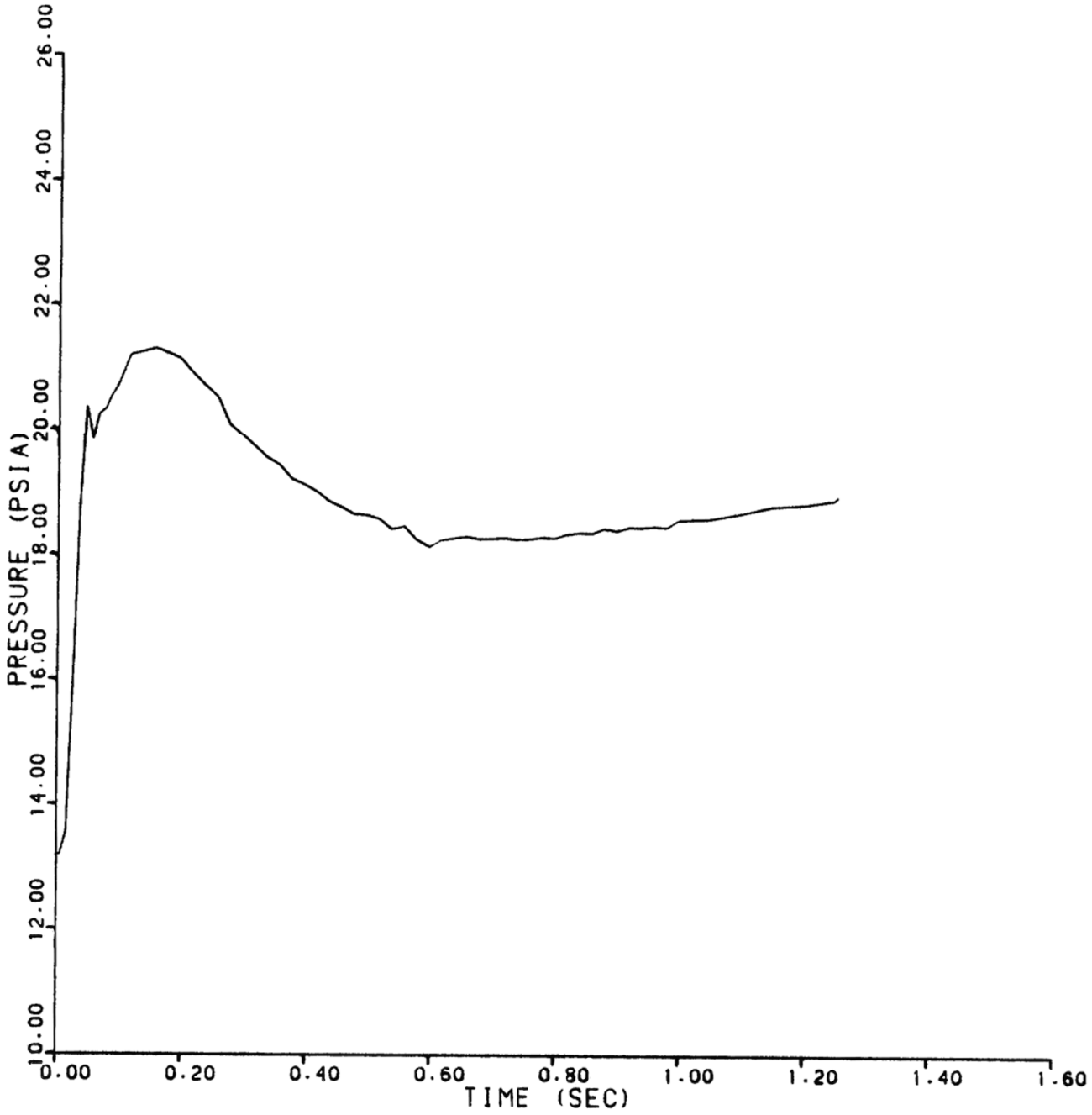
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E14
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 14 OF 74)



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SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E15 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 15 OF 74)

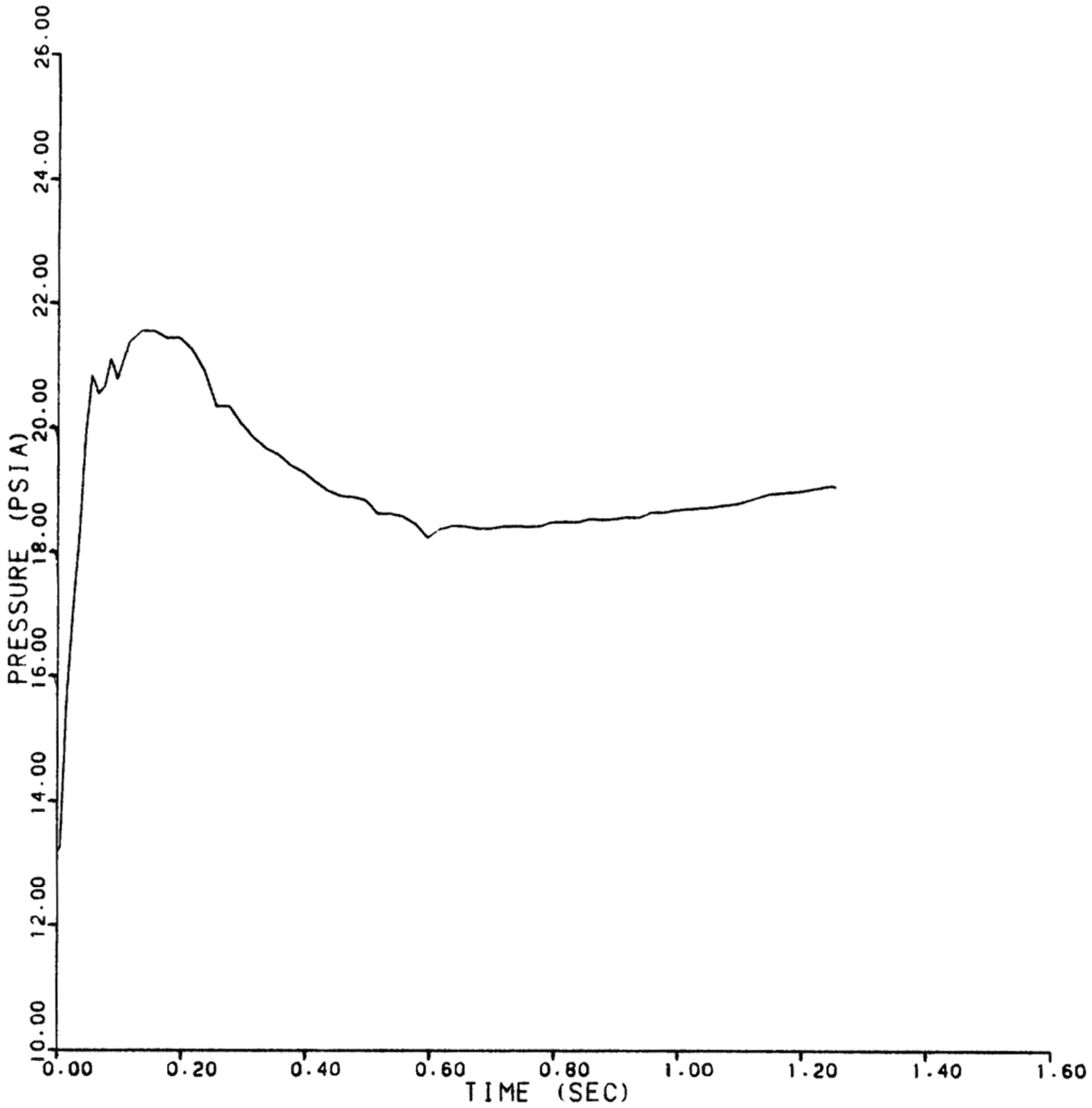

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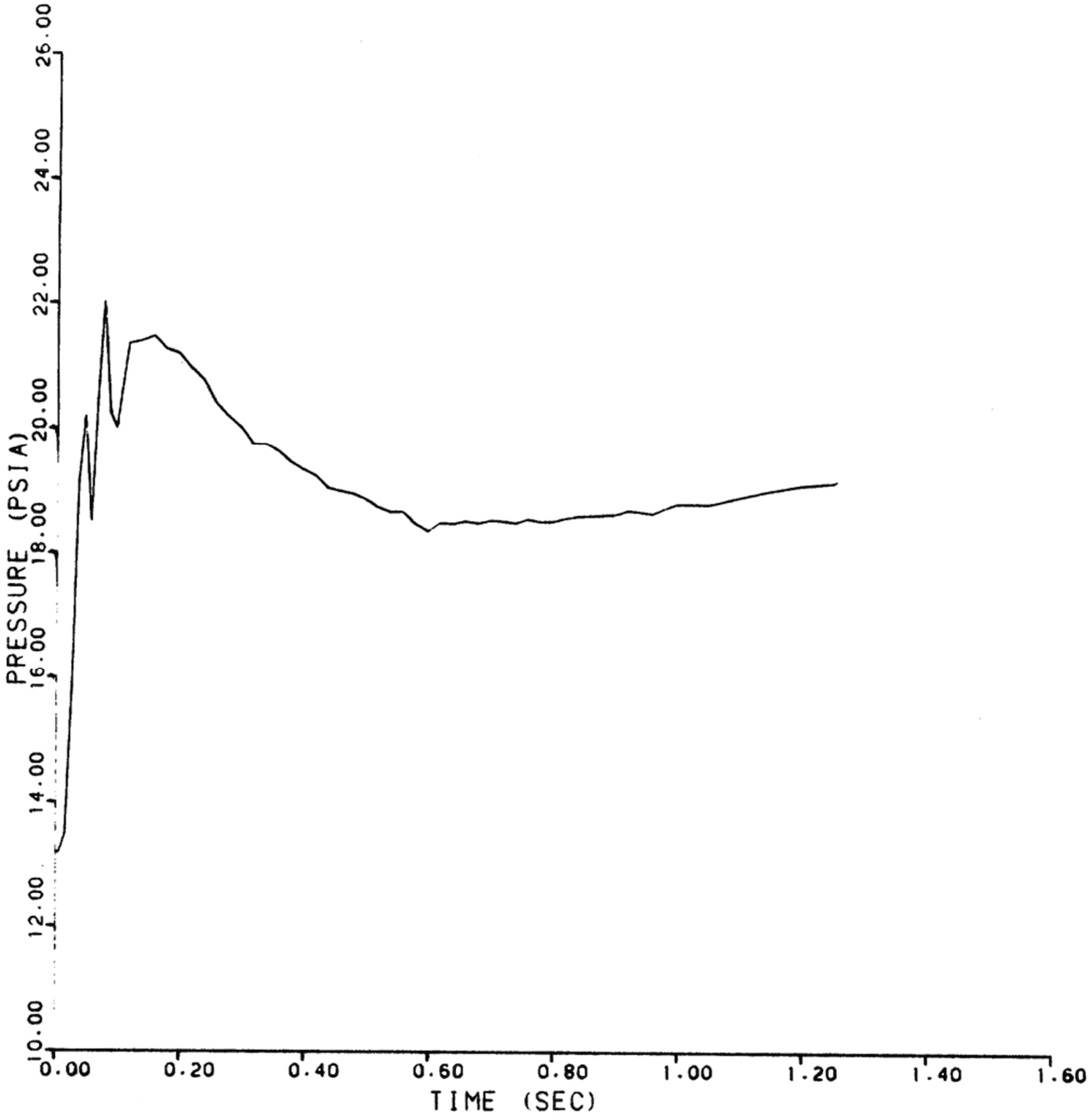
	FIGURE 6.2.1–19 (SHEET 16 OF 74)



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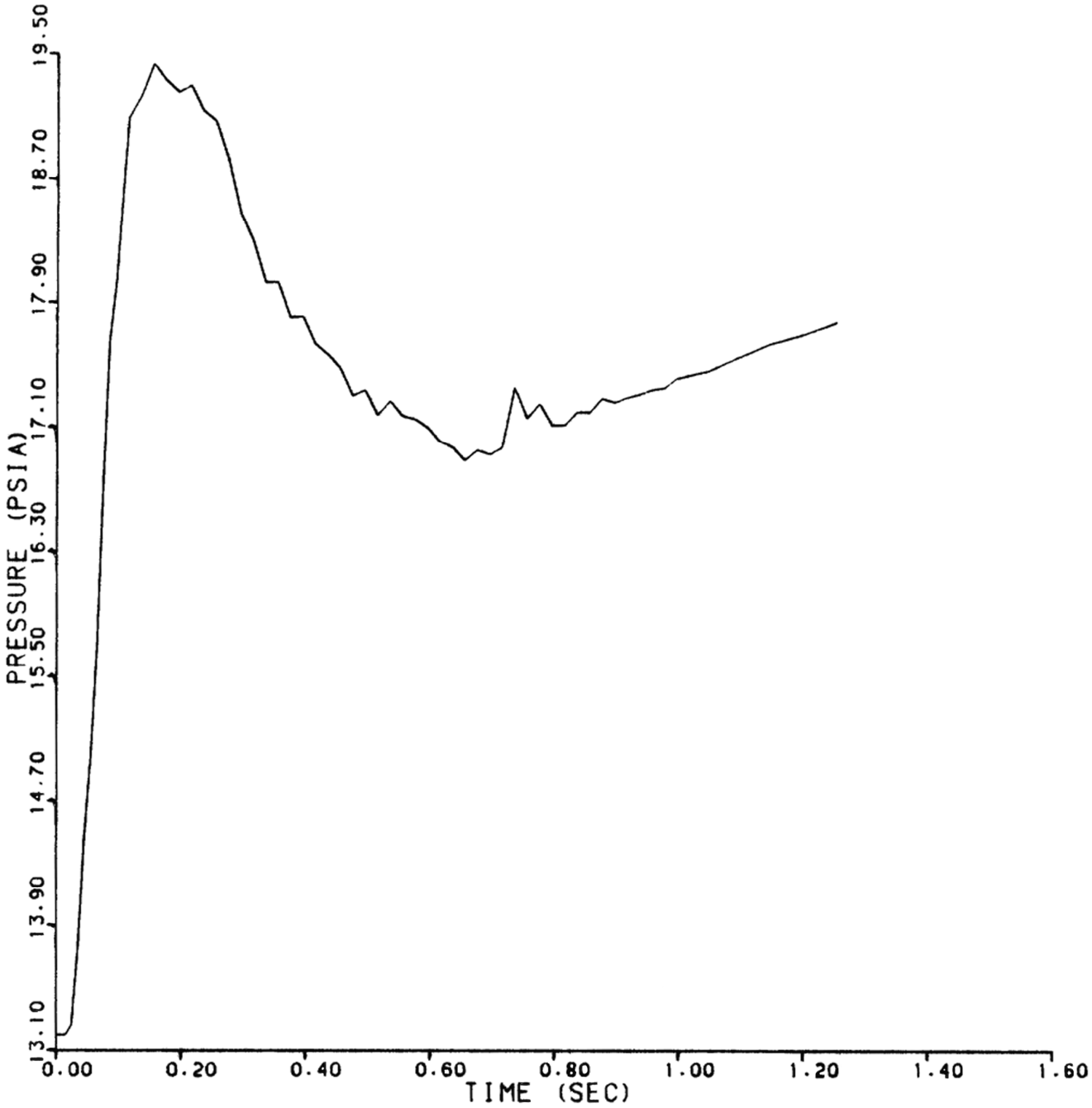
SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E17 (306–in.^{2} BREAK AREA)
		FIGURE 6.2.1–19 (SHEET 17 OF 74)



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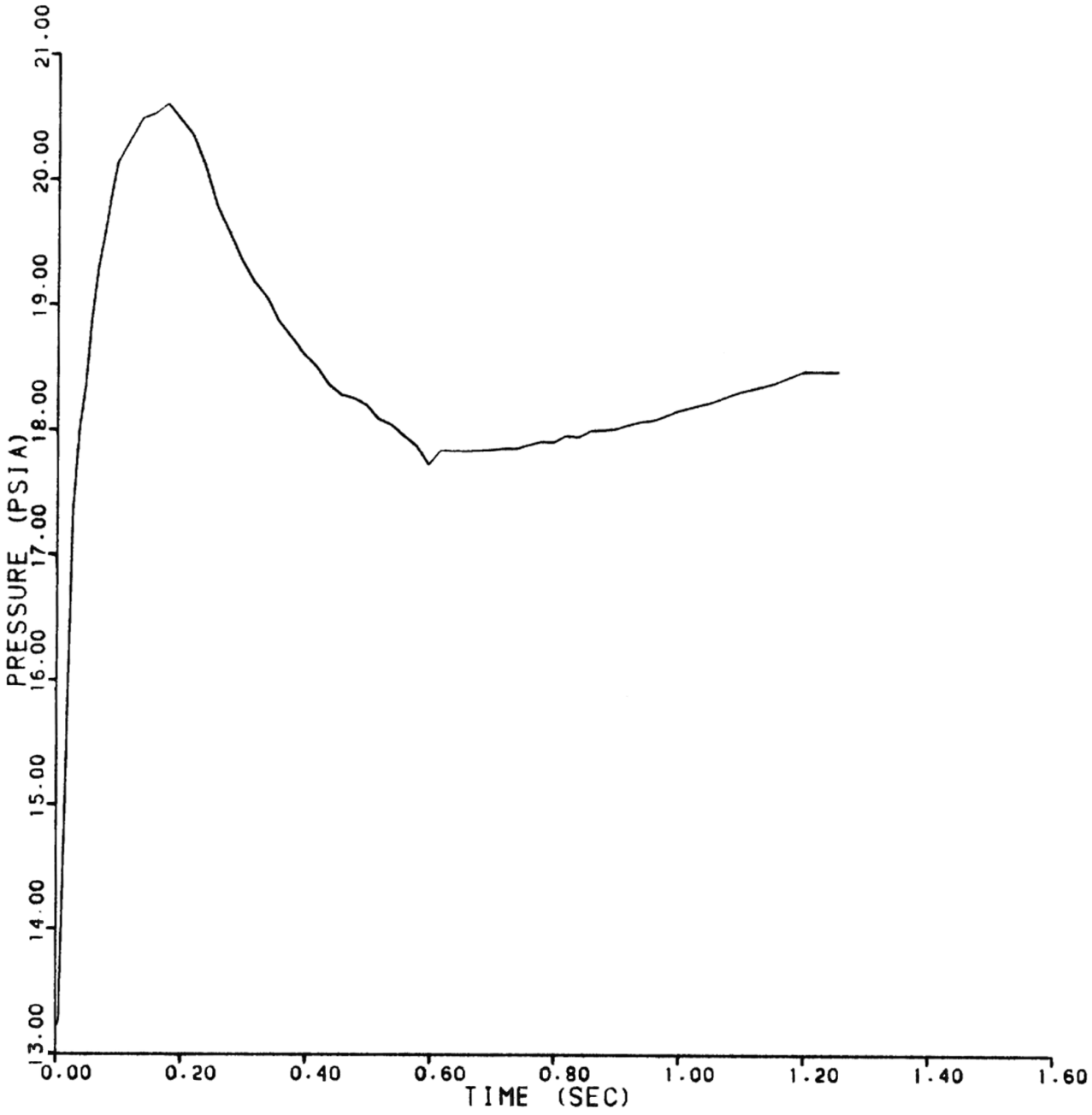
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E18 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 18 OF 74)



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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E19
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 19 OF 74)